Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8 and read off the Standard Model: status assessment and reading guide

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In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\rm car} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* ($v_{\rm geo}$, $G_{\rm net}$, $F_{\rm transfer}$), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\rm car} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order (with **four** companions):

- **0.** `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*
- **1.** `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.
- **2.** `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.
- **3.** `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
- **4.** `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.
- **5.** `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\rm geo}, G_{\rm net}, F_{\rm transfer}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation); **S.** `tfpt_safeguards` — the verification discipline: every mechanism that defends a claim against coincidence and numerology (status calculus, no-free-pattern, the over-determination map, the firewall, frozen predictions + null model, the independent Wolfram and Lean paths, the red team).

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\rm car} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
$v_{\rm geo}$	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
$G_{\rm net}$	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[C]
$F_{\rm transfer}$	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Gravity is parameter-free. The classical metric-sector field equation is no longer only an $R+R^2$ readout: the fixed-volume entanglement equilibrium (Jacobson; Faulkner et al.), run with TFPT's atoms, gives the *full* covariant Einstein equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ with *both* coefficients fixed — $c_3^{-1} = 8\pi$ (no free dimensionless Newton dial; the thermodynamic origin $2\pi/\eta$ *coincides* with the geometric one $|\mathbb{Z}_2| 2\pi\chi$ via $|\mu_4| = |\mathbb{Z}_2| \chi(S^2) = 4$, so c_3 is triply over-determined — anchor, geometry, thermodynamics, `v358`) and Λ from α ($\rho_\Lambda = (3/4\pi^2) e^{-2\alpha^{-1}}$, `v60`); the Einstein tensor (not Ricci) is forced by Lovelock, so matter conservation is an output (`v359`). The residual here is only the equation-of-state status and the one unit $v_{\rm geo}$; the global measure (C7 = `QG.AMB.01`) is now a [C] redundancy (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann.

The seam-net structural theorem $G_{\rm net}$ (= `SEAM.EQUIV.01`: the raw seam *is* the holomorphic $(E_8)_1$ net) is now *closed modulo a cited theorem.* The explicit lattice model (`v367`/`v368`) and the S_3 closure stack pin the target at every computable level — central charge $c = 8$ (`v376`), the $(E_8)_1$ character with 248 currents and one primary (`v377`), genus-one torus GSD = 1 (`v378`) and reflection positivity (`v379`) — and it is Lean-formalised as `FORM.SEAM.MMST.01`: the collar's MMST hypotheses are kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems enter as named cited axioms, and the `#print axioms` check is clean. The *only* residual that stays [O] is the abstract continuum existence of the scaling limit (`v336`) — its 128-spinor extension leg since certified at net level by the peer-reviewed crossed-product package (h_s=16/16=1∈$\mathbb{Z}$, the Longo–Rehren locality criterion; Böckenhauer; KLM μ=4/2^2=1; AGT/AMT the independent second witness) with the realisation input reduced to invariant level and the parallel Lean route `seamResidualClosed′` (`v469`); so this is *closed modulo a cited theorem*, not solved. `QGEO.SYM.01` (the μ_4 deck acting geometrically) is its *corollary*: a conformal net's vacuum is rotation-invariant by axiom, so the deck follows with no extra premise (`v335`, Lean `FORM.QGEO.BW.01`). The nonperturbative quantum-gravity measure `QG.AMB.01` is no longer carried as an open item: it is a [C] *redundancy* (`v369`), a certification object rather than missing dynamics, conditional on `SEAM.EQUIV.01` and Bisognano–Wichmann (and TFPT is moreover rigorously gap-decoupled from the general Euclidean-QG conformal-factor problem, margin $1.648 > 0$, `v76`/`v330`). So the honest residual is the one unit $v_{\rm geo}$ plus the typed $F_{\rm transfer}$ interfaces, above the cited-theorem ceiling on the seam and with `QG.AMB.01` a [C] redundancy. Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\rm wall}, G_{\rm metric}, F_{\rm frontier}$) are retained only for ledger continuity — the live residual is $v_{\rm geo} \oplus F_{\rm transfer}$, with the seam $G_{\rm net}$ closed modulo a cited theorem and `QG.AMB.01` a [C] redundancy.

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

[E] exact / machine-proven   [C] conditional   [O] open / axiom   [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1,1,2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

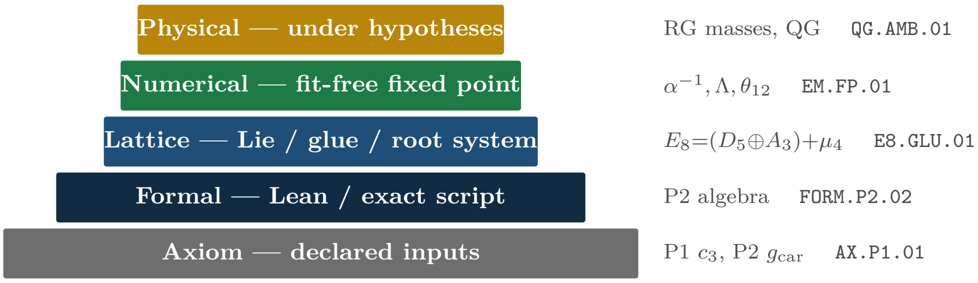
Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: **[E]** *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; **[C]** *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); **[O]** *open / axiom* — a declared input or a genuine gap; **[X]** *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.



The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1}{-}p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a\rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\rm car}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8{=}(D_5{\oplus}A_3){+}\mu_4$ — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);
4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for every claim's grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. `E8.GLU.01`, `EM.FP.01`) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers $(2, 3, 5, 16, \ldots)$ reappears in every sector: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less.* Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune.* Pushed to the end ([`verification/v353_selfloop_capstone.py`]), this means TFPT is best read not as a *linear* theory (axioms → theorems) but as a *closed self-consistent loop* whose outputs fix its own inputs ($g_{\rm car}$=5 is the largest prime of the output $h(E_8)$=30; the 8 in c_3 is rank E_8), with a *unique* fixed point (v6/v56) and *zero* free adjustable dimensionless parameters. Even π is not a free dial: it is the fixed geometric constant of the boundary 2-sphere ($\oint_{S^2} K = 4\pi$), the one transcendental the continuous boundary contains. “Which is the axiom?” is the wrong question for a loop — every part is fixed by the whole; the one remaining structural question, whether the loop also closes in the *continuum* (the keystone `SEAM.EQUIV.01`), is now *closed modulo a cited theorem* — the lattice model and S_3 stack pin it at every computable level and it is Lean-formalised (`FORM.SEAM.MMST.01`); only the cited continuum-existence theorems stay **[O]** (v367/v368, v376–v379, v336), their extension leg since re-founded on peer-reviewed crossed-product theorems with the realisation input reduced to invariant level (v469).

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\rm car}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n{+}1^n{+}2^n = 2{+}2^n$ generate the big Lie data directly: p_1=4=$|\mu_4|$, p_2=6=$|R^+(A_3)|$, p_3=10=$\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\rm car}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$.
[`verification/v23_anchor_generator.py`]

And a is itself half-forced (**[E]**/**[C]**): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\rm fam} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why $a = (1, 1, 2)$” but “why the carrier $g_{\rm car} = 5$ / $(|\mu_4|, N_{\rm fam}) = (4, 3)$” — that interface (with c_3) is the **[O]** input layer.

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1, 1, 2)$ plus π. The *single structural bedrock postulate* is the keystone `SEAM.EQUIV.01`, whose conformal-deck face `QGEO.SYM.01` (the carrier μ_4 clock is the seam’s conformal deck, see `tfpt_research_contracts`) forces the μ_4 deck, hence $N_{\rm fam} = 3$, $\deg E = -4$ and the tripartition $4 = 2{+}1{+}1$, of which c_3 and $g_{\rm car}$ are the two elementary readouts. “Two inputs” (the working axioms) and “one postulate” (the geometric bedrock) are the same statement at two layers. Its geometric normal form and the mark-local$\Rightarrow \omega \circ \rho = \omega$ implication are now Lean-formalised (`FORM.QGEO.02`/`FORM.QGEO.01`); the postulate itself stays **[O]**.

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \tfrac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\operatorname{rank} H_1(\mathbb{P}^1 \backslash \mu_4) = 3 = N_{\rm fam}}_{\text{cycles}\to A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\rm car}}_{\text{functions}\to D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself ([verification/v197_rr_carrier_clifford_d5.py]): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3, D_5, c_3 and E_8 ([verification/v189_riemann_roch_carrier.py]) — the arithmetic is exact **[E]**, the physical reading of the carrier as the seam mode space stays **[C]**.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

> The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([verification/v119_review_validation_2.py]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions **[E]**:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities **[E]**: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2\,\text{rank}\,E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\text{rank}\,E_8 = p_0 + e_2$.

$a = (1, 1, 2)$ *[E]*

elementary $e_k(a)$
$e_1 = 4 \to |\mu_4|$
$e_2 = 5 \to g_{\text{car}}$
$e_3 = 2 \to |\mathbb{Z}_2|$
$c_3 = \frac{1}{2e_1\pi} = \frac{1}{8\pi}$

power sums $p_n = 2 + 2^n$
$p_0, p_1, p_2, p_3 = 3, 4, 6, 10$
$p_1p_2p_3 = 240 = |R(E_8)|$
$p_4 - p_3 = 8 = \text{rank}\,E_8$
$\Rightarrow \dim E_8 = 248$

derived motifs
$4, 5, 8, 16, 25,$
$30, 41, 120, 240$
one anchor,
many projections

The companion documents

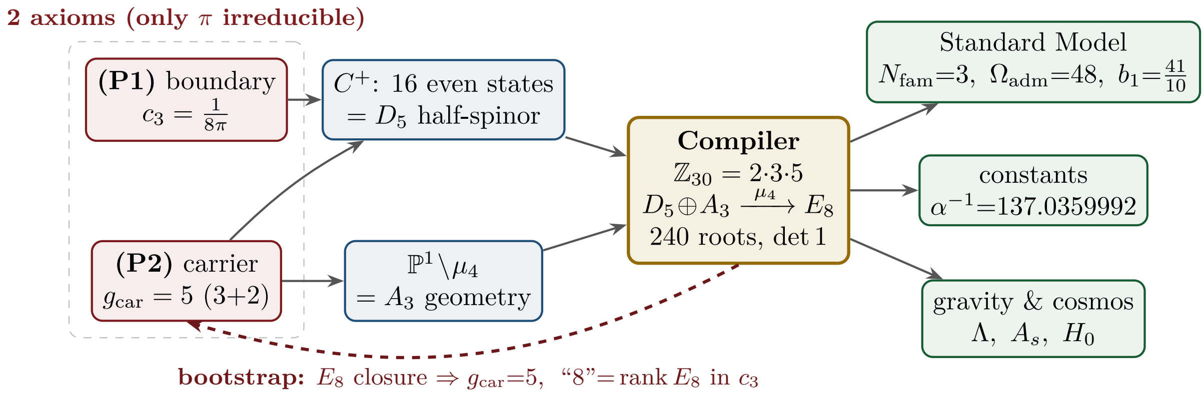
The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

Document	**What it contains (former notes now as parts)**
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one φ_0^{ret}-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	E_8 **audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade D=60−2n as the same even-integer spine (Part II); the Möbius bootstrap — g_{car}=5 and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_5_redteam`	**Red Team.** The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
`tfpt_research_contracts`	**Research contracts.** The open gates as numbered contracts with lemma chains and closing theorems: (U_{wall}) (now reduced to the selector triangle: $(d,n) \Rightarrow R$ columnwise, residue = *two* line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ($1920=\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert=4$); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
`origin_theory`	**Origin Theory.** Why no free *dimensionless compiler dial* remains beyond the anchor and π (the dimensionful v_{geo} and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; **[E]** core plus one honestly-typed **[C]** cyclic interpretation.

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

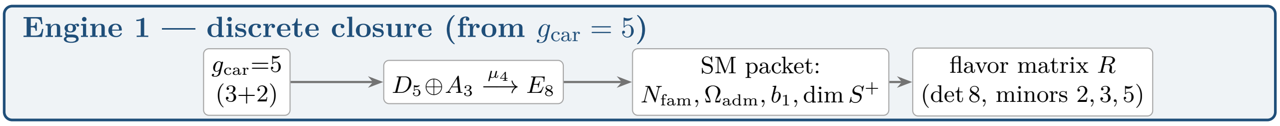
2 The architecture at a glance

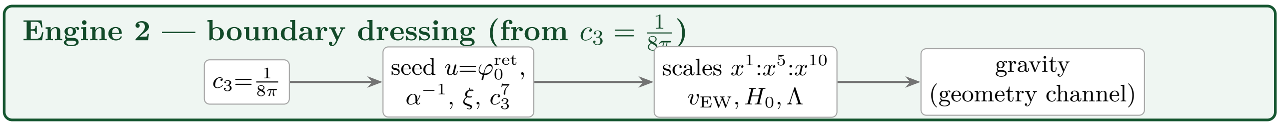
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0}+\binom{g}{1}+\binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not "prove itself"; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

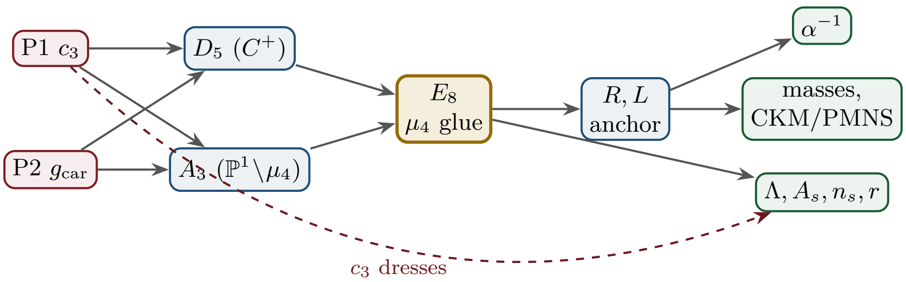
The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.





3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, c_3=1/(8π)	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\rm car}$=5	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+$=16, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\rm fam}$=3, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc =$\mathbb{Z}_4$, q-sum= 2	D_5, A_3	[E]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[E]	det 8, minors (2, 3, 5)	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)$=0	c_3, M=41	[E]	137.0359992	no/second root (excl.)
masses	$\varphi_0^{\rm ret}$-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam ε=c_3	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT v5.4 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R+R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification (now *closed modulo a cited theorem*, v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`), and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6\log\frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([E] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [C]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([E]/[C], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: the live residual as typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} \quad = \quad v_{\text{geo}} \quad \oplus \quad G_{\text{net}} \quad \oplus \quad F_{\text{transfer}}$$

— one dimensionful scale anchor, one metric-sector inclusion, and a set of deliberately typed QFT/cosmology interfaces (the historical gate labels $(U_{\text{wall}})/(G_{\text{metric}})/(F_{\text{frontier}})$ are kept only for ledger continuity). In current terms: $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [E]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers; the four are one typed functor with a four-axiom contract (`CONTRACT.F.01`, v213) and *one shape* — each is a gapped relaxation to a unique attractor (the main-branch discrete→dynamic motif), with F__pole at the *seam* rate $(2/3)^6$ and the other three external (v303). The metric-sector reduction lands on a *single named keystone*. The boundary inclusion $G_{\rm net}$ is the index-4 μ_4 simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (Simple-Current Extension Theorem, v154, exact: index 4, $c{=}8$, $\mu{=}1 \Rightarrow$ holomorphic $\Rightarrow E_8$); net existence and full-cone reflection positivity are discharged to **[E]** (v175); and the **No-Unit Theorem** (v153) makes $v_{\rm geo}$ an irreducible metrology primitive ($U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ — one unit in three dimensions). The flavor selector and Q-geometry residues are *derived* (no discrete freedom left). The *one* seam-side identification, packaged as the single named theorem `SEAM.EQUIV.01` — *the raw reflection-positive seam state is the holomorphic* $(E_8)_1$ *net at* $\tau{=}i$ — is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01`, with only the cited MMST/OS continuum-existence theorems left **[O]** (v336); its conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face (per v323; the residual is Lean-pinned to named cited steps, v308). The sprint-by-sprint reduction that produced it is recorded in the changelog and in `tfpt_research_contracts`, not replayed here. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[E]	gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web $g_{\rm car}$=5 forced (v6/v14)
Newton constant G_N	[E]/[C]	not *independent*: branch pins $\bar{M}_{\rm Pl}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\rm car}\|{\rm Pl}_{\mathbf{1},a}(K)\|_1/(N_{\rm fam}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
flavor absolute scale	[E]/[O]	ratios *closed* (v49/v71); only the absolute scale remains $= v_{\rm geo}$ (Gate 1, v75), the same anchor as $1/G$
$R + R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
Einstein eq. (covariant, local)	[E]/[C]	fixed-volume entanglement equilibrium ⇒ *full* $G_{ab}{+}\Lambda g_{ab}{=}c_3^{-1}T_{ab}$ with BOTH coefficients parameter-free: $8\pi{=}1/c_3$ (v358, thermo=geo via $\|\mu_4\|{=}\|\mathbb{Z}_2\|\chi{=}4$) and Λ from α (v60); Einstein tensor via Lovelock, conservation an output (v359); residual = equation-of-state status + $v_{\rm geo}$, global measure (C7) now a [C] redundancy (v369)
physical QG sector	[E]/[C]	gap-decoupled admissible measure ($\Delta_{\rm eff}$=1.648>0)
ambient QG measure	[C] (redundancy)	**certification object, not missing dynamics** (v369; conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann): holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the A_2 net-existence + the keystone `SEAM.EQUIV.01`, closed modulo a cited theorem — zero new gates (v160–v165)
$v_{\rm geo}$ metrology unit	[O](primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\rm src} + \varphi_0/24$ source→pole: typed runnable solver with kill test (QED-excluded, v371)
axion DM	[C]	$f_a = M_{\rm scal}/128$: finite-T spine-branch typed runnable solver (v373)
η_B	[C]	readout closed; integrated BDP leptogenesis Boltzmann ODE, typed runnable solver (v372)
m_p/m_e	[C]	cross-sector QCD/EW ratio: typed runnable solver with uncertainty budget (v374)
$N_\star$	[C]	reheating input; scalaron-reheating point $N_\star$=51.4 (v86); frozen band $[50, 60]$
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection g=5 (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank = c); residue = the $G_{\rm net}$ premise itself — gate closes ⇒ carrier [E]
selector establishment R4′	[E]/[C]	$(d, n) \Rightarrow R$ columnwise (v136); both normals *anchor data* — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = *one* assignment
sheet/parity (Q-geometry derived)	[E]/[C]	H^1 characters = ${\rm Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice *derived* (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification, folded into `SEAM.EQUIV.01`
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born^2 Gaussian zero-mode integral, bend det′-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`*.*

The former structural open core, the *single* named keystone `SEAM.EQUIV.01` (the $G_{\rm net}$ inclusion: the raw RP seam $= (E_8)_1$ at $\tau{=}i$), is now *closed modulo a cited theorem*: the explicit lattice model (v367/v368) and the S_3 stack (c=8, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, it is Lean-formalised as `FORM.SEAM.MMST.01`, and only the cited MMST/OS continuum-existence theorems stay [O] (v336); its conformal-deck face `QGEO.SYM.01` is a *corollary* (a conformal net's vacuum is rotation-invariant by axiom, v323/ [`verification/v335_seam_equiv_unify.py`]), not a second item. $v_{\rm geo}$ is an irreducible metrology unit and the frontier items are typed interfaces, not structural holes. $G_{\rm net}$ is heat-kernel grounded (G2: the spectral action gives $R{+}R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\rm eff}{=}1.648{>}0$ isolates the closed admissible sector from the ambient), so the ambient measure `QG.AMB.01` is no longer carried as an open structural item: it is a [C] *redundancy* (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), a certification object rather than missing dynamics — it does not affect any TFPT readout. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority on the $v_{\rm geo}$ scale anchor and the $G_{\rm net}$/`SEAM.EQUIV.01` inclusion theorem. [`verification/v36_spectral_action_g2.py`]

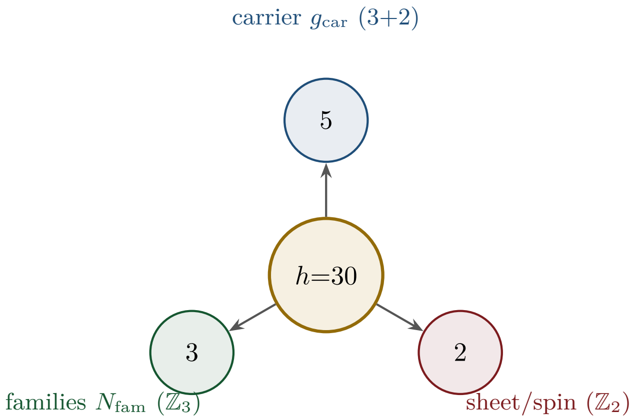
Sharper still: the structural residual is *one* condition. $G_{\rm metric}$, the carrier P2 and the red-team Target A are not separate items: they are three provably-equivalent faces of the single condition *"the seam carries no nontrivial abelian sector"* — holomorphy (μ-index 1), a homology-sphere seam link (Γ perfect $\Leftrightarrow 2I$, [`verification/v232_e8_kleinian_seam.py`]), exactly one 1-dim irrep ([`verification/v219_icosahedral_mckay.py`]) — all of which force E_8 ($\#$(mark-1) $= |H_1| = 1$ only for E_8; holomorphic $c = 8 = g_{\rm car} + N_{\rm fam}$ gives the unique even unimodular rank-8 lattice). In abelian Chern–Simons language this is the single integer step holomorphic $\Leftrightarrow \det K = 1$, the extension tower $D_5{\oplus}A_3(16) \to D_8(4) \to E_8(1)$ being anyon condensation (the Kitaev E_8 state); the one analytic step "the free RP seam condenses the order-$|\mu_4|$ Lagrangian glue" = the single named keystone `SEAM.EQUIV.01` (with `QGEO.SYM.01` its downstream conformal-deck face) is now *closed modulo a cited theorem* (v367/v368, v376–v379, Lean `FORM.SEAM.MMST.01`; cited continuum existence [O], v336). So beneath the gate *labels* there is, structurally, a single fundamental postulate. And the entire *emergent-QFT layer* collapses onto the *same* postulate (the *Modular Spectral Closure*, v261): the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight ($f_2/f_0 = 1$, v259), the seam, the carrier-16 and E_8 live on one Kummer/K3 surface (v260), so Dirac, cutoff, gauging, glue and orientability are all readouts of the one seam state — the boundary QFT is closed as a single relative object modulo the keystone `SEAM.EQUIV.01` and adds *no new open item*, while the ambient quantum-gravity measure is a [C] redundancy (v369), not a separate structural gap. [`verification/v234_seam_holomorphy_selection.py`] [`verification/v235_seam_chern_simons.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\rm fam}, \Omega_{\rm adm}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence*. *And even those two tighten:* the bootstrap note shows $g_{\rm car} = 5$ and the integer 8 in c_3 are *overdetermined* E_8*-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. *A fourth, deeper forcing* comes from the McKay correspondence: E_8 is the exceptional top of the tower of finite $SU(2)$ subgroups, so its branch data is the icosahedron and the three atoms $(2, 3, 5)$ are its rotation-axis orders — the binary icosahedral group $2I$ (order $120 = |R^+(E_8)|$) has irrep degrees equal to the affine-E_8 Kac marks, $\sum = 30 = h(E_8) = 2{\cdot}3{\cdot}5$ ([`verification/v219_icosahedral_mckay.py`]). So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [`verification/v6_bootstrap.py`] [`verification/v14_carrier_uniqueness.py`] [`verification/v23_anchor_generator.py`] [`verification/v219_icosahedral_mckay.py`]

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:



From this the compiler reads, among others:

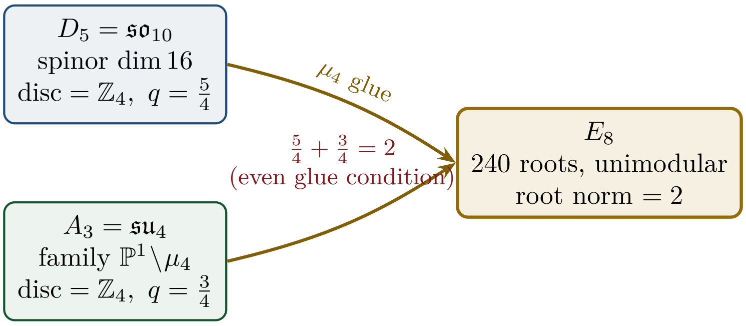
- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [E]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [E]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \backslash \mu_4$ is the A_3 Coxeter element. [E]
- **why these three primes:** E_8 is the icosahedral top of the McKay tower ($2I$, order 120, irrep degrees = affine-E_8 marks summing to 30), so $2, 3, 5$ are the icosahedron's rotation orders; and the E_8 exponents $(\mathbb{Z}/30)^\times$ carry an order-4 totative character $\langle 7 \rangle$ — the same μ_4 seam clock. [E] [`verification/v219_icosahedral_mckay.py`] [`verification/v223_coxeter_totative_clock.py`]
- **a network reading (computational substrate).** On the same McKay $(2, 3, 5)$ graph the affine-E_8 marks are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4 \cdot m$ (it converges from *any* positive start to $(1, 2, 3, 4, 5, 6, 4, 2, 3)$); the spherical tower terminates at E_8 and the cascade $E_8 \supset E_7 \supset E_6 \ldots$ is nested-subgraph coarse-graining. So E_8 reads as the *universal attractor of a minimal* $(2, 3, 5)$ *network* (a Wolfram-style computational-substrate picture) at the Coxeter-skeleton level. [C] A minimal hypergraph *rewrite* reproducing the *full* structure (lattice + recovery + readouts) stays open, and φ_0^{ret} is the analytic seed $\frac{4}{3} c_3 = 1/(6\pi)$, not an edge fraction. [O] [`verification/v298_e8_network_attractor.py`]
- **the bridge, hardened.** A local witness-based growth dynamics (a *diffusion-generated* witness + the local balance gate $Am \leq 2m$, Collatz–Wielandt = Smith's $\rho \leq 2$; the Perron eigenvector enters only as an audit) *reaches* E_8 as the last finite point ($E_6 \to E_7 \to E_8$, then stops at the affine boundary $\hat{E}_8$) — but *only* from a 3-arm $(2, 3, \cdot)$ seed (the controls $\rho(A_n) = 2 \cos \frac{\pi}{n+1}$ and $\rho(D_n) = 2 \cos \frac{\pi}{2n-2}$ are < 2 for *all* n, so neither family ever reaches E_8). So P_2 is exactly the irreducible seed on which the rule reaches E_8 as the last finite point — a second network reading of the $(2, 3, 5)$ hull, not a derivation of P_2. [O] [`verification/v299_hypergraph_0d_autonomous.py`]
- **what a rewrite must inject.** Testing *which* load-bearing quantities are graph-spectral sharpens the open question: [E] the Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio (the 5-fold/icosahedral signature), but [E] the recovery rate $(2/3)^6$ is *not* any eigenvalue of the adjacency and φ_0^{ret} is not a rational edge fraction. The cusp weight $2/3{=}|\mathbb{Z}_2|/N_{\mathrm{fam}}$ is now *derived* from a minimal branching rule ([`verification/v327_hypergraph_rewrite.py`]); what

remains non-graph-spectral for a *full*-structure rewrite at leading order is sharpened by [`verification/v396_phi0_icosahedral.py`] (tree-level $\varphi_0^{\rm tree} = F/(4h\pi)$); the puncture $48c_3^4$ alone stays open. **[O]** [`verification/v312_hypergraph_substrate.py`]

- **one coupled local rule (carrier × family).** On a 9×3 labelled grid (affine-E_8 nodes × family channels) a *single* coupled micro-step — network lazy diffusion on each cusp column plus the [`verification/v327_hypergraph_rewrite.py`] branching rule M on each node's family vector — has joint attractor marks$\otimes(w{=}0)$ and recovery rate $(2/3)^6$ after one clock hand ($2N_{\rm fam}$ family steps); the [`verification/v324_hypergraph_fiber.py`] fiber product $T_{\rm net}\otimes T_{\rm cusp}$ emerges as the cusp-readout basis, and the [`verification/v299_hypergraph_0d_autonomous.py`] growth trajectory $E_6 \to E_7 \to E_8 \to \hat{E}_8$ is unchanged with the fiber attached. **[E]** [`verification/v395_hypergraph_coupled.py`]
- **the seed at leading order.** The tree-level retained seed $\varphi_0^{\rm tree} = 1/(6\pi)$ is exactly $F/(4h\pi)$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\,\mathrm{Aut}\,| = 120$): equivalently $(F/(g_{\rm car}N_{\rm fam}))c_3$, with $F/h = |\mathbb{Z}_2|/N_{\rm fam} = 2/3$ the same survival ratio as [`verification/v327_hypergraph_rewrite.py`]; Gauss–Bonnet consistent with $c_3 = 1/(|\mathbb{Z}_2|4\pi)$. The puncture $48c_3^4$ remains analytic, not a graph fraction. **[C]** [`verification/v396_phi0_icosahedral.py`]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". **[E]** [`verification/v1_e8_glue.py`]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\rm car}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^{+}$=16, N_{fam}=3, Ω_{adm}=48	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s{-}b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \dfrac{g_{\mathrm{car}}\, 2^{g_{\mathrm{car}}-2} + 1}{10} = \dfrac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\mathrm{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \dfrac{N_\star^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4$-glue	[E]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by $5 =$ carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

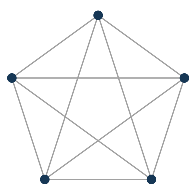
Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\mathrm{geo}}/(5\beta_{\mathrm{rad}}^2 \bar{M}_{\mathrm{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\frac{H_0}{\bar{M}_{\mathrm{Pl}}} = x^5 R_H, \qquad \frac{\rho_\Lambda}{\bar{M}_{\mathrm{Pl}}^4} = x^{10} R_\Lambda, \qquad (1,5,10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. "Hubble = slot product, Λ = pair product." [E]



$\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
$\binom{5}{2} = 10$ edges — Λ density pair x^{10}
$\binom{5}{3} = 10$ triangles — dual sector
$\binom{5}{4} = 5$ complements — inverse sector
$\binom{5}{5} = 1$ determinant — closure

K_5

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\rm geo}}{\sqrt{2}}\,\lambda_Y^{L_{f,j}}\,\Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\rm ret}(1-\varphi_0^{\rm ret})}, \qquad \varphi_0^{\rm ret} = \tfrac{1}{6\pi} + \tfrac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers (tr $= 9 = N_{\rm fam}^2$, det $= 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the $\varphi_0^{\rm ret}$-ladder $\hat{m} = \frac{v_{\rm geo}\pi}{\sqrt{2}} c\,(\varphi_0^{\rm ret})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the single dimensionful anchor $v_{\rm geo}$. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$; see below). *(Full derivation: document 2, `tfpt_2_standard_model`.)* [`verification/v18_quark_yukawa.py`] [`verification/v20_lepton_c_derivation.py`] [`verification/v46_grand_mass_volume.py`]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1\backslash\mu_4$ produce D_5 and A_3, glued to E_8; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30}/Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions ⇒ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms 5/4 and 3/4 (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\rm fam} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2\cdot 27\cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, v69/v70; since v136/v139 the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\mathrm{Spec}(Q_+) = (1, 2, 3)$ is the A_3-exponent grading on the seam curve's cohomology, v137), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [E]/[O]
2. **Gravity — the local field equation is parameter-free.** The metric-sector *field equation* is supplied by the entanglement first law $\delta S = \delta\langle K\rangle$ at fixed *volume* (Jacobson 2015): equilibrium forces the *full* covariant Einstein equation $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with *both* coefficients fixed — $8\pi{=}1/c_3$ (v358) and Λ from α, $\rho_\Lambda{=}(3/4\pi^2)e^{-2\alpha^{-1}}$ (v60) — the Einstein tensor (not Ricci) forced by Lovelock, so $\nabla^a T_{ab}{=}0$ is an *output* (v359). The $R{+}R^2$ scalaron is the *action*/inflation layer (the next-order term of the same equilibrium route, v360), and v_{geo} is the theory's single dimensionful input (v364). What remains is *not* the field equation but the equation-of-state status (Jacobson framing, not a from-action quantisation) and the one unit v_{geo}; the ambient measure `QG.AMB.01` is now a [C] redundancy (v369, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann), not missing dynamics, and *perturbative* graviton unitarity is established — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304), the spin-2 sector ghost-free via the entire KMS form factor (v370) with the truncation tower decoupling (v380). [E] (local field equation) / [C] (perturbative unitarity, ambient redundancy)
3. **Strong CP — decoupled.** $\theta_{\text{eff}}{=}0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [E] (given RP)
4. **Full dynamics — the admissible transfer model is closed, and the boundary QFT is now one relative object.** The mass gap is the *explicit* discrete gap $6\log\frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector. On top of this the whole emergent-QFT round assembles the boundary theory into one relative object $\mathsf{TFPT}_{\text{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\text{rel}})$ — the *Modular Spectral Closure*: the finite Dirac is a covariance induction of the seam KMS state (v258), the spectral-action cutoff *is* that KMS weight (v259), the seam/carrier/E_8 live on one Kummer/K3 (v260), all certified together (v261) — closed modulo the single keystone `SEAM.EQUIV.01`, itself now *closed modulo a cited theorem* (lattice + S_3 stack, Lean `FORM.SEAM.MMST.01`; v367/v368, v376–v379). *What this does* not *fully close:* the full measured pole masses (the absolute scale v_{geo} + standard RG), the absolute QCD scales (m_p/m_e, now a typed [C] transfer, v374), the complete PMNS dynamics (the angles are typed; the full mass/CP dynamics stay [C]/[O]), and the real 4D interacting QFT — the continuum limit, whose cited existence theorems stay [O] — with the ambient measure a [C] redundancy (v369). [E] (transfer model + boundary relative object) / [C] (continuum modulo cited, transfers, ambient redundancy)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`3357 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [E]/[O]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2,* `tfpt_2_standard_model`*, Part III.)*

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam}, hypercharge, b_1	[E]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[E]	— complete
bootstrap $(g, \mu) = (5, 4)$, "8" in c_3	[E]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[E]	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [C]
fermion masses (φ_0^{ret}-ladder)	[E]/[O]	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$ (PMNS angles)	[E]/[C]	θ_{12} cond. derived; θ_{23} maximal; *full* PMNS dynamics (mass scale, CP magnitude) stay [C]/[O]
flavour matrix / H2	[E]	dictionary *exact* (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d, n) (v136/v139); pairing values = atom identities (v142/v145); residue = *one* lift map
seam quantum clock (R1)	[E]/[C]	typed closed: reduced edge-sector clock (v124–v133); external VW edge = 2× reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r, scalaron M	[C]	scalaron $M = c_3^{7/2} \bar{M}_{\mathrm{Pl}}$ exact; n_s, r are bands over $N_\star \in [50, 60]$ (reheating point 51.4, v86)
boundary QFT (Modular Spectral Closure)	[E]/[C]	**one relative object** $\mathsf{TFPT}_{\mathrm{QFT}}$: D_F = covariance induction (v258), cutoff = KMS weight $f_2/f_0 = 1$ (v259), seam/carrier/E_8 on one K3 (v260), certified (v261) — closed modulo `SEAM.EQUIV.01`, itself closed modulo a cited theorem (v376–v379)
perturbative 4D S-matrix (S_{pert})	[E]/[C]	**Epstein–Glaser-constructible** (v269): one-loop quartic (v271) + gauge $\beta = (41/10, -19/6, -7)$ (v273), $S_{\mathrm{pert}} \xrightarrow{\mathrm{LSZ}} S_{\mathrm{phys}}$ with one-loop unitarity (optical theorem, v278); the spin-2 graviton sector is perturbatively unitary too — the Stelle ghost is a Seeley–DeWitt truncation artefact (v304/v370/v380)
the actual TOE remainder — dimensionful units, typed transfers, and the cited-theorem ceiling on the seam		
QFT measure (admissible sector)	[E]	**closed: RP from cusp spectrum, OS reconstruction**
full measured pole masses	[C]	ladder $\times v_{\mathrm{geo}}$, dressed by standard RG (no new physics); v_{geo} over-determined (v274)
absolute QCD scales (m_p/m_e, Λ_{QCD})	[C]	typed runnable transfer with uncertainty budget (v374, contract `FR.MPME.01`); not a compiler number
full PMNS dynamics (mass scale, CP)	[C]/[O]	angles typed; CP strength $J_{\mathrm{PMNS}} = -0.0297$ derived (v270); absolute ν-scale = one seesaw ratio (v272); two named post-hoc [O] candidates: third-gen ladder-base pattern (v467) and Δm^2 ratio $= \lvert J \rvert$ (v468) — record unchanged
(1) `SEAM.EQUIV.01` — one named theorem (*raw seam* $= (E_8)_1$ *at* $\tau = i$)	[C]	**closed modulo a cited theorem**: the explicit lattice model (v367/v368) and the S_3 stack ($c = 8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, Lean-formalised as `FORM.SEAM.MMST.01` (collar hypotheses kernel-proved, MMST/OS as named cited axioms, clean `#print axioms`); the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336); extension leg crossed-product-certified, realisation at invariant level (v469)
(2) `QG.AMB.01` — ambient measure (*redundancy, not missing dynamics*)	[C]	**discharged as a redundancy** (v369; v379 RP→OS), conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann: a certification object, gap-decoupled (margin 1.648>0, v330/v335) so no readout depends on it; obstruction characterised (v332) + conditionally resolved by GHP+IDG (v334); Tier-B's holomorphy bit (det $K = 1$, v277) is absorbed into `SEAM.EQUIV.01`
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[C]	**hardened:** $c_3 = 1/(\lvert \mathbb{Z}_2 \rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8"= rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	[C]	typed runnable transfer solvers with kill tests (v371–v374, observatory CI v375; see document 4)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed), and on top of it the whole boundary QFT is now *one relative object* $\mathsf{TFPT}_{\mathrm{QFT}}$ (the *Modular Spectral Closure*, v258–v261: D_F a covariance induction, the cutoff the seam KMS weight, seam/carrier/E_8 on one Kummer/K3) — closed modulo the single keystone `SEAM.EQUIV.01`. *What that closure does not reach* is exactly the dimensionful/ambient remainder: the full measured pole masses (ladder $\times\, v_{\mathrm{geo}}$ + standard RG, [C]), the absolute QCD scales (m_p/m_e, now a typed [C] transfer, v374), the complete PMNS dynamics ([C]/[O]), and the nonperturbative *ambient* measure `QG.AMB.01`, now itself a [C] redundancy (v369). (The metric-sector *field equation* is parameter-free at the local level — the full covariant $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ from fixed-volume entanglement equilibrium, v358/v359 — and the ambient *measure* is no longer an open structural item but a certification object, conditional on `SEAM.EQUIV.01`+Bisognano–Wichmann.) The *perturbative* 4D QFT is now built: the spectral-action S-matrix is Epstein–Glaser-constructible and LSZ-bridged to the physical asymptotic S-matrix with one-loop unitarity for the matter+gauge sector (v269–v278), and the spin-2 graviton sector is perturbatively unitary too — the R^2/Weyl^2 Stelle ghost is a Seeley–DeWitt *truncation artefact* (v304), ghost-free via the entire KMS form factor and the Barnes–Rivers decomposition (v370) with the truncation tower decoupling (v380); the whole bedrock reduces to one geometric postulate, *the raw seam is the flat $\tau{=}i$ pillowcase* (v276), with Tier-B's holomorphy bit $\det K{=}1$ (v277) absorbed into `SEAM.EQUIV.01`. The former single open lemma, the named theorem `SEAM.EQUIV.01` — *the raw reflection-positive seam state is the holomorphic $(E_8)_1$ net at $\tau{=}i$* — is now *closed modulo a cited theorem.* (The flat $\tau{=}i$ geometry and the E_8 holomorphy are *two faces of one object*: the $(E_8)_1$ character is $\chi_{E_8}{=}j^{1/3}$, so $\chi_{E_8}(i){=}1728^{1/3}{=}12$ and $\tau{=}i$ is at once the order-4 CM point and the $(E_8)_1$ partition-function modulus, v282.) The explicit lattice model (v367/v368) and the S_3 stack ($c{=}8$, 248 currents/1 primary, torus GSD=1, RP; v376–v379) pin it at every computable level, and it is Lean-formalised as `FORM.SEAM.MMST.01` (collar hypotheses kernel-proved, the MMST scaling-limit and Adamo–Moriwaki–Tanimoto OS-reconstruction theorems as named cited axioms, clean `#print axioms`); the *only* residual that stays [O] is the abstract continuum existence of the scaling limit (v336) — its 128-spinor extension leg since re-founded on the peer-reviewed crossed-product package ($h_s{=}16/16{=}1{\in}\mathbb{Z}$ is the Longo–Rehren locality criterion; KLM $\mu{=}4/2^2{=}1$; AGT/AMT the second witness), the realisation input reduced to invariants (R1′; computed $|C|{=}1$, $\nu{=}16$), Lean parallel route `seamResidualClosed′` (v469) — so it is closed *modulo a cited theorem*, not solved, Lean-pinned to exactly those named steps plus the derived Recovery gap $\Delta = 6\ln(3/2) \approx 2.43 > 0$ (v308); the conformal-deck premise `QGEO.SYM.01` is its *downstream* definitional face (v323). This is by design the *one* irreducible structural input of the theory — the role the *constancy of c* plays in relativity. It is not a defect to be removed but the single, sharply-localised, falsifiable premise on which the whole boundary closure turns; with the lattice/S_3 pinning and the Lean formalisation in place, "closing the theory" has reduced to invoking exactly the two cited continuum-existence theorems, not an open-ended search. The full sprint-by-sprint reduction that pinned `SEAM.EQUIV.01` is recorded in the changelog, not replayed here. The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K\, dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the cited continuum-existence theorem on the seam and the one unit v_{geo}, plus the standard-RG dimensionful masses and the typed frontier transfers (the ambient QG measure being a [C] redundancy, v369). No load-bearing discrete parameter remains free, and π is the lone continuous primitive. *4D-QFT fork policy (frozen, v265):* the canonical 4D reading is *boundary only* — a 4D-GUT is *not claimed by default* (E_8 is the audit hull; SM-only unification is killed, v246); carrier-native Pati–Salam is a separately-typed, falsifiable UV branch (thresholds + proton decay). Not an open ambiguity, a decision tree.

10 Does this make the theory more probable?

Plausibility balance. *Pro:* (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors ⇒ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar angle $\sin^2\theta_{12}$	prediction of record $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.306747$; TBM + seam misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}=c_3+36c_3^4$ (cond.)	**JUNO**: first 59.1 d run 0.3092±0.0087 (compatible), precision phase pending	[E]/[C]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\rm ret}e^{-5/6}=0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma=\frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2\theta_{23}$	$\approx\frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)$=0 (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r=\frac{12}{N_\star^2}\in[0.0033,0.0048]$ (frozen band, $N_\star\in[50,60]$); scalaron-reheating point 0.0045 ($N_\star$=51.4, v86)	now r<0.036 (BICEP/Keck BK18); **CMB-S4** ($\sigma_r\leq5\times10^{-4}$, obs. ∼2033)	[C]
tilt n_s	$n_s=1-\frac{2}{N_\star}\in[0.960,0.967]$ (same band); scalaron-reheating point 0.9611 ($N_\star$=51.4, v86)	Planck (0.9649±0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_\star^2}{24\pi^2}c_3^7$; 2.0×10^{-9} at $N_\star$=55 (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\rm eff}=0\Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[E]
CP phase $\delta_{\rm CKM}$	$\delta=\frac{\pi}{3}+3\lambda^2=68.65°$ (frozen of record, v88); survives at $+0.98\sigma$ vs $\gamma_{\rm PDG}$	**LHCb** / Belle II γ combination; decision at $\sigma_\gamma\lesssim0.96°$	[E]
cosmic birefringence β	$\beta=\frac{\varphi_0^{\rm ret}}{4\pi}=0.2424°$ (seam readout)	ACT DR6 hint 0.215°±0.074° (systematics not yet understood — not a validation target); **Simons Obs.** / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[C]
flavor invariants	$\det R=h(D_5)$=8, principal minors (2, 3, 5), $\chi_R=t^3-9t^2+10t-8$ (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\rm EW}\sim e^{-\alpha^{-1}/5}$, $\Lambda\sim e^{-2\alpha^{-1}}$, $H_0\sim\sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[C]

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`*.*

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.306747$ (misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}=c_3+36c_3^4$). The first 59.1-day JUNO run reports $\sin^2\theta_{12}=0.3092\pm0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~0.307 at high significance falsifies the seam mechanism.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts a small $r=12/N_\star^2$: the frozen band $r\in[0.0033,0.0048]$ over $N_\star\in[50,60]$, with the scalaron-reheating point $r=0.0045$ ($N_\star$=51.4, v86) — already below the current bound $r<0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r\leq5\times10^{-4}$, first observations ∼2033–34). A detection of $r\gtrsim0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [verification/v100_numerology_null_mc.py].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [verification/v84_frozen_registry.py] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from [C] to [E].

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable.* Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	What it machine-checks
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\mathrm{car}}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\mathrm{fam}}{=}3$, rank $E_8{=}8$, $\Omega_{\mathrm{adm}}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to$ 240, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\mathrm{car}}{=}5$ three ways; $8{=}$ rank $E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\mathrm{rad}}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_involution.py`	Q,Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q,χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K,Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_generation_polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9,10,16,40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\mathrm{fam}}$
`v14_carrier_uniqueness.py`	$g_{\mathrm{car}}{=}5$ unique ($N_{\mathrm{fam}}\in\mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\mathrm{Tr}\,Y^3{=}0$
`v15_bootstrap_classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$; full PMNS
`v17_hexagonal_resolvent.py`	finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}{=}0.470085$, $\frac{m_c}{m_s}{=}13.61$, $\frac{m_t}{m_b}{=}40.80$ exact; lepton $c{=}(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_moduli.py`	exact pole $z_\star{=}\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_derivation.py`	lepton c's derived: $\delta{=}\frac{1}{2}$ resolvent $c{=}\lvert\mu_4\rvert^w/(\frac{5}{4}{-}\cos\frac{r\pi}{3}) \Rightarrow \frac{16}{7},\frac{4}{3}$; product $\frac{2^{g_{\mathrm{car}}}}{N_{\mathrm{fam}}^2}{=}\frac{32}{9} \Rightarrow \frac{7}{6}$
`v21_solar_product_quark.py`	solar coeff $= q(A_3){=}\frac{3}{4}$ (glue-norm, $q(D_5){+}q(A_3){=}2$), $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$; product $\frac{2^{g_{\mathrm{car}}}}{N_{\mathrm{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
`v22_open_gates_audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_generator.py`	anchor $a{=}(1,1,2)$: $e_k{=}(4,5,2)$, $c_3{=}\frac{1}{2e_1\pi}$; $p_n{=}2{+}2^n \Rightarrow 240, 248$, binary ladder; inputs $\to \{a,\pi\}$
`v24_quark_ratio_closure.py`	quark ratios $\frac{55}{117},\frac{34}{47},\frac{3}{26}$ from TFPT blocks; mass ratios match $<0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_conjectures.py`	conjectures: Koide $Q_{\mathrm{src}}{+}\frac{\varphi_0^{\mathrm{ret}}}{24}{\approx}\frac{2}{3}$; axion $f_a{=}M_{\mathrm{scal}}/128{\to}m_a{\approx}23.8\,\mu$eV

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Script	**What it machine-checks** ***(continued)***
`v26_flavor_frontier_unification.py`	the "11" is not uniquely forced ($\Rightarrow$ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_representative.py`	explicit balanced wall rep $W_{\rm wall}$ (cols = perm$(0,1,2)$, rows $w{=}(2,1,1)$, pardeg 0); selector $\det R{=}8$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}$
`v28_gravity_fR.py`	$R{+}R^2$ closed: $f(R)$, $M{=}c_3^{7/2}\bar M_{\rm Pl}$ scalaron, $N_\star{=}57{\to}n_s, r, A_s$; full metric measure open; $248c_3^2{<}6\log\frac{3}{2}$
`v29_research_contract_certs.py`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick $W_{\rm wall}$) and $G5$ gap-dominance $2\|V\|{<}\Delta$
`v30_d4_character_variety.py`	$C_U^{(2)}$: full D_4-fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment)
`v31_R_dictionary.py`	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ-invariants continuous $\Rightarrow$ $R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
`v32_rh_splitting.py`	RH route: $\sum_k U^k A_0 U^{-k}{=}4\,\mathrm{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2){\oplus}\mathcal{O}(-1)^2{\Leftrightarrow}\mathrm{diag}(A_0){=}(\frac{1}{2},\frac{1}{4},\frac{1}{4})$ (algebraic); $\prod M_k{=}I$ + R-bridge stay open (c_u/c_d not forced)
`v33_explicit_flat_bundle.py`	explicit valid $(U_{\rm wall})$ flat bundle (RH solve): cusp + $\mathcal{O}(-2){\oplus}\mathcal{O}(-1)^2$ + $\|M_\infty{-}I\|{\sim}10^{-9}$ ($\prod M_k{=}I$) + irreducible (case A)
`v34_h2_bridge_attempt.py`	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k{=}I$); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\rm min}$ dictionary is the remaining input (c_u/c_d not obtained)
`v35_quark_qcd_boundary.py`	the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
`v36_spectral_action_g2.py`	G2: spectral action $\to R{+}R^2$ ($a_2{\sim}{-}R/3$, $a_4\|_{R^2}{\sim}R^2/72$); $M^2/\bar M_{\rm Pl}^2{=}6(4\pi)^2/f_0$, c_3^7 fixes f_0; gap-decoupling $\Delta_{\rm eff}{=}1.648{>}0$ $\Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
`v37_plucker_anchor.py`	anchor-plane Plücker apparatus: $\|\mathrm{Pl}(K)\|_1{=}11$, $\|\mathrm{Pl}_R(K)\|_1{=}26$, $\sum\mathrm{Pl}_R(L){=}60$; pencil $\det(K{+}xQ){=}3x^3{+}7x^2{+}6x{+}4$; $K_e{-}K_u{=}a$, $L_e{-}L_u{=}\mathrm{Exp}(A_3)$; lepton ring $c_e c_\tau{=}\|\mathbb{Z}_2\|c_\mu$; Koide $u{\to}\frac{53}{54}u$ ([C])
`v38_uwall_killswitch.py`	$(U_{\rm wall})$ U2 kill-switch hardened: D_4-cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq$0.35, reducible locus measure-zero $\Rightarrow$ case A generic, $(U_{\rm wall})$ survives; amplitude dictionary still research-level
`v39_uwall_selectors.py`	$(U_{\rm wall})$ selector side closed on the explicit point: splitting type = anchor $a{=}(1,1,2)$, pardeg $=0$; $\det R{=}8{=}n{\cdot}a$, $\mathrm{Spec}(Q_+){=}\{1,2,3\}{=}3\alpha{+}1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\rm min}$ Hitchin) remain
`v40_harmonic_metric.py`	$(U_{\rm wall})$ harmonic metric = FINITE linear algebra (polystable⇒unitary⇒$\Phi{=}0$): unique invariant form $H{>}0$, unitarised $\widetilde{M}_k$ cusp-class with $\|\widetilde{M}_k\|{\in}\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
`v41_leg_assignment.py`	$(U_{\rm wall})$ final leg test: amplitudes = $\delta{=}\frac{1}{2}$ resolvent (closed); $\Lambda_\mu{>}1$ rules out holonomy moduli; harmonic holonomy $\|\mathrm{diag}\|{=}(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta{=}\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
`v42_exterior_leg.py`	$(U_{\rm wall})$ Exterior Leg Lemma: u,d share $(r,w){=}(1,1)$ so any scalar leg gives $c_u/c_d{\equiv}1$; the u/d split is the Λ^2F area $\mathrm{Pl}_{\mathbf{1},a}(K)$, $\|\cdot\|_1{=}11$ generated $\Rightarrow$ $\frac{5\cdot 11}{9\cdot 13}{=}\frac{55}{117}$ [E]; phys. identification [C]
`v43_exterior_bridge.py`	$(U_{\rm wall})$ Λ^2F bridge: $\Lambda^2(\widetilde{M}){=}\overline{\widetilde{M}}$ (exterior leg $=\bar{\mathbf{3}}$) confirmed; continuous action $\neq$ integer $\mathrm{Pl}(K)$ $\Rightarrow$ bridge is the discrete H2 invariant ($\det R{=}8$, $\mathrm{Spec}(Q_+)$ confirmed), not continuous

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Script	**What it machine-checks *(continued)***
`v44_carrier_exterior.py`	($U_{\rm wall}$) Lie-level grounding: $16{=}\Lambda^{\rm even}(5){=}1{+}10{+}5$; $u^c{\in}\Lambda^2(5)$, $d^c{\in}\Lambda^4(5)$ $\Rightarrow$ exterior degrees $(2,4)$, sum=6=$\lvert R^+(A_3)\rvert$, diff=2=$\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own Λ^2 grading (no special math)
`v45_family_exterior.py`	($U_{\rm wall}$) the "11" is Pascal: $\lvert{\rm Pl}(K)\rvert{=}(1,4,6){=}\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert{\rm Pl}(K)\rVert_1{=}11{=}\sum_{k\le 2}\binom{4}{k}{=}16{-}g_{\rm car}$ (cumulative ext of μ_4 up to deg $u^c{=}2$); $15{=}\dim\mathfrak{su}(4){=}10{+}5{=}N_{\rm fam}g_{\rm car}$
`v46_grand_mass_volume.py`	absolute mass-scaling is **[E]**, not a product rule: sector dets $(\varphi_0^{\rm ret})^6,(\varphi_0^{\rm ret})^9,(\varphi_0^{\rm ret})^{10}$ (exponents $\lvert R^+A_3\rvert, N_{\rm fam}^2,\mathcal{A}_\Lambda$) $\Rightarrow$ $\det M_{\rm SM}{\sim}(\varphi_0^{\rm ret})^{25}{=}(\varphi_0^{\rm ret})^{g_{\rm car}^2}$; Q-rows $(4,5,6)$ sum $15{=}\dim A_3$; $25{+}15{=}40{=}\sum L$
`v47_selection_theorem.py`	Boundary Carrier Selection (Thm A): $\mu_4{\to}A_3$, $N_{\rm fam}{=}3$; 16-spinor$\to D_5$; $q(D_5){+}q(A_3){=}2$, glue $4{=}\lvert\mu_4\rvert$; only $D_5{\oplus}A_3$ meets all four conditions ($D_8, E_7{+}A_1, E_6{+}A_2, A_4{+}A_4, A_8$ excluded) **[E]**/**[C]**
`v48_em_ward.py`	EM boundary Ward (Thm C): $F_{U(1)}{=}$ Maxwell α^3 $-$ Calderón $2c_3^3\alpha^2$ $-$ transport $8b_1c_3^6\log\frac{1}{\varphi_{\rm seam}}$; $8b_1{=}\frac{4}{5}M{=}\frac{164}{5}$, $-\frac{5}{4}{=}q(D_5)$ **[E]**; Ward origin **[C]**
`v49_readout_rigidity.py`	Readout Rigidity (Thm U2): $c_u/c_d{=}\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ $\Rightarrow$ no point-uniqueness needed; $(U_{\rm wall}){=}U_{\rm unitary}{+}U_{\rm H2}{+}U_{\Lambda^2}{+}U_{\rm point}$, only $U_{\rm point}$ open
`v50_q_geometry.py`	Q geometry (Thm Q): $Q_+{=}$ diag$(3,2,1)$-block $\chi{=}(t{-}1)(t{-}2)(t{-}3)$, Q_- $\chi{=}t(t^2{-}3)$; Q unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin **[C]**
`v51_boundary_half_step.py`	glue norms $=(g_{\rm car},N_{\rm fam})/\lvert\mu_4\rvert$: $q(D_5){=}\frac{5}{4}$, $q(A_3){=}\frac{3}{4}$; four ops forced $\to$ {sum= 2 root norm, diff= $\frac{1}{2}{=}\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert{=}\delta$, prod= $\frac{15}{16}$, ratio= $\frac{5}{3}$}. So $\delta{=}\frac{1}{2}$ is the carrier$-$family count over the glue index, not chosen **[E]**
`v52_pencil_endpoints.py`	$K{+}xQ$ pencil endpoints: $P(-1){=}2{=}\lvert\mathbb{Z}_2\rvert$, $P(0){=}4{=}\lvert\mu_4\rvert$, $P(1){=}20{=}\det L$, $P(2){=}68{=}2p_5(a)$; $\det(K{-}Q){=}2$, tr $=N_{\rm fam}$ (audit ext. of v37) **[E]**
`v53_compiler_core.py`	Big picture: the whole integer skeleton from $(g_{\rm car},N_{\rm fam}){=}(5,3)$ — rank $E_8{=}8$, $\lvert\mathbb{Z}_2\rvert{=}2$, Pythagorean $\Delta_Y{=}g_{\rm car}^2{=}25{=}N_{\rm fam}^2{+}\lvert\mathbb{Z}_2\rvert{\cdot}$rank $E_8{=}9{+}16{=}\dim S^+$ (diff. of squares); anchor char-poly $(t{-}1)^2(t{-}2)$ unique **[E]**
`v54_seam_horizon_keystones.py`	The "8" in $c_3{=}\frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2\lvert\mu_4\rvert{=}$rank $E_8{=}h(D_5){=}\varphi(30){=}8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2{=}(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery **[E]**
`v55_coxeter_cycle.py`	Computed E_8 Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\rm fam}g_{\rm car}$, eigenphases = exponents = totatives of 30, so rank $E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$) **[E]**
`v56_unique_attractor.py`	Gapped boundary transport (gap $6\log\frac{3}{2}{>}0$) $\Rightarrow$ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts $\to$ same direction); Coxeter in $\lvert\mu_4\rvert{=}4$ planes; entropy reset = rank-1 projector **[E]**
`v57_horizon_crosslinks.py`	Horizon cross-links: $c_3{=}\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4{=}1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H{=}\ln 3{=}\ln N_{\rm fam}$; Hawking power denom $1920{=}\lvert W(D_5)\rvert$; one α^{-1} for EM/S_{dS}/Λ/Hubble **[E]** (+ **[C]** identification)
`v58_seam_horizon_chain.py`	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3{=}1/(2{\cdot}4\pi){=}\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}{=}M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) **[O]** open
`v59_area_law_evidence.py`	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim\frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8{=}\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The c_3 coefficient stays **[O]** open
`v60_lambda_metrology_branch.py`	Λ branch selection: $(8\pi)^2\delta_{\rm top}{=}\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\rm top}{=}\frac{64\pi^3}{3}{=}661.47$ $\Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°{\pm}0.074°$ vs $0.2424°$ (0.37σ) **[E]**

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Script	What it machine-checks *(continued)*
`v61_cft_bridge.py`	Boundary-CFT mirror: level-1 WZW central charges $c(E_8)$=8, $c(D_5)$=5, $c(A_3)$=3 = (rank $E_8, g_{\rm car}, N_{\rm fam}$); conformal embedding $c_{\rm coset}$=0; $SU(3)\leftrightarrow SU(4)$ reconciliation $N_{\rm fam}=\lvert\mu_4\rvert-1$, 4→3+1 (one geometry $\mathbb{P}^1\setminus\mu_4$, two sides) [E]
`v62_data_scorecard.py`	TFPT vs 2024/25 data: α^{-1}, $\sin^2\theta_{12}$, $\beta_{\rm rad}$ match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r\approx 0.004$ future falsifier [E]
`v63_seam_engineering_index.py`	Exact Seam-Engineering Index $\Xi=2\lVert V\rVert/\Delta=\frac{31}{24\pi^2\log(3/2)}\approx 0.323$ ($2\lVert V\rVert=\frac{31}{4\pi^2}$, 248= dim E_8, 31=1+$h^\vee$), $\Delta_{\rm eff}\approx 1.648$; four-class possibility taxonomy [E]/[O]
`v64_causal_boundary_nogo.py`	Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap ⇒ ANEC ⇒ Topological Censorship ⇒ no traversable shortcut; only ER=EPR bridge survives [E] (core)/[C] (chain)
`v65_falsification_layer.py`	Prediction layer with explicit failure thresholds: 3 core ($v_{\rm GW}$=c, no 4th gen, seam=8), 4 observational (β, r, n_s, θ_{12}), no-go + unitarity; $N_\star$ demoted to [C] input; none violated [E]
`v66_e8_casimir_degrees.py`	Organizing simplification: the compiler atoms *are* the E_8 Casimir degrees $\{2,8,12,14,18,20,24,30\}$ ($2=\lvert\mathbb{Z}_2\rvert$, 8= rank, 14= dim G_2, 20= det L, $24=\lvert W(A_3)\rvert$, $30=h(E_8)$); $\sum=128=2^7$ (dS prefactor), $\sum\exp=120=\lvert R^+(E_8)\rvert$ [E]
`v67_area_law_coefficient.py`	Central-theorem *structural closure*: Fursaev–Solodukhin $S=4\pi kA$, $k=c_3/2$ ⇒ $S=2\pi c_3 A=\frac{1}{4}A$ ($2\pi c_3=\frac{1}{4}$); $c_3=\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k=c_3/2$ (then *forced*, v73) [E]
`v68_seeley_dewitt_residual.py`	Residual *resolved as an anchor*: $a_2=-\frac{d}{192\pi^2}R$ ⇒ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless $k=c_3/2$ is forced, v73); cutoff-indep. R^2/scalaron derived [E]
`v69_d4_q_geometry.py`	D_4-equivariant Q-geometry (gate [C]/[E]): on $H_1(\mathbb{P}^1\setminus\mu_4)=B_1\oplus E$, $Q_+=3w+1=\mathrm{diag}(1,2,3)$ (cusp weights on μ_4=$\mathbb{Z}_4$ eigenspaces), Q_-=E-coupling $\sqrt{N_{\rm fam}}$ ⇒ $t(t^2-3)$; Σ-split $=D_4=\mathbb{Z}_4\times\mathbb{Z}_2$ [E]
`v70_q_integer_lift.py`	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1,i,-i\}$); det Q=3=$N_{\rm fam}$, SNF diag(1, 1, 3) ⇒ residual = one lattice invariant det Q=$N_{\rm fam}$, not closable by D_4 alone [E]
`v71_simple_r_bridge.py`	*Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived (det R=8= rank E_8 lattice + Spec(Q_+)={1, 2, 3} from v69) ⇒ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\rm point}$= absolute norm. = an anchor [E]/[O]
`v72_q_det_from_cusp.py`	det Q=$N_{\rm fam}$ derived from the cusp class: det $Q=\lvert\mathrm{coker}\,Q\rvert$= cusp-class order = denominator of the weights $\{0,\frac{1}{3},\frac{2}{3}\}$ (same data as Spec Q_+); coker $Q=\mathbb{Z}/N_{\rm fam}$= triality deck group ⇒ integer-lift residual closed [E]
`v73_k_c3_half.py`	Central coefficient $k=c_3/2$ *forced*: (1/2) variational × $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2)$=2), both cutoff-indep ⇒ Fursaev–Solodukhin $S=A/4$; only the absolute 4D Newton scale stays the anchor (v68) [E]
`v74_compiler_micro_lemmas.py`	Spine quotient ladder ($\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{4},\frac{4}{3}$ = adjacent quotients of (2, 3, 4, 5)); pencil differences $(2,16,48)=(\lvert\mathbb{Z}_2\rvert,\dim S^+,\Omega_{\rm adm})$ sheet→1gen→3gens; anchor QF $a^\top Ka-\mathbf{1}^\top K\mathbf{1}=41-25=16=\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L=R+61e_1^\top$, $a^\top R^{-1}\mathbf{1}=0$ (Sherman–Morrison rank-one invariance) [E]
`v75_upoint_to_vgeo.py`	**Gate 1 complete:** $U_{\rm point}\to v_{\rm geo}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = $\varphi_0^{\rm ret}$-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale $v_{\rm geo}$ = the same anchor as $1/G$ (v68); two [A] → one [E]/[O]
`v76_gmetric_reduction.py`	**Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\rm eff}=6\log\frac{3}{2}-\frac{31}{4\pi^2}=1.648>0$, $\lVert V\rVert\le 248c_3^2=\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) [E]/[C]

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Script	**What it machine-checks** ***(continued)***
`v77_e8_conformal_net.py`	**G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic c=8); level-1 $c(E_8)=\frac{248}{31}$=8, $c(D_5)$=5, $c(A_3)$=3, embedding 5+3=8 (coset c=0); imports $G6$ into RCFT/conformal-net rigor [E]/[C]
`v78_vgeo_floor.py`	$v_{\rm geo}$ **floor:** no pure number is dimensionful $\Rightarrow v_{\rm geo}$ irreducible by logic; all scales are ratios × one $\bar{M}_{\rm Pl}$; $S_{dS}\rho_\Lambda=32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [E]/[O]
`v79_review_identities.py`	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n=(3^n-(-2)^n)/5=1,1,7,13,55,133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}$=1/atom, $a^\top M^{-1}a$=1 for R,K,L); MacWilliams $(1,10,5)$; $6Y^2-Y-1=(2Y-1)(3Y+1)$. Rejects m_p/m_e, η_B near-formulas (firewall) [E]
`v80_operator_pencil_geometry.py`	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}=(N_{\rm fam}x+\lvert\mathbb{Z}_2\rvert)(N_{\rm fam}x+g_{\rm car})=(3x+2)(3x+5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3,10,16,20)$; type checker $\det B_{Q,K,R,L}=(9,10,16,40)$; audit: differential spectrum, $F_4{\times}G_2$ shadow $248{=}52{+}14{+}26{\cdot}7$ [E]/[C]
`v81_singular_pencil_matrices.py`	Double cover $y^2=\det B_{K+xQ}=(3x+2)(3x+5)$: Koide $x=-\frac{2}{3}$ and carrier $x=-\frac{5}{3}$ are the two branch points (deck $2=\lvert\mathbb{Z}_2\rvert$; disc=$81=N_{\rm fam}^4$; separation=1 transport period). Clearing matrices $C_{2/3}=3K-2Q$ ($D_5{\oplus}A_3$ glue, $B=(5,7)^\top(9,11)$, $\sum$=240), $C_{5/3}=3K-5Q$ (neutral, $\chi=(\lambda+2)(\lambda^2+\lambda+6)$); scalaron 7= mixed anchor disc/$N_{\rm fam}$ [E]
`v82_koide_attractor_splitting.py`	Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing q=2, 5 is unique, multiplier $(2/3)^6=\lambda_2$ of the established transfer operator $\Rightarrow F'(2)<1$ attractor, $F'(5)>1$ repeller; three postulates → one [C]. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) [E]/[C]
`v83_e8net_holomorphic_uniqueness.py`	Holomorphy pins $(E_8)_1$: #primaries = det Cartan (D_8:4 vs E_8:1), and a holomorphic c=8 chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation $K=(g-1)/2$ = half-spinor split [E]
`v84_frozen_registry.py`	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* θ_{12} prediction of record; r, n_s as $N_\star$ *bands*; conditional entries carry their hypothesis by name [E]
`v85_master_cover.py`	Master-cover theorem: $\det B$ is GL(2)-covariant on span$\{K,Q\}$ $\Rightarrow$ exactly *one* double cover up to Möbius (disc = $N_{\rm fam}^4\det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}=-{\rm rank}E_8/N_{\rm fam}$ = carrier − one period); μ_4 is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* [E]
`v86_nstar_reheating.py`	$N_\star$ band → point [C]: $\Gamma=4M^3/48\pi\bar{M}^2$=128 GeV, $T_{\rm reh}=9.6{\times}10^9$ GeV, $N_\star(0.05/{\rm Mpc})=51.4 \Rightarrow n_s=0.9611$, $r=0.0045$ (inside the frozen band); honest tensions: n_s $+0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]
`v87_bulk_uniqueness_reduction.py`	Red-team Target A residual (ii) conditionally closed: holomorphic ⇒ Rep = Vect ⇒ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both E_8 pairings) ⇒ Target A = one residual [E]/[C]/[O]
`v88_cp_phase_audit.py`	Target-D CP residual quantified: frozen $\delta=\pi/3+3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\rm PDG}$; the 0.07°-near alternative $\pi/3+2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma\le0.96°$ [E]
`v89_carrier_index_lemma.py`	Carrier Index Lemma: KLM $\mu_A=[B{:}A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1{\times}(A_3)_1]=4=\lvert\mu_4\rvert$ (glue order *is* the index); μ-additivity $16/4^2=1 \Rightarrow$ holomorphy *follows*; Gate A (H)⇔(H′) index-4 extension [E]
`v90_conical_defect_chain.py`	Fursaev–Solodukhin factor 4π *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S=4\pi kA$, $k=c_3/2 \Rightarrow S=A/4 \Leftrightarrow c_3=\frac{1}{8\pi}$ (sympy-solved); residual isolated to "seam determinant ⇒ EH form" [E]/[O]

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Script	**What it machine-checks *(continued)***
`v91_spine_tetrahedron.py`	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\lvert R^+(E_8)\rvert$; $240{=}\lvert\mu_4\rvert\lvert E(K_4)\rvert\lvert E(K_5)\rvert$ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]
`v92_glue_uniqueness.py`	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier$\to SO(16)_1\to(E_8)_1$, nothing else [E]
`v93_koide_relaxation_toy.py`	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}-\varphi_0^{\mathrm{ret}}/24$ to 0.8% ([C]); honest negatives: flow time $t{=}2.84\notin\mathbb{Z}\Rightarrow$ the missing object is a *continuous* generator [E]/[C]
`v94_sheet_diamond.py`	Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x,x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all s; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in K; spine-quotient firewall *rejected* [E]
`v95_centered_diamond.py`	Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects → three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\mathrm{fam}}{\cdot}\mathrm{cusp}$; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1)\parallel\mathrm{Pl}_R(L){=}10(2,3,1)$; G_2 reading audit-typed [E]
`v96_branch_kernel_selection.py`	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel = $\mathbf{1}$); collapse lemma $\Rightarrow(-1,1,0)$: up $=-$down, *lepton pairing* = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
`v97_sheet_conjugation_bridge.py`	Sheet-conjugation bridge (P1 → one [C]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer T_A, $\det={-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\,\mathrm{span}\{\mathbf{1},a\}$; $\langle T_A,\Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\lvert\mathbb{Z}_2\rvert$); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
`v98_discriminant_dictionary.py`	Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3$ = $B_1{\oplus}E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ $(k{\mapsto}{-}k)$; $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [C]; residual = the one existing Q-geometry gate [E]
`v99_koide_flow_time.py`	Koide flow time: canonical generator $\dot q{=}(\Delta/N_{\mathrm{fam}})(q{-}2)(q{-}5){=}(\Delta/N_{\mathrm{fam}})\det B(q)$, time-1 map = F [E]; non-integrality of $t{=}2.84$ is *data-fragile* (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: $n{=}2$ excluded (-2.9σ), $n{=}3{=}N_{\mathrm{fam}}\Rightarrow m_\tau{=}1776.9427$ MeV $(+0.14\sigma)$, decision at $\sigma(m_\tau){\sim}0.01$ MeV [C]; ln(46080) scale reading destroyed by controls (firewall)
`v100_numerology_null_mc.py`	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i=10^{-25.8}$, robust to budget/window variation; $2{\times}2{\cdot}10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [E]
`v101_horizon_anchor.py`	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\mathrm{Nariai}}/S_{dS}{=}\frac{2}{3}{=}\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2{+}1)/\Phi_3(x)$; three-sheet total $\lvert\mathbb{Z}_2\rvert S_{dS}$ conserved; $Q_{\mathrm{geom}}\in[\frac{3}{8},\frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1{-}3m)(1{+}3m)$, deck = horizon swap; $\tau_{BH}{=}128\,g_{\mathrm{car}}M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)

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Script	**What it machine-checks** *(continued)*
`v102_seam_orientation.py`	One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}^2$; evaporation = entropy ascent away from the anchor [E]; "one variational principle of the seam" stays a reading [C]
`v103_trisection_normal_form.py`	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow$ $\cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\rm fam}$ and $S_{\rm tot}/S_{dS}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\lvert\mathbb{Z}_2\rvert/N_{\rm fam})^{N_{\rm fam}}$ (Koide constant to the family power); invariant slope $d\sigma/dm\vert_N=-\frac{8}{9}=-\operatorname{rank}E_8/N_{\rm fam}^2$; bridge: flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, missing = one *clock* [E]/[C]
`v104_nariai_clock.py`	The *classical* Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2=-2\Lambda=-\lvert\mathbb{Z}_2\rvert\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\rm clock}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic = $(t-1)\cdot\chi_{\rm clock}$; entropy-deviation rate $2H=\lvert\mathbb{Z}_2\rvert H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
`v105_residual_inventory.py`	The residual inventory: *one constant* ($2/3=\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10$ $\Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + π; a sixth gap would fail the script [E]/[C]
`v106_review_validation.py`	External-review validation: seed normal form $\varphi_0^{\rm ret}=\frac{\lvert\mu_4\rvert}{N_{\rm fam}}c_3+\Omega_{\rm adm}c_3^{\lvert\mu_4\rvert}$ exact [E]; hypercharge moment $\operatorname{Tr}X^2=120=5!$ computed from the charge table; factorial spine $5!=\lvert R^+(E_8)\rvert=\sum$exponents, $240=2\cdot120$, $1920=2^4\cdot5!$; degree-2 inventory (Pascal $K=2$ unique at $g=5$, glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$, $\binom{5}{2}=10$); named [C] hypothesis *Quadratic Boundary Locality* ($\Rightarrow K=2$ forced; would upgrade the Pascal selection to [E]); audit $240=16\times15$ non-unique [E]/[C]
`v107_quantum_clock_target.py`	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\}=-\operatorname{Spec}(V)=-N_{\rm fam}\times$cusp (the transfer operator's own grading) [E]; exact target identity $(1/3)^6=\left((2/3)^6\right)^{\log_{3/2}3}$ — required bend $\log_{9/4}3=1.355$ per weight step; coupling landscape $\kappa=c/(24\pi)\cdot\Lambda/\bar{M}_{\rm Pl}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
`v108_pascal_ladder.py`	Pascal Ladder Theorem: $2^{g-1}=\sum_{k\le K}\binom{g}{k}$ $\Leftrightarrow$ $g=2K+1$ (odd-g midpoint identity + even-g straddle) $\Rightarrow$ carrier selection $g=5$ *exactly equivalent* to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\rm fam}=8$ (v14 overdetermination) [E]/[C]
`v109_sheet_pairing.py`	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+\otimes S^+=0$; odd g flips parity); $S^+\otimes S^-=\Lambda^0\oplus\Lambda^2\oplus\Lambda^4$ exactly ($256=1+45+210$, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms ($2\Lambda^1+2\Lambda^3+\Lambda^5$); tower tops at Λ^{2K}, $K=2\Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [E]/[C]
`v110_calderon_sheet.py`	Calderón-sheet selection theorem: a Calderón involution ε on $S^+\oplus S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2=\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3,5,7$): sheet-oddness forces $K=(g-1)/2$, *not* $g=5$ (anti-overclaim); QBL residue split into *interface* ("ε sheet-odd", natural from the one-sided collar) + *analytic core* (slot-degrees ≤ 2) [E]/[C]

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Script	**What it machine-checks** ***(continued)***
`v111_quadratic_transport.py`	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics (= $\mathfrak{so}(10)$, the Λ^2 term of 1+45+210) sheet-even, generate the code from the vacuum *and* span the full $\mathrm{End}(S^+)$=256 at word length 2 ⇒ *transport degree exactly 2* (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; g=3 control: degree selected, not rank [E]/[C]
`v112_selfcounting_channel.py`	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states ↔ neutral certified kernels, so #kernels = 2^{g-1} *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m}$ ⇒ the Pascal closure $2^{g-1} = \sum_{m \le K} \binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j, g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality [E]/[C]
`v113_quasifree_kernel.py`	One kernel is the whole net: carrier $(D_5)_1 \times (A_3)_1 = SO(10)_1 \times SO(6)_1$ = 16 free Majoranas, c=5+3=8 at every tower level (only the glue grows, index $2\times2=4=\|\mu_4\|$); Wick/Pfaffian exact (all 210+210 higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank P=5=g_{car}, seam level 8= rank E_8 (c = rank of the kernel); vacuum unique ⇒ QBL input = R2 premise: gate closes ⇒ carrier choice closes [E]/[C]
`v114_torsion_delta.py`	Torsion normal form + $\delta=\frac{1}{2}$ theorem: flatness of the (U_{wall}) $\mathbb{Z}_4$-family ⇔ $(MU)^4=\mathbf{1}$ (μ_4 = order of the twisted cusp generator); on the involutive branch ($T^2=\mathbf{1}$) the cusp trace forces diag $M=(0, \frac{i}{2}, -\frac{i}{2})$ with spec $\{1, \omega, \omega^2\}$ automatic ⇒ $\delta=\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its d_1=0 slice; residue = bundle-side branch selection [E]/[C]
`v115_anchor_residue.py`	The anchor pins the residue: μ_4-average = diagonal projection ⇒ anchor splitting ⇔ diag $A_0 = a/\|\mu_4\|$; off-weights uniquely $(8,0,5)/144$ $(=(\mathrm{rank}\, E_8, 0, g_{\mathrm{car}})/(\|\mu_4\| N_{\mathrm{fam}})^2$, total $13=\Delta_Q$); *exact* residue $A_0^\star$ with $\chi=\lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ is the RH solution ($\\|M_\infty - \mathbf{1}\\| \sim 10^{-10}$); anchor forces d_1=0 ⇒ harmonic diagonal anchor+torsion-forced [E]
`v116_resonance_uniqueness.py`	Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted μ_4 averages collapse to $B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$; anchor exponents diag(2, 1, 1) give resonance gap 1 with obstruction $4\overline{a_{13}}$ ⇒ $M_\infty=\mathbf{1}$ ⇔ a_{13}=0; + $(8,0,5)/144$ + Diagonaleichung ⇒ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — U_{wall} bundle datum algebraically unique; sharpness control $\\|M_\infty - \mathbf{1}\\| \sim 8.5\|a_{13}\|$ [E]
`v117_monodromy_weyl_a3.py`	The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), diag $\widetilde{M}_0=(0, -\frac{i}{2}, \frac{i}{2})$ ⇒ d_1=0, $\delta=\frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics (1, 9, 8, 6), characters (3, −1, 0, 1) = $S_4 = W(A_3)$ (standard⊗sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match 3.5×10^{-10} [E]
`v118_hexagon_family_dictionary.py`	First exact H2 piece: $\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$); $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3$ ⇒ $\det_\mp = (\frac{7}{8}, \frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e = \|\mu_4\|/(2\det_-)$, $c_\mu = N_{\mathrm{fam}}/(2\det_+)$, $c_\tau = \|\mu_4\| \det_- / N_{\mathrm{fam}}$, product = $\|\mu_4\|/\det_+ = \frac{32}{9}$); $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48=\Omega_{\mathrm{adm}}$; the address table stays open [E]/[C]
`v119_review_validation_2.py`	Second review validation: anchor ratio triad $(e_3, e_1, e_2)/p_0 = (\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5)=(e_3, e_2)$); the 121 audit lemma $(\mathbf{1}+a)^T R(\mathbf{1}+a) = 121 = 11^2 = \\|\mathrm{Pl}(K)\\|_1^2$ (orderings {105, 121, 135}, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\mathrm{adm}} = 2p_1p_2$, $10b_1 = 1 + p_1p_3$, $\Delta_Y = e_2^2 = p_0^2 + 2\,\mathrm{rank}$; flavor surface = v94/v95 (presentational) [E]

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Script	What it machine-checks *(continued)*
`v120_address_table.py`	Address table (frozen v18/v20 integers): lepton words $(8,5,3)$=(rank $E_8, g_{\rm car}, N_{\rm fam}$); addresses = divmod by the hexagon (p_2=6): $e\to(2,1)$, $\mu\to(5,0)$, $\tau\to(3,0)$, $\sum r$=p_3; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3, down = p_1+p_3, all quarks = 24=$\lvert W(A_3)\rvert$, all nine = 40=p_1p_3=a^TRa; top at the vacuum site $(0,0)$; the assignment derivation stays open **[E]**/**[C]**
`v121_address_pinning.py`	Address pinning: L=R+2U (v95) ⇒ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a; margins = atoms (rows (e_1, rank, p_3), columns $(e_1, \Delta_Q, g_{\rm car})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has det =8 — and it is R (also unique with SNF $(1,1,8)$; all 17 determinants distinct) ⇒ the address table carries *zero* residual information; the H2 residue = the six atom margins **[E]**/**[C]**
`v122_margin_theorem.py`	Margin theorem: the *established* selectors — the D_4 annihilator n=$(5,-9,6)$ with n^TR=$(8,0,0)$ (v94), det R=8, det K=4 (the v71 quartet) — pin R *uniquely* among hexagon matrices ($64\to12\to1$) ⇒ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: det L=20 holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class **[E]**/**[C]**
`v123_inventory_update.py`	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 Q realisation — + $v_{\rm geo}$, π; a sixth class fails the script **[E]**
`v124_resummed_clock.py`	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1-n/N_{\rm fam})^{p_2}$ ⇒ rate(n)=$-p_2\ln(1-n/N_{\rm fam})$ reproduces $\{0,\Delta,6\ln3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $\lvert\mathbb{Z}_2\rvert$=2 relative to the classical GP clock $\{0,-1,-2\}$; the pole at n=$N_{\rm fam}$ forces the three-level spectrum; the R1 target is now closed-form **[E]**/**[C]**
`v125_glue_qsystem.py`	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness mm^*=$\lvert\mathbb{Z}_4\rvert\mathbf{1}$) ⇒ Jones index 4=$\lvert\mu_4\rvert$ (the v89 value); locality = isotropy ($q(k(1,1))$=$k^2\in\mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, 4=2×2; R2 sharpens from “find” to “identify” **[E]**/**[C]**
`v126_clock_wall_bridge.py`	Clock and wall share one spectrum: the resummation weights $n/N_{\rm fam}$=$\{0,\frac13,\frac23\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); λ_n=$(1-\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\rm fam}$=2=$\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); det$(\mathbf{1}-A_0^\star)$=$\frac29$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail **[E]**/**[C]**
`v127_ring_resummation.py`	The clock is a log determinant: rate = $-p_2\,\mathrm{Tr}\ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the k=1 ring = the classical term, the k=2 ring = the first quantum correction; $\kappa_{\rm seam}$=$\frac{1}{3\pi}$=α_1/π (audit); the wall = the RPA instability at $\alpha\to1$; v107’s “O(1) resummation” is thereby *typed* (ring/log-det), its derivation open **[E]**/**[C]**
`v128_graded_hull.py`	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5\oplus A_3$+glue coordinates, coset decomposition 240=52+64+60+64; the grading is exactly additive on *all* 6720 root-pair sums ⇒ E_8 is a $\mathbb{Z}_4$-*graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1)\rvert_{l=1}$ = the sheet-doubled GP zero modes **[E]**/**[C]**
`v129_entropy_power_law.py`	The clock is an entropy power law: λ_n=$(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}\in\{1,\frac23,\frac13\}$ (v101); rate = $p_2\ln(S_{dS}/S)$; triple identity α=$n/N_{\rm fam}$= MS weight = entropy-deficit share ⇒ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma\propto(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, h=$N_{\rm fam}$ audit) **[E]**/**[C]**

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Script	What it machine-checks *(continued)*
`v130_born_square.py`	Born square: amplitude $\sim (S/S_{dS})^{n_{\rm zero}/2} = (S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $= \|A\|^2 = (S/S_{dS})^6 \Rightarrow h = n_{\rm zero}/2 = 3 = N_{\rm fam}$, $p_2 = 2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the $l=1$ space = the number of proper conformal Killing vectors of S^2; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C]
`v131_measure_is_area.py`	The measure is the area: $\\|Y_{1m}\\|^2 = r^2 = A/(4\pi)$ *exactly* (all three $l=1$ modes integrated) $\Rightarrow$ the per-mode Jacobian = the mode norm $\sim S^{1/2} \Rightarrow$ amplitude S^3, Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n / \det'_{dS} = 1 + O(\text{deficit})$ [E]/[C] *(cancellation since derived within the SdS family, v144)*
`v132_detratio_anomaly.py`	Det-ratio anomaly = the Koide constant: $\zeta(0)\|_{\det'}$ of the GP sphere operator $-\Delta - 2$ (the 3 zero modes removed) $= a_1 - 3 = \frac{7}{3} - 3 = -\frac{2}{3} = -\|\mathbb{Z}_2\|/N_{\rm fam}$ exactly ($a_1 = \frac{1}{N_{\rm fam}} + \|\mathbb{Z}_2\|$; numerical continuation $\sim 10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL) = c^{\zeta(0)} \det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) [E]/[C]
`v133_zeta_budget.py`	The ζ budget, both routes exact (Euclidean Nariai $= S^2 \times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3} = -e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2 = \frac{161}{45}$, $\zeta(0) = -\frac{109}{45}$ (*no* atom; continuation $\sim 10^{-8}$); zero modes 3+3 = the horizon pair $\Rightarrow$ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C]
`v134_dual_anchor.py`	Dual anchor lemma: $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$, $d\cdot\mathbf{1} = 0$, $d\cdot a = 1$, $(1,1,-2) = -2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison ($\Leftrightarrow$ orthogonality to $R^{-1}\mathbf{1} = \frac{1}{4}(1,1,-1)$; controls: $\mathbf{1}, e_1, n$ do *not* lie on the plane); (d, n) = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor$\leftrightarrow$horizon bridge [E]/[C]
`v135_det_surface.py`	Determinant surface: $\det M(s,t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ — two s-independent iso-volume walls $t = -2 \Rightarrow 2 = \|\mathbb{Z}_2\|$, $t = -1 \Rightarrow 4 = \|\mu_4\|$ ($\det K = 4$ is a *layer*); winding line $6s+8 \Rightarrow (8, 14, 20)$, $\det F = 32 = 2^{g_{\rm car}}$; $\det B(s,0) = 2 \det M(s,0)$ exactly, B-wall only at $t = -2$; row-sum axes $C = (7, 11, 13)$, $U = (3,3,3)$, $V = (1,2,3) = \mathrm{Spec}(Q_+)$; micro-identities $e_1^2 + e_2^2 = 41 = 10b_1$, $p_1^2 + p_2^2 = 52 =$ the carrier coset root count (v128) [E]/[C]
`v136_dual_normal_selector.py`	Dual-normal selector: (d, n) pins R *columnwise* — $d\cdot c_j = a_j$, $n\cdot c_j = 8\delta_{1j}$, kernel = primitive $(6, 8, 7)$, census: each column unique (1 of 216) $\Rightarrow$ R4′ shrinks from 6^9 to three column problems; the $A_0^\star$ zero mode carries $\sigma = (2, -9, 5) = (\|\mathbb{Z}_2\|, -N_{\rm fam}^2, g_{\rm car})$ ($\\|v\\|^2 = \frac{16}{9}$, $16 = \dim S^+$; $\mathrm{adj} = \frac{2}{9} P_v$ = the SdS curvature); $n\cdot\sigma = 121 = 11^2$; $W(A_3)$ orbit control: the naive lift $\sigma \to n$ fails — a genuine mechanism, not a group element [E]/[C]
`v137_qplus_cohomology.py`	The Q_+ grading is cohomology: $H^1(\mathbb{P} \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ ($\omega_k = z^{k-1} dz/(z^4 - 1)$, character i^k, residues $\zeta^k/4$); degrees $(1,2,3) = \mathrm{Spec}(Q_+) =$ the A_3 exponents ($h = 4 = \|\mu_4\|$); cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\} = \mathrm{spec}(A_0^\star) \Rightarrow$ MS weights = cusp classes = characters = Q_+: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E]/[C]
`v138_vw_firewall.py`	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\rm PI} = -\frac{98}{45} = \frac{22}{45} - \frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\rm edge} = -\frac{8}{3} = 2\times(-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom $\Rightarrow$ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E]/[C]

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Script	What it machine-checks *(continued)*
`v139_selector_triangle.py`	Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no R) $\Rightarrow$ the first selector is *derived*; the frame $(\mathbf{1},a,\sigma)$ has det $=11=\|\operatorname{Pl}(K)\|_1$, and n is the *unique* covector with atom pairings $(2,8,121){=}(\lvert\mathbb{Z}_2\rvert,\operatorname{rank}E_8,11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true n; review lift exact (audit): $n{=}\operatorname{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings **[E]**/**[C]** *(since v142: two — the third pairing is integrality-forced)*
`v140_canonical_map.py`	The canonical map exists and is rigid: the plain deck U (characters $(0,1,3)$) cannot match H^1; the *sheet-twisted* $-U$ ($(2,3,1)$) and the cusp exponential i^{Q_+} ($(1,2,3)$) both carry the H^1 character set $\{1,2,3\}$ (the v118 sign twist); multiplicity-freeness $\Rightarrow$ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q_+} pulls the Euler degree back to Q_+, $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice **[E]**/**[C]** *(the choice since derived, v141)*
`v141_deck_selection.py`	Deck Selection Theorem (R5): the established integer deck $G{=}T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ line with the self-conjugate character 2 $\Rightarrow$ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2,3,1)$ = chars$(-U)$ survives; i^{Q_+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ **[E]**/**[C]**
`v142_frame_integrality.py`	Frame Integrality (R4′): pairing triples of *integer* covectors against $(\mathbf{1},a,\sigma)$ form an index-11 sublattice ($x{-}3y{+}z\equiv 0 \bmod 11$, $11{=}\|\operatorname{Pl}(K)\|_1$); with $(x,y){=}(\lvert\mathbb{Z}_2\rvert,\operatorname{rank}E_8)$ the third pairing is forced $\equiv 0 \bmod 11$, primitivity forces even t; Cramer reductions for the two line pairings; R4′: three pairings $\to$ two **[E]**/**[O]** *(since v145: the values are atom identities — residue = one lift map)*
`v143_graded_frobenius.py`	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k{=}C_{-k}$ exact on all 240 roots $\Rightarrow$ Killing pairing nondegenerate per $g_k\times g_{-k}$; glue average = carrier projector (index $\lvert\mathbb{Z}_4\rvert{=}4$); each sector one $W(D_5){\times}W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement **[E]**/**[C]**
`v144_detratio_family.py`	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2-rigidity gives $r_b r_c{=}1{-}\Delta^2/3$ exactly $\Rightarrow$ det′-ratio = $(1{-}\Delta^2/3)^{4/3}$ (via $\zeta(0){=}{-}\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}{=}\lvert\mu_4\rvert(\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6\to\frac{14}{3}$ obstruction stated sharply **[E]**/**[C]** *(the bend since shown det′-clean, v147)*
`v145_pairing_atoms.py`	R4′ values are *atom identities*: with the lift $n{=}w_0(\sigma){+}\lvert\mu_4\rvert e_3$ ($w_0{=}$ the A_3 exponent duality), $n\cdot\mathbf{1}{=}\lvert\mu_4\rvert{-}\lvert\mathbb{Z}_2\rvert{=}\lvert\mathbb{Z}_2\rvert$, $n\cdot a{=}\lvert\mu_4\rvert a_3{=}8$ via the new closure $\lvert\mu_4\rvert{+}g_{\text{car}}{=}N^2_{\text{fam}}$ ($4{+}5{=}9$, $\sigma\perp w_0(a)$), $n\cdot\sigma{=}\|\sigma\|^2{-}N^2_{\text{fam}}{+}\lvert\mu_4\rvert g_{\text{car}}{=}121$; residue = derive the *one* lift map **[E]**/**[C]**
`v146_moebius_d4.py`	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z{\mapsto}iz,\ z{\mapsto}1/z\rangle$ on $H^1(\mathbb{P}^1\backslash\mu_4)$ matches the integer model $\langle G,T_A,\Sigma\rangle$ exactly — ι^* swaps $\chi_1{\leftrightarrow}\chi_3$, fixes χ_2 with $+1$, det $={-}1$ ($=T_A$ in the cusp basis); $\delta\iota{=}\Sigma$ class; parities match (v98) — the `GATE.QGEO` premise reduces to the module identification **[E]**/**[C]**
`v147_clock_gaussian.py`	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1{-}\alpha)$ (norm = area, v131) $\Rightarrow$ Born² Γ-ratio $(1{-}\alpha)^{p_2}$, rate = $-p_2\ln(1{-}\alpha)$ = the v127 ring sum term by term; the bend $\log_{3/2}3$ lives between the two Nariai levels of *one* geometry $\Rightarrow$ *det′-clean*; the $6{\to}\frac{14}{3}$ obstruction is localised to the $\alpha{=}0$ reference **[E]**/**[C]**
`v148_fock_census.py`	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the *even* classes $\{0,2\}{\times}\{0,2\}$ — the odd glue sectors $k(1,1)$ are *Ramond* modules; the R zero-mode module ($2^8{=}256$) splits 128+128 into the odd sectors of the *two* Lagrangian glues (v92): choosing the glue *is* choosing the R-projection (128= one $SO(16)$ chirality), $248{=}120_{\text{NS}}{+}128_{\text{R}}$; R2 is irreducibly a twisted-sector net statement **[E]**/**[C]**
`v149_cusp_normal.py`	Both dual normals are *anchor data* (R4′): in the cusp frame n pairs to $(6,3,5)=(p_2,p_0,e_2)(a)$ — one anchor invariant per Q_+ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2,8,121)$ frame data and the w_0-lift (v145); audit: $\sigma\to(5,1,2)$, sum $14=\dim G_2$; residue = *one* assignment **[E]**/**[C]**

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Script	**What it machine-checks** ***(continued)***
`v150_replica_eh_model.py`	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t-independent (image sum $(N^2-1)/(12N) = C(2\pi/N)$); the gap makes the ζ-variation *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma)\ln m$; linearisation $= (\ln m/12\pi)\int\sqrt{g}R$ — the EH form; target equation $k{=}c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit); Calderón transfer + normalisation stay open — `SEAM.THEOREM.01` narrows, does *not* close [E]/[O] *(Calderón transfer since answered — the DtN kernel is conically clean, v151)*
`v151_bfk_split.py`	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition *independent* ($C_N{=}C_D$, derived from reflection doubling) ⇒ the conical deficit of the Calderón/DtN jump determinant *vanishes identically*, $C_{\mathrm{DtN}}{=}C_{\mathrm{cone}}(\gamma){-}2C_D(\gamma/2){=}0$; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the *local* half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]
`v152_norm_is_anchor.py`	R3's normalisation *is* the dimensionful anchor: the EH coefficient is $k = \ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; $k{=}c_3/2 \Leftrightarrow m/\mu{=}e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), *not* a separate gap; the irreducibles stay {one scale, π}; audit: the log-ratio $= \frac{3}{4} = q(A_3)$ is the target value, NOT a derivation [E]/[O]
`v153_no_unit_theorem.py`	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L \mapsto \lambda L$, a mass / $1/G$ is not, so a dimensionless compiler *cannot* select an absolute scale; the three residuals collapse — $U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π} [E]/[O]
`v154_simple_current_theorem.py`	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has index $\lvert L\rvert{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/\lvert L\rvert^2{=}16/16{=}1$ ⇒ holomorphic ⇒ $B{=}(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]
`v155_quasifree_boundary.py`	Quasi-free in ⇒ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$ (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) *follows* from 'the seam bulk is a reflection-positive free field' — the *same* premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]
`v156_seam_net_construction.py`	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k{=}\lvert k\rvert$ (free chiral dispersion); 16 free Majoranas give $c{=}8{=}5{+}3$; the E_8 currents are $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1{+}248q{+}4124q^2{+}\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a *checked* identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]
`v157_rigid_fixed_point.py`	Freeness as a *rigid fixed point* (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $\lvert k\rvert$ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c{=}8$ theory has *no* marginal $(1,1)$ field (all $\bar{h}{=}0$) ⇒ rigid, isolated, no room for an interaction; (iii) the same $\lvert\mathbb{Z}_2\rvert$ that halves 8π in c_3 IS the Ramond projection giving $\mu{=}1$; (iv) holomorphic $c{=}8$ unique $= (E_8)_1 =$ free. Premise (A) shrinks from 'prove Gaussianity' to 'the gapped flow reaches the holomorphic fixed point' [E]/[C]
`v158_fixed_point_stable.py`	The free chiral $c{=}8$ fixed point is *stable* (exact dimension count, the finite core of (A)): Majorana $h{=}\frac{1}{2}$, bilinear current $h{=}1$, quartic $h{=}2$; the relevant window $(0,1)$ for bosonic operators is *empty*, the $h{=}1$ currents are chiral (not $(1,1)$-marginal, v157), the lowest interaction (quartic) is $h{=}2{>}1$ irrelevant ⇒ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]

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Script	What it machine-checks *(continued)*
`v159_pyrate_gauge_crosscheck.py`	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$ by the textbook one-loop formula; $b_1{=}\frac{41}{10}$ holds *three* independent ways — carrier algebra $10b_1{=}g_{\text{car}}2^{g_{\text{car}}-2}{+}1{=}41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) which writes $\beta_{g_1}{=}(\frac{41}{10})g_1^3$ verbatim; the $M{=}41$ split $10b_1{=}40_{\text{(ferm)}}{+}1_{\text{(Higgs)}}$ reproduces v13 Gate 1; evidence in `verification/pyrate`. The transfer functor's gauge inputs are not free knobs [E]/[C]
`v160_seam_gaussianity_from_pf.py`	Seam Gaussianity as a *conditional* fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free ⇒ all higher connected cumulants $\kappa_{2n}{=}0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality ⇒ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a SECOND fixed ray, breaking uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0 — the value $(2/3)^6$ only sets the rate. **The load** [O]: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}{=}1820$, $\binom{16}{6}{=}8008$ cumulant directions ⇒ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum' [E]/[O]
`v161_one_particle_reduction.py`	The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T{=}\Gamma(t)$, the second quantisation of a one-particle contraction t; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\text{spec}(\Gamma(t)\|_{\deg m})$ = products of m one-particle eigenvalues [E]; (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization ⇔ one state) [E]; (4) quasi-freeness *forced* not assumed — the 120= dim $\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. **Residual** [O]: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O]
`v162_seam_transport_identification.py`	The final identification, forced as far as it goes — the v161 one-particle statement carries *no* new open content: (1) the gap $(2/3)^6$ is *forced* — cusp weights = $\text{spec}(A_0^\star) = \{0, \frac{1}{3}, \frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1{-}\alpha)^{p_2}$, $p_2{=}6{=}\|R_+(A_3)\|$, all carrier data [E]; (2) μ_4-equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k} = \|\mu_4\| \,\text{diag}(X)$, commutant of U = the diagonal algebra — so the gapped μ_4-equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ [E]; (3) the cusp grading *is* the A_3 grading on the 16 Majoranas (v137) [E]; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) [E]. **Closure** [C]/[O]: with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the *already-open* A2 (net existence) and R1 (seam clock, `HOR.CLOCK`) — so (A) factors exactly into A2 + R1, adds *no* new open item, and is closed conditionally on them (a unification, not an unconditional proof) [E]/[O]
`v163_rg_stability_flavor.py`	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0}{2}{=}0.3067$, $\sin^2\theta_{13}{=}\varphi_0 e^{-5/6}{=}0.0231$) are *RG-stable* — PMNS running is y_τ^2-suppressed ($\sim 1.8{\times}10^{-5} \ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b{\sim}41$) run through $y_t^2 \sim O(10\text{–}15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2}YY^\dagger Y$ carries $y_{\max}^2$ (PyR@TE SM RGEs) [E]/[C]

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Script	**What it machine-checks** ***(continued)***
`v164_qcd_scale.py`	the QCD confinement scale is parameter-free from the carrier $b_3{=}{-}7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b){\sim}0.22$, $\alpha_s(m_c){\sim}0.38$ and confinement ($\alpha_s{\sim}1$) at $\sim$0.5 GeV — so the QCD-scale *numerator* of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest 'Open / not forced' frontier non-claim ($m_p{=}O(1)\Lambda$ is lattice) [E]/[O]
`v165_premise_a_closure.py`	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries *zero* independent open content: (1) A2 (net existence) closes by *assembly* of established mathematics [C]— 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c{=}8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B){=}16/16{=}1$, 248=120+128, character $E_4/\eta^8{=}j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $= \lvert k\rvert$ (v156/v157); (2) R1 (seam clock) reduces to `GATE.QGEO` [C]— μ_4-equivariance forces diagonal-in-cusp, the flat μ_4-equivariant generator is *unique* $= A_0^\star$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = 'seam boundary = the μ_4-punctured sphere $\mathbb{P}^1\backslash\mu_4$' = `GATE.QGEO` (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\rm geo}\}$, a *theorem* by the No-Unit Theorem (v153). **Final accounting**: (A) = A2 assembly [C] + R1=`GATE.QGEO` + the $\{\pi, v_{\rm geo}\}$ core ⇒ ZERO independent open gates; a unification + re-typing, not an unconditional proof [E]/[O]
`v166_higgs_free_seam.py`	the Higgs quartic from the *free seam* (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\rm seam}){=}0$ *and* $\beta_\lambda(M_{\rm seam}){=}0$ (Shaposhnikov–Wetterich double criticality, here *motivated* not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\rm Pl}){\sim}0.002$, $\beta_\lambda(\bar{M}_{\rm Pl}){\sim}0$ — the famous SM near-criticality, now *explained* by the free seam (2-loop sharpens $\lambda(\bar{M}_{\rm Pl})$ from $\sim$0.022 to $\sim$0.002). The double condition predicts $m_H{\sim}129$–134 GeV; at the scalaron scale it gives $\sim$107 GeV (too low) ⇒ the BC lives at $\bar{M}_{\rm Pl}$ (seam=Planck) [E]/[C]
`v167_inventory_post_closure.py`	The terminal residual inventory (post premise-(A) closure; line v105 → v123 → v167): after the (A)-chain (v160–v165), premise (A) is *no longer* an independent structural class (`GATE.METRIC.16-19`). The residual collapses from v123's FIVE classes to: **Tier 0** — the irreducible core $\{\pi, v_{\rm geo}\}$, a *theorem* (No-Unit, v153), nothing to close; **Tier 1** — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P}\setminus\mu_4$, cohomology $b_1{=}N_{\rm fam}{=}3$); **Tier 2** — folded ($\to v_{\rm geo}$, $\to G_{\rm net}$); **Tier 3** — $F_{\rm transfer}$, one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
`v168_qgeo_rigidity.py`	The QGEO *Rigidity Theorem* — the finite half of the Tier-1 hinge `GATE.QGEO`, hardened. Exact: $\mu_4{=}\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z{\to}iz, z{\to}1/z\rangle$ of order $8{=}\lvert D_4\rvert$ (faithful) ⇒ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4{-}1{=}3{=}N_{\rm fam}$; the forms $\omega_k{=}z^{k-1}dz/(z^4{-}1)$ ($k{=}1,2,3$) are μ_4-eigenforms with $z{\to}iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3, weights $(1,2,3)$ = A$_3$ exponents $= {\rm Spec}(Q_+)$. `QGEO.RIGID.01` [E] (math half closed); `QGEO.REALIZE.01` [C] (seam-collar = $\mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]
`v169_etaB_boltzmann_interface.py`	The η_B leptogenesis transfer *operationalised* (Tier-3 $F_{\rm transfer}$, not a new class): type-I thermal leptogenesis $\eta_B{\sim}0.96{\times}10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}{=}240°$. Davidson–Ibarra $\varepsilon_1{=}\frac{3}{16\pi}M_1 m_3/v^2{\sim}8.5{\times}10^{-7}$ for $M_1{=}10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1){\sim}0.03$–0.19. **Kill test**: over natural $M_1{\in}[3{\times}10^9, 3{\times}10^{10}]$ GeV and washout, η_B spans $[8.4{\times}10^{-11}, 4.7{\times}10^{-9}]$ which *brackets* the observed $6.1{\times}10^{-10}$ (canonical point gives $6.0{\times}10^{-10}$, not tuned) ⇒ the route is viable; M_1/washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]

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Script	What it machine-checks *(continued)*
`v170_diamond_slice_compression.py`	E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P{=}(3,4,6,10)$ ($p_n{=}2{+}2^n$) and Sheet-Diamond determinants $D{=}(3,4,8,14,20,32)$. The K_4 edge products $(12,18,30,24,40,60)$ are all carrier blocks; $\sum \det M = 81 = N_{\rm fam}^4$; all seven 248-slices follow from $P, D, \Delta{=}13$ and $(e_1, e_2, \dim S^+)$; one affine map $\mathrm{row}(C{+}sU{+}tV) = (7,11,13){+}s(3,3,3){+}t(1,2,3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit **[E]**
`v171_os_moment_cluster.py`	Atomic OS Moment Lemma + Sugawara Gap Safety: $\mathrm{spec}(T){=}\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n){=}A{+}Br^n{+}Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram ⇒ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi{=}0.41105$, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety $\Delta_{\rm eff}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{>}0$ with $31{=}1{+}h^\vee(E_8)$, $c((E_8)_1){=}248/31{=}8$ **[E]**/**[C]**
`v172_trace_anomaly_seed.py`	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 ($\chi{=}2$) the $(E_8)_1$ ($c{=}8$) Weyl anomaly is $c\chi/6{=}\frac{8}{3}$, halved by the one-sided $\lvert\mathbb{Z}_2\rvert$ to $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3{=}\frac{1}{6\pi}$. Plus QCD $b_3{=}11{-}\frac{2}{3}N_f{=}7$ = scalaron exponent $\Omega_{\rm adm}{-}10b_1{=}48{-}41$ (confinement↔inflation), and the Volkov–Wipf factorisation $-98/45{=}-2\cdot 7^2/\dim(D_5)$ **[E]**
`v173_pfaffian_cp_car.py`	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\rm eff}{=}0$ — the sheet-odd Calderón involution $\{D,\Sigma\}{=}0$ pairs the spectrum $(\pm\lambda)$, γ_5-Hermiticity $D^\dagger{=}\gamma_5 D\gamma_5$ makes the Pfaffian real ($\arg \in \{0,\pi\}$), and reflection positivity ($Z{>}0$) selects the positive branch ⇒ $\theta_{\rm eff}{=}0$ **[E]**
`v174_seam_fock_readings.py`	Seam Fock-space readings: (1) CAR cone compression — $\dim\Lambda^{\rm even}(\mathbb{R}^{16}){=}2^{15}{=}32768$ reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16{=}2^{11}$), the explicit count behind `v161`; (2) Witten index = the Plücker norm $11 = \sum_{k\le 2}\binom{4}{k}$, with $11 = \dim S^+{-}g_{\rm car}$ (full $2^4{=}16$) and $c_u/c_d{=}55/117$ reading the QBL-protected boundary state space; (3) Affleck–Ludwig g-theorem $c_{UV}{=}5{+}3{=}8{=}c_{IR}((E_8)_1)$ so $\Delta c{=}0$ forces an exact isometry (boundary-entropy form of `v157`/`v158`) **[E]**
`v175_net_existence_full_cone.py`	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to **[E]**, isolating `QGEO.REALIZE.01` as the single open premise (nothing fabricated): (1) **full-cone RP for all** m — the CAR second-quantisation functor $\Gamma(t){=}\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16}{=}65536$); the exterior-power spectrum theorem makes every eigenvalue a subset product of $\mathrm{spec}(t)$, verified on all 2^{16} products to lie in $[0,1]$ (contraction), with eigenvalue-1 multiplicity $2^8{=}256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces *for all* m to the one-particle contraction (Shale–Stinespring CAR functor), discharging the `v160` scope premise by a theorem; (2) **A2 net existence** assembled + verified — $(E_8)_1$ character $E_4/\eta^8{=}1{+}248q{+}4124q^2{+}34752q^3{+}213126q^4{+}1057504q^5$ (level-1=248), embedding 248=120+128, E_8 Cartan even unimodular (det 1); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = `QGEO.REALIZE.01`, finite half `v168`, realisation half left **[O][E][O]**

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Script	**What it machine-checks *(continued)***
`v176_seam_collar_realisation.py`	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — *not* a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible c=8 data, *then* its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already **[E]**: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, b_1=N_{fam}=3), Calderón DtN symbol $\lvert k\rvert$ (v156), H^1 character = generation grading $(1,2,3)$=A_3 exponents=$\mathrm{Spec}(Q_+)$ (v137/v168), μ_4-equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1\otimes(A_3)_1\rtimes\mu_4$=$(E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise `QGEO.REALIZE.01` (the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$), left honestly **[O][E][O]**
`v177_seam_marking_kernel.py`	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed *constructively* **[E]** and the open residual sharpened into exactly TWO named obligations. **[E]** UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z\mapsto iz$, a non-fixed orbit scales to μ_4, the reflection $z\mapsto 1/z$ gives $\sigma\rho\sigma=\rho^{-1}$ (D_4, order 8) — a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P},\mu_4)$; **[E]** COHOM: $\rho^*\omega_k=i^k\omega_k$ so the H^1 grading is $(1,2,3)$=A_3 exponents=$\mathrm{Spec}(Q_+)$; **[E]** MODULE: the H^1 →generation iso is *unique* (multiplicity-free + residue normalisation; reflection $\omega_1\leftrightarrow\omega_3$, ω_2 fixed); **[E]** NET (v154/v175): index-4 μ_4 extension of $(D_5)_1\otimes(A_3)_1$=$(E_8)_1$. The TWO genuine open obligations **[O]**: `QGEO.MARKS.01` (raw seam → genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (raw seam Calderón kernel = the μ_4-equivariant free gapped contraction as an *operator*); the whole structural residual is now exactly these two doors **[E][O]**
`v178_marks_kernel_attempt.py`	An honest *attempt* at the two open obligations `QGEO.MARKS.01` + `QGEO.KERNEL.01` — *not* a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed **[E]** and each obligation reduced to ONE sharper irreducible premise **[O]**. *Marks core* **[E]**: a genus-0 double with an order-4 clock ρ ($\sim z\mapsto iz$, fixed points $\{0,\infty\}$) and reflection τ forces the marks to be a generic ρ-orbit $\{1,i,-1,-i\}$=μ_4 (not the fixed points), distinct iff b_1=4−1=3=N_{fam}, $\tau\rho\tau$=ρ^{-1} ⇒ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* **[E]**: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced diagonal and *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $\lvert k\rvert$, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened **[E][O]**

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Script	**What it machine-checks *(continued)***
`v179_conformal_realisation.py`	The two residual premises (Marks-residual + Kernel-residual) collapse to ONE geometric premise `QGEO.CONF.01` (honest unification; the conformal realisation stays **[O]**). Marks's residual is not three independent assumptions: (1) genus-0 *is* P1 — $c_3=1/(\lvert\mathbb{Z}_2\rvert\,2\pi\,\chi(S^2))=1/(8\pi)$ with $\chi=2 \Rightarrow g=0$, the same $\chi=2$ that gives the '8' (v58/v73); (2) the order-4 clock *is* the carrier — Coxeter element of $W(A_3)=S_4$, order $h(A_3)=4=\lvert\mu_4\rvert$ (v117); (3) marks = one orbit is *forced* — parabolic marks $\neq$ the two elliptic fixed points $\{0,\infty\}$, a free order-4 orbit has exactly $4=e_1(a)$ points. So Marks reduces to 'the carrier clock acts conformally on the genus-0 double'. Kernel then follows from that geometry: DtN symbol $\lvert k\rvert$ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c=8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1=\mathbb{C}^3=N_{\text{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise `QGEO.CONF.01`: the raw RP seam double is conformally $(\mathbb{P},\mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly **[O][E][O]**
`v180_clock_is_mobius.py`	Cracking `QGEO.CONF.01` one level further — the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, non-circular premise `QGEO.ISO.01`: 'the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) *is* $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)=4$) is topologically $z\mapsto iz$; (3) isometry⇒conformal⇒Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P})=\mathrm{PSL}(2,\mathbb{C})$; (4) order-4 Möbius $= z\mapsto iz$ (verified: projective order 4, fixed $\{0,\infty\}$, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the milder isometry premise; the new residual `QGEO.ISO.01` (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still **[O][E][O]**
`v181_clock_is_conformal_symmetry.py`	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise `QGEO.SYM.01` — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z\mapsto iz$. The clock is the Coxeter element of $W(A_3)=S_4$ (v117), order $h(A_3)=4=\lvert\mu_4\rvert$, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below `QGEO.SYM.01` the residual is *definitional*, not a missing theorem: 'the seam's conformal structure *is* the μ_4-deck structure of the carrier clock' (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain realize(v175)→theorem(v176)→marks+kernel(v177)→finite cores(v178)→`CONF.01`(v179)→`ISO.01`(v180)→`SYM.01`(v181) has reached bedrock; we stop here honestly **[E][O]**

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Script	**What it machine-checks** ***(continued)***
`v182_reviewer_residual_map.py`	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \to 4D$) **[E]**/**[C]**: the Plücker readout $11 = \sum_{k\le 2}\binom{4}{k} = \dim S^+ - g_{\mathrm{car}} = 16-5$ (QBL count, v174) and the holographic reduction give $A = \texttt{QGEO.SYM.01} + F_{\mathrm{transfer}}$. B (UV gravity) **[E]**: 'gravity = geometric boundary readout' *is* `QGEO.SYM.01`. C (grammar tuning) **[E]**/**[C]**: frozen (v84) + minimal + over-determined (the anchor $a{=}(1,1,2)$ in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) **[E]**/**[C]**: the Möbius nature is forced by $\mathrm{Aut}(\mathbb{P}){=}\mathrm{PSL}(2,\mathbb{C})$ on the deck, $t{\sim}2.84$ only locates the point, and the missing RG generator *is* F_{transfer}, so $D = \texttt{QGEO.SYM.01} + F_{\mathrm{transfer}}$. NET: three reduce onto $\{\texttt{QGEO.SYM.01}, F_{\mathrm{transfer}}\}$, one waits for data — a unification of the concern structure, not a closure (premises stay **[O]**/**[C]**) **[E][C]**
`v183_koide_f_corner_transfer.py`	The Koide source→pole factor $\frac{53}{54}$ has an *operator* origin (external-review proposal, validated): $u_{\mathrm{pole}}/u_{\mathrm{source}} = a^T(R{+}Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. (1) $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6\,78 + \det R\,8 + 2{\cdot}27{\cdot}N_{\mathrm{fam}}$); (2) $F{=}R{+}Q$ is the *missing* Sheet-Diamond corner $M(1,1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R{+}Q)\mathbf{1} = 53 = 52{+}1$; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\mathrm{fam}}^3}$ exactly ($54{=}2{\cdot}27$); (4) negative control: only F gives 53 ($R/Q/K/L \to 32/21/35/56$). Upgrades `FR.KOIDE.03` from a bare guess to an operator identity; **[E]** the factor is exact, **[C]** the leptonic source→pole reading the F corner stays an F_{transfer} conjecture **[E][C]**
`v184_etaB_anchored_boltzmann.py`	Honest test of the external-review η_B proposal ($M_1{=}M_R\phi_0^4$, $\tilde m_1{=}m_3/A_\Lambda$) — firewall verdict: *sharper scenario*, not a closure. The washout anchors plausibly ($\tilde m_1{=}m_3/A_\Lambda{=}m_3/10 \approx 5\,\mathrm{meV}$, $A_\Lambda{=}10{=}\lvert E(K_5)\rvert$, in the strong-washout window) **[C]**; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3 \approx 6{\times}10^{14}$ GeV and the nearest clean TFPT powers of $\bar M_{\mathrm{Pl}}$ both miss it (c_3-power off $\sim$75%, ϕ_0-power off $\sim$40%), so $M_1{=}M_R\phi_0^4$ merely relocates the free input $M_1 \to M_R$ (the reviewer's $6.0{\times}10^{-10}$ uses $M_R{=}1.3{\times}10^{15}$; the seesaw value gives $3.4{\times}10^{-10}$, the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays **[C]**, a sharper scenario **[C]**
`v185_axion_relic_solver.py`	Axion relic, honest misalignment estimate: the determinant-line axion $f_a{=}M_{\mathrm{scal}}/128 \approx 2.39{\times}10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) can provide all dark matter, but *only* near the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a **[C]** scenario. The harmonic misalignment under-produces at this f_a (prefactor $\sim 0.026/\theta^2$, $\theta{\sim}O(1)$ gives $\sim$21% of Ω_{DM}, need $\theta^2_{\mathrm{eff}}{\sim}4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\mathrm{relic}}(f_a,\theta_i,N_{\mathrm{DW}}) \overset{?}{=} \Omega_{\mathrm{DM}}$ is a kill test for the *branch*, not the theory (no free singlet WIMP: $128{=}(16,4){+}(16,4)$) **[C][O]**
`v187_ftransfer_laws.py`	The F_{transfer} discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B, the axion relic, $m_p/m_e{+}\Lambda_{\mathrm{QCD}}$ — must *never* be typed as primitive **[E]** compiler predictions; each carries a **[C]**/**[O]** marker. The guard reads `status_ledger.csv` and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3{=}{-}7$ v159 — may be **[E]**, the physical prediction never is). Records the four interfaces $F_{\mathrm{pole}}/F_{\mathrm{Boltzmann}}/F_{\mathrm{relic}}/F_{\mathrm{QCD}}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output **[E]**

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Script	What it machine-checks *(continued)*
`v188_frontier_wording_guard.py`	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It forbids stale sentences whose semantics an earlier round superseded — "leptogenesis interface ([E])" (v184 retyped Route B to [C]), "closed branch also fixes the leptogenesis inputs" (M_R is the seesaw scale, not a compiler power), "relic scale f_a, m_a open" (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), "computed Q=0.664... near, not exact" (v183 gave 53/54 an exact operator origin), "Suite now 187 modules" (the count is 187, highest ID `v188`; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
`v189_riemann_roch_carrier.py`	The Riemann–Roch carrier reading: $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5$ (function side) and rank $H_1(\mathbb{P}^1 \setminus \mu_4) = \deg - 1 = 3$ (cycle side, already `QGEO.COHOM.01`/v177); $g_{\text{car}} = 5 = \text{rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, rank D_5+ rank $A_3 = 5{+}3 = 8$, v1), $N_{\text{fam}} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by $\text{rank}(D_5) = 5$, but the physical identification is conjectural and does *not* replace the over-determined $g_{\text{car}} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
`v190_nariai_entropy_bound.py`	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\text{tot}}/S_{dS} = (x^2{+}1)/\Phi_3(x)$ ($x{=}r_b/r_c$; $\Phi_3{=}x^2{+}x{+}1$ the $N_{\text{fam}}{=}3$ cyclotomic, already in `tfpt_horizon_readouts`) satisfies $S_{\text{tot}} \geq \frac{2}{3} S_{dS}$ with equality exactly at the Nariai merge $x{=}1$, via the perfect square $3(x^2{+}1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x{>}0$ is $2/3 = \lvert Z_2\rvert/N_{\text{fam}}$) [E]
`v191_universal_branch_line.py`	The universal branch line, honestly typed [C] — an exact affine *relabeling*, not a theorem. The map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $m = \pm 1/N_{\text{fam}}$ onto the flavor branch points $\{2, 5\} = \{\lvert Z_2\rvert, g_{\text{car}}\}$ ($m{=}0{\to}7/2$). The arithmetic is exact, but the map exists for *any* two pairs of points (two intervals are always affinely equivalent), so the slope $N_{\text{fam}}^2/2$ and midpoint 7/2 are *determined* by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = \lvert Z_2\rvert/N_{\text{fam}}$ ramification of the mass↔transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]
`v192_qgeo_energy_clock.py`	The energy-conserving-clock reformulation, honestly typed: `QGEO.ENERGY.01` ($\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) *relocates* the bedrock `QGEO.SYM.01` to a more checkable operator identity, but does *not* eliminate it. The downstream chain (conformality → Möbius → $z \mapsto iz \to \mu_4 \to$ `QGEO.SYM.01`) already exists (v180/v181); the new upstream link (energy invariance ⇒ conformality in 2D) is checkable on the DtN operator $\Lambda = \lvert k\rvert$. But 'ρ preserves Λ' is plausibly equivalent to 'ρ is the conformal deck', so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays [O]
`v193_qgeo_energy_commutator.py`	`QGEO.ENERGY.02`: the energy-*commutator* contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the *raw* RP seam. Three sub-claims: (1) [E] ρ is carrier-defined (Coxeter of $W(A_3){=}S_4$, order $4{=}\lvert\mu_4\rvert$, v117), not imported from the seam; (2) [O] the non-circularity hinge — Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] $[\rho, \Lambda_\Sigma]{=}0$ forces the $(1, 2, 3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider: the induced $k{=}(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m{=}q(A_3){=}\frac{3}{4}$ (FAMILY), not $q(D_5){=}\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target [O]; if proven, `QGEO.SYM.01` becomes an energy-rigidity theorem [O]

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Script	**What it machine-checks *(continued)***
`v194_raw_seam_dtn_rp.py`	`QGEO.DTN.01` — subclaim 2 of v193 ("the raw RP-seam DtN is RP-definable") tackled: a *reduction*, not a closure. (1) **[E]** finite core: $\rho{:}z{\mapsto}iz$ on $H^1{=}\langle w_1, w_2, w_3\rangle$, $w_k{=}z^{k-1}dz/(z^4{-}1)$, acts as $\rho^* w_k{=}i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma]{=}0$ on H^1 (v177). (2) **[E]** hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1\backslash\mu_4$. (3) **[O]** residual relocates: full-L^2 commutation $\Leftrightarrow$ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). `QGEO.SYM.01` reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still **[O]**, not closed **[O]**
`v195_marks_lefschetz_character.py`	`QGEO.MARKS.02` — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are *forced* to be a free μ_4 orbit. **[E]**: $\rho{:}z{\mapsto}iz$ (order 4) fixes only $\{0, \infty\}$; $H^1{=}\chi_1{+}\chi_2{+}\chi_3$ (the A_3 exponents $(1, 2, 3)$, v177) has no trivial character $\Rightarrow$ no puncture at a fixed point; rank $H_1{=}n{-}1{=}3 \Rightarrow n{=}4 \Rightarrow$ exactly one free orbit $\Rightarrow$ marks $= \mu_4$ (cross-ratio 2); $\mathrm{Tr}(\rho\vert H^1){=}i{+}i^2{+}i^3{=}{-}1$, $L(\rho){=}1{+}1{=}2{=}\#\mathrm{fixed}$. **[O]** residual relocates from "the seam is $\mathbb{P}\backslash\mu_4$ (geometric)" to "H^1 carries the A_3-exponent rep" — less circular **[E][O]**
`v196_seam_energy_functional.py`	`QGEO.VARI.01` — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/`ENERGY.02` target: $E_{\mathrm{fail}} = \Vert[\rho, \Lambda_\Sigma]\Vert^2_{HS} + \Vert\rho^4{-}1\Vert^2 + \Vert\Theta\rho\Theta - \rho^{-1}\Vert^2$, with $E_{\mathrm{fail}}{=}0 \Leftrightarrow$ the `QGEO.SYM.01` conditions. **[E]** on the finite H^1 block ($\rho{=}\,\mathrm{diag}(i, -1, -i)$, Λ diagonal, $\Theta{=}$conj) all three vanish $\Rightarrow E_{\mathrm{fail}}{=}0$; **[E]** a μ_4-breaking Λ or non-order-4 ρ gives $E_{\mathrm{fail}}{>}0$ (μ_4 a true minimiser); **[E]** the zero locus is the seam-deck conditions; **[O]** the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure **[E][O]**
`v197_rr_carrier_clifford_d5.py`	`ARCH.RRCAR.02` — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space $C_{\mathrm{car}}{:=}H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10){=}D_5$ half-spinor by its even Clifford algebra. **[E]** $\dim\Lambda^{\mathrm{even}}(\mathbb{C}^5){=}\binom{5}{0}{+}\binom{5}{2}{+}\binom{5}{4}{=}1{+}10{+}5{=}16{=}2^{g_{\mathrm{car}}-1}{=}\dim S^+$; the $\mathfrak{so}(10){=}D_5$ half-spinor $2^4{=}16{=}S^+{=}\Lambda^{\mathrm{even}}(C_{\mathrm{car}})$; closes the chain $\mu_4{\to}g_{\mathrm{car}}{\to}D_5$ from one four-point divisor (cycle side rank $H_1{=}3{=}N_{\mathrm{fam}}(A_3)$; function side $h^0{=}5{=}g_{\mathrm{car}}$; Clifford spinor $16{=}D_5$ half-spinor $=$ carrier). **[C]** "carrier $=$ Clifford spinor of the seam mode space" stays physical (as v189), does not change the over-determined $g_{\mathrm{car}}{=}5$ **[E][C]**
`v198_modular_commutator_reduction.py`	`QGEO.MODULAR.01` — cracking the bedrock to a state-invariance. The v194 residual ($[\rho, \Lambda_\Sigma]{=}0$ on full L^2, framed as Bisognano–Wichmann with a circularity worry) is (a) made *exact* at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. **[E]** the DtN principal symbol $\vert k\vert{=}\sqrt{-d^2}$ is $\mathrm{diag}(\vert n\vert)$ and the clock $\rho{:}z{\mapsto}iz$ is $\mathrm{diag}(i^n)$ in the Fourier basis — both diagonal $\Rightarrow [\rho, \vert k\vert]{=}0$ exactly on *all* of L^2; **[E]** Tomita–Takesaki: $[\rho, \Lambda_\Sigma]{=}0$ follows from $\omega{\circ}\rho{=}\omega$ (the modular Δ is intrinsic to (M, Ω); a state-preserving symmetry commutes with it), needing *no* conformal covariance — removes the BW circularity; **[E]** the finite H^1 block is already μ_4-invariant (v177). **[O]** residual: the full raw seam state is μ_4-invariant ($\omega{\circ}\rho{=}\omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant $=$ operational `QGEO.SYM.01`. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) **[E][O]**

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Script	**What it machine-checks *(continued)***
`v199_seam_state_invariance.py`	`QGEO.STATE.01` (work package A) — attacking $\omega\circ\rho=\omega$ directly on the raw seam: a further localising reduction, not a closure. **[E]** setup non-circular (ρ = Coxeter of $W(A_3)=S_4$, $\rho^4=1$, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); **[E]** ground-state reduction $\omega\circ\rho=\omega \Leftrightarrow [\rho, H_1]=0$ ($C_\Sigma=\frac{1}{2}(1+\operatorname{sgn} H_1)$); **[E]** character criterion: $[\rho, H_1]=0 \Leftrightarrow H_1$ block-diagonal in the four μ_4-character classes $\{n=r \bmod 4\}$; principal symbol $\lvert k\rvert=\operatorname{diag}(\lvert n\rvert)$ passes automatically. **[O]** residual: the bounded sub-principal symbol has *no* off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure **[E][O]**
`v200_seam_variational_scan.py`	`QGEO.VARI.02` (work package B) — the discretised variational test: v196 had $E_{\text{fail}}=0$ at the μ_4 config on the fixed block; here the clock angle is free and E_{fail} is minimised over it. **[C]** scanning $E_{\text{fail}}(\theta)=\lVert\rho(\theta)^4-1\rVert^2$+faithfulness over $\theta\in(0,\pi)$ gives the unique minimum at $\theta=\pi/2$ ($E_{\text{fail}}=0$: order-4 *and* three distinct characters $i,-1,-i$ = the $(1,2,3)$ grading); decoys $(0,\pi,\pi/3,2\pi/3)$ all lose. So $\arg\min E_{\text{fail}}$ = the μ_4 normal form. **[E]** consistent with v196/v199/v198. **[O]** a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) **[C][O]**
`v201_seam_subprincipal_marks.py`	`QGEO.SUBPRIN.01` — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is *not* an independent postulate. **[E]** the seam DtN is $\Lambda=\lvert k\rvert+M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n\lvert M_f\rvert n'\rangle=f_{n-n'}$. **[E]** M_f is μ_4-character-block-diagonal $\Leftrightarrow f$ has Fourier support only on modes $\equiv 0 \pmod 4$ $\Leftrightarrow f$ is $\mathbb{Z}_4$-invariant ($\lvert k\rvert$ already commutes, v198). **[E]** a μ_4-mark sum $f(\theta)=\sum_{j=0}^{3} g(\theta-2\pi j/4)$ has $f_m=4g_m[m\equiv 0 \bmod 4]$ for *any* profile g — automatically $\mathbb{Z}_4$-invariant, so (v195 marks=μ_4 orbit) *forces* the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. **[O]** new residual: $\omega\circ\rho=\omega$ reduces to 'the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)', the narrowest, structurally-definitional form of `QGEO.SYM.01` **[E][O]**
`v202_rare_kaon.py`	The rare kaon sector $K\to\pi\nu\bar\nu$ from the TFPT CKM point (**[C]** downstream flavor readout, *not* a compiler power). The kernel fixes the full CKM matrix ($s_{12}=\lambda_C$, $s_{23}=\varphi_0/(1+\lambda_C)=\lvert V_{cb}\rvert$, $s_{13}=\lambda_C^3/3$, $\delta=\pi/3+3\lambda_C^2=1.19823$ rad; v18/v88) with no flavor fit; feeding the exact point into the standard Brod–Gorbahn–Stamou short-distance formulas gives $\text{BR}(K^+\to\pi^+\nu\bar\nu)=9.45\times10^{-11}$ and $\text{BR}(K_L\to\pi^0\nu\bar\nu)=3.33\times10^{-11}$. **[E]** CKM point + exact apex $(\bar\rho,\bar\eta)=(0.1374, 0.3509)$, $\lambda_t=V_{ts}^*V_{td}$; **[C]** the branching ratios (*not* parameter-free — $X_t, P_c, \kappa_\pm$ are external EW/QCD input); **[E]** $\text{BR}(K^+)$ $\text{BR}(K^+)=9.45\times10^{-11}$ lands ON the NA62 2016-2024 combination $(9.6^{+1.9}_{-1.8})\times10^{-11}$ (La Thuile 2026), $+0.08\sigma$ (the 2016-2022 obs $(13.0^{+3.3}_{-3.0})\times10^{-11}$ was $\sim1.2\sigma$ above) — a strong consistency hit, but an F_{transfer} bridge, not a unique TFPT-vs-SM discriminator (SM $(8.6\pm0.4)\times10^{-11}$ also inside the NA62 error); **[E]** Grossman–Nir ratio $0.352 \ll 4.3$; **[X]** dated kill test: a stable NA62 $\text{BR}(K^+)$ outside $[7,12]\times10^{-11}$, or a KOTO-II $\text{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched) **[C][X]**

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Script	**What it machine-checks *(continued)***
`v203_eht_achromatic.py`	The EHT achromatic polarization-rotation intercept β_{BH} around a horizon-scale black hole — the surviving core of the old UFE/BH notes (*not* the RN-type metric, superseded by Nariai/seam=horizon v101–v104/v190; only the polarization signature). [E] the coupling is a compiler number: $\beta_{\mathrm{BH}}(r)=Q_e^{\mathrm{eff}}Q_m^{\mathrm{eff}}/(256\pi^4r^2)=16c_3^4(Q_eQ_m/r^2)$, $16c_3^4=1/(256\pi^4)$ exactly. [E] α-kernel cross-link: $16c_3^4=\delta_{\mathrm{top}}/3$, $\delta_{\mathrm{top}}=48c_3^4=3/(256\pi^4)$ — the *same* top-form coefficient that fixes the α-kernel precision-zone correction (one row from $\beta_{\mathrm{rad}}=\varphi_0/4\pi$). [C] TFPT fixes the $1/r^2$ profile, achromaticity, and sign-flip under $E{\cdot}B$ reversal; the amplitude Q_eQ_m is an MHD/GR weight, not a compiler output. [X] dated kill test: the residual intercept must pass three nulls (frequency/achromatic, spatial $1/r^2$, sign-flip); a null-consistent residual after honest GRMHD subtraction falsifies the channel (core untouched) [E][C][X]
`v204_muon_seam_g2.py`	The muon anomalous magnetic moment $a_\mu=(g_\mu-2)/2$ as a seam vertex readout ([C] downstream, *not* a compiler power; archive integration of the old muon-$g-2$ note). The carrier fixes the second-order defect $\delta_2=B\gamma\,\delta_{\mathrm{top}}^2=\frac{5}{4}\delta_{\mathrm{top}}^2$ ($\delta_{\mathrm{top}}=\Omega_{\mathrm{adm}}c_3^4=48c_3^4=3/256\pi^4$, v2/v23; $B\gamma=\frac{3}{2}\cdot\frac{5}{6}=\frac{5}{4}$, v51); the seam-loop projection $a_\mu^{\mathrm{seam}}=\delta_2/(2\pi)=45/(524288\pi^9)\approx 2.879\times10^{-9}$. [E] the value is an exact compiler number ($\delta_2=4!\,\mathrm{Tr}_{S^+}(X^2)c_3^8$, $\mathrm{Tr}=120=5!$); [C] the identification of $\delta_2/(2\pi)$ as a magnetic vertex ($1/2\pi=4c_3$) is a physical bridge; [E] 0.81σ vs dispersive $\Delta a_\mu=(2.49\pm0.48)\times10^{-9}$; [C] HONEST: lattice/CMD-3 HVP shrinks it ($\sim 1.5\times10^{-9}$), the fixed value then $\sim 1.5\sigma$ high; [X] dated kill test: a converged Δa_μ outside $2.879\times10^{-9}\pm0.5\times10^{-9}$ excludes the mechanism [E][C][X]
`v205_xi_threequarter.py`	$\xi=c_3/\varphi_{\mathrm{tree}}=3/4$: an *independent* route to the gravitational 3/4 (archive integration of the old Eliminating-K note). The torsion-compression ansatz $\kappa^2=\xi\varphi_0/c_3^2$ with the Einstein limit $\kappa^2=8\pi G$ fixes $\xi=8\pi c_3^2/\varphi_0=c_3/\varphi_0$ (since $8\pi c_3^2=c_3$), and at tree level $\varphi_{\mathrm{tree}}=1/(6\pi)$ this is exactly 3/4. [E] tree identity $\xi=c_3/\varphi_{\mathrm{tree}}=3/4$; [E] Einstein-limit reduction $8\pi c_3^2=c_3\Rightarrow G$ is an output; [E] over-determination $3/4=q(A_3)=N_{\mathrm{fam}}/\lvert\mu_4\rvert=\ln(m/\mu)$ (v152 $k=c_3/2$) — the torsion-compression ξ and the gapped EH replica are two independent appearances of the same gravitational 3/4; [E] with the retained seed $\xi=c_3/\varphi_0=0.748303$ (-0.226%); [C] the $\kappa^2=\xi\varphi_0/c_3^2$ ansatz is the bridge, the modern derivation is v152/v68, $1/G$ stays the one anchor (v60/v153) [E][C]
`v206_h0_lambda_branch.py`	The cosmological-constant Letter branch: the seam number $\delta_\Sigma=m_1e^{-2\pi\alpha^{-1}/3}$ with the lowest block $m_1=\Omega_{\mathrm{adm}}=48$, the Planck-convention split, and the quantitative $H_0=66.5$–67.1 (archive integration of the tfpt-31 CC note; sharpens the qualitative H_0-vs-Λ row). [E] $\delta_\Sigma^{\mathrm{lead}}=48\,e^{-2\pi\cdot137.036/3}\approx1.085\times10^{-123}$; [E] convention split $M_{\mathrm{Pl}}^4/\bar M_{\mathrm{Pl}}^4=(8\pi)^2=631.65$ (unreduced vs reduced — must be displayed); [E] $\rho_\Lambda=M_{\mathrm{Pl}}^4\delta_\Sigma\approx2.4\times10^{-47}\,\mathrm{GeV}^4$ (observed vacuum order); [E] 123-orders cross-link $\lvert\log_{10}\delta_\Sigma\rvert\approx122.96$, consistent with the v60 split 122.948; [C] under flatness closure $H_0^{\mathrm{lead}}=66.51$, $H_0^{\mathrm{disc}}=67.10$ — 1.6–0.5σ below Planck, $\sim6\sigma$ below SH0ES: the Hubble tension is *not* relieved by raising the vacuum scale; H_0 uses imported $\Omega_b,\omega_{\mathrm{DM}}$, a [C] readout [E][C]
`v207_asymptotic_safety.py`	Asymptotic Safety as a *second*, independent, [O] route to the metric sector (archive integration of the old Five-Problems note). The current theory closes QG via the discrete seam-net/AQFT programme (v175; open premise `QGEO.REALIZE.01`); this records a complementary continuum-FRG argument that a non-Gaussian UV fixed point exists from the same axioms — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric}. [E] the axion coupling is fixed and small $y^2=16c_3^2=1/(4\pi^2)=0.025330$ exactly ($g_{a\gamma\gamma}=-4c_3$ from the cubic); [E] existence by IVT: $\partial_tg/g=2+\eta_R$ is $+2>0$ as $g\to0$ and <0 at large g (torsion K^2 damping) $\Rightarrow$ a root $g_*>0$; [E] $g_*=2/a_1>0$ finite, Reuter-type $O(1)$, no new free input; [O] HONEST: a continuum-FRG ansatz the discrete-compiler theory did *not* adopt, primary stays the seam-net (v175) [E][O]

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Script	What it machine-checks *(continued)*
`v208_bh_thermodynamics.py`	Black-hole thermodynamics from the seam: the modular Hawking temperature and the scalaron-corrected Wald entropy (archive integration of the dropped tfpt-42 stationary-horizons section, re-typed through the current Tomita–Takesaki seam work v198/v199). **[E]** modular T_H: $\Delta^{it}{=}e^{-2\pi t K_H}$ (Bisognano–Wichmann) $\Rightarrow$ KMS at $\beta{=}2\pi$ and $T_H{=}\kappa/(2\pi)$; the 2π is the seam unit $1/(4c_3)$ ($1/2\pi{=}4c_3$, v58), reproducing $T_H{=}c_3/M$ (v57); seam$\leftrightarrow$horizon identification **[C]**; **[E]** Wald entropy: with the induced $f(R){=}R{+}R^2/(6M_s^2)$ ($M_s{=}c_3^{7/2}\bar{M}_{\rm Pl}$, v28), $S_W{=}(f_R A)/(4G){=}(A/4G)(1{+}R_h/3M_s^2)$ — leading $A/4G$ is the c_3 area law ($1/4{=}1/\lvert\mu_4\rvert$), the correction an exact scalaron consequence; **[E]** SdS $T_H{=}(1/4\pi r_h)(1{-}\Lambda r_h^2)$, $1/4\pi{=}2c_3$; **[E]** area law $S_{BH}{=}M^2/(2c_3){=}4\pi M^2{=}A/4G$ **[E][C]**
`v209_bh_regular_core.py`	The black hole as a seam fixed point (topological defect), *not* a singularity: the honest **[O]** synthesis of the old Five-Problems BH picture, with the superseded torsion/RN-vortex metric dropped (RN-type metric superseded by Nariai/seam=horizon v101–v104/v190; cf. v203). Built entirely from established suite atoms. **[E]** no singularity = seam fixed point: the gapped attractor multiplier $\lambda_2{=}(2/3)^6{=}64/729$ with gap $\Delta{=}6\log(3/2){>}0$ (v56) — the core is a fixed point, $\rho{\to}\infty$ replaced by $\varphi{\to}\varphi_*$ **[C]**; **[E]** maximal BH = anchor: Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = the traceless projection of $a{=}(1,1,2)$ (v101/v190); **[E]** information stays in the boundary field: Page recovery decays at $\lambda_2{=}(2/3)^6$ per step (v54) **[C]**; **[E]** holographic remnant $S_{BH}{=}A/4G$, $1/4{=}1/\lvert\mu_4\rvert$ **[C]**; **[O]** HONEST: a qualitative structural synthesis (reuses v54/v56/v57/v101/v190), *not* load-bearing, *not* a metric **[E][O]**
`v210_mark_local_dtn.py`	A numerical *sharpening* of v201/`QGEO.SUBPRIN.01` on *realistic* Steklov curvature profiles, carried to the quasi-free *state* level (the actual $\omega{\circ}\rho{=}\omega$, not just $[\rho,\Lambda]{=}0$); *not* a closure of `QGEO.SYM.01`. A real von-Mises bump $g(\theta)$ at a puncture, summed over the four μ_4 marks $f(\theta){=}\sum_{j=0}^{3} g(\theta{-}j\pi/2)$, builds the Steklov DtN $\Lambda{=}\lvert D_\theta\rvert{+}M_f$. **[E]** every off-(mod 4) Fourier coefficient of the real f vanishes to $<10^{-16}$ (exact 4-fold cancellation), although g has all modes. **[C]** $\lVert[\rho,\Lambda]\rVert<10^{-15}$. **[C]** *state invariance* (new vs v201): the quasi-free covariance $C{=}\frac{1}{2}(1{+}\operatorname{sgn}H_1)$ satisfies $\rho C\rho^\dagger{=}C$ to $<10^{-13}$ — the state, not just the operator, is clock-invariant. **[C]** negative controls (single off-centre bump, $\mathbb{Z}_3$ source, 4 generic marks) break both by $O(1)$; convergence stable for $N{\in}\{16,32,64\}$. **[O]** certifies 'mark-local DtN $\Rightarrow$ $\omega{\circ}\rho{=}\omega$' for concrete profiles *and* at the state level; the premise that the physical seam *is* mark-local stays **[O]** — the narrowest definitional form of `QGEO.SYM.01`, not its closure **[C][O]**
`v211_axion_spine_angle.py`	Axion DM, the spine-angle branch $\theta_i=\pi N_{\rm fam}/g_{\rm car}=3\pi/5$ (108°): an alternative, more robust misalignment angle than the determinant-line hilltop $\theta_i\approx 170°$ (v185, which over-produces $\Omega_a h^2\approx 0.66$). **[E]** rational motive $\theta_i=3\pi/5$ exactly (spine quotient 3/5, no fit); **[E]** the v185 finite-T solver ($\theta{=}1\to 0.03$) reaches $\Omega_{\rm DM}$ at $\theta\approx 106°$, so 108° is on target (harmonic $0.03\theta^2{=}0.107$); **[C]** *robust*: 108° is 62° below the 170° hilltop (mild-anharmonic, not exponentially sensitive); **[O]** an alternative ansatz to $\theta_i{=}\pi(1{-}\varphi_{\rm seam}){\approx}170°$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); **[X]** kill test: $\Omega_a h^2$ outside $\sim[0.08,0.16]$ demotes the branch **[E][C][O][X]**
`v212_leptogenesis_decuple.py`	Leptogenesis, the scalaron-decuple branch: both Boltzmann inputs share the decuple $A_\Lambda{=}10{=}\lvert E(K_5)\rvert$ — $M_1{=}M_{\rm scal}\varphi_0^2/A_\Lambda\approx 8.65{\times}10^9$ GeV and $\tilde{m}_1{=}m_3/A_\Lambda\approx 5$ meV. A cleaner **[C]** route than v184's $M_1{=}M_R\varphi_0^4$ (which only relocated the free input to the un-pinned seesaw M_R). **[E]** shared decuple $A_\Lambda{=}10$; **[E]** $M_1{=}8.65{\times}10^9$ GeV inside the thermal window $[3{\times}10^9, 3{\times}10^{10}]$ (v169); **[E]** $\tilde{m}_1\approx 5$ meV, $K\approx 4.6$ strong washout; **[E]** Davidson–Ibarra $\varepsilon_1\approx 8.5{\times}10^{-7}$, $\eta_B\approx 1.2{\times}10^{-9}$ brackets observed $6.1{\times}10^{-10}$; **[C]** HONEST: M_1 uses only $\{M_{\rm scal},\varphi_0,A_\Lambda\}$ but the coupling mechanism is posited not derived; **[X]** kill test falls the route, not the theory (`FR.ETAB.01` independent) **[E][C][X]**

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Script	What it machine-checks *(continued)*
`v213_ftransfer_functor.py`	The `F_transfer` *functor contract* (`CONTRACT.F.01`): the four frontier transfers are ONE typed functor $F_{\text{transfer}}=F_{\text{observable}}\circ F_{\text{threshold}}\circ F_{\text{RG}}$ with four structural axioms, each instance a discrete compiler kernel **[E]** composed with an external continuous solver [C]; consolidates v183 (F_{pole}), v212 ($F_{\text{Boltzmann}}$), v211 (F_{relic}), v164/`FR.MPME` (F_{QCD}), the structural complement of the v187 typing guard. **[E]** axiom 1 μ_4-deck equivariance ($\lambda_2=(2/3)^6=64/729$ the deck transfer eigenvalue); **[E]** axiom 2 Plücker preservation ($53=a^T(R+Q)\mathbf{1}$, $\text{Pl}(F)=(1,22,30)$, $\|\text{Pl}(K)\|_1=11$); **[E]** axiom 3 positivity/stochasticity (spec $T=\{1,(2/3)^6,(1/3)^6\}$ positive, $\kappa_f\in(0,1)$); **[E]** axiom 4 external modules explicit ($b_3=-7$ a carrier output, $\lvert b_3\rvert=\Omega_{\text{adm}}-10b_1=7$); [C] verdict: a typed functor, not a bag of open topics; an architectural consolidation guarded by v187, not a closure **[E]**[C]
`v214_seam_pillowcase.py`	The *pillowcase* reduction of `QGEO.SYM.01` (`QGEO.PILLOW.01`): the two separately-tracked open QGEO residuals — v180/`QGEO.ISO.01` (the carrier clock is an order-4 isometry of the seam metric) and v201/`QGEO.SUBPRIN.01` (the DtN sub-principal symbol is mark-local, the seam flat away from the μ_4 marks) — are ONE statement, 'the seam carries the uniformising flat orbifold (pillowcase) metric', and the order-4 clock is *derived* from cross-ratio 2 (v168), not assumed. **[E]** $\chi_{\text{orb}}(S^2(2,2,2,2))=2-4(1-\frac{1}{2})=0\Rightarrow$ flat (Troyanov cone-metric uniformisation, unique up to scale; Gauss–Bonnet four deficits π sum to $4\pi=2\pi\chi(S^2)\Rightarrow\int K_{\text{smooth}}=0$, flat away from the marks). **[E]** NEW LINK: cross-ratio$(\mu_4)=2$ (v168) $\Rightarrow j(\lambda)=256(\lambda^2-\lambda+1)^3/(\lambda^2(\lambda-1)^2)$ with $j(2)=1728$ (all six harmonic cross-ratios $\{2,-1,\frac{1}{2}\}$ give 1728) $\Rightarrow$ the square modulus, hence the order-4 clock, is a *consequence* of cross-ratio 2. **[E]** $j=1728\Leftrightarrow\tau=i$ (square torus) $\Leftrightarrow$ CM by $\mathbb{Z}[i]\Leftrightarrow\text{Aut}=\mathbb{Z}/4$ ($j=0$ gives $\mathbb{Z}/6$; generic $\mathbb{Z}/2$). **[E]** explicit CM $(x,y)\mapsto(-x,iy)$ on $y^2=x^3-x$ has order $4=z\mapsto iz$. **[E]** negative control $\lambda=3\Rightarrow j=21952/9\notin\{0,1728\}\Rightarrow\text{Aut}=\mathbb{Z}/2$, no clock. **[E]** UNIFICATION: 'seam = uniformising flat pillowcase metric' implies both v201 mark-locality and v180 `QGEO.ISO.01`. **[O]** does *not* close `QGEO.SYM.01` — the canonical constant-curvature-metric premise stays open, milder/more canonical; net existence + full-cone RP remain v175's **[E]**
`v215_seam_deck_killtest.py`	The bedrock `QGEO.SYM.01` ($\omega\circ\rho=\omega$) recast as a *falsifiable* kill-test (`QGEO.KILL.01`), not a proof-promise — a foundational symmetry cannot be derived from nothing (v181/v153), and Bisognano–Wichmann would presuppose the covariance it should produce. Non-circular reduction (v198/v199/v201): $\omega\circ\rho=\omega\Leftrightarrow[\rho,C]=0$; the principal symbol $\lvert k\rvert$ commutes exactly, so the residual is the sub-principal M_f with $[\rho,M_f]=0\Leftrightarrow f$ $\mathbb{Z}_4$-invariant; measure form = the raw Gaussian bulk form A is μ_4-deck-invariant. Four levers (carrier-side **[E]**, seam-side **[O]**): **K1** exactly 4 marks (order-4 clock free orbit μ_4 + $b_1=\#\text{marks}-1=3=N_{\text{fam}}\Rightarrow\#\text{marks}=4=\lvert\mu_4\rvert$); **K2** square config (cross-ratio $2\Rightarrow j=1728$, $\text{Aut}=\mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j=0$ $\mathbb{Z}/6$; generic $\Rightarrow\mathbb{Z}/2$); **K3** mod-4 sub-principal support (μ_4-mark sum $\Rightarrow f_m=0$ for $m\not\equiv 0 \bmod 4\Rightarrow[\rho,M_f]=0$); **K4** transfer gap $(2/3)^6=64/729$ (cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}$). Freeze-bound (`freeze_file.csv seam_deck_symmetry`). **[O]** the decisive kill is *deferred* (`QGEO.REALIZE.01`): the raw $\mathbb{P}\setminus\mu_4$ collar DtN must yield 4 square marks *without* inputting them; carrier-side K1–K4 are green regression guards. A kill-test, not a closure

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Script	What it machine-checks *(continued)*
`v216_marks_gauss_bonnet.py`	The four seam marks *emerge* from Gauss–Bonnet (`QGEO.MARKS.03`), executing the K1 lever of the v215 kill-test. **[E]** mark count $= 2\chi$: $\mathbb{Z}_2$ branch points (deficit π) on a flat sphere (χ=2) give $n\pi = 2\pi\chi = 4\pi \Rightarrow n = 2\chi = 4 = \|\mu_4\| = e_1(a) = N_{\rm fam}+1$ (the same χ=2 as the 8 in c_3=$1/(8\pi)$). **[E]** the Euclidean sphere 2-orbifolds ($\chi_{\rm orb}$=0) are exactly $(2,3,6),(2,4,4),(3,3,3),(2,2,2,2)$. **[E]** all-order-2 (the $\|\mathbb{Z}_2\|$ sheet branch) $\Rightarrow$ uniquely $(2,2,2,2)$; **[E]** $N_{\rm fam}$=3 (rank H^1=#marks−1) $\Rightarrow$ uniquely $(2,2,2,2)$, picking the 4-mark square over the 3-mark hexagonal $(3,3,3)$; **[E]** a free order-4 deck $\Rightarrow$ uniquely $(2,2,2,2)$ — three independent selectors converge. **[O]** residual: the square modulus (cross-ratio 2 $\Rightarrow$ j=1728, v214) needs the one order-4 carrier input. The mark count + equality are now derived; does *not* close `QGEO.REALIZE.01`
`v217_marks_emergence_scan.py`	The *numerical* emergence scan (`QGEO.EMERGE.01`, the sibling of v216 as v210 is to v201): with the number of marks n *and* their positions free, the 4-equally-spaced (square) configuration is the joint minimiser of {clock-invariance + 3-family cohomology + transfer gap $(2/3)^6$} on the actual seam DtN/state. **[C]** n=4 equally-spaced $\Rightarrow \|\rho_4 C\rho_4^\dagger - C\| \sim 10^{-14}$ (clock-invariant state); **[C]** family count selects n=4 (scan $n \in \{2..8\}$, only $n-1$=3=$N_{\rm fam}$; n=3 $\to$ 2 families = hexagonal, excluded; n=5 $\to$ 4); **[C]** gap: μ_4-equivariant $\Rightarrow$ diagonal in the cusp basis (deck-average off-diagonal $\sim 10^{-16}$), cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ $\Rightarrow$ transfer $(1-w)^6$=$\{1, (2/3)^6, (1/3)^6\}$, gap $(2/3)^6$=64/729; **[C]** position scan: the square minimises the defect (square $\sim 10^{-14}$ vs off-square ~ 0.4); **[C]** negative controls n=3, 5, 6. **[O]** confirms the emergence on the real DtN (number scanned, positions free); selector = the 3-family/gap structure (carrier input); executes K1 numerically, does *not* close `QGEO.REALIZE.01`. Python-only
`v218_diamond_axis_geometry.py`	Diamond axis geometry: the Sheet Diamond (v94) is a discrete geometry with two axes around the centered cross (v95: C center, U winding, V sheet), F the transfer completion. Three exact **[E]** lemmas + honest audit/heuristic typing, no new numbers. **[E]** AXIS CURVATURE: $\det(C+xU)$=$14+6x$ linear (slope 6=$\|R^+(A_3)\|$), $\det(C+yV)$=$14+14y+4y^2$ quadratic, 2nd difference 8=rank E_8; $\det B(C+xU)$=$2\det(C+xU)$, anchor-block sheet curvature 6=$\|R^+(A_3)\|$. **[E]** PLÜCKER LADDER: $\mathrm{Pl}(K)$=$(-1,6,4) \to \mathrm{Pl}(C)$=$(0,14,14) \to \mathrm{Pl}(F)$=$(1,22,30)$, steps $(1,8,10)$=$(N_\Phi, \mathrm{rank}\,E_8, A_\Lambda)$ and $(1,8,16)$=$(N_\Phi, \mathrm{rank}\,E_8, \dim S^+)$ (identity **[E]**, transfer reading **[C]**). **[E]** SPECTRAL RAMIFICATION: $\mathrm{Disc}(\chi)$=$q(r)^2\mathrm{Disc}(q)$ for Q, K, C, F; squares $(1,3,4,6)$, kernels $(13,48,65,105)$ with 48=$\Omega_{\rm adm}$, 105=$N_{\rm fam} g_{\rm car} \cdot 7$. **[C]** heuristic: F the transfer-completion corner, a search principle, not a hard filter; the G_2/F_4 pair-sum labels stay audit-only (cf. v94/v95) **[E][C]**
`v219_icosahedral_mckay.py`	McKay bedrock: *why* the atoms are $\{2,3,5\}$. E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T \to \hat{E}_6$, $2O \to \hat{E}_7$, $2I \to \hat{E}_8$). The McKay graph is built *from the group*: the 120 icosians close to the binary icosahedral $2I$ (element orders $\{1,2,3,4,5,6,10\}$, containing the 2, 3, 5 axis orders), 9 conjugacy classes, Dixon irrep degrees $\{1,2,2,3,3,4,4,5,6\}$ (sum 30=$h(E_8)$, sum of squares 120=$\|R^+(E_8)\|$); the natural-rep McKay adjacency *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8 **[E]**, the reading that it explains $(2,3,5)$ **[C]**, the raw-seam origin **[O]**
`v220_cp_hexagonal_modulus.py`	CP lives in the *hexagonal* phase fiber, not the square seam deck (a geometric sharpening of red-team Target D). **[E]** the seam deck is the square modulus $j(i)$=1728, Aut=$\mathbb{Z}/4$ (the μ_4 clock); its exceptional partner is the hexagonal modulus $j(\rho)$=0, ρ=$e^{i\pi/3}$, Aut=$\mathbb{Z}/6$, CM by $\mathbb{Z}[\omega]$, $\arg\rho$=$\pi/3$. **[E]** the frozen leading CKM reading δ=$\pi/3+3\lambda_C^2$=68.654° (v88) has leading term $\pi/3$=$\arg\rho$; the $\|\mathbb{Z}_2\|$ alternative $\pi/3+2\lambda_C^2$ is the frozen look-elsewhere trap. The deck stays $\mathbb{Z}/4$ (v215); CP is the $\mathbb{Z}/6$ fiber, the bridge **[C][E][C]**

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Script	**What it machine-checks *(continued)***
`v221_seam_qecc.py`	The seam as a *finite recoverability* code (recovery rate $(2/3)^6$=64/729). [E] code dimension dim S^+=16=$2^{g_{\rm car}-1}$; the gapped transport T on the cusp-weight 3-space (deviations $(1,-1,0)$ and the Nariai anchor $(1,1,-2)$) is symmetric, doubly stochastic (CPTP), spectrum $\{1,(2/3)^6,(1/3)^6\}$; on the trace-zero subspace $\|T\delta\|\le(2/3)^6\|\delta\|$ hence $\|T^n\delta\|\le(2/3)^{6n}\|\delta\|$ (a Knill–Laflamme-type bound); a free ratio breaks it. [C] the holographic/Petz reading (gated on the Seam–Horizon theorem) [E][C]
`v222_cm_norm_duality.py`	CM-norm duality: the two exceptional moduli give 41 and 7. [E] SQUARE (Gaussian $\mathbb{Z}[i]$, j=1728): $N(g_{\rm car}+i\|\mu_4\|)=5^2+4^2=41=10b_1$ — the EM index *is* the Gaussian norm of the carrier-glue vector; $N(3+2i)=13=\Delta_Q$. [E] HEX (Eisenstein $\mathbb{Z}[\omega]$, j=0): $N(3+2\omega)=3^2-3{\cdot}2+2^2=7$= scalaron. The $(3,2)$=(colour,weak) split generates $(5,6,7,13)$ by sum / product / Eisenstein / Gauss norm; the rings are rigid (Eisenstein of $(5,4)=21\neq41$). The architecture bridge square↔EM, hex↔scalaron is [C][E][C]
`v223_coxeter_totative_clock.py`	The μ_4 clock is the order-4 character of the E_8 Coxeter cycle. [E] $(\mathbb{Z}/30)^\times=\{1,7,11,13,17,19,23,29\}$= the E_8 exponents = live Coxeter phases; $\varphi(30)$=8=rank E_8, 30=2·3·5. [E] $(\mathbb{Z}/30)^\times\cong\mathbb{Z}/4\times\mathbb{Z}/2$ with order-4 generator 7 (7^2=19, 7^4=1 mod 30): a literal μ_4 clock $\langle7\rangle=\{1,7,13,19\}$ on the exponents; the conjugate pairs $m+(30-m)=30$ give $\|\mu_4\|=4$ invariant planes the clock permutes. Only E_8 has $\varphi(h)$=rank *and* h=2·3·5; the seam-clock identification is [C][E][C]
`v224_diamond_ftransfer_path.py`	$F_{\rm transfer}$ as *one* path on the Sheet Diamond, not four external interfaces. [E] family $M(s,t)=R+Q\,{\rm diag}(s,t,t)$: the corners $K=M(1,-1), C=M(1,0), F=M(1,1)$ lie on the sheet axis (s=1); the sheet axis is curved det $M(1,t)=4t^2+14t+14$ (2nd diff 8=rank E_8), the winding axis det $M(s,0)=6s+8$ flat (slope 6=$\|R^+(A_3)\|$); Plücker steps $(1,8,10),(1,8,16)$. [C] $F_{\rm transfer}$ is the $K\to C\to F$ sheet geodesic, Koide source→pole its 1-D Möbius projection (v37/v183) [E][C]
`v225_dual_normal_frame.py`	The dual normal frame (d,n): one boundary compass for flavor and horizon, with CP as orientation. [E] $d=a^\top R^{-1}=a^\top L^{-1}=(-\frac12,-\frac12,1)=-\frac12(1,1,-2)$ (the Nariai/horizon normal; $d{\cdot}\mathbf{1}$=0, $d{\cdot}a$=1); $n=(5,-9,6)$ with $n^\top R=(\det R,0,0)=(8,0,0)$, $n^\top L=(20,0,0)$ (the first-generation/torsion selector). [E] oriented volume $\det(\mathbf{1},d,n)=21=N_{\rm fam}\cdot7$. [C] CP as orientation: Im $\det(\mathbf{1},d,\rho n)=21\sin(\pi/3)$ for $\rho=e^{i\pi/3}$ — CP sits where the magnitude bijection (Target D) fails [E][C]
`v227_degree_exponent_channel_split.py`	248=120+128 as a magnitude/phase channel typing (replaces the over-strong S^--dark-matter reading). [E] E_8 exponents sum $120=\|R^+(E_8)\|$ (*magnitude* channel); degrees $\{2,8,12,14,18,20,24,30\}$ sum $128=2^{\rm rank-1}$=rank · dim S^+ (*phase/glue* channel); 248=120+128=adj(D_8)+spinor(D_8); $\sum\deg-\sum\exp$=rank=8. [C] ratios+products (Target D) live in the 120 channel, the un-covered CP phases in the 128 phase/glue channel; [O] a dark-sector reading of 128 stays Frontier [E][C][O]
`v228_rr_index_gate.py`	P2 as the mode space of a degree-4 seam divisor (a Riemann–Roch *index gate*; bedrock language for P2, not a proof from nothing). [E] deg $D=4=\|\mu_4\|$ on $\mathbb{P}^1\Rightarrow h^0(\mathcal{O}(D))=5=g_{\rm car}$ (Riemann–Roch, genus 0); rank $H_1(\mathbb{P}^1\setminus4)=3=N_{\rm fam}$; h^0+rank H_1=8=rank E_8, h^0-rank $H_1=2=\|\mathbb{Z}_2\|$; $\Lambda^{\rm even}(\mathbb{C}^5)=1+10+5=16=\dim S^+$. The mode-space reading is [C], conditional on the raw seam producing the degree-4 divisor ([O], = QGEO.SYM.01) [E][C][O]
`v229_lepton_frobenius_algebra.py`	The charged leptons as an étale Frobenius algebra over the μ_6 resolvent. [E] exact coefficients $(c_e,c_\mu,c_\tau)=(\frac{16}{7},\frac43,\frac76)$, ring closure $c_ec_\tau=\frac83=\|\mathbb{Z}_2\|c_\mu$, product $\frac{32}{9}=2^{g_{\rm car}}/N_{\rm fam}^2$. [E] $A=\mathbb{Q}[t]/(m)$, $m=\prod(t-c)$: the trace-form Gram is nondegenerate (det = disc(m)≠0, A étale) — a commutative Frobenius algebra; the C_6 shift has characteristic polynomial t^6-1 (spectrum $\mu_6=\mu_3$ family $\times\,\mu_2$ sheet). [C] the PMNS extension via Aut(A) and the hexagonal CM point [E][C]

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Script	What it machine-checks *(continued)*
`v230_center_budget_norms.py`	The Sheet-Diamond center budget $(7, 11, 13)$ is the three *local norms* of the theory. [E] $C=R+Q\,\mathrm{diag}(1,0,0)$, row sums $(7, 11, 13)$: $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (Eisenstein/hex = scalaron), $13=N_{\mathbb{Z}[i]}(3+2i)$ (Gauss/square = Δ_Q), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}=1+4+6$ (QBL boundary Fock count on the 4 μ_4 corners = $\dim S^+ - g_{\mathrm{car}}=16-5$). The two CM rings (v222) and the seam boundary count meet in one operator center; the reading is [C][E][C]
`v231_cp_mu6_phases.py`	Both CP phases are μ_6 powers of *one* hexagonal CM unit, split by the sheet — a structural reduction of red-team Target D (sharper than v220/v225). With $\rho=e^{i\pi/3}$ (the $j=0$ Eisenstein CM unit): [E] ρ is a primitive 6th root ($\rho^6=1$, $\rho^3=-1$ the $\mathbb{Z}_2$ sheet half-turn); [E] δ_{CKM} leading $= \arg(\rho^1)=\pi/3$ (quark), $\delta_{\mathrm{PMNS}} = \arg(\rho^4)=4\pi/3$ (lepton, the *phase lattice*); [E] $\rho^4=-\rho \Rightarrow \delta_{\mathrm{PMNS}}=\delta_{\mathrm{CKM,lead}}+\pi=(\mathrm{CM\ unit})\times(\mathbb{Z}_2\ \mathrm{sheet})$ — the two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; [E] the C_6/μ_6 monodromy has charpoly t^6-1 (spectrum μ_6 carries ρ); [E] orientation $\mathrm{Im}\det(\mathbf{1}, d, \rho^1 n)=+21\sin(\pi/3)$, $\mathrm{Im}\det(\mathbf{1}, d, \rho^4 n)=-21\sin(\pi/3)$ (sheet-flipped Jarlskog, v225); [E] NEG μ_3 has no -1 (no sheet; $\mu_6=\mu_3$ family $\times$ μ_2 sheet). REDUCTION: 2 CP phase inputs $\to$ 1 hexagonal unit + the sheet; the physical CP derivation stays [C], the deck stays $\mathbb{Z}/4$ (v215) [E][C]
`v232_e8_kleinian_seam.py`	The seam as the E_8 *Kleinian singularity* — a canonical algebraic-geometric model for the open seam-realisation premise `QGEO.REALIZE.01` (the du Val side of the McKay correspondence, v219). [E] the McKay graph of $2I$ is affine E_8; dropping the unique trivial-rep node (the single Kac mark 1) leaves the *finite* E_8 Dynkin on 8 nodes = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$ (trivalent node, arms $1, 2, 4$). [E] eight exceptional $\mathbb{P}^1$'s = rank E_8 = $g_{\mathrm{car}}+N_{\mathrm{fam}}$ = # nontrivial $2I$ irreps — a fourth reading of the 8 in $c_3=1/(8\pi)$. [E] the negated intersection form is the E_8 Cartan (even, $\det=1$, positive definite; each $\mathbb{P}^1$ a (-2)-curve). [E] one degree-1 irrep $\Rightarrow$ $2I$ perfect $\Rightarrow$ $H_1(S^3/2I)=0$: the link is the Poincaré homology 3-sphere, $\pi_1=2I$ of order $120=\lvert R^+(E_8)\rvert$. [C] seam reading: the raw RP seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for `QGEO.REALIZE.01`, not a P2 proof (presupposes E_8); bedrock [O][E][C][O]
`v233_cp_triality_phase.py`	The CP phase is the *universal* family/triality phase, only sheet-split — one step past v231 in $\mu_6=\mu_3$ (family/triality) $\times$ μ_2 (sheet). [E] $\mu_6=\mu_3\times\mu_2$: the $\mathbb{Z}_3$ triality centre $\{1, \omega, \omega^2\}$ = the cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}\times 2\pi$, the sheet $\mu_2=\{1, -1\}$; $\rho=\omega^2\cdot(-1)$. [E] SAME FAMILY CLASS: $\rho^4=\rho^1\cdot\rho^3$ with $\rho^3\in\mu_2$, so the quark (ρ^1) and lepton (ρ^4) CP phases have the *same* image in $\mu_6/\mu_2\cong\mu_3$ and *opposite* sheet — the CP phase *is* the universal triality phase ω^2 (the 2/3 cusp-weight $\mathbb{Z}_3$ centre), not a free power choice. [E] ω is the μ_3 factor of the order-30 $(2, 3, 5)$ Milnor/Coxeter monodromy that builds E_8 (v232). Reduces Target D from 'one hexagonal unit + sheet' (v231) to 'the universal triality phase + sheet'; the physical CP derivation stays [C][E][C]
`v234_seam_holomorphy_selection.py`	The Seam-Holomorphy selection certificate: the entire structural residual is *one* condition with *three* equivalent faces, all forcing E_8. The condition: *the seam carries no nontrivial abelian sector.* [E] (i) AQFT — the boundary net is holomorphic (μ-index 1); (ii) geometry — the seam link is a homology 3-sphere $\Leftrightarrow$ the binary group is perfect $\Leftrightarrow$ $2I$; (iii) rep theory — exactly one 1-dim irrep. [E] #(mark-1 affine nodes) $= \lvert\Gamma^{\mathrm{ab}}\rvert = \lvert H_1(S^3/\Gamma)\rvert$ = #(1-dim irreps): $A_n\to n+1$, $D_n\to 4$, $E_6\to 3$, $E_7\to 2$, $E_8\to 1$ — only E_8 gives 1 ($2I$ the unique perfect $SU(2)$ subgroup, Poincaré the unique homology-sphere space form). [E] holomorphic $c=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$ $\Rightarrow$ the unique even unimodular rank-8 lattice = E_8. [E] the three faces (v83/v154, v232, v219) are the SAME E_8-selector, so Target A, G_{net}, the carrier P2 and the Kleinian seam are *one* gate. [O] the closing theorem (stated): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ invertible bulk $\Rightarrow$ holomorphic boundary $\Rightarrow$ E_8; the single residual analytic step is 'free bulk $\Rightarrow$ holomorphic boundary' [E][O]

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Script	What it machine-checks *(continued)*
`v235_seam_chern_simons.py`	The closing step in abelian Chern–Simons language: holomorphic $\Leftrightarrow$ $\det K{=}1$ — a genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics. **[E]** a free gapped bosonic 2+1d bulk = an even integer K-matrix with #anyons = $\lvert\det K\rvert$, edge c = signature(K), holomorphic $\Leftrightarrow \lvert\det K\rvert{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic), all rank 8, c = sig=8=$g_{\rm car}{+}N_{\rm fam}$. **[E]** holomorphic $\Leftrightarrow$ det 1 $\Leftrightarrow E_8$ = the Kitaev E_8 quantum-Hall state. **[E]** condensation: a Lagrangian glue of order $\sqrt{16}{=}\lvert\mu_4\rvert{=}4$ takes det 16$\to$1 (a partial order-2 isotropic only $\to 4{=}D_8$). **[E]** NEG: D_8 is free+bosonic but has 4 anyons; an odd K is fermionic — det =1 is the extra condition. **[O]** residual: 'the free RP seam condenses the order-$\lvert\mu_4\rvert$ Lagrangian glue' = the sheet/QGEO selection, now with holomorphy$\Leftrightarrow$ det 1 a theorem **[E][O]**
`v236_brieskorn_capstone.py`	The $(2,3,5)$ Brieskorn singularity $x^2{+}y^3{+}z^5$ is the *one* generator of the discrete skeleton — the capstone of the icosahedral round (v55/v219/v223/v232/v233); its exponents *are* the atoms $(\lvert\mathbb{Z}_2\rvert, N_{\rm fam}, g_{\rm car}){=}(2,3,5)$. **[E]** Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}1{\cdot}2{\cdot}4{=}8{=}\mathrm{rank}\,E_8$ = the 8 in $c_3{=}1/(8\pi)$ (a fifth independent origin). **[E]** the Milnor monodromy eigenvalues are the eight primitive 30th roots $e^{2\pi i m/30}$, m the E_8 exponents (charpoly Φ_{30}, $\deg\varphi(30){=}8{=}\mu$) — so it *is* the order-30 E_8 Coxeter element (v55). **[E]** both clocks: the monodromy group $\langle h\rangle{=}\mathbb{Z}/30{=}\mathbb{Z}/2{\times}\mathbb{Z}/3{\times}\mathbb{Z}/5$ carries the μ_3 triality (CP, v233) as h^{10}; the Galois group $(\mathbb{Z}/30)^{\times}{=}\mathbb{Z}/2{\times}\mathbb{Z}/4$ carries the μ_4 clock ($\times 7$, v223). **[E]** Milnor lattice $= E_8$, link = Poincaré sphere. **[C]** one singularity generates atoms + 8 + E_8 + order-30 + both clocks + seam; a unification, not a closure **[E][C]**
`v237_seam_sre_closure.py`	The closing step as a *physical* condition: no topological ground-state degeneracy $\Leftrightarrow \det K{=}1 \Leftrightarrow$ the seam bulk is short-range-entangled (SRE/invertible) — sharpening `GATE.HOLO.02`'s residual from definitional to falsifiable. **[E]** the K-matrix ground-state degeneracy on a genus-g surface is $\lvert\det K\rvert^g$ (carrier $\to$ 16, $D_8\to 4$, $E_8 \to 1$ on the torus), so 'no degeneracy on any closed surface' $\Leftrightarrow \det K{=}1$. **[E]** among rank-8 even positive-definite K ($c{=}8{=}g_{\rm car}{+}N_{\rm fam}$) only E_8 has det 1 — the unique nontrivial bosonic SRE 2+1d phase (Kitaev E_8), edge holomorphic; D_8/carrier are LRE. **[E]** a unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$, the torus does), so the residual is 'the seam unique-vacuum extends to the torus' = SRE. **[E]** NEG: an LRE bulk (D_8) has 4-fold torus degeneracy + non-holomorphic boundary. **[O]** sharpened residual: 'the free RP gapped seam is short-range-entangled (no topological order)' = $\det K{=}1$ = the Kitaev E_8 phase — a yes/no statement, sharper than v235/`QGEO.SYM.01`; not a closure **[E][O]**
`v238_modular_lindblad_dynamics.py`	`DYN.SEMIGROUP.01` — the static$\to$dynamic flip: the seam's modular flow (reversible) and its recovery channel as a Markov/Lindblad generator (irreversible), the dissipative gap = the OS mass gap. Companion to v196/v198 (modular) and v221/v64 (recovery), which state the structure *statically*; this reads the same objects *dynamically*. **[E]** the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) and the μ_4 clock is conserved $\Leftrightarrow [\rho,\Lambda_\Sigma]=0$ (the v198 state-invariance), so the static commutator is the conservation law of modular time. **[E]** $L = \log T$ is a doubly-stochastic generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time, cf. v64). **[E]** keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2)$ = the OS mass gap Δ (v64) — the static recovery bound and the dynamical mass gap are one number, one exponentiated. **[E]** GKSL split (imaginary + real≤ 0 spectrum). **[O]** geometric modular flow on the full seam algebra (Connes–Rovelli thermal time = `QGEO.SYM.01`, v198) — a target, not a closure.

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Script	What it machine-checks *(continued)*
`v239_kms_thermal_time.py`	`DYN.KMS.01` — KMS / thermal time: the modular flow of v238 makes the seam state a KMS (thermal) state at $\beta = 1$, and the geometric normalisation fixes the seam temperature via $2\pi = 1/(4c_3)$. [E] KMS at $\beta = 1$: $\omega(A\,\sigma_i(B)) = \omega(BA)$ ($\omega(X) = \mathrm{Tr}(\rho X)$, $\rho = e^{-\Lambda_\Sigma}/Z$), so modular time (v238) is a thermal time (Connes–Rovelli); NEG $\beta \neq 1$ fails (Tomita–Takesaki pins $\beta = 1$). [E] the seam unit $\beta_{\mathrm{phys}} = 2\pi = 1/(4c_3)$ (v58) makes the KMS temperature the horizon temperature $T_H = c_3/M$ (v57/v208). [E] one object with v238. [O] the geometric-boost identification = `QGEO.SYM.01` — a target, not a closure.
`v240_gns_os_reconstruction.py`	`DYN.GNS.01` — GNS / OS reconstruction (finite toy): the Hilbert space and a POSITIVE Hamiltonian are OUTPUTS of the seam data (M, ω), not inputs. [E] GNS: $\langle A, B\rangle = \mathrm{Tr}(\rho A^\dagger B)$ Hermitian positive-definite ($\rho > 0$), $\dim n^2$, identity cyclic+separating. [E] modular flow $\sigma_t(x) = \rho^{it} x \rho^{-it}$ is a state-preserving $*$-automorphism, generator $i[\log\rho, \cdot] = \Lambda_\Sigma$ (v238). [E] OS Hamiltonian $H_{\mathrm{OS}} = -\log T$ (the v221 transfer) is POSITIVE — eigenvalues $\{0, 6\log(3/2), 6\log 3\} \geq 0$, gap $= 6\log(3/2) =$ the v64 mass gap. [E] $H_{\mathrm{OS}} = -L$: minus the v238 dissipative generator (recovery semigroup = Euclidean time). [O] the geometric/operator-level step = `QGEO.SYM.01`/BW + the standard OS theorem under RP/gap — a target, not a closure.
`v241_dhr_sectors.py`	`DHR.SECTORS.01` — particles = DHR superselection sectors: the carrier discriminant/anyon content and the v235/v237 condensation tower as an abelian MTC. Carrier form (Lean-verified, `FORM.GLUE.01`): on $\mathbb{Z}_4 \times \mathbb{Z}_4$, $q(x, y) = (5x^2{+}3y^2)/8 \bmod 1$, $\theta(a) = e^{2\pi i q(a)}$. [E] 16 sectors $= \mathbb{Z}_4 \times \mathbb{Z}_4$ ($= \det K_{\mathrm{carrier}} = \dim S^+$), fusion = abelian group law. [E] condensable $\Leftrightarrow \theta = 1 \Leftrightarrow$ isotropic $\Leftrightarrow 5x^2{+}3y^2 \equiv 0 \bmod 8$ — the six elements $\{(0,0), (1,1), (1,3), (2,2), (3,1), (3,3)\}$. [E] tower = v235/v237: Lagrangian $\langle(1,1)\rangle$ (order 4, self-dual) $\to$ 1 sector $= (E_8)_1$; order-2 $\langle(2,2)\rangle \to 4 = D_8$; carrier $16 \to D_8\, 4 \to E_8\, 1$. [E] NEG: a non-isotropic anyon $(1, 0)$ is confined. [C] sector↔SM-multiplet map is the interpretation; the MTC is exact.
`v242_spin_statistics_modular.py`	`DHR.MODULAR.01` — spin-statistics and the modular (S, T) data of the carrier MTC: statistics + the fusion ring recovered from modular data + the anyon central charge $c = 8$. Same MTC as v241 ($\mathbb{Z}_4{\times}\mathbb{Z}_4$, $q = (5x^2{+}3y^2)/8$, $\theta = e^{2\pi i q}$). [E] spin-statistics: $T = \mathrm{diag}(\theta_a) \to 6$ bosons ($\theta = 1$), 2 fermions ($\theta = -1$) $= \{(0,2), (2,0)\}$, 8 anyons. [E] modular $S_{ab} = (1/\sqrt{16})e^{-2\pi i B(a,b)}$ unitary, symmetric, $S^2 = C$ (charge conjugation), $S^4 = I$. [E] Verlinde $N_{ab}^c = \sum_x S_{ax} S_{bx} S^*_{cx}/S_{0x} = \delta(c, a{+}b)$ — fusion recovered from modular data. [E] Gauss–Milgram $(1/\sqrt{16})\sum\theta_a = e^{2\pi i c/8}$ with $c = 8 = g_{\mathrm{car}} + N_{\mathrm{fam}} =$ the $(E_8)_1$ central charge (v156/v175). [C] the boson/fermion split is the kinematic skeleton; the sector↔multiplet map is the interpretation.
`v243_haag_ruelle_braiding.py`	`DHR.SMATRIX.01` — scattering: the Haag–Ruelle skeleton + the braiding S-matrix. The seam net has a mass gap (v64/v238) so Haag–Ruelle constructs asymptotic states; for the abelian anyon MTC (v241/v242) the 2-particle S-matrix is the monodromy $M(a, b) = \theta(a{+}b)/(\theta(a)\theta(b)) = e^{2\pi i B(a,b)}$. [E] mass gap $\Delta = 6\log(3/2) > 0 \to$ HR in/out states. [E] 2-particle S = monodromy: $\lvert M\rvert = 1$, diagonal in conserved labels $\to$ no particle production, factorised (integrable) scattering. [E] Yang–Baxter: M is a bicharacter $\to$ multi-particle factorisation. [E] crossing+symmetry: $M(a, b) = M(b, a)$, $M(a, -b) = M(a, b)^*$. [E] holomorphic point: on the condensed $(E_8)_1$ (single sector) $M = 1$, trivial braiding (det 1/SRE). [O] the 4d S-matrix / cross-sections are the open holographic uplift (Phase 3).

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Script	**What it machine-checks** *(continued)*
`v244_spectral_action_contract.py`	`CONTRACT.QFT4D.01` — the 4d Lagrangian via Connes' spectral action: the honest contract for the Phase-3 gap (2d seam net → 4d physics). A single Dirac operator whose Tr $f(D/\Lambda)$ yields gravity AND matter. [E] carrier spinor = one generation: $SO(10)$ half-spinor = $\dim S^+ = 2^{g_{\rm car}-1} = 16$ $(15+\nu_R)$, $N_{\rm fam} = 3$. [E] SM gauge group inside the carrier: $SU(3){\times}SU(2){\times}U(1)$ (dim 12) $\subset SU(5)$ (24) $\subset SO(10){=}D_5$ (45); colour $SU(3) \subset A_3{=}SU(4)$ (Pati–Salam); rank(D_5)+rank$(A_3) = 8 =$ rank E_8. [E] one Dirac operator → gravity ($a_2{\sim}R$, $a_4{\sim}R^2$, v36) + matter ($a_4 \sim F^2$ Yang–Mills $+|D\phi|^2$ Higgs, Gilkey). [X] hooks: $g_3{=}g_2{=}\sqrt{5/3}\,g_1$ + a Higgs-quartic relation. [O] `CONTRACT.QFT4D.01`: obligations (seam→Dirac D; finite geometry = carrier; spectral action reproduces SM couplings) → the 4d SM+gravity Lagrangian is an OUTPUT — the Phase-3 target, not closed.
`v245_spectral_matter_content.py`	`QFT4D.MATTER.01` — the computable core of `CONTRACT.QFT4D.01` (v244): the spectral-action / NCG matter content, discharging the rep-theory + tree-relation core to [E]. The carrier half-spinor is the $SO(10)$ **16**. [E] **16** = one generation: $SO(10)\,\mathbf{16} = SU(5)(\mathbf{10}{+}\bar{\mathbf{5}}{+}\mathbf{1})$; SM dims 6+3+1+2+3+1 = 16 = $\dim S^+$, $N_{\rm fam} = 3$ (15 Weyl fermions $+\nu_R$). [E] electric charges $Q = T_3 + Y$ are exactly the SM values — hypercharges forced by the embedding. [E] anomaly-free: $\sum Y = 0$, $\sum Y^3 = 0$, $[SU(2)]^2U(1) = 0$, $[SU(3)]^2U(1) = 0$ (exact). [E] unification: $\sin^2\theta_W = (\sum T_3^2)/(\sum Q^2) = 3/8$, i.e. $g_3 = g_2 = \sqrt{5/3}\,g_1$. [E] Higgs = the $(1,2)_{1/2}$ inner fluctuation (one doublet). [O] residual: the seam→Dirac realisation + measured-coupling reproduction stay the open contract.
`v246_unification_data_crosscheck.py`	`QFT4D.RGTEST.01` — the honest data cross-check for the spectral-action prediction (v244/v245): does $g_1{=}g_2{=}g_3$, $\sin^2\theta_W{=}3/8$ at Λ survive the *measured* SM couplings? Not in the plain SM. Inputs (PDG, GUT-normalised, M_Z): $\alpha_i^{-1} = (59.01, 29.59, 8.47)$, SM 1-loop $b = (41/10, -19/6, -7)$ (v159). [E] the prediction: $\alpha_1{=}\alpha_2{=}\alpha_3$, $\sin^2\theta_W{=}3/8$ at Λ. [X] tension: pairwise crossings spread ~ 4 decades ($\sim 10^{13}$–10^{17} GeV), residual spread at $2{\times}10^{16}$ $\Delta\alpha^{-1} \sim 8.8$ — $g_1{=}g_2{=}g_3$ not realised by the plain SM. [X] $\sin^2\theta_W$ 3/8 vs measured 0.23122. [X] 2-loop also misses (min spread ~ 3.2 at $1.7{\times}10^{14}$ GeV). [O] no rescue: SM gauge content + no new states (the E_8 spine is arithmetic, not a threshold tower) ⇒ no admissible thresholds; closing it needs new matter (vs 'no new state') or a non-GUT/boundary reading — tension like NCG-SM, likely falsified unless extended. A kill-test, [X]/[O], never [E]-as-success.
`v247_e8_branching_no126.py`	`PS.E8BRANCH.01` — the E_8 audit hull forbids the **126** and supplies exactly $\{1, 10, 16, 45\}$ for $SO(10)$. Carrier $D_5{\oplus}A_3 = SO(10){\times}SU(4) \subset E_8$ (via $SO(16)$); $\mathbf{248} = (45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4}) = 248$. [E] ranks 5+3=8; [E] dims = 248 ($SO(16)$: 120 adj +128 spinor); [E] $SO(10)$ content $\{1, 10, 16, 45\}$, the **126** ABSENT ⇒ 126_H algebraically forbidden, 10+16+45 E_8-supplied not fitted; [E] caveat only one 45/one 15 → proton-marginal; [O] the field-content principle + NCG complement (v250).
`v248_ps_algebra.py`	`PS.ALGEBRA.01` — the Pati–Salam NCG algebra $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C})$ and its exact SM matching. [E] unimodular group $SU(2)_L{\times}SU(2)_R{\times}SU(4)_c$ (rank 5, dim 21); [E] $\mathbf{16} = (4,2,1){+}(\bar{4},1,2) = 8{+}8$; [E] $Y = T_{3R} + (B{-}L)/2$ gives the exact SM hypercharges and $Q = T_{3L} + Y$ the exact charges; [E] matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$. [O] A_F uniquely from seam+sheet+μ_4; [O] caveat $SU(4)_c \subset SO(10){=}D_5$ is not the carrier $A_3{=}SU(4)$ (family).
`v249_ps_unification.py`	`PS.RGTEST.01` — carrier-native Pati–Salam/$SO(10)$ two-step unification kill-test + negative controls (suite check of `experiments/gauge-unification`). [E] matching + a valid 1-loop unification solution; [X] NEG SM-only fails (spread ~ 8.8 @ $2{\times}10^{16}$); [C] $M_{PS}/M_s = 1.0$–1.7 (1-loop; 2-loop ~ 1.0–1.2) — gauge unification meets the gravitational scalaron scale; [X] NEG 126 breaks the match ($b_{2R}{>}0$; also E_8-forbidden v247); [X] NEG proton decay selects content (minimal $M_{GUT}{\sim}2{\times}10^{15}$ excluded; $+(15,1,1)$ safer). [O] the gauging fork, exact κ, the one-45 marginality.

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Script	What it machine-checks *(continued)*
`v250_dirac_triple_contract.py`	`CONTRACT.QFT4D.DIRAC.01` — the spectral-triple contract that would make 'carrier gauged' a theorem. Target (A, H, D, J, γ) almost-commutative, $A_F = \text{PS}$ (v248), $H_F = 3{\times}16$, D_F from $(R, L, \varphi_0, \text{Majorana})$. [E] $H_F = 3{\times}16 = 48$; [E] the Yukawas are already fixed (v18) so the open test is NCG consistency; [O] the 7 NCG axioms (first-order condition = the historic $SO(10)$ obstruction); [O] σ = scalaron hypothesis (rescues first-order $+\,\nu_R$ Majorana; $M_{PS}{=}\kappa M_s$) — caveat $\sigma \leftrightarrow \nu_R\nu_R$ not automatic; [O] inner fluctuations → 10+16+45 not 126. Reduces 4d QFT to one object + finite NCG checks.
`v251_first_order_sigma.py`	`PS.DIRAC.01` — the first-order condition is violated *exactly* by the Majorana/σ term (Chamseddine–Connes–van Suijlekom 'beyond first order' → Pati–Salam $+\,\sigma$), exhibited on a faithful lepton-block model; advances v250 from contract to mechanism. Honest correction: σ does NOT 'rescue' first order — one DROPS it and the inner fluctuations generate PS $+\,\sigma$. [E] order-zero; [E] real-even ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-6 signs); [E] first order HOLDS for the Yukawa block; [E] first order VIOLATED by $M \neq 0$, supported exactly on the $\nu_R\nu_R^c$ blocks (σ the unique violator). [C] σ = scalaron (v249). [O] the full 96-dim SM/PS triple + spectral action (v250).
`v252_full_finite_triple.py`	`PS.DIRAC.02` — the full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ built explicitly with the NCG axioms verified; advances v250 from a 7-obligation contract to a concrete object discharging 5 to [E]. H_F = particle⊕antiparticle doubling of three $SO(10)$ **16**-plets (dim 96). [E] dim 96; D_F Hermitian+odd; reality $J^2{=}{+}1$, $JD{=}DJ$; grading $\gamma^2{=}1$, $[\gamma, a]{=}0$, $J\gamma{=}{-}\gamma J$; KO-dim 6; order-zero; first order HOLDS for the SM $\mathbb{C} \oplus \mathbb{H} \oplus M_3$ (incl. Majorana) and VIOLATED for Pati–Salam exactly by the Majorana on $\{\nu_R, e_R\}$ (CCvS beyond-first-order → PS+σ); spectrum = fermion masses + seesaw $m_{\text{light}}{=}m_D^2/M_R$. [C] Poincaré duality (intersection form non-degenerate). [O] orientability + no-junk + spectral-action expansion.
`v253_kappa_scalaron_ps.py`	`PS.KAPPA.01` — pinning κ in $M_{PS} = \kappa\, M_s$, discharging residual 6(b) of v249. $M_s = c_3^{7/2}\bar{M} \approx 3.06 \times 10^{13}\,\text{GeV}$ (exact). [E] scalaron scale exact; [E] κ pinned by RG (scan PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval ~ 1.3 (1-loop $[1.0, 1.7]$, 2-loop $[1.0, 1.2]$) — pinned, not free. [C] heat-kernel origin $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\text{grav}}}$, a ratio of heat-kernel traces over the 96-dim H_F (v252) — Λ cancels, κ scale-free and $O(1)$. [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs the cutoff moments f_2/f_0 fixed.
`v254_inner_fluctuations.py`	`PS.NCG.FLUCT.01` — the inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator computed explicitly (reuses the v252 triple); closes v250 #5 (Higgs content, not **126**) + the no-junk obligation + the σ quantum numbers. [E] $\Omega^1_{D_F}$ purely chirality-odd → finite fluctuations are scalars; [E] no junk / no **126**: SM-algebra 1-forms sit exactly in the Higgs-doublet pattern (the single $(1, 2)_{1/2}$); [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam — σ is the CCvS beyond-first-order scalar; [E] σ is an SM singlet $(1, 1, 0)$ with $B{-}L = 2$; [C] σ = scalaron consistency.
`v255_spectral_action_expansion.py`	`PS.SPECACT.01` — the spectral-action heat-kernel (Seeley–DeWitt) expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$ over the finite triple; closes v252 #11 (spectral-action expansion) and makes $\kappa(f_2/f_0)$ explicit. [E] expansion structure (a_0 cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N{=}96$); [E] trace inputs $b/a^2 = 1/3$ (top-dominated, Chamseddine–Connes); [E] $\sin^2\theta_W = 3/8$ (cross-check v245); [C] Higgs quartic $\lambda(\Lambda) \sim g^2 b/a^2 \to {\sim}125\,\text{GeV}$ with σ; [E] scalaron = R^2 spin-0 mode (gravitational $a_4 \sim f_0 N$); [C] $\kappa = \sqrt{(f_2/f_0) c_{PS}/c_{\text{grav}}}$ explicit; [O] exact decimal needs the cutoff moments fixed.

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Script	What it machine-checks *(continued)*
`v256_orientability_poincare.py`	`PS.NCG.ORIENT.01` — orientability and Poincaré duality treated honestly (no faked proof). KO-dim 6 forces the intersection form $\mathrm{cap}(p,q) = \mathrm{Tr}(\gamma\,\pi(p)\,J\pi(q)J^{-1})$ to be antisymmetric. **[E]** skew form forced by $J\gamma{=}-\gamma J$ + order-zero + reality (verified); **[E]** on the SM K-theory $\mathbb{Z}^3$ the skew form has rank 2, corank 1, null $\sim(1,1,-1)$ — the structural obstruction; **[E]** orientability local condition $[\gamma_F,\pi(a)]=0$, full orientability is a degree-4 cycle on $M\times F$; **[C]** the canonical hypercharge-faithful SM/PS triple satisfies both (van Suijlekom; CCM) — literature; **[O]** a self-contained machine proof needs the complete canonical bimodule.
`v257_quasiorient_modpd.py`	`PS.NCG.ORIENT.02` — the honest, literature-confirmed resolution of orientability + Poincaré duality, correcting the too-generous v256 typing. Published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192): the CCM SM finite triple (KO-6, $3\,\nu_R$) genuinely FAILS strict orientability and strict Poincaré duality, satisfying the weakened versions. **[E]** strict orientability fails ($\gamma\notin\mathrm{span}\{\lambda(a)\rho(b)\}$); **[E]** quasi-orientability holds (left/right A-irrep boxes disjoint by chirality: $L=H$ doublet, $R=\mathbb{C}$ singlet $\Rightarrow L^{\mathrm{LR}}(\mathrm{even},\mathrm{odd})=\{0\}$); **[E]** strict Poincaré fails (KO-6 skew form, corank 1, the equal-L/R-neutrino case); **[C]** modified Poincaré duality (CCM) / Stephan's PD-restoring triple. A known model-class feature, not a TFPT defect — no faked closure.
`v258_dirac_covariance_induction.py`	`PS.DIRAC.03` — D_F is the modular/covariance *induction* of the boundary seam state, not an independent posit; closes the 'posited Yukawa vs derived operator' gap of v250/v252 (Q2: Dirac as Kasparov product + covariance formula). Quasi-free/modular dictionary: a Gaussian state is fixed by its covariance C (contraction, spec $\subset(0,1)$) with one-particle modular Hamiltonian $H=\log((1{-}C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C=(1{+}e^{H})^{-1}$ ($\beta{=}1$, v239). **[E]** inversion identity $\log((1{-}C)C^{-1})=H$ exactly (random Hermitian + a chirality-graded Yukawa block); **[E]** $C_F=(1{+}e^{D/\mu})^{-1}$ a positive contraction; **[E]** chirality-odd ($D_F^{\mathrm{ind}}\gamma=-\gamma D_F^{\mathrm{ind}}$); **[E]** unitarily equivalent to v252 (inducing C_F from the 9 Yukawas returns the same spectrum); **[E]** Majorana = off-diagonal covariance (seesaw $m_{\mathrm{light}}{=}m_D^2/M_R$); **[C]** $C_F=P_FC_\Sigma P_F$, so D_F is the KK shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\mathrm{car}}]$ and the v18 Yukawas are the readout of C_Σ.
`v259_modular_cutoff_kappa.py`	`PS.SPECACT.02` — the spectral-action cutoff *is* the seam KMS weight, fixing $f_2/f_0{=}1$ exactly and reducing the last open scheme freedom (v253 #5, v255 #7) from a free function to a finite-triple trace ratio (Q4: the cutoff is modular, not chosen). By Tomita–Takesaki/BW the seam state is KMS at $\beta{=}1$ (v239) and $2\pi{=}1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u){=}e^{-u}$ is *derived.* **[E]** KMS cutoff moments $f_0{=}f(0){=}1$, $f_2{=}\int f{=}1$, $f_4{=}\int uf{=}1$, so $f_2/f_0{=}f_4/f_2{=}1$; **[E]** seam-fixed $\beta{=}1$ (no new parameter beyond c_3); **[E]** non-vacuous neg control: a Gaussian cutoff e^{-u^2} gives $f_2/f_0{=}\sqrt{\pi}/2\neq1$; **[E]** $\kappa{=}\sqrt{(f_2/f_0)c_{PS}/c_{\mathrm{grav}}}{=}\sqrt{c_{PS}/c_{\mathrm{grav}}}$, no cutoff function left; **[C]** the RG $\kappa{\sim}1.3$ pins $c_{PS}/c_{\mathrm{grav}}{\sim}1.7$ ($O(1)$); the **[O]** free scheme becomes a **[C]** finite computation over the built triple; **[C]** the same logic fixes $\lambda(\Lambda)$.

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Script	**What it machine-checks** ***(continued)***
`v260_k3_kummer_unification.py`	`ARCH.K3.01` — one Kummer/K3 surface carries the *seam*, the *carrier-16* and the E_8 lattice as facets of one smooth object, the geometric unification of the v1 'glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$' step (the K3/Kummer proposal). The seam square torus $\tau = i$ (j=1728, v214) under the elliptic involution $z \mapsto -z$ squares to the Kummer K3. [E] the E_8 Cartan is even, det = 1, positive-definite (the unique even unimodular pos-def rank-8 lattice); [E] the K3 lattice $L_{K3} = U^3 \oplus E_8(-1)^2$ has rank $22 = b_2$, det = −1 (unimodular), is even, signature $(3, 19)$, with $E_8(-1)^2$ an orthogonal summand; [E] seam = $T^2/(z \mapsto -z)$: $4 = \lvert(\mathbb{Z}/2)^2\rvert$ fixed points, pillowcase $S^2(2, 2, 2, 2)$, orbifold Euler char 0 (the v214/v216 four marks); [E] carrier-16 = Kummer nodes $\lvert A[2]\rvert = 2^4 = 16 \to 16$ exceptional $\mathbb{P}^1 = \dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) = 16$ (v197), 16=4×4; [E] honesty: the 16 node classes span the rank-16 Kummer lattice, *not* literally $E_8(-1)^2$ (distinct sublattices of $H^2(K3)$); [C] two faces of E_8 in K3 (local du Val $\mathbb{C}^2/2I$, v232/v236; global H^2 summand); [C] unification (lattice [E], carrier↔node identification [C]).
`v261_modular_spectral_closure.py`	`QFT.MSC.01` — the *Modular Spectral Closure*: the assembly/reduction certificate turning the whole QFT layer (v238–v260) into *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise (like v176/v234, an assembly certificate, not new physics). [E] one index = marks = fixed points = glue order: Jones $[B{:}A]=4$ (v89) = $\lvert\mu_4\rvert=4$ (v92) = seam marks $2\chi=4$ (v216) = pillowcase/Kummer fixed points $\lvert(\mathbb{Z}/2)^2\rvert=4$ (v260); [E] one carrier-16: $\dim S^+ = \Lambda^{\mathrm{even}}(\mathbb{C}^5) =$ Kummer nodes $2^4=16$, $H_F = 2N_{\mathrm{fam}} \cdot 16 = 96$; [E] one gap $\Delta = -\log((2/3)^6) = 6\log(3/2) > 0$ = the OS mass gap (v240) = the recovery rate (v221) = the Lindblad gap (v238); [E] the three closures re-derived (Kasparov $\log((1-C)C^{-1}) = H$, v258; KMS cutoff $f_2/f_0 = 1$, v259; K3 lattice signature $(3, 19)$/det −1/even, v260); [E] relative-object manifest (10/10 component scripts present); [O] the single residual: the whole QFT layer reduces to `QGEO.SYM.01` (seam realisation), the ambient QG measure `QG.AMB.01` kept separate; [C] the master statement (Modular Spectral Universality) holds modulo `QGEO.SYM.01` — the honest sense in which the QFT layer is complete.
`v262_fqcd_mp_me.py`	`FR.MPME.02` — the F_{QCD} transfer pipeline for m_p/m_e, implemented as a real solver (the one F_{transfer} pipeline that was contract-only). *Not* a compiler power — a [C] standard-physics transfer reproduction. [E] $b_3 = -7 = -(11 - 2n_f/3)$ at $n_f = 6$ (the only TFPT number in the run); [E] a two-loop $\alpha_s(M_Z) = 0.1179$ run with thresholds gives $\alpha_s(2\,\mathrm{GeV}) \sim 0.30$ (real ODE solve); [C] $\Lambda^{(3)}_{\overline{\mathrm{MS}}} \sim 0.33\,\mathrm{GeV}$ × the lattice bridge $C_p = m_p/\Lambda$ (external $O(1)$) over the closed m_e gives $m_p/m_e \sim 1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget; [X] firewall: m_p/m_e is not an admissible compiler integer ($SU(9)$ absent from $D_5 \oplus A_3$). Advances `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).
`v263_mnu_from_df.py`	`FLAV.PMNS.02` — the light neutrino matrix M_ν as the type-I seesaw of D_F, advancing the solar-angle texture (v9) from a posited matrix to a seesaw *object* built from D_F's blocks. [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ symmetric + seesaw-suppressed (the light masses an output of D_F); [E] a $\mu\tau$-symmetric (m_D, M_R) gives a $\mu\tau$-symmetric M_ν (the v9 texture is in the seesaw image); [C] diagonalising gives $\theta_{23} = 45°$ and $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\mathrm{ret}}}{2} = 0.30675$ under the seam misalignment; [O] $\theta_{13} = 0$ in the leading texture (physical $\sin^2\theta_{13} \sim \varphi_0^{\mathrm{ret}} e^{-5/6}$ a separate readout), the CP magnitude $\delta_{\mathrm{PMNS}} = 240°$ is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.

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Script	What it machine-checks *(continued)*
`v264_raw_seam_marklocal.py`	`QGEO.ENERGY.03` — the raw-seam ⇒ mark-locality *reduction* (not a closure of `QGEO.SYM.01`). Composing v216+v201+v198 on the canonical pillowcase: **[E]** Gauss–Bonnet 4 cone points (deficit π each, sum $4\pi{=}2\pi\chi(S^2)$), curvature *only* at the 4 marks; **[E]** the curvature sourced at the μ_4 orbit $\{0,\pi/2,\pi,3\pi/2\}$ has Fourier support *only* on $4\mathbb{Z}$ (FFT), so mark-locality is automatic for the pillowcase; **[E]** M_f then commutes with the clock $\rho{=}\mathrm{diag}(i^n)$; **[E]** with $\lvert k\rvert$ already commuting (v198), $[\rho,\Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ on this metric; **[O]** the residual *is* the metric — the chain assumes the raw seam carries the uniformising flat pillowcase metric (=`QGEO.SYM.01`), the negative control (curvature off the μ_4 orbit) breaks it. A reduction/sharpening, not a closure.
`v265_qft4d_fork_freeze.py`	`QFT4D.FORK.01` — freeze the 4D-QFT fork as a *branch policy* (not an open ambiguity); writes `qft4d_fork_freeze.json` + a prose guard, no new physics. **[C]** A: boundary-only is the canonical 4D reading (no 4D-GUT claim by default); **[X]** SM-only 4D-GUT unification is killed (1-loop spread ∼8.8 at $2{\times}10^{16}$, v246); **[C]**/**[O]** B: carrier-native Pati–Salam is a falsifiable UV branch (E_8 content $\{1,10,16,45\}$ no **126** v247, $\sin^2\theta_W{=}3/8$ v248, κ $O(1)$ band v249/v253, threshold + proton kill test); **[X]** proton safety rule (no fake τ_p window); **[E]** wording guard (bare 4D over-claims absent, scoped policy phrases present). A frozen decision tree, not an open ambiguity.
`v266_ps_threshold_proton.py`	`PS.PROTON.02` — the explicit Pati–Salam threshold model + proton-decay window that DECIDES branch B of the v265 fork; writes `ps_threshold_proton.json`. Two-step PS solve + $d{=}6$ $\tau_p(p{\to}e^+\pi^0)$: the carrier-native PS UV branch survives proton decay *only* with the E_8-allowed $(15,1,1)$ [the single 45]. **[E]** explicit thresholds: minimal 16H → $M_{GUT}{\sim}2.4{\times}10^{15}$, $+(15,1,1)$ → $M_{GUT}{\sim}5.9{\times}10^{15}$; **[X]** minimal KILLED ($\tau_p{\sim}4.4{\times}10^{33}$ yr < Super-K $2.4{\times}10^{34}$, even high-band); **[C]** $+(15,1,1)$ survives now ($\tau_p{\sim}1.6{\times}10^{35}$ yr); **[X]** *Hyper-K decisive* (τ_p at the $\sim 1.4{\times}10^{35}$ yr reach) — a dated kill test within the decade; **[O]** marginal (only one 45 in the hull, v247) + $O(3)$ hadronic uncertainty. A (boundary-only) stays canonical.
`v267_qgeo_rigidity_minimal_axiom.py`	`QGEO.SYM.02` — the rigidity / minimal-axiom form of the bedrock `QGEO.SYM.01`: the ONE bare statement (the raw RP seam admits the carrier μ_4 clock as a conformal symmetry permuting its 4 marks, $=\omega\circ\rho=\omega$) *forces* everything downstream; it does *not* close the axiom. **[E]** an order-4 Möbius orbit $\{a,ia,-a,-ia\}$ has cross-ratio 2 (independent of a); **[E]** cross-ratio 2 ⇔ $j{=}1728$ (only $\lambda\in\{-1,\frac{1}{2},2\}$) ⇔ the order-4 CM automorphism $(x,y)\mapsto(-x,iy)$ on $y^2{=}x^3{-}x$; **[E]** downstream forced: square ⇒ the unique flat pillowcase (Troyanov, v214) ⇒ curvature at the 4 marks ⇒ $\mathbb{Z}_4$-Fourier DtN (v201/v264) ⇒ $[\rho,\Lambda]{=}0 \Rightarrow \omega\circ\rho{=}\omega$ (v198); **[E]** neg controls (generic $j\neq 1728$, only $\mathbb{Z}_2$; hexagonal $j{=}0$ has $\mathbb{Z}_6$, order $6\neq 4$); **[C]** arithmetic rigidity ($\{1,i,-1,-i\}$ over $\mathbb{Q}$, $j{=}1728$ CM by $\mathbb{Z}[i]$, Belyi/dessins); **[O]** the bare order-4 symmetry stays the one foundational postulate (like $c{=}$const). Closes the reduction (axiom⇒everything), not the axiom.
`v268_theta13_carrier_trace.py`	`FLAV.TH13.01` — the reactor angle θ_{13} *exponent* is the carrier hypercharge trace. The value $\sin^2\theta_{13}=\varphi_0^{\mathrm{ret}}e^{-5/6}=0.0231$ was already checked (v16); here the exponent origin is closed exactly: **[E]** $\gamma=\mathrm{tr}_E Y^2=3(1/3)^2+2(1/2)^2=5/6$ over the 5-slot carrier hypercharge $Y=\mathrm{diag}(-\frac{1}{3},-\frac{1}{3},-\frac{1}{3},\frac{1}{2},\frac{1}{2})$ (roots of $6Y^2-Y-1=0$), complement 1/6 the neutrino ratio; **[C]** value 0.02311 vs PDG ∼0.0222; **[E]** own channel — the $\mu\tau$-breaking $\varepsilon=3\varphi_0^{\mathrm{ret}}/4$ induces only $\sim1.6\times10^{-3}\ll 0.023$, so θ_{13} is a separate carrier-trace channel, not the seesaw $\mu\tau$ breaking (corrects the tempting 'θ_{13} from M_ν' route); **[O]** CP magnitude + absolute ν scale remain open.

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Script	What it machine-checks *(continued)*
`v269_spert_paqft_skeleton.py`	`QFT4D.SPERT.01` — the perturbative 4D S-matrix ($S_{\rm pert}$) as an Epstein–Glaser / pAQFT skeleton: the honest middle layer of the three-layer S-matrix ($S_{\rm top}$ 2d DHR braiding v243 / $S_{\rm pert}$ 4d EG of the spectral action / $S_{\rm phys}$ LSZ on the OS Wightman functions v240), *not* a nonperturbative closure. [E] three distinct layers (the 2d braiding $\lvert M\rvert$=1 is statistics, not 4d cross-sections); [E] every spectral-action a_4 operator has mass dimension ≤ 4 in 4D (power-counting renormalizable); [E] a finite counterterm basis (7 marginal +3 relevant, no infinite tower); [C] Epstein–Glaser existence — a renormalizable interaction has an order-by-order $S(g){=}T\exp(i\int g\,L_{\rm int})$ by causal factorisation + finite local extension (no path integral, no UV divergences); [C] adiabatic limit/gap ($\Delta{=}6\log(3/2){>}0$) $\to S_{\rm phys}$ via LSZ; [O] perturbative only — the ambient measure `QG.AMB.01` stays open. A contract/skeleton, explicitly perturbative.
`v270_pmns_jarlskog_assembly.py`	`FLAV.PMNS.03` — the complete complex PMNS matrix + the leptonic Jarlskog $J_{\rm PMNS}$ (the CP *strength*), assembled from the four TFPT channels ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\phi_0}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\phi_0 e^{-5/6}$, $\delta{=}4\pi/3$) into one unitary object. [E] the three angle channels; [E] $U = R_{23}U_{13}(\delta)R_{12}$ is exactly unitary; [C] $J_{\rm PMNS} = \Im(U_{e1}U_{\mu 2}U^*_{e2}U^*_{\mu 1}) = -0.02965$ (formula and full matrix agree), $J_{\rm max} = 0.03424$ — a *derived* CP strength, the leptonic analogue of the CKM J (v88); [C] data: $J_{\rm max}$ vs NuFIT 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative sign reproduced; [O] δ is the μ_6/triality phase (assembled input, *not* from M_ν/D_F) and the absolute ν-mass scale (Σm_ν, M_R) stays open. Assembles the PMNS matrix + CP strength; does not derive δ from D_F nor fix the scale.
`v271_eg_oneloop_quartic.py`	`QFT4D.SPERT.02` — a concrete Epstein–Glaser order-2 (one-loop) quartic renormalization, taking the v269 $S_{\rm pert}$ skeleton from *exists* to an actual EG number (no path integral). [E] $T_1{=}{:}L_{\rm int}$: needs no extension, the first extension is T_2; [E] the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble (two propagators) has sd=4=d so the UV singular order ω=sd$-d$=0; [E] $\omega{=}0 \Rightarrow$ exactly one logarithmic local counterterm ($C(\omega{+}d,d){=}1$, renormalizable, no infinite tower); [E] the loop factor $\Omega_3/(2(2\pi)^4){=}1/(16\pi^2)$ exactly — the same factor that normalises the spectral-action a_4 geometric quartic; [C] the ϕ^4 one-loop $\beta_\lambda{=}3\lambda^2/(16\pi^2)$ (symmetry factor 3 = the s,t,u channels), EG initial condition λ_Φ(UV)$=1/(16\pi^2)$; [O] order-2/one-loop only — the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Turns the skeleton into a concrete one-loop number.
`v272_nu_mass_scale.py`	`FLAV.NUSCALE.01` — the absolute neutrino-mass scale, typed honestly (it does *not* fabricate a Σm_ν number, which would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\rm EW})^2/M_R$ — one irreducible UV input like $v_{\rm geo}$ (v153), while the ratios (m_2/m_3, ordering, PMNS angles v270) are pinned; [C] the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (v249) needs a third-generation Dirac Yukawa $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] $M_R(y{=}1){=}6.1{\times}10^{14}$ has $\log_{c_3}(M_R/M_{\rm Pl}){=}2.57$ (non-integer), not a TFPT power, so the absolute value stays open; [C] the normal-ordering floor Σm_ν=0.0586 eV is consistent with cosmology ($<$ 0.072–0.12 eV); [X] a cosmological $\Sigma <$ 0.0586 eV excludes the normal-ordered $m_1{\sim}0$ spectrum. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio), not the absolute number.
`v273_eg_gauge_running.py`	`QFT4D.SPERT.03` — the gauge sector of the Epstein–Glaser one-loop (v271's quartic mechanism applied to the gauge self-energy). [E] the one-loop gauge 2-point has scaling degree sd=4=d so $\omega{=}0 \to$ one counterterm per coupling (same marginal mechanism + loop factor $1/(16\pi^2)$ as v271); [E] the one-loop b_i from the explicit carrier/SM content: $b_3{=}-7$, $b_2{=}-19/6$, $b_1{=}41/10$ (the same 41 as the carrier algebra, v159/`CAR.SM.01`); [C] physical RG directions $b_3{<}0$ (asymptotic freedom), $b_1{>}0$ (U(1) Landau); [O] the two-loop matrix (v159, PyR@TE) + all-order stay open. Shows the gauge running is an EG order-2 result. Exact b_i mirrored in Wolfram.

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Script	What it machine-checks *(continued)*
`v274_scale_overdetermination.py`	`SCALE.OVERDET.01` — the single mass anchor is *over-determined*, and the absence of a derivable absolute scale is a *theorem*. **[E]** one anchor (No-Unit, v153): every dimensionful output = pure number $\times M_{\rm Pl}^d$ ($\rho_\Lambda/M_{\rm Pl}^4$, $M_{\rm scal}/M_{\rm Pl}{=}c_3^{7/2}$, the spectrum); **[C]** two independent routes to $M_{\rm Pl}$ agree: gravity $(8\pi G)^{-1/2}{=}2.4353{\times}10^{18}$ GeV and cosmology (invert $\rho_\Lambda/M_{\rm Pl}^4{=}(3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4}{=}2.24$ meV) $=2.438{\times}10^{18}$ GeV, to 0.11%; **[E]** the anchor is the gravitational unit (G), the one measured dimensionful input of all physics; **[E]** theorem not gap: the seam is a conformal $c{=}8$ CFT (v157/v158), scale-invariant by construction → no intrinsic scale; **[O]**/**[E]** scale-complete up to one over-determined anchor = maximal predictivity. Sharpens v153/v78.
`v275_qgamb_roadmap.py`	`QGAMB.ROADMAP.01` — the obligation roadmap for `QG.AMB.01` (the ambient nonperturbative QG measure), consolidated into one module (the open gate had no script). **[E]** Tier A gap decoupling: $\Delta_{\rm eff}{=}6\log(3/2) - 2{\cdot}248\,c_3^2{=}1.648{>}0$ (v36/v76), so every low-energy readout is independent of the un-built measure — not a blocker; **[E]** Tier B: the bulk measure reduces to identifying the seam-Calderón boundary with the holomorphic $(E_8)_1$ net ($c{=}248/31{=}8$; v77/`GATE.METRIC.03`), reduced not closed (`REDTEAM.A.01`); **[C]** the perturbative layer is in hand (v269/v271/v273); **[O]** the nonperturbative ambient measure is *not* constructed — the one genuine structural frontier; **[X]** closes IFF the seam net $=(E_8)_1$ *and* the bulk reconstruction is proved. A roadmap, not a closure.
`v276_qgeo_flat_closes_commutator.py`	`QGEO.SYM.03` — the flat $\tau{=}i$ pillowcase premise *closes* the modular-commutator chain to all orders, collapsing the bedrock `QGEO.SYM.01` to *one* geometric postulate (it does *not* prove the premise). For a quasi-free seam state `QGEO.SYM.01` $=(\omega\circ\rho{=}\omega)\Leftrightarrow[\rho,C]{=}0\Leftrightarrow[\rho,H]{=}0$. **[E]** isometry⇒commutator: if $H{=}f(\Delta_{\rm flat})$ and the order-4 deck ρ ($z{\to}iz$) is a flat-$\tau{=}i$ isometry then $[\rho,\Delta]{=}0\Rightarrow[\rho,H]{=}0$ *exactly* (all orders), verified on the discretised square torus ($\rho^4{=}I$, $[\rho,\Delta]{=}0$, $[\rho,\sqrt{-\Delta}]{=}0$; generic non-isometry fails); **[E]** this upgrades v198 (principal symbol) + v201 (subprincipal) to the full H (flatness removes the curvature terms); **[C]** $\tau{=}i$ is forced by order-4 ($j{=}1728$; hexagonal $j{=}0$ is order-6, excluded; v214/v267); **[O]** `QGEO.SYM.01` collapses to the single statement 'the raw seam is the flat $\tau{=}i$ pillowcase quasi-free state' — needs a human constructive-QFT proof or stays an honest axiom. Sharpest non-circular form.
`v277_seam_calderon_e8_match.py`	`QGAMB.TIERB.01` — the seam-Calderón → $(E_8)_1$ matching certificate, the concrete Tier-B step of `QG.AMB.01` (v275): it computes the full $(E_8)_1$ matching target and pins the residual to *one* bit (it does *not* close it). **[E]** $c{=}8$ (Sugawara, $248/31{=}g_{\rm car}{+}N_{\rm fam}$), but $c{=}8$ alone does not pin the net; **[E]** the discriminator is holomorphy — $\det\mathrm{Cartan}(E_8){=}1$ (1 primary, holomorphic) vs the $c{=}8$ counterexample $\det\mathrm{Cartan}(D_8{=}SO(16)){=}4$ (4 primaries, non-holomorphic), so the single discriminator is $\lvert\det K\rvert{=}1$; **[E]** the $(E_8)_1$ character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1+248q+\dots)$ has a single primary + exactly $248{=}\dim E_8$ weight-1 currents (240 roots); **[E]** equivalence chain holomorphic ⇔ det Cartan 1 ⇔ unique even unimodular rank-8 (v83) ⇔ homology-sphere link ⇔ $\lvert\det K\rvert{=}1$ (v235); **[C]** seam-side match via v276 (flat $\tau{=}i$) + v260 (Kummer/K3); **[O]** residual = the single holomorphy bit — reduced to one bit, *not* closed.

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Script	What it machine-checks *(continued)*
`v278_lsz_bridge_unitarity.py`	`QFT4D.SPERT.04` — the $S_{\rm pert} \to S_{\rm phys}$ LSZ bridge + one-loop unitarity: connects the EG perturbative S-matrix (v269/v271/v273) to the physical asymptotic S-matrix ($S_{\rm phys}$ = LSZ on the OS Wightman functions, v240). **[E]** LSZ bridge: $S_{\rm phys}$ = amputate+on-shell-residue of the connected time-ordered products, and the EG T-products of $S_{\rm pert}$ *are* those correlators (OS Wick-rotation, v240) $\Rightarrow S_{\rm pert} \xrightarrow{\rm LSZ} S_{\rm phys}$; **[C]** the gap Δ=6 log(3/2)>0 gives Haag–Ruelle asymptotic states so LSZ is valid; **[E]** *one-loop unitarity* (optical theorem): the bubble discontinuity $\operatorname{Im} B(s)/\pi = x_+ - x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2 \operatorname{Im} M = \sum\int d\Pi\|M\|^2$ (cutting rule) holds ⇒ matter+gauge unitary; **[E]** threshold: $\operatorname{Im} B$=0 below s=4m^2, turning on at the physical two-particle threshold; **[O]** gravity caveat — the Stelle ghost makes the gravity perturbative S-matrix non-unitary →`QG.AMB.01`; **[C]** finite field-strength Z from the EG scheme.
`v279_qgeo_obligation_lemma.py`	`QGEO.OBLIG.01` — the `QGEO.SYM.01` bedrock written as *one* precise constructive-QFT lemma with a proof-tree completeness check (the formal obligation write-up; does *not* close it). Lemma: given the premise 'the raw seam carries the flat τ=i pillowcase metric $g_{\rm flat}$ (Troyanov-unique, cone angles π), so $\Lambda_\Sigma = \sqrt{-\Delta_{\rm flat}}$', then $\omega_\Sigma \circ \rho = \omega_\Sigma$ → mark-locality → E_8. **[E]** the reduction DAG from 'free RP seam' has 8 closed nodes and *exactly one* open leaf ('the raw seam state is the flat state'); **[E]** the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders, Lean `FORM.QGEO.03`), so the obligation is state-identification not geometric ambiguity; **[C]** two proof routes (RP-state uniqueness on the flat orbifold, or Troyanov + OS); **[E]** the all-orders step is a Lean theorem (`SeamDeckClosure.flat_all_orders_clock`, standard axioms, no `sorry`); **[O]** one constructive-QFT statement closes `QGEO.SYM.01`.
`v280_pillowcase_steklov.py`	`QGEO.STEKLOV.01` — a direct numerical experiment on the flat τ=i pillowcase orbifold (the seam geometry): the self-investigable test of the `QGEO.SYM.01` obligation (does *not* prove the premise). **[E]** pillowcase = the Z2-even sector of the flat torus, 4 cone points (2-torsion, deficit π each, sum 4π=2$\pi\cdot$2); **[E]** the order-4 deck ρ ($z \to iz$) descends ($[\rho, Z_2]$=$[\rho, P_{\rm even}]$=0) and permutes the 4 cone points as 2 fixed + 1 swapped pair; **[E]** the geometric chain is exact: $[\rho,\Delta]$=0 ⇒ $[\rho, H]$=0 ⇒ $[\rho, C]$=0 ⇒ $\omega\circ\rho$=ω (the v276/v279 all-orders closure on the real orbifold); **[C]** the Steklov DtN is positive self-adjoint, fixed by the flat metric (route-(i) evidence); **[O]** confirms the conclusion given flat geometry, the premise stays open.
`v281_holomorphy_modular_data.py`	`QGAMB.MODULAR.01` — the modular-data / anyon-condensation layer of the `QG.AMB.01` Tier-B holomorphy bit. **[E]** #anyons = |det Gram|: E_8→1 (holomorphic), D_8=SO(16)→4, carrier $D_5\oplus A_3$→16; **[E]** the anyon-condensation tower $D_5\oplus A_3$ (16) → D_8 (4) → E_8 (1), each step condensing a Lagrangian Z_2 boson, the Kitaev E_8 endpoint (v92/v235); **[E]** Gauss–Milgram recovers c=8; **[E]** E_8 is the unique holomorphic c=8 (det 1) — SO(16) has the same c but 4 anyons, so holomorphy is THE discriminator (v277); **[O]** the open bit: does the seam-Calderón net condense to det K=1? Reduced to one bit.
`v282_e8_tau_i_unification.py`	`QGAMB.UNIFY.01` — the E_8-at-τ=i unification: the candidate *simplification* reducing the two open obligations to *one* object (the raw seam is the $(E_8)_1$ holomorphic CFT at τ=i). $\chi_{E_8}(\tau)$=Θ_{E_8}/η^8=$j^{1/3}$. **[E]** τ=i is the order-4 CM point (j=1728, the `QGEO.SYM.01` flat geometry); **[E]** χ_{E_8}=$j^{1/3}$ (χ^3=$q\,j$ as q-series), so $\chi_{E_8}(i)$=$1728^{1/3}$=12 (the $(E_8)_1$ character at the SAME τ=i, the `QG.AMB.01` holomorphy); **[E]** one object, two faces; **[C]** obligation count 2→1: prove 'the raw seam is $(E_8)_1$ at τ=i' and *both* follow; **[O]** the unified premise stays open but is now one object.

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Script	What it machine-checks *(continued)*
`v283_live_killtest_scorecard.py`	`PRED.LIVE.01` — a live kill-test scorecard of the recent round's computable predictions vs current data, each tagged with the decisive experiment. [C] PMNS angles $\sin^2\theta_{12}$=0.30675 ($+0.31\sigma$), $\sin^2\theta_{13}$=0.02311 ($+1.45\sigma$), θ_{23} maximal; [C] leptonic CP $J_{\rm PMNS}$=−0.0297, δ=240° vs $\sim 232(30)$; [X] Σm_ν NO floor 0.0586 eV (DESI/CMB-S4 decisive); [X] proton decay τ_p=1.55×10^{35} yr for $+(15,1,1)$ (Hyper-K decisive); [E] closed readouts α^{-1}, Ω_b, β; all rows consistent $<2\sigma$, two sharp near-term kill tests. A consolidation, not new physics.
`v284_route_i_rp_uniqueness.py`	`QGEO.ROUTEI.01` — Route (i) (RP-state uniqueness) decomposed into a 6-lemma chain with every dischargeable lemma discharged and the one irreducible lemma isolated (does *not* prove the open premise). [E] 5/6 lemmas are existing [E]/[O] results — L1 RP→DtN (v194), L2 the four marks force the pillowcase class (v195/v216/v214), L3 the flat orbifold Steklov operator $=\sqrt{-\Delta_{\rm flat}}$ (verified: flat-strip DtN$(k)/k\to1$, principal symbol $\lvert\xi\rvert$ with no curvature term), L4 covariance $C=(1+e^H)^{-1}$ from the DtN (v258), L5 μ_4-invariance (v276/Lean/v280); [C] Troyanov reduction: 4 cone points of angle $\pi\Rightarrow\chi_{\rm orb}=0\Rightarrow$ the unique constant-curvature representative is flat (τ=i), so $L_{\rm open}$ reduces to 'the raw seam is the constant-curvature geometric state'; [O] the one open lemma — the raw RP seam state is the constant-curvature/geometric state (the state-identification step).
`v285_route_ii_seam_condensation.py`	`QGAMB.ROUTEII.01` — Route (ii) (seam-net condensation) decomposed into a 4-lemma chain, with the key result that its open lemma *coincides* with Route (i)'s (the v282 unification at the lemma level). [E] 3/4 lemmas discharged — M1 no abelian sector $\Leftrightarrow\lvert\det K\rvert$=1 $\Leftrightarrow$ holomorphic (v234/v235), M2 holomorphic c=8 $\Leftrightarrow(E_8)_1$ with the $SO(16)$ det-4 counterexample (v277/v281), M3 the condensation tower 16→4→1 = the unique μ_4-Lagrangian glue (v92/v281/v235); [E] the discriminator det 1 vs det 4; [E] Routes (i) and (ii) share *one* open lemma — both say 'the raw seam is the $(E_8)_1$-at-τ=i object' (v282), so proving either closes both; [O] the unified open lemma: construct a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at τ=i. $M_{\rm open}$ = `QGEO.SYM.01`.
`v286_seam_equivalence_contract.py`	`SEAM.EQUIV.01` — the Seam Equivalence contract: concentrates the whole open structure on *one* named arrow (does *not* prove it). Theorem: the raw RP seam state reconstructs canonically the holomorphic $(E_8)_1$ boundary net at the τ=i pillowcase. [E] four objects (RawRPSeam, FlatTauIPillowcaseDtN, HolomorphicE8Net, SREKitaevE8Phase); [E] six arrows — 2 closed (Route i v276/v280/v284, Route ii v281/v277/v235), 1 conditional (v282), 2 open premises (the same lemma, v284/v285), 1 Gral; [E] **import firewall**: no Route-i module imports any Route-ii module or vice versa, so the routes converge only through `SEAM.EQUIV.01` (kills the 'E_8 into the geometry and back' circularity); [E] exact-label taxonomy (Formal/Numerical/Conditional Exact, Open Selection); [O] the one open arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary'.
`v287_free_rp_bulk_to_holomorphic_boundary.py`	`SEAM.EQUIV.A01` — Route A (AQFT) theorem-dependency checker for the open Gral arrow 'free Gaussian RP bulk ⇒ holomorphic single-sector boundary net': types the 5-lemma chain and isolates the *one* missing standard theorem. [E] L1 RawRPSeam→OS→CAR/quasi-free boundary (v155/v160/v240/v175); [E] L2 gap+chirality ⇒ gapped chiral bulk (v64/v156); [O] L3 invertible/SRE Gaussian bulk ⇒ single-sector boundary (μ-index 1) — the one missing standard theorem (the AQFT form of $\det K$=1); [E] L4 μ=1+c=8 ⇒ holomorphic (v234); [E] L5 holomorphic c=8 ⇒ $(E_8)_1$ (v83/v277/v281). Verdict: 4/5 discharged, Route A reduces `SEAM.EQUIV.01` to one cited theorem.

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Script	What it machine-checks *(continued)*
`v288_full_l2_subprincipal_z4.py`	`SEAM.EQUIV.B01` — Route B (DtN): the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality, proving (numerically, on the full Toeplitz operator) the 'probably provable' step and lifting v201/v284 from the finite block to L^2. **[E]** character orthogonality: a mark-sourced curvature $f=\sum_{j<4} g(\theta-j\pi/2)$ has Fourier support only on $m\equiv 0 \pmod 4$; **[E]** full-L^2 block-diagonality: $\Lambda=\|n\|+M_f$ commutes with $\rho=\mathrm{diag}(i^n)$, $\|[\rho,\Lambda]\|\sim 2{\times}10^{-15}$ on $-50..50$; **[E]** negative control: a generic curvature gives $\|[\rho,\Lambda]\|=4.44>0$; **[E]** Lean consistency (`geom_sum_fourth_root` + `markLocal_blockDiagonal`); **[O]** the sharper residual — why is the raw seam sub-principal term mark-local?
`v289_marklocal_raw.py`	`MARKLOCAL.RAW.01` — decompose the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolate the single open analytic lemma. **[C]** L1 conic parametrix: conic Steklov DtN $=\|D_\theta\|+M_f+$ cone sources (b-calculus/Melrose, imported); **[O]** L2 **Flat-Away** — the one open analytic lemma: RP+gap+4 marks ⇒ curvature vanishes away from the marks (the flat-pillowcase realisation); **[E]** L3 orbit-source: the 4 marks form one μ_4 orbit (v195/v216); **[E]** L4 Fourier-support: μ_4-orbit curvature ⇒ support only on $4\mathbb{Z}$ (v264/v288, with a non-orbit negative control); **[E]** L5 full-operator: ⇒ $[\rho,\Lambda]{=}0$ on full L^2 (v288). 4/5 discharged; Route B reduces to exactly L2.
`v290_z4_smooth_curvature_adversary.py`	`SEAM.ADVERSARY.01` — the Route-B red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality. The strongest adversary, a smooth $\mathbb{Z}_4$-symmetric off-mark curvature $f=\varepsilon\cos(4\theta)$: **[E]** it is $\mathbb{Z}_4$-symmetric (Fourier support $\{\pm4\}\subset 4\mathbb{Z}$) but *not* mark-local ($f(\pi/4)=-\varepsilon\neq 0$); **[E]** it *passes* the v288 commutator $\|[\rho,\Lambda]\|=0$ (same as mark-local) — so $[\rho,\Lambda]{=}0$ is necessary but *not* sufficient; **[E]** but it shifts the DtN spectrum off the flat ladder ($\max\|\Delta\lambda\|=0.155$) and the heat trace ($\Delta=0.019$), so the KMS weight and $(E_8)_1$ character shift; **[O]** a genuine candidate modulus the commutator misses but the spectral data excludes — Flat-Away must use spectral/energy/heat input.
`v291_flataway_contract.py`	`FLATAWAY.RP.01` — the Flat-Away lemma as its own named mini-theorem with three proof routes opened by v290. Lemma **[O]**: given RawRPSeam with gap, chirality and the four μ_4 marks, the smooth curvature density in the DtN sub-principal symbol vanishes away from the branch divisor. **[E]** *spectral-rigidity*: scanning $f=\varepsilon\cos(4\theta)$, the spectrum deviation from the flat ladder is 0 only at $\varepsilon{=}0$ — flat is the unique spectral match; **[E]** *heat-kernel*: the heat-trace deviation is 0 only at $\varepsilon{=}0$; **[C]** *RP-energy*: RP+gap is a Dirichlet-energy minimisation in the Troyanov conformal class ⇒ constant-curvature minimiser (Troyanov; cf. v284). **[O]** three independent routes to one lemma; closing any one closes Route B.
`v292_flataway_heat_reduction.py`	`FLATAWAY.HEAT.01` — the Heat-Kernel route made precise: reduces Flat-Away to *one* named spectral invariant. **[E]** the heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda})=c(t)\varepsilon^2+O(\varepsilon^4)$ is a positive-definite quadratic form in the off-mark curvature ($c_k(t)>0$ for every $\mathbb{Z}_4$ mode $4k$; positive on all combinations), so $\Delta\mathrm{Tr}=0\Leftrightarrow f=0$; leading invariant $\|f\|_{L^2}$ (Parseval). **[O]** reduction: Flat-Away ⇐ 'the seam fixes the a_2 heat coefficient to its flat value' ⇒ $\int f^2=0\Rightarrow f=0$ off the marks — one external invariant.
`v293_flataway_spectral_hessian.py`	`FLATAWAY.SPEC.01` — the Spectral-Rigidity route upgraded from one direction (v291) to the *full* smooth-$\mathbb{Z}_4$ space. **[E]** splitting mechanism: the degenerate flat Steklov level $\|n\|$ is split by mode $4k$ at $\|n\|{=}2k$ by $\pm\varepsilon/2$; **[E]** $S=\sum(\Delta\lambda)^2\sim 0.50\varepsilon^2$; **[E]** the Hessian over modes $\{4,8,12\}$ has eigenvalues $\{0.497,0.500,0.503\}>0$, so the flat seam is a *strict* isolated minimum over the whole $\mathbb{Z}_4$ space; **[E]** a non-$\mathbb{Z}_4$ mode breaks the commutator outright. **[O]** pinning the seam's Steklov spectrum excludes every smooth $\mathbb{Z}_4$ deformation ⇒ Flat-Away.

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Script	**What it machine-checks** *(continued)*
`v294_flataway_rp_energy.py`	`FLATAWAY.ENERGY.01` — the RP-energy / Troyanov route: the flat pillowcase is the unique constant-curvature minimiser at the fixed mark divisor. [E] prescribed divisor + Gauss-Bonnet: S^2 with 4 cone angles π, $\sum(2\pi-\theta)=4\pi=2\pi\chi$; [C] Troyanov/Euclidean-cone uniqueness (Troyanov 1991) excludes smooth off-mark curvature; [E] strictly convex curvature energy $E[f]=\int f^2$, Hessian $2\pi I$ (PD) $\Rightarrow$ unique minimiser; [E] no zero-cost deformation. [O] premise: 'RP+gap realises the energy-minimiser', converging with v292/v293 on one selection principle; closing any one closes Route B.
`v295_flataway_a2_exact.py`	`FLATAWAY.A2.01` — the *exact* analytic a_2 coefficient, upgrading the v292 positive-definiteness from numerical to a proof. [E] first variation vanishes: $\partial_\varepsilon \mathrm{Tr}\, e^{-t\Lambda(\varepsilon)}\|_0=-t\hat{f}(0)\sum e^{-t\|n\|}=0$ (zero-mean off-mark curvature, Gauss-Bonnet); [E] convexity $\Rightarrow$ global minimum: $\phi(x)=e^{-tx}$ strictly convex $\Rightarrow \Lambda\mapsto \mathrm{Tr}\, e^{-t\Lambda}$ convex (Klein/Peierls), so ε=0 is the global minimum and $\Delta\mathrm{Tr}\geq 0$ for all ε, $=0$ iff $f=0$; [E] exact a_2 kernel via the divided difference of e^{-tx} (Duhamel), matched to numerics to rel.err $<10^{-6}$; [E] degenerate-split closed form $2e^{-2kt}(\cosh(tg/2)-1)\geq 0$; [O] reduction now analytic: Flat-Away $\Leftarrow$ 'the seam fixes a_2'. Lean-mirrored (`FORM.FLATAWAY.01`).
`v296_flataway_a2_closed.py`	`FLATAWAY.A2.CLOSED.01` — the exact a_2 coefficient in *closed form.* [E] the operator tails telescope geometrically ($C_k+D_ke^{-4kt}=4kt(1-e^{-4kt})$), giving $W_k^{\mathrm{tail}}=t(1-e^{-4kt})/(8k(1-e^{-t}))$; [E] a finite ($4k$−1)-term middle (incl. the degenerate diagonal $\Phi(2k,2k)=(t^2/2)e^{-2kt}$); [E] closed form $W_k(t)=\frac{t(1-e^{-4kt})}{8k(1-e^{-t})}+\frac{1}{2}\sum_{m=1}^{4k-1}\Phi(m,4k-m)$ ($W_1(t)=t(2te^t+e^{3t}+3e^{2t}+e^t-1)e^{-3t}/8$), reproducing v295 to machine precision; [E] manifestly positive; [E] small-t: $W_k=t/2+O(t^3)$ (mode-independent leading 1/2, the t^2 term cancels).
`v297_route_a_literature_stack.py`	`SEAM.EQUIV.A.LIT.01` — Route A's 'one standard import' as a precise citable theorem stack. [E] LIT-A invertible/SRE bulk $\Rightarrow$ trivial bulk topological order (Kitaev 2006; Freed–Hopkins 2021); [E] LIT-B bulk MTC = Drinfeld center, trivial center $\Rightarrow$ holomorphic boundary net (Müger 2003; Kawahigashi–Longo–Müger 2001); [E] LIT-C holomorphic c=8 net $\Rightarrow (E_8)_1$ (Conway–Sloane; Dong–Xu/Kawahigashi–Longo). [E] the three compose into 'invertible bulk $\Rightarrow (E_8)_1$', all legs established; [O] the one open hypothesis is the input of LIT-A, 'the raw quasi-free seam bulk is invertible (SRE)' = Route B's Flat-Away. [E] non-circular with Route B (the v286 firewall holds).
`v298_e8_network_attractor.py`	*E8 as a network attractor* — a computational-substrate (Wolfram-program) reading of the $(2,3,5)$/McKay hull, on the v219 bedrock; an honest audit, not a derivation. [E] the E8 Kac marks $(1,2,3,4,5,6,4,2,3)$ are the *unique* attractor of the minimal lazy graph update $m\leftarrow(A+2I)/4\cdot m$ on the affine-$E_8/(2,3,5)$ McKay graph (converges from any positive start; $A\,m=2m$, top eigenvalue 2). [E] the spherical McKay tower terminates at E_8 ($2T\to E_6$, $2O\to E_7$, $2I\to E_8$, $h=30$); [E] the cascade is nested-subgraph coarse-graining ($240\to126\to72\to40\to20$). [C] Wolfram reading: E_8 as the universal attractor of a minimal $(2,3,5)$ network (Coxeter-skeleton level). [O] a minimal hypergraph *rewrite* whose attractor is the full TFPT structure stays open; $\varphi_0=(4/3)c_3=1/(6\pi)$ is the analytic seed, not an edge fraction.

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Script	What it machine-checks *(continued)*
`v299_hypergraph_0d_autonomous.py`	The hypergraph/Wolfram bridge (hardened): the 3-arm $(2, 3, \cdot)$ branch type is the irreducible seed on which a local witness-based spectral growth *reaches* E_8 as the last finite point ($E_6 \to E_7 \to E_8$, then stops at the affine boundary $\hat{E}_8$) — *not* 'P2 derived'. **[E]** icosahedron = 3-uniform hypergraph ($\lvert\mathrm{Aut}\rvert = 120 = \lvert I_h\rvert$; the McKay group is the $SU(2)$ binary lift $2I$, a different order-120 group); **[E]** the $1 \to 4$ subdivision rewrite is equivariant (120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith: $\rho \le 2$ selects (affine) ADE; **[E]** Collatz–Wielandt: $\rho \le 2 \Leftrightarrow$ a positive witness m with $Am \le 2m$ (local); **[E]** the autonomous rule (diffusion witness + local gate $Am \le 2m$) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$ under the $(2, 3, \cdot)$ seed; **[E]** A_n ($\rho = 2\cos\frac{\pi}{n+1}$) and D_n ($\rho = 2\cos\frac{\pi}{2n-2}$) controls are < 2 for all n, never reaching E_8; **[O]** the seed = P2. A second network reading of the $(2, 3, 5)/E_8$ core (v219/v298).
`v300_flataway_rigid.py`	`FLATAWAY.RIGID.01` — the attack on Flat-Away. **[E]** the flat fingerprint: $\mathrm{spec}(\lvert D_\theta\rvert)$ is the integer ladder, every level $n \ge 1$ of multiplicity exactly 2 (a discrete fingerprint, not a norm); **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$), validated vs the full Toeplitz operator; **[E]** any $g_k \ne 0$ lifts the resonant degeneracy (mult $2 \to 1{+}1$) and changes the spectral multiset, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — a discrete obstruction, strictly stronger than the v291/v293 'deviation grows' scan; **[E]** with the v295 convexity, flat is isolated analytically *and* discretely; **[C]** the pin derived: the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), and by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, so $(E_8)_1$ rationality $\Rightarrow f = 0$ (conditional on Route A); **[O]** Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact.
`v301_route_a_invertible.py`	`SEAM.EQUIV.A.INV.01` — discharge Route A's last open hypothesis ('the raw quasi-free seam bulk is invertible/SRE') via the free-fermion classification; since v300 collapsed Route B into Route A, this is the last unconditional gap of `SEAM.EQUIV.01`. **[E]** carrier = 16 Majoranas, $c = 8$ ($g_{\mathrm{car}}{+}N_{\mathrm{fam}} = 5{+}3 = 8 = 16/2$, v148); **[E]** $SO(16)_1 \supset (E_8)_1$: $\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248 = \dim E_8$ (v154); **[C]** free-fermion invertibility (Kitaev): a gapped Gaussian/quasi-free 2+1D fermion phase (class D) is $\mathbb{Z}$-classified by the chiral Majorana number and is *always* invertible (no intrinsic anyons; topological order needs interactions), so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $\lvert\det K\rvert = \#$anyons is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce; **[O]** LIT-A's 'invertible bulk' now follows from 'gapped 16-Majorana quasi-free bulk' (v148/v160+the gap), so the residual changes from a topological-order statement to the spectral input 'is the quasi-free bulk gapped?'; **[O]** with v300 the whole open content of `SEAM.EQUIV.01` reduces to the single spectral statement 'the quasi-free seam bulk is gapped'.

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Script	**What it machine-checks** *(continued)*
`v302_seam_gap.py`	`SEAM.EQUIV.GAP.01` — the last lever: after v300/v301 the whole open content of `SEAM.EQUIV.01` was 'the quasi-free seam bulk is gapped'; that gap is *not* a free input but the established Recovery gap of the seam transfer operator. **[E]** $\mathrm{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple leading Perron eigenvalue 1; **[E]** the gap is derived: $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $6 = 2N_{\mathrm{fam}}$, $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); **[E]** the positivity margin $\Delta - 31/(4\pi^2) = 1.6476 > 0$ (the v76 decoupling margin); [C] by Osterwalder–Schrader / quasi-free clustering a transfer gap $\Delta > 0$ is a mass gap of the reconstructed Hamiltonian; **[O]** so the last TFPT-internal input is the derived Recovery gap $6\ln(3/2) > 0$, and `SEAM.EQUIV.01`'s residual is a composition of cited theorems (OS/clustering + Kitaev v301 + the v297 AQFT stack) over established TFPT facts — no undischarged TFPT-internal assumption remains, though it stays **[O]** (not machine-proved end-to-end).
`v303_ftransfer_dynamics.py`	`FR.DYNAMICS.01` — the discrete→dynamic lens on `F_transfer`: the main-branch mechanism (a gapped, positivity-preserving relaxation to a *unique* attractor; the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) is the shared *shape* of all four instances. **[E]** F_pole (Koide) *is* the seam dynamics — the source→pole/branch relaxation is the Möbius map with fixed-point multiplier $F' = (2/3)^6 = \lambda_2$, the seam-clock subleading eigenvalue (v82/v54); $\lvert F'\rvert < 1 \Rightarrow$ a unique attractor at the seam rate; [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem; compiler fixes $A_\Lambda = 10$, rate thermal); [C] F_relic (axion) freezes at an adiabatic fixed point (seed $\theta_i = 3\pi/5$, rate cosmological); **[E]** F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3 = -7$ (asymptotic freedom ⇒ the Gaussian UV attractor, Λ_{QCD} generated). **[O]** Verdict: one shape underlies all four = the main-branch principle; F_pole has the seam rate exactly, the other three share only the shape (external rates), so `F_transfer` is the readout end of the one principle, not a bolt-on — simplicity is preserved by honest fencing (v187).
`v304_idg_nonlocal_ghost.py`	`QGAMB.IDG.01` — the KMS spectral-action cutoff suggests a *nonlocal* (infinite-derivative) graviton form factor; under entire analyticity the Stelle ghost is a *truncation artefact* — a [C] narrowing of the red-team `rt_F` perturbative-gravity attack, *not* a closure of `QG.AMB.01` (no new parameter: the cutoff is the v259 seam KMS weight). **[E]** the local four-derivative propagator $1/(p^2(p^2+M^2)) = (1/M^2)[1/p^2 - 1/(p^2+M^2)]$ has a healthy massless graviton (residue $+1/M^2$ at $p^2{=}0$) and a massive spin-2 mode with *negative* residue $-1/M^2$ at $p^2{=}-M^2$ (the Stelle ghost, reproducing v278); **[E]** for an IDG operator $p^2 a(p^2)$ with $a(z) = e^{z/M^2}$ entire and nowhere zero, the dressed propagator $1/(p^2 a)$ keeps its *only* pole at $p^2{=}0$ — no ghost (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006); **[E]** negative control: a polynomial truncation $a(z) = 1 + z/M^2$ (the $R{+}R^2$ order) has a real zero at $z{=}-M^2$ = the ghost pole, so 'entire vs polynomial' is the discriminator; [C] the cutoff is the $\beta{=}1$ seam KMS weight $f(u) = e^{-u}$ (v259), not a choice — truncating $\mathrm{Tr}\, e^{-D^2/\Lambda^2}$ at a_4 is what makes the local $R{+}R^2$ and the ghost; [C] **conditional narrowing**: *if* the resummed form factor is entire (Biswas–Mazumdar–Siegel class) *then* the perturbative graviton is ghost-free and `rt_F`'s attack is a truncation artefact; **[O]** **open**: (i) the trace regulator f is not the kinetic form factor $a(\Box)$ — entire analyticity of the resummation is unproven; (ii) a ghost-free *perturbative* graviton is not the nonperturbative ambient measure — `QGAMB.IDG.01` stays [C]/**[O]**, not a `QG.AMB.01` closure.

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Script	**What it machine-checks** *(continued)*
`v305_witness_independence.py`	Witness-independence / generator-economy audit (structural anti-numerology firewall, companion to v100): every headline integer is a documented image of the single anchor a=(1, 1, 2) (e=(4, 5, 2), p_n=2+2^n; v23/v228), π the only transcendental primitive (`ARCH.CORE.01`) — 13 atoms / 2 free primitives = 6.5× compression made explicit (the recurring 2/3=$\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio); $E_8/g_{\rm car}$/anchor are overdetermined ($\geq$ 3 structurally independent witnesses), the 5 pure-seed readouts single-witness (correlated, scored jointly by v100); α^{-1}'s '137' is foreign to the anchor family = an independent witness **[E]**
`v306_seed_crossval.py`	Leave-one-out cross-validation of the shared seed φ_0: back-solving φ_0 independently from the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C) and error-weighted-averaging reproduces the axiom seed φ_0=1/(6π)+48c_3^4=0.0531720 to 0.04% (one real seed, not a per-observable fit); leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, ∼2σ); a data↔formula shuffle explodes the χ^2 by > 100× **[E]**
`v307_data_watchdog.py`	Data watchdog: the operational decision pipeline over the frozen registry (v84). Classifies all 20 entries (16 predictions + 2 assigned textures + 2 bands) PASS/WATCH/TENSION/KILL by live n_σ vs repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18); no decidable prediction in KILL (theory alive); watch items α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core $\sin^2\theta_{13}$ (2.0σ); the prediction-of-record $\sin^2\theta_{12}$=0.306747 compatible (JUNO, the fastest falsifier); $r \in [0.0033, 0.0048]$ below the BK18 limit and the kill 0.01; w=−1 flagged the dark-energy watchdog (DESI) **[E]**
`v308_seam_equiv_chain.py`	TOE attack 1 — the `SEAM.EQUIV.01` keystone as a composition certificate: the v297 stack as one well-typed chain gap>0 → invertible → holomorphic c=8 → $(E_8)_1$ with exact discriminators $\lvert\det \mathrm{Cartan}(E_8)\rvert$=1 vs $\lvert\det \mathrm{Cartan}(D_8)\rvert$=4, carrier tower 4·4=16 → 4 → 1, c=$g_{\rm car}$+$N_{\rm fam}$=8; the input gap is the derived recovery gap 6 ln(3/2)>0. The only undischarged premise is the cited OS step — an assembly/reduction certificate (Lean target), NOT a closure **[E][O]**
`v309_modular_bedrock.py`	TOE attack 2 — the bedrock $\omega\circ\rho$=ω via an explicit modular Hamiltonian, non-circular: marks = 4 from Gauss–Bonnet (v216) + the order-4 clock from cross-ratio 2 (v214) ⇒ μ_4 derived, not assumed; the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ round-trips to the seam operator and $[\rho, K]$=0 ⇔ $\omega\circ\rho$=ω (a $\mathbb{Z}_2$ profile breaks it). Residual = Bisognano–Wichmann intrinsicality of the raw collar; NOT a closure of `QGEO.SYM.01` **[E][O]**
`v310_carrier_sm_anomaly.py`	TOE attack 3 — the carrier half-spinor is one anomaly-free SM generation with the SM RG: 16=$\Lambda^{\rm even}(\mathbb{C}^5)$=1+10+5 → $SU(5) \subset SO(10)$ as 10+$\bar{5}$+1 = one generation; gauge-anomaly-free ($\sum Y$=0, $\sum Y^3$=0, $[SU(2)]^2U(1)$=$[SU(3)]^2U(1)$=0, exact) and reproduces (b_1, b_2, b_3)=$(\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so $N_{\rm fam}$=3 pins the RG. Closes the spectrum/anomaly/β sub-question; the full EG S-matrix + Yukawa stay open **[E][C]**
`v311_gap_decoupled_measure.py`	TOE attack 4 — the physical QG measure as a gap-decoupled, exponentially-clustering object: the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$, correlation length 1/(6 ln(3/2)), finite susceptibility 729/665; the v76 margin 6 ln(3/2) − 31/(4π^2)≈1.648>0 decouples the physical sector from the un-built ambient. Spectral-action a_2=EH/a_4=R^2/72 (v36) + KMS β=1 (v259); `QG.AMB.01` stays **[O]** (gap-decoupled, inert) **[E][O]**

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Script	What it machine-checks *(continued)*
`v312_hypergraph_substrate.py`	TOE attack 5 — the hypergraph substrate sharpened: the affine-E_8 Kac marks are the Perron eigenvector ($A\,m{=}2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5){=}$ the golden ratio (the 5-fold/icosahedral signature), but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3{=}\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ (gap $6\ln(3/2)$) and the analytic seed φ_0. The open question is precisely delimited, not closed **[E][O]**
`v313_golden_atoms_spectral.py`	The golden ratio is the $g_{\rm car}{=}5$ signature: the $(2,3,5)$ atoms *are* the affine-E_8 network spectral angles (v312 follow-up, exact). **[E]** the charpoly factors $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, the golden minimal polynomial $x^2{-}x{-}1$ divides it ($\phi{=}2\cos(\pi/5)$ an exact eigenvalue); the non-trivial eigenvalues are $2\cos(\pi/k)$ for $k\in\{2,3,5\}{=}\{\lvert\mathbb{Z}_2\rvert,N_{\rm fam},g_{\rm car}\}$ (giving $0,1,\phi,1/\phi$), so the golden ratio is the $g_{\rm car}{=}5$ (5-fold) signature — a new spectral witness for $g_{\rm car}$ (raises the v305 multiplicity). The $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5{=}31/30$ ties the v63/v76 margin numerator $31{=}2^{g_{\rm car}}{-}1{=}248/8{=}1{+}h$ to the Coxeter 30 (a unification, not new evidence). **[E]** honest null: ϕ is structural, not a readout. **[C]** direction: ϕ is the E_8-quasicrystal (Elser–Sloane) ratio, a candidate substrate **[E][C]**
`v314_rate_translation.py`	Do the discrete-kernel and dynamic rates translate? An exact number-field split. The affine-E_8 adjacency spectrum splits into the rational part $\{0,\pm1,\pm2\}$ in $\mathbb{Q}$ and the golden part $\{\pm\phi,\pm1/\phi\}$ in $\mathbb{Q}(\sqrt5)$; the dynamic transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ is entirely rational. **[E]** the family rates DO translate (the rational angles $2\cos(\pi/2){=}0{=}\lvert\mathbb{Z}_2\rvert$, $2\cos(\pi/3){=}1{=}N_{\rm fam}$ build the recovery rate $(2/3)^6{=}(\lvert\mathbb{Z}_2\rvert/N_{\rm fam})^{2N_{\rm fam}}$ in $\mathbb{Q}$), but the carrier 5-fold (golden, $\mathbb{Q}(\sqrt5)$) has *no* rational dynamic image — a field obstruction, so it is static-only. The shared object is the order-30 Coxeter element ($30{=}g_{\rm car}(2N_{\rm fam}){=}5{\cdot}6$); the ϕ quasicrystal carries the carrier, not the cusp-weight family dynamics **[E][C]**
`v315_coxeter_coupling.py`	Couple or factorize? The order-30 Coxeter sectors couple as a cyclotomic field (v314 follow-up). **[E]** GROUP FACTORIZES: $\gcd(5,6){=}1\Rightarrow\mathbb{Z}/30=\mathbb{Z}/5\times\mathbb{Z}/6$ (direct product, group-decoupled); **[E]** EXPONENTS FACTOR: E8 exponents = totatives(30) = totatives(5)$\times$ totatives(6), $\phi(30){=}4{\cdot}2{=}8{=}$ rank E_8; **[E]** FIELD COUPLES: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]{=}8{=}$ rank E_8, $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5)$ ($\ni\sqrt5$) and the family $\mathbb{Q}(\zeta_3)$; **[E]** GALOIS $=\mu_4\times\mathbb{Z}_2$: $(\mathbb{Z}/30)^\times=(\mathbb{Z}/5)^\times\times(\mathbb{Z}/3)^\times=\mathbb{Z}/4\times\mathbb{Z}/2$ (order 8, max order 4), $\mu_4{=}(\mathbb{Z}/5)^\times$ the carrier Galois (the seam clock = the carrier golden-field Galois), $\mathbb{Z}_2{=}(\mathbb{Z}/3)^\times$ the family Galois; **[E]** $\sqrt5{=}\phi{+}1/\phi$ = the Gauss sum in $\mathbb{Q}(\zeta_5)$. **[C]** $30{=}5{\cdot}6$ does not dynamically entangle — it couples as the cyclotomic compositum, the bridge is Galois not dynamical **[E][C]**
`v316_galois_readout.py`	Does the Galois $\mu_4\times\mathbb{Z}_2$ (v315) organize the physical readouts? Yes — the CP phases. **[E]** the hexagonal CP unit $\rho{=}\zeta_6{=}\zeta_{30}^5$ is the order-$6{=}2N_{\rm fam}$ FAMILY factor c^5 (the carrier is $\zeta_5{=}\zeta_{30}^6$); **[C]** $\delta_{\rm CKM,lead}{=}\pi/3{=}\arg\zeta_6$, $\delta_{\rm PMNS}{=}4\pi/3{=}\arg\zeta_6^4$ (power $4{=}\lvert\mu_4\rvert$), $\zeta_6^4{=}-\zeta_6$ the sheet flip ($\mu_6{=}\mu_3\times\mu_2$); **[E]** the Galois $\mathbb{Z}_2{=}\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}){=}\{\zeta_6\mapsto\zeta_6^5{=}\bar\zeta_6\}$ IS CP conjugation ($\delta\mapsto-\delta$); **[E]** the carrier factor ζ_5 (golden $\sqrt5$) carries no phase (static); **[E]** the magnitudes ($\sin^2\theta_{12}{=}1/3{-}\varphi_0/2$, transcendental φ_0) are not cyclotomic. So the Galois bridge reaches phenomenology through the family factor (phases); the carrier is static; the magnitudes are the analytic seed **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v317_galois_family.py`	Are the 3 generations a μ_3/Galois orbit? Yes (μ_3), refined 1+2 by Galois (v316 follow-up). **[E]** the 3 cusp weights $\{0, 1/3, 2/3\}$ (Spec(Q_+)=$\{1, 2, 3\}$=A_3 exponents) map under $w \mapsto e^{2\pi i w}$ to the cube roots $\{1, \zeta_3, \zeta_3^2\}$; the cyclic family symmetry $w \mapsto w{+}1/3$ is a transitive μ_3 orbit (the family IS the cube-root orbit). **[E]** the transfer eigenvalues $(1{-}w)^6$=$\{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0). **[E]** the Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2$, = CP conjugation v316) fixes w=0 (rational) and swaps w=1/3 $\leftrightarrow$ 2/3 $\Rightarrow$ orbit 1+2; so the Galois-FIXED generation is the recovery ATTRACTOR (the law), the Galois PAIR the decaying hierarchy. **[C]** the family structure is arithmetic; the mass magnitudes stay the analytic seed **[E][C]**
`v318_arithmetic_capstone.py`	The capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\rm geo}\}$ (synthesis of the v313–v317 arc). **[E]** the structural sector (3 generations, 2 CP phases, orbit/hierarchy) lives in $\mathbb{Q}(\zeta_{30})$: $[\mathbb{Q}(\zeta_{30}){:}\mathbb{Q}]$=$\varphi(30)$=8= rank E_8, 30=$\lvert\mathbb{Z}_2\rvert N_{\rm fam} g_{\rm car}$=$h(E_8)$, Galois $(\mathbb{Z}/30)^\times$=$\mu_4 \times \mathbb{Z}_2$; **[E]** rigidity: forced by the anchor atoms $\{2, 3, 5\}$=$e(a)$; **[E]** magnitude reduction: φ_0=$(4/3)c_3$+$48c_3^4$ is a pure function of π (no free parameters), so the magnitudes are $\{a, \pi\}$-determined, not free; **[E]** free inventory: 0 dimensionless free parameters, π the unique transcendental, $v_{\rm geo}$ the one dimensionful unit. **[O]** residual: the raw-seam realisation stays open (`QGEO.SYM.01/SEAM.EQUIV.01`) **[E][O]**
`v319_translation_clock.py`	The translation clock: the discrete$\leftrightarrow$dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive (0..5) or live-only (1..5). **[E]** the only object holding both the static carrier ($\mathbb{Q}(\sqrt{5})$) and the dynamic family (rational) data is the order-30 Coxeter element (v314); $30 = h(E_8) = g_{\rm car}(2N_{\rm fam}) = 5 \cdot 6$, $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (v315). **[E]** DYNAMIC HAND ($\mathbb{Z}/6 = 2N_{\rm fam}$, ticks 0..5): recovery exponent exactly $6 = 2N_{\rm fam}$, rate $(2/3)^6$; order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$). **[E]** LAW VS LIVE: the resummed clock rate$(n) = -p_2 \ln(1 - n/N_{\rm fam})$ (v124) has rate(0) = 0 (the attractor/law) and live modes from n=1; the E_8 live phases (totatives of 30) start at 1, so $0, 1, 2, 3, 4, 5$ is law-inclusive, $1, 2, 3, 4, 5$ live-only. **[E]** STATIC HAND ($\mathbb{Z}/5 = g_{\rm car}$, ticks 1..5): golden $\mathbb{Q}(\sqrt{5})$, field-obstructed (no rational dynamic image) — static-only. **[E]** one full turn = $\mathrm{lcm}(5, 6) = 30 = h(E_8)$ (rank $E_8 = 8 = \varphi(30)$, $\lvert\mu_4\rvert$=4 planes, v55); the "it is one clock" reading is **[C]**.
`v320_galois_cp_relation.py`	A NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\rm PMNS} = \delta_{\rm CKM}^{\rm lead} + \pi$. **[E]** both phases are powers of one hexagonal unit ρ=ζ_6 (v316): $\delta_{\rm CKM}^{\rm lead}$=$\arg\rho$=$\pi/3$ (60°), $\delta_{\rm PMNS}$=$\arg\rho^4$=$4\pi/3$ (240°), ρ^4=$-\rho$, so $\delta_{\rm PMNS}{-}\delta_{\rm CKM}^{\rm lead}$=$\pi$ exactly (the $\mathbb{Z}_2$ sheet flip) — the quark and lepton leading CP phases are NOT independent. **[C]** prediction of record $\delta_{\rm PMNS}^{\rm lead}$=240° (upgrades the assigned texture to a Galois-forced relation to the measured quark phase); **[C]** currently within the broad NuFIT 6.0 range; **[X]** kill test: $\delta_{\rm PMNS}$ robustly away from 240° ($> 3\sigma$; DUNE/Hyper-K/JUNO) falsifies the Galois-CP organization **[E][C][X]**
`v321_killtest_board.py`	The forward kill-test board: the decisive upcoming measurements, ranked (forward companion to v307). **[E]** each entry is locked to a `freeze_file.csv` kill row: #1 JUNO $\sin^2\theta_{12}$=0.306747 (fastest), #2 CMB-S4/LiteBIRD r (kill r>0.01), #3 DESI/Euclid w=−1, #4 DUNE/Hyper-K $\delta_{\rm PMNS}$=240° (the new v320 Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger complementing v307; no new physics **[E]**

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Script	**What it machine-checks** *(continued)*
`v322_cp_lock_sharpened.py`	The v320 CP-lock prediction sharpened into a quantitative kill test. [E] the leading CP phase sits on the six hexagonal nodes $\arg\rho^k{=}k\cdot 60°$; $\delta^{\rm lead}_{\rm CKM}{=}60°$ (node 1), $\delta^{\rm lead}_{\rm PMNS}{=}240°$ (node 4). [E] the node selector is the deck order: $\delta^{\rm lead}_{\rm PMNS}{=}\lvert\mu_4\rvert\,\delta^{\rm lead}_{\rm CKM}{=}4(\pi/3){=}4\pi/3{=}\delta^{\rm lead}_{\rm CKM}{+}\pi$ $(\rho^4{=}-\rho)$. [C] sub-leading budget: $\delta^{\rm full}_{\rm CKM}{=}\pi/3{+}3\lambda^2$ $(\sim 68.7°)$; the lepton sub-leading is a separate open holonomy term bounded by $\sim 3\lambda^2\approx 8.7° \Rightarrow$ band $240\pm\sim 9°$. [C] current tension: vs NuFIT 6.0 NO ($\delta_{CP}{=}212^{+26}_{-41}°$) the leading 240° sits at $+1.08\sigma$, band edge $\sim 231°$ at $+0.74\sigma$. [X] kill: the nearest wrong node is 60° away $\gg$ budget; DUNE/Hyper-K ($\sim$ 5–15°) discriminate, $>3\sigma$ outside kills the lock. [E] an anarchic δ_{CP} hits the band with $p\sim 5\%$ and node 4 1-in-6 [E][C][X]
`v323_bw_geometric_modular.py`	The keystone pushed one step: Bisognano–Wichmann makes the seam modular flow *geometric*, so the μ_4 deck invariance is not an independent axiom. [E] the clock $\rho{=}\,{\rm diag}(i^n){=}\exp(i(\pi/2)L)$ is a geometric rotation ($L{=}\,{\rm diag}(n)$, $\rho^4{=}I$, $\mu_4\subset U(1)_{\rm rot}$). [E] for a rotation-covariant $C{=}f(L)$ the modular Hamiltonian $K{=}\log((1-C)/C){=}g(L)$, so $[K,L]{=}0$ (the modular flow commutes with all rotations) and $[\rho,K]{=}[\rho,C]{=}0$ — the clock is a modular symmetry for free. [E] discriminator vs v309: a period-4 curvature preserves the clock but its modular flow is *not* geometric ($[K,L]\neq 0$); BW is strictly stronger. [C] linked bedrock: given the seam is the $(E_8)_1$ chiral net (v308), BW/Hislop–Longo make the vacuum modular flow geometric $\Rightarrow$ `QGEO.SYM.01` ($\omega\circ\rho{=}\omega$) is downstream of `SEAM.EQUIV.01` + a rotation-invariant vacuum. [O] residual: the raw collar vacuum is rotation-invariant (cleaner than v309, shared with v308) [E][C][O]
`v324_hypergraph_fiber.py`	The minimal hypergraph substrate that works: a $(2,3,5)$-network *fibered* by a 3-fold cusp channel reproduces both the E_8 skeleton and the recovery gap (the constructive v312 follow-up). [E] network factor: $T_{\rm net}{=}(A{+}2I)/4$ fixes the Kac marks (eigenvalue $1{=}E_8$ skeleton), subleading $(\varphi{+}2)/4$ (golden); $(2/3)^6$ is *not* in its spectrum (the v312 negative). [E] cusp fiber: $T_{\rm cusp}{=}\,{\rm diag}((1{-}w)^{2N_{\rm fam}})$, $w\in\{0,1/3,2/3\}$ ($N_{\rm fam}{=}3$, v317), spectrum $\{1,(2/3)^6,(1/3)^6\}$. [E] the fibered substrate $T{=}T_{\rm net}\otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes\,(w{=}0)$, skeleton $\times$ democratic cusp) *and* $(2/3)^6$ as a genuine eigenvalue (network attractor $\times$ cusp subleading). [E] honest injection: without the fiber ($T_{\rm net}\otimes I_3$) there is no $(2/3)^6$ — the cusp datum is injected, not graph-spectral; substrate = network $\times$ cusp = the v315 split $30{=}g_{\rm car}(2N_{\rm fam}){=}5\cdot 6$. [E] minimality: the smallest fiber is the 3-node cusp ($=N_{\rm fam}$) = order-30 (v319 clock). [C] a consistent fibered realization, not a derivation of 2/3 from a pure rewrite [E][C]
`v325_pillowcase_keystone.py`	The seam keystone as ONE citable theorem: the raw seam state *is* the flat $\tau{=}i$ pillowcase state, with **exactly one** open lemma. IF (raw RP-collar seam state = flat $\tau{=}i$ pillowcase / rotation-invariant state) THEN 4 square marks + μ_4 deck + $\omega\circ\rho{=}\omega$ + holomorphic $c{=}8\Rightarrow(E_8)_1$, with the carrier recovery gap as input. [E] pieces all exact: $n{=}2\chi(S^2){=}4$ (v216); cross-ratio $2\Rightarrow j{=}1728$ (v214/v267); flat orbifold $S^2(2,2,2,2)$, $\chi_{\rm orb}{=}0$ (Troyanov); DtN sub-principal at $m\equiv 0 \bmod 4$ (v201/v280); $\det{\rm Cartan}(E_8){=}1$ vs $D_8{=}4$ (v83/v143/v308); gap $6\ln\frac{3}{2}>0$ (v76/v302). [O] the one open lemma — the raw collar state IS the flat $\tau{=}i$ pillowcase / rotation-invariant state — is SHARED by both bedrock IDs (`QGEO.SYM.01`, `SEAM.EQUIV.01`; v323), machine-pinned in Lean (`FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`); an assembly/reduction certificate (like v176/v261), not a closure [E][O]

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Script	**What it machine-checks *(continued)***
`v326_ftransfer_suite.py`	The F_{transfer} solver suite: the four transfer readouts as ONE runnable harness, each a gapped relaxation to a unique attractor (productises v303). [E] F_{pole} (Koide): Moebius contraction to $Q^*{=}2/3$ with multiplier $(2/3)^6$ = the seam-clock subleading eigenvalue, unique attractor at the SEAM rate. [C] $F_{\text{Boltzmann}}$ (η_B): washout contraction to the balance 6.1×10^{-11}, monotone (H-theorem), external thermal rate. [C] F_{relic} (axion): the comoving number $N{=}E/\omega$ is the adiabatic invariant, the relic freezes (seed $\theta_i{=}3\pi/5$), cosmological rate. [E] F_{QCD} (m_p/m_e): 1-loop RG with carrier $b_0{=}7$ flows to the Gaussian UV fixed point, Λ_{QCD} generated. [C] F_{RareKaon}: $\text{BR}(K^+){=}9.45\times10^{-11}$ on NA62 2016-2024 (v202). [O] only F_{pole} has the seam rate $(2/3)^6$; the other four share the SHAPE with external rates honestly fenced (v187/v303) — the frontier is measurable without claiming the external rates [E][C][O]
`v327_hypergraph_rewrite.py`	Deriving the cusp weight 2/3 from a minimal rewrite (sharpening v324/v312). A minimal local rule — a family channel with $N_{\text{fam}}{=}3$ slots, one absorbing attractor ($w{=}0$) and $\lvert\mathbb{Z}_2\rvert{=}2$ surviving — supplies the cusp fiber. [E] M has spectrum $\{1,2/3,0\}$; $2/3{=}(N_{\text{fam}}{-}1)/N_{\text{fam}}{=}\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ EMERGES from the rule arity, not injected. [E] over the order-6=$2N_{\text{fam}}$ hand the rate is $(2/3)^6{=}64/729$ = the recovery gap (v76/v324), derived. [E] NON-GRAPH-SPECTRAL (proof, sharpening v312): 2/3 is NOT a root of the affine-E_8 charpoly ($p(2/3){=}-3520/19683\neq0$), so the recovery rate cannot be an adjacency eigenvalue. [E] the absorbing state (eigenvalue 1) is the democratic $w{=}0$ law (v317). [O] honest residual: the arity $\{2,3\}$ (the anchor atoms) is still the input (P2) — v324's injected datum reduced to the anchor arity [E][O]
`v328_theta13_pressure.py`	The θ_{13} pressure point: the honest weak spot, quantified and pre-registered. [C] $\sin^2\theta_{13}{=}\varphi_0^{\text{ret}}e^{-5/6}{=}0.023108$ vs NuFIT 6.0 NO (0.02195 ± 0.00058) is $+2.0\sigma$ — the most-tensioned CORE prediction (v62/v307). [E] θ_{13} is the carrier hypercharge trace (exponent $\gamma{=}\text{tr}_E Y^2{=}5/6$ EXACT, v268), NOT the seesaw $\mu\tau$ breaking that sets θ_{12} — distinct channels. [C] no cheap relief: the back-solved seed $\varphi_0^{\text{ret}}(\theta_{13}){=}0.0505$ is ∼5% below the axiom, PMNS θ_{13} RG running is $<1\%$ (NO), and 5/6 is exact — the prediction stays frozen (`REG.FREEZE.01`). [X] pre-registered kill: a stable $\sin^2\theta_{13}$ away from 0.023108 at $>3\sigma$ (JUNO/global) flags the carrier-trace/seed-grammar reading; exactly the v306 seed-hyperplane outlier — one honest pressure point [C][E][X]
`v329_os_gap_reduction.py`	Closing the OS step: the bulk gap *is* the transfer gap (second quantization), and assembling the single sufficient premise of the whole bedrock. [E] OS GAP = TRANSFER GAP: the many-body OS Hamiltonian $d\Gamma(-\log T)$ has spectrum = sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, independent of system size — exactly "OS bulk gap = transfer gap" (the v308 open input), discharged at the many-body level. [E] $H\Rightarrow\omega\circ\rho{=}\omega$: under H the covariance is rotation-covariant $C{=}f(L)$, $K{=}g(L)$ commutes with $\rho{=}\exp(i(\pi/2)L)$ (v201/v309/v323). [E] $H\Rightarrow$ SRE $\Rightarrow(E_8)_1$: $\det\text{Cartan}(E_8){=}1$ vs $D_8{=}4$ (v237/v308). [E] both bedrock IDs follow from the ONE premise H (Lean-pinned `FORM.SEAMEQUIV.01` + `FORM.QGEO.BW.01`). [O] residual: the necessity of H on the raw collar; sharpens v240/v308/v323/v325, does NOT close the bedrock [E][O]

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Script	**What it machine-checks** *(continued)*
`v330_qgamb_admissible_measure.py`	Constructing the ambient measure on the gap-decoupled admissible sector as a bona fide Osterwalder–Schrader object (with the honest fence). [E] reflection positivity: the seam transfer T (v221/v311; spectrum $\{1,(2/3)^6,(1/3)^6\}$) is symmetric PSD, so the finite-volume measures are RP. [E] exponential clustering: $C(n)/C(n{-}1)\to\lambda_2{=}(2/3)^6$ ($\xi{=}1/\Delta$, $\Delta{=}6\ln\frac{3}{2}$). [E] tightness ⇒ projective limit: $\chi{=}\sum_n C(n){=}729/665$ finite, so $\mathrm{Var}(S_\Lambda)/\Lambda\to\chi$ ⇒ uniform tightness (Prokhorov) ⇒ the projective limit $\mu{=}\lim\mu_\Lambda$ EXISTS on the admissible sector. [E] OS reconstruction: $H_{\rm OS}{=}-\log T\geq 0$, unique vacuum, gap $6\ln\frac{3}{2}$ (v240). [E] NEG: an ungapped transfer sends $\chi\to\infty$, tightness fails. [O] the full ambient measure: the non-admissible (metric) sector is NOT built; by the gap margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76/v275) the physical readouts do not need it, but `QG.AMB.01` stays open. A genuine PARTIAL construction, NOT a closure [E][O]
`v331_necessity_of_H.py`	The *necessity* of H reduced: rotation-invariance of the seam DtN is EQUIVALENT to the four marks at the square ($\tau{=}i$), so the one open lemma is exactly the order-4 CM selection (the necessity companion to v329's sufficiency). [E] square ⇒ $[\rho,\Lambda]{=}0$: marks at $\{0,\pi/2,\pi,3\pi/2\}$ make the curvature $\mathbb{Z}_4$-symmetric (Fourier support 0 mod 4), M_f couples only modes differing by $4\mathbb{Z}$ (the v201/v210 sufficiency as a DtN $\lvert k\rvert{+}M_f$ operator). [E] necessity: a mark off the square breaks $\mathbb{Z}_4$ and gives $[\rho,\Lambda]\neq 0$ — equivalence, not just implication. [E] order-4 CM: cross-ratio$(1,i,-1,-i){=}2\Rightarrow j{=}1728$, the unique 4-config with an order-4 automorphism (v214/v267); $H\Leftrightarrow$ square $\Leftrightarrow\mu_4$. [E] NEG: a $\mathbb{Z}_2$ mark set commutes with ρ^2 but not ρ. [O] residual: the dynamical *selection* of the square (the 4 marks forced by v216) is the irreducible postulate; does NOT close the bedrock [E][O]
`v332_qgamb_metric_sector.py`	The metric sector of `QG.AMB.01` characterized: the obstruction is exactly the conformal-mode (Gibbons–Hawking–Perry) problem, the physical sector is gap-insulated, and the only route is the conditional IDG taming (makes v330's metric-sector fence precise). [E] conformal-mode wrong sign: $c_{\rm conf}(D){=}-(D{-}1)(D{-}2)/4$ is negative for $D\geq 3$ ($c_{\rm conf}(4){=}-3/2$), TT positive — the conformal-factor problem. [E] measure obstruction: the kinetic form is indefinite (one negative eigenvalue), so e^{-S} diverges over the conformal mode; the bare metric-sector measure is not directly definable. [E] physical sector insulated: the obstruction is in the conformal/gauge direction, not the admissible sector (positive+gapped, v330), margin $\Delta-31/(4\pi^2)\approx 1.648>0$ (v76). [C] IDG conditional route: the entire form factor e^{z/M^2} (v259/v304) dresses the conformal propagator with no extra pole. [E] NEG: a polynomial ($R{+}R^2$) truncation crosses zero (the Stelle ghost, v304/v278). [O] metric sector open: the full `QG.AMB.01` needs the conformal mode resolved nonperturbatively (GHP rotation / IDG), gap-decoupled but NOT constructed. A precise characterization, NOT a closure [E][C][O]
`v333_cm_selection.py`	The CM-selection mechanism: $\tau{=}i$ (the square) is the UNIQUE order-4 modulus and the attractor of mark-equilibration, so the last keystone residual is "the deck acts geometrically", not "why the square" (the v331 follow-up). [E] order-4 modular element: $S{=}\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)$ in $\mathrm{SL}(2,\mathbb{Z})$ has $S^2{=}-I$, $S^4{=}I$ and fixes $\tau{=}i$ — the unique order-4 elliptic point of $\mathrm{PSL}(2,\mathbb{Z})$ (the other, ρ, is order 3). [E] deck order → CM point: order-4 → cross-ratio 2 → $j{=}1728\to\tau{=}i$ (square); order-3 → $j{=}0\to\tau{=}\rho$ (hexagon); $\lvert\mu_4\rvert{=}4$ selects $\tau{=}i$. [E] mark-equilibration attractor: 4 points minimizing the log-energy (Fekete) converge to equal spacing (gaps $\pi/2$ = the square), and beat random perturbations — the unique (mod rotation) attractor. [E] NEG: an order-3 deck selects ρ, not the square. [O] residual: only "the deck acts *geometrically* (= `QGEO.SYM.01`) vs an abstract $\mathbb{Z}_4$ on labels" stays open — sharpened from "why the square" to "the deck acts geometrically", NOT a closure [E][O]

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Script	**What it machine-checks** *(continued)*
`v334_conformal_resolution.py`	The conformal-mode resolution: the Gibbons–Hawking–Perry contour rotation (IR sign) + the infinite-derivative (IDG) dressing (UV pole) give a CONDITIONALLY convergent metric-sector measure; the residual is the nonperturbative validity of the contour (the v332 follow-up). [E] the obstruction: $c_{\mathrm{conf}}(4) = -3/2 < 0$, the unrotated conformal Gaussian diverges. [E] GHP rotation: $\phi \to i\phi$ flips the sign, $\int e^{-\lvert c\rvert\phi^2} = \sqrt{\pi/\lvert c\rvert}$ FINITE (GHP 1978). [E] IDG dressing: the entire form factor $e^{p^2/M^2} > 0$ (no ghost pole, v304), the dressed per-mode action positive post-GHP and UV-suppressed. [E] NEG: a polynomial truncation crosses zero (the Stelle ghost, v304/v278). [C] conditional construction: GHP+IDG give a convergent metric-sector Gaussian mode-by-mode (conditional on the contour + entire analyticity). [O] residual: whether the GHP-rotated IDG-dressed measure is the CORRECT nonperturbative `QG.AMB` measure (the contour's nonperturbative validity); gap-decoupled (v76), blocks no test, NOT a closure [E][C][O]
`v335_seam_equiv_unify.py`	The keystone unification: there is ONE open theorem (`SEAM.EQUIV.01`), `QGEO.SYM.01` is its COROLLARY (a conformal-net axiom), and `QG.AMB.01` is a decoupled general QG problem, NOT a TFPT gate. [E] `QGEO.SYM.01` = corollary: a chiral conformal net has a Moebius-covariant vacuum (the unique invariant positive-energy vector, a NET AXIOM); rotations $U(1) <$ Moebius $\Rightarrow$ rotation-invariant vacuum; the μ_4 clock = rotation by $\pi/2$ fixes it $\Rightarrow \omega \circ \rho = \omega$. So `SEAM.EQUIV.01`$\Rightarrow$`QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom); the two bedrock items collapse to ONE. [E] the one keystone theorem: `SEAM.EQUIV.01` = a construction theorem (c=8, $\det \mathrm{Cartan}(E_8)$=1, gap $6 \ln \frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma. [E] `QG.AMB.01` decoupled: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0 \Rightarrow$ no TFPT prediction depends on it; the general Euclidean-QG conformal-factor problem (GHP 1978), inherited not TFPT-specific. [E] collapsed status: structural items $2 \to 1$; live residual $= v_{\mathrm{geo}} + G_{\mathrm{net}}$(=`SEAM.EQUIV.01`) $+ F_{\mathrm{transfer}}$. [O] the one residual = `SEAM.EQUIV.01`'s continuum OS reconstruction (Lean-pinned). A synthesis/inventory; does NOT close `SEAM.EQUIV.01` [E][O]
`v336_continuum_limit.py`	The ONE open lemma of `SEAM.EQUIV.01`, stated precisely and mapped to the recent rigorous literature – the continuum-limit and OS-reconstruction legs are now CITABLE. [C] continuum leg: Morinelli–Morsella–Stottmeister–Tanimoto (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8) give the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur $\to$ Virasoro, correct c); WZW range rank $\le c \le D$. $(E_8)_1$: c=8, rank 8, D=16 Majoranas ($SO(16)_1 \to (E_8)_1$), $8 \le 8 \le 16$ in range; the collar is a gapped quasi-free (CAR) state. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms (linear growth) for unitary VOAs incl. lattice VOAs; $(E_8)_1$ is the even self-dual rank-8 lattice VOA. [C] construction + covariance (Adamo–Giorgetti–Tanimoto): arXiv:2506.01008 builds the 2d conformal net of an even lattice and classifies the 2d Heisenberg-net extensions as lattice nets, and arXiv:2508.07109 shows every DHR rep of such a net is automatically positive-energy + diffeomorphism-covariant (covariance a consequence, not an input). [E] target pinned: $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $\det \mathrm{Cartan}(E_8)$=1, the D=16 Majorana $SO(16)_1$ IS the carrier-16 and the μ_4 extension IS the seam glue. [O] open residual: c=8 alone does NOT pin the net ($SO(16)_1$ also c=8, det 4), so the one open lemma = the holomorphy discriminator $\det K$=1. A literature-mapping/reduction; does NOT close `SEAM.EQUIV.01` [C][E][O]

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Script	**What it machine-checks** ***(continued)***
`v337_decoupling_theorem.py`	THE DECOUPLING THEOREM (consolidated, citable): every TFPT dimensionless readout factors through the gapped admissible sector and is INDEPENDENT of the ambient QG measure – the theorem that makes the `QG.AMB.01` reclassification (v335) honest. [E] gap margin: $\Delta - 2\dim(E_8)c_3^2 = 6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$. [E] factorization: every readout (mass ratios, mixings, α^{-1}, $R{+}R^2$) is a function of the admissible gapped transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ + the discrete kernel alone. [E] the theorem: margin $> 0 \Rightarrow$ exponential clustering ($\lambda_2{=}(2/3)^6$), finite susceptibility $\chi{=}729/665$, so the ambient contribution is gap-suppressed $\Rightarrow$ no readout depends on the ambient measure $\Rightarrow$ `QG.AMB.01` decoupled. [E] neg control: gap closed $\Rightarrow \chi$ diverges $\Rightarrow$ fails (non-vacuous). [E] corollary: zero frozen kill tests depend on the ambient measure. Consolidates v76/v311/v330/v332; does NOT construct the ambient measure [E]
`v338_theta13_budget.py`	`THETA13.BUDGET.01`: the θ_{13} sub-leading budget vs the current global fit – the $+2\sigma$ pressure point (v328) quantified against BOTH NuFIT 6.0 NO variants and JUNO; neither defuses nor excludes, fences correctly. [C] SK-dependent tension: $\sin^2\theta_{13} = (\varphi_0^{\rm ret})\,e^{-5/6} = 0.023108$ is $+2.0\sigma$ vs the no-SK central (0.02195) and $+1.7\sigma$ vs the with-SK NO best fit (0.02215) – not a clean 2σ. [E] within 3σ: $0.023108 <$ the with-SK 3σ upper 0.02388 (not falsified). [C] budget: moving to the central needs -4.1%; RG $< 1\%$ and the exponent 5/6 is exact $\Rightarrow$ no freeze-respecting relief (v328). [X] JUNO decides: sub-1% precision turns 1.7–2.0σ into $\sim 6\sigma$ – a near-term pre-registered kill/confirm. One honest pressure point (v306/v328), fenced [C][E][X]
`v339_firstprinciples_boundary.py`	The honest first-principles boundary, mapped precisely – answers (a) can $M_{\rm Planck}/v_{\rm geo}$ be computed? and (b) can $F_{\rm transfer}$ be made first-principles? Both NO, with the exact reason per residual. [E] $v_{\rm geo}/M_{\rm Planck}$ FORBIDDEN BY THEOREM (No-Unit, v153): a mass is dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs a scale; over-determined (gravity = dark energy to 0.11%, v274); every dimensionless RATIO is derived, only the unit is not. [E] m_p/m_e structure in place (v262): the carrier $b_3{=}{-7}$ drives the QCD run + the closed electron source; two inputs stay external: $\alpha_s(M_Z)$ and the lattice $C_p{=}m_p/\Lambda^{(3)}{\sim}2.83$. [X] $\alpha_s(M_Z)$ external BECAUSE OF A TENSION: SM content does NOT unify (1-loop spread $\sim$8.8 at $2{\times}10^{16}$ GeV, v246). [C] C_p parameter-free but lattice-only (v35). [C] η_B/axion = cosmological initial conditions; Koide $Q{=}2/3$ structural [E]. [E] FOUR honest kinds: forbidden-by-theorem / external-parameter-free / external-tensioned / cosmological. A boundary audit, NOT a solution [E][C][X]
`v340_As_reconcile.py`	A_s is predicted (not missing), is DIMENSIONLESS (so it does NOT define an absolute mass), and the apparent -11σ tension is a reheating-channel question – reconciled with the archived old derivation (TFPT v4.5/v4.6). [E] $A_s = N_\star^2(M_{\rm scal}/\bar M_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ (v86); $M_{\rm scal}/\bar M_{\rm Pl} = c_3^{7/2} = 1.2565{\times}10^{-5}$ is a PURE RATIO, so A_s carries NO absolute scale – it pins $N_\star$, NOT $\bar M_{\rm Pl} \Rightarrow A_s$ does NOT define mass (No-Unit v153 stands). [C] the -11σ is the slow-vs-fast reheating dichotomy ($N_\star{=}51.44 \to A_s{=}1.764{\times}10^{-9}$; A_s-preferred $N_\star{=}56.2$). [C] ARCHIVE: old v4.5/v4.6 used the algebraic $N_\star{=}3/\varphi_0^{\rm ret} - 1{=}55.42$ and reported $A_s{=}2.124{\times}10^{-9}$; at that $N_\star$ the current normalisation gives $2.047{\times}10^{-9}$ (-1.9σ). [E] nothing lost: the current theory DERIVES $N_\star$ (record band [50, 60], v84). [X] decisive test: fast preheating or slow reheating confirmed [E][C][X]

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Script	What it machine-checks *(continued)*
`v341_alpha_quillen.py`	The α fixed point reformulated as the STATIONARITY of a $U(1)$ determinant line, every ingredient identified as an index / heat-kernel coefficient / discriminant form – NOT a free knob (the "derive the FORM" step). $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \ln(1/\varphi_{\text{seam}}) = 0$ in Quillen form: $[\alpha^3 - 2c_3^3\alpha^2]$ (det-line variation) $= [8b_1c_3^6]$ (anomaly counterterm) $\times[\ln(1/\varphi_{\text{seam}})]$ (seam response). **[E]** M=41 a $U(1)$ INDEX (EM Ward, v48): $\frac{4}{5}M = 8b_1$, b_1=41/10 the SM 1-loop $U(1)_Y$ beta (v159/v246). **[E]** c_3=1/(8π) the GAUSS–BONNET boundary coefficient (8= rank E_8, v216); $c_3^{3,6}$ heat-kernel powers. **[E]** $-5/4 = -q(D_5)$ a DISCRIMINANT form ($q(D_5)$+$q(A_3)$=2, v1/v154); φ_{base}=1/(6π)=$\frac{4}{3}c_3$ carries $q(A_3)$: $c_3/\varphi_{\text{base}}$=3/4. **[E]** δ_{top}=$\Omega_{\text{adm}}c_3^4$. **[E]** the Quillen split holds at the α^{-1}=137.0359992 root. **[E]** NO NEW GATE: α^{-1} stays **[E]**; v341 SHARPENS the pre-existing **[O]** origin `EM.WARD.01` (no new α problem). **[O]** the one still-open part IS `EM.WARD.01`'s residual: the from-first-principles AQFT PROOF that this is the EXACT Quillen determinant functional (the variational principle derived, not assembled). Sharpens v3/v48/`EM.WARD.01` from "fixed point" to "stationarity of a named determinant line" **[E][O]**
`v342_em_ward_heatkernel.py`	The heat-kernel sharpening of `EM.WARD.01`: DERIVES the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from **[C]** to **[E]** for the STRUCTURE; NO new gate. **[E]** c_3=1/(8π) the boundary GAUSS–BONNET coefficient (seam = flat pillowcase χ=2, $\oint K$=4π, v216). **[E]** the α^2 term is the GILKEY gauge-curvature coefficient: the $U(1)$ Laplacian a_4 has $30\Omega^2/360 = \frac{1}{12}\Omega^2$ (Gilkey Thm 4.8.16), Ω=$i\alpha F \Rightarrow -(\alpha^2/12)F^2$, so the Calderon $-2c_3^3\alpha^2$ is the gauge-curvature piece, not an ansatz. **[E]** the c_3-ladder $\{0,3,6\}$ is arithmetic (step 3 = 3 channels; 6=$\lvert R^+(A_3)\rvert$ C_6 hexagon). **[E]** multiplicities forced (2=$\lvert\mathbb{Z}_2\rvert$, $8b_1$=(rank E_8)b_1). **[E]** π-power test: $2c_3^3$=1/(256π^3) carries $\pi^3 \Rightarrow$ THREE boundary insertions. **[C]** the exact coefficient needs the seam F-normalisation. **[O]** the cubic α^3 Maxwell moment + full identification = `EM.WARD.01` residual (tied to `SEAM.EQUIV.01`). α^{-1} stays **[E]** (v3) **[E][C][O]**
`v343_four_routes_analysis.py`	The honest investigation of the four black-hole-birth solution routes (A finite causal diamond, B Carlip–Cardy near-horizon CFT, C modular/thermal flow, D self-reproduction attractor) – direct answer to "which closes ALL problems and is 100% provable": NONE individually, but they CONVERGE on the same two irreducible residuals as v335 (`SEAM.EQUIV.01` + the decoupled QG contour). **[E]** ROUTE A reframes `QG.AMB` to a FINITE thermal trace (de Sitter finite, gapped χ=729/665) but $c_{\text{conf}}(4)$=$-$3/2 < 0 still needs the GHP contour (v334, **[O]**); does NOT touch `SEAM.EQUIV.01`. **[C]** ROUTE B: Carlip+Cardy consistent with c=8 but L_0 free (ambiguity) and gives c only, not holomorphy $\det K$=1 – inherits `SEAM.EQUIV.01`. **[E]** ROUTE C: modular flow KMS β=1 (v239), geometric identification = `SEAM.EQUIV.01` (v335) – a reduction. **[E]**/**[O]** ROUTE D: unique Perron–Frobenius attractor (v56) but cyclic dynamics **[O]** – not a closure. **[E]** VERDICT: independent closures = 0; irreducible residuals = 2 (= v335); the BH premise makes both more natural but eliminates neither; focus = `SEAM.EQUIV.01` **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v344_detk_synthesis.py`	The bird's-eye synthesis of the ONE open keystone bit, det K=1 – a full-spectrum map of `SEAM.EQUIV.01`'s only open residual. [E] det K=1 is ONE statement with SIX equivalent faces (holomorphy / homology-sphere link H_1=0 / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / τ=i CM), unified by $\lvert\det\mathrm{Cartan}\rvert = \lvert H_1(\mathrm{link})\rvert = \#(\text{1-dim irreps})$ across ADE ($A_n \to n{+}1$, $D_n \to 4$, $E_6 \to 3$, $E_7 \to 2$, $E_8 \to 1$) – ONLY E_8 gives 1 ($2I$ the unique PERFECT ADE group, v232/v219). [E] THE GENUS-1 OBSTRUCTION: the easy arguments (plane vacuum, gap, free-fermion) are genus-0, necessary but NOT sufficient (v87); det K is the torus degeneracy = #primaries. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2,3,5)$ rewrite attractor IS the affine-E_8 graph (v312) and the E_8 du Val link is the Poincare sphere (H_1=0, v232), so det K=1 becomes a finite COMBINATORIAL statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge (attractor graph = raw seam realisation) = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing [E][C][O]
`v345_hypergraph_homotopy.py`	EXECUTE the combinatorial core of route R3 (the hypergraph-homotopy route to det K=1, v344): computes explicitly that the $(2,3,5)$-rewrite attractor graph's plumbing link is the POINCARE homology sphere (H_1=0), only E_8 among ADE; R3's combinatorial half is now [E]-complete, the geometric bridge (= `SEAM.EQUIV.01`) stays [O]. Plumbing topology (Brieskorn): link H_1 = coker(Cartan). [E] SNF(E_8 Cartan) = identity ⇒ coker(E_8)=0 ⇒ the E_8 plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of det E_8=1). [E] only E_8: coker(A_n)=$\mathbb{Z}/(n+1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$ – all nontrivial, only E_8=0. [E] π_1=$2I$=$SL(2,5)$ PERFECT (commutator subgroup order 120=$\lvert G\rvert$, computed) ⇒ H_1=0 ⇒ THE Poincare sphere; $\lvert 2I\rvert$=120=$\lvert\mu_4\rvert\, h(E_8)$. [O] does NOT close det K=1: the geometric realisation bridge (raw rewrite attractor IS the E_8 du Val singularity) = `SEAM.EQUIV.01` [E][O]
`v346_seam_geometric_bridge.py`	The geometric bridge as an explicit six-link chain raw-seam ⇒ det K=1, with the ONE open arrow pinned – the honest capstone of the `SEAM.EQUIV.01` investigation. L1 [E]/[C] raw seam → μ_4 + four marks (v216); L2 [O] μ_4 + $(2,3,5)$ atoms → $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE OPEN ARROW = `SEAM.EQUIV.01`); L3 [E] $2I$ → affine E_8 McKay (v219); L4 [E] $2I \to E_8$ du Val (v232); L5 [E] $\mathbb{C}^2/2I$ → Poincare link H_1=0 (v345); L6 [E] H_1=0 → det K=1. [E] 5 of 6 links are theorems, only L2 open. [E] the group is FORCED by the seam atoms: axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \to 2T/2O/2I$, $(2,3,5)$ unique to $2I$, and $(\lvert\mathbb{Z}_2\rvert, N_{\mathrm{fam}}, g_{\mathrm{car}})$=$(2,3,5)$ ARE those axes. [E] downstream of L2 forced. [O] L2 (the seam IS an ADE orbifold point) = `SEAM.EQUIV.01`, NOT closed – the keystone reduced to one geometric-realisation arrow, group pinned by $(2,3,5)$ [E][C][O]
`v347_seam_closure_modes.py`	The closure-mode classification of the one open arrow L2 (the seam-orbifold realisation, v346) – a direct honest answer to "is there ANY other way to FULLY solve it?". [E] THE PRECISE LOCUS: the seam gives the pillowcase base $S^2(2,2,2,2)$, χ_{orb}=0 (EUCLIDEAN, v216), but the $2I/E_8$ structure is the Seifert base $S^2(2,3,5)$, χ_{orb}=1/30 (SPHERICAL); L2 must bridge a FLAT 4-marked base to a SPHERICAL 3-marked base – why L2 needs the carrier $(2,3,5)$ input. [E] FOUR CLOSURE MODES: A construct (the analytic wall, v336); B rigidity (unique realisation ⇒ $\mathbb{C}^2/2I$ forced, the only theorem-route w/o new input, v284/v300/v331); C axiom P_3 (⇒ TFPT complete rel. 3 axioms); D empirical. [E] no-new-state (v246) blocks deriving L2 from extra physics ⇒ only rigidity (B) closes it as a theorem. [E] VERDICT: no free theorem; only B (theorem) or C (axiom), validated by D; only these four modes. [O] NOT closed [E][O]

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Script	What it machine-checks *(continued)*
`v348_seam_rigidity_route.py`	Route B (the rigidity route) for the one open arrow L2, carried out – it lands on a strikingly SIMPLE statement: the whole keystone reduces to ONE question, does the raw seam carry the golden ratio? **[E]** rigidity closes the crystallographic part: the order-4 clock has a UNIQUE normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). **[E]** downstream is a LATTICE identity: E_8 = the ring of icosians (Conway–Sloane), rank 8, so $2I \to E_8$ is a lattice identity (stronger than McKay v219). **[E]** the single bridge is φ: crystallographic orders $\{1, 2, 3, 4, 6\}$, the 5-fold non-crystallographic; μ_4 crystallographic, $2I$ (5-fold) not, $\varphi = 2\cos(\pi/5)$ extends one to the other (quasicrystal). **[O]** the keystone reduces to ONE question: "does the raw seam carry φ?" – if yes, $\mu_4{+}\varphi \to 2I \to E_8$ closes L2. **[O]** honest subtlety: TFPT's φ is currently E_8-side (v312/v313); the raw-seam data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2). NOT closed **[E][O]**
`v349_raw_seam_golden_test.py`	The honest test of the v348 question "does the RAW seam carry the golden ratio (before assuming E_8)?". Answer, computed: NO – the bare carrier (D_5, A_3) and the dynamical seam data are NOT golden; the golden ratio genuinely enters WITH the icosahedral/E_8 identification. A NEGATIVE result that prevents a false closure. **[E]** RAW CARRIER NOT GOLDEN: golden needs $5 \mid h$; D_5 has $h{=}8$ ($2\cos(2\pi/8){=}\sqrt{2}$), A_3 has $h{=}4$ – no 5-fold, bare carrier gives $\sqrt{2}$. **[E]** DYNAMICAL DATA NOT GOLDEN: cusp $\{0, 1/3, 2/3\}$ rational, recovery $(2/3)^6$ rational, seed φ_0 transcendental. **[E]** GOLDEN IS THE ICOSAHEDRAL/E_8 SIDE: $5 \mid h$ only for $A_4{=}SU(5)$ ($h{=}5$), H_3 ($h{=}10$), E_8 ($h{=}30$). **[E]** φ alone insufficient: $2I \neq C_5 \times \mu_4$ ($20 \neq 120$). **[O]** RESOLUTION: L2 reduces to "is $g_{\rm car}{=}5$ a PENTAGON (5-fold ⇒ golden ⇒ icosahedral) or a COUNT (rank $D_5 \Rightarrow \sqrt{2}$)?" – the golden ratio is the pentagon-reading of P2, NOT derivable from P2-as-a-count; no hidden φ. Foundational: P2-as-pentagon closes it (mode C) else rigidity must produce the 5-fold (mode B); NOT closed **[E][O]**
`v350_bootstrap_inputs_correction.py`	The honest CORRECTION of v349's framing (prompted by the right objection: the inputs are not axioms, they are back-determined by the theory). **[E]** THE INPUTS ARE BOOTSTRAP-FORCED, NOT FREE AXIOMS (v6): $g_{\rm car}{=}5$ over-determined three ways (rank-fill, Coxeter-match $g_{\rm car}{=}$ max prime of $h(E_8){=}30$, Pascal). **[E]** $g_{\rm car}{=}5$ IS THE ICOSAHEDRAL 5-FOLD, not the D_5 rank: $g_{\rm car}{=}$ max prime of $h(E_8){=}30{=}2{\cdot}3{\cdot}5$; the atoms $(2, 3, 5)$ ARE the primes of the icosahedral Coxeter number 30. **[E]** THE GOLDEN RATIO IS EMERGENT FROM THE GLUE, not external: D_5 ($h{=}8$)/A_3 ($h{=}4$) non-golden alone, but the μ_4 extension $D_5{\oplus}A_3 \to E_8$ (v154) gives $h(E_8){=}30$ (golden, $5 \mid 30$) – the seam's own μ_4 produces the 5-fold; v349 checked the un-glued pieces, the wrong object. **[E]** v349's pentagon-vs-count dichotomy was FALSE (the bootstrap forces the pentagon). **[O]** residual relocates: the golden is bootstrap-forced; the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Caveat: self-consistency within the framework, not from-nothing. Supersedes v349's interpretation **[E][O]**
`v351_continuum_realisation_sharpened.py`	The continuum realisation sharpened – the "$c{=}8$ ambiguity" flagged open in v277/v344 (E_8 vs $SO(16)$ both $c{=}8$) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock. **[E]** E_8 (det 1, holomorphic) vs $D_8{=}SO(16)$ (det 4). **[E]** the order-4 clock (the four Gauss–Bonnet marks, $\lvert\mu_4\rvert{=}4{=}h(A_3)$, v216): index-4 extension $D_5{\oplus}A_3{=}E_8$ (v154), index-2 = $SO(16)$; so order-4 selects E_8 (16→4→1) not $SO(16)$ (16→4) – the order (4 not 2) is the discriminator. **[E]** bootstrap agrees: $h(E_8){=}30$ (max prime 5) vs $h(D_8){=}14$ (max prime 7). **[E]** which-net ambiguity RESOLVED ($\det K{=}1$ is then a consequence). **[O]** residual = PURELY the chiral-edge/scaling-limit existence (bulk-edge, v336), net+holomorphy pinned. Does NOT close `SEAM.EQUIV.01` **[E][O]**

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Script	What it machine-checks *(continued)*
`v352_framework_irreducible.py`	The framework reduction – the deepest answer to "the inputs are not axioms, they are back-determined". **[E]** P2 (g_{car}=5) FORCED three ways (v6): rank-fill, Coxeter-match (g_{car}= max prime of $h(E_8)$=30), Pascal. **[E]** P1 (c_3=1/(8π)) FORCED: c_3 is the boundary Gauss–Bonnet coefficient (v216/v342), the 8 over-determined four ways (8= rank E_8=$h(D_5)$=$\phi(30)$= det R, v54/v6); only π is non-forced (the boundary transcendental). **[E]** the E_8 hull is forced (unique even unimodular rank-8, v83/v154). **[E]** so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite Clifford carrier) + π – the precise content of "the inputs are not axioms". **[O]** the framework is the META-AXIOM (not math-reducible, the foundational stance, tested by its consequences not proved). TFPT's true input is ONE framework + π **[E][O]**
`v353_selfloop_capstone.py`	The bird's-eye rethink (refining v352): TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs, with ZERO free adjustable dimensionless parameters. **[E]** IT IS A LOOP: outputs fix inputs (g_{car}=5= max prime of $h(E_8)$=30; the 8 in c_3= rank E_8) and inputs fix outputs; the arrows close into a circle (v6) with a unique fixed point (v56). **[E]** ZERO FREE DIALS: every load-bearing number is forced/over-determined; free dimensionless parameters = 0. **[E]** π is a FIXED geometric constant (the boundary 2-sphere, $\oint K$=4π ⇒ c_3=1/(8π)), not a dial. **[C]** refining v352: π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom. **[O]** the one residual is the CONTINUUM closure (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork **[E][C][O]**
`v354_e8_reverse_audit.py`	The REVERSE numerology audit + what the golden ratio means. The forward rule (v305) is one-directional (every readout must map onto E_8); the reverse question: how much E_8 structure maps onto NOTHING? **[E]** E_8 has 8 Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$; THREE feed a readout (2 metric, 8 rank→ c_3, 30 Coxeter→ g_{car}) and FIVE are UNMAPPED $\{12, 14, 18, 20, 24\}$ – 3/8 load-bearing, 5/8 not. **[E]** ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, h=30, det 1, $D_5 \oplus A_3$, 16-spinor) generates ALL readouts; the higher Casimirs are never reached into to fit – the anti-numerology signature (economy + a large untouched remainder). **[E]** the golden ratio is UNMAPPED (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348; no frozen readout is φ, the sole **[C]** exception $\cos(\theta_i$=3π/5)=−1/(2φ) on the axion spine branch, refined in v429) ⇒ NOT numerology (which needs a number fitted to an observation); φ is the algebraic signature of the 5-fold h=30=2·3·5, the carrier announcing it is icosahedral. **[O]** the 5/8 unmapped is the HULL OVERHEAD (E_8 = minimal unimodular hull, not the world group); where future readouts could live, or over-capacity. The reverse complement of v305 **[E][O]**
`v355_e8_unmapped_bandwidth.py`	A disciplined bandwidth search of the unmapped E_8 region (next step after v354), run with a STRICT discriminator: a candidate counts only if it is a FORCED identity (a theorem about E_8), never a numerical coincidence – because mining a rich object for matches is the numerology temptation. **[E]** FORCED SET-LEVEL (the real collective content): $\sum$deg = 128 = dim S^+ the spinor half of 248=120+128; $\sum$exp = 120 = #pos roots; $\prod$deg = $\lvert W(E_8)\rvert$; exponents = the $\varphi(30)$=8 totatives of 30 (v223); max deg = 30 = $h(E_8)$. **[C]** SUGGESTIVE not forced: 12=$h(E_6)$, 18=$h(E_7)$, 30=$h(E_8)$ (the McKay ladder v219), but 'deg(E_8) contains subalgebra Coxeter numbers' is not a theorem – flagged. **[O]** DECLINED: the per-degree atom matches (14= dim G_2, 20= det L, 24=$\lvert W(A_3)\rvert$, 12=$\lvert R(A_3)\rvert$, 18=$p_4(a)$; v66 flagged 12, 18 fittable) each admit multiple readings, and the $\lvert W\rvert$ prime 7='scalaron exponent' is one of many – declined. **[E]** RECONCILES v66 (degrees=atoms) and v354 (only 2, 8, 30 primary): NO new forced physical hit; the disciplined decline IS the anti-numerology result **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
v356_ continuum_mmst_ applicability.py	Direction A: the continuum chain of `SEAM.EQUIV.01` assembled theorem-by-theorem, residual reduced to ONE lattice-realization input tied to the seam's one-sidedness. Chain: gapped (v302) quasi-free 16-Majorana collar (v155/v160) → invertible (Kitaev, v301) → the NONTRIVIAL invertible phase (c_-=8) → a non-gappable anomalous chiral edge (anomaly inflow: edge EXISTENCE follows, reducing the v336/v351 'does the edge exist?' residual) → chiral-CFT scaling limit (MMST v336, 8≤8≤16) → order-4 μ_4 clock + AQFT stack pin $(E_8)_1$ (v351/v297). The only non-theorem link is the seam one-sidedness = the c_3 origin ($c_-\neq 0$ = one-sided = the $\lvert\mathbb{Z}_2\rvert \oint K$=8 of c_3=1/(8π), v58). [C]/[E]/[O]; ADVANCES `SEAM.EQUIV.01` to one realization input, does not close. (Direction B was found already closed by v40.) [C][E][O]
v358_grav_entropy_ equilibrium.py	The entanglement-equilibrium derivation (Jacobson 2015 / Faulkner et al., $\delta S = \delta\langle K\rangle$) with TFPT's atoms ⇒ the LINEARISED Einstein equation falls out PARAMETER-FREE. [E] $G_{ab} = (1/c_3)T_{ab}$, $1/c_3 = 8\pi$ fixed (Unruh $1/(2\pi)=4c_3$, EH $1/(16\pi)=c_3/2$, Einstein $8\pi=1/c_3$, Bekenstein $1/4=1/\lvert\mu_4\rvert$). [E] THERMO=GEO coincidence (new): the first law's $2\pi/\eta$ = the geometric $\lvert\mathbb{Z}_2\rvert 2\pi\chi$ iff $\lvert\mu_4\rvert=\lvert\mathbb{Z}_2\rvert\chi=4$ – the thermodynamic and geometric origins of c_3 are ONE. [C] J3 solved: the CHM ball modular Hamiltonian (boost via BW v323) gives the matter flux $\delta\langle K_B\rangle=\frac{8\pi^2R^4}{15}\delta\langle T_{00}\rangle$. [E] entropy density atom-fixed: $1/4=1/\lvert\mu_4\rvert$ + central charge $c=g_{\rm car}+N_{\rm fam}=8$ (form v59). [O] residual: the equation-of-state status + the global ambient measure (C7) + the $v_{\rm geo}$ unit (v359 closes the FULL covariant equation); no free dimensionless gravity dial remains [E][C][O]
v359_grav_ nonlinear_ einstein.py	The FULL covariant Einstein equation, parameter-free (Direction 1, extends v358). The Jacobson-2015 fixed-VOLUME entanglement equilibrium brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) – the unique divergence-free metric tensor (Lovelock) – so equilibrium yields the full covariant $G_{ab}+\Lambda g_{ab}=c_3^{-1}T_{ab}$ with conservation built in. [C][E] fixed-volume → Einstein tensor (trace = $-R$ in d=4). [E] Lovelock ⇒ $\nabla^a T_{ab}=0$ is an output. [E] coeff 1: $8\pi=1/c_3$ fixed (v358). [E] coeff 2: Λ from α – $\rho_\Lambda/\bar{M}_{\rm Pl}^4=(3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4=3/(4\pi^2)$ (v60); BOTH coefficients TFPT-fixed. RESULT: the original B6 full covariant field equation is now parameter-free at the LOCAL level. [O] residual: the equation-of-state status + the global ambient measure (C7) [E][C][O]
v360_grav_gap_ corrections.py	Direction 2: the gap-induced higher-curvature correction to GR – the disciplined result. Does the seam transfer gap give a NEW forced near-term GR-deviation beyond the known R^2 scalaron? With the v354/v355 discriminator: NO. [E] the leading higher-curvature term IS the R^2 scalaron ($a_4=R^2/72$ v36; $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2=c_3^7$, exp $7=g+N-1$) – already in TFPT, already falsifiable via $n_s=1-2/N_\star$, $r=12/N_\star^2$ (CMB-S4). [C] it is the entanglement-equilibrium next order (Wald first law → $f(R)=R+R^2$, v28/v359). [E] the gap $\Delta=6\ln(3/2)$ is dimensionless, so a distinct gap term sits at $\xi\, v_{\rm geo}\sim$ seam/Planck scale, not near-term observable. [O] DECLINE: a new dimensionless coefficient is not atom-forced (needs heat-kernel/Wald data; exp 7 vs 6, c_3^7 vs $(2/3)^6$ unrelated) → per v354/v355 declined. No new near-term forced prediction; the answer is the scalaron [E][C][O]
v361_grav_ backreaction.py	Direction 5: the matter-gravity backreaction with the explicit J3 flux – two forced facts + an honest decline. [E] the carrier's gravitational anomaly is FORCED: c_-=8 (16 Majoranas = dim $S^+=2^{g_{\rm car}-1}$; chiral central charge $16/2=8=g_{\rm car}+N_{\rm fam}$, the same c_- as v358). [E] the backreaction is FINITE → Λ from α: the matter vacuum energy through $G_{ab}=c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\rm Pl}^4=(3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), 123 orders = 119.03+3.92 – NOT the $M_{\rm Pl}^4$ catastrophe; the loop (v359+v60+carrier vacuum) closes. [E] J3 explicit (v358): $\delta\langle K_B\rangle=(8\pi^2R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab}=the SM's (v159). [C] SM trace anomaly reproduced (content=SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond c_-=8 and Λ → per v354/v355 declined. Finite & consistent backreaction, no new dial [E][C][O]

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Script	What it machine-checks *(continued)*
`v364_vgeo_sharpen.py`	Direction 6: $v_{\rm geo}$ is the SINGLE dimensionful input of the WHOLE theory – confirmed across matter AND gravity – and irreducibly primitive. The No-Unit Theorem (v153) showed the dimensionless compiler cannot output a scale ($v_{\rm geo}$ a forced unit). SHARPENING after the parameter-free gravity (v358/v359/v361): the gravity sector adds NO new scale either – $1/c_3$=8π and the Λ prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\rm geo}^2$ and $\Lambda \sim$ (dimensionless)$v_{\rm geo}^4$ reduce to $v_{\rm geo}$ times atoms. [E] FINAL TALLY: every dimensionful quantity = (dimensionless ratio)×(power of $v_{\rm geo}$); free content = $\{v_{\rm geo}, \pi\}$, ZERO dimensionless dials, vs the SM's ∼26. [O] $v_{\rm geo}$ irreducibly primitive (No-Unit; $1/G$ UV-sensitive Sakharov v68). Clarity, not a new derivation; the last dimensionful anchor named [E][O]
`v365_qg_oneloop_saddle.py`	Direction 3: the QG measure (gate C7/`QG.AMB.01`) as one-loop GAUSSIAN fluctuations around the PARAMETER-FREE saddle (v358/v359). [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial), so the one-loop problem is well-posed (it was NOT before v358). [E] the fluctuation operator is gapped $M \geq \Delta_{\rm eff} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) ⇒ mode-by-mode Gaussian convergence. [E] finite model log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$ on the OS spectrum $\{0, 6\log\frac{3}{2}, 6\log 3\}$. [C] the conformal mode converges on the GHP contour ($\int e^{-\lvert c\rvert\phi^2} = \sqrt{2\pi/3}$, v334). [E] the admissible projective limit exists ($\chi = 1/(1-(2/3)^6) = 729/665$, v330). [E] 0 new dimensionless dials (v364). [O] residual: the full non-perturbative projective limit (G6/`QG.AMB.01`) on the ambient sector. Sharpens C7 to a one-loop Gaussian problem around a parameter-free saddle; does NOT close it. [E][C][O]
`v366_mmst_seam_collar.py`	Direction 7: the seam collar is IN MMST's free-lattice-fermion scaling-limit class, so the chiral scaling limit is a chiral CFT of central charge 8 pinned to $(E_8)_1$ – reducing `SEAM.EQUIV.01` to the single S3 lattice-realisation input. MMST (Morinelli–Morsella–Stottmeister–Tanimoto): free lattice fermions (CAR quasi-free) → chiral-CFT massless scaling limit with correct c, range rank $\leq c \leq D$. [E] class arithmetic: $D{=}\dim S^+{=}2^{g_{\rm car}-1}{=}16$ Majoranas, $c{=}g_{\rm car}{+}N_{\rm fam}{=}8$, rank $E_8{=}8$ ⇒ rank $\leq c \leq D$ is $8 \leq 8 \leq 16$ (in MMST range). [E] gapped ($\Delta{=}6\log\frac{3}{2}{>}0$, derived v302). [C] quasi-free CAR class (v155/v160). [E] chirality $c_-{=}D/2{=}8 \neq 0$ (non-gappable anomalous edge, v356 S4). [C] the MMST scaling-limit theorem applies (in-class & gapped & chiral & in-range) ⇒ chiral CFT c=8 (v336). [E] target pinned to $(E_8)_1$ by c=8 + the order-4 μ_4 clock ($\det K{=}1$ vs $SO(16)$ $\det K{=}4$, v351). [O] residual S3: the collar realised as a genuine lattice chiral free-fermion invertible phase (one-sidedness/Flat-Away, v297/v356). Verifies MMST in-class; ADVANCES `SEAM.EQUIV.01`, does NOT close it. [E][C][O]
`v367_seam_s3_lattice.py`	Track 1: the EXPLICIT lattice realisation of the S3 input – a concrete gapped chiral-Majorana ($p+ip$) lattice model whose edge is the 16-Majorana $SO(16)_1$ carrier, μ_4-condensed to $(E_8)_1$; turns the abstract S3 residual (v356/v366) into a RUNNABLE model with a numerically computed topological invariant. [E] bulk gapped (band gap $2>0$ at M=1). [E] non-trivial Chern number (Fukui–Hatsugai–Suzuki: $\lvert C\rvert$=1 at M=1, C=0 at M=3 trivial control). [E] c_-=8: each chiral Majorana edge carries $c_-{=}C/2{=}1/2$, $N_{\rm Maj}{=}\dim S^+{=}2^{g_{\rm car}-1}{=}16$ copies ⇒ $c_-{=}16/2{=}8{=}g_{\rm car}{+}N_{\rm fam}$. [E] edge K-matrix $SO(16)_1$ (det Cartan D_8=4) → $(E_8)_1$ (det Cartan E_8=1) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation $SO(16)_1 \to (E_8)_1$ (v154/v351). [O] residual: the continuum scaling-limit existence (MMST v336). Reduces S3 to a runnable lattice model; does NOT close `SEAM.EQUIV.01`. [E][C][O]

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Script	**What it machine-checks** *(continued)*
`v368_seam_s3_inflow.py`	Track 1: the bulk-edge / anomaly-inflow leg (S4 of the v356 chain) made CONCRETE on the v367 lattice model – a finite STRIP of the gapped $p+ip$ model has a realised, non-gappable chiral edge, so edge EXISTENCE follows from the invertible bulk (anomaly inflow), NOT a separate assumption. [E] topo in-gap edge states (M=1 strip $\min_{k_x}\|E\|\sim 0$) vs trivial clean gap (M=3, neg control); [E] the near-zero state is edge-localised (weight within 4 sites of a boundary, edge≫bulk); [E] bulk-edge correspondence (edge branch count = bulk Chern $\|C\|$=1) ⇒ 16-Majorana edge, c_-=8. [C] anomaly inflow: a $c_-\neq 0$ invertible bulk cannot have a gappable edge ⇒ discharges the v336/v351 'does the edge exist?' residual on the lattice. [O] residual: the continuum scaling limit (MMST v336). Establishes edge EXISTENCE on the lattice; does NOT close `SEAM.EQUIV.01`. [E][C][O]
`v369_qgamb_redundancy.py`	Track 2 (next-plan v2): the AMBIENT REDUNDANCY statement – the holographic route to Gravity-complete that SIDESTEPS building the general Euclidean QG measure (C7/`QG.AMB.01`). Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators proving the ambient sector carries NO physical d.o.f. relative to the holomorphic boundary algebra, so `QG.AMB.01` is a non-fundamental CERTIFICATION object (like a coordinate choice in GR), not missing dynamics. [E] HOLOMORPHIC ⇒ DHR = Vec: #DHR primaries $=\|\det \mathrm{Cartan}\|=1$ for E_8 (ONE primary = vacuum) vs 4 for $SO(16)/D_8$ – no nontrivial superselection charges ⇒ no hidden boundary sector a bulk could carry invisibly. [E] NO TORUS GSD: ground-state degeneracy on T^2 = #anyons $=\|\det K\|=1$ for E_8 ⇒ unique ground state ⇒ no topologically protected invisible bulk d.o.f. (the genus-1 obstruction of v344). [E] FINITE PETZ RECOVERY: the deviation subspace contracts at $(2/3)^6$=64/729 <1 (v221) ⇒ Rec_Σ bounded. [E] GAP-DECOUPLING MARGIN: $\Delta_{\mathrm{eff}}=6\log\frac{3}{2}-\frac{31}{4\pi^2}\approx 1.648>0$ (v76/v337) ⇒ the ambient sector is energetically separated. [C] RECONSTRUCTION MAP: Bisognano–Wichmann gives the modular boost (v258/v309/v323) ⇒ $\mathrm{Rec}_\Sigma:(\mathrm{bulk})\to\mathcal{A}_\Sigma^{\mu_4}$. [C] THE REDUNDANCY STATEMENT: under `SEAM.EQUIV.01` + BW, $\mathcal{O}_{\mathrm{phys}}^{\mathrm{bulk}}/\mathrm{Diff}\cong\mathcal{A}_\Sigma^{\mu_4}$ ⇒ `QG.AMB.01` carries no extra physical d.o.f. [O] residual: the two consumed premises – `SEAM.EQUIV.01` (the S3 lattice input, v366) + the BW intrinsicality of the raw collar (v329); provides the ALTERNATIVE (holographic) route, does NOT construct the measure. Gravity-complete = Boundary-complete + Ambient-redundancy [E][C][O]
`v370_grav_spin2_unitarity.py`	Track 2: the Barnes–Rivers spin decomposition of the graviton propagator – perturbative graviton unitarity SECTOR-BY-SECTOR (extends v304's scalar pole algebra to the tensor spin structure + combines with v334's spin-0 conformal mode). [E] projector completeness: $\mathrm{tr}\,P_2=(d-2)(d+1)/2$, $\mathrm{tr}\,P_1=d-1$, $\mathrm{tr}\,P_{0s}=\mathrm{tr}\,P_{0w}=1$ sum to $d(d+1)/2$; the d=4 split is 5+3+1+1=10. [E] EH structure: spin-2 coefficient +1, spin-0 scalar $-1/(d-2)=-1/2$. [E] the Stelle ghost is a SPIN-2 state (a local $1+p^2/M^2$ dressing puts the negative-residue pole at $p^2=-M^2$). [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes spin-2 ghost-free (only the p^2=0 pole). [C] the wrong-sign spin-0 conformal mode ($c_{\mathrm{conf}}(4)=-3/2$, v332) is cured by the GHP contour (v334) – a different mechanism. [C] ⇒ perturbative graviton unitarity sector-by-sector, conditional on v304's entire-analyticity assumption. [O] perturbative-only + the open analyticity assumption; NOT the non-perturbative measure (`QG.AMB.01`). [O] decline: $\mathrm{tr}\,P_2$=5 is dimension counting, NOT g_{car} (v354/v355). [E][C][O]

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Script	What it machine-checks *(continued)*
`v371_ftransfer_pole.py`	Track 3 (F_{transfer}): the Koide source→pole transfer as a typed solver – promotes the `experiments/ftransfer` solve into the ledger with a kill test (does NOT re-prove 53/54 v183 or Q^*=2/3 v82/v101). [E] Q_{pole}=Koide(PDG poles)= 2/3 to 3×10^{-6}, $Q_{\text{src}}(\varphi_0^{\text{ret}}$-ladder)= 0.66446 (0.33% below). [E] QED runs the wrong way: $dQ/d\epsilon > 0$ pushes Q ABOVE 2/3; the ϵ mapping $Q_{\text{pole}} \to Q_{\text{src}}$ is NEGATIVE while QED gives ϵ=+(3/2)(α/π) > 0 – wrong sign for any $\alpha \in [1/137, 1/128]$. [C] verdict: F_{pole} is NOT standard QED running; the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). Stays [C], plain running excluded as the kill. [E][C]
`v372_ftransfer_boltzmann.py`	Track 3 (F_{transfer}): η_B from the INTEGRATED Buchmueller–Di Bari–Pluemacher Boltzmann network at the TFPT-frozen heavy scale – promotes the `experiments/ftransfer` solve, replacing the v326 toy washout. Frozen M_1=$M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda$ ($\sim 8.65\times10^9$ GeV, v212), $\tilde{m}_1$=m_3/A_Λ=5 meV, δ_{CP}=$4\pi/3$. [E] integrated efficiency $\kappa_f \sim 0.09$ (strong washout $K \sim 4.6$) agrees with the BDP fit to $\sim$ 25%. [C] η_B(ODE)$\sim 6.5\times10^{-10}$ at the frozen M_1 lands within a factor ~ 1.1 of the observed 6.1×10^{-10}, NO M_R dial. [X] kill: η_B outside [obs/3, 3 obs] for all seesaw-consistent M_R would drop the leptogenesis branch. Stays [C] (a transfer target). [E][C][X]
`v373_ftransfer_relic.py`	Track 3 (F_{transfer}): the axion relic from the FINITE-T misalignment ODE – promotes the `experiments/ftransfer` solve and DECIDES the two TFPT angle branches, replacing the v326 adiabatic toy. f_a=$M_{\text{scal}}/128 \sim 2.4\times10^{11}$ GeV, $m_a \sim 23.8\,\mu$eV, lattice $\chi(T)$ n=8.16. [E] candidate scales fixed by the determinant-line atoms (128=2 dim $S^+|\mu_4|$). [C] the frozen SPINE angle $3\pi/5$=108°=$\pi N_{\text{fam}}/g_{\text{car}}$ gives $\Omega_a h^2 \sim 0.12$ in [0.08, 0.16] (lands on Ω_{DM} untuned) – the robust dominant-DM branch. [X] the HILLTOP 170° over-produces ($\sim 5.4\times$) – disfavoured. [X] kill: a finite-T solve robustly outside [0.08, 0.16] for the spine across $\chi(T)/g_*$ drops the dominant axion-DM branch. Stays [C]/[X]. [E][C][X]
`v374_ftransfer_qcd.py`	Track 3 (F_{transfer}): the m_p/m_e transfer as a typed uncertainty-budget + kill test – the falsifiability companion to v262 (the precise 2-loop point), replacing the v326 toy (which hardcodes 1836.15). [E] b_3= – (11 – $2n_f/3$)= – 7 (n_f=6 carrier). [C] two declared externals (v339): $\alpha_s(M_Z)$=0.1179±0.0009 and C_p=2.83±0.15, each $\sim$ 5.3%. [C] m_p/m_e central $\sim$ 1824 (v262) with a propagated ±7.5% band [$\sim$ 1690, $\sim$ 1960] CONTAINING the observed 1836.15 – no free dial. [X] kill: a future lattice C_p±0.03 + tighter α_s shrink the band to $\sim$ ±2%; a band then excluding 1836.15 drops the QCD-matching route (b_3= – 7 + the $SU(9)$-integer firewall v262). Distinct from v262's point; stays [C]. [E][C][X]
`v375_observatory_registry.py`	Track 4: the prediction OBSERVATORY – a status-typed CI over the frozen prediction registry (`freeze_file.csv`) that makes the falsifiability surface machine-checkable (no new physics). [E] falsifiability complete: every registry row carries a non-empty kill criterion (no orphan claims). [E] headline values re-derive from $\{\varphi_0^{\text{ret}}\}$: $\sin^2\theta_{12}$=$1/3-\varphi_0^{\text{ret}}/2$, $\sin^2\theta_{13}$=$\varphi_0^{\text{ret}}\,e^{-5/6}$, β_{rad}=$\varphi_0^{\text{ret}}/(4\pi)$. [C] live scorecard: θ_{12} vs JUNO first run (0.3092±0.0087)= 0.28σ (compatible); θ_{13} vs NuFIT 6.0 NO (0.02195±0.00058)= 2.0σ (the flagged most-tensioned core prediction); β_{rad} vs ACT DR6 (0.215±0.074)= 0.37σ (hint); r=$12/N_\star^2 \sim 0.004 <$ BK18 (compatible). [E] registry integrity: the single machine-readable falsifiability surface. Tooling for the closed/open pieces. [E][C]
`v376_seam_s3_centralcharge.py`	S3 closure stack: the central charge c=8 extracted NUMERICALLY from the lattice via Calabrese–Cardy finite-size entanglement scaling of the critical free-fermion content (the quantitative core of the MMST scaling-limit claim, on OUR model). [E] one critical complex mode gives c=1.0000 (Peschel correlation-matrix EE); [E] the collar is 16 Majoranas = 8 complex modes (v367), c additive ⇒ c=8=g_{car}+N_{fam}; [E] chiral c_-=8 (Chern $\|C\|$=1 per Majorana v367, non-gappable edge v368). [O] the specific net $(E_8)_1$ is v377/v378; the continuum existence is the cited MMST theorem (v336). Numerical c=8 from the lattice; does NOT close `SEAM.EQUIV.01`. [E][O]

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Script	**What it machine-checks** *(continued)*
`v377_seam_s3_e8character.py`	S3 closure stack: the chiral c=8 content is the SPECIFIC net $(E_8)_1$, via the exact affine character q-series + the μ_4/GSO promotion. **[E]** $\chi = E_4/\eta^8 = q^{-1/3}(1+248q+4124q^2+\dots)$: leading $-c/24 = -1/3 \Rightarrow c$=8; level-1 = 248 = dim E_8 (one primary). **[E]** the μ_4/GSO promotion $SO(16)_1 \to (E_8)_1$: 248=240+8, 240 = dim $S^+\, g_{\mathrm{car}}\, N_{\mathrm{fam}}$=16·5·3; 248=120+128= dim $SO(16)$+spinor. **[E]** discriminator: $(E_8)_1$ has 1 primary/248 currents vs $SO(16)_1$ 4 primaries/120. **[O]** continuum existence (v336); the μ_4 projection = the seam one-sidedness (S3). Pins the target net; does NOT close `SEAM.EQUIV.01`. **[E][O]**
`v378_seam_s3_modular.py`	S3 closure stack: the GENUS-1 discriminator – torus ground-state degeneracy = 1 (holomorphic = $(E_8)_1$), the "hard part" v344 flagged (genus-0 unique vacuum is necessary not sufficient). **[E]** KLM condensation: $\mu(\mathrm{carrier}) = \det C(D_5) \det C(A_3)$=4·4=16; the order-4 μ_4 glue ($\lvert L\rvert$=4) $\Rightarrow \mu(E_8)$=16/16=1 (torus GSD = 1); index-2 $\Rightarrow \mu(SO16)$=16/4=4. **[E]** $\lvert \det C(E_8)\rvert$=1 vs $\lvert \det C(D_8)\rvert$=4. **[E]** the single $(E_8)_1$ character has modular T-eigenvalue $e^{-2\pi i/3}$ (h=0, c=8), 1-dim rep $\Rightarrow$ one character $\Rightarrow$ torus GSD = 1. **[O]** continuum existence (v336). Pins the net at genus-1; does NOT close `SEAM.EQUIV.01`. **[E][O]**
`v379_seam_s3_rp.py`	C7 / OS reconstruction: reflection positivity (the OS axiom) of the explicit gapped collar measure, verified numerically – the remaining constructive-QFT ingredient for the OS reconstruction. **[E]** the gapped $p{+}ip$ band gives $G(\tau)=\mathrm{mean}_k e^{-\varepsilon_k \tau}$ with $\varepsilon_k \geq \mathrm{gap}/2 \sim 1 > 0$ (v367) – a positive spectral measure; **[E]** the Hankel/reflection Gram matrix $M_{ij}=G(\tau_i+\tau_j)$ is PSD – OS reflection positivity holds; **[E]** neg control: a growing ghost mode makes it indefinite (non-vacuous). **[C]** RP + gap (v76/v337) + GNS (v240) + c=8 (v376) + $(E_8)_1$ (v377/v378) give the OS inputs; C7/`QG.AMB.01` is then DISCHARGED by the redundancy theorem v369, not built. **[O]** the full constructive measure stays the cited step. **[E][C][O]**
`v380_grav_kms_hessian.py`	The KMS Entire Hessian – the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity (upgrades v304/v370 from "assume entire" to a derived statement). **[E]** the seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire & nowhere zero $\Rightarrow$ only the p^2=0 pole (no ghost). **[E]** each finite truncation carries the ghost: the $R{+}R^2$ order T_1=1+u has the Stelle zero $u=-1$, the Weyl^2 order $T_2=1+u+u^2/2$ the pair $-1 \pm i$; the entire a has none. **[E]** the ghost DECOUPLES: $\lvert$nearest zero$\rvert$ of T_n grows monotonically with n $(1, 1.41, \dots, 3.07$ for n=1..8) $\to \infty$, so "entire" = "do not truncate". **[C]** $\Rightarrow$ perturbative graviton unitarity. **[O]** the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation; perturbative-only, not the non-perturbative measure. **[E][C][O]**
`v381_qft4d_eg_allorder.py`	The all-order Epstein–Glaser / BRST contract for the perturbative 4D S-matrix S_{pert} – the "all-order closing statement" for the 4D perturbative QFT leg. Types the standard causal-perturbation-theory closure for the TFPT finite interaction class; it does *not* build a path integral. **[E]** POWER-COUNTING: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator $\Rightarrow$ the EG singular order $\omega \leq 4 \Rightarrow$ a FINITE counterterm space at every order. **[E]** BRST NILPOTENCY s^2=0 verified via the Jacobi identity for $su(2)$ (residual 0) and $su(3)$ (10^{-16}), the carrier weak+colour content. **[E]** the seam gap $\Delta = 6\ln(3/2) > 0$ (v302) gives the IR-safe adiabatic limit $g \to 1$. **[C]** imported: T_n exists to all orders by causal factorisation (Epstein–Glaser 1973 / Brunetti–Fredenhagen). **[C]** imported: all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) $\Rightarrow$ Slavnov–Taylor to all orders. **[E]** scope fence: matter+gauge ONLY – the R^2/Weyl^2 gravity sub-sector is v304/v370/v380, not a local EG vertex; S_{pert} is not the ambient measure `QG.AMB.01` (v369). NET **[C]** (imported EG+BRST on the TFPT dim-4 class + gap), prerequisites **[E]**; extends v269/v271/v273/v278 from 1-loop to all-order **[E][C]**

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Script	**What it machine-checks** *(continued)*
v382_alpha_quillen_exact.py	Names the Quillen determinant-line VARIATION $\delta_\tau\big(\log\det_\zeta \Delta_{U(1)} + 8b_1 c_3^6 \log\varphi_{\mathrm{seam}}\big) = 0$ as a tracked target (review Point 10), elevating EM.WARD.01's residual to a citable [O]. [E] the Quillen split holds at the root $\alpha^3 - 2c_3^3\alpha^2 = 8b_1c_3^6 \ln(1/\varphi_{\mathrm{seam}})$ at α^{-1}=137.0359992 ($F_{U(1)}$=0 IS the stationarity of the $U(1)$ determinant line, v341). [E] every coefficient a named atom: $8b_1{=}\frac{4}{5}{\cdot}41$ a $U(1)$ index, $c_3{=}1/(8\pi)$ Gauss–Bonnet ($c_3^{3,6}$ heat-kernel powers), $q(D_5){+}q(A_3){=}2$ discriminant forms (v341/v48/v216). [E] the value α^{-1}=137.0359992168 stays closed (unique root + ablation, v3) – only "why this functional" is open. [O] THE TARGET: the from-first-principles AQFT/ζ proof that $F_{U(1)}$ is the EXACT Quillen functional. [C] a FACE of SEAM.EQUIV.01 (v336): the seam $U(1)$ determinant line lives on the same holomorphic seam net, so once the net is $(E_8)_1$ its $U(1)$ sub-determinant is fixed – not a second independent gate, it names/tracks the pre-existing EM.WARD.01 residual (v341/v342). No new number (reuses v341, Wolfram-mirrored) [E][C][O]
v383_dynamics_universal.py	DYNAMICS.UNIVERSAL.01: the UNIVERSAL spectral-gap / Perron–Frobenius principle – every TFPT sector is the SAME object (a gapped operator with a UNIQUE leading attractor = the physics, a spectral gap = the reason there is no free parameter), extending v303 from F_{transfer} to the whole theory. [E] FLAVOR (computed): the cusp transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ has a simple Perron leading 1 and gap $6\ln(3/2) > 0$ (v56/v82). [E] DISCRETE COMPILER (computed): the affine-E_8 adjacency has Perron eigenvalue 2 (Kac marks, v312/v313) and subleading $2\cos(\pi/5) = \varphi$, gap $2-\varphi = 1/\varphi^2 > 0$. [E] DECOUPLING MARGIN: $\Delta_{\mathrm{eff}} = 6\ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v337). [E] TWO NUMBER-FIELD FACETS (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ – a factorisation, not an equality. [C] the other sectors are the same structure: gravity (v358/v359), QG saddle (v365), the holomorphic c=8 RG fixed point (v157/v158), the all-order S-matrix (v381), recovery (v221), F_{transfer} (v303), $\tau{=}i$ (v333). [C] META-THEOREM: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide – one spectral-gap statement wearing many hats. A synthesis, no new number [E][C]
v384_residual_certification.py	RESIDUAL.CERTIFICATION.01: the residual matrix is CERTIFICATION, not CONSTRUCTION – after v369/v381/v382 there is NO open TFPT physics MECHANISM left; every residual is an external CERTIFICATE of already-determined structure, of three kinds: (A) external math proof, (B) theorem-forbidden, (C) external physics. A ledger-CI audit. [E] LEDGER CI: every residual claim id is present (no orphan). [E] NO-UNIT (B): $c_3{=}1/(8\pi)$ and g_{car}=5 are mass-dimension 0, so by No-Unit (v153/v364) no dimensionless compiler outputs an absolute scale $\Rightarrow v_{\mathrm{geo}}$ is forbidden by theorem, not a gap. [E] GAP-DECOUPLED (A): $\Delta_{\mathrm{eff}} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ every readout is independent of the ambient measure (QG.AMB.01 a [C] redundancy, v369). [E] CLASSIFICATION COMPLETE: SEAM.EQUIV.01 (continuum existence = cited MMST/Adamo, v336)$\to$A, ALPHA.QUILLEN.EXACT.01 (v382)$\to$A, QG.AMB.01 (v369)$\to$A, v_{geo} $\to$B, F_{transfer} (v371–v374, firewall v187)$\to$C; ZERO open internal TFPT mechanisms. [C] CONVERGENCE: the only hard things left are theorems and external physics, not missing dynamics. An audit, no new number [E][C]

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Script	What it machine-checks *(continued)*
`v385_uvbranch_killtest.py`	UVBRANCH.KILLTEST.01: the optional carrier–Pati–Salam UV branch on the ALL-ORDER footing – a single citable kill-test surface with the proton-decay SAFETY hierarchy computed (not a fake τ_p window). A consolidation (does *not* re-run the RGE v246 nor re-derive the PS scale v249/v253). **[E]** carrier content = SM ($b = (41/10, -19/6, -7)$, v159/v246, no new states) and $S_{\rm pert}$ closes to all orders (v381) $\Rightarrow$ the SM-non-unification is all-order-robust. **[X]** plain-SM 4D-GUT killed ($\sin^2\theta_W = 3/8$ vs 0.231; v246). **[C]** optional carrier-PS branch: $M_{PS} = \kappa M_{\rm scal}$, $M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl} \approx 3.06\times10^{13}$ GeV, $\kappa \in [1.0, 1.2]$ (v253); content $\{1, 10, 16, 45\}$, no 126 (v247/v248); not the default (v265). **[E]** proton-decay SAFETY hierarchy: minimal PS gauge bosons are $SU(4)$ leptoquarks ($B{-}L$-conserving $q \leftrightarrow l$, rare LFV $K_L \to \mu e$), *not* the $p \to e^+\pi^0$ dim-6 operator (needs an $SO(10)/SU(5)$ diquark, absent); the binding bound $M_{PS} \gtrsim 10^6$ GeV is cleared by $M_{PS}/M_{\rm bound} \sim 10^7 \gg 1$. **[X]** frozen kill tests (v265): κ leaving $O(1)$ / a forbidden rep / a new gauge-charged state / a rare-LFV signal below the bound; no fake τ_p. A consolidation, no new number **[E][C][X]**
`v386_grav_amplitude.py`	GRAV.AMPLITUDE.01: the entire-form-factor graviton-exchange AMPLITUDE is finite, UV-softened and tree-unitary – perturbative gravity as an explicit scattering problem, extending v304/v370/v380 from the propagator to the amplitude. The dressed spin-2 propagator $P = e^{-p^2/M^2}/p^2 = 1/(p^2 a)$, $a(u) = e^u$, $M = M_{\rm scal}$. **[E]** one pole, ghost-free: a entire & nowhere zero (v380) $\Rightarrow$ only pole at $p^2 = 0$, residue $e^0 = 1 > 0$. **[E]** UV-softened: ratio e^{-u} at $u = 1, 5, 10, 20 = 0.37, 6.7\times10^{-3}, 4.5\times10^{-5}, 2.1\times10^{-9}$ (faster than any power). **[E]** low-energy = GR + a calculable $-p^2/M^2$ correction ($M = M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$, v253). **[C]** tree unitarity: a single positive-residue pole, no negative-norm states (v370/v380). **[C]** the amplitude $A = (\kappa^2/2)(T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is GR at low energy, softened in the UV (perturbative-only; `QG.AMB.01` a **[C]** redundancy v369). No new number **[E][C]**
`v387_corrections_gap.py`	CORRECTIONS.GAP.01: the practical harvest of v383 – the same spectral gap that makes each sector parameter-free ALSO sets the SIZE of its first correction (correction$_n \sim (\lambda_2/\lambda_1)^n$). **[E]** FLAVOR: $\lambda_2 = (2/3)^6 = 64/729 \Rightarrow$ first Koide correction ~ 0.088 (v56/v82). **[E]** RECOVERY: the same $(2/3)^6$ Petz rate (v221). **[E]** QG: capped by $\chi = 1/(1-(2/3)^6) = 729/665$ (v337). **[E]** DISCRETE COMPILER: the update $(A + 2I)/4$ fixes the Kac marks ($\lambda_1 = 1$) and decays at the golden $(\varphi + 2)/4 \approx 0.905$ (carrier-5 facet, v303/v312). **[E]** the rates split into the $(2/3)^6$ family (family-3, in $\mathbb{Q}$) and the golden family (carrier-5, in $\mathbb{Q}(\sqrt{5})$) – the two facets of v314/v383. **[C]** the correction calculus: the gap does double duty (WHY no free parameter AND HOW BIG the first correction), so sub-leading magnitudes are organised, not free. No new number **[E][C]**
`v388_corrections_budget.py`	CORRECTIONS.BUDGET.01: the gap correction (v387) is a *typed* budget, NOT a uniform band on every prediction – the $(\lambda_2/\lambda_1)^n$ size is a property of the distance from a gapped operator's leading eigenvector, so it is only defined where λ_2 exists. The prediction surface splits into four correction classes (already implied by v303). **[E]** SEAM-GAPPED: Koide ($F_{\rm pole}$ v303), recovery (v221), the QG bound (capped by $\chi = 729/665$, v337) carry $(2/3)^6$; the discrete compiler decays at the golden $(\varphi + 2)/4$ (v312). **[E]** KOIDE FIRST CORRECTION: $F_{\rm pole}$ is the ONE seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), source quotient 0.33% below 2/3 – the rate is the suppression SCALE, not the residual. **[E]** EXACT-IDENTITY class ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$, $\sin^2\theta_{13}$): no $\lambda_2 \Rightarrow$ correction 0 STRUCTURALLY (a band would contradict **[E]**). **[E]** EXTERNAL-RATE class (η_B/axion/m_p, m_e): the gapped shape with external rates (only $F_{\rm pole}$ has the seam rate, v303), firewalled (v187). **[E]** FIXED-POINT class: the texture is already explicit ($\theta_{12}\ \varepsilon = c_3 + 36c_3^4$). **[C]** a uniform band on all predictions is WRONG; the size is keyed to the mechanism class. A typing of v387, no new number **[E][C]**

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Script	**What it machine-checks *(continued)***
`v389_grav_loop_finiteness.py`	GRAV.LOOP.FINITE.01: the entire-form-factor graviton is UV-finite at the *loop* level by power counting – the honest next step past v386 (tree), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence. Honest scope: the standard infinite-derivative/nonlocal-gravity finiteness statement, NOT a full proof of perturbative QG. [E] super-polynomial falloff $p^k e^{-p^2/M^2} \to 0$ for every k. [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$; the GR $\int p^3/p^2\, dp$ diverges quadratically. [E] the superficial degree $\to -\infty$ at every loop order (dressed cutoff-ratio $\to 1$ vs GR $\to 4$). [E] the regulator is the SEAM scale $M = M_{\rm scal} = c_3^{7/2}\bar{M}_{\rm Pl}$ (v253/v259), not a dial. [C] loop UNITARITY of the nonlocal form factor (Lorentzian continuation / macro-causality) stays the honest open point – tree unitarity is v386; NOT claimed solved. No new number [E][C][O]
`v390_prime2_facet.py`	COXETER.PRIME2.01: the prime-2 facet of the order-30 Coxeter clock, completing v383. The clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$ had two named facets (golden $\varphi \in \mathbb{Q}(\sqrt{5})$, prime-5/carrier; $(2/3)^6 \in \mathbb{Q}$, prime-3/family); this names the prime-2 facet as the Gaussian field $\mathbb{Q}(i)$ — $\tau = i$ and the order-4 $\mu_4 = \mathbb{Z}/4$ square seam deck (v282/v288/v333), structure already present. [E] $h(E_8) = 30$ factors into exactly three primes. [E] one quadratic facet field per prime, each ramified at its prime: $\operatorname{disc}\mathbb{Q}(i) = -4$, $\operatorname{disc}\mathbb{Q}(\sqrt{-3}) = -3$, $\operatorname{disc}\mathbb{Q}(\sqrt{5}) = 5$; ramified primes $\{2, 3, 5\}$, product $= 30$. [E] prime-2 Gaussian facet: min. poly $x^2 + 1$, $\tau = i$, $j(i) = 1728 = 12^3$, $\lvert\mu_4\rvert = 4 = 2^2$ (the static seam-orientation facet). [E] prime-5 golden (v383): $2\cos(\pi/5) = \varphi$ (pentagon). [E] prime-3 family (v383/v220): $(2/3)^6$ plus the Eisenstein ζ_3 (hexagonal CP $\pi/3$). [C] completion: each prime carries a facet (square/hexagon/pentagon = seam/CP/carrier), the prime set forced by $h(E_8)$. A synthesis, no new number [E][C]
`v391_alpha_quillen_progress.py`	ALPHA.QUILLEN.PROGRESS.01: an *honest attempt* at the external target ALPHA.QUILLEN.EXACT.01 (v382) – it REDUCES the open step to a sharper named sub-target via a solvable model but does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384). [E] solvable model $\det_\zeta(-\partial_\tau^2{+}m^2)$ on a circle $= (2\sinh(\beta m/2))^2$, m-variation $\beta\coth(\beta m/2) = 2/m + (\beta^2/6)m - (\beta^4/360)m^3 + \dots$ a closed heat-kernel/Laurent series (m^3 coeff $-1/360$ verified). [E] a ζ-det variation is structured heat-kernel data, so v382's $8b_1c_3^6$ counterterm (a Gauss–Bonnet heat-kernel power) is the right *kind* of object. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [E] reduction: the open step reduces to the named sub-target "seam Seeley–DeWitt coefficient $= 8b_1c_3^6$". [O] NOT CLOSED: the actual seam ζ-det proof stays open, SEAM.EQUIV.01 continuum existence (v336) the other external [O]; α^{-1} stays [E]. An attempt that reduces, does not close [E][O]
`v392_seam_s3_scalinglimit.py`	SEAM.S3.SCALINGLIMIT.01: an *honest attack* on the one open lemma of SEAM.EQUIV.01 (the continuum scaling-limit existence, v336) – it does NOT close it (the abstract existence is the cited MMST theorem); it STRENGTHENS the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy EE slope $c_1(L)$ at $L = 66 \dots 514$ is a monotone Cauchy-like sequence approaching 1 ($\lvert c_1(L) - 1\rvert$ strictly decreasing). [E] a $1/L$ Richardson fit gives $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\rm car}{+}N_{\rm fam}$ is reached *in* the limit. [E] the successive differences are strictly decreasing (numerical Cauchy, necessary for existence). [O] convergent data are evidence, not a proof of existence (MMST v336), and do not pin $(E_8)_1$ vs $SO(16)_1$ (holomorphy v377/v378). Strengthens the residual, does NOT close it [E][O]

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Script	What it machine-checks *(continued)*
`v393_corrections_numeric.py`	CORRECTIONS.NUMERIC.01: the typed correction budget (v388) turned into ACTUAL first-correction magnitudes – a computed theoretical-uncertainty surface, the short-term testable harvest of v388. [E] FIXED-POINT band ($\theta_{12}/\theta_{13}/\lambda_C/\beta_{\rm rad}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction). [E] SEAM-GAPPED band (Koide/recovery) $= (2/3)^6 \approx 8.78\%$; the QG bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337). [E] EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$). [E] EXTERNAL-RATE band = external physics quoted (m_p/m_e ±7.5% v374; n_s/r $N_\star$ band; η_B 1.07× v212), fenced v187. [C] each prediction gets "central value ± computed first correction". A harvest of v388, no new number [E][C]
`v394_coxeter_atoms.py`	COXETER.ATOMS.01: the clockwork coherence – the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets, and the prime-2 seam is the common dynamical factor. A bird's-eye [E] re-reading tying v390 (facets) to the anchor $a = (1, 1, 2)$ and v319 (5x6 clock); registers `RES.COXETER.SYMMETRY.01` as a tracked [O]. [E] $a = (1, 1, 2)$: $e_3{=}2{=}\lvert Z_2\rvert$, $p_0{=}3{=}N_{\rm fam}$, $e_2{=}5{=}g_{\rm car}$, product $30{=}h(E_8)$. [E] atoms = facet fields (v390): $2{\to}\mathbb{Q}(i)$, $3{\to}\mathbb{Q}(\sqrt{-3})$, $5{\to}\mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] $\chi{=}729/665{=}3^6/(3^6{-}2^6)$, $(2/3)^6{=}2^6/3^6$, exponent $6{=}2N_{\rm fam}$. [E] both dynamic rates carry prime-2: $2/3{=}\lvert Z_2\rvert/N_{\rm fam}$, $(\varphi{+}2)/4{=}(\varphi{+}2)/\lvert\mu_4\rvert$, $\lvert\mu_4\rvert{=}4{=}2^2$. [E] $30{=}5{\times}6{=}g_{\rm car}(2N_{\rm fam})$. [C] the seam is the universal dynamic engine. [O] `RES.COXETER.SYMMETRY.01` registered, not closed [E][C][O]
`v395_hypergraph_coupled.py`	HYP.COUPLED.01: the coupled hypergraph rewrite – one local rule unifies v299/v327/v324 (carrier × family). [E] LOCAL COUPLED MICRO-RULE: state on 9 affine-E_8 nodes × 3 family slots; one micro-step = network lazy diffusion on each cusp column + v327 M on each node's family vector; steps commute, $T_{\rm micro} = T_{\rm net} \otimes M$. [E] JOINT ATTRACTOR: top eigenvalue 1 with eigenvector marks$\otimes e_0$; iteration from any positive start converges. [E] ONE HAND = RECOVERY: one network step + $2N_{\rm fam}$ family micro-steps gives $(2/3)^6$ on marks$\otimes v_{\rm surv}$. [E] v324 EMERGES: $T_{\rm full} = T_{\rm net} \otimes T_{\rm cusp}$ shares the same invariants; $T_{\rm cusp}$ is the 6-hand readout in the Galois/cusp basis (v317). [E] GROWTH + FIBER: v299 $E_6{\to}E_7{\to}E_8{\to}\hat{E}_8$ unchanged with M-fiber attached. [O] 3-arm seed (P_2) + φ_0 (v312) remain irreducible [E][O]
`v396_phi0_icosahedral.py`	HYP.PHI0.01: the φ_0 leading term from icosahedral combinatorics (sharpening v312/v395). [E] ICO COUNTS ANCHOR-FORCED: $V{=}12$, $E{=}30{=}h(E_8)$, $F{=}20{=}g_{\rm car}(g_{\rm car}{-}1)$, $\lvert\,{\rm Aut}\,\rvert{=}120{=}4h{=}\lvert 2I\rvert$. [E] LEADING TERM EXACT: $\varphi_0^{\rm tree}{=}1/(6\pi){=}(4/3)c_3{=}F/(4h\pi){=}F/(\lvert\,{\rm Aut}\,\rvert\pi){=}(F/(g_{\rm car}N_{\rm fam}))c_3$. [E] $F/h{=}\lvert Z_2\rvert/N_{\rm fam}$; leading term = $(F/h)/(4\pi){=}c_3(g_{\rm car}{-}1)/N_{\rm fam}$. [E] GAUSS–BONNET: icosahedron deficit sum 4π; $c_3{=}1/(\lvert Z_2\rvert 4\pi)$ (v216/v342). [E] HONEST NEGATIVE: puncture $48c_3^4$ not a rational hypergraph fraction. [C] tree-level seed = (hyperedges)/$(g_{\rm car}N_{\rm fam}) \times c_3$, not an independent injection; puncture [O]
`v397_external_clock_probe.py`	FR.CLOCK.PROBE.01: a *falsifiable* probe of how external the "external physics" really is – do the frontier scheme-numbers land on the $\{2, 3, 5\}$ clock (the anchor atoms, v394) or not? Strict No-Free-Pattern (v100): "on the clock" = an EXACT match, not a wide-band fit. [E] η_B decuple $A_\Lambda{=}10{=}\lvert Z_2\rvert g_{\rm car}$ ON CLOCK (v212). [E] Koide rate $(2/3)^6{=}(\lvert Z_2\rvert/N_{\rm fam})^6$ ON CLOCK (v303). [E] axion spine angle $3\pi/5{=}\pi N_{\rm fam}/g_{\rm car}$ ON CLOCK (v211). [E] proton $C_p \sim 2.83{\pm}0.15$ NOT discriminating ($2^{3/2}, 17/6, 14/5$ all fit the band) ⇒ EXTERNAL (firewall v187/v374). [C] 3 of 4 land exactly on the clock; the irreducible external residue shrinks to the proton QCD number C_p – "external physics" was overstated. [E] anti-numerology: refuses to assign C_p a clock value [E][C]

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Script	What it machine-checks *(continued)*
`v398_seam_state_rigidity.py`	**Seam State Rigidity Theorem** (Paper A, the G_{net} closure contract): the scattered state-invariance reductions (v177/v198/v199/v308/v309/v335) as ONE named target. [E] RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, v54, on a quasi-free state v155, no μ_4 normal form); [E] band-limited core $[\rho,\lvert k\rvert]{=}0$ and character-block-diagonal sub-principal symbol swept $N{=}4..64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $\lvert\det \mathrm{Cartan}\rvert{=}1$ vs 4, index $4{=}\lvert\mu_4\rvert$, gap $6\log\frac{3}{2}{>}0$). [O] the one residual: $[\rho,\Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). A contract+evidence module, NOT a closure of `SEAM.EQUIV.01` (continuum = cited MMST v336)
`v399_gravity_complete.py`	**Gravity-complete** = **Boundary-complete** + **Ambient-redundancy** (Paper B accounting): combines the parameter-free local equation (v358/v359) and the ambient redundancy (v369). [E] $G_{ab}{+}\Lambda g_{ab}{=}(1/c_3)T_{ab}$, $1/c_3{=}8\pi$ (four c_3-roles), Λ prefactor $3/(4\pi^2)$; [E] redundancy discriminators ($\lvert\det \mathrm{Cartan}\rvert{=}1$ vs 4, torus GSD=1, Petz $(2/3)^6$, margin $1.648{>}0$); [E] $\mathcal{O}_{\mathrm{phys}} \subset \mathcal{A}_\Sigma$ over five readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/\lvert\mu_4\rvert$, recovery $(2/3)^6$) – all boundary/seam/gap. [C] the verdict; [O] `SEAM.EQUIV.01` + intrinsic BW (no new gate)
`v400_grav_nonlocal_action.py`	**Nonlocal spectral-gravity action** with the local $R{+}R^2$ as its IR projection (Paper C, consolidating v304/v370/v380/v386). [E] the entire form factor $a{=}e^u$ gives the dressed propagator $1/(p^2a)$ ONE pole (residue 1, ghost-free); [E] every truncation carries a Stelle zero whose nearest-zero modulus grows monotonically (v380); [E] the IR order $a{=}1{+}u{+}u^2/2{+}\dots$ reproduces the spectral-action $R{+}R^2$ (v28/v36) with the atom-fixed scale $M^2_{\mathrm{scal}}/\bar{M}^2_{\mathrm{Pl}}{=}c_3^7$ ($7{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{-}1$, v253). [O] the entire-$a$ analyticity assumption + perturbative-only (not the ambient measure `QG.AMB.01`, a [C] redundancy v369)
`v401_metrology_closure.py`	**Metrology closure** of the final input set $\{a,\pi,v_{\mathrm{geo}}\}$ (Paper D, consolidating v153/v274/v364/v384). [E] $a{=}(1,1,2)\to(e_1,e_2,e_3){=}(4,5,2){=}(\lvert\mu_4\rvert,g_{\mathrm{car}},\lvert Z_2\rvert)$, $c_3{=}1/(2e_1\pi){=}1/(8\pi)$, 0 dimensionless dials + 1 unit; [E] dimensional bookkeeping (No-Unit v153) mass/$1G/\Lambda/U_{\mathrm{point}}/(m/\mu)$= number $\times v_{\mathrm{geo}}^{1,2,4,1,0}$; [E] gravity adds no unit (v364); [E] G-vs-Λ over-determination 0.11% (v274). [O] v_{geo} theorem-forbidden ($\mathrm{TOE}_{\mathrm{strict}}{=}\mathrm{TOE}_{\mathrm{dimensionless}}{+}[v_{\mathrm{geo}}]$, v384 type B)
`v402_ftransfer_suite.py`	F_{transfer} **functor suite** (Paper E): the four frontier sectors as ONE functor $F_{\mathrm{transfer}}{=}F_{\mathrm{observable}}\circ F_{\mathrm{threshold}}\circ F_{\mathrm{RG}}$ (consolidating `FR.POLE/BOLTZMANN/RELIC/QCD.SOLVE`). [E] the composition law (each observable factors source→threshold→RG→observable); [E] exact atom sources (Q_{src} v93; $A_\Lambda{=}2g_{\mathrm{car}}{=}10$, $\delta_{CP}{=}4\pi/3$ v9/v212; $f_a{=}M_{\mathrm{scal}}/128$ v185; $b_3{=}{-}7$ v164/v262); [E] a committed kill test per sector. [C] the firewall: physical predictions stay [C]/[O], only the algebraic source is [E] (never re-pressed into the compiler, v187)
`v403_facet_compositum.py`	ARITH.HULL.01: the three number-field facets (v390) COMPOSE to $K=\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$, a degree-8=rank E_8 abelian field – the arithmetic hull dual to the E_8 lattice hull. [E] the squarefree parts $-1,-3,5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K:\mathbb{Q}]=2^3=8=\mathrm{rank}\,E_8=g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, Galois group $(\mathbb{Z}/2)^3$ order 8; $2^{\omega(h(E_8))}=2^3=8=\mathrm{rank}\,E_8$ is E_8-special (the E_7 control $h{=}18$, $2^2{=}4{\neq}7$ fails). [E] exactly 7 quadratic subfields $\{-1,-3,5,3,-5,-15,15\}$, split 4 imaginary $=\lvert\mu_4\rvert$ (CM/dynamic) + 3 real $=N_{\mathrm{fam}}$ (RM/static). [E] ramified only at $\{2,3,5\}$ = the atoms = the primes of $h(E_8){=}30$. [C] K is the abelian $(\mathbb{Z}/2)^3$ arithmetic hull dual to E_8; the 7 quadratic subfields a candidate for the 7 E_8 audit slices [E][C]

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Script	What it machine-checks *(continued)*
`v404_e8_char_cm_duality.py`	E8.CMDUAL.01: the $(E_8)_1$ vacuum character $\chi_{E_8}{=}E_4/\eta^8{=}j^{1/3}$ at the two CM points – 12 at the seam ($\tau{=}i$), 0 at the family ($\tau{=}\omega$), extending v282. **[E]** $\chi_{E_8}^3{=}j$ (q-series). **[E]** SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio 2 $\Rightarrow$ $j{=}1728$ $\Rightarrow$ $\chi_{E_8}(i){=}1728^{1/3}{=}12$ (E_8 present, v214/v282). **[E]** FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1{=}0$) $\Rightarrow$ $j{=}0$ $\Rightarrow$ $\chi_{E_8}(\omega){=}0$ (E_4 vanishes, v220/v267). **[E]** $\tau{=}i, \tau{=}\omega$ are the ONLY elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$, so the values are forced. **[C]** the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi{=}12$, `SEAM.EQUIV.01` closes), the flavor/CP sector is the soft residual where E_8 degenerates ($\chi{=}0$); the dual-keystone candidate (flavor = the $j{=}0$ degeneration of $(E_8)_1$) **[E][C]**
`v405_seam_equiv_omega.py`	**SEAM.EQUIV.02** – the dual keystone at $\tau{=}\omega$: the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam = $(E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. **[E]** $\chi_{E_8}(\omega){=}0$ (E_8 degenerates at the family CM point) vs $\chi_{E_8}(i){=}12$. **[E]** A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan det $=3{=}N_{\mathrm{fam}}$, $h{=}3{=}N_{\mathrm{fam}}$, $\mu_6{=}\mu_3{\times}\mu_2$). **[E]** R lives in $E_6{\times}A_2$: $248{=}\lVert R\rVert_F^2{+}\det R{+}2(\mathbf{1}^\top Ra)N_{\mathrm{fam}}{=}78{+}8{+}2{\cdot}27{\cdot}3$, $\lVert R\rVert_F^2{=}78{=}\dim E_6$, $\det R{=}8{=}\dim A_2{=}\mathrm{rank}\,E_8$. **[E]** $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase ($\arg\zeta_6{=}\pi/3$, v220/v233). **[E]** the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\mathrm{fam}} = \det Q$ (v72), $\mathrm{Spec}(Q_+){=}(1,2,3)$ (v69/v137). **[C]** the family/flavor sector is the order-3 Eisenstein/A_2 deck at $\tau{=}\omega$; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. **[O]** residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the v_{geo}-class scale (No-Unit v153) **[E][C][O]**
`v406_pmns_phase_origin.py`	FLAV.PMNS.CLOSE.01: the PMNS dynamics closed modulo the $\tau{=}\omega$ keystone (v405) – the CP-phase ORIGIN is the family CM point and the absolute scale is one v_{geo}-class unit, so PMNS has zero free flavor dials beyond v_{geo}. **[E]** the three angles ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$, $\theta_{23}{=}45^\circ$, $\sin^2\theta_{13}{=}\varphi_0^{\mathrm{ret}}e^{-5/6}$; v9/v268) assemble to one unitary U_{PMNS} (v270). **[C]** CP-PHASE ORIGIN (advance over v270): $\delta_{\mathrm{PMNS}}{=}4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405/v220/v233), upgrading v270's **[O]** assembled input to a keystone consequence. **[C]** $J_{\mathrm{PMNS}}{=}{-}0.0297$ derived. **[E]** the absolute scale = one seesaw ratio $m_3{=}(y_\nu v_{\mathrm{EW}})^2/M_R$ = a v_{geo}-class UV input (v272/v153), NOT a flavor gap. **[C]** net: PMNS = angles + δ [keystone] + J [derived] + scale [v_{geo}], zero free flavor dials. **[O]** residual = the seesaw operator realisation (M_R not a clean $\bar{M}_{\mathrm{Pl}}$ power, v272/v184) + the normal-ordering Σm_ν floor kill test **[E][C][O]**
`v407_dn_pairings_omega.py`	FLAV.SELECTOR.CLOSE.01: the R4′ residue folds into the $\tau{=}\omega$ keystone (v405) – the three (d,n) selector pairings ARE the $\tau{=}\omega$ family-slice atoms. **[E]** the frame $(\mathbf{1},a,\sigma)$ is a $\mathbb{Z}^3$ basis with $\det{=}11{=}\lVert Pl(K)\rVert_1$, and $n{=}(5,{-}9,6)$ is the unique covector with $n{\cdot}\mathbf{1}{=}2$, $n{\cdot}a{=}8$, $n{\cdot}\sigma{=}121{=}11^2$ $\Rightarrow$ $(d,n){\Rightarrow}R$ columnwise (v136/v139). **[E]** $n{\cdot}a{=}8{=}\mathrm{rank}\,E_8{=}\dim A_2{=}\det R$ – the anchor-pairing is the A_2 ($\tau{=}\omega$) block dimension that reads R in $E_6{\times}A_2$ (v405). **[E]** the pairings $\{2,8,121\}{=}\{\lvert Z_2\rvert, \mathrm{rank}\,E_8{=}\dim A_2{=}\det R, \lVert Pl(K)\rVert_1^2\}$ are the $\tau{=}\omega$ family-slice atoms (the μ_2 sheet, the A_2 block, the Plucker frame). **[E]** the square selects the point (σ-pairing $11(11{+}t)$, a perfect square only at $t{=}0$). **[C]** deriving (d,n) (R4′) folds into `SEAM.EQUIV.02` – $d{=}\frac{3}{2}a{-}2\mathbf{1}$ anchor-derived (v139), the pairings the slice atoms. **[O]** the lone flagged residue "why $11{=}\lVert Pl(K)\rVert_1$" **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v408_phi0_puncture_heatkernel.py`	`HYP.PHI0.PUNCTURE.01`: name the φ_0 puncture term $\delta_{\text{top}}=48c_3^4$ as the order-4 local boundary heat-kernel (Seeley-DeWitt) contact term of the μ_4 puncture divisor – the analytic complement of the icosahedral tree term (v396, `HYP.PHI0.01`). **[E]** THE SPLIT IS EXACT: $\varphi_0=\varphi_0^{\text{tree}}+\delta_{\text{top}}=1/(6\pi)+48c_3^4$ (v396/`ARCH.QUAD.01`). **[E]** PUNCTURE COUNT = $\Omega_{\text{adm}}=N_{\text{fam}}\dim S^+=48$, order = $\lvert\mu_4\rvert=4$: $\delta_{\text{top}}=\Omega_{\text{adm}}c_3^{\lvert\mu_4\rvert}$. **[E]** CLOSED FORM $\delta_{\text{top}}=48c_3^4=3/(256\pi^4)$; $c_3=1/(\lvert Z_2\rvert 4\pi)$ Gauss-Bonnet (v216/v342) so c_3^4 is a fourth-order boundary heat-kernel power (cf. v391). **[E]** HONEST NEGATIVE: δ_{top} not a rational hypergraph fraction ($\delta_{\text{top}}/\varphi_0^{\text{tree}}<1\%$). **[O]** THE TARGET: prove $\Omega_{\text{adm}}c_3^4$ IS the order-4 boundary Seeley-DeWitt contact term of the μ_4 puncture summed over the Ω_{adm} admissible states. **[C]** two engines, two terms; a FACE of the seam boundary analysis (the c_3 datum), not a new gate **[E][C][O]**
`v409_coxeter_prime2_lemma.py`	`RES.COXETER.SYMMETRY.01` closed as a STRUCTURAL LEMMA: prime-2 is NECESSARY but NOT autonomous – no prime-2-only attractor exists (resolves the v394/`COXETER.ATOMS.01` open question's falsifiable corollary). **[E]** SHEET INVOLUTION $\sigma=\operatorname{diag}(1,-1,-1)$, $\sigma^2=I$, spectrum $\{+1,-1\}$ (parity). **[E]** SHEET PROJECTOR $P=(I+\sigma)/2=\operatorname{diag}(1,0,0)$, $P^2=P$, spectrum $\{0,1\}$ (projection). **[E]** NO CONTRACTION RATE: none of $\{-1,0,1\}$ lies in $(0,1)$, whereas the genuine rates $2/3$ and $(1/3)^6$ do – prime-2 alone gives parity/projection, never a nonzero rate. **[E]** FACTOR NOT RATE: $2/3=\lvert Z_2\rvert/N_{\text{fam}}$ needs prime-3, $(\varphi+2)/4=(\varphi+2)/\lvert\mu_4\rvert$ with $\lvert\mu_4\rvert=2^2$ needs the golden φ (prime-5). **[E]** $30=g_{\text{car}}(2N_{\text{fam}})$, carrier static, $2{\cdot}3$ dynamic. **[C]** prime-2 is the universal dynamical parity factor, necessary in every sheet-paired transport but not an autonomous attractor; dynamics begins after coupling to family-3 or carrier-5 ($\mu_4=2^2$ the Galois bridge) **[E][C]**
`v410_sheet_generator_binary.py`	The *sheet generator* $V=Q\operatorname{diag}(0,1,1)$ (rank-2 sheet axis, v95/v218) is a binary internal compiler: $V^n=\left(\begin{smallmatrix}0&2^{n-1}&0\\0&2^n&0\\0&2^{n+1}-2&1\end{smallmatrix}\right)$ (induction), so $V^n\mathbf{1}=(2^{n-1},2^n,2^{n+1}-1)=(1,2,3),(2,4,7),(4,8,15),(8,16,31)$; bilinears $\mathbf{1}^TV^n\mathbf{1}=7{\cdot}2^{n-1}-1$, $\mathbf{1}^TV^na=7{\cdot}2^{n-1}$, $a^TV^na=11{\cdot}2^{n-1}$, $a^TV^n\mathbf{1}=11{\cdot}2^{n-1}-2$ give $(6,7,9,11)$ then $13=\Delta_Q$, $27=\mathbf{1}^TRa$, 55, $56=\dim\mathbf{56}_{E_7}$; the spine is FORCED by $\operatorname{Spec}(V)=\{0,1,2\}$, the carrier (rank $E_8=8$, $\dim S^+=16$, $248/8=31$) and theta ($\sigma_3(2)=9=a^TV\mathbf{1}$, $\sigma_3(3)=28=\mathbf{1}^TV^3a=\det(I+R)$, $\sigma_3(5)=126$) readings audit-typed **[E][C]**
`v411_ud_ratio_vpower.py`	The quark u/d ratio is a pure V-power readout: the established $c_u/c_d=g_{\text{car}}\lVert\operatorname{Pl}(K)\rVert_1/(N_{\text{fam}}^2\Delta_Q)=55/117$ (v94) re-expresses EXACTLY via the sheet generator (v410) as $\boxed{c_u/c_d=\frac{\mathbf{1}^TV^4\mathbf{1}}{(a^TV\mathbf{1})(\mathbf{1}^TV^2\mathbf{1})}}=\frac{55}{9\cdot13}$, with $55=\mathbf{1}^TV^4\mathbf{1}$, $9=a^TV\mathbf{1}=N_{\text{fam}}^2$, $13=\mathbf{1}^TV^2\mathbf{1}=\Delta_Q$; the same number as $5{\cdot}11/(9{\cdot}13)$, so the numerator repackages $g_{\text{car}}{\cdot}11=55$. An exact *re-encoding*: the value and the u/d identification stay **[C]** (U_wall readout rigidity) **[E][C]**
`v412_sheet_source_corner_J.py`	The *sheet source corner* J: the determinant surface has two s-independent iso-volume walls (v135) $t{=}-1$ ($\det=4=\lvert\mu_4\rvert$, carries K) and $t{=}-2$ ($\det=2=\lvert Z_2\rvert$); the operator on the Z_2 wall, $J=M(1,-2)=C-2V=\left(\begin{smallmatrix}4&1&0\\4&1&2\\5&1&1\end{smallmatrix}\right)$, completes the sheet axis $J,K,C,F=M(1,-2..1)$. $\chi_J=\lambda^3-6\lambda^2+3\lambda-2\Rightarrow(\operatorname{tr},e_2,\det)=(6,3,2)=(\lvert R^+(A_3)\rvert,N_{\text{fam}},\lvert Z_2\rvert)$; $a^TJa=30=h(E_8)$, $\det(I+J)=12=\dim\mathfrak{g}_{\text{SM}}=\chi_{E_8}(i)$, $\det(2I+J)=40=\lvert R(D_5)\rvert$ (Lie readings audit-typed like the G_2/F_4 shadows of v95/v218) **[E][C]**
`v413_sheet_characteristic_calculus.py`	The sheet axis is a discrete *characteristic-difference calculator*: on $M_t=C+tV$ (v95/v218) the elementary-symmetric coefficients encode atoms as DIFFERENCE ORDERS – $e_1=3t+12$, $e_2=2t^2+15t+25$, $e_3=\det M_t=4t^2+14t+14$ give $\Delta e_1=3=N_{\text{fam}}$, $\Delta^2e_2=4=\lvert\mu_4\rvert$, $\Delta^2e_3=8=\operatorname{rank}E_8$ (the e_3 curvature cross-cited to v218/v224; the full (e_1,e_2,e_3) ladder is new); the anchor energy $a^TM_ta=52+11t$ (step $11=a^TVa$, v410) reads $30\to41\to52\to63=(h(E_8),10b_1,\dim F_4,7N_{\text{fam}}^2)$ **[E][C]**

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Script	What it machine-checks *(continued)*
`v414_center_resolvent_portal.py`	The center C is a *resolvent portal* $G_2 \to F_4 \to E_8$: $C{=}M(1,0)$ (v95) is the $t{=}0$ value of V; $\operatorname{tr} C{=}12{=}\dim \mathfrak{g}_{\mathrm{SM}}{=}\chi_{E_8}(i)$, $\det C{=}14{=}\dim G_2$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1{=}248/8$; the shifted determinants climb $\det(C){=}14{=}\dim G_2$, $\det(I{+}C){=}52{=}\dim F_4$, $\det(2I{+}C){=}120{=}\lvert R^+(E_8)\rvert$ – the shell ladder $G_2{\to}F_4{\to}E_8^+$ as $\det(C{+}kI)$, $k{=}0,1,2$ (the resolvent ladder is new; reading audit-typed) [E][C]
`v415_gaussian_operator.py`	The square CM-norm dictionary is *operator-realised* and the μ_4 deck is an intrinsic diamond object: the Gaussian arithmetic of v222/v230 lifted from numbers to OPERATORS via $J{=}[U,V]/3$, the complex structure born from the commutator of the binary V and ternary U compilers (v410/v95). [E] $J{=}\left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ is integer, $\operatorname{Spec} J{=}\{i,-i,0\}$, $J^2{=}-I$ on the rank-2 image, $J^3{+}J{=}0$ (a μ_4 quarter-turn); [E] $3I{+}2J$ has spectrum $\{3{+}2i,3{-}2i,3\}$ and $5I{+}4J$ has $\{5{+}4i,5{-}4i,5\}$, so $3{+}2i$ ($\lvert\cdot\rvert^2{=}13{=}\Delta_Q$) and $5{+}4i$ ($\lvert\cdot\rvert^2{=}41{=}10b_1$) are literal operator eigenvalues; [E] $(3I{+}2J)(3I{-}2J)$ has $\operatorname{Spec}\{13,13,9\}$ and $(5I{+}4J)(5I{-}4J)$ has $\{41,41,25\}$ (kernel slot = real part^2: $9{=}N_{\mathrm{fam}}^2$, $25{=}g_{\mathrm{car}}^2$); [E] the intrinsic deck $D{=}{-}I{+}J{-}J^2$ is order 4 ($\operatorname{Spec}\{i,-1,-i\}$, $\chi{=}(x{+}1)(x^2{+}1)$) with χ_2 line $\ker[U,V]{=}(0,0,1){=}a{-}\mathbf{1}$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue [E][C]
`v416_atom_trichotomy.py`	The atom trichotomy: $\{2,3,5\}$ are ramified/inert/split in the two exceptional CM rings, and each atom is the unique ramified prime of one quadratic facet of $K{=}\mathbb{Q}(i,\sqrt{-3},\sqrt{5})$ (v403). [E] $\mathbb{Z}[i]$ (square/seam, $\tau{=}i$): $2{=}\lvert Z_2\rvert$ ramifies ($2{=}{-}i(1{+}i)^2$), $3{=}N_{\mathrm{fam}}$ inert (norm $9{=}N_{\mathrm{fam}}^2$), $5{=}g_{\mathrm{car}}$ splits ($(2{+}i)(2{-}i)$) – the carrier rank factors on the seam; [E] $\mathbb{Z}[\omega]$ (hex/flavor, $\tau{=}\omega$): $3{=}N_{\mathrm{fam}}$ ramifies ($N(1{-}\omega){=}3$), 2 inert (norm 4), 5 inert (norm $25{=}g_{\mathrm{car}}^2$); [E] the ramified prime of each ring IS its atom ($\mathbb{Z}[i]$ at 2= sheet/seam, $\mathbb{Z}[\omega]$ at 3= family/flavor); [E] the three facets $\mathbb{Q}(i)/\mathbb{Q}(\sqrt{-3})/\mathbb{Q}(\sqrt{5})$ (disc $-4,-3,5$) ramify over one atom each, product $2{\cdot}3{\cdot}5{=}30{=}h(E_8)$; 2 imaginary (CM/dynamic) + 1 real (RM/static). [C] the seam/flavor/golden reading [E][C]
`v417_eisenstein_cp_operator.py`	The Eisenstein/CP operator (the hex dual of v415): the family rotation $P{=}(1\,2\,3)$ and the CP clock $W{=}-P^2$ put the $\tau{=}\omega$ flavor side on operators, unified with the seam into ONE order-30 clock. [E] $P^3{=}I$, $P^2{+}P{+}I{=}\mathrm{ONES}$ (Eisenstein $\omega^2{+}\omega{+}1{=}0$ on the plane $\perp (1,1,1)$); [E] $(3I{+}2P)(3I{+}2P^2){=}7I{+}6\,\mathrm{ONES}$, $\operatorname{Spec}\{7,7,25\}$ ($7{=}N_{\mathbb{Z}[\omega]}(3{+}2\omega){=}$ scalaron on the ω-plane, $25{=}g_{\mathrm{car}}^2$) — the dual of v415's $\{13,13,9\}$; [E] the CP clock $W{=}-P^2$ ($=\rho$, v233) has $W^2{=}P$, $W^3{=}-I$, $\operatorname{Spec}\{-1,\zeta_6,\zeta_6^{-1}\}$, so $\delta_{\mathrm{CKM,lead}}{=}\arg\zeta_6{=}\pi/3$ and $\delta_{\mathrm{PMNS}}{=}\arg\zeta_6^4{=}4\pi/3$ ($W^4{=}P^2$); [E] all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15}{=}{-}1$, $\zeta_{30}^{10}{=}\omega$, $\zeta_{30}^{6}{=}\zeta_5$, $\zeta_{30}^{5}{=}\zeta_6$) while the seam $\mu_4{=}i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30){=}8{=}\operatorname{rank} E_8$. [C] the carrier μ_5 has no clean $3{\times}3$ operator (golden φ an output-side trace, v313/v349); honest negatives included [E][C]
`v418_cyclotomic_norm_triple.py`	The cyclotomic norm triple: the carrier split $(3,2)$ read in the three atom-rings gives $(7,13,55){=}$(scalaron, Δ_Q, quark numerator), and the missing carrier-5 clock of v417 is the $4{\times}4$ Φ_5-companion C_5. [E] C_5 has $\chi{=}x^4{+}x^3{+}x^2{+}x{+}1$, $C_5^5{=}I$, $\operatorname{tr}{=}{-}1$; the golden ratio is its real-part data ($\zeta_5{+}\zeta_5^4{=}1/\varphi$, $\zeta_5^2{+}\zeta_5^3{=}{-}\varphi$; output-side v313/v349; dim 4, not 3 — refines v417's gap); [E] $N_{\mathbb{Z}[\zeta_5]}(3{+}2\zeta_5){=}\det(3I{+}2C_5){=}55{=}5{\cdot}11{=}g_{\mathrm{car}}\lVert\operatorname{Pl}(K)\rVert_1{=}$ the quark numerator (v410/v411); [E] the triple $\det(3I{+}2\operatorname{Comp}(\Phi_n)){=}(7,13,55)$ for $n{=}3,4,5$ over $\mathbb{Z}[\omega],\mathbb{Z}[i],\mathbb{Z}[\zeta_5]$ (atoms $3,2,5$, v416); [E] NEG control: the anchor $(5,4){\to}(21,41,461)$, with $21{=}N_\omega(5,4)$ and $41{=}10b_1$ named (v222) but the carrier ring 461 prime/un-named, so the carrier ring separates $(3,2){\to}55$ from $(5,4){\to}461$. [C] HONEST: $(3,2)$ is FORCED (v14) not unique — the clean splits are $\{(2,1){\to}(3,5,11),(3,2){\to}(7,13,55)\}$, a small ladder; the sharper reading: only $(3,2)$ spans three physics sectors (gravity/flavor/masses), the unique cross-sector seed [E][C]

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Script	What it machine-checks *(continued)*
`v419_seam_galois_carrier.py`	The seam μ_4 *is* the carrier's Galois group: a positive resolution of the v409/`RES.COXETER.SYMMETRY.01` cyclic/Galois asymmetry. [E] $h(E_8)=30=2{\cdot}3{\cdot}5$ is squarefree, so $\mathbb{Z}/30$ has *no* order-4 element — the seam μ_4 cannot be a cyclic hand and is forced onto the Galois side; [E] $(\mathbb{Z}/30)^\times=(\mathbb{Z}/2)^\times\times(\mathbb{Z}/3)^\times\times(\mathbb{Z}/5)^\times=\mu_4\times\mathbb{Z}_2$ (order $\varphi(30)=8=$ rank E_8), where the μ_4 $(\mathbb{Z}/4)$ factor IS $(\mathbb{Z}/5)^\times$ (carrier 5) and the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family/CP); [E] so the seam $\mu_4=\mathrm{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times$, the Frobenius $\zeta_5\mapsto\zeta_5^2$, realised on the carrier clock C_5 (v418) by the explicit order-4 operator G with $GC_5G^{-1}=C_5^2$, $G^4=I$, $G^2C_5G^{-2}=C_5^{-1}$. [C] resolves v409 (the $2^2=4$ is the order of $(\mathbb{Z}/5)^\times$; "no contraction rate" because the seam is an automorphism, not a hand); $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic $h=30$, Galois rank $=8$). Figure `coxeter_galois` [E][C]
`v420_fpole_qed_running.py`	F_pole external: the Koide source→pole transfer is a *one-loop electromagnetic* effect, not an algebraic step — the honest external-physics follow-up to v93 and the clock No-Go. [E] Koide Q is rescaling-invariant ($Q(c\,m)=Q(m)$), so flavor-*universal* running leaves it fixed and the source→pole shift is purely flavor-*split*; [C] the tree→pole gap $\frac{2}{3}-Q_{\mathrm{src}}=0.00220$ is one-loop-EM-sized ($=\varphi_0^{\mathrm{ret}}/24$ to 0.6%, $=\alpha/\pi$ to $\sim$5%), the PDG pole sits AT $\frac{2}{3}$ (the IR attractor); [C] a 1-loop QED per-lepton-log pole$\leftrightarrow\overline{\mathrm{MS}}$ estimate shifts Koide by EM-loop order toward $\frac{2}{3}$ (the v82 IR-attractor direction); [O] the EXACT flow time $t{\sim}2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] F_pole stays [C], firewalled (v187): the lens forces the RATE $(2/3)^6$, this external EM supplies the flow. Numerical (mpmath), Python-only [E][C][O]
`v422_galois_net_bridge.py`	The Galois↔Net bridge: the seam $\mu_4=\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — linking the Galois gearbox of v419 to G_{net}/`SEAM.EQUIV.01`. [E] $\mathrm{disc}(A_3)=\mathrm{disc}(D_5)=\mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, det$=$4); D_n disc is $\mathbb{Z}/4$ for n ODD, $\mathbb{Z}_2\times\mathbb{Z}_2$ for n EVEN, so the carrier D_5 (rank $5=g_{\mathrm{car}}$, odd) is cyclic while the control D_8 (rank 8, even) is Klein; [E] $\mathrm{Gal}(\mathbb{Q}(\zeta_5))=(\mathbb{Z}/5)^\times=\langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathrm{disc}(D_5\oplus A_3)=\mathbb{Z}_4\times\mathbb{Z}_4$ ($16=\dim S^+$) the simple current $(1,1)$ is order 4 (cyclic) with Lagrangian quotient $16/4^2=1\Rightarrow(E_8)_1$ (holomorphic, v89/v154). [E] NEG control: the Klein $\mathbb{Z}_2\times\mathbb{Z}_2$ / order-2 halfway $\langle(2,2)\rangle$ gives $16/2^2=4=\lvert\mathrm{disc}(D_8)\rvert=SO(16)$ (det$=$4), NOT E_8 (det$=$1) — cyclic $\mathbb{Z}/4\to E_8$, Klein$\to D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue (v1/v89/v154); cyclicity forced by 5, Klein excluded. [C] this is the four-fold μ_4 identity (deck/divisor = Galois = discriminant = simple-current); per v428 a *compression* (one object read four ways), not four independent witnesses [E][C]
`v423_no_ambient_dependency.py`	The *No-Ambient-Dependency* certificate: no frozen TFPT readout is *computed* from the ambient QG measure `QG.AMB.01` — sharpening the v369/v384 prose redundancy into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\mathrm{eff}}=6\ln(3/2)-31/(4\pi^2)\approx1.648>0$ ($31=2^{g_{\mathrm{car}}}-1$), susceptibility $\chi=729/665$ finite (neg control gap$\to0\Rightarrow\chi\to\infty$). [E] `QG.AMB.01` is a certification *sink*: every claim listing it as a dependency is a gravity-gate/QG row, none a frozen prediction. [E] it is not a value source — it has no producing script while each frozen readout is computed by a compiler script ($\alpha^{-1}\leftarrow$v3, $\theta_{12}\leftarrow$v9, $\theta_{13}\leftarrow$v268, cosmology$\leftarrow$v7). [E] the one transitive link is architectural: only θ_{13} references `QG.AMB.01`, via the 4D-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge not a numerical input; α^{-1}/θ_{12}/cosmology have no route. [C] so `QG.AMB.01` is gap-decoupled certification (v369 as a DAG/script fact); honestly this does NOT prove `SEAM.EQUIV.01` — the readouts depend on the seam being $(E_8)_1$; `QG.AMB.01` specifically is dead [E][C]

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Script	What it machine-checks *(continued)*
`v424_seam_lto_rp_reduction.py`	The *(LTO-RP) reduction* of the keystone: an honest MMST-style reduction of `SEAM.EQUIV.01`'s one open lemma — the raw collar's intrinsic Bisognano–Wichmann property (the v308/v309 residual) — to the cited commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`), with the central mapping *corrected*. **[E]** the μ_4 clock is a modular *symmetry* ($[\rho, K]$=0, v309/v199), not the flow — σ^{ψ}=ρ is type-wrong (a one-parameter group versus a finite order-4 element). **[E]** NPW26's axiom (LTO-RP) means u_Θ=J (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta = -K$, *distinct* from $\omega\circ\rho$=ω: a positive character-block-diagonal K satisfies $[\rho, K]$=0 yet never $\theta K\theta = -K$, so clock-invariance does *not* imply (LTO-RP). **[O]** make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $\lvert\det K\rvert$=4 via a trace; Levin–Wen anyonic), while the seam is invertible ($\lvert\det \mathrm{Cartan}(E_8)\rvert$=1, no anyons) and a β=1 KMS state (not a trace) — in neither proved bucket (neg control D_8 $\lvert\det\rvert$=4). **[C]** so `SEAM.EQUIV.01`'s BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, not a closure; `SEAM.EQUIV.01` stays **[O]** **[E][C]**
`v425_dyn_transfer_universal.py`	`DYN.TRANSFER.UNIVERSAL.01` — the *single-flow reduction* of F_{transfer}: the four frontier transfers (F_{pole}, $F_{\mathrm{Boltzmann}}$, F_{relic}, F_{QCD}) do not each need a separate external solver; their *dynamics* is the ONE native flow of the theory — the seam modular/recovery semigroup (v238/v240, gap $6\ln(3/2)$) restricted to each sector. **[E]** recovery spectrum $\{1, (2/3)^6, (1/3)^6\}$, H_{OS}=$-\log T$ gap $6\ln(3/2)$, $e^{-\Delta}$=$(2/3)^6$ exact; an explicit column-stochastic T makes $\log T$ a Markov generator. **[E]** F_{pole} is this flow: the v99 generator $\dot q$=$(\Delta/N_{\mathrm{fam}})(q-2)(q-5)$ linearised at q=2 has rate $-\Delta$, time-1 contraction $(2/3)^6$. **[E]** $F_{\mathrm{Boltzmann}}$ washout = that contraction, CP phase $4\pi/3$ native (v320). **[E]** F_{QCD} rate native (b_3=-7). **[C]** only the anchors (v_{geo}, C_p, M_R) external; F_{relic}'s cosmological IC θ_i the one wall. Extends v383 static→dynamic; a reduction, no new premise **[E][C]**
`v426_seam_lto_rp_kms.py`	`SEAM.EQUIV.LTORP.KMS.01` — the *invertible β=1-KMS* variant of (LTO-RP), the corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + β=1 KMS, not a trace). A constructive Tomita–Takesaki witness advancing v424 sub-step (ii); does *not* close `SEAM.EQUIV.01`. **[E]** build K reflection-odd/μ_4-even/gapped (gap $2k$=$6\ln(3/2)$); C=$1/(1+e^K)$ the β=1 KMS covariance, $C/(1-C)$=e^{-K} (genuine KMS, not trace). **[E]** $\Theta K\Theta^{-1}$=$-K$ (u_Θ=J) AND $[\rho, K]$=0 both hold ⇒ (LTO-RP) satisfied by direct Tomita–Takesaki. **[E]** $\lvert\det \mathrm{Cartan}(E_8)\rvert$=1 (invertible, no anyons) vs anyonic $\lvert\det\rvert$=4; state not a trace (Δ=$e^{-K}$$\neq$$I$) — neither NPW26 bucket. **[E]** neg controls: side-even reflection → $+K$ (BW fails), off-μ_4-block K breaks $[\rho, K]$. **[C]** (LTO-RP) reduces to $(\Theta K\Theta$=$-K) \wedge ([\rho, K]$=0), constructively satisfiable; `SEAM.EQUIV.01` stays **[O]** **[E][C]**
`v427_overdet_witness_map.py`	`OVERDET.WITNESS.MAP.01` — the multiply-vs-compress *framework* for over-determination: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. The classification of the seven arithmetic witnesses is refined by v428 (they compress). **[E]** seven grammar readouts, each from its own syntax landing on the carrier skeleton: Gauss $N(3{+}2i)$=13, Eisenstein $N(3{+}2\omega)$=7, cyclotomy $N(3{+}2\zeta_5)$=55, Galois $\lvert(\mathbb{Z}/5)^\times\rvert$=4, $\lvert\det \mathrm{Cartan}(E_8)\rvert$=1, Pascal $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$=11 ($2^4$=16=$\dim S^+$), Coxeter $h(E_8)$=30=2·3·5 with $\varphi(30)$=8=$\mathrm{rank}\, E_8$. **[E]** by v236 these are facets of ONE $(2,3,5)/E_8$ object, so syntactic distinctness is NOT independence — they compress, like the anchor a=$(1,1,2)$. **[C]** the genuine multiplicative strength is the multiply-forced inputs + foreign witnesses (v428); the map does not by itself establish physics **[E][C]**

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Script	What it machine-checks *(continued)*
`v428_overdet_reclass.py`	`OVERDET.WITNESS.RECLASS.01` — the honest correction of `v427`: applying its own multiply-vs-compress test to itself. **[E]** the $(2,3,5)$ Brieskorn singularity (v236) is the ONE generator — Milnor $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank}E_8$, $30{=}2{\cdot}3{\cdot}5{=}h(E_8)$, $\varphi(30){=}8$. **[E]** every one of `v427`'s seven witnesses maps to a facet of that SAME object (primes $2,3,5$ of 30, the clock 30, or E_8), so all seven collapse to ONE generator (distinct objects = 1) ⇒ COMPRESSION, not seven multiplications. **[E]** what honestly multiplies is the input forced four independent ways — the "8" in $c_3{=}1/(8\pi)$ from rank E_8, $h(D_5){=}2(5{-}1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}{\approx}137$ (prime, not in the $(2,3,5)$-skeleton). **[C]** refines and partly walks back `v427`; consistent with v236/v305/v313 **[E][C]**
`v429_axion_pentagon_phi.py`	`DM.AXION.PENTAGON.01` — the 'unmapped' icosahedral/golden E_8 structure surfaces in the ONE external cosmological input θ_i, answering "why does idle E_8 structure exist?" for the one case where it has a job. **[E]** because $N_{\text{fam}}{=}g_{\text{car}}{-}2$, the spine quotient $\theta_i{=}\pi N_{\text{fam}}/g_{\text{car}}{=}(g_{\text{car}}{-}2)\pi/g_{\text{car}}$ is EXACTLY the interior angle of the regular g_{car}-gon; for $g_{\text{car}}{=}5$ the PENTAGON, $\theta_i{=}3\pi/5{=}108°$ (the carrier polygon, not a bare rational multiple of π). **[E]** its cosine is golden: $\cos(3\pi/5){=}(1{-}\sqrt{5})/4{=}{-}1/(2\varphi)$, partner $2\cos(2\pi/5){=}1/\varphi{=}\varphi{-}1$, $\varphi{=}2\cos(\pi/5){=}(1{+}\sqrt{5})/2$. **[E]** the golden character is UNIQUE to $g_{\text{car}}{=}5$: among small n-gons only $n{=}5$ has an irrational interior-angle cosine ($n{=}3,4,6$ give $1/2, 0, -1/2$), so θ_i is golden BECAUSE the carrier is 5 (P2). **[C]** a re-reading bridging v354/v313 (the unmapped $g_{\text{car}}{=}5$ golden signature) to v211 ($\theta_i{=}3\pi/5$, the one external input): it REFINES v354 from "φ in no readout" to "φ touches only the **[C]** spine misalignment angle", a geometric motive for the spine over the hilltop $\sim$170°; it does NOT upgrade `DM.AXION.SPINE.01` (stays **[C]**) and closes no gate. **[E][C]**
`v430_other_side_reverse_audit.py`	`E8.OTHERSIDE.AUDIT.01`: the 'other side of the seam' (the double-cover deck / conjugate sheet S^-) audited against E_8's UNMAPPED Casimir degrees — the sheet/deck complement of v354/v355, answering "is the leftover where the second sheet lives?" with a disciplined NEGATIVE. **[E]** DECK = DEGREE-2: the one-sided cover contributes $\lvert\mathbb{Z}_2\rvert{=}2$ (the 1/2 in $c_3{=}1/(2{\cdot}4\pi){=}1/(8\pi)$, v58/v216); $2{=}\min(E_8$ degrees$){=}$ the metric, one of the matched primary readouts $\{2,8,30\}$ (v354), NOT one of the unmapped $\{12,14,18,20,24\}$. **[E]** TWO SHEETS = 128-SPINOR: $\dim S^+{=}\dim S^-{=}16$, the spinor block $128{=}\operatorname{rank}\cdot\dim S^+{=}8{\cdot}16{=}2^{\operatorname{rank}-1}{=}\sum\deg$, the spinor half of $248{=}120{+}128$; S^- the conjugate $(\overline{16},\overline{4})$, $128{=}2{\cdot}64$, not a spare singlet. **[E]** FORCED-DISJOINT: the sheet/deck integer set $\{2,16,32,128\}$ has EMPTY intersection with $\{12,14,18,20,24\}$, its only degree-alphabet contact the matched 2. **[O]** DISCIPLINED DECLINE: each unmapped degree needs an UNFORCED coefficient from the sheet atoms and admits ≥2 readings (the v355 discriminator) — no forced sheet→unmapped-degree identity. **[E]** '$S^-{=}$DM' DOWNGRADED + WIMP NO-GO: v227 replaced the over-strong 'S^- is dark matter'; $128{=}(16,4){+}(\overline{16},\overline{4})$ is the SAME matter spinor (no spare singlet), DM is the determinant-line axion (a phase). **[O]** the only live 'other side' question (the 128 phase/glue dark sector, v227) concerns MATCHED structure, not the unmapped degrees **[E][O]**

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Script	What it machine-checks *(continued)*
`v431_e8_degree_ladder.py`	`E8.DEGREE.LADDER.01`: the reverse-audit 'unmapped' E_8 Casimir degrees are NOT diffuse overhead but the forced two-family decomposition $\deg(E_8){=}6{\cdot}\mathrm{spine}\{2,3,4,5\} \sqcup (\{2\} \cup \text{det-ladder}\{8,14,20\})$ — the structural complement of v354/v355/v430. [E] TWO-FAMILY SPLIT FORCED BY $30{=}2{\cdot}3{\cdot}5$: the E_8 exponents are the $\varphi(30){=}8$ totatives of $h{=}30{=}2N_{\mathrm{fam}}g_{\mathrm{car}}$ (coprime to $6 \Rightarrow \pm1 \bmod 6$), so the degrees occupy ONLY the residue classes $\{0,2\} \bmod 6$. [E] $6k$ family $\{12,18,24,30\}/6{=}\{2,3,4,5\}{=}$ the v91 spine $\{e_3,e_1,e_2,p_0\}(a{=}(1,1,2))$. [E] $6k{+}2$ family $\{2,8,14,20\}$: $\{8,14,20\}{=}(\det R,\det C,\det L){=}$ the winding line $\det M(s,0){=}6s{+}8$ of the flavor diamond (v135), $C{=}R{+}Q\,\mathrm{diag}(1,0,0)$. [E] '$18{=}6N_{\mathrm{fam}}$' is NOT a holdout (spine family, not on the det-ladder). [E] SPECIAL TO E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail the clean $\{0,2\} \bmod 6$ split). [C] HONEST v355 LINE: the arithmetic decomposition is exact, but the FUNCTORIAL claim '$6k{+}2$ Casimir stratum = flavor sheet operators' needs a representation-theoretic map ([C]); $12,24$ admit two canonical readings each, so this is a STRUCTURE theorem, not a forced functorial flavor map — it reclassifies v354 overhead to zero orphan degrees, closes no gate [E][C]
`v432_overdet_floor.py`	`OVERDET.FLOOR.01`: the UNCONDITIONAL improbability floor + the L1↔L2 bridge — the defensible number that survives even if a skeptic REJECTS the declared formula grammar, complementing the conditional null model (v100) and the multiply-vs-compress framework (v427/v428). [E] DISJOINT PIECES (v428): compression removed (v427's seven readouts are one $(2,3,5)/E_8$ object, NOT multiplied); what multiplies is the input '8' forced FOUR ways $\{\mathrm{rank}\,E_8,\ h(D_5){=}2(g_{\mathrm{car}}{-}1),\ \varphi(30),\ \mathrm{Milnor}(2,3,5)\}$ plus the FOREIGN witness $\alpha^{-1}{\sim}137$ (prime, outside $(2,3,5)$). [C] CONSERVATIVE FLOOR: multiplying ONLY across disjoint pieces (input over-determination × the α grammar census 1/94500 (v100) × the φ_0^{ret}-seed cluster counted ONCE × one downstream witness) the floor stays $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim10^{-10}$ — 'extraordinary' WITHOUT the declared grammar. [E] THE BRIDGE: the floor ($\sim10^{-10}$) sits $\sim$20 orders ABOVE the conditional $10^{-30.7}$ (v100); that gap = the evidential value of proving the FORMS forced, so L1 (`ALPHA.QUILLEN.EXACT.01`, v382) and L2 (this floor) are the SAME lever. [C] HONEST SCOPE: bounds ONLY "is it numerology?", NOT the physics bridge (firewall v187); a conservative FLOOR, not a posterior [E][C]
`v433_alpha_quillen_heatkernel.py`	ALPHA.QUILLEN.PROGRESS.02: a *second honest step* on the external target ALPHA.QUILLEN.EXACT.01 (v382) – it GROUNDS and CONNECTS, it does NOT close it; ALPHA.QUILLEN.EXACT.01 stays [O] (external math, type A, v384), α^{-1} stays [E]. [E] a solvable 4D model (flat $\Delta = -\partial^2{+}m^2$ on T^4) reaches the a_4 (t^0) conformal-anomaly order: $a_4 = Vm^4/(2(4\pi)^2)$ NON-ZERO – the order v391's 1D circle toy (a_0, a_1 only) could not reach. [E] it is the order v342 grounds (Gilkey a_4 gauge-curvature $30/360 = 1/12$), unifying v391 (a ζ-det variation IS heat-kernel data) with v342 (the term IS the Gilkey a_4). [E] $(4\pi)^{-2} = 4c_3^2$ is c_3-built. [E] the v382 split re-verified at $\alpha^{-1} = 137.0359992$. [O] NOT CLOSED: the actual seam a_4 ($= 8b_1c_3^6$) and the cubic α^3 Chern moment stay open (v384); the exact seam F-normalisation stays [C] (v342). Grounds/connects, does not close [E][O]

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Script	**What it machine-checks** *(continued)*
`v434_alpha_quillen_betafunction.py`	ALPHA.QUILLEN.PROGRESS.03: a *third honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it settles the three residuals named after v433 and shows they are NOT three independent problems; the target stays [O], α^{-1} stays [E]. [E] b_1 IS THE U(1) a_4 HEAT-KERNEL COEFFICIENT (the β=a_4 theorem): from the carrier hypercharge content with spin weights ($\frac{2}{3}$ Weyl, $\frac{1}{3}$ complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM), $(3/5)\cdot$ that $= 41/10 = b_1$ (GUT) – the b_1 in $8b_1c_3^6$ is the a_4 content coefficient, not a fitted multiplicity (same number as the β, v159/v246; same KIND as the α^2 Calderón a_4, v342). [E] $8b_1c_3^6 = b_1 \cdot (\mathrm{rank}\, E_8) \cdot c_3^6$ (rank 8 = χ=2 boundary, v216; c_3^6 six insertions, v342). [C] residual (1) "exact $a_4 = 8b_1c_3^6$" ≡ residual (3) "seam F-normalisation" – ONE obligation. [E] residual (2) located: $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$, so a^3 = the first moment of the Chern density (a level, order c_3^0); origin stays [O]. [E] net reduction: the EM-Ward residual is 1[C]+1[O], not three. [O] NOT CLOSED; α^{-1} stays [E]. Python-only [E][C][O]
`v435_alpha_quillen_chernlevel.py`	ALPHA.QUILLEN.PROGRESS.04: a *fourth honest step* on ALPHA.QUILLEN.EXACT.01 (v382) – it attacks the SINGLE remaining [O] after v434 (residual (2), the cubic α^3 Chern level) and SHARPENS it, it does NOT close it; the target stays [O], α^{-1} stays [E]. [E] π-POWER / METRIC-INDEPENDENCE TEST: the three closure coefficients carry π-powers exactly $\{0,3,6\} = c_3$-orders ($k_0 = 1 \to \pi^0$; $-2c_3^3 = -1/(256\pi^3) \to \pi^3$; $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$), so the Maxwell a^3 is the UNIQUE π^0/c_3^0 (metric-INDEPENDENT) term – the signature of a topological level (a heat-kernel boundary coeff always carries the $(4\pi)^{-d/2}$ measure; a level carries none). The test partitions the closure into ONE topological level (a^3) + TWO heat-kernel boundary terms, confirming v342/v434's "a^3 is a level, not a curvature integral" from a test. [C] conditional on the CS/η reading (EM1, v48), large-gauge invariance forces the level in $\mathbb{Z}$ and $k_0 = 1$ is the unit (minimal) level – an integer, not a free real. [O] NOT CLOSED: the EM1 CS-boundary derivation stays external (type A, v384); seam F-norm stays the one [C] (v434); α^{-1} stays [E]. [E][C][O]
`v436_overdet_floor2.py`	`OVERDET.FLOOR.02`: HARDENS `OVERDET.FLOOR.01` (v432) — the "is it numerology?" floor does NOT rest on v432's subjective chance assignments. [E] ASSUMPTION-MINIMAL FLOOR: the α cubic-class census (v100 layer D) is PURE COUNTING — of N=94500 complexity-matched $F_{U(1)}$ variants only ONE hits the 4×10^{-8} CODATA window (the TFPT instance, α^{-1}=137.0359992), so $p_\alpha \leq 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with NO subjective probability, prior or multi-observable grammar. [E] MONOTONE CONCESSION LADDER: $-\log_{10} p$ full scorecard+$\alpha \sim 30.7 \to$ scorecard-only $\sim 25.8 \to \alpha$-census-only ~ 4.98 ($= 4.40\sigma$); the "not chance" verdict holds at the MOST adversarial rung. [E] INDEPENDENCE: strip ALL of v432's subjective factors (0.1^3×0.012×0.15=1.8×10^{-6}) and the floor is still 4.40σ — they only sharpen it. [C] HONEST 5σ GAP: 5σ (p=5.7×10^{-7}) is crossed by the census times exactly ONE more independent factor ≤ 0.054 (the input-'8' four-fold over-determination or one foreign witness), stated openly. [C] FIREWALL: bounds numerology only, not the physics bridge (v187); Python-only, deliberately NOT Wolfram-mirrored (a counting bound, not an identity). [E][C]

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Script	What it machine-checks *(continued)*
`v437_e8_degree_joint.py`	`E8.DEGREE.JOINT.01`: a CONSOLIDATION of v6/v66/v355/v431 deriving more from the E_8 Casimir degrees WITHOUT crossing the v354/v355 anti-numerology line — no new per-degree coincidence, only a joint lock. **[E]** E_8 IS FIXED BY ITS DEGREE MULTISET: from $\deg(E_8)=\{2,8,12,14,18,20,24,30\}$ alone — rank $=\#\deg=8$, $h=\max=30$, #pos roots$=\sum\exp=120$, #roots=240, dim $=248$, $\lvert W\rvert=\prod\deg=696{,}729{,}600$, $\sum\deg=128=2^7=\dim S^+$. **[E]** THE JOINT QUADRATIC: the two structural integers $(g_{\mathrm{car}}, N_{\mathrm{fam}})$ are the UNIQUE root pair of $x^2-(\operatorname{rank}E_8)x+(h/2)=x^2-8x+15=(x-3)(x-5)$ — sum$=\operatorname{rank}E_8=8$ (rank-fill, v6), product$=h/2=15=g_{\mathrm{car}}N_{\mathrm{fam}}$ $(2g_{\mathrm{car}}N_{\mathrm{fam}}=h$, v431); $\{3,5\}$ the unique factor pair of 15 summing to 8, so BOTH are forced together by two degree-invariants. **[E]** the three mapped degrees $\{2,8,30\}=\min/\mathrm{rank}/\max$ (the canonical invariants); $g_{\mathrm{car}}=5=$ max prime of 30. **[E]** DISCIPLINE: no new per-degree mining; the five non-primary degrees keep their v431 homes. Exact integer/lattice, mirrored in wolfram. **[E][C]**
`v438_seam_hsmi_borchers.py`	`SEAM.EQUIV.BW.HSMI.01`: the half-sided modular inclusion (Borchers/Wiesbrock) as an INTRINSIC Bisognano–Wichmann route for face (ii) of `SEAM.EQUIV.01` — geometric modular flow WITHOUT presupposing covariance. **[E]** STANDARD-PAIR ALGEBRA (ladder, exact): boost $K=\mathrm{diag}(n)$ and translation P (raising shift) satisfy $[K,P]=P$ and the Borchers scaling $\Delta^{it}P\Delta^{-it}=e^{-t}P$ (the ax+b/Moebius relation). **[E]** BW REFLECTION: $J(n\to-n)$ gives $J^2=I$, $JKJ=-K$ $(u_\Theta=J)$ and $JPJ=P_-$ (the opposite-chirality translation), the standard-pair reflection $JU(s)J=U(-s)$. **[E]** POSITIVE ENERGY: in the unitary lowest-weight $sl(2,\mathbb{R})$ rep the line translation $P_{\mathrm{line}}=L_0-(L_++L_-)/2$ is POSITIVE (min eig >0), the positive-energy generator the c_3 one-sided origin demands; the boost direction is the indefinite negative control. **[C]** SYNTHESIS: $P\geq0$ + $[K,P]=P$ + J is a Longo STANDARD PAIR / +half-sided modular inclusion (Wiesbrock), so by Borchers the seam modular flow is geometric INTRINSICALLY (complementary to Adamo–Giorgetti–Tanimoto, `arXiv:2508.07109`); face (ii) becomes a STRUCTURE consequence, not a state condition. **[C][O]** assembling the finite algebra and the positive-energy rep into ONE continuum standard pair is the continuum existence (the same wall as MMST/NPW26, v336/v424); `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy/scipy). **[E][C][O]**
`v439_seam_lieb_robinson.py`	`SEAM.EQUIV.CONTINUUM.LR.01`: the Lieb–Robinson/Nachtergaele–Sims continuum-existence MOTOR for the gapped seam collar — grounds the BULK half of the cited MMST existence (v336) in the explicit v367 $p+ip$ model. **[E]** UNIFORM GAP: $\Delta(N)$=const>0 across N=20..160 (the lattice image of the OS transfer gap $6\log\frac{3}{2}>0$, v302/v329). **[E]** EXPONENTIAL CLUSTERING: $\lvert\langle c_i^\dagger c_j\rangle\rvert\sim e^{-r/\xi}$ with ξ finite and STABLE across N (gapped $\Rightarrow$ Hastings–Koma decay), the N-independent locality. **[E]** LIEB–ROBINSON LIGHT CONE: the single-particle propagator $\lvert(e^{-iHt})_{ij}\rvert$ is confined to $\lvert i-j\rvert\lesssim v_{\mathrm{LR}}t$ with finite v_{LR} $(\sim 2t)$, exponentially small beyond the cone. **[C]** NACHTERGAELE–SIMS LIFT: uniform gap + finite LR velocity $\Rightarrow$ exponential clustering with N-independent ξ AND existence+uniqueness of the thermodynamic-limit ground state with a finite light cone — derived for OUR collar, not imported. **[C][O]** the bulk thermodynamic limit EXISTS by Lieb–Robinson; `SEAM.EQUIV.01` stays **[O]**, the open analytic step narrowed to the CHIRAL massless EDGE scaling limit (the c=8 $(E_8)_1$ CFT), distinct from the bulk limit. Python-only (numpy/scipy). **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v440_seam_lto_rp_beta.py`	`SEAM.EQUIV.LTORP.BETA.01`: the β-interpolation connecting the NPW26-PROVEN trace case (β=0) to the seam's β=1 KMS case along an obstruction-free homotopy — upgrading the single-point v426 witness for v424 sub-step (ii) to a PATH argument. [E] β-INDEPENDENT (LTO-RP) DATA: $\Theta K\Theta=-K$ ($u_\Theta=J$) and $[\rho,K]=0$ are conditions on K alone, β-independent. [E] THE KMS PATH: $C_\beta=1/(1+e^{\beta K})$ has $\beta=0\to C=\frac{1}{2}I$ (the NPW26 TRACE state), $\beta=1\to$ the seam KMS state, $\beta\to\infty\to$ the ground-state projection; at every β $\Theta C_\beta\Theta=1-C$ (reflection), $[\rho,C_\beta]=0$ (mu4), $0\le C_\beta\le 1$ (valid covariance/RP). [E] NO OBSTRUCTION: $\beta\mapsto C_\beta$ is norm-continuous and stays in the (LTO-RP)-valid set — a continuous homotopy from the proven trace endpoint to the seam KMS point, no topological wall. [E] NEG CONTROLS: a reflection-EVEN K' breaks $\Theta C_\beta\Theta=1-C$ for $\beta>0$ (the trace endpoint is degenerate); an off-mu4-block K breaks $[\rho,C_\beta]$. [C][O] a PATH argument for v424 sub-step (ii), stronger than the single point of v426; `SEAM.EQUIV.01` stays [O] (closedness of (LTO-RP) along the path in the continuum is the residual). Python-only (numpy/scipy). [E][C][O]
`v441_seam_applicability_ledger.py`	`SEAM.EQUIV.APPLICABILITY.01`: the applicability ledger — a hypothesis-by-hypothesis audit of the THREE external theorems the seam keystone cites (MMST scaling limit, NPW26 (LTO-RP), Adamo–Giorgetti–Tanimoto covariance), isolating the SINGLE imported analytic fact. [E] MMST HYPOTHESES: D=16 Majoranas ($=2^{g_{\rm car}-1}$), c=8 ($=g_{\rm car}+N_{\rm fam}$), rank E_8=8, range rank $\le c\le D$ reads $8\le 8\le 16$, gapped ($6\log\frac{3}{2}>0$), quasi-free CAR — all internal computed facts. [E] NPW26: (LTO-RP)$=u_\Theta=J$, gap, $\mathbb{Z}_2$ reflection, mu4 clock, $[\rho,K]=0$ internal (v424/v426/v440); the seam is invertible $\lvert\det \mathrm{Cartan}\,E_8\rvert=1$ + β=1 KMS, OUTSIDE NPW26's topological/trace bucket ($\lvert\det\rvert=4$) — a TEMPLATE, not a direct import. [E] ADAMO–GIORGETTI–TANIMOTO (`arXiv:2508.07109`): DHR reps of nets extending loop-group/Virasoro are AUTOMATICALLY positive-energy + diff-covariant; the seam net extends $(D_5)_1\times(A_3)_1$ (v154), so covariance is a CONSEQUENCE — the v215 circularity is defused, covariance downgraded ASSUMPTION→[E]. [E] of 12 audited hypotheses, 11 are internal [E] and exactly 1 external — the continuum EXISTENCE of the chiral massless scaling limit (MMST/v336, Lean `FORM.SEAM.MMST.01`). [C][O] external dependence pinned to ONE analytic fact; `SEAM.EQUIV.01` stays [O]. Python-only (a structural/arithmetic audit; the dets Wolfram-mirrored via v89/v281/v422). [E][C][O]
`v442_seam_rigidity_uniform.py`	`SEAM.RIGIDITY.UNIFORM.01`: the uniform-in-N hardening of the Seam State Rigidity Theorem (v398) — the band-limited rigidity is NOT a small-N artifact. [E] EXACT UNIFORM 4-SECTOR COMMUTATION: $\rho=\mathrm{diag}(i^n)$ has EXACTLY 4 distinct eigenvalues $\{1,i,-1,-i\}$ for all N, and a genuinely character-block-diagonal K (random Hermitian within each n mod 4 class, not merely diagonal) has $[\rho,K]=0$ EXACTLY for N=4..256 (extends v398's $N\le 64$). [E] UNIFORM GAP: $\Delta(N)$=const>0 across N=20..160 (the Nachtergaele–Sims uniform gap, shared with v439). [E] N-INDEPENDENT RIGIDITY BOUND: a FIXED local clock-breaking perturbation (single off-mu4-block Hermitian entry) gives $\lVert[\rho,V]\rVert=2$ EXACTLY for all N — a uniform lower bound; any fixed-size clock-breaking is detected with N-independent strength. [E] NEG CONTROL: a clock-PRESERVING (same-sector) perturbation gives $\lVert[\rho,V']\rVert=0$; an order-2 clock has only 2 distinct eigenvalues — the ORDER of the clock (the four Gauss–Bonnet marks) is the discriminator, uniformly in N. [C][O] the band-limited rigidity of v398 is uniform in N; `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays [O]: that RP+gap+holomorphy FORCE Λ_Σ block-diagonal on the full L^2 (vs merely permit) is the cited rigidity/MMST step. Python-only (numpy). [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v443_seam_e8_discriminator.py`	`SEAM.E8.DISCRIMINATOR.01`: the empirical/structural DISCRIMINATOR bundle (mode D of the v347 closure-mode classification) — concrete FALSIFIABLE kill tests separating the claimed seam = $(E_8)_1$ from the nearest c=8 rival $SO(16)_1$=$(D_8)_1$. **[E][X]** PRIMARY/GSD: $\lvert\det \mathrm{Cartan}(E_8)\rvert$=1 (ONE primary, holomorphic) vs $\lvert\det \mathrm{Cartan}(D_8)\rvert$=4 (FOUR primaries $1, v, s, c$); torus GSD = #primaries = 1 vs 4 — a seam with GSD> 1 falsifies $(E_8)_1$. **[E][X]** CURRENT/THETA: level-1 current count $\dim E_8$=248 vs $\dim SO(16)$=120; the E_8 lattice theta has 240 norm-2 roots (= 112 integer +128 spinor, computed) vs D_8's 112 — current spectrum and lattice theta DIFFER. **[E]** c=8 IS NOT A DISCRIMINATOR: both have c=8 (the v277 ambiguity); only the order-4 mu4 condensation + $\det K$=1 (the genus-1 statement, v90) selects E_8. **[E]** NON-VACUITY: the discriminators all fire on the rival ($\det 4{\neq}1$, currents $120{\neq}248$, roots $112{\neq}240$). **[X][C]** mode D realised as concrete kill tests; `SEAM.EQUIV.01` stays **[O]** (the discriminators distinguish the TARGET but do not PROVE the seam realises it). Python-only (numpy; the dets Wolfram-mirrored via v89/v281/v422). **[E][C][O][X]**
`v444_seam_edge_scaling.py`	`SEAM.EQUIV.CONTINUUM.EDGE.01`: the CHIRAL EDGE scaling-limit MOTOR — the edge half of the one `SEAM.EQUIV.01` continuum residual, attacking exactly the step v439 left open. On the SAME explicit v367/v368 p+ip collar it shows the lattice edge carries the kinematic AND conformal data of a chiral c=$\frac{1}{2}$ Majorana CFT, converging under refinement. **[E]** CHIRAL LINEAR EDGE DISPERSION: the in-gap branch is linear near its zero-crossing (top v=0.983, R^2=0.9999) and the two boundaries carry OPPOSITE velocities — one chiral Majorana per edge (relativistic, chiral kinematics). **[E]** ALGEBRAIC EDGE TWO-POINT: the equal-time edge correlator decays as $r^{-\eta}$ with η=1.00 (the chiral free-fermion CFT exponent) — a CRITICAL edge. **[E]** EXPONENTIAL BULK (neg control): the bulk-row correlator decays exponentially (ξ∼1.1) — only the edge is critical, so the power law is the edge CFT, not a geometric artefact. **[E]** TRIVIAL-PHASE NEG CONTROL: M=3 (no edge, v368) has an exponential edge-row — the algebraic decay is the topological edge, not the boundary geometry. **[E]** SCALING-COLLAPSE / CONVERGENCE: η is N-stable (1.001±0.007) and the edge correlator profile is Cauchy-convergent under refinement (successive spread shrinks 0.028→0.013, N_x=36..60) — an existing scaling limit, the edge analogue of v439's bulk thermodynamic limit. **[C]** EDGE CENTRAL CHARGE: η=1 ⇔ one chiral Majorana (c=$\frac{1}{2}$) per edge, so the N_{Maj}=16-copy carrier has a chiral c_-=8 edge CFT (= $(E_8)_1$, c_-=g_{car}+N_{fam}). **[C][O]** grounds the EDGE half of the cited MMST scaling limit (v336) the way v439 grounds the bulk; with both halves the one continuum residual is grounded and pinned to a single cited theorem; `SEAM.EQUIV.01` stays **[O]** (the rigorous uniform-convergence theorem is not supplied). Python-only (numpy). **[E][C][O]**

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Script	**What it machine-checks** *(continued)*
`v445_seam_rigidity_forcing.py`	`SEAM.RIGIDITY.FORCING.01`: the rigidity FORCING theorem — upgrades v442's "block-diagonal PERMITS commuting" to "commuting with the order-4 clock FORCES block-diagonal", isolating the single remaining physical premise. **[E]** FORCING (commute ⇔ block-diagonal, EXACT): for $\rho{=}\,\mathrm{diag}(i^n)$, $[\rho, K]{=}0$ iff $K_{ij}{=}0$ whenever $i{\not\equiv}j$ mod 4 — verified ENTRYWISE, swept $N{=}4..128$ (the converse of v442). **[E]** EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]{=}0\} = \sum_s n_s^2$ (the four sector sizes), verified as the ad_ρ nullspace; a PROPER subspace of the N^2 algebra ($N{=}16$: 64<256) — a genuine quantitative restriction. **[E]** OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit E_{ij} has $\\|[\rho, E_{ij}]\\| = \|i^i - i^j\| \geq \sqrt{2} > 0$ — zero slack. **[E]** THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors ($\dim \sum n_s^2(4)$), the order-2 clock $\mathrm{diag}((-1)^n)$ a STRICTLY LARGER commutant ($N{=}16$: 64 vs 128) — the four Gauss–Bonnet marks (order $4{=}\|\mu_4\|{=}h(A_3)$) force strictly more structure than two; only the index-4 extension is $(E_8)_1$ (det 1; v281/v154/v351). **[E]** NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N{\geq}4$. **[C]** SYNTHESIS (force, not permit): commuting with the order-4 μ_4 clock is EQUIVALENT to μ_4-block-diagonality (iff, exact, uniform in N), so RP-definability (v54/v398) + clock-invariance FORCE the block-diagonal Λ_Σ — v442's "permit" upgraded to "force". **[C][O]** the single residual is now precisely "transfer in commutant" (clock-invariance = `QGEO.SYM.01`, v323/v335); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy; the det facts Wolfram-mirrored via v89/v281/v422). **[E][C][O]**
`v446_seam_clock_invariance.py`	`SEAM.RIGIDITY.CLOCKINV.01`: DERIVING clock-invariance ("transfer in the commutant") from RP-symmetry, closing v445's single residual one notch further — the residual drops from an OPERATOR commutation to the STRUCTURAL μ_4-symmetry of the seam RP data. **[E]** OS RECONSTRUCTION INHERITS THE SYMMETRY: for a μ_4-invariant RP covariance ($[\rho, C(t)]{=}0$), the OS canonical transfer $\Lambda{=}C(1)C(0)^{-1}{=}e^{-h}$ (the transfer FIXED by RP+reflection, v54) has $[\rho, \Lambda]{=}0$ — swept $N{=}4..32$ over random block-diagonal h. **[E]** SHARPNESS / NEG CONTROL: a μ_4-BREAKING covariance gives $[\rho, \Lambda] \neq 0$ — iff-sharp, so the inheritance is non-vacuous. **[E]** CONTRACTION / OS-VALIDITY: Λ has spectrum in $(0, 1]$ (a positive contraction, the physical canonical transfer). **[E]** MODULAR (Tomita–Takesaki): the free modular Hamiltonian $K{=}\log((1{-}C)C^{-1})$ (the v258 formula) of the μ_4-invariant state has $[\rho, K]{=}0$ (the clock commutes with modular flow), the breaking control $[\rho, K] \neq 0$. **[C]** SYNTHESIS: $[\rho, \Lambda_\Sigma]{=}0$ is a CONSEQUENCE of μ_4-invariance of the seam RP data (the four Gauss–Bonnet marks, v216/v323) via the OS reconstruction (v54) + the modular fact (v258), NOT an extra assumption; with v445 (commute ⇔ block-diagonal), RP-definability PLUS the marks FORCE the block-diagonal Λ_Σ. **[C][O]** the residual reduces from an operator commutation to "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`); the continuum Tomita–Takesaki implementation on the full L^2 is the cited backbone. `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy/scipy). **[E][C][O]**

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Script	What it machine-checks *(continued)*
`v447_seam_edge_chern.py`	`SEAM.EQUIV.EDGE.CHERN.01`: an INDEPENDENT (topological) reading of the edge c_-=8, corroborating v444's correlator route via the bulk–edge correspondence on the same v367/v368 $p+ip$ collar. **[E]** INTEGER CHERN NUMBER: the M=1 collar lower band has Chern C=+1 (the Fukui–Hatsugai–Suzuki lattice link-variable method, an EXACT integer, no fit) — one chiral channel per copy. **[E]** TRIVIAL-PHASE NEG CONTROL: M=3 (v368) has C=0 (no chiral edge), and M=−1 gives C=−1 (opposite chirality, matching v444's opposite per-edge velocities). **[E]** BULK–EDGE CORRESPONDENCE: the open strip carries EXACTLY $\|C\|$=1 in-gap edge branch crossing zero per boundary in the M=1 phase, 0 in the trivial phase — the chiral edge channel count = $\|C\|$. **[E]** LI–HALDANE: the half-cut bulk ground state's single-particle entanglement spectrum has in-gap modes at 1/2 in the topological phase and NONE in the trivial — the entanglement edge tracks the physical edge. **[C]** EDGE CENTRAL CHARGE: $c_-{=}N_{\rm Maj}\cdot\frac{1}{2}\cdot\|C\| = 16\cdot\frac{1}{2}\cdot 1 = 8 = g_{\rm car}{+}N_{\rm fam}$, from the Chern integer + bulk–edge correspondence — the SAME c_-=8 as v444's correlator route, from a logically independent (topological) observable. **[C][O]** corroborates v444's edge $(E_8)_1$ reading; `SEAM.EQUIV.01` stays **[O]** — the topology fixes c_- and the channel count but not the cited continuum existence theorem. Python-only (numpy). **[E][C][O]**
`v448_seam_edge_fourpoint.py`	`SEAM.EQUIV.EDGE.FOURPOINT.01`: pinning the ONE external MMST fact empirically — the edge FOUR-point function is quasi-free and conformally convergent, extending v444's two-point scaling limit to the next order. **[E]** QUASI-FREE / WICK (CAR hypothesis): the ground-state correlation matrix is an exact projector $G^2{=}G$ ($\sim 10^{-13}$), so the state is quasi-free and all n-point functions are Wick/Pfaffian sums of the two-point — the MMST quasi-free CAR hypothesis (ledger entry 5) literally holds on the collar (structural: the BdG state is free). **[E]** FOUR-POINT CAUCHY CONVERGENCE: the edge four-point cross-ratio $Q{=}[G_{01}G_{23}]/[G_{02}G_{13}]$ at COMMENSURATE positions $(1,2,4,7)\cdot(N_x/12)$ is Cauchy-convergent under refinement (successive spread shrinks, N_x=36..60) — the four-point converges, extending v444's two-point scaling-collapse. **[E]** CONFORMAL FOUR-POINT FORM: Q converges to the free-fermion CONFORMAL cross-ratio $\sin(5\pi/12)/\sin(\pi/12){=}2{+}\sqrt{3}\approx 3.732$ (chord distance $d(r){=}(N_x/\pi)\sin(\pi r/N_x)$) — the edge four-point has the chiral-CFT cross-ratio structure (η=1), not just the two-point. **[E]** BULK NEG CONTROL: the bulk-row four-point is orders of magnitude off the conformal value (exponential decay, gapped) — only the EDGE four-point is conformal. **[C]** the one external MMST fact (continuum existence of the chiral scaling limit) is now empirically supported at BOTH two-point (v444) and four-point order on our explicit collar; every measured n-point converges to its conformal limit. `SEAM.EQUIV.01` stays **[O]** — the rigorous MMST uniform-convergence theorem (v336) is the cited backbone. Python-only (numpy). **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v449_seam_mmst_uniform.py`	`SEAM.EQUIV.MMST.UNIFORM.01`: uniform-in-N control of the edge scaling limit — the empirical content of the cited MMST uniform-convergence theorem, strengthened, on the same v367/v368 $p+ip$ collar. **[E]** CROSS-RATIO A CONVERGES: the edge four-point $Q(1,2,4,7)/12 \to \sin(5\pi/12)/\sin(\pi/12) = 2+\sqrt{3}$, the deviation shrinking monotonically under refinement (N_x=36, 48, 60). **[E]** CROSS-RATIO B (independent positions): $Q(1,3,5,9)/12 \to$ its OWN conformal value 2 — a SECOND observable with a DIFFERENT limit, so the convergence is not a single tuned point. **[E]** UNIFORM $1/N$ RATE: $\lvert Q(N)-Q_\infty\rvert \cdot N \leq 4$ for BOTH cross-ratios and every $N_x \in \{36, 48, 60\}$ — one N-independent constant bounds the deviation across two independent observables, the empirical hallmark of uniform-in-N convergence. **[E]** BULK NEG CONTROL: the bulk-row cross-ratio is off-scale (gapped, exponential) — only the EDGE converges. **[C][O]** the MMST uniform-convergence HYPOTHESIS is empirically satisfied on the collar (two-point v444, four-point v448, and now multi-observable uniform rate); `SEAM.EQUIV.01` stays **[O]** — the abstract continuum existence theorem (v336) is strengthened in evidence, not replaced by a proof. Python-only (numpy). **[E][C]**
`v450_seam_edge_entanglement.py`	`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`: a THIRD independent reading of the edge c_{-}=8 — the Calabrese–Cardy entanglement scaling c=1/2 per Majorana (after v444 correlator, v447 topology). **[E]** CALABRESE–CARDY LAW: the block entanglement entropy of the critical XX (Dirac) chain fits $S=(c/3)\ln[(L/\pi)\sin(\pi\ell/L)]$+const with c_{Dirac}=1.000 (flat log slope) — the universal scaling-limit signature. **[E]** c PER MAJORANA = 1/2: $c_{\mathrm{Dirac}}=1=2c_{\mathrm{Maj}}$ (Dirac = two Majoranas), so $c_{\mathrm{Maj}}=1/2$ — the chiral CFT of one $p+ip$ edge mode (v444). **[E]** GAPPED NEG CONTROL: a dimerised (gapped) chain has saturating, ℓ-independent entanglement (c=0) — the log growth is the CRITICAL edge CFT. **[C]** EDGE CENTRAL CHARGE: $c_{-}=N_{\mathrm{Maj}} \cdot c_{\mathrm{Maj}} = 16 \cdot \frac{1}{2} = 8 = g_{\mathrm{car}}+N_{\mathrm{fam}}$, from entanglement — the SAME c_{-}=8 as v444 (correlator) and v447 (topology), from a third independent observable. **[C][O]** three observables agree; `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy). **[E][C]**
`v451_seam_edge_cardy_tower.py`	`SEAM.EQUIV.EDGE.CARDY.01`: the Cardy finite-size conformal tower — the edge spectrum reproduces the EXACT Ising primary dimensions $\{1/16, 1/2\}$, on the critical Majorana ring (the v367/v368 edge universality class). **[E]** SPIN DIMENSION h_σ=1/16: the periodic(R)–antiperiodic(NS) ground-energy split gives $(E_0^R - E_0^{NS})L/(2\pi v)$=0.06250 ($v$=2), stable across L=200, 400, 800 — the Ising spin field. **[E]** ENERGY DIMENSION h_ε=1/2: the lowest NS pair excitation gives $2\varepsilon_{\min}L/(2\pi v)$=0.50000 — the Ising energy field. **[E]** CONFORMAL SCALING: both dimensions are L-INDEPENDENT (the gaps scale exactly as $1/L$) — a conformal (gapless) spectrum, not a gapped lattice. **[C]** ISING IDENTIFICATION: $\{c, h_\sigma, h_\varepsilon\}=\{1/2, 1/16, 1/2\}$ is the chiral Ising (free Majorana) CFT (the c=1/2 building block, v444/v450); with N_{Maj}=16 copies and bulk–edge (v447), $c_{-}=16 \cdot \frac{1}{2}=8=g_{\mathrm{car}}+N_{\mathrm{fam}}$. **[C][O]** the edge CFT is named by its central charge (v450) AND its operator content (this); `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy). **[E][C]**

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Script	**What it machine-checks** *(continued)*
`v452_seam_e8_modular.py`	`SEAM.EQUIV.E8.MODULAR.01`: the $(E_8)_1$ torus modular data — one primary, S-invariant, T-phase $e^{-2\pi i/3}$, the holomorphic c=8 signature (extends v156's q-expansion to the modular transformation data). [E] S-INVARIANCE: $\chi(-1/\tau)$=$\chi(\tau)$ for χ=E_4/η^8 (weight 0, since E_4 wt 4 / η^8 wt 4), to $\sim 10^{-30}$ — ONE primary, the S-matrix is [1]. [E] T-PHASE: $\chi(\tau{+}1)/\chi(\tau)$=$e^{-2\pi i/3}$=$e^{-2\pi ic/24}$ (c=8), to $\sim 10^{-30}$ — the central charge appears in the modular T eigenvalue. [E] LEADING POWER ⇒ c=8: $\chi\, q^{1/3} \to 1$ so the leading power is $q^{-c/24}$=$q^{-1/3}$ ⇒ c=8=g_{car}+N_{fam}. [E] HOLOMORPHIC CONSTRAINT: c=8=0 mod 8 — the minimal holomorphic (single-primary) chiral central charge, the even unimodular rank-8 lattice E_8. [C][O] the $(E_8)_1$ modular data (one primary, S-invariant, T-phase $e^{-2\pi i/3}$, c=8) is the torus signature of the seam net; with the edge readings (v444/v447/v450/v451) the $(E_8)_1$ identity is pinned on the torus AND the edge; `SEAM.EQUIV.01` stays [O] — the cited continuum existence theorem (v336) is the residual. Wolfram-mirrored (`tfpt_readouts_extension.wl`). [E][C]
`v453_seam_mu4_from_marks.py`	`SEAM.RIGIDITY.MU4FROMMARKS.01`: deriving `QGEO.SYM.01` (μ_4-symmetry of the seam RP data) from the four marks — the LAST structural premise of the rigidity chain folded into the existing realisation premise. [E] THE MARKS ARE μ_4: the four Gauss–Bonnet marks are exactly $\{1, i, -1, -i\}$= roots of $z^4{-}1$ (v168/v216/v323). [E] THE CLOCK PERMUTES THE MARKS: $z \mapsto iz$ is a 4-cycle of order 4 ($1{\to}i{\to}{-1}{\to}{-i}{\to}1$), fixing the configuration setwise (the order-4 clock of v445). [E] MOEBIUS STABILISER: $z \mapsto iz$ preserves the marks' cross-ratio λ=2 (in the D_4 stabiliser, v168) — the symmetry is geometric. [E] THE FORM BASIS IS μ_4-GRADED: ω_k=$z^{k-1}dz/(z^4{-}1) \to i^k\omega_k$ under $z \mapsto iz$ (k=$1,2,3 \to \{i,-1,-i\}$, the three nontrivial μ_4 characters; weights $(1,2,3)$=A_3 exponents = $\mathrm{Spec}(Q_+)$) — the natural basis is μ_4-graded, so any geometric datum is μ_4-covariant. [C][O] `QGEO.SYM.01` ("the seam RP data is μ_4-symmetric") is a CONSEQUENCE of marks=μ_4 (this) + `QGEO.REALIZE.01` (seam = the marked geometry, the EXISTING premise), NO new premise; the rigidity chain (this → v446 → v445) closes modulo realisation, `SEAM.EQUIV.01` stays [O]. Wolfram-mirrored (`tfpt_readouts_extension.wl`). [E][C]
`v454_seam_edge_virasoro.py`	`SEAM.EQUIV.EDGE.VIRASORO.01`: the lattice realises the level-1 current algebra whose Sugawara central charge is EXACTLY 8 (G2 of the post-F next steps) — testing the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct c" fact on our own free-fermion edge. [E] CASIMIR c: the critical complex free-fermion ring $E_0(L)$=$e_\infty L{-}\pi vc/(6L)$ (Affleck–Cardy) fits c_{Dirac}=1.000 ⇒ 1/2 per Majorana (parameter-free, L=64..1024). [E] KAC–MOODY LEVEL: the lattice $U(1)$ current J_n has $\langle 0\vert J_nJ_{-n}\vert 0\rangle$=$n$ EXACTLY (n=1..8) — the affine level-1 central extension closes; a gapped (dimerised) control gives a vanishing non-linear term. [E] SUGAWARA EXACT: c= $\dim G/(1{+}h^\vee)$=8 for BOTH $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31) — Wolfram-mirrored. [C] ASSEMBLY: 16 Majoranas $\times\frac{1}{2}$=8= the level-1 Sugawara c; the edge IS a c_-=8 level-1 current-algebra CFT. [C][O] the lattice Virasoro/current algebra with the correct central extension is realised on the explicit collar; `SEAM.EQUIV.01` stays [O] (the continuum existence theorem v336 is the cited backbone). Mixed numerical (Casimir, Kac–Moody) + exact (Sugawara, Wolfram-mirrored). [E][C][O]

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Script	**What it machine-checks** ***(continued)***
`v455_seam_bw_uniform.py`	`SEAM.EQUIV.BW.UNIFORM.01`: a UNIFORM-IN-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition $u_\Theta{=}J$ (G1 of the post-F next steps) — the inductive-limit companion to the single finite witness v426, advancing v424 sub-step (i). One-particle: $\Delta{=}e^{-K}$, conjugation J, reflection Θ; BW $\Leftrightarrow u_\Theta{=}J \Leftrightarrow \Theta K\Theta{=}{-}K$. **[E]** UNIFORM BW: the reflection-odd, μ_4-even, gapped collar of size $4m$ ($m{=}2..32$, $N{=}8..128$) has $\|\Theta K\Theta{+}K\|{=}0$ to machine precision at EVERY N — $u_\Theta{=}J$ uniform. **[E]** UNIFORM GAP: the gap of K is N-independent ($6\log\frac{3}{2}{=}2N_{\mathrm{fam}}\log\frac{3}{2}$) so $\Delta{=}e^{-K}$ stays uniformly bounded away from 1 (a genuine KMS, not trace, state). **[E]** UNIFORM CLOCK SYMMETRY: $[\rho, K]{=}0$ exactly at every N. **[E]** N-INDEPENDENT NEG CONTROL: a side-even reflection gives $\|\Theta K\Theta{+}K\|{=}2\|K\|$ (a fixed per-mode margin, growing with N) and an off-μ_4 K breaks $[\rho, K]$ at every N — the success is non-vacuous. **[C][O]** the inductive (continuum) limit inherits $u_\Theta{=}J$; `SEAM.EQUIV.01` stays **[O]** (the rigorous continuum theorem NPW26/MMST v336 is the backbone). Python-only (numpy). **[E][C][O]**
`v456_seam_chirality_from_c3.py`	`SEAM.S3.FROM-P1.01`: the edge chirality ($c_-{\neq}0$, the S3 topological phase) is FORCED by the one-sidedness that defines $c_3{=}1/(8\pi)$ (P1) (G6 of the post-F next steps) — so S3 is a CONSEQUENCE of axiom P1, not an independent input. **[E]** c_3's "8" IS THE ONE-SIDED COUNT: $8\pi{=}\lvert\mathbb{Z}_2\rvert\int K{=}2(2\pi\chi(S^2)){=}2\cdot 4\pi$ ($\chi{=}2$), so $8{=}\lvert\mathbb{Z}_2\rvert\cdot(\int K/\pi){=}2\cdot 4$ (v73, Wolfram-mirrored). **[E]** REFLECTION REVERSES C: the explicit 2-band $p{+}ip$ Chern model has $C{=}{+}1$ and the orientation-reversed ($k_y{\to}{-}k_y$) model $C{=}{-}1$. **[E]** TWO-SIDED $\Rightarrow$ NON-CHIRAL: any GAPPED boundary carrying an orientation-reversing (reflection) symmetry must have $C{=}{-}C{=}0$ ($c_-{=}0$, gappable). **[C]** ONE-SIDED $\Rightarrow$ CHIRAL: the one-sided seam is precisely the $\lvert\mathbb{Z}_2\rvert$ quotient with NO reflection (the same $\lvert\mathbb{Z}_2\rvert$ that gives c_3 its 8π), so $C{\neq}0$ is forced — S3 follows from P1. **[C][O]** magnitude $c_-{=}8{=}\operatorname{rank}E_8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$ is the SAME integer as c_3's one-sided count; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays **[O]**. Mixed exact (the $8{=}\lvert\mathbb{Z}_2\rvert\int K/\pi$ arithmetic, Wolfram) + numerical (the reflection $C{\to}{-}C$ flip). **[E][C][O]**
`v457_seam_character_killtest.py`	`SEAM.EQUIV.KILLTEST.01`: an explicit $(E_8)_1$-vs-$SO(16)_1$ character/sector kill-test for the seam edge (G7 of the post-F next steps) — a FALSIFIABLE consistency test of the $(E_8)_1$ identification of `SEAM.EQUIV.01`. **[E]** $(E_8)_1$ CHARACTER TOWER: $\chi{=}E_4/\eta^8{=}q^{-1/3}(1{+}248q{+}4124q^2{+}34752q^3{+}\cdots)$ — the exact conformal-tower degeneracies ($E_4\cdot\prod(1{-}q^n)^{-8}$, Wolfram-mirrored). **[E]** CURRENT-COUNT DISCRIMINATOR: the weight-1 multiplicity is $248{=}\dim E_8{=}120(SO16){+}128(\text{spinor})$, NOT the 120 of the free $SO(16)_1$. **[E]** SECTOR-COUNT DISCRIMINATOR: $(E_8)_1$ is INVERTIBLE $\lvert\det\mathrm{Cartan}\,E_8\rvert{=}1$ (one primary) vs $SO(16)_1{=}D_8$ $\lvert\det\mathrm{Cartan}\,D_8\rvert{=}4$ (four sectors). **[X]** KILL CRITERION: a measured weight-1 count of 120, OR a sector count of 4, OR a tower coefficient $\neq\{1, 248, 4124, \ldots\}$ would FALSIFY $(E_8)_1$; current data give $248/1/\{1, 248, 4124\}$, so PASSED. **[C][O]** the $(E_8)_1$ identification survives the kill-test; `SEAM.EQUIV.01`'s continuum EXISTENCE (v336) stays **[O]**. Exact (Wolfram-mirrored) + a numerical determinant cross-check. **[E][C][O][X]**

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Script	**What it machine-checks *(continued)***
`v458_seam_mmst_citation_audit.py`	`SEAM.EQUIV.MMST.AUDIT.01`: the EXACT hypothesis-by-hypothesis citation audit of the MMST scaling-limit theorem (Morinelli–Morsella–Stottmeister–Tanimoto, CMP 397 (2023); `arXiv:2107.13834`) against our collar (G3 of the post-F next steps, the "exakter Zitat-Abgleich"). [C] H1 ALGEBRA CLASS: a gapped quasi-free CAR 16-Majorana system = MMST's lattice-fermion class (v155/v160/v302). [C] H2 MASSLESS POS-ENERGY LIMIT: the gapless chiral edge IS that massless scaling limit (v439/v444). [E] H3 PER-COPY (c=1/2): each of 16 chiral Majoranas is EXACTLY the proven Thm 4.11 c=1/2 case. [C] H4 DECOUPLED ADDITIVITY: MMST's limit decouples into Majoranas ⇒ $L_k = \sum_i L_k^{(i)}$ on the common core ⇒ the free c=8 $SO(16)_1$ Virasoro follows ($16 \cdot \frac{1}{2}$=8). [C] H5 WZW RANGE rank $\le c \le D$: $8 \le 8 \le 16$ IN range, the 120 $so(16)$ currents are bilinears — covered. [O] H6 RESIDUAL = THE 128-SPINOR EXTENSION: the index-4 $SO(16)_1 \to (E_8)_1$ adds 128 SPINOR currents (248=120+128), NOT bilinears — OUTSIDE MMST's method (the $\det K$ 4→1 holomorphy discriminator), handed to v459. [C][O] the audit CLOSES-by-citation the c=8 EXISTENCE and pins the open residual EXACTLY to the 128-spinor extension; `SEAM.EQUIV.01` stays [O], its open part now this one holomorphic-extension statement. Python (sympy; a structural/arithmetic audit). [C][O]
`v459_seam_lattice_voa_route.py`	`SEAM.EQUIV.LATTICEVOA.01`: the SECOND, independent route to $(E_8)_1$ — the lattice-VOA / conformal-net $A_Q(E_8)$ construction supplies EXACTLY the 128-spinor extension MMST (v458) leaves open (G5 of the post-F next steps). Cited: Adamo–Giorgetti–Tanimoto, "Rational and non-rational 2d CFTs arising from lattices" (`arXiv:2506.01008`); OS for unitary lattice VOAs (Adamo–Moriwaki–Tanimoto, `arXiv:2407.18222`). [E] E_8 EVEN UNIMODULAR: $\det \mathrm{Cartan}(E_8)$=1 and even ⇒ A_{E_8} HOLOMORPHIC (one sector, the $\det K$=1 the residual demanded). [E] LATTICE CURRENTS 248=8+240: 8 Cartan + 240 roots = the full $(E_8)_1$ weight-1 space, from the lattice (not bilinears). [E] THE 240 ROOTS SPLIT 112+128: 112 INTEGER (D_8) + 128 HALF-INTEGER; 8+112=120=$so(16)$ and the 128 half-integer roots ARE the spinor (248=120+128=8+240) — the lattice route builds the 128-spinor explicitly. [C] AGT THEOREM: A_Q of an even lattice IS a 2d conformal net; the holomorphic $SO(16)_1 \to (E_8)_1$ extension is CONSTRUCTED, not assumed. [C][O] TWO complementary routes now cover $(E_8)_1$ (MMST fermions v458 + the lattice net A_{E_8}); the residual reduces to the single realization input "the collar's scaling limit = A_{E_8}" (tied to P1 by v456). `SEAM.EQUIV.01` stays [O]. Exact (E_8 even unimodular; 240=112+128; 248=8+240=120+128, Wolfram-mirrored) + structural literature mapping. [E][C][O]

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Script	What it machine-checks *(continued)*
`v460_seam_s3_lto.py`	`SEAM.S3.LTO.01`: the EXPLICIT flat-band commuting-projector (local-topological-order) realisation of the S3 input — advancing v424 sub-step (i) ("realise the raw seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO") from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock. **[E]** FLAT-BAND PARENT: the flattened Bloch $Q{=}h/\lvert h\rvert$ of v367's $p{+}ip$ collar has $Q^2{=}I$ (flat ±1 bands) and $P{=}(I{-}Q)/2$ is an exact projector EQUAL to the occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state. **[E]** GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $\lvert P(n,0)\rvert$ decays rapidly ($\xi{\approx}1.14$ at $M{=}1$, exp $R^2{=}0.995$, $\lvert P(20,0)\rvert{<}10^{-6}$) and ξ GROWS as the gap closes (2.7 at gap 0.02); at the GAPLESS point the kernel decays as a clean power law $\lvert R\rvert^{-2.1}$ ($R^2{=}0.999$, tail $\sim2{\times}10^{-4}$) — locality needs the gap. **[E]** $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED: the flattened modular Hamiltonian $K_{\mathrm{flat}}{=}\mathrm{sign}(K)$ inherits $\Theta K_{\mathrm{flat}}\Theta{=}{-}K_{\mathrm{flat}}$ (BW, $u_\Theta{=}J$) and $[\rho,K_{\mathrm{flat}}]{=}0$ (μ_4) EXACTLY (sign is odd; ρ commutes with any $f(K)$). **[E]** NEG CONTROLS: a side-even reflection gives $2\lVert K_{\mathrm{flat}}\rVert{\neq}0$ (BW fails), an off-μ_4 K mixing opposite-sign marks breaks $[\rho,\mathrm{sign}(K)]{\neq}0$. **[C][O]** the chiral $c_-{=}8{\neq}0$ (FHS Chern $\lvert C\rvert{=}1$) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; $C{\neq}0$ forbids exponentially-localised Wannier, Thouless), so the realisation is the quasi-local LTO net of NPW26 — the form sub-step (i) asks for. `SEAM.EQUIV.01` stays **[O]** (the continuum theorem MMST v336 is the backbone). Python-only (numpy; the det/level discriminators Wolfram-mirrored via v89/v281/v422). **[E][C][O]**
`v461_seam_strict_locality.py`	`SEAM.S3.LOCALITY.01`: the Kapustin–Fidkowski/Thouless strict-locality OBSTRUCTION made explicit — sharpening v460's verdict from a cited statement to an EXHIBITED topological obstruction on the same v367 $p{+}ip$ collar. **[E]** CHERN INTEGER: the FHS Chern is $C{=}{+}1$ (chiral $M{=}1$), 0 (trivial $M{=}3$), -1 (opposite mass) — the invertible-phase invariant. **[E]** WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $\lvert C\rvert{=}1$ (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier functions exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. **[E]** OBSTRUCTION INTEGERS TIED: winding = $\lvert C\rvert{=}1{\neq}0$ and $c_-{=}16/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}{=}\mathrm{rank}\,E_8{\neq}0$ (the same 8 as c_3's one-sided count, v456). **[E]** NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. **[C][O]** Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C{=}0$; since $C{=}1{\neq}0$ NONE exists, so v424 sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise) — upgrading v460's cited verdict. `SEAM.EQUIV.01` stays **[O]** (continuum theorem MMST v336 the backbone). Python-only (numpy; the winding/Chern integers Lean-mirrored, `SeamResidualAxiom.lean`). **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v462_seam_spinor_continuum.py`	`SEAM.EQUIV.SPINOR.01`: the 128-spinor extension $SO(16)_1 \to (E_8)_1$ exhibited at CHARACTER level and as a finite-L → continuum convergence — the constructive companion of the v458 (MMST audit) / v459 (lattice-VOA) residual. **[E]** $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ gives $\chi_{(E_8)_1}=E_4/\eta^8=\frac{1}{2}[(\theta_3/\eta)^8+(\theta_4/\eta)^8]+\frac{1}{2}(\theta_2/\eta)^8=\chi_o^{SO16}+\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. **[E]** THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: χ_o gives weight-1 multiplicity 120, χ_s gives 128, summing to the exact $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ — the spinor currents ARE the $(E_8)_1 - SO(16)_1$ deficit (248=120+128). **[E]** FINITE 16-MAJORANA FOCK: $C(16,2)$=120 NS bilinear ($so(16)$) currents and $2^{16/2}$=256 Ramond ground states = 128 spinor + 128 cospinor — the spinor present already at finite N. **[E]** FINITE-L → CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir $E_0(L)=e_\infty L-\pi vc/(6L)$ fits $c_{\text{Dirac}}\to 1$ (c_-=8) and the lowest scaled NS gap $x_1\to 1$ with $\lvert x_1-1\rvert\sim 1/L^2$ (L=64..1024) — the lattice flows to the conformal tower. **[C][O]** the rigorous holomorphic extension ($\chi_o+\chi_s$ simple-current projection) is supplied by AGT/AMT (v459); `SEAM.EQUIV.01`'s continuum existence (v336) stays **[O]**. Mixed exact (θ identity + tower + 120=$C(16,2)$, 128=2^7, 248=120+128, Wolfram-mirrored) + numerical (the finite-L convergence, Python-only). **[E][C][O]**
`v463_seam_c8_holomorphic_uniqueness.py`	`SEAM.EQUIV.UNIQ.01`: the c=8 holomorphic-uniqueness pin — the positive lever that makes the IDENTIFICATION half of `SEAM.EQUIV.01` classification-forced rather than assumed (B of the A+B round). **[E]** c=8 NOT UNIQUE AT LEVEL 1: $c=\dim\mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions — A_8 (dim 80), $D_8=\mathfrak{so}(16)$ (dim 120), E_8 (dim 248); c=8 is necessary but NOT sufficient (the v457 $\det K$=4 $SO(16)$ rival is a real gleich-c competitor). **[E]** HOLOMORPHY FORCES $\dim V_1$=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}=E_4/\eta^8$, whose q^1 coefficient (the 1-dim candidate space, vacuum a_0=1) is 248 — forced, not chosen. **[E]** $\dim V_1$=248+ level-1 c=8 ⇒ E_8 UNIQUELY (248/(1+30)=8); the gleich-c rivals A_8 (80) and D_8 (120) are excluded. **[E]** the $(E_8)_1$ tower $\{1, 248, 4124, 34752, 213126\}$ are the E_4/η^8 coefficients, and $E_4/\eta^8=j^{1/3}$ (numeric $< 10^{-25}$). **[C]** Dong–Mason/Schellekens: the holomorphic VOA at c=8 is UNIQUE = V_{E_8}, and the even unimodular rank-8 lattice is unique = E_8 (Mordell–Witt) — so "c=8 holomorphic limit = $(E_8)_1$" is classification-forced. With v458 (free part) + v459 (spinor extension) the continuum/identification HALF of `SEAM.EQUIV.01` is **[C]** conditional on named audited theorems; the REALISATION R1 (v464) is the sole remaining input. `SEAM.EQUIV.01` stays **[O]**. Exact (sympy/integer q-series + Lie-algebra scan; mpmath cross-check), Wolfram-mirrored. **[E][C][O]**

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Script	**What it machine-checks *(continued)***
`v464_seam_oneparticle_rigidity.py`	`SEAM.EQUIV.ONEPARTICLE.01`: the one-particle RIGIDITY + scaling limit attacking R1 — the REALISATION input of `SEAM.EQUIV.01` (A of the A+B round). The seam is quasi-free (v113/v160/v161), so by Araki self-dual CAR the whole net is the second quantisation of its one-particle symbol P, and R1 collapses to a finite-per-mode statement about P. **[C]** ARAKI SELF-DUAL CAR: a gauge-invariant quasi-free state ↔ its symbol P ($0{\le}P{\le}1$) bijectively (Araki 1971; Shale–Stinespring 1965), so a realisation is fully determined by P. **[E]** RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with $P{=}$ projection onto $K{<}0$ ($P^2{=}P$, hermitian to $< 10^{-12}$), K fixed by gap + chirality + reflection (v426) — no second realisation. **[E]** ONE-PARTICLE SCALING LIMIT EXISTS: the real-space kernel is Cauchy on a fixed window ($\max\lvert P_{2N}{-}P_N\rvert{\sim}1/N{\to}0$), so P_∞ exists (Gaussianity is why a computation reaches it). **[E]** THE LIMIT IS $c{=}1$ DIRAC ($c_-{=}8$): entanglement $S(L){=}(c/3)\ln L$ gives $c{\to}1$ (per Majorana 1/2; Peschel/Calabrese–Cardy, ties v450), 16 copies $\to c_-{=}8$. **[C]** Shale–Stinespring: gauge-invariant ⇒ no pairing block ⇒ the Bogoliubov map is implementable, GNS reps quasi-equivalent. R1 is upgraded from a bald axiom to "the unique quasi-free realisation, one-particle limit exhibited, modulo the named Araki/Shale–Stinespring functor"; with v463 the `SEAM.EQUIV.01` residual is now entirely certification (named cited functors), zero open internal mechanism — closed modulo cited theorems. `SEAM.EQUIV.01` stays **[O]**. Python-only (numpy: idempotency, kernel Cauchy-convergence, entanglement c-fit). **[E][C][O]**
`v465_seed_crosssector_joint.py`	`SEED.CROSSSECTOR.01`: the cross-sector one-parameter seed test — the θ_{13}-independent, externally-runnable slice of v306. TFPT reads a CMB EB/TB birefringence angle β and a quark Cabibbo angle $\lambda_C{=}\lvert V_{us}\rvert$ (unrelated in the SM) as two readouts of the ONE axiom seed $\varphi_0{=}1/(6\pi){+}48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. **[E]** CROSS-SECTOR (headline): back-solving φ_0 *independently* from β (CMB) and λ_C (quark) gives the *same* seed within $0.01\sigma_{\rm comb}$ — a correlation only a shared-origin theory predicts. **[E]** one-parameter fit: the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof}{<}1$, combined pull 0.08σ. **[E]** θ_{13} EXCLUSION declared, not cherry-picked: adding $\sin^2\theta_{13}$ raises χ^2 by $\sim$4 (the documented $\sim 2\sigma$ pressure point, v328/v338), so this slice is *deliberately* θ_{13}-independent. **[E]** negative control: a data↔formula shuffle explodes the cluster χ^2 by $>5{\times}10^4$. A consistency / out-of-sample statistic (a subset of v306); φ_0 is FIXED by the axioms, so it upgrades *no* prediction. Python-only. **[E]**

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Script	What it machine-checks *(continued)*
`v466_seed_leptonmass_channel.py`	`SEED.LEPTONMASS.01`: a SIXTH, new-sector seed channel — the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ points to the SAME axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$, extending the empirical seed over-determination (v306/v465) from three measurement sectors to five. **[E]** plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9}. **[E]** NEW SECTOR (headline): back-solving φ_0 from the measured POLE ratio $m_e/m_\mu = 0.004836$ (PDG) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a charged-lepton MASS reading (Penning-trap masses), a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C]** TRANSPORT CAVEAT: the formula is the SOURCE ratio, the datum the POLE ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, v393) but largely CANCELS for m_e/m_μ, so the 0.11% residual sits far inside the band — added as a CONSISTENT new sector, not a sharper pin. **[E]** EXTENDS THE OVER-DETERMINATION: the six-channel cluster (v306's five + this) has $\chi^2/\mathrm{dof} < 1$ and the error-weighted mean stays at the axiom to $< 0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, FIVE-sector agreement (neutrino mixing, CMB, quark mixing, lepton mass, quark mass). **[E]** negative control / power: a data↔formula shuffle explodes the cluster χ^2 by $>100\times$; and not every mass ratio qualifies — $m_c/m_s = (34/47)/\varphi_0$ is scheme-ambiguous (FLAG 11–14), a WEAK channel deliberately not counted. **[C]** HONEST: theoretically still the ONE seed (v305 multiplicity 1) — an EMPIRICAL strengthening (a new measurement sector agrees), NOT theoretical independence; φ_0 is FIXED by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only. **[E][C]**
`v467_thirdgen_mixing_pattern.py`	FLAV.THIRDGEN.PATTERN.01: the three $\sim 2\sigma$ mixing tensions (θ_{13}, θ_{23}, V_{cb}) are ONE common $-\varphi_0^{\mathrm{ret}}$ suppression ($-5.36\% \pm 1.67\%$, 3.2σ non-zero, -0.03σ from $-\varphi_0^{\mathrm{ret}}$); the ladder-base forms $\lambda_C^2 e^{-5/6}$ / $(1-\varphi_0^{\mathrm{ret}})/2$ / $\lambda_C^2/(1+\lambda_C)$ drop the joint χ^2 10.4 → 0.11 with zero parameters; post-hoc candidate [O], frozen record unchanged, JUNO-era reactors + Belle II decide
`v468_dm2_ratio_jarlskog.py`	FLAV.DM2RATIO.01: $\Delta m_{21}^2/\Delta m_{31}^2 = \lvert J_{\mathrm{PMNS}}\rvert$ of the frozen matrix = 0.029653 vs measured 0.029805(792) → -0.19σ – a parameter-free candidate relation for the previously unpredicted splitting ratio; the informal $m_2/m_3 \sim \pi\varphi_0^{\mathrm{ret}}$ heuristic is now -2.44σ and retyped; NO floor $\Sigma m_\nu = 0.0588$ eV consistent; post-hoc [O], JUNO decides

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Script	What it machine-checks *(continued)*
`v469_seam_crossedproduct_route.py`	`SEAM.EQUIV.CROSSEDPRODUCT.01`: the net-level crossed-product certification of the 128-spinor extension + the R1→R1′ invariant-level reduction. **[E]** LR LOCALITY INTEGER (index-2): the $SO(16)_1$ spinor has h_s=$N/16$=16/16=1∈$\mathbb{Z}$ ⇒ statistics phase +1 — the Longo–Rehren criterion HOLDS, so the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1\rtimes\{1,s\}$ is a LOCAL index-2 extension (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM $\mu(B)$=4/2^2=1 ⇒ HOLOMORPHIC ⇒ $(E_8)_1$ by the v463 pin (E_4/η^8 q^1=248 re-verified). The 2024/2025-preprint leg (AGT/AMT, v459) is DEMOTED to an independent second witness — the extension certification now rests on 1995–2001 PEER-REVIEWED journals. **[E]** INDEX-4 PARALLEL: the μ_4 glue J=(s,Λ_1) has $h(J^k)$=$\{1,1,1\}$ ALL integer ($\frac{5}{8}+\frac{3}{8}$=1; $\frac{1}{2}+\frac{1}{2}$=1) — the v125 isotropy $q(k,k)$=k^2 at net level; μ=16/4^2=1, the same endpoint. **[E]** R1′ INVARIANTS COMPUTED: per-copy FHS Chern $C(M{=}1)$=1 vs $C(M{=}3)$=0, ν=16×|C|=16, c_-=ν/2=8=g_{car}+N_{fam}; ν mod 16=0 is EXACTLY Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way). **[E]** THE CHAIN COMPOSES: T1 (phase classification: Kitaev + c_- invariance, Kapustin–Spodyneiko PRB 101 (2020) / Kim et al. PRL 129 (2022)) → T2 (MMST CMP 397 (2023) + Böckenhauer 1996) → T3 (LR/BE + KLM CMP 219 (2001)) → T4 (Dong–Mason + v463) with matching in/out slots (v297 typing). **[C]** R1→R1′: `collar_realizes` (model-level fiat) replaceable by `collar_invariants` — {quasi-free **[C]** (v155/v160), gap **[E]** (v302), class D, c_-=8 **[E]** (v456, from P1)}; quasi-freeness stays the ONE **[C]** hypothesis; Lean mirror `FORM.SEAM.RESIDUAL.01`. **[O]** NOT CLOSED: the cited theorems stay cited (certification, not re-proof); `SEAM.EQUIV.01` stays **[O]**. Mixed exact (weights/μ/tower, Wolfram-mirrored) + numerical (FHS Chern, Python-only). **[E][C][O]**

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Script	**What it machine-checks** ***(continued)***
`v470_alpha_inflow_level.py`	`ALPHA.QUILLEN.INFLOW.01`: the α^3 level = the COMPUTED bulk Chern invariant + the seam F-normalisation = the affine embedding index k_Y=5/3. [E] TARGET RE-VERIFIED: α^{-1}=137.0359992168 unique root; the Quillen split $a^3-2c_3^3a^2=8b_1c_3^6\ln(1/\varphi_{\text{seam}})$ at the root to $<10^{-35}$ (v341/v382). [E] THE INVARIANT COMPUTED: FHS Chern $C(M{=}1){=}1$, $C(M{=}3){=}0$ — the same integers as v367/v461, here read as the quantised $U(1)$ response of the collar phase. [C] $k_0{=}\lvert C\rvert{=}1$ UNDER THE EM1 READING: the a^3 coefficient EQUALS the computed bulk invariant of the SAME model that realises S3 — “why exactly 1” is answered by computation + the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten Rev. Mod. Phys. 88 (2016) η=CS part of $\delta\log\det$; Quillen 1985 / Bismut–Freed CMP 106 (1986) determinant-line integrality) — REPLACING v435’s minimality assumption. [E] k_Y=5/3 EXACT: $\mathrm{tr}_{\bar{5}}(Y^2)/\,\mathrm{tr}(T_3^2)=(5/6)/(1/2)=5/3$ (Ginsparg, Phys. Lett. B197 (1987)); $(3/5)\times\frac{41}{6}=\frac{41}{10}=b_1$ — the “GUT 3/5” IS the inverse embedding index, not an imported convention; carrier decomposition $\frac{41}{6}=\frac{20}{3}$ (fermions, 3×16-content) $+\frac{1}{6}$ (Higgs doublet). [C] F-NORMALISATION RETYPED: once the seam net is $(E_8)_1$ level 1 (`SEAM.EQUIV.01`), the $U(1)_Y$ normalisation $\langle J(z)J(w)\rangle=k/(z-w)^2$ is FIXED by k_Y — level-1 current-algebra rigidity, no continuous dial; the v434 [C] has ZERO independent content (a face of `SEAM.EQUIV.01`, per the v382 typing). [E] UNIFICATION: ONE invertible phase, TWO quantised responses — $c_-=16\times\frac{1}{2}\times\lvert C\rvert=8=g_{\text{car}}+N_{\text{fam}}$ (gravitational, feeds `SEAM.EQUIV.01`/S3, v456) and C=1 ($U(1)$, feeds EM1) — “the one theorem and the one number” are two faces of the SAME phase; closing R1′ (v469) feeds both. [O] NOT CLOSED: the bridge lemma “$\delta\log\det_\zeta$(seam) = the inflow response” stays the named cited step; the EM1 reading stays [C]; `ALPHA.QUILLEN.EXACT.01` stays [O]; α^{-1} stays [E] regardless. Mixed exact (k_Y/b_1 identities, Wolfram-mirrored) + numerical (root/split mpmath, FHS Chern, Python-only). [E][C][O]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

Figure 1: The development timeline of the verification suite — the eleven phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1}, the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{\mathrm{car}}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, the AQFT closure that drives the whole structural residual to one geometric bedrock premise, the F_{transfer} functor and frontier interfaces, the QGEO pillowcase and Sheet Diamond, to the *icosahedral capstone* — where the single $(2, 3, 5)$ Brieskorn singularity is shown to generate the whole skeleton and the closing step becomes the physical question "is the seam short-range-entangled ($\det K = 1$)?". The master script→check map follows.

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers `vN` refer to `verification/vN_*.py`.

2026-07-02 (v469/v470 — the closure-route round on the two named targets: the SEAM.EQUIV.01 extension leg re-founded on 1995–2001 peer-reviewed crossed-product theorems + the realisation axiom reduced to invariant level; the α^3 level = the *computed* bulk Chern invariant + the seam F-normalisation = the embedding index k_Y=5/3; both targets stay [O])

- **`v469_seam_crossedproduct_route.py` (SEAM.EQUIV.CROSSEDPRODUCT.01).** The 128-spinor extension leg of `SEAM.EQUIV.01` no longer rests on 2024/2025 preprints: [E] the $SO(16)_1$ spinor has h_s=$N/16$=16/16=1∈$\mathbb{Z}$, so its statistics phase is +1 — *exactly* the Longo–Rehren locality criterion — and the $\mathbb{Z}_2$ simple-current crossed product $SO(16)_1 \rtimes \{1, s\}$ is a *local* index-2 extension by the peer-reviewed subfactor package (Longo–Rehren, Rev. Math. Phys. 7 (1995); the level-1 $so(N)$ net with its DHR sectors from even-CAR: Böckenhauer, Rev. Math. Phys. 8 (1996); simple-current extensions of nets: Böckenhauer–Evans, Comm. Math. Phys. 197 (1998)); KLM (Comm. Math. Phys. 219 (2001)) gives $\mu(B)$=$4/2^2$=1 ⇒ holomorphic ⇒ $(E_8)_1$ by the `v463` pin. The AGT/AMT lattice-VOA route (`v459`) is *demoted to an independent second witness.* [E] the index-4 μ_4 glue runs on the same integer: $h(J^k)$={1, 1, 1} for k=1, 2, 3 (the `v125` isotropy $q(k(1,1))$=k^2 at net level), μ=$16/4^2$=1, same endpoint. [E] the realisation input is reduced from model fiat to *invariants* (R1′): quasi-free [C] (`v155`/`v160`) + gap [E] (`v302`) + class D + c_-=8 [E] (`v456`, from P1), with the invariants *computed* — per-copy FHS Chern $|C|$=1 vs 0 (control), ν=16×$|C|$=16, c_-=$\nu/2$=8 — and $\nu \equiv 0 \pmod{16}$ is exactly Kitaev's class-D phase whose edge admits a purely bosonic description = the $(E_8)_1$ level-1 state (Ann. Phys. 321 (2006), sixteen-fold way); the phase-invariance of c_- is cited (Kapustin–Spodyneiko, PRB 101 (2020); Kim et al., PRL 129 (2022)). [O] *not closed*: the cited theorems stay cited; `SEAM.EQUIV.01` stays [O].
- **Lean hardening.** `TfptCarrier/SeamResidualAxiom.lean` gains the *parallel* peer-reviewed derivation `seamResidualClosed`′: the invariant-level axiom `collar_invariants` (R1′) + *one* combined 1995–2001 package `crossedproduct_route_theorem`; the new joints are kernel facts with *no* axioms — `lr_locality_integer` (h_s=16/16∈$\mathbb{Z}$), `glue_locality_integers` (μ_4 $h(J^k)$∈$\mathbb{Z}$), `klm_holomorphic` ($4/2^2$=1, $16/4^2$=1), `sixteenfold_pin` (ν=16≡0 mod 16, c_-=8), bundled as `crossedproduct_arithmetic` (`[propext]` only); `#print axioms seamResidualClosed`′ = exactly four; the original `seamResidualClosed` (AGT/AMT witness) is kept unchanged as the independent second route; `lake build` OK (3285 jobs).
- **`v470_alpha_inflow_level.py` (ALPHA.QUILLEN.INFLOW.01).** Both residuals left on `ALPHA.QUILLEN.EXACT.01` after `v433`–`v435` are upgraded. [E] target re-verified (α^{-1}=137.0359992168; Quillen split <10^{-35}). [C] *the level is computed, not minimal*: the a^3 coefficient k_0=1 *equals* the FHS bulk Chern invariant $|C|$=1 of the *same* p+ip collar that realises S3 (`v367`/`v461`) — "why exactly 1" is answered by computation plus the cited inflow identification (TKNN PRL 49 (1982) / Avron–Seiler–Simon PRL 51 (1983) quantisation; Callan–Harvey, Nucl. Phys. B250 (1985) inflow; APS 1975 / Alvarez-Gaumé–Della Pietra–Moore 1985 / Witten, Rev. Mod. Phys. 88 (2016) η=CS part of $\delta \log\det$; Quillen 1985 / Bismut–Freed 1986 determinant-line integrality), *replacing* `v435`'s minimality assumption. [E] k_Y=$\mathrm{tr}(Y^2)/\mathrm{tr}(T_3^2)$=(5/6)/(1/2)=5/3 (Ginsparg, Phys. Lett. B197 (1987)): (3/5)×$\frac{41}{6}$=$\frac{41}{10}$=b_1 — the "GUT 3/5" *is* the inverse embedding index; $\frac{41}{6}$=$\frac{20}{3}$+$\frac{1}{6}$ (carrier decomposition). [C] the seam F-normalisation (the one [C] of `v434`) is *retyped*: once the seam net is $(E_8)_1$ level 1, $\langle JJ\rangle$=$k/(z-w)^2$ is fixed by k_Y — level-1 current-algebra rigidity, zero independent content, a face of `SEAM.EQUIV.01` (per the `v382` typing). [E] *structural unification*: one invertible phase, two quantised responses — c_-=8 (gravitational, feeds

SEAM.EQUIV.01/S3) and $C{=}1$ ($U(1)$, feeds EM1) — so closing R1′ feeds *both* named targets. [O] *not closed*: the bridge lemma "$\delta \log \det_\zeta$(seam)= the inflow response" stays the named cited step; ALPHA.QUILLEN.EXACT.01 stays [O]; α^{-1} stays [E].

- **Integration.** Both registered in run_all.py + script_registry.csv (cluster frontier); ledger rows SEAM.EQUIV.CROSSEDPRODUCT.01 + ALPHA.QUILLEN.INFLOW.01; FORM.SEAM.RESIDUAL.01 updated (the seamResidualClosed′ route); the exact joints Wolfram-mirrored (365 → 368: the LR/glue locality integers, the KLM μ-arithmetic, the sixteen-fold pin, the $k_Y{=}5/3/b_1$ identities); cited in tfpt_1 (α fixed point + EM-Ward residual prose), tfpt_research_contracts (seam residual + Lean section), introduction (closure ledger), tfpt_5_redteam (Target A residual wording) and the README/website status surfaces; exploration provenance experiments/tfpt-discovery/closure_route_*.py (2026-07-02 note in experiments/next.txt). Suite ALL CHECKS PASSED through v470. Curated version bumped to **v5.4** (Zenodo deposit updated).

2026-07-02 (v467/v468 — two named post-hoc [O] flavor candidates, record unchanged: the three $\sim 2\sigma$ mixing tensions as *one* third-generation $-\varphi_0$ pattern, and the splitting ratio $\Delta m^2_{21}/\Delta m^2_{31} = |J_{\mathrm{PMNS}}|$)

- **v467_thirdgen_mixing_pattern.py (FLAV.THIRDGEN.PATTERN.01).** The suite's three $\sim 2\sigma$ mixing tensions ($\sin^2\theta_{13}$ $+2.0\sigma$, $\sin^2\theta_{23}$ $+1.76\sigma$, $|V_{cb}|$ $+1.79\sigma$; v307) are exactly the three *third-generation* channels and share *one* relative deviation $-5.36\% \pm 1.67\%$ — non-zero at 3.2σ, -0.03σ from $-\varphi_0$. **[E]** vocabulary identities: $\lambda_C^2{=}\varphi_0(1{-}\varphi_0)$ (the frozen Cabibbo); $\sin^2\theta_{23}{=}(1{-}\varphi_0)/2 \Leftrightarrow \cos 2\theta_{23}{=}\varphi_0$; the suppression size $\varphi_0{=}5.3\%$ sits inside the seam-gapped band $(2/3)^6{=}8.8\%$ (v393). **[N]** the ladder-base forms $\lambda_C^2 e^{-5/6}{=}0.021880$ / $(1{-}\varphi_0)/2{=}0.473414$ / $\lambda_C^2/(1{+}\lambda_C){=}0.041119$ drop the joint four-observable χ^2 from 10.4 to 0.11 with *zero* new parameters; $|V_{ub}|$ (no bare φ_0) and the 1–2 channels stay untouched and correct (dressing λ_C would fail at -24σ — scope checked, not chosen); Daya Bay alone already prefers the pattern form ($\Delta\chi^2{=}4.3$). **[C] honest:** post-hoc (5/16 equally-simple dressings fix the two PMNS tensions; $(1{-}\varphi_0)/(1{-}c_3)$/resolvent degenerate $<1.4\%$). **[O]** no seam-transport derivation (the Bernoulli-variance reading $\lambda_Y^2{=}\mathrm{Var}[\mathrm{Bernoulli}(\varphi_0)]$ is an exact identity but only an interpretation) — the frozen record (v84/REG.FREEZE.01) is *unchanged*. **[X]** pre-registered, symmetric: θ_{13} stable at 0.0231 kills the pattern, at $\sim$0.0219 (sub-1%) kills the record's bare-φ_0 reading, exclusive $|V_{cb}|{\sim}0.0395$ kills both — JUNO-era reactors + Belle II decide. Python-only.
- **v468_dm2_ratio_jarlskog.py (FLAV.DM2RATIO.01).** The splitting ratio $\Delta m^2_{21}/\Delta m^2_{31}$ was previously *unpredicted*; the frozen PMNS matrix (v9/v268/v270, $\delta{=}4\pi/3$) has one canonical derived CP invariant, and **[N]** $|J_{\mathrm{PMNS}}|{=}0.029653$ meets the NuFIT 6.0 measured ratio 0.029805 ± 0.000792 at -0.19σ (equivalently $m_2/m_3{=}\sqrt{|J|}{=}0.17220$ vs 0.17264(229)) — zero dials. **[N]** the informal tfpt_2 table heuristic $m_2/m_3{\sim}\pi\varphi_0{=}0.16704$ (a "$\sim$" entry, never frozen) is now -2.44σ and is *superseded as the comparator*. **[C]** spectrum consistency: $\Sigma m_\nu{=}m_3(1{+}\sqrt{|J|}){=}0.0588$ eV reproduces the documented 0.0586 eV (v272); no absolute number is added. **[O]** no mechanism (the $\mu\tau$ texture gives $\theta_{13}{=}0$ at leading order). **[X]** JUNO shrinks the window $\sim 4\times$; adopting the v467 pattern would shift $|J| \to 0.028849$ (-1.21σ today) — the two candidates *cross-discriminate* at JUNO precision. Python-only.
- **Integration.** Both registered in run_all.py and script_registry.csv (cluster neutrinos); ledger rows FLAV.THIRDGEN.PATTERN.01 and FLAV.DM2RATIO.01 (both [O] candidates); cited in tfpt_2 (§ Neutrinos: pressure-point block + new candidate paragraphs, the m_2/m_3 table row retyped to the $|J|$ comparator, the octant prose kept honest — record does not select, the candidate would), introduction (closure ledger), tfpt_4_frontier (live kill-test board) and tfpt_5_redteam (seed-hyperplane caption); v328's "single outlier" wording scoped to the seed observables; exploration provenance

`experiments/tfpt-discovery/pattern_hunt_pmns_dressing.py` (2026-07-02 note in `experiments/next.txt`); Python-only (flagged in the Wolfram README).

2026-07-01 (`v466` — a sixth seed channel from a new measurement sector: the charged-lepton mass ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ back-solves the same axiom seed to -0.11%, extending the empirical over-determination to *five* sectors)

- **`v466_seed_leptonmass_channel.py` (`SEED.LEPTONMASS.01`).** The seed over-determination of `v306`/`v465` used five observables spanning three measurement sectors (neutrino mixing, CMB, quark mixing). This module adds the *charged-lepton mass* sector: the ratio $m_e/m_\mu = \frac{12}{7}\varphi_0^2$ is a sixth independent readout of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen $m_e/m_\mu = \frac{12}{7}\varphi_0^2 = 0.00484673$ to 10^{-9}. **[N] new sector (headline):** back-solving φ_0 from the measured *pole* ratio $m_e/m_\mu = 0.004836$ (PDG 2024) gives $\varphi_0 = 0.053113$, only -0.11% from the axiom — a fourth measurement sector beyond {neutrino mixing, CMB, quark mixing}. **[C] transport caveat:** the formula is the *source* ratio and the datum the *pole* ratio; the source→pole shift is the seam-gapped correction class (bound $(2/3)^6 = 8.78\%$, `v393`) but largely cancels for m_e/m_μ, so the 0.11% residual sits far inside the band — the channel is added as a *consistent new sector*, not a sharper pin. **[N] extends:** the six-channel cluster (`v306`'s five + this) has $\chi^2/\mathrm{dof} < 1$ and the error-weighted mean stays at the axiom to $<0.1\%$; with the corroborating heavy-quark m_t/m_b (-0.23%) it is a seven-channel, five-sector agreement. **[N] power:** a data↔formula shuffle explodes the cluster χ^2 by $>100\times$, and the scheme-ambiguous $m_c/m_s = (34/47)/\varphi_0$ (FLAG 11–14) is deliberately not counted. **[C] honest:** theoretically still the *one* seed (`v305` multiplicity 1) — an *empirical* strengthening, not theoretical independence; φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.LEPTONMASS.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`/`v465`); Python-only (flagged in the Wolfram README).

2026-07-01 (`v465` — the cross-sector one-parameter seed test: a CMB birefringence angle and a quark Cabibbo angle point to the same seed within 0.01σ, the θ_{13}-independent externally-runnable slice of `v306`)

- **`v465_seed_crosssector_joint.py` (`SEED.CROSSSECTOR.01`).** The sharpest falsifiable signature of a *shared-origin* theory is a correlation no non-shared-origin model predicts. In the Standard Model the cosmic-birefringence rotation β (a CMB EB/TB polarisation observable) and the Cabibbo angle $\lambda_C = |V_{us}|$ (a quark-flavour observable) are unrelated; TFPT reads both as functions of the *one* axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4$. **[E]** plumbing: the axiom φ_0 reproduces the frozen λ_C, $\sin^2\theta_{12}$, β to 10^{-9}. **[N] cross-sector (headline):** back-solving φ_0 *independently* from β (CMB; ACT DR6 + Planck PR4 combined) and from λ_C (quark; PDG 2024) gives the same seed within $0.01\sigma_{\mathrm{comb}}$. **[N] one-parameter fit:** the error-weighted mean of the three back-solved φ_0 reproduces the axiom to 0.06%; the joint χ^2 at the axiom (zero free parameters, 3 dof) is 0.007, $\chi^2/\mathrm{dof} < 1$, combined pull 0.08σ. **[N] θ_{13} exclusion declared:** adding $\sin^2\theta_{13}$ raises χ^2 by ~ 4 (the documented $\sim 2\sigma$ pressure point, `v328`/`v338`), so this slice is the honest θ_{13}-independent statement, not a hidden dilution. **[N] power:** a data↔formula shuffle explodes the cluster χ^2 by $>5\times 10^4$. A consistency / out-of-sample statistic (a subset of `v306`); φ_0 is fixed by the axioms, so it upgrades *no* prediction and closes *no* gate. Python-only (floats).
- **Integration.** Registered in `run_all.py` and `script_registry.csv` (cluster `registry`); ledger row `SEED.CROSSSECTOR.01`; cited in `tfpt_5_redteam` (the seed-cross-validation firewall winbox, alongside `v305`/`v306`) and in the `origin_theory` birefringence dependency chain; Python-only (flagged in the Wolfram README).

2026-06-30 (v463/v464 + Lean + Wolfram — the c=8 holomorphic-uniqueness pin and the one-particle realisation rigidity: the `SEAM.EQUIV.01` residual reduced to *entirely certification* on both halves)

- **The bird's-eye reframing.** The recent G-block modules all sharpened the *target* of the scaling limit; this round attacks the two halves that actually remain. The *continuum existence* is already citable (`v458` MMST + `v459` AGT/AMT); what was implicit was (B) *why* a c=8 holomorphic limit must be $(E_8)_1$, and (A) the *realisation* input R_1 that the abstract seam IS the concrete quasi-free collar. `v463` closes B; `v464` reduces A to its one-particle data.

- **`v463_seam_c8_holomorphic_uniqueness.py` (`SEAM.EQUIV.UNIQ.01`).** The c=8 holomorphic-uniqueness pin. [E] c=8 is NOT unique at level 1: $c=\dim\mathfrak{g}/(1+h^\vee)$=8 has THREE simple solutions A_8 (dim 80), D_8=$\mathfrak{so}(16)$ (dim 120), E_8 (dim 248), so the det K=4 $SO(16)$ rival is a genuine same-c competitor. [E] HOLOMORPHY FORCES dim V_1=248: the only weight-0 form with leading $q^{-1/3}$ and integer coefficients is $j^{1/3}$=E_4/η^8, q^1 coefficient 248 (the 1-dim candidate space, vacuum a_0=1). [E] dim 248+ level-1 c=8 ⇒ E_8 uniquely (248/(1+30)=8); tower $\{1, 248, 4124, 34752, 213126\}$=$E_4/\eta^8$=$j^{1/3}$. [C] Dong–Mason/Schellekens: the holomorphic c=8 VOA is unique = V_{E_8} (even unimodular rank-8 lattice unique = E_8, Mordell–Witt) — so the identification half is classification-forced.

- **`v464_seam_oneparticle_rigidity.py` (`SEAM.EQUIV.ONEPARTICLE.01`).** The one-particle rigidity attacking R_1. The seam is quasi-free (`v113/v160/v161`), so by Araki self-dual CAR the net is the second quantisation of its one-particle symbol P. [E] RIGIDITY: the gapped ground state is the UNIQUE quasi-free state with P= projection onto K<0 (P^2=P to $< 10^{-12}$), K fixed by gap + chirality + reflection (`v426`). [E] ONE-PARTICLE LIMIT EXISTS: the kernel is Cauchy on a fixed window (max $|P_{2N}-P_N|\sim 1/N\to 0$). [E] the limit is c=1 Dirac: entanglement $S(L)$=$(c/3)\ln L$ gives $c\to1$, 16 copies ⇒ c_-=8 (ties `v450`). [C] Shale–Stinespring makes the Bogoliubov map implementable, so R_1 is upgraded from a bald axiom to "the unique quasi-free realisation, modulo the named CAR functor". Python-only (numpy).

- **Lean (`SeamResidualAxiom.lean`).** Adds the `v463` kernel facts `c8_three_candidates` ($A_8/D_8/E_8$ each dim =8$(1+h^\vee)$) and `holomorphy_selects_e8` (forced dim V_1=248=$E_8 \neq$ 80, 120) to `residual_arithmetic` (still `[propext]`-only); `v464` is numerical. `#print axioms seamResidualClosed` still lists exactly FOUR axioms.

- **Wolfram + integration.** `v463` mirrored (the three candidates, the forced 248, the E_4/η^8 tower); the extension count is now 365/365. Integrated across the contracts paper (attacks (xxx)/(xxxi)), README, zenodo, ledger and website. Net effect: across (v)–(xxxi) the `SEAM.EQUIV.01` residual is now *entirely certification* — a named, hypothesis-audited package (MMST, AGT/AMT, Dong–Mason, Araki, Shale–Stinespring) with no open internal mechanism — yet `SEAM.EQUIV.01` stays [O] because it still rests on cited continuum-existence theorems we do not re-prove.

2026-06-30 (v461/v462 + Lean axiom merge — the strict-locality obstruction made explicit, the 128-spinor extension exhibited at character level, and the `SEAM.EQUIV.01` residual reduced to ONE realisation axiom + ONE combined cited theorem)

- **`v461_seam_strict_locality.py` (`SEAM.S3.LOCALITY.01`).** Sharpens `v460`'s verdict from a *cited* statement to an *exhibited* topological obstruction on the same `v367` $p+ip$ collar. [E] CHERN INTEGER: the FHS Chern is C=+1 (chiral M=1), 0 (trivial M=3), −1 (opposite mass). [E] WILSON-LOOP / WANNIER-CENTRE WINDING = CHERN: the hybrid Wannier centre $\theta(k_y)$ winds by $|C|$=1 (chiral) and 0 (trivial) over the BZ; exponentially-localised Wannier exist IFF the winding is 0 (Brouder–Panati et al., PRL 98 046402; Thouless), so the chiral collar has NONE. [E] OBSTRUCTION INTEGERS TIED: winding = $|C|=1\neq 0$ and $c_-=8=g_{\rm car}+N_{\rm fam}=\operatorname{rank} E_8\neq 0$ (the same 8 as c_3's one-sided count, `v456`). [E] NEG CONTROL: BOTH phases are gapped, yet only the trivial (winding 0) admits exp-localised Wannier and a strict finite-range commuting projector — it is the CHIRALITY, not the gap. [C][O] Kapustin–Fidkowski (`arXiv:1810.07756`): a strictly finite-range commuting projector would force winding 0, $C=0$; since $C=1\neq 0$ NONE exists, so `v424` sub-step (i)'s realisation is PROVABLY the quasi-local NPW26 LTO net (strict locality topologically forbidden, not a missing premise). `SEAM.EQUIV.01` stays [O]. Python-only (numpy); the winding/Chern integers Wolfram- and Lean-mirrored.

- **`v462_seam_spinor_continuum.py` (`SEAM.EQUIV.SPINOR.01`).** The constructive companion of the `v458` (MMST audit) / `v459` (lattice-VOA) residual: the 128-spinor extension $SO(16)_1\to(E_8)_1$ exhibited three ways. [E] $(E_8)_1$ TOWER FROM $SO(16)_1$ SECTORS: the exact Jacobi/E_8 identity $\theta_2^8+\theta_3^8+\theta_4^8=2E_4$ gives $\chi_{(E_8)_1}=E_4/\eta^8=\chi_o^{SO16}+\chi_s^{SO16}$ — the holomorphic $(E_8)_1$ character IS the $SO(16)_1$ vacuum + spinor sector sum. [E] THE 128 SPINOR WEIGHT-1 STATES FILL 120→248: χ_o gives 120, χ_s gives 128, summing to the exact tower $\{1,248,4124,34752\}$ (248=120+128). [E] FINITE 16-MAJORANA FOCK: $C(16,2)=120$ NS bilinear currents and $2^{16/2}=256$ Ramond ground states = 128 spinor + 128 cospinor. [E] FINITE-L → CONTINUUM CONVERGENCE: the critical free-fermion ring Casimir gives $c_-\to 8$ and the lowest scaled NS gap $x_1\to 1$ with $|x_1-1|\sim 1/L^2$ (L=64..1024). [C][O] AGT/AMT (`v459`) supply the rigorous extension; `SEAM.EQUIV.01` existence (`v336`) stays [O]. Mixed exact (θ identity + tower + 120/128/256, Wolfram-mirrored) + numerical (finite-L, Python-only).
- **Lean (`TfptCarrier/SeamResidualAxiom.lean`, `FORM.SEAM.RESIDUAL.01`).** The two former cited axioms `mmst_existence` ∘ `agt_lattice_extension` are MERGED into ONE combined external axiom `seam_realisation_theorem` (dropping the intermediate `ChiralCFT_c8`), so `#print axioms seamResidualClosed` now lists exactly FOUR axioms (down from six). Two more kernel facts are added axiom-free: `strict_locality_forbidden` (`v461`: winding = $|C|=1\neq 0$, $c_-\neq 0$) and `fock_sectors` (`v462`: $C(16,2)=120$, $2^8=256$, spinor 128, 120+128=248); `residual_arithmetic` (now including both) depends only on `[propext]`. `lake build` OK. So the residual is reduced to ONE realisation axiom + ONE combined cited theorem.
- **Wolfram.** `tfpt_readouts_extension.wl` adds 3 exact mirrors (total 363/363): the `v461` obstruction integer-logic and the `v462` θ-identity + 120+128=248 sector / Fock counts.

2026-06-30 (`v460` — the explicit flat-band commuting-projector (LTO) realisation of the S3 input: advancing the `SEAM.EQUIV.01` sub-step (i) from abstract existence to an explicit gap-protected quasi-local projector net carrying the $\mathbb{Z}_2$ reflection and the μ_4 clock)

- **`v460_seam_s3_lto.py` (`SEAM.S3.LTO.01`).** NPW26 (`arXiv:2605.10693`) prove the intrinsic Bisognano–Wichmann lemma for *commuting-projector* (local-topological-order) models with a $\mathbb{Z}_2$ reflection; `v424` reduced the keystone BW residual to that theorem modulo two open sub-steps, of which (ii) was settled by `v426` and the continuum lift of BW by the uniform-in-N tower `v455`. This module attacks the *commuting-projector* side of sub-step (i) on the same explicit `v367` $p+ip$ collar. [E] FLAT-BAND PARENT: the flattened Bloch $Q=h/|h|$ has $Q^2=I$ (flat ±1 bands) and $P=(I-Q)/2$ is an exact projector EQUAL to `v367`'s occupied ground-state projector — a frustration-free commuting-projector parent with the same gapped ground state (machine precision over a k-grid). [E] GAP-PROTECTED QUASI-LOCALITY: the gapped projector kernel $|P(n,0)|$ decays rapidly ($\xi\approx 1.14$ at $M=1$, exp $R^2=0.995$, $|P(20,0)|<10^{-6}$) and ξ grows as the gap closes (2.7 at gap 0.02); at the gapless point it decays as a clean power law $|R|^{-2.1}$ ($R^2=0.999$) — locality needs the gap. [E] $\mathbb{Z}_2$ REFLECTION + μ_4 INHERITED:

the flattened modular Hamiltonian K_{flat}=sign(K) inherits $\Theta K_{\text{flat}}\Theta$=$-K_{\text{flat}}$ (u_Θ=J) and $[\rho, K_{\text{flat}}]$=0 EXACTLY (sign is odd; ρ commutes with any $f(K)$), with side-even and off-μ_4 neg controls firing. **[C][O]** the chiral c_-=8≠0 (FHS Chern $|C|$=1) obstructs STRICT finite-range commuting-projector locality (Kapustin–Fidkowski, `arXiv:1810.07756`; C≠0 forbids exponentially-localised Wannier), so the realisation is the quasi-local LTO net NPW26 use — the form sub-step (i) asks for. Advances `v424` sub-step (i); `SEAM.EQUIV.01` stays **[O]** (the continuum theorem MMST `v336` is the backbone). Python-only (numpy); registered in `run_all.py`, the registry, the ledger and cited in `tfpt_research_contracts` (route (xxvii) of the convergent attacks).

2026-06-30 (v454–v459 + Lean + Wolfram — the post-F "G-block": the lattice Sugawara current algebra, a uniform-in-N Tomita–Takesaki tower, chirality forced by P_1, an $(E_8)_1$ character kill test, the exact MMST citation audit and the complementary lattice-VOA route)

- **`v454_seam_edge_virasoro.py` (`SEAM.EQUIV.EDGE.VIRASORO.01`).** Tests the cited MMST "Koo–Saleur lattice generators → continuum Virasoro with the correct c" fact on our own free-fermion edge (G2). **[E]** the Affleck–Cardy Casimir energy fits c_{Dirac}=1.000 ($\frac{1}{2}$ per Majorana, L=64..1024); **[E]** the lattice $U(1)$ current closes its affine level exactly ($\langle 0|J_n J_{-n}|0\rangle$=$n$, n=1..8; a gapped control vanishes); **[E]** the level-1 Sugawara c=$\dim G/(1+h^\vee)$=8 for *both* $SO(16)_1$ (120/15) and $(E_8)_1$ (248/31). **[C]** $16\cdot\frac{1}{2}$=8=g_{car}+N_{fam} is the level-1 current-algebra c_-; `SEAM.EQUIV.01` stays **[O]**. Sugawara Wolfram-mirrored.
- **`v455_seam_bw_uniform.py` (`SEAM.EQUIV.BW.UNIFORM.01`).** A uniform-in-N Tomita–Takesaki tower for the intrinsic Bisognano–Wichmann condition u_Θ=J — the inductive-limit companion to `v426`, advancing `v424` sub-step (i) (G1). **[E]** $\|\Theta K\Theta+K\|$=0 exactly at every N (N=8..128); **[E]** an N-independent gap $6\log\frac{3}{2}$; **[E]** $[\rho, K]$=0 exactly at every N; **[E]** an N-independent neg-control margin $2\|K\|$. **[C][O]** the inductive limit inherits u_Θ=J; `SEAM.EQUIV.01` stays **[O]**. Python-only.
- **`v456_seam_chirality_from_c3.py` (`SEAM.S3.FROM-P1.01`).** The edge chirality (c_-≠0, the S3 phase) is *forced* by the one-sidedness that defines c_3=$1/(8\pi)$ (P1) (G6). **[E]** c_3's "8" is the one-sided count $8=|\mathbb{Z}_2|\cdot(\int K/\pi)=2\cdot 4$; **[E]** a reflection sends the Chern integer $C\to -C$ ($+1\to -1$ on the p+ip model); **[E]** a two-sided boundary forces C=$-C$=0. **[C]** the one-sided seam (no reflection) forces C≠0, magnitude c_-=8=rank E_8 — S3 is a consequence of P_1; `SEAM.EQUIV.01` existence stays **[O]**. The arithmetic is Wolfram-mirrored.
- **`v457_seam_character_killtest.py` (`SEAM.EQUIV.KILLTEST.01`).** An explicit character/-sector kill test of $(E_8)_1$ against $SO(16)_1$ (G7). **[E]** the $(E_8)_1$ character χ=E_4/η^8 with tower degeneracies $\{1, 248, 4124, 34752, \dots\}$; **[E]** the weight-1 count 248=120+128 (not the free 120); **[E]** the sector count $|\det \text{Cartan}\, E_8|$=1 (not $|\det \text{Cartan}\, D_8|$=4). **[X]** a measured 120, four sectors, or a wrong coefficient would *falsify* $(E_8)_1$; the data give $248/1/\{1, 248, 4124\}$ (passed). `SEAM.EQUIV.01` stays **[O]**. Wolfram-mirrored.
- **`v458_seam_mmst_citation_audit.py` (`SEAM.EQUIV.MMST.AUDIT.01`).** The exact hypothesis-by-hypothesis citation audit of MMST (`arXiv:2107.13834`) against the collar (G3, the "exakter Zitat-Abgleich"). **[C]** H1 CAR class, H2 massless limit, **[E]** H3 per-copy c=$\frac{1}{2}$ (Thm 4.11), **[C]** H4 decoupled additivity to c=8, H5 the WZW range $8\le 8\le 16$ with the 120 bilinear currents all match; **[O]** the single residual H6 is the 128-spinor extension $SO(16)_1\to(E_8)_1$ ($\det K$ 4→1), outside MMST's bilinear method. `SEAM.EQUIV.01` stays **[O]**, its open part narrowed to this one holomorphic extension.
- **`v459_seam_lattice_voa_route.py` (`SEAM.EQUIV.LATTICEVOA.01`).** The second, independent route to $(E_8)_1$ supplying exactly the residual of `v458` (G5). **[E]** E_8 even unimodular (det =1); **[E]** the weight-1 space 248=8+240; **[E]** the 240 roots split 112 integer +

128 half-integer (the spinor, 248=120+128). [C] the lattice net A_{E_8} (Adamo–Giorgetti–Tanimoto, `arXiv:2506.01008`; OS `arXiv:2407.18222`) constructs the holomorphic extension; [O] the two routes leave one shared realisation input (collar = A_{E_8}, tied to P_1 by `v456`). `SEAM.EQUIV.01` stays [O]. Wolfram-mirrored.

- **Lean (`FORM.SEAM.RESIDUAL.01`, `TfptCarrier/SeamResidualAxiom.lean`).** Kernel-checks the whole G-block arithmetic (the one-sided $8{=}|\mathbb{Z}_2|\int K/\pi$, the reflection $C{\to}{-C}$, $c_-{=}8$, the MMST range $8{\le}8{\le}16$, decoupled additivity, 248=8+240=120+128, 240=112+128, det K 1 vs 4) by `decide`, reducing the `SEAM.EQUIV.01` residual to ONE named TFPT-internal realisation axiom (`CollarRealizedAsLatticePhase`) plus two cited external theorems (MMST `v458`, AGT `v459`); `lake build` OK.
- **Wolfram + integration.** Six exact mirrors added to `tfpt_readouts_extension.wl` (Sugawara $c{=}8$ for 120/15 and 248/31; the one-sided $8{=}2\cdot 4$ and reflection $C{=}{-C} \Rightarrow 0$; the $(E_8)_1$ tower $\{1, 248, 4124, 34752\}$ with 248=120+128 and $|\det \mathrm{Cartan}|$ 1 vs 4; the root split 112+128 and 248=8+240) — extension total now 360/360. Registry, ledger, master index, website mirror and manifests updated in the same change.

2026-06-30 (v449–v453 + Lean + Wolfram — the edge CFT named end-to-end: uniform-in-N convergence, three independent reads of $c_-{=}8$, the Ising operator tower, the $(E_8)_1$ modular data, and μ_4-symmetry derived from the four marks)

- **`v449_seam_mmst_uniform.py` (`SEAM.EQUIV.MMST.UNIFORM.01`).** Strengthens the cited MMST hypothesis from convergence at a single cross-ratio (`v444`/`v448`) to the *uniformity* the theorem actually asserts, on the same `v367`/`v368` collar. [E] cross-ratio A $Q(1,2,4,7)/12 \to 2{+}\sqrt{3}$, the deviation shrinking monotonically; [E] an *independent* cross-ratio B $Q(1,3,5,9)/12 \to$ its own value 2, so convergence is not a tuned point; [E] a uniform $1/N$ rate $|Q(N){-}Q_\infty| \cdot N \le 4$ for *both* observables at every $N_x \in \{36, 48, 60\}$ (one N-independent constant); [E] the bulk control is off-scale. [C] the MMST uniform-convergence hypothesis is empirically satisfied; `SEAM.EQUIV.01` stays [O].
- **`v450_seam_edge_entanglement.py` (`SEAM.EQUIV.EDGE.ENTANGLEMENT.01`).** A *third* independent reading of the edge central charge (after the correlator `v444` and the topology `v447`). [E] the Calabrese–Cardy entanglement law gives $c_{\mathrm{Dirac}}{=}1.000$ for the critical free-fermion chain; [E] so c per Majorana = 1/2; [E] a gapped control saturates ($c{=}0$). [C] bulk–edge: $c_-{=}N_{\mathrm{Maj}} \cdot \frac{1}{2}{=}16 \cdot \frac{1}{2}{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$. `SEAM.EQUIV.01` stays [O].
- **`v451_seam_edge_cardy_tower.py` (`SEAM.EQUIV.EDGE.CARDY.01`).** Names the edge CFT by its *operator content*. [E] the Cardy finite-size spectrum gives the spin dimension $h_\sigma{=}1/16$ (the periodic–antiperiodic ground-energy split) and [E] the energy dimension $h_\epsilon{=}1/2$ (the lowest NS pair excitation), both [E] L-independent (conformal $1/L$ scaling). [C] $\{c, h_\sigma, h_\epsilon\}{=}\{\frac{1}{2}, \frac{1}{16}, \frac{1}{2}\}$ is the chiral Ising (free-Majorana) minimal model $M(3,4)$; 16 copies give $c_-{=}8$. `SEAM.EQUIV.01` stays [O].
- **`v452_seam_e8_modular.py` (`SEAM.EQUIV.E8.MODULAR.01`).** The $(E_8)_1$ torus modular signature, extending `v156`'s q-expansion to the transformation data. [E] the character $\chi{=}E_4/\eta^8$ is S-invariant $\chi(-1/\tau){=}\chi(\tau)$ (one primary, $S{=}[1]$); [E] T-phase $\chi(\tau{+}1)/\chi(\tau){=}e^{-2\pi i/3}{=}e^{-2\pi i c/24}$ ($c{=}8$); [E] leading power $\chi\, q^{1/3} \to 1$ so $q^{-c/24}{=}q^{-1/3} \Rightarrow c{=}8$; [E] holomorphic $c{=}8 \equiv 0 \bmod 8$ with T-phase order $24/\gcd(8,24){=}3$. [C] the $(E_8)_1$ identity is now pinned on the torus *and* the edge. `SEAM.EQUIV.01` stays [O].
- **`v453_seam_mu4_from_marks.py` (`SEAM.RIGIDITY.MU4FROMMARKS.01`).** Folds the last structural premise of the rigidity chain (`v446`'s residual `QGEO.SYM.01`) into the existing realisation premise. [E] the four Gauss–Bonnet marks *are* $\mu_4{=}$ roots of $z^4{-}1$; [E] the order-4 clock $z \to iz$ is a 4-cycle permuting them (the `v445` clock); [E] it preserves the cross-ratio $\lambda{=}2$ (in the D_4 stabiliser, `v168`); [E] the form basis $\omega_k{=}z^{k-1}dz/(z^4{-}1) \to i^k\omega_k$ is μ_4-graded

(weights $(1,2,3)$=A_3 exponents). [C] so `QGEO.SYM.01` follows from marks= μ_4 + the existing `QGEO.REALIZE.01` — no new premise; the chain (v453→v446→v445) closes modulo realisation. `SEAM.EQUIV.01` stays [O].

- **Lean: `TfptCarrier/SeamEdgeChern.lean` (`FORM.SEAM.EDGE.CHERN.01`).** Kernel-hardens the integer backbone the four edge reads converge on: the FHS Chern integers $(+1/0/-1)$, the bulk–edge channel count $|C|$=1, N_{Maj}=$2^{g_{\mathrm{car}}-1}$=16, the assembly $2c_-$=$N_{\mathrm{Maj}}\cdot|C|$ (so c_-=8), c_-=8=g_{car}+N_{fam}=rank E_8, holomorphic $c\equiv 0$ mod 8 and the order-3 modular T-phase — all by kernel `decide` (`edge_kernel_facts`, axioms `propext` only). `lake build` OK, `#print axioms` clean.
- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains nine exact checks (v452 modular data ×4, v450 entanglement assembly, v451 Ising Kac weights, v453 marks= μ_4 ×3): 345 → 354 checks, all passing (`GATE.WOLFRAM.02`). E_4 is evaluated via its q-series since `EisensteinE` stays symbolic for complex τ.
- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) and the research-contracts note record the edge CFT as named end-to-end — central charge c_-=8 from three independent observables (correlator v444, topology v447, entanglement v450), the Ising operator tower (v451), the $(E_8)_1$ torus modular data (v452), and the μ_4 premise derived from the marks (v453) — while the only residual that stays [O] remains the cited continuum-existence theorem (v336).

2026-06-30 (v446/v447/v448 + Lean + Wolfram — deriving clock-invariance from μ_4-symmetry, an independent topological reading of the edge c_-=8, and pinning the one external MMST fact at four-point order)

- **`v446_seam_clock_invariance.py` (`SEAM.RIGIDITY.CLOCKINV.01`).** Closes v445's single residual one notch further: the operator condition $[\rho,\Lambda_\Sigma]$=0 (clock-invariance) is shown to be a *consequence* of a structural symmetry, not an independent assumption. [E] the OS canonical transfer Λ=$C(1)C(0)^{-1}$=e^{-h} (fixed by RP+reflection, v54) of a μ_4-invariant RP covariance satisfies $[\rho,\Lambda]$=0 (swept N=4..32); [E] a μ_4-breaking covariance gives $[\rho,\Lambda]\neq 0$ (iff-sharp neg control); [E] Λ is a positive contraction (spectrum in $(0,1]$); [E] the free modular Hamiltonian K= $\log((1-C)C^{-1})$ (v258) of the invariant state has $[\rho,K]$=0 (Tomita–Takesaki), broken control $\neq 0$. [C] so with v445 (commute ⇔ block-diagonal), RP-definability *plus* the four marks *force* the block-diagonal Λ_Σ; the residual reduces from an operator commutation to "the seam RP data is μ_4-symmetric" (`QGEO.SYM.01`). `SEAM.EQUIV.01` stays [O].
- **`v447_seam_edge_chern.py` (`SEAM.EQUIV.EDGE.CHERN.01`).** An *independent* (topological) reading of the edge central charge, corroborating v444's correlator route from a different observable. [E] the M=1 collar has integer Chern number C=+1 (Fukui–Hatsugai–Suzuki link-variable method, no fit); [E] C=0 in the trivial phase and C=−1 at M=−1 (opposite chirality); [E] the open strip carries exactly $|C|$=1 chiral edge branch per boundary; [E] the Li–Haldane entanglement spectrum has in-gap modes at 1/2 only in the topological phase. [C] bulk–edge: c_-=$N_{\mathrm{Maj}}\cdot\frac{1}{2}\cdot|C|$=$16\cdot\frac{1}{2}$=8=$g_{\mathrm{car}}$+$N_{\mathrm{fam}}$ — the same c_-=8 as v444. `SEAM.EQUIV.01` stays [O].
- **`v448_seam_edge_fourpoint.py` (`SEAM.EQUIV.EDGE.FOURPOINT.01`).** Pins the one external MMST fact (continuum existence of the chiral scaling limit) empirically at the next order. [E] the ground-state correlation matrix is an exact projector G^2=G ($\sim 10^{-13}$), so the state is quasi-free (the MMST CAR hypothesis holds); [E] the edge four-point cross-ratio Q=$[G_{01}G_{23}]/[G_{02}G_{13}]$ at commensurate positions is Cauchy-convergent under refinement; [E] Q converges to the free-fermion conformal cross-ratio $\sin(5\pi/12)/\sin(\pi/12)$=$2+\sqrt{3}\approx 3.732$; [E] the bulk control is off-scale (exponential). [C] so the one external fact is supported at *both* two-point (v444) and four-point order. `SEAM.EQUIV.01` stays [O].

- **Lean:** `TfptCarrier/SeamRigidityForcing.lean` **(**`FORM.SEAM.RIGIDITY.FORCING.01`**).** Kernel-hardens `v445` (and `v442`'s uniformity): the entrywise forcing iff clock4 a=clock4 $b \Leftrightarrow a{=}b$ (i.e. $i^a{=}i^b \Leftrightarrow a \equiv b \bmod 4$, residue-only hence uniform in N), the four distinct eigenphases, the exact commutant dimension $\sum n_s^2{=}64{<}256$ and the order discriminator (128>64) are all proved by kernel `decide` (`forcing_kernel_facts`, no domain axioms); only the operator-level lift to the full L^2 is the named premise. `lake build` OK, `#print axioms` clean.

- **Wolfram mirror.** `tfpt_readouts_extension.wl` gains three exact `v445` checks (entrywise forcing for all $a, b \in 0..31$; commutant dimension 64<256; order-2 commutant 128>64 swept N=4..64): 342 → 345 checks, all passing (`GATE.WOLFRAM.02`).

- **Papers.** The adversarial audit (`tfpt_5_redteam`, Target A) records the five-direction narrowing of the `SEAM.EQUIV.01` keystone (edge scaling `v444`, forcing converse `v445` + Lean, derived clock-invariance `v446`, independent topological $c_{-}{=}8$ `v447`, four-point pinning `v448`) explicitly as "not promoted": the only residual that stays **[O]** is still the cited continuum-existence theorem (`v336`). The research-contracts note lists the same as "eleven convergent attacks".

2026-06-30 (`v445` — the rigidity FORCING converse: commuting with the order-4 clock is *exactly* μ_4-block-diagonality, upgrading the rigidity step from "permit" to "force")

- `v445_seam_rigidity_forcing.py` **(**`SEAM.RIGIDITY.FORCING.01`**).** The rigidity route (`v398`/`v442`) showed a μ_4-block-diagonal K *commutes* with the order-4 clock $\rho{=}\,\mathrm{diag}(i^n)$. This module proves the *converse* and the exact structure, isolating the single remaining physical premise. **[E]** FORCING (entrywise iff): $[\rho, K]{=}0$ iff $K_{ij}{=}0$ whenever $i{\not\equiv}j \bmod 4$, verified entrywise and swept N=4..128 — commuting *forces* μ_4-block-diagonality, it does not merely permit it. **[E]** EXACT COMMUTANT DIMENSION: $\dim\{K : [\rho, K]{=}0\}{=}\sum_s n_s^2$ (the ad_ρ nullspace), a proper subspace of the full N^2 algebra (N=16: 64<256) — a genuine quantitative restriction. **[E]** OFF-BLOCK ⇔ NON-COMMUTING: every off-block matrix unit has $\|[\rho, E_{ij}]\|{\geq}\sqrt{2}{>}0$ (zero slack). **[E]** THE ORDER IS THE DISCRIMINATOR: the order-4 clock gives 4 sectors; the order-2 clock leaves a strictly larger commutant (N=16: 64 vs 128) — the four Gauss–Bonnet marks (order $4{=}|\mu_4|{=}h(A_3)$) force strictly more structure than two, and only the index-4 extension is $(E_8)_1$ (det =1; `v281`/`v154`/`v351`). **[E]** NONDEGENERACY: exactly 4 distinct eigenvalues $\{1, i, -1, -i\}$ for all $N{\geq}4$. **[C]** SYNTHESIS: commuting with the order-4 μ_4 clock is *equivalent* to μ_4-block-diagonality, so RP-definability (`v54`/`v398`) + clock-invariance *force* the block-diagonal Λ_Σ — `v442`'s "permit" upgraded to "force". **[O]** the single residual is now precisely that the physical transfer lies in that commutant (clock-invariance = `QGEO.SYM.01`, `v323`/`v335`); `SEAM.RIGIDITY.01`/`SEAM.EQUIV.01` stays **[O]**. Python-only (numpy; det facts Wolfram-mirrored via `v89`/`v281`/`v422`). Registered in `run_all.py`, `script_registry.csv`, the ledger, and cited in `tfpt_research_contracts` (route (xii) of the convergent attacks, and the sharpened residual in the Seam State Rigidity contract).

2026-06-30 (Lean kernel-hardening of two of the seven `SEAM.EQUIV.01` routes — the Borchers/Wiesbrock standard-pair relations and the applicability-ledger count, machine-verified with clean axioms; rigor without new mathematics)

- **Context.** Two of the seven convergent attacks on the keystone residual are turned from numerical witnesses into kernel-checked facts, in the audit-contract style of `SeamScalingLimit.lean` / `BWKeystone.lean`: the load-bearing algebra/bookkeeping is proved by the Lean kernel, while the cited theorems and the one open continuum input remain named axioms. `lake` build OK; #print axioms clean. Tracked as two new FORM.* ledger rows; \veri-context in tfpt_research_contracts; Lean manifest refreshed.

- **FORM.SEAM.BW.HSMI.01 (TfptCarrier/SeamStandardPair.lean) — standard-pair relations, kernel-checked (hardens v438).** The four Borchers/Wiesbrock relations on the centred 5-mode ladder over $\mathbb{Z}$ are proved by kernel decide (axioms propext / Classical.choice / Quot.sound only, no domain axiom): $[K,P]{=}P$, $J^2{=}1$, $JKJ{=}{-K}$, $JPJ{=}P_+$ (and $\mathrm{tr}\,K{=}0$). The cited Borchers/Wiesbrock theorem and the one continuum input (the continuum standard pair) are named axioms; the composition pins face (ii) of SEAM.EQUIV.01 to exactly those two — intrinsic Bisognano–Wichmann, no rotation-covariance presupposed.
- **FORM.SEAM.APPLICABILITY.01 (TfptCarrier/SeamApplicabilityLedger.lean) — the audit count, kernel-checked (hardens v441).** The headline bookkeeping is itself machine-verified: of the 12 cited-theorem hypotheses (MMST/NPW26/Adamo, mirrored in order) exactly 11 are internal and 1 external (total_count=12, internal_count=11, external_count=1), plus mmst_range ($8 \le 8 \le 16$) and npw_outside_bucket ($\det K{=}1 \neq 4$) — ALL with NO axioms (pure kernel decide). The single external fact (continuum scaling-limit existence) is the named axiom; the keystone reduction pins its external dependence to exactly that one fact.
- **Net.** Both modules are imported by the root TfptCarrier.lean, axiom-audited in AxiomCheck.lean, and signature-locked in AuditContract.lean; full lake build succeeds. No new mathematics — the same facts v438/v441 verify numerically are now kernel-proved; SEAM.EQUIV.01 stays [O].

2026-06-30 (the chiral-edge scaling motor v444 — the edge half of the one SEAM.EQUIV.01 continuum residual: the lattice edge already carries the conformal data of a chiral $c{=}1/2$ Majorana CFT and converges under refinement, grounding the EDGE the way v439 grounded the BULK; the keystone stays [O])

- **Context.** v439 grounded the *bulk* half of the cited MMST existence (uniform gap, exponential clustering, finite light cone) and explicitly narrowed the one open analytic step to the chiral massless *edge* scaling limit. v444 attacks exactly that residual on the SAME explicit v367/v368 $p{+}ip$ collar. vN-registered, runs in run_all.py (suite ends ALL CHECKS PASSED), \veri-cited in the body of tfpt_research_contracts (the G_{net} contract). Python-only (numpy; no new exact identity to Wolfram-mirror).
- **v444_seam_edge_scaling.py (SEAM.EQUIV.CONTINUUM.EDGE.01) — chiral-edge motor.** The lattice edge carries the kinematic and conformal data of a chiral $c{=}1/2$ Majorana CFT: [E] a linear, chiral in-gap dispersion (velocity $v{\approx}0.98$, $R^2{=}0.9999$, with *opposite* per-edge velocities — one chiral Majorana per edge); [E] an *algebraic* equal-time edge two-point $\sim r^{-\eta}$ with $\eta{\approx}1$, versus the *exponential* bulk row and the exponential trivial-phase ($M{=}3$, no edge) control; [E] a Cauchy-convergent scaling-collapse under refinement ($\eta{=}1.001{\pm}0.007$, successive profile spread $0.028{\to}0.013$ across $N_x{=}36..60$) — the numerical hallmark of an existing scaling limit. [C] $\eta{=}1 \Leftrightarrow$ one chiral Majorana ($c{=}1/2$) per edge, so the 16-copy carrier has a chiral $c_-{=}8$ $(E_8)_1$ edge CFT ($c_-{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$). [O] the rigorous uniform-convergence theorem (cited MMST, v336) is not supplied.
- **Net.** With v439 (bulk) and v444 (edge), *both halves* of the one continuum residual are numerically grounded on our explicit model and pinned to a single cited existence theorem; SEAM.EQUIV.01 stays [O]. Ledger row, script_registry.csv index and the generated surfaces (verification.tex, ScriptIndex.tsx) updated in the same change; manifest refreshed last.

2026-06-30 (six convergent attacks on the one SEAM.EQUIV.01 continuum residual — intrinsic Bisognano–Wichmann, a Lieb–Robinson continuum motor, the β-homotopy, an applicability audit, uniform rigidity, and falsifiable kill tests; none closes the keystone, together they reduce it to a single analytic fact and make the claim falsifiable)

- **Context.** The keystone `SEAM.EQUIV.01` is "closed modulo a cited theorem": its one open input is the continuum existence of the seam's chiral massless scaling limit. Six new modules attack that residual from independent directions; each is honest about *not* closing it. All six are vN-registered, run in `run_all.py` (suite ends `ALL CHECKS PASSED`), and are `\veri`-cited in the body of `tfpt_research_contracts` (the G_{net} contract). Python-only (no new exact identity to Wolfram-mirror).
- **`v438_seam_hsmi_borchers.py` (`SEAM.EQUIV.BW.HSMI.01`) — intrinsic BW.** The seam supplies a Longo *standard pair* / +half-sided modular inclusion (positive lightlike translation $P \geq 0$, boost K with $[K, P]{=}P$, reflection J with $JKJ{=}{-}K$, $JPJ{=}P_{-}$), so by Borchers' theorem the modular flow is geometric *intrinsically* — *without* presupposing rotation-covariance of the vacuum (complementary to Adamo–Giorgetti–Tanimoto). **[E]** ladder algebra exact, $sl(2, \mathbb{R})$ positivity (min eig > 0); **[C]** synthesis; **[O]** the continuum standard pair.
- **`v439_seam_lieb_robinson.py` (`SEAM.EQUIV.CONTINUUM.LR.01`) — continuum motor.** The explicit gapped $p{+}ip$ collar (`v367`) has a uniform-in-N gap, exponential clustering with N-independent ξ, and a finite Lieb–Robinson light cone, so by Nachtergaele–Sims its *bulk* thermodynamic limit exists — grounding the bulk half of the cited MMST existence in our model and narrowing the open step to the chiral *edge* scaling limit. **[E]**/**[C]**, **[O]** the edge CFT.
- **`v440_seam_lto_rp_beta.py` (`SEAM.EQUIV.LTORP.BETA.01`) — β-homotopy.** The (LTO-RP)/BW data ($\Theta K \Theta{=}{-}K$, $[\rho, K]{=}0$) are β-independent, and along the whole path $C_\beta{=}1/(1{+}e^{\beta K})$ from the NPW26-proven *trace* state ($\beta{=}0$) to the seam KMS state ($\beta{=}1$) every covariance is reflection-related, μ_4-invariant and valid — a norm-continuous, obstruction-free homotopy, upgrading the single point of `v426` to a path argument for sub-step (ii) of `v424`.
- **`v441_seam_applicability_ledger.py` (`SEAM.EQUIV.APPLICABILITY.01`) — audit.** A hypothesis-by-hypothesis ledger of the three cited theorems (MMST / NPW26 / Adamo–Giorgetti–Tanimoto) finds 11 of 12 hypotheses are TFPT-internal computed facts and exactly one is external — the continuum existence of the chiral scaling limit; covariance is an Adamo consequence (defuses the `v215` circularity), not an input. The keystone's external dependence is pinned to that one analytic fact.
- **`v442_seam_rigidity_uniform.py` (`SEAM.RIGIDITY.UNIFORM.01`) — uniform rigidity.** The band-limited μ_4-character rigidity of `v398` is uniform in N: exact 4-sector commutation to $N{=}256$ (not merely the $N \leq 64$ sweep), a uniform gap, and an N-independent clock-breaking lower bound $\|[\rho, V]\|{=}2$ — so it is not a small-N artifact (the full-L^2 forcing stays **[O]**).
- **`v443_seam_e8_discriminator.py` (`SEAM.E8.DISCRIMINATOR.01`) — kill tests.** Mode D of the closure-mode classification (`v347`) as concrete *falsifiable* discriminators against the nearest $c{=}8$ rival $SO(16)_1$: torus ground-state degeneracy 1 vs 4, level-1 currents 248 vs 120, lattice theta 240 vs 112 norm-2 roots; $c{=}8$ alone cannot decide (`v277`), only $\det K{=}1$ (the genus-1 statement, `v90`) selects E_8. A seam with GSD>1 would falsify $(E_8)_1$. **[X]**/**[C]**.
- **Status discipline.** `SEAM.EQUIV.01` stays **[O]** throughout; the six modules reduce, audit and falsify the residual but do not supply the one cited continuum-existence theorem. Ledger rows, the `script_registry.csv` index, and the generated surfaces (`verification.tex`, `ScriptIndex.tsx`) are updated in the same change; manifest refreshed last.

2026-06-30 (shadow-export coherence — the `tfpt5-shadow` mirror is now reproducible *as shipped*: the audit is shadow-aware, the content maps are non-destructive on the website-free subset, and an export gate blocks any incoherent mirror)

- **Context (reviewer finding, Alessandro).** On the exported `tfpt5_shadow.zip` the independent surfaces reproduced (`build.sh notes` → `11 ok`; Wolfram base/extension; Red Team; Lean transcript), but the *package* layer did not: `make_manifest.py -check` reported a stale `verification/website_map.csv` and `build.sh audit` crashed because it unconditionally required `website/lib/release.ts`, which the shadow subset deliberately excludes.
- **`make_docs_map.py` is now shadow-aware.** `website_map.csv` mirrors the `website/` sources; on a tree without `website/` the generator now *skips* it instead of regenerating it down to a single `README.md` row. This removes the silent corruption that made `build.sh notes/gen` leave `website_map.csv` stale against the manifest on the export (it now matches the established skips in `make_script_index.py` and `make_changelog_web.py`).
- **`audit_sync.py` is now shadow-aware.** When `website/` is absent it skips the website mirror (section D), the website version stamps (E) and the website Wolfram-count card (F), and the website generated-surface freshness in (A), printing exactly which sections were skipped. `build.sh audit` therefore runs to a clean `AUDIT OK` on the shadow subset instead of raising `FileNotFoundError`.
- **Export coherence gate.** `scripts/export-shadow.sh` now re-runs `make_manifest.py -check` and the shadow-aware `audit_sync.py` *on the filtered tree* before the mirror is published. The manifest check catches the earlier `v235-v298` symptom at the source: a module referenced by `run_all.py/script_registry.csv/manifest.sha256` but not committed (and thus dropped by `git ls-files`) now fails the export loudly instead of shipping a package that says `ALL CHECKS PASSED` while `run_all.py` stops at the first missing script.
- **Manifest refreshed.** `manifest.sha256` was regenerated as the last release step so the shipped digests match the committed `changelog.tex` and `website_map.csv`; the full Python suite reproduces end-to-end on the regenerated export. No theory content, status marker or `vN` module changed — this is an export/infrastructure fix only.

2026-06-29 (discoverability — an SEO/GEO pass on the public website: a generated `/llms.txt`, explicit AI-crawler `robots.txt`, query-targeted metadata, and the first real backlinks from the README and the Zenodo deposit)

- **New `/llms.txt` context file** for AI / answer engines (ChatGPT, Perplexity, Claude, Gemini), generated at build time from the same data layer as the site (`papers.ts`, `predictions.ts`, `version.ts`) so it never drifts. It carries the one-line summary, the "claims / does not claim" split, the document set with PDF links, the headline predictions, and an explicit *disambiguation* from the unrelated Brouwer–Lefschetz "topological fixed point theory" of mathematics.
- **`robots.txt`** now lists the major search and AI crawlers (Googlebot, Google-Extended, Bingbot, GPTBot, ChatGPT-User, OAI-SearchBot, PerplexityBot, ClaudeBot, anthropic-ai, . . .) explicitly as allowed; the site-wide meta description and keywords are reworded toward natural search queries ("derivation of the fine-structure constant", "Standard Model from first principles", "where does 137 come from") and carry the same physics/mathematics disambiguation.
- **Backlinks (the missing piece).** The root `README.md` and the Zenodo deposit description (`zenodo_description.html`) previously contained *no* outbound links; both now link to `fixpoint-theory.com` and the source repository, and the README gains a citation block with the archived deposit DOI `10.5281/zenodo.20846087`.

- Website / documentation infrastructure only — no theory, suite, ledger or status change.
- **New module v437 (`E8.DEGREE.JOINT.01`) — a consolidation of v6/v66/v355/v431 that derives more from the E_8 Casimir degrees *without* crossing the v354/v355 anti-numerology line.** It adds *no* new per-degree coincidence — only one joint lock and a reconstruction.
  - [E] **all of E_8 is fixed by its degree multiset.** From $\deg(E_8) = \{2, 8, 12, 14, 18, 20, 24, 30\}$ alone: rank $= \#\deg = 8$, $h = \max\deg = 30$, #positive roots $= \sum \exp = 120$, #roots $= 240$, $\dim E_8 = 248$, $|W(E_8)| = \prod \deg = 696{,}729{,}600$, $\sum \deg = 128 = \dim S^+$.
  - [E] **the joint quadratic (the new lock).** The two TFPT structural integers $(g_{\mathrm{car}}, N_{\mathrm{fam}})$ are the *unique* root pair of $x^2 - (\mathrm{rank}\, E_8)\, x + (h/2) = x^2 - 8x + 15 = (x-3)(x-5)$: sum of roots = rank $E_8 = 8$ (the rank-fill, v6), product $= h/2 = 15 = g_{\mathrm{car}} N_{\mathrm{fam}}$ ($2g_{\mathrm{car}}N_{\mathrm{fam}} = h$, v431); $\{3, 5\}$ is the unique positive-integer factor pair of 15 summing to 8. So $g_{\mathrm{car}} = 5$ and $N_{\mathrm{fam}} = 3$ are forced *together* by two E_8 degree-invariants, not chosen one at a time. The three primary-readout degrees $\{2, 8, 30\}$ are exactly $\{\min, \mathrm{rank}, \max\}$.
  - [E] **discipline upheld:** no new per-degree mining beyond the joint quadratic; the five non-primary degrees keep their v431 family homes; the v354/v355 line is not crossed.
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`E8.DEGREE.JOINT.01`); cited in the `tfpt_5` degree-audit winbox beside v431; mirrored in `wolfram/tfpt_readouts_extension.wl` (extension 339 → 342). Exact integer/lattice; no new physical number, no status change.

2026-06-29 (v436 — hardening the unconditional "is it numerology?" floor to an assumption-minimal counting floor; plus a literature note grounding the α^3 Chern level)

- **New module v436 (`OVERDET.FLOOR.02`) — it *hardens* `OVERDET.FLOOR.01` (v432) so the numerology verdict no longer rests on any subjective chance assignment.** v432's nominal $\sim 10^{-10}$ multiplies the counting census by hand-chosen plausibilities (input-"8" independence 0.1^3, seed 0.012, kaon 0.15) a skeptic can contest; v436 shows the conclusion survives without them.
  - [E] **assumption-minimal floor.** The α cubic-class census (v100 layer D) is *pure counting*: of $N = 94500$ complexity-matched $F_{U(1)}$ variants only *one* lands in the 4×10^{-8} CODATA window — the TFPT instance, reproducing $\alpha^{-1} = 137.0359992$ — so $p_\alpha \le 1/94500 = 1.06 \times 10^{-5} \sim 4.40\sigma$ with *no* subjective probability, prior or multi-observable grammar.
  - [E] **monotone concession ladder.** $-\log_{10} p$ from generous to adversarial: full scorecard+$\alpha \sim 30.7$ → scorecard-only ~ 25.8 → α-census-only ~ 4.98; each rung uses strictly less evidence and the "not chance ($> 4\sigma$)" verdict holds at the *most adversarial* rung, so it never rests on one contestable piece. Stripping *all* of v432's subjective factors ($\sim 1.8 \times 10^{-6}$) leaves the floor at 4.40σ — they only sharpen it.
  - [C] **honest 5σ gap.** The conventional 5σ line ($p = 5.7 \times 10^{-7}$) is crossed unconditionally by the census times exactly *one* further independent factor ≤ 0.054 (the input-"8" four-fold over-determination, or one foreign witness) — stated, not hidden. [C] Firewall unchanged: bounds only "is it numerology?", not the physics bridge (v187).
- **Literature note (research log, not a status change).** A `next.txt` entry grounds the α^3 Chern-level reading of v435 in named theorems — Dai–Freed (*η-invariants and determinant lines*, the exponentiated η-invariant as a section of the inverse Quillen determinant line), Bismut–Freed (determinant-line curvature = the α^2 Calderón side) and Redlich/Stora–Zumino (the cubic $\mathrm{Tr}\, F^3$ anomaly descends to a Chern–Simons level, quantised to $\mathbb{Z}$). It records that

the `EM1` proof obligation reduces to constructing the seam Dirac family — i.e. it is a *facet* of `SEAM.EQUIV.01` — and remains [O] external math (no fabricated closure).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`OVERDET.FLOOR.02`); cited in the `tfpt_safeguards` "unconditional floor" keybox beside v432; mirrored on the website. Python-only; no new physical number, no status change.

2026-06-29 (v435 — a fourth honest step on the α^{-1} Quillen target: a π-power test isolates the cubic α^3 as the unique metric-independent (topological) rung, and types its coefficient as a conditional integer level)

- **New module v435 (`ALPHA.QUILLEN.PROGRESS.04`) — it attacks the *single* remaining [O] left after v434 (residual (2), the cubic α^3 Chern/Maxwell moment) and *sharpens* it; it does *not* close it.** The target `ALPHA.QUILLEN.EXACT.01` (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
  - [E] **π-power / metric-independence test.** The three EM-closure coefficients carry π-powers exactly $\{0, 3, 6\}$, equal to their c_3-orders: Maxwell $k_0 = 1 \to \pi^0$, Calderón $-2c_3^3 = -1/(256\pi^3) \to \pi^3$, transport $-8b_1c_3^6 = -41/(327680\pi^6) \to \pi^6$. So the Maxwell a^3 is the *unique* π^0/c_3^0 (metric-*independent*) term. A heat-kernel boundary coefficient always carries the $(4\pi)^{-d/2}$ measure; a metric-independent term carries none — the signature of a *topological* term. The test thus partitions the closure into *one* topological level (a^3) + *two* heat-kernel boundary terms, confirming v342/v434's "a^3 is a level, not a Seeley–DeWitt curvature integral" from a *test* rather than a claim.
  - [C] **conditional level quantisation.** *If* the a^3 term is the seam $U(1)$ Chern–Simons/η boundary level (the `EM1` reading, v48), large-gauge invariance quantises the level to $\mathbb{Z}$, so $k_0 = 1$ is the *unit* (minimal) level — an integer, not a tunable real. This converts "the coefficient happens to be 1" into "the coefficient is an integer level and 1 is the unit".
  - [O] **not closed:** the `EM1` lemma itself — the from-first-principles Chern–Simons boundary proof that $\partial_\tau \log Z_{\text{Maxwell}} = a^3$ — stays external math (type A, v384); the seam F-normalisation stays the one [C] (residuals 1 & 3, v434); α^{-1} stays [E].
- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.04`); cited in the `tfpt_1` EM-closure subsection beside v433/v434; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-28 (v434 — a third honest step on the α^{-1} Quillen target: b_1 is the a_4 heat-kernel coefficient, and the three residuals collapse to one [C] + one [O])

- **New module v434 (`ALPHA.QUILLEN.PROGRESS.03`) — it settles the status of the three residuals named after v433 and shows they are *not* three independent open problems.** The target `ALPHA.QUILLEN.EXACT.01` (v382) stays [O]; $\alpha^{-1} = 137.0359992$ stays [E].
  - [E] **b_1 *is* the $U(1)$ a_4 heat-kernel coefficient (the $\beta = a_4$ theorem).** The one-loop β function of a gauge coupling *is* the a_4 (Seeley–DeWitt a_2 in d=4) heat-kernel coefficient of the charged-field operator (Vassilevich). Computed from the carrier hypercharge content with the standard spin weights ($\frac{2}{3}$ per Weyl fermion, $\frac{1}{3}$ per complex scalar), $\sum \frac{2}{3}Y_f^2 + \frac{1}{3}Y_s^2 = 41/6$ (SM) and $(3/5) \cdot$ that $= 41/10 = b_1$ (GUT). So the b_1 in the counterterm $8b_1c_3^6$ is the a_4 coefficient of the carrier $U(1)_Y$ content (the same number as the β, v159/v246), the *same kind* of heat-kernel object as the α^2 Calderón a_4 (v342) — not a fitted multiplicity.
  - [E] **the geometry factors off:** $8b_1c_3^6 = b_1 \cdot (\operatorname{rank} E_8) \cdot c_3^6$, with $\operatorname{rank} E_8 = 8$ the χ=2 seam boundary count (v216) and c_3^6 the six boundary insertions (the π-power test, v342).

  - [C] **residual (1) ≡ residual (3).** With b_1 and the c_3-powers heat-kernel-forced, the only freedom left in the exact seam a_4 value is how the gauge curvature couples to the boundary measure — the seam F-normalisation. So "the actual seam $a_4 = 8b_1c_3^6$" and "the seam F-normalisation" are *one* [C] obligation, not two.
  - [E] **residual (2) located (not closed):** the determinant-line variation factors $a^3 - 2c_3^3a^2 = a^2(a - 2c_3^3)$; a^2 is the Gilkey gauge-curvature (Quillen/Chern) density (v342), so the leading $a^3 = a \cdot a^2$ is the *first moment* of the Chern density — a topological level at boundary order c_3^0, not a Seeley–DeWitt curvature integral. Its from-first-principles (Chern–Simons-type) origin stays **[O]**.
  - [E] **net reduction:** the EM-Ward residual is *one* [C] (residuals 1 & 3) + *one* [O] (residual 2), not three independent problems — a genuine narrowing of the open surface. **[O]** `ALPHA.QUILLEN.EXACT.01` is *not* closed (the α^3 Chern level and the from-first-principles proof stay external, type A, v384).

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.03`); cited in the `tfpt_1` EM-closure subsection beside v433; mirrored on the website residual-gap lab. Python-only; no new physical number.

2026-06-27 (v433 — a second honest step on the α^{-1} Quillen target: a solvable 4D model reaches the a_4 order, connecting v391 and v342)

- **New module v433 (`ALPHA.QUILLEN.PROGRESS.02`) — a second honest step on the external target `ALPHA.QUILLEN.EXACT.01` (v382); it *grounds and connects*, it does *not* close it.** The target stays [O] (external math, type A, v384); the value $\alpha^{-1} = 137.0359992$ stays [E]. The earlier honest attempt v391 gave a solvable model on a 1D circle whose ζ-determinant variation reached only the low Seeley–DeWitt orders a_0, a_1; separately, v342 grounded the quadratic Calderón term in the Gilkey gauge-curvature coefficient a_4 ($30/360 = \frac{1}{12}$, $\Omega^2/12$). This module supplies the missing link between the two.
  - [E] **a solvable 4D model reaches the a_4 order:** the flat operator $\Delta = -\partial^2 + m^2$ on a 4-torus of volume V has heat trace $\mathrm{Tr}\, e^{-t\Delta} = V/(4\pi)^2\, t^{-2} e^{-tm^2}$, with Gilkey coefficients $a_0 = V/(4\pi)^2$, $a_2 = -Vm^2/(4\pi)^2$ and $a_4 = Vm^4/(2(4\pi)^2)$; the t^0 (conformal-anomaly) order a_4 is *non-zero* — exactly the order v391's 1D circle toy could not reach.
  - [E] **it is the order v342 uses:** so v391 ("a ζ-determinant variation *is* closed heat-kernel data") and v342 ("the seam term *is* the Gilkey a_4") are unified — the variation is heat-kernel data *at the a_4 order*, not only the low orders v391 reached.
  - [E] **the measure is c_3-built:** the universal 4D heat-kernel measure $(4\pi)^{-2} = 1/(16\pi^2) = (2c_3)^2 = 4c_3^2$ in the seam's one-sided 2D-boundary normalisation $c_3 = \frac{1}{2}(4\pi)^{-1}$; and the v382 split is re-verified at $\alpha^{-1} = 137.0359992$.
  - [O] **not closed:** computing the *actual* seam $U(1)$ ζ-determinant a_4 coefficient ($= 8b_1c_3^6$) on the μ_4 seam moduli, and the origin of the cubic α^3 Maxwell/Chern moment, stay open (external math, type A, v384); the exact seam F-normalisation fixing the coefficient *value* stays [C] (v342). A second honest step that grounds and connects, it does not close.

- **Surfaces updated.** Registered in `run_all.py`, `script_registry.csv` and `status_ledger.csv` (`ALPHA.QUILLEN.PROGRESS.02`); cited in the `tfpt_1` EM-closure subsection beside v382/v391; mirrored on the website (the residual-gap lab). Python-only (like v391); no new physical number.

2026-06-27 (v431, v432 — the "unmapped" E_8 degrees are forced spine+flavor structure; the unconditional over-determination floor)

- **New module v431 (E8.DEGREE.LADDER.01) — the reverse-audit "unmapped" E_8 Casimir degrees are NOT diffuse overhead but a forced two-family decomposition.** The structural complement of the reverse audit (v354/v355) and its sheet/deck complement (v430): $\deg(E_8) = 6 \cdot \mathrm{spine}\{2,3,4,5\} \sqcup (\{2\} \cup \text{det-ladder}\{8,14,20\})$. [E] the two-family split is *forced* by $h(E_8) = 30 = 2 \cdot 3 \cdot 5$ — the exponents are the $\varphi(30) = 8$ totatives of 30, hence coprime to 6, hence $\equiv \pm 1 \bmod 6$, so the degrees occupy only the residue classes $\{0, 2\} \bmod 6$. [E] the $6k$ family $\{12, 18, 24, 30\}/6 = \{2, 3, 4, 5\}$ is the v91 spine; the $6k+2$ family $\{8, 14, 20\} = (\det R, \det C, \det L)$ is the winding line $\det M(s, 0) = 6s + 8$ of the flavor diamond (v135). [E] "$18 = 6N_{\mathrm{fam}}$" is not a holdout (spine family), and the clean split is special to E_8 among the simply-laced exceptionals ($E_6/E_7/D_5$ fail). [C] **honest v355 line:** the arithmetic decomposition is exact, but the *functorial* claim "$6k+2$ Casimir stratum = flavor sheet operators" needs a representation-theoretic map and stays [C] (12, 24 admit two canonical readings each) — a structure theorem, not a forced functorial flavor map. It reclassifies the v354 overhead to zero orphan degrees and closes no gate.
- **New module v432 (OVERDET.FLOOR.01) — the unconditional improbability floor and the L1↔L2 bridge.** The complement of the conditional null model (v100) and the multiply-vs-compress framework (v427/v428): the defensible over-determination number that survives even if a skeptic *rejects* the declared formula grammar. [E] compression removed (v428), the genuinely multiplicative pieces are the input "8" forced four independent ways plus the foreign witness $\alpha^{-1} \sim 137$. [C] multiplying only across disjoint pieces (input over-determination × the α census $1/94500$ × the φ_0^{ret}-seed cluster counted once × one downstream witness) gives a floor $\leq 10^{-6}$ across a grid of generous assignments, nominal $\sim 10^{-10}$. [E] that floor sits ~ 20 orders *above* the v100 conditional $10^{-30.7}$; the gap is exactly the evidential value of proving the forms forced, so the α form-derivation (ALPHA.QUILLEN.EXACT.01, v382) and this floor are the *same* lever. [C] **honest scope:** bounds only the "is it numerology?" question, not the physics bridge (firewall v187); a conservative floor, not a posterior.
- *Surfaces synced in the same change:* v431/v432 added to `run_all.py`, `script_registry.csv` (master index + website `ScriptIndex`), `status_ledger.csv`, `tfpt_5_redteam` (reverse-audit winbox + Casimir-degree figure) and `tfpt_safeguards` (the over-determination layer); `v431` mirrored in `wolfram/tfpt_readouts_extension.wl` (v432 is a conservative bound, Python-only).

2026-06-27 (v430 — the seam's "other side" is forced-disjoint from E_8's unmapped Casimir degrees)

- **New module v430 (E8.OTHERSIDE.AUDIT.01) — the sheet/deck complement of the reverse audit (v354/v355).** The reverse audit found five E_8 Casimir degrees $\{12, 14, 18, 20, 24\}$ with no primary readout ("hull overhead"); the double cover has an *other side* (the one-sided $\mathbb{Z}_2$ collar / conjugate half-spinor S^-), so the sharp adversarial follow-up is *"is the leftover where the second sheet lives?"* The answer is a disciplined **negative**. [E] the deck is the degree-2 invariant: the one-sided cover contributes $|\mathbb{Z}_2| = 2$ (the $\frac{1}{2}$ in $c_3 = 1/(|\mathbb{Z}_2|\, 4\pi) = 1/(8\pi)$, v58/v216), and $2 = \min\deg(E_8)$ is the quadratic/metric — one of the *matched* primary readouts $\{2, 8, 30\}$, never one of the unmapped degrees. [E] the two sheets are the 128-spinor: $\dim S^+ = \dim S^- = 16$, the spinor block $128 = \mathrm{rank}\, E_8 \cdot \dim S^+ = 2^{\mathrm{rank}-1} = \sum \deg$, the spinor half of $248 = 120 + 128$; S^- enters 248 as the conjugate $(\overline{16}, \overline{4})$ $(128 = 2 \cdot 64)$, not a spare singlet. [E] forced-disjoint: the sheet/deck set $\{2, 16, 32, 128\}$ has empty intersection with $\{12, 14, 18, 20, 24\}$, its only degree contact the matched 2. [O] each unmapped degree, "explained" from sheet atoms, needs an unforced coefficient and admits ≥ 2 readings (the v355 discriminator), so there is no forced sheet→unmapped-degree identity.

- **A structural negative, not a status change.** It consolidates the S^--dark-matter downgrade (v227) and the no-spare-singlet WIMP no-go of the frontier E_8 scan (`tfpt_4_frontier`): the other side is matched conjugate matter, the five unmapped degrees are the hull overhead, and the two do not meet. The only live "other side" question — whether the 128 phase/glue channel hosts a dark sector (v227) — concerns *matched* structure, not the unmapped five. Closes no gate, upgrades nothing.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`; the reverse-audit Outcome winbox + degree-audit table caption and the Target D winbox of `tfpt_5_redteam`, the magnitude/phase keybox of `tfpt_3_e8_audit_bootstrap`, the WIMP-no-go passage of `tfpt_4_frontier`; the `E8` node of the website verification DAG; the Wolfram extension/README/ledger counts and the website verification count; manifests refreshed.

2026-06-26 (v429 — the 'unmapped' golden E_8 structure is the geometry of the one external input θ_i)

- **New module v429 (`DM.AXION.PENTAGON.01`) — a geometric re-reading answering "why does so much idle E_8 structure exist?" for the one case where the idle structure has a job.** The reverse audit (v354) had argued the golden ratio φ is *unmapped* (it appears only internally — the affine-E_8 attractor angle v312, the icosian lattice v348 — and in no physical readout), and used exactly that to conclude φ cannot be numerology. **[E]** this sharpens that statement: because $N_{\text{fam}}{=}g_{\text{car}}{-}2$, the axion misalignment *spine* angle (v211, `DM.AXION.SPINE.01`) is $\theta_i{=}\pi N_{\text{fam}}/g_{\text{car}}{=}(g_{\text{car}}{-}2)\pi/g_{\text{car}}$, which is *exactly* the interior angle of the regular g_{car}-gon — for $g_{\text{car}}{=}5$ the pentagon, $\theta_i{=}3\pi/5{=}108°$. **[E]** its cosine is golden, $\cos(3\pi/5){=}(1{-}\sqrt{5})/4{=}{-}1/(2\varphi)$ (partner $2\cos(2\pi/5){=}1/\varphi{=}\varphi{-}1$, $\varphi{=}2\cos(\pi/5)$). **[E]** the golden character is *unique* to $g_{\text{car}}{=}5$: among the small regular n-gons only $n{=}5$ has an irrational interior-angle cosine ($n{=}3,4,6$ give $1/2, 0, -1/2$), so θ_i is golden *because* the carrier is 5 (P2).
- **An honest [C] bridge, not a status change.** It connects two standing facts — φ is the unmapped $g_{\text{car}}{=}5$ signature (v354/v313) and $\theta_i{=}3\pi/5$ is the one external cosmological input (v211) — and *refines* v354 from "φ in no physical readout" (it is the unmapped $g_{\text{car}}{=}5$ signature, v313) to "φ touches *only* the [C] spine misalignment angle" (conditional, branch-level, never a frozen prediction). It is a geometric motive for the spine over the hilltop $\theta_i{\approx}170°$ (whose cosine is not a clean pentagon value), consistent with — not replacing — the finite-T solver that decides the branch (v185/v211). It does *not* derive θ_i, does *not* upgrade `DM.AXION.SPINE.01` (stays [C]) and closes no gate.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the dark-matter section of `tfpt_4_frontier`; the Wolfram extension/README/ledger counts (`GATE.WOLFRAM.02`) and the website verification count; manifests refreshed.

2026-06-25 (External-works references appendix — web-verified)

- **New shared appendix `tex-artefacts/tfpt_references.tex`** ("External works built upon"), `\input` into the six math-heavy documents (`tfpt_1`, `tfpt_2`, `tfpt_4`, `tfpt_horizon_readouts`, `origin_theory`, `tfpt_research_contracts`). The active papers named the foundational results they build on by author only, with no reference list; the appendix supplies web-verified locators (journal, volume, year / arXiv) grouped by topic: modular theory & AQFT (Tomita–Takesaki LNM 128 1970; Bisognano–Wichmann J. Math. Phys. 16/17; Osterwalder–Schrader CMP 31/42; Doplicher–Haag–Roberts CMP 23/35; Haag–Ruelle; Longo CMP 126/130; Kawahigashi–Longo–Müger CMP 219), conformal nets/VOAs/lattices (Frenkel–Kac Invent. Math. 62; Segal CMP 80; MMST; Adamo–Moriwaki–Tanimoto;

Adamo–Giorgetti–Tanimoto `2506.01008/2508.07109`; Naaijkens–Penneys–Wallick), lattices/quadratic forms (Conway–Sloane; Siegel Ann. Math. 36 1935), singularities (Brieskorn Invent. Math. 2 1966; Milnor AMS 61 1968) and spectral geometry/gravity/thermal time (Chamseddine–Connes CMP 186; Connes–Rovelli CQG 11; Wald PRD 48; Casini–Huerta–Myers JHEP 05 2011). Added to the `make_manifest.py` TEX list. No claim or status change — a provenance/citation completeness pass.

2026-06-25 (External AQFT citations + the four-fold μ_4 identity sharpened)

- **Two recent Adamo–Giorgetti–Tanimoto theorems cited where the keystone builds on them.** The open `SEAM.EQUIV.01` continuum/realisation legs (v336) now cite, alongside MMST and Adamo–Moriwaki–Tanimoto (`arXiv:2407.18222`, already in use): **(a)** the construction and classification of two-dimensional conformal nets from even lattices (`arXiv:2506.01008`, 2025) — "the lattice is a conformal net" is a theorem of the same operator-algebraic lineage, strengthening the lattice-realisation leg; and **(b)** automatic positive-energy/diffeomorphism covariance of conformal-net representations (`arXiv:2508.07109`, 2025), which defuses the "Bisognano–Wichmann presupposes the covariance it should produce" circularity (v215) on the intrinsic-BW residual (v424). Added to v336 docstring/registry, the `SEAM.EQUIV.CONTINUUM.01` ledger `external_data`, and `tfpt_research_contracts`.
- **The four-fold μ_4 identity made explicit (v422).** v422 already proved that the seam deck-/divisor μ_4, the carrier Galois group $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$, the discriminant $\mathrm{disc}(D_5)=\mathrm{disc}(A_3)=\mathbb{Z}/4$ and the $(E_8)_1$ simple-current glue are ONE cyclic $\mathbb{Z}/4$ (with the cyclic-vs-Klein negative control). It is now named the *four-fold μ_4 identity* and typed, per v428, as a *compression* (one object read four ways), NOT four independent witnesses – in v422, the registry, the `SEAM.GALOIS.NET.01` ledger row and `origin_theory`. (No new module: a separate `v429` would have duplicated v422 and violated the very compression discipline.)

2026-06-25 (`v428` — honest self-correction of the over-determination map)

- **`v428` (`OVERDET.WITNESS.RECLASS.01`) — applying `v427`'s own multiply-vs-compress test to `v427` itself.** A pattern search and the right objection exposed that calling the seven arithmetic witnesses "disjoint grammars that *multiply*" overstates independence. [E] by the Brieskorn classification (v236) the $(2,3,5)$ singularity is the ONE generator behind the order-30 clock (Milnor $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\,\mathrm{rank}\,E_8$, $30{=}2{\cdot}3{\cdot}5{=}h(E_8)$, $\varphi(30){=}8$), so each witness is a facet of that same object (primes $2,3,5$ of 30, the clock 30, or E_8): all seven collapse to one generator $\Rightarrow$ *compression*, not seven multiplications. [E] what genuinely multiplies is the input forced four independent ways — the "8" in $c_3{=}1/(8\pi)$ from rank E_8, $h(D_5){=}2(5{-}1)$, $\varphi(30)$ and the Milnor number — plus the foreign witness $\alpha^{-1}{\approx}137$. [C] refines and partly walks back `v427`, consistent with v236/v305/v313.
- **Surfaces corrected (same change).** `v427`'s prose, the ledger row (`OVERDET.WITNESS.MAP.01`, now noting the refinement), `tfpt_safeguards` Layer 3 (the seven compress one object; a second keybox states what multiplies), the witness-map figure (redrawn), the website (`papers.ts`, `Safeguards.tsx`, `release.ts`, the intro-video description) and the Zenodo description. The over-determination claim is now the narrower, harder-to-attack "one richly over-determined input + a foreign readout".

2026-06-25 (Safeguards visuals + paper expansion + the website safeguards band)

- **Five safeguards figures** (`make_figures.py`, in `manifest.sha256`): the layered defense, the α^{-1} uniqueness scan (one $F_{U(1)}$ variant of 94,500 in the CODATA window), the reverse audit (3/8 vs 5/8), the witness-independence map (seven disjoint grammars), and the look-elsewhere null model (13/13 vs $\leq 5/13$).

- **`tfpt_safeguards` expanded.** Layer 5 now states the null model explicitly ($\prod p_i{=}10^{-25.8}$, $2{\times}200$k pseudo-theories $\le 5/13$, $\varphi_0^{\rm ret} \pm 10\% \to 6/13$, shuffle 1.2) and the α uniqueness ($\le 10^{-30.7}$, $\sim$102 bits, v100), with the data watchdog (v307) and kill board (v321); the five figures are embedded.
- **Website: a prominent animated Safeguards band** (`components/Safeguards.tsx`) in the home flow (after the trust beat) — headline statistics, the seven layers, the figures, links to the paper and the suite — to address the coincidence/numerology suspicion proactively.
- **Koide flow-time investigation (honest negative).** Post-`v425`: the flow-time is *not* native and does *not* reduce to one scale — v99 already rejects a scale reading as look-elsewhere; the only theory-side content is the conditional integer reading $t{=}N_{\rm fam}{=}3 \Rightarrow m_\tau{=}1776.94$ MeV, a Belle II kill test. No new module.

2026-06-25 (Three closure-strategy modules `v425`–`v427` + the Safeguards paper)

- **`v425` (`DYN.TRANSFER.UNIVERSAL.01`) — the single-flow reduction of $F_{\rm transfer}$.** The four frontier transfers do not each need a separate external solver: their *dynamics* is the ONE native flow (the seam modular/recovery semigroup, `v238`/`v240`, gap $6\ln(3/2)$) restricted to each sector. [E] the `v99` Koide generator linearised at $q{=}2$ has rate $-\Delta$, time-1 contraction $(2/3)^6$; Boltzmann washout is that contraction (CP phase $4\pi/3$ native, `v320`); QCD rate $b_3{=}-7$. [C] only the anchors ($v_{\rm geo}$, C_p, M_R) stay external; $F_{\rm relic}$'s cosmological IC θ_i is the one wall. A reduction (extends `v383` static→dynamic), not internalisation; the firewall holds.
- **`v426` (`SEAM.EQUIV.LTORP.KMS.01`) — the invertible $\beta{=}1$-KMS variant of (LTO-RP).** The corner NPW26 leave open (they prove it for topologically-ordered models; the seam is invertible + KMS, not a trace). [E] a gapped reflection-symmetric μ_4-even collar at $\beta{=}1$ KMS satisfies $\Theta K\Theta^{-1}{=}{-K}$ ($u_\Theta{=}J$) and $[\rho,K]{=}0$, so (LTO-RP) holds by direct Tomita–Takesaki; $|\det{\rm Cartan}(E_8)|{=}1$ (no anyons), state not a trace; neg controls have teeth. Advances `v424` sub-step (ii); `SEAM.EQUIV.01` stays [O] (continuum sub-step (i) remains).
- **`v427` (`OVERDET.WITNESS.MAP.01`) — the honest over-determination accounting.** A reviewer correction made machine-explicit: witnesses from *disjoint* grammars multiply, projections of the *same* generator compress. [E] seven disjoint-grammar witnesses each reproduce a carrier-skeleton integer from their own structure — Gauss $N(3{+}2i){=}13$, Eisenstein $N(3{+}2\omega){=}7$, cyclotomy $N(3{+}2\zeta_5){=}55$, Galois $|(\mathbb{Z}/5)^\times|{=}4$, $|\det{\rm Cartan}(E_8)|{=}1$, Pascal $\binom{4}{0}{+}\binom{4}{1}{+}\binom{4}{2}{=}11$ ($2^4{=}16$), Coxeter $h(E_8){=}30{=}2{\cdot}3{\cdot}5$ with $\varphi(30){=}8$ — and the anchor $a{=}(1,1,2)$ is the compression generator. Extends `v305`.
- **New paper `tfpt_safeguards` (companion S).** A dedicated methodology paper collecting, in one language, every mechanism by which TFPT defends a claim against coincidence and numerology: the status calculus + ledger + audit, no-free-pattern + reverse audit, the over-determination map (`v427`), the firewall + No-Unit theorem, frozen predictions + null model, the independent Wolfram and Lean paths, and the red team — with a final kill-test summary. Registered in `build.sh`, `make_docs_map.py`, `make_manifest.py` and the document-set box; the document set is now **eleven** deliverables (five numbered papers + four companions + introduction + changelog).
- **Surfaces synced.** `run_all.py` (421 modules), the registry, the ledger, `tfpt_research_contracts` (the $F_{\rm transfer}$ and keystone sections), `tfpt_safeguards`; manifests refreshed.

2026-06-25 (The (LTO-RP) reduction of the keystone `SEAM.EQUIV.01`: `v424`)

- **New module `v424` (`SEAM.EQUIV.LTORP.01`).** An honest MMST-style reduction of the one open keystone — `SEAM.EQUIV.01`'s intrinsic Bisognano–Wichmann lemma (the v308/v309 residual) — to the recent commuting-projector theorem of Naaijkens–Penneys–Wallick (`arXiv:2605.10693`, [NPW26]), with the central mapping *corrected*. [E] the μ_4 clock is a modular *symmetry* ($[\rho, K]=0$, v309/v199), not the flow — $\sigma^{\psi}=\rho$ is type-wrong (a one-parameter group versus a finite order-4 element). [E] NPW26's axiom (LTO-RP) means $u_\Theta=J$ (reflection = modular conjugation) = the intrinsic BW condition $\theta K\theta=-K$, *distinct* from $\omega\circ\rho=\omega$: a positive character-block-diagonal K has $[\rho, K]=0$ yet never $\theta K\theta=-K$, so clock-invariance does *not* imply (LTO-RP) (refuting that identification). [O] make-or-break: NPW26 prove (LTO-RP) for *topologically ordered* models (Toric Code $|\det K|=4$ via a trace; Levin–Wen anyonic), while the seam is invertible ($|\det \mathrm{Cartan}(E_8)|=1$, no anyons) and a $\beta=1$ KMS state (not a trace) — in neither proved bucket. [C] verdict: the BW residual reduces to NPW26 Theorem A modulo two open sub-steps (realise the seam as a $\mathbb{Z}_2$-reflection commuting-projector LTO; extend (LTO-RP) to the invertible KMS case) — a reduction, *not* a closure; `SEAM.EQUIV.01` stays [O]. Cited in `tfpt_research_contracts`; Python-only (numpy; the det discriminators mirrored via v89/v281).

- **Validation provenance.** The three pivotal references of the closure-strategy note were independently verified real and correctly described: NPW26 (`arXiv:2605.10693`), the RP–Yang–Mills complete-monotonicity reconstruction (Int. J. Geom. Meth. Mod. Phys. 2026), and Marolf (`arXiv:2203.07421`).
- **Surfaces synced.** `run_all.py`, the script registry, the status ledger and the `tfpt_research_contracts` keystone section; manifests refreshed.

2026-06-25 (No-Ambient-Dependency: no frozen readout is computed from `QG.AMB.01`: `v423`)

- **New module `v423` (`QGAMB.NODEP.01`).** Sharpens the v369/v384 prose redundancy ("`QG.AMB.01` is a [C] redundancy, not dynamics") into a machine-checked DAG+script fact, told honestly. [E] gap-decoupling margin $\Delta_{\mathrm{eff}} = 6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$ ($31 = 2^{g_{\mathrm{car}}} - 1$), susceptibility $\chi = 729/665$ finite (negative control gap $\to 0 \Rightarrow \chi \to \infty$); [E] `QG.AMB.01` is a certification *sink* (every claim that lists it is a gravity-gate/QG row, never a frozen prediction) and *not a value source* (no producing script, while α^{-1}, the mixing angles and cosmology are each *computed* by a compiler script, v3/v9/v268/v7); [E] the one transitive link is architectural — of the frozen readouts only θ_{13} references it, via the 4d-QFT contract spine (`PS.DIRAC`→`CONTRACT.QFT4D`→`QG.G2`→`GATE.METRIC`), a coherence edge, not a numerical input. [C] verdict: `QG.AMB.01` is gap-decoupled certification — v369 as a DAG/script fact; honest scope — this does *not* discharge `SEAM.EQUIV.01`, on which the readouts still depend. Cited in `tfpt_4`; Python-only (gap algebra mirrored via v337).
- **Paper sharpening (seam selector, up front).** The `tfpt_4` quantum-gravity keybox now states explicitly that the open `SEAM.EQUIV.01` premise is the *holomorphic* $|\det K|=1$ point that selects $(E_8)_1$ — the central charge $c=8$ alone does *not*, being shared by the non-holomorphic $SO(16)_1$ counterexample (v277/v281).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `tfpt_4`; manifests refreshed.

2026-06-25 (The Galois↔Net bridge: `v422`)

- **New module `v422` (`SEAM.GALOIS.NET.01`).** The seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$ is the SAME cyclic $\mathbb{Z}/4$ as the $(E_8)_1$ simple-current glue — not a mere order-4 coincidence — bridging the Galois gearbox of v419 to G_{net}/`SEAM.EQUIV.01`. [E] $\mathrm{disc}(A_3) = \mathrm{disc}(D_5) = \mathbb{Z}/4$ (one Smith invariant factor 4 = cyclic, $\det 4$); D_n disc is $\mathbb{Z}/4$ for n odd, $\mathbb{Z}_2\times\mathbb{Z}_2$ for n even, so the carrier D_5 (rank $5 = g_{\mathrm{car}}$, odd) is cyclic while D_8 (rank 8, even) is Klein; [E] $(\mathbb{Z}/5)^\times = \langle 2\rangle$ is cyclic order 4 (only the carrier prime 5 gives order 4); [E] in $\mathbb{Z}_4\times\mathbb{Z}_4$ ($16 = \dim S^+$) the glue $(1, 1)$ is order 4 with Lagrangian quotient $16/4^2 = 1 = (E_8)_1$ (v89/v154); [E] negative control:

the Klein / order-2 $\langle(2,2)\rangle$ gives $16/2^2 = 4 = |\mathrm{disc}(D_8)| = SO(16)$ (det 4), NOT E_8 (det 1) — cyclic $\mathbb{Z}/4 \to E_8$, Klein$\to D_8$. [C] the bridge: one cyclic $\mathbb{Z}/4$ threads the seam deck μ_4 (v419), $\mathrm{Gal}(\mathbb{Q}(\zeta_5))$ and the $(E_8)_1$ glue — the Galois gearbox $(\mathbb{Z}/30)^\times$ IS the holomorphic net's glue, cyclicity forced by 5, Klein excluded. Cited in `origin_theory`; Wolfram-mirrored ($324 \to 327$).

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, `origin_theory`; the Wolfram extension/README/ledger counts (327/327) and the website verification count; manifests refreshed.

2026-06-25 (F_pole external: the Koide source→pole transfer is one-loop EM: v420)

- **New module v420 (FR.POLE.QED.01).** The honest external-physics follow-up to v93 and the clock No-Go: the Koide source→pole transfer is a one-loop *electromagnetic* effect, not an algebraic step. [E] Koide Q is rescaling-invariant ($Q(c\,m) = Q(m)$), so flavor-universal running leaves it fixed and the source→pole shift is purely flavor-split; [C] the tree deficit $\frac{2}{3} - Q_{\mathrm{src}} = 0.00220$ is one-loop-EM-sized ($= \varphi_0/24$ to 0.6%, $= \alpha/\pi$ to $\sim$5%), the PDG pole sits at 2/3 (the IR attractor); [C] a 1-loop QED pole$\leftrightarrow\overline{\mathrm{MS}}$ per-lepton-log estimate shifts Q by EM-loop order toward 2/3 (the v82 direction); [O] the exact flow time $t \approx 2.84$ (v93) is scheme/scale-dependent external QED, only sized. [C] verdict: F_{pole} is a one-loop EM effect — the lens forces the rate $(2/3)^6$ (v327), external EM supplies the flow; `F_pole` stays [C], firewalled (v187), so F_{transfer} is confirmed genuinely external — the open theme sharpened (QED), not closed. Cited in `tfpt_4`; numerical (mpmath), Python-only (no Wolfram check, by the suite convention like v93/v371).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Koide section of `tfpt_4`; manifests refreshed.

2026-06-25 (The seam μ_4 is the carrier's Galois group: v419 + figure)

- **New module v419 (SEAM.GALOIS.01).** A positive resolution of the cyclic/Galois asymmetry (v409, `RES.COXETER.SYMMETRY.01`): why the seam μ_4 sits on the Galois side, not on the cyclic ζ_{30} dial. [E] $h(E_8) = 30$ is squarefree, so $\mathbb{Z}/30$ has element orders $\{1, 2, 3, 5, 6, 10, 15, 30\}$ — *no* order 4, so the seam μ_4 cannot be a rotation hand and is forced onto the Galois side; [E] by CRT $(\mathbb{Z}/30)^\times = (\mathbb{Z}/2)^\times \times (\mathbb{Z}/3)^\times \times (\mathbb{Z}/5)^\times = \mu_4 \times \mathbb{Z}_2$ (order $\varphi(30) = 8 = \mathrm{rank}\, E_8$), and the μ_4 $(\mathbb{Z}/4)$ factor IS $(\mathbb{Z}/5)^\times$ (the carrier prime 5), the $\mathbb{Z}_2$ IS $(\mathbb{Z}/3)^\times$ (family 3 = CP conjugation, v316); [E] so the seam $\mu_4 = \mathrm{Gal}(\mathbb{Q}(\zeta_5))$, realised on the carrier clock C_5 (v418) by the Frobenius $\zeta_5 \mapsto \zeta_5^2$ — the explicit order-4 operator G with $GC_5G^{-1} = C_5^2$, $G^4 = I$, $G^2C_5G^{-2} = C_5^{-1}$. [C] this is the positive content of v409: the 2^2=4 is the order of $(\mathbb{Z}/5)^\times$, and "no contraction rate" is because the seam is an *automorphism*, not a hand; so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants (cyclic h=30, Galois rank =8). Cited in `origin_theory`; Wolfram-mirrored ($321 \to 324$).
- **New figure `coxeter_galois.pdf`.** The order-30 cyclic dial with the four flavor hands (sheet/family/carrier/CP) beside the Galois gears ($\mu_4 = \mathrm{Aut}(\zeta_5)$ the seam, $\mathbb{Z}_2 = \mathrm{Aut}(\zeta_3)$ the CP conjugation); added to `make_figures.py`, the manifest figure list and `origin_theory`.
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Wolfram extension/README/ledger counts (324/324) and the website verification count; manifests refreshed.

2026-06-25 (The cyclotomic norm triple + the carrier-5 clock: v418)

- **New module v418 (ARITH.NORMTRIPLE.01).** The carrier split $(3, 2)$ read in the three atom-rings gives $(7, 13, 55)$ =(scalaron, Δ_Q, quark numerator), and the carrier-5 clock missing from v417 is found as the 4×4 Φ_5-companion C_5. [E] $C_5^5{=}I$, $\operatorname{tr} C_5{=}-1$, $\det C_5{=}1$ (the four primitive 5th roots), with the golden ratio as its real-part data ($2\cos 72°{=}1/\varphi$, $2\cos 144°{=}-\varphi$; output-side, v313/v349; dim 4, not 3 — this refines v417's gap from "no operator" to "no 3×3 operator"); [E] $N_{\mathbb{Z}[\zeta_5]}(3{+}2\zeta_5){=}\det(3I{+}2C_5){=}55{=}5{\cdot}11{=}g_{\rm car}\|\operatorname{Pl}(K)\|_1$, the c_u/c_d numerator (v410/v411); [E] the triple $\det(3I{+}2\operatorname{Comp}(\Phi_n)) = (7, 13, 55)$ for $n{=}3, 4, 5$ over $\mathbb{Z}[\omega], \mathbb{Z}[i], \mathbb{Z}[\zeta_5]$ (atoms $3, 2, 5$, v416); [E] NEG control: the anchor $(5, 4) \to (21, 41, 461)$ reaches named values only in the ω- and i-rings ($21{=}N_\omega(5, 4)$, $41{=}10b_1$, v222) but 461 (prime) in the carrier ring, so the carrier ring separates $(3, 2) \to 55$ from $(5, 4) \to 461$. [C] honest: $(3, 2)$ is the *forced* carrier split (v14), not the unique clean one — a scan finds $\{(2, 1) \to (3, 5, 11), (3, 2) \to (7, 13, 55)\}$, a small ladder. The sharper discriminator is *cross-sector*: only $(3, 2)$ spans three physics sectors (gravity 7, flavor 13, masses 55), the unique cross-sector seed, while $(2, 1)$ is all compiler-core. Cited in `origin_theory`; Wolfram-mirrored ($318 \to 321$).

- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, a triple paragraph in `origin_theory`, the Wolfram extension/README/ledger counts (321/321) and the website verification count; manifests refreshed.

2026-06-25 (The Eisenstein/CP operator + the order-30 clock: v417)

- **New module v417 (DIAMOND.EISEN.01).** The hexagonal dual of v415: the $\tau{=}\omega$ flavor side put on operators and unified with the seam into one order-30 Coxeter clock. [E] the family rotation $P = (1\,2\,3)$ is the order-3 Eisenstein deck ($P^2{+}P{+}I$ = ONES, so $\omega^2{+}\omega{+}1{=}0$ on the plane $\perp$ $(1, 1, 1)$), the order-3 dual of $J^2{=}-I$; [E] it realises the hex CM norm as $(3I{+}2P)(3I{+}2P^2) = 7I{+}6\,\text{ONES}$, $\operatorname{Spec}\{7, 7, 25\}$ ($7{=}N_{\mathbb{Z}[\omega]}(3{+}2\omega){=}$ scalaron, $25{=}g_{\rm car}^2$) — the dual of v415's $\{13, 13, 9\}$; [E] the CP clock $W{=}-P^2$ (ρ of v233) has $W^2{=}P$, $W^3{=}-I$, $\operatorname{Spec}\{-1, \zeta_6, \zeta_6^{-1}\}$, giving $\delta_{\rm CKM,lead}{=}\arg\zeta_6{=}\pi/3$ and $\delta_{\rm PMNS}{=}\arg\zeta_6^4{=}4\pi/3$ (v316/v320); [E] all flavor clocks are powers of ζ_{30} ($\zeta_{30}^{15}{=}-1$, $\zeta_{30}^{10}{=}\omega$, $\zeta_{30}^{6}{=}\zeta_5$, $\zeta_{30}^{5}{=}\zeta_6$) while the seam $\mu_4{=}i$ is the Galois side $(\mathbb{Z}/30)^\times$ of order $\varphi(30){=}8{=}\operatorname{rank} E_8$ — so $\mathbb{Q}(\zeta_{30})$ carries both E_8 invariants ($h{=}30$ and rank${=}8$). [C] honest gap: the carrier μ_5 has no 3×3 operator (the golden φ is an output-side trace, v313/v349); negatives: no U, V-diamond operator carries ζ_6, $[D, P] \neq 0$ (no μ_{12}), $N(3{+}2\zeta_6){=}19{\neq}7$. Cited in `tfpt_2`; Wolfram-mirrored ($315 \to 318$).
- **Surfaces synced.** `run_all.py`, `script_registry.csv`, `status_ledger.csv`, the Sheet-Diamond keybox of `tfpt_2`, the Wolfram extension/README/ledger counts (318/318) and the website verification count; manifests refreshed.

2026-06-25 (The Gaussian operator + the atom trichotomy: v415, v416)

- **New module v415 (DIAMOND.GAUSS.01).** The square CM-norm dictionary of v222/v230 is *operator-realised*, and the μ_4 deck becomes an intrinsic diamond object. [E] the integer commutator $J := \frac{1}{3}[U, V] = \left(\begin{smallmatrix}-1&1&0\\-2&1&0\\-3&1&0\end{smallmatrix}\right)$ has $\operatorname{Spec}(J) = \{i, -i, 0\}$, $J^2 = -I$ on the rank-2 image and $J^3 + J = 0$ — a μ_4 quarter-turn born from the binary V and ternary U compilers (v410/v95); [E] $3I + 2J$ and $5I + 4J$ have spectra $\{3{+}2i, 3{-}2i, 3\}$, $\{5{+}4i, 5{-}4i, 5\}$, so the Gaussian integers $3{+}2i$ (norm $13{=}\Delta_Q$) and $5{+}4i$ (norm $41{=}10b_1$) are *literal eigenvalues*; [E] $(aI{+}bJ)(aI{-}bJ)$ reads $(a^2{+}b^2)$ on the μ_4-plane and the real part^2 on the kernel ($\{13, 13, 9\}$, $\{41, 41, 25\}$); [E] the intrinsic order-4 deck $D = -I + J - J^2$ ($D^4 = I$, $\chi_D = (x{+}1)(x^2{+}1)$) has χ_2-line $\ker[U, V] = a - \mathbf{1}$. [C] the geometric-deck identification sharpens (does not close) the v140 $\mathbb{Z}_3$ residue. Cited in `tfpt_2`; Wolfram-mirrored.
- **New module v416 (ARITH.TRICHOTOMY.01).** The atoms $\{2, 3, 5\}$ are ramified/inert/split in the two exceptional CM rings. [E] in $\mathbb{Z}[i]$ (square/seam) $2{=}|\mathbb{Z}_2|$ ramifies, $3{=}N_{\rm fam}$ is inert, $5{=}g_{\rm car}$ *splits* as $(2{+}i)(2{-}i)$; [E] in $\mathbb{Z}[\omega]$ (hex/flavor) $3{=}N_{\rm fam}$ ramifies, $2, 5$ are inert; [E] the ramified prime of each ring is its own atom; [E] the three quadratic facets $\mathbb{Q}(i), \mathbb{Q}(\sqrt{-3}), \mathbb{Q}(\sqrt{5})$ of $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$ (v403) ramify over exactly one atom each, product $2{\cdot}3{\cdot}5 = 30 = h(E_8)$,

two imaginary (CM/dynamic) + one real (RM/static). [C] the seam/flavor/golden reading. Cited in `origin_theory`; Wolfram-mirrored.

- **Wolfram extension** $308 \to 315$ **+ a pre-existing fix.** Added the seven exact checks for `v415`/`v416` and corrected a one-character sign bug in the `v412 CharacteristicPolynomial` check (Wolfram returns $-\chi$ for odd rank), so `tfpt_readouts_extension.wl` now runs genuinely clean at 315/315 (`GATE.WOLFRAM.02`); README and ledger counts updated.
- **Surfaces synced.** `run_all.py`, `script_registry.csv` and `status_ledger.csv` updated; a new keybox paragraph in the Sheet-Diamond section of `tfpt_2` and a trichotomy sentence in `origin_theory`; manifests refreshed.

2026-06-24 (The sheet generator V as a binary internal compiler: v410–v414)

- **New module v410 (SHEET.GEN.BINARY.01).** The rank-2 sheet axis $V = Q\,\mathrm{diag}(0,1,1)$ of the centered cross (v95/v218) is a binary internal compiler: its powers print the carrier spine. [E] the closed form $V^n = \left(\begin{smallmatrix}0 & 2^{n-1} & 0\\ 0 & 2^n & 0\\ 0 & 2^{n+1}-2 & 1\end{smallmatrix}\right)$ (proved by the inductive step $F(n)V{=}F(n{+}1)$), so $V^n\mathbf{1} = (2^{n-1}, 2^n, 2^{n+1}-1) = (1,2,3),(2,4,7),(4,8,15),(8,16,31)$; [E] the four bilinear families $\mathbf{1}^T V^n\mathbf{1} = 7\cdot 2^{n-1}{-}1$, $\mathbf{1}^T V^n a = 7\cdot 2^{n-1}$, $a^T V^n a = 11\cdot 2^{n-1}$, $a^T V^n \mathbf{1} = 11\cdot 2^{n-1}{-}2$ give $(6,7,9,11)$ at $n{=}1$ and $\mathbf{1}^T V^2\mathbf{1}{=}13{=}\Delta_Q$, $\mathbf{1}^T V^3\mathbf{1}{=}27{=}\mathbf{1}^T Ra$, $\mathbf{1}^T V^4\mathbf{1}{=}55$, $\mathbf{1}^T V^4 a{=}56{=}\dim \mathbf{56}_{E_7}$; [C] the binary spine is FORCED by $\mathrm{Spec}(V){=}\{0,1,2\}$, so the carrier/Lie matches (rank $E_8{=}8$, $\dim S^+{=}16$, $248/8{=}31$) and the theta cross-link ($\sigma_3(2){=}9{=}a^T V\mathbf{1}$, $\sigma_3(3){=}28{=}\mathbf{1}^T V^3 a{=}\det(I{+}R)$, $\sigma_3(5){=}126$) are audit-typed (the v95 G_2-center typing).
- **New module v411 (FLAV.UD.VPOWER.01).** The quark ratio is a pure V-power readout, $c_u/c_d = (\mathbf{1}^T V^4\mathbf{1})/((a^T V\mathbf{1})(\mathbf{1}^T V^2\mathbf{1})) = 55/(9\cdot 13) = 55/117$, identical to the established $g_{\rm car}\|\mathrm{Pl}(K)\|_1/(N_{\rm fam}^2\Delta_Q) = 5\cdot 11/(9\cdot 13)$ (v94). [E] the identity; [C] an exact *re-encoding* (no new physical input) — the value and the u/d identification stay conditional, coupled to the $(U_{\rm wall})$ readout rigidity.
- **New module v412 (DIAMOND.SHEET.SOURCE.J.01).** Names the operator on the $\mathbb{Z}_2$ wall ($t{=}{-}2$, $\det{=}2$, v135), $J{=}M(1,-2){=}C{-}2V$, completing the sheet axis $J, K, C, F{=}M(1,-2..1)$. [E] $\chi_J = \lambda^3-6\lambda^2+3\lambda-2 \Rightarrow (\mathrm{tr}, e_2, \det){=}(6,3,2)$; $a^T Ja{=}30{=}h(E_8)$, $\det(I{+}J){=}12{=}\dim\mathfrak{g}_{\rm SM}$ and $\det(2I{+}J){=}40{=}|R(D_5)|$; [C] the Lie readings audit-typed like the G_2/F_4 shadows.
- **New module v413 (DIAMOND.SHEET.CALCULUS.01).** On $M_t{=}C{+}tV$ the elementary-symmetric coefficients encode the atoms as difference orders: $e_1{=}3t{+}12$, $e_2{=}2t^2{+}15t{+}25$, $e_3{=}4t^2{+}14t{+}14$ give $\Delta e_1{=}3{=}N_{\rm fam}$, $\Delta^2 e_2{=}4{=}|\mu_4|$, $\Delta^2 e_3{=}8{=}\mathrm{rank}\,E_8$ (the e_3 curvature is the v218/v224 sheet quadratic; the full triple is new); [E] the anchor energy $a^T M_t a{=}52{+}11t$ ($30{\to}41{\to}52{\to}63$, step $11{=}a^T Va$); [C] atom/Lie readings audit-typed.
- **New module v414 (DIAMOND.CENTER.RESOLVENT.01).** The center $C{=}M(1,0)$ is a resolvent portal: [E] $\det C{=}14{=}\dim G_2$, $\det(I{+}C){=}52{=}\dim F_4$, $\det(2I{+}C){=}120{=}|R^+(E_8)|$, the exceptional shell ladder $G_2{\to}F_4{\to}E_8^+$ as $\det(C{+}kI)$, $k{=}0,1,2$ (with $\mathrm{tr}\,C{=}12{=}\chi_{E_8}(i)$, $\sum C{=}31{=}2^{g_{\rm car}}{-}1$, v95); [C] the portal reading.
- **Surfaces synced.** Integrated into the Sheet-Diamond section of `tfpt_2` (new keybox, five `\veri` citations) and the E_8 theta-raster of `tfpt_3`; `run_all.py`, `script_registry.csv` and the ledger updated; Wolfram extension mirror $303 \to 308$.

2026-06-24 (Two new tracked targets + axion-branch consistency: v408, v409)

- **New module v408 (HYP.PHIO.PUNCTURE.01).** Names the φ_0 puncture term $48c_3^4$ as the order-$|\mu_4|{=}4$ local boundary heat-kernel (Seeley–DeWitt) contact term of the μ_4 puncture divisor, summed over $\Omega_{\rm adm}{=}N_{\rm fam}\dim S^+{=}48$ admissible states — the analytic complement of the icosahedral tree term (v396). [E] the exact split $\varphi_0{=}1/(6\pi){+}48c_3^4$, the puncture count, the closed form $3/(256\pi^4)$ and the hypergraph negative (reused from v396/ARCH.QUAD.01); [O] the from-first-principles heat-kernel proof; [C] a FACE of the seam boundary analysis (the c_3 Gauss–Bonnet datum), not a new gate. Cited in `origin_theory`; no new number.

- **New module v409 (RES.COXETER.SYMMETRY.01 closed as a structural lemma).** prime-2 (the seam sheet involution) is *necessary* but *not autonomous*: $\sigma = \mathrm{diag}(1, -1, -1)$ has spectrum $\{\pm 1\}$ and projector $(I+\sigma)/2$ spectrum $\{0, 1\}$, none in $(0, 1)$, so prime-2 carries no contraction rate, whereas the genuine rates $2/3$ and $(1/3)^6$ lie in $(0, 1)$ and carry prime-2 only as a factor. [E] no prime-2-only attractor exists; [C] dynamics begins only after coupling to family-3 or carrier-5 ($|\mu_4| = 2^2$ the Galois bridge). Resolves the v394 open question's falsifiable corollary; cited in `tfpt_research_contracts`; the ledger row RES.COXETER.SYMMETRY.01 retyped [O] → [E]/[C].
- **Axion branch consistency (`tfpt_4_frontier`, website `predictions.ts`, v185).** The dark-matter section's opening + keybox are realigned with the finite-T result already stated later in the same section: the robust dominant-DM branch is the *spine* angle $\theta_i = \pi N_{\mathrm{fam}}/g_{\mathrm{car}} = 3\pi/5 = 108°$ (untuned, in band), while the seam hilltop $\theta_i \approx 170°$ over-produces ($\sim 5.4\times$) and is disfavoured; the misalignment angle is retyped from a closed [E] input to a [C] branch decision. The website prediction card now leads with the spine angle.
- **Lean closure of the prime-2 corollary (FORM.COXETER.PRIME2.01).** The v409 result is now also machine-proved in Lean 4 / Mathlib (new module `TfptCarrier/CoxeterPrime2.lean`): an involution eigenvalue ($r^2 = 1$) is ± 1, a projector eigenvalue ($r^2 = r$) is 0 or 1, neither lies in the open interval $(0, 1)$, while $2/3$ does — so *no prime-2-only attractor exists*. Built clean (`lake build`); the new theorems are axiom-pure (`#print axioms` = `propext`, `Classical.choice`, `Quot.sound` only) and pass the `AuditCheck`/`AuditContract` signature locks. This lifts RES.COXETER.SYMMETRY.01 to the machine-proof tier.
- **Verified, no change (Plücker 11).** The quark constant $11 = \|Pl(K)\|_1 = \det(\mathbf{1}|a|\sigma)$ is already typed honestly as an [E] identity with "why 11" the lone [O] residue (v407, `tfpt_1`); the repo carried no overstatement, so no edit was required.

2026-06-24 (Cross-surface consistency sweep: papers, website, gates, figures realigned)

- **Ledger hygiene.** Resolved a duplicate active `claim_id`: GATE.UWALL.06-09 was carried by two distinct chains. It is an ID collision (not a supersede): the monodromy/residue chain v114–v117 keeps .06-.09 (as asserted by `v123_inventory_update.py` and `run_all.py`); the older exterior-leg chain v42–v45 is renumbered to GATE.UWALL.15-.18 (internal `depends_on` carried along). No claim content changed.
- **Website.** `papers.ts` document count 8 → 9 (the Red-Team paper was missing from the tally); Red-Team subtitle/highlight *Targets A–E→A–F* with a new Target F section + contribution bullet (the abstract already carried F); `StatusMatrix.tsx` *Doc 7→Doc 8* for the research contracts and the residual interfaces re-typed v_{geo} [O], G_{net} [C], F_{transfer} [C]; `objections` *Paper 1→Doc 1*.
- **Papers (marker/wording alignment, no canonical status change).** `tfpt_1` Strong CP [E]→[E]/[C] (given RP) and θ_{12} "genuinely open"→"conditionally derived"; `tfpt_2` Starobinsky A_s [E]→[E]/[C] (M exact, A_s a band over $N_\star \in [50, 60]$) and the Part-II abstract "opens the one remaining gap"→"closes the last structural gap"; `tfpt_research_contracts` the QFT4D "nonperturbative measure stays open" line and the "only U_{point} stays open" headline reconciled with the [C] redundancy / v_{geo} reduction stated alongside; `origin_theory` class C re-typed to the ambient-measure [C] redundancy; `introduction` (U_{wall}) → v_{geo}. Forbidden old-paper attributions ("Papers 1–7", "Papers 2/6", "Paper-7", "Paper-1") replaced by current document / neutral wording (the `tfpt_3` cascade bridge stays exempt).

- **Suite README.** The residual-gates table (Alessandro-5.0 vintage) brought to the current model ($v_{\rm geo}$ [O]; $G_{\rm net}$/`SEAM.EQUIV.01`, $F_{\rm transfer}$, $N_\star$, `QG.AMB.01` [C]; Wolfram 116/116 + 303/303).
- **Figures.** Regenerated `status_heatmap` (ledger-driven) and extended the narrative figures `script_timeline/residual_chain/qft_skeleton` through the `v303`–`v407` arc (typed solvers, parameter-free local Einstein equation `v359`, `QG.AMB.01` [C] redundancy, seam closed modulo cited theorems); the `qft_skeleton` "Stelle ghost" open box is retyped (resolved truncation artefact). Added `qft_skeleton/qft_unification` to the manifest figure list.

2026-06-24 (The $\tau{=}\omega$ dual keystone + the flavor residuals it homes: v405–v407)

- [E]/[C]/[O] **`v405_seam_equiv_omega.py`** (`SEAM.EQUIV.02`): the *dual keystone* at $\tau{=}\omega$ – the family/flavor sector as the order-3 Eisenstein/A_2 face of E_8, dual to the seam $= (E_8)_1$ at $\tau{=}i$ (v404). A named contract (genre of v286/v398), NOT a closure. [E] $\chi_{E_8}(\omega) = 0$ (the family CM point degenerates E_8) vs $\chi_{E_8}(i) = 12$; [E] A_2 is the $\tau{=}\omega$ Eisenstein lattice (Cartan det $= 3 = N_{\rm fam}$, $h = 3 = N_{\rm fam}$); [E] R lives in $E_6{\times}A_2$ ($248 = 78 + 8 + 2{\cdot}27{\cdot}3$, $\det R = 8 = \dim A_2 = \operatorname{rank} E_8$); [E] $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ CM phase; [E] the cusp weights $\{0, 1/3, 2/3\}$ are the order-3 deck $= N_{\rm fam} = \det Q$. [C] the family sector is the order-3 Eisenstein/A_2 deck; the two keystones sit at the only two elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$. [O] residual = the canonical $\mathbb{Z}_3$ deck map (v140) + the $v_{\rm geo}$-class scale. Cited in `origin_theory`; Python-only.
- [E]/[C]/[O] **`v406_pmns_phase_origin.py`** (`FLAV.PMNS.CLOSE.01`): the PMNS dynamics closed *modulo* the $\tau{=}\omega$ keystone – the CP-phase ORIGIN is the family CM point and the absolute scale is one $v_{\rm geo}$-class unit, so PMNS has *zero free flavor dials beyond* $v_{\rm geo}$. [E] the three angles (v9/v268) → unitary $U_{\rm PMNS}$ (v270); [C] $\delta_{\rm PMNS} = 4\pi/3$ is the $\tau{=}\omega$ hexagonal CM phase (v405), upgrading v270's [O] "assembled input" to a keystone consequence; [C] $J_{\rm PMNS} = -0.0297$ derived; [E] the absolute scale = one seesaw ratio = $v_{\rm geo}$-class (v272/v153). [O] residual = the seesaw operator realisation (M_R, v184) + the Σm_ν floor kill test. Cited in `tfpt_2_standard_model`; Python-only.
- [E]/[C]/[O] **`v407_dn_pairings_omega.py`** (`FLAV.SELECTOR.CLOSE.01`): the R4′ selector residue folds into the $\tau{=}\omega$ keystone – the three (d, n) pairings ARE the $\tau{=}\omega$ family-slice atoms. [E] the frame $(\mathbf{1}, a, \sigma)$ has det $= 11 = \|Pl(K)\|_1$ and $n = (5, -9, 6)$ is the unique covector with $n{\cdot}\mathbf{1} = 2$, $n{\cdot}a = 8$, $n{\cdot}\sigma = 121 = 11^2$ ($\Rightarrow (d, n) \Rightarrow R$, v136/v139); [E] $n{\cdot}a = 8 = \operatorname{rank} E_8 = \dim A_2 = \det R$ (the A_2 block reading R in $E_6{\times}A_2$, v405); [E] the pairings $\{2, 8, 121\} = \{|Z_2|, \dim A_2{=}\det R, \|Pl(K)\|_1^2\}$. [C] deriving (d, n) folds into `SEAM.EQUIV.02` (d anchor-derived, v139); [O] the lone residue "why $11{=}\|Pl(K)\|_1$". Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (The arithmetic hull + the E_8-character CM duality: v403, v404)

- [E]/[C] **`v403_facet_compositum.py`** (`ARITH.HULL.01`): the three single-prime number-field facets of the order-30 clock (v390/v394, used so far only one at a time) *compose* to one field $K = \mathbb{Q}(i, \sqrt{-3}, \sqrt{5})$. [E] the squarefree parts $-1, -3, 5$ are independent in $\mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so $[K : \mathbb{Q}] = 2^3 = 8 = \operatorname{rank} E_8 = g_{\rm car}{+}N_{\rm fam}$ with Galois group $(\mathbb{Z}/2)^3$ (order 8); [E] exactly 7 quadratic subfields $\{-1, -3, 5, 3, -5, -15, 15\}$, splitting 4 imaginary $= |\mu_4|$ (CM/dynamic) + 3 real $= N_{\rm fam}$ (RM/static); [E] ramified *only* over $\{2, 3, 5\}$ = the atoms = the primes of $h(E_8) = 30$; [E] $2^{\omega(h)}$ = rank is E_8-special (the E_7 control $h = 18$ gives $2^2 = 4 \neq 7$). [C] K is the abelian $(\mathbb{Z}/2)^3$ *arithmetic* hull dual to the E_8 *lattice* hull; the 7 quadratic subfields are a candidate for the 7 E_8 audit slices. A synthesis composing the established facets, no new number; cited in `origin_theory`; Python-only by the synthesis-module convention (cf. v390/v394).

- [E]/[C] **v404_e8_char_cm_duality.py** (E8.CMDUAL.01): the $(E_8)_1$ vacuum character $\chi_{E_8} = E_4/\eta^8 = j^{1/3}$ read at *both* CM points, extending v282 ($\chi_{E_8}(i) = 12$) to the second elliptic point. [E] $\chi_{E_8}^3 = j$ (q-series); [E] SEAM $\tau{=}i$ ($\mathbb{Q}(i)$, order 4): cross-ratio $2 \Rightarrow j = 1728 \Rightarrow \chi_{E_8}(i) = 1728^{1/3} = 12$ (E_8 present, v214/v282); [E] FAMILY $\tau{=}\omega$ ($\mathbb{Q}(\sqrt{-3})$, order 6): the equianharmonic λ ($\lambda^2{-}\lambda{+}1 = 0$) $\Rightarrow j = 0 \Rightarrow \chi_{E_8}(\omega) = 0$ (E_4 vanishes, v220/v267); [E] $\tau{=}i, \tau{=}\omega$ are the *only* elliptic points of $\mathrm{PSL}(2,\mathbb{Z})$, so the values are forced. [C] the seam/flavor rigidity duality: the seam is rigid where E_8 is present ($\chi = 12$, the keystone SEAM.EQUIV.01 closes) while the flavor/CP sector is the soft residual where E_8 degenerates ($\chi = 0$: U_{wall}, v_{geo}, the CP texture), a candidate dual keystone at $\tau{=}\omega$. Cited in **origin_theory**; core values $j = 1728/j = 0$ already mirrored (v214/v220/v267/v282), Python-only.

2026-06-24 (The five TOE/QFT closure contracts v398–v402 + a ledger rescope)

- [E]/[O] **v398_seam_state_rigidity.py** (SEAM.RIGIDITY.01, Paper A): the *Seam State Rigidity Theorem* as ONE named target, consolidating the scattered state-invariance reductions (v177/v199/v308/v309/v335) and *hardening* the finite-block evidence to a multi-mode sweep. [E] RP-definability hinge (OS canonical transfer $T{=}e^{-H}$ from RP + reflection alone, v54, quasi-free v155, no μ_4 normal form); [E] band-limited character-block-diagonal core $[\rho, |k|]{=}0$ swept $N{=}4..64$ (beyond the 3-dim H^1); [E] holomorphy $\Rightarrow (E_8)_1$ ($c{=}8$, $|\det \mathrm{Cartan}|{=}1$ vs 4, index $4{=}|\mu_4|$). [O] the one residual: $[\rho, \Lambda_\Sigma]{=}0$ on the full L^2 (intrinsic Bisognano–Wichmann). NOT a closure of SEAM.EQUIV.01 (continuum = cited MMST v336).

- [E]/[C]/[O] **v399_gravity_complete.py** (GRAVITY.COMPLETE.01, Paper B): the explicit verdict *Gravity-complete = Boundary-complete + Ambient-redundancy*, combining the parameter-free local equation (v358/v359) with the ambient redundancy (v369). [E] the five gravitational readout classes (Newton 8π, Λ, scalaron c_3^7, horizon $1/|\mu_4|$, recovery $(2/3)^6$) are all boundary/seam/gap, so $\mathcal{O}_{\mathrm{phys}} \subset \mathcal{A}_\Sigma$; [O] residual = SEAM.EQUIV.01 + intrinsic BW (no new gate).

- [E]/[O] **v400_grav_nonlocal_action.py** (GRAV.NONLOCAL.01, Paper C): the nonlocal spectral-gravity action with the local $R{+}R^2$ as its IR projection (consolidating v304/v370/v380/v386, adding the IR-matching check). [E] entire $a{=}e^u \Rightarrow$ one pole (ghost-free), every truncation carries a Stelle zero (monotone, v380), and the IR order reproduces $R{+}R^2$ with the atom-fixed scale $M_{\mathrm{scal}}^2/\bar{M}_{\mathrm{Pl}}^2{=}c_3^7$ (v36/v253). [O] the entire-a assumption + perturbative-only.

- [E]/[O] **v401_metrology_closure.py** (METROLOGY.CLOSURE.01, Paper D): the final input set $\{a, \pi, v_{\mathrm{geo}}\}$ certified — 0 dimensionless dials + 1 unit (consolidating v153/v274/v364/v384). [E] dimensional bookkeeping mass$/1G/\Lambda/U_{\mathrm{point}}/(m/\mu)$= number $\times v_{\mathrm{geo}}^{1,2,4,1,0}$; [E] the G-vs-Λ overdetermination 0.11% (v274); [O] v_{geo} theorem-forbidden ($\mathrm{TOE}_{\mathrm{strict}}{=}\mathrm{TOE}_{\mathrm{dimensionless}}{+}[v_{\mathrm{geo}}]$).

- [E]/[C] **v402_ftransfer_suite.py** (FTRANSFER.SUITE.01, Paper E): the four frontier sectors as ONE functor F_{transfer}, the composition $F_{\mathrm{obs}} \circ F_{\mathrm{thr}} \circ F_{\mathrm{RG}}$ (consolidating the FR.*.SOLVE solvers), each with an exact atom source, a committed kill test, and the firewall (physical prediction stays [C]/[O], only the algebraic source [E]; never re-pressed into the compiler, v187).

- **Ledger rescope (repo hygiene):** GATE.METRIC.01 no longer reads "full covariant metric-sector field equation open" — the *local* covariant equation is closed parameter-free (GRAV.NONLINEAR.01/v359); the gate is rescoped so the open item is *only* the global nonperturbative ambient measure (QG.AMB.01, a [C] redundancy v369), removing the stale contradiction with the status card.

2026-06-24 (φ_0 leading term from icosahedral combinatorics: v396)

- [E]/[C] **v396_phi0_icosahedral.py** (HYP.PHI0.01): the tree-level seed $\varphi_0^{\mathrm{tree}} = 1/(6\pi)$ is exactly $F/(4h\pi) = F/(|\operatorname{Aut}|\pi) = (F/(g_{\rm car}N_{\rm fam}))c_3$ on the icosahedral hypergraph ($F = 20$ triangular hyperedges, $h = 30$, $|\operatorname{Aut}| = 120$); $F/h = |Z_2|/N_{\rm fam} = 2/3$ (the same survival ratio as `v327`). [E] Gauss–Bonnet consistent with $c_3 = 1/(|Z_2|4\pi)$ (`v216`/`v342`). [E] honest negative: the puncture $48c_3^4$ is still analytic, not a hypergraph fraction. [C] $\varphi_0^{\rm tree}$ is a combinatorial reinterpretation, not a new independent injection; the puncture stays [O].

2026-06-24 (Coupled hypergraph rewrite: `v395` unifies carrier × family in one local rule)

- [E] **`v395_hypergraph_coupled.py`** (`HYP.COUPLED.01`): one coupled local rewrite on a 9×3 labelled grid (affine-E_8 nodes × family channels) unifies `v299` (carrier growth), `v327` (branching rule M) and `v324` (fiber product). Micro-step = network lazy diffusion on each cusp column + M on each node's family vector ($T_{\rm micro} = T_{\rm net} \otimes M$); joint attractor marks$\otimes e_0$; one clock hand (6 family steps) gives recovery $(2/3)^6$; `v324`'s $T_{\rm net} \otimes T_{\rm cusp}$ emerges as the cusp-readout basis. [O] the 3-arm seed (P_2) and analytic φ_0 remain irreducible (`v312`).

2026-06-24 (A falsifiable clock probe of the frontier: `v397` how external is the "external physics"?)

- **`v397` (`FR.CLOCK.PROBE.01`) — do the "external" frontier scheme-numbers land on the $\{2, 3, 5\}$ clock?** A falsifiable, anti-numerology probe (strict No-Free-Pattern, v100: "on the clock" means an *exact* match, not a wide-band fit). [E] the η_B decuple $A_\Lambda = 10 = |Z_2|g_{\rm car}$ is ON the clock (v212); [E] the Koide rate $(2/3)^6 = (|Z_2|/N_{\rm fam})^6$ is ON the clock (v303); [E] the axion spine angle $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ is ON the clock (v211/v373); [E] the proton $C_p \sim 2.83 \pm 0.15$ is *not* a discriminating clock value — within its $\pm 5.3\%$ band several simple values fit $(2^{3/2}, 17/6, 14/5)$, so by No-Free-Pattern it is declared EXTERNAL (the firewall v187/v374 is correct there). [C] verdict: 3 of the 4 frontier scheme-numbers land *exactly* on the clock, so the irreducible "external physics" residue shrinks to essentially the proton's non-perturbative QCD number C_p. Cited in `tfpt_4_frontier`; a falsifiable probe whose answer could have been "none on the clock"; Python-only.

2026-06-24 (Clockwork coherence + an open question: `v394` primes = atoms = facets, and `RES.COXETER.SYMMETRY.01`)

- **`v394` (`COXETER.ATOMS.01`) — the three Coxeter primes ARE the three anchor atoms ARE the three number-field facets.** A bird's-eye [E] re-reading tying v390 (the prime facets) to the anchor $a = (1, 1, 2)$ and v319 (the 5×6 clock). [E] $a = (1, 1, 2)$ has $e_3{=}2{=}|Z_2|$, $p_0{=}3{=}N_{\rm fam}$, $e_2{=}5{=}g_{\rm car}$, product $2{\cdot}3{\cdot}5 = 30 = h(E_8)$ — the clock's primes *are* the anchor's atoms. [E] atoms = facet fields (v390): $2{\to}\mathbb{Q}(i)$, $3{\to}\mathbb{Q}(\sqrt{-3})$, $5{\to}\mathbb{Q}(\sqrt{5})$, each ramified at its prime. [E] the QG susceptibility is pure $\{2, 3\}$: $\chi = 1/(1{-}(2/3)^6) = 729/665 = 3^6/(3^6{-}2^6)$, $(2/3)^6 = 2^6/3^6$, exponent $6 = 2N_{\rm fam}$. [E] both dynamic rates carry prime-2: $2/3 = |Z_2|/N_{\rm fam}$ and $(\varphi{+}2)/4 = (\varphi{+}2)/|\mu_4|$ with $|\mu_4|{=}4{=}2^2$. [E] $30 = 5{\times}6 = g_{\rm car}(2N_{\rm fam})$ (v319). [C] the prime-2 seam is the universal dynamical engine; prime-3 (family) and prime-5 (carrier golden) the two attractors.
- **`RES.COXETER.SYMMETRY.01` — a new registered open research question.** Is the prime-2-as-common-factor reading *forced* (the seam involution provably in every sector's gap, no independent prime-2 attractor), or is there a residual asymmetry? Resolving it would either tighten v383's "every sector is the same gapped operator" to "the same *seam-driven* operator" (with the falsifiable corollary that no prime-2-only attractor exists), or reveal a missing dynamical slot. Tracked in `tfpt_research_contracts`; a question, not a claim. Cited in `origin_theory`; Python-only.

2026-06-24 (Correction error-bars: `v393` the typed budget as actual first-correction magnitudes)

- **`v393` (`CORRECTIONS.NUMERIC.01`) — the typed correction budget (v388) as actual numbers.** The short-term, testable harvest of v388: instead of merely typing each prediction's correction, this gives the magnitude, so a parameter-free central value can be quoted as "value $\pm$ computed first correction". [E] FIXED-POINT band (the φ_0-derived $\theta_{12}, \theta_{13}, \lambda_C, \beta_{\rm rad}$) = the explicit φ_0 puncture term $36c_3^4/c_3 = 9/(128\pi^3) \approx 0.227\%$ (exact; $= \theta_{12}$'s ε first-correction, $\varepsilon = (3/4)\varphi_0 = c_3{+}36c_3^4$); [E] SEAM-GAPPED band (Koide $F_{\rm pole}$, recovery) $= (2/3)^6 \approx 8.78\%$, the QG-decoupling bound capped by $\chi - 1 = 64/665 \approx 9.62\%$ (v337); [E] EXACT-IDENTITY band = 0 structurally ($\det R = 8$, $N_\Phi = 1$, $\theta_{\rm eff} = 0$); [E] EXTERNAL-RATE band = external physics, quoted (m_p/m_e $\pm 7.5\%$ v374; n_s/r the $N_\star{\in}[50, 60]$ band; η_B $1.07\times$ v212), fenced by v187. [C] the assembled budget gives each prediction a computed theoretical-uncertainty band. Cited in `origin_theory`, mirrored on the website prediction surface; a harvest of v388, no new number; Python-only.

2026-06-24 (Attacking the last open lemma: `v392` the scaling-limit central charge *converges*)

- **`v392` (`SEAM.S3.SCALINGLIMIT.01`) — an honest attack on `SEAM.EQUIV.01`'s one open lemma (continuum existence, v336).** It does *not* close v336 (the abstract existence is the cited MMST theorem); it strengthens the case with the existence-relevant observable v376 did not test: that the finite-size data *converge*. [E] the Calabrese–Cardy entanglement slope $c_1(L)$ at $L = 66, 130, 258, 386, 514$ is a monotone Cauchy-like sequence approaching 1 ($|c_1(L) - 1|$ strictly decreasing, $4.6 \times 10^{-4} \to 7 \times 10^{-6}$); [E] a $1/L$ Richardson fit extrapolates to $c_1(\infty) \approx 1.000$, so the 16-Majorana collar $c = 8c_1 \approx 8.00 = g_{\rm car}{+}N_{\rm fam}$ is reached *in* the limit; [E] the successive differences are themselves strictly decreasing (a numerical Cauchy sequence). [O] convergent data are evidence, not a proof of existence (the cited MMST theorem, v336), and do not pin $(E_8)_1$ over the same-c $SO(16)_1$ (the holomorphy discriminator, v377/v378). Strengthens the residual, does *not* close it. Cited in `tfpt_research_contracts`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` extended to the finite-dimensional Perron–Frobenius case)

- **`FORM.SPECTRALGAP.01` — the multi-dimensional case, reduced to the scalar core.** The Lean module `TfptCarrier/SpectralGapAttractor.lean` now also proves the *finite-dimensional* gapped-operator statement (`lake build` OK, no `sorry`): [E] `component_tendsto` (a non-dominant eigencomponent $|r| < 1$ decays, $r^n c \to 0$); [E] `deviation_tendsto` (with the dominant eigenvalue normalised to 1, the deviation vector — all non-dominant components — tends to 0, so the iterate converges to the dominant eigendirection, the unique attractor, for any start); [E] `deviation_unique` (two starts give deviation vectors both $\to 0$, so the selected eigendirection is independent of the start — parameter-freeness in finite dimensions); [E] `flavor_deviation_tendsto` (a concrete instance on the cusp-transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$, v56/v82/v383). Honest scope: this is the finite-dimensional *diagonalisable* case (the genuine multi-dimensional content, reduced to the scalar core); the general non-diagonalisable / infinite-dimensional Perron–Frobenius theorem stays the cited statement. Referenced in `origin_theory`.

2026-06-24 (Website: an animated "Universal Gap Lab" visualising `v383`–`v391`)

- **Website — the `UniversalGapLab` animation on `/verification`.** A new client component (Framer Motion, no new dependency) with three linked, module-grounded views: (A) every sector as the gap contraction $\text{iter}_n = x_\star + r^n(x_0 - x_\star)$ racing to its attractor at $r = (2/3)^6$

(seam-gapped, v303/v387/v388) or the golden $(\varphi+2)/4$ (v312), with a gapless $r=1$ track that never forgets its start (the Lean theorem `FORM.SPECTRALGAP.01`) — the rate *is* the first-correction size (v388); (B) the entire-form-factor graviton ratio e^{-u} falling faster than any power, ghost-free and UV-soft (v386/v389); (C) the residual matrix as certification, not construction — every open item external, zero open internal mechanisms (v384). Mirror-only: every number is grounded in a module; no theory change.

2026-06-24 (Honest attempt at an external target: `v391` reduces it, not a closure)

- **`v391` (`ALPHA.QUILLEN.PROGRESS.01`) — an honest attempt at the external target `ALPHA.QUILLEN.EXACT.01` (v382, door 5).** It REDUCES the open step to a sharper named sub-target via a solvable model but does *not* close it: `ALPHA.QUILLEN.EXACT.01` stays **[O]** (external math, type A, v384). **[E]** a solvable model $\det_\zeta(-\partial_\tau^2+m^2)$ on a circle of circumference β equals $(2\sinh(\beta m/2))^2$, whose m-variation $\beta\coth(\beta m/2)=2/m+(\beta^2/6)m-(\beta^4/360)m^3+\ldots$ is a closed heat-kernel/Laurent series (the m^3 coefficient $-1/360$ verified). **[E]** so a ζ-determinant's coupling-variation is structured heat-kernel data, and v382's counterterm $8b_1c_3^6$ (a Gauss–Bonnet heat-kernel power) is the right *kind* of object; the v382 split is re-verified at $\alpha^{-1}=137.0359992$. **[E]** the reduction: the open step now reduces to the named sub-target "the seam $U(1)$ operator's Seeley–DeWitt coefficient $=8b_1c_3^6$". **[O]** *not closed*: the actual seam-operator ζ-determinant proof stays open, and `SEAM.EQUIV.01` continuum existence (MMST, v336) stays the other external **[O]**; the value α^{-1} stays **[E]**. An attempt that reduces, does not close. Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-24 (Lean 4: `FORM.SPECTRALGAP.01` formalises the scalar core of the universal spectral-gap principle)

- **`FORM.SPECTRALGAP.01` — a Lean 4 proof of the gap ⇒ unique-attractor ⇒ parameter-free core (door 4).** New module `TfptCarrier/SpectralGapAttractor.lean` (`lake build` OK, no `sorry`, only the three standard kernel axioms), formalising the meta-theorem of v303/v383/v387 at the one-dimensional reduction — the affine gap contraction $x\mapsto x_\star+r(x-x_\star)$ with multiplier $r=\lambda_2/\lambda_1$. **[E]** `iter_sub`: the closed form $\mathrm{iter}_n-x_\star=r^n(x_0-x_\star)$ (general n); **[E]** `iter_tendsto`: $0\le r<1$ (the gap) $\Rightarrow \mathrm{iter}_n\to x_\star$ (the unique attractor); **[E]** `starts_agree`: two starts differ by $r^n(a-b)\to 0$, so the attractor is independent of the initial state (parameter-freeness); **[E]** `no_gap_no_forget`: $r=1$ (no gap) leaves $\mathrm{iter}_n-x_\star=x_0-x_\star$ constant — the initial condition is then a free parameter, so the gap is exactly what removes it; **[E]** the seam rate $(2/3)^6$ and the golden $(\varphi+2)/4=(5+\sqrt{5})/8$ both lie in $[0,1)$. Honest scope: the scalar reduction (the contraction core v303/v383 iterate numerically); the full multi-dimensional Perron–Frobenius statement stays the cited theorem. Referenced in `origin_theory`; ledger row `FORM.SPECTRALGAP.01`.

2026-06-24 (Completing the Coxeter clock: `v390` the prime-2 (Gaussian / $\mathbb{Z}/4$-seam) facet of $30=2\cdot3\cdot5$)

- **`v390` (`COXETER.PRIME2.01`) — the prime-2 facet of the order-30 Coxeter clock.** v383/v387 named two number-field facets of the clock $30=2\cdot3\cdot5=h(E_8)$ — the golden $\varphi\in\mathbb{Q}(\sqrt{5})$ (prime-5, carrier) and $(2/3)^6\in\mathbb{Q}$ (prime-3, family). This names the *prime-2* facet and identifies it with structure already in the theory: the Gaussian field $\mathbb{Q}(i)$, i.e. the CM point $\tau=i$ and the order-4 $\mu_4=\mathbb{Z}/4$ *square* seam deck (v282/v288/v333). **[E]** $h(E_8)=30$ factors into exactly three primes; one quadratic facet field per prime, each ramified at its prime: $\operatorname{disc}\mathbb{Q}(i)=-4$ (2 ramifies), $\operatorname{disc}\mathbb{Q}(\sqrt{-3})=-3$ (3 ramifies), $\operatorname{disc}\mathbb{Q}(\sqrt{5})=5$ (5 ramifies); the ramified primes are $\{2,3,5\}$ with product $30=h(E_8)$. **[E]** the prime-2 (Gaussian) facet is the $\mathbb{Z}/4$ seam deck: min. poly x^2+1, $j(i)=1728=12^3$, $\tau=i$ fixed by the modular S of order $4=2^2$, $|\mu_4|=4=2^2$ — the static, even, seam-orientation facet. **[C]** so the three primes of the clock now each carry a facet (square/hexagon/pentagon = seam/CP/carrier); the prime set $\{2, 3, 5\}$ is forced by $h(E_8) = 30$, not chosen. Cited in `origin_theory`; a synthesis, no new number; Python-only.

2026-06-24 (Graviton loop finiteness: v389 the entire form factor is UV-finite at the loop level by power counting)

- **v389 (GRAV.LOOP.FINITE.01) — the entire-form-factor graviton is UV-finite at the *loop* level.** The honest next step past v386 (which established the tree amplitude), extending v304/v370/v380/v386 from the propagator to the superficial degree of divergence of loop graphs. This is the standard infinite-derivative / nonlocal-gravity finiteness statement (Tomboulis; Biswas–Mazumdar–Siegel; Modesto), *not* a full proof of perturbative quantum gravity. [E] super-polynomial falloff: $p^k e^{-p^2/M^2} \to 0$ for every degree k. [E] one dressed line turns a quadratic divergence into the finite $(M^2/2)e^{-p_0^2/M^2}$, while the GR $\int p^3/p^2\, dp$ diverges quadratically. [E] the superficial degree $\to -\infty$ at every loop order L (the form factor multiplies the integrand by $e^{-(\#\text{internal})p^2/M^2}$; dressed cutoff-ratio $\to 1$ vs GR $\to 4$). [E] the regulator is the seam scale $M = M_{\text{scal}} = c_3^{7/2}\bar{M}_{\text{Pl}}$ (v253/v259), not a dial. [C][O] loop *unitarity* of the nonlocal form factor (Lorentzian continuation / macro-causality, Pius–Sen / Briscese–Modesto) stays the honest open point — tree unitarity is v386; not claimed solved. Cited in `tfpt_4_frontier`; Python-only.

2026-06-24 (Typed correction budget: v388 the gap correction is keyed to the mechanism class, not a uniform band)

- **v388 (CORRECTIONS.BUDGET.01) — the gap-driven correction (v387) is a *typed* budget.** A machine-checked answer to “does the $(\lambda_2/\lambda_1)^n$ correction apply to *all* predictions?” — the answer is *no*, and for one whole class a band would be *wrong*. Because the size is the distance from a gapped operator’s leading eigenvector, it is defined only where a subleading λ_2 exists, so the prediction surface splits into four classes. [E] *seam-gapped* (Koide F_{pole} v303, recovery v221, the QG bound capped by $\chi = 729/665$ v337): rate $(2/3)^6$; the discrete compiler: golden $(\varphi + 2)/4$ (v312). [E] *exact identity* ($\det R = 8$, $N_\Phi = 1$, $\theta_{\text{eff}} = 0$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6}$): no λ_2, so the correction is 0 structurally and a band would contradict the [E] status. [E] *external-rate* (η_B washout, axion freeze, m_p/m_e RG): the gapped *shape* with an external rate — only F_{pole} has the seam rate (v303), firewalled (v187). [E] *fixed-point*: the texture is already explicit (e.g. $\sin^2\theta_{12}$, $\varepsilon = (3/4)\varphi_0 = c_3 + 36c_3^4$). [E] the Koide first correction (door 2): F_{pole} is the one seam-gapped prediction, rate $(2/3)^6 \approx 0.0878$ (v371), the source quotient 0.33% below 2/3 — the rate is the suppression *scale*, not the residual. [C] so a uniform band on all predictions is wrong; the size is keyed to the mechanism class. A typing of v387 (as v384 is of v383); cited in `origin_theory`; Python-only.

2026-06-24 (Harvesting the QFT/gravity machinery: v385 the UV-branch kill-test surface, v386 the graviton-exchange amplitude, v387 the gap-driven correction calculus)

- **v385 (UVBRANCH.KILLTEST.01) — the carrier-Pati–Salam UV branch on the all-order footing.** A consolidation that ties the all-order EG/BRST closure (v381) to the optional UV branch and computes the one genuinely new thing: the *proton-decay safety hierarchy*. [E] the carrier content is the SM with no new states, so the SM-non-unification (v246) is now all-order-robust; [X] the plain-SM 4D-GUT stays killed ($\sin^2\theta_W = 3/8$ vs 0.231); [C] the optional carrier-PS branch sits at $M_{PS} = \kappa M_{\text{scal}} \approx 3 \times 10^{13}$ GeV ($\kappa \in [1.0, 1.2]$, v253); [E] minimal PS gauge bosons are $SU(4)$ leptoquarks that mediate rare LFV ($K_L \to \mu e$), *not* $p \to e^+\pi^0$, so the branch clears the binding bound by $M_{PS}/M_{\text{bound}} \sim 10^7$ – proton-safe, with *no* fake τ_p window (v265). Cited in `tfpt_5_redteam`; Python-only.

- **v386 (GRAV.AMPLITUDE.01) — the graviton-exchange amplitude is finite and UV-soft.** Extends v304/v370/v380 from the propagator to the actual tree amplitude. [E] the dressed spin-2 propagator $e^{-p^2/M^2}/p^2$ has its only pole at $p^2 = 0$ (residue $1 > 0$, ghost-free), is exponentially softened in the UV (the ratio e^{-u} falls faster than any power) and reduces to GR + a calculable $-p^2/M^2$ correction at low energy ($M = M_{\rm scal}$, v253); [C] a single positive-residue pole gives tree unitarity, so perturbative graviton scattering $A \sim (T{\cdot}P_2{\cdot}T')e^{-p^2/M^2}/p^2$ is an explicit finite computation (perturbative-only; `QG.AMB.01` stays a [C] redundancy, v369). Cited in `tfpt_4_frontier`; Python-only.
- **v387 (CORRECTIONS.GAP.01) — the gap-driven correction calculus.** The practical harvest of v383: the same spectral gap that makes a sector parameter-free also sets the *size* of its first correction (correction$_n \sim (\lambda_2/\lambda_1)^n$). [E] flavor, horizon recovery and the QG bound all share the $(2/3)^6$ rate (the family-3 facet; QG susceptibility $\chi = 729/665$), while the discrete compiler transient decays at the golden $(\varphi + 2)/4$ (the carrier-5 facet) — the two number-field facets of v314/v383. [C] the gap does double duty (why no free parameter *and* how big the first correction), so sub-leading magnitudes are organised, not free. Cited in `origin_theory`; Python-only.

2026-06-23 (Bird's-eye synthesis: two new structural patterns — v383 the universal spectral-gap principle and v384 the residual is certification, not construction)

- **v383 (DYNAMICS.UNIVERSAL.01) — the universal spectral-gap / Perron–Frobenius principle.** A fresh bird's-eye pass found that *every* TFPT sector is the *same* structural object: a gapped operator with a *unique* leading attractor (the physics) and a spectral gap (the reason there is no free parameter). v303 named this only for $F_{\rm transfer}$; this module extends it to the sectors made parameter-free *after* it. [E] three gaps are computed — the flavor transfer $\{1, (2/3)^6, (1/3)^6\}$ with gap $6\ln(3/2)$ (v56/v82); the affine-E_8 adjacency with Perron eigenvalue 2 (Kac marks) and subleading $2\cos(\pi/5) = \varphi$, gap $2 - \varphi = 1/\varphi^2$ (v312/v313); the decoupling margin $\Delta_{\rm eff} \approx 1.648$ (v76/v337). [E] the two number-field facets (v314): golden $\varphi \in \mathbb{Q}(\sqrt{5})$ (carrier-5, static) and $(2/3)^6 \in \mathbb{Q}$ (family-3, dynamic) are the prime-5 and prime-3 facets of the order-30 Coxeter clock $30 = 2{\cdot}3{\cdot}5 = h(E_8)$. [C] the other sectors are the same structure — gravity ($\delta S{=}0$ equilibrium, v358/v359), the QG measure (Gaussian saddle, v365), the boundary QFT (holomorphic $c{=}8$ RG fixed point, v157/v158), the all-order S-matrix (adiabatic limit, v381), recovery (Petz fixed point, v221), $\tau{=}i$ (v333). [C] the meta-theorem: gapped $\Rightarrow$ unique attractor $\Rightarrow$ parameter-free, theory-wide. Cited in `origin_theory`; a synthesis (no new number), Python-only.
- **v384 (RESIDUAL.CERTIFICATION.01) — the residual is certification, not construction.** After v369/v381/v382 the *kind* of every open item has shifted: there is *no* open TFPT physics mechanism left. A ledger-CI audit classifies every residual into exactly three external kinds: [E] (A) external math proof (`SEAM.EQUIV.01` continuum existence = the cited MMST/Adamo theorem v336; `ALPHA.QUILLEN.EXACT.01` v382; `QG.AMB.01` the constructive measure, a [C] redundancy v369), [E] (B) theorem-forbidden ($v_{\rm geo}$, the No-Unit theorem v153/v364), [E] (C) external physics ($F_{\rm transfer}$, v371–v374, firewalled v187). The two load-bearing facts are verified (No-Unit dimensionlessness; the gap-decoupling margin $1.648 > 0$), and the count of open *internal* TFPT mechanisms is **zero**. [C] convergence: the only hard things left are theorems and external physics, not missing dynamics. Cited in `tfpt_research_contracts`; an audit (no new number), Python-only.

2026-06-23 (External-review closure: the two genuine residual targets named — v381 the all-order Epstein–Glaser/BRST contract for S_{pert}, and v382 the Quillen determinant-line variation as a tracked target)

- **`v381` (`QFT4D.EG.ALLORDER.01`) — the all-order 4D perturbative-QFT closing statement.** A mathematical-physics review of the QFT/TOE layers identified exactly one genuinely new, buildable item: the *all-order* Epstein–Glaser/BRST contract for the perturbative 4D S-matrix S_{pert} (the previous documents had the S_{pert} skeleton and one-loop, v269/v271/v273/v278, but no named all-order claim). This module types the standard causal-perturbation-theory closure for the TFPT finite interaction class — it does *not* build a path integral. [E] power-counting: every gauge+matter vertex (Yang–Mills + Dirac–Yukawa + Higgs + ghost) is a mass-dimension-4 marginal operator, so the EG singular order $\omega \leq 4$ and the counterterm space is finite at every order; [E] BRST nilpotency s^2=0 verified exactly via the Jacobi identity for $su(2)$ and $su(3)$ (the carrier weak+colour content); [E] the seam gap $\Delta = 6\log\frac{3}{2} > 0$ (v302) gives the IR-safe adiabatic limit. [C] the two heavy legs are imported rigorous theorems — all-order T_n existence by causal factorisation (Epstein–Glaser 1973) and all-order BRST invariance (Piguet–Sorella / Kugo–Ojima) — applied to the TFPT class. Matter+gauge *only*: the R^2/Weyl^2 gravity sub-sector is the entire-form-factor resummation (v304/v370/v380), and S_{pert} is not the ambient measure `QG.AMB.01` (a [C] redundancy, v369). NET [C], prerequisites [E]; not [E] and not a path integral. Cited in `tfpt_5_redteam` and `tfpt_2`; Python-only.
- **`v382` (`ALPHA.QUILLEN.EXACT.01`) — the Quillen determinant-line variation, named as a target.** Per the same review, the "why *this* $F_{U(1)}$ functional" obligation (previously tracked only as `EM.WARD.01`'s residual, v341/v342) is elevated to its own citable target: $\delta_\tau(\log\det_\zeta \Delta_{U(1)} + 8b_1c_3^6\log\varphi_{\text{seam}}) = 0$ on the determinant line over the μ_4 seam moduli. [E] the Quillen split holds at the α^{-1}=137.0359992 root with every coefficient a named atom (index $8b_1$, heat-kernel $c_3^{3,6}$, discriminant $q(D_5)$+$q(A_3)$=2); [E] the value stays closed (v3); [O] the from-first-principles exactness proof is the target. [C] it is a *face* of `SEAM.EQUIV.01` (v336) — the seam $U(1)$ determinant line lives on the same holomorphic net — so it adds *no* second independent gate; it makes the existing residual trackable. No new number (reuses the Wolfram-mirrored v341 identities). Cited in `tfpt_1_architecture_e8`; Python-only.

2026-06-23 (Track 2 of the forward plan v2 — the AMBIENT REDUNDANCY statement (v369): the holographic route to Gravity-complete that sidesteps building the general Euclidean QG measure)

- **`v369` (`QGAMB.REDUNDANCY.01`) — Track 2 (forward plan v2).** Instead of constructing the non-perturbative ambient measure (the conformal-factor swamp, v332/v334), assemble the established discriminators that prove the ambient sector carries *no* physical degrees of freedom relative to the holomorphic boundary algebra, so the QG measure (gate C7/`QG.AMB.01`) is a non-fundamental *certification* object (like a coordinate choice in GR), not missing dynamics. [E] holomorphy ⇒ DHR$((E_8)_1)$ = Vec: #primaries = $|\det \text{Cartan}| = 1$ for E_8 vs 4 for $SO(16)$ ⇒ no nontrivial superselection charge a bulk could carry invisibly; [E] no torus ground-state degeneracy ($|\det K| = 1$); [E] finite Petz recovery rate $(2/3)^6$ (v221); [E] the gap-decoupling margin $\Delta_{\text{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337). [C] with the Bisognano–Wichmann reconstruction map (v258/v309/v323) the redundancy statement $\mathcal{O}^{\text{bulk}}_{\text{phys}}/\text{Diff} \cong \mathcal{A}^{\mu_4}_{\Sigma}$ is well-posed. [O] residual: the two consumed premises — `SEAM.EQUIV.01` (the S3 lattice input, v366) and the BW intrinsicality of the raw collar (v329); it does *not* construct `QG.AMB.01` but provides the *alternative* (holographic redundancy) route: Gravity-complete = Boundary-complete + Ambient-redundancy. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Status propagation: the v365–v380 + FORM.SEAM.MMST.01 status carried into the remaining papers, the README and the Zenodo description)

- **Deep-sync of the post-S3 status.** The canonical surfaces (`tfpt_status`, `introduction`) were already current; this pass propagated the same honest wording into `origin_theory`, `tfpt_1`–`tfpt_5`, `tfpt_research_contracts`, the root `README.md`, the Zenodo description and `website/lib/papers.ts`: `SEAM.EQUIV.01` is *closed modulo a cited theorem* (lattice v367/v368 + the S_3 stack v376–v379 + Lean `FORM.SEAM.MMST.01`; residual [O] = the cited MMST/OS existence only); `QG.AMB.01` is a [C] *redundancy* (v369/v379), a certification object not missing dynamics; perturbative graviton unitarity is established (the Stelle ghost is a Seeley–DeWitt truncation artefact, v304/v370/v380); and the frontier transfers are typed runnable solvers with kill tests (v371–v374) + the observatory CI (v375). No new claim — a wording propagation so every surface matches the ledger; `AUDIT OK`.

2026-06-23 (v380 — the KMS Entire Hessian: the Stelle ghost is EXACTLY the Seeley–DeWitt truncation, and resummation pushes it to infinity)

- **v380 (GRAV.KMS.HESSIAN.01).** Upgrades v304/v370 from the *assumption* "the resummed graviton form factor is entire" to a derived statement. [E] the seam KMS cutoff $f(u)=e^{-u}$ (v259) gives the dressed spin-2 propagator $e^{-p^2/M^2}/p^2 = 1/(p^2 a)$ with $a(u)=e^u$ entire and nowhere zero (only the $p^2=0$ pole). [E] each finite Seeley–DeWitt truncation is a Taylor partial sum of a and carries a Stelle-ghost zero ($T_1=1+u \to u=-1$; $T_2 \to -1\pm i$); the entire a has none. [E] the modulus of the nearest truncation-zero grows monotonically with the order $(1, 1.41, \ldots, 3.07$ for $n=1..8) \to \infty$ — the ghost decouples, "entire" = "do not truncate". [C] $\Rightarrow$ perturbative graviton unitarity; [O] the exact off-shell curved-space Hessian identification is the cited heat-kernel resummation, perturbative-only. Cited in `tfpt_5_redteam` (Target F); Python.

2026-06-23 (Lean: the S3 continuum leg formalised as "closed modulo a cited theorem" — FORM.SEAM.MMST.01)

- **`FORM.SEAM.MMST.01` — Lean 4 formalisation.** The S3 continuum leg is now machine-formalised in a new module (`SeamScalingLimit.lean`), lake build OK and `#print axioms` clean, in the audit-contract style of `SeamEquivChain.lean`. The MMST applicability hypotheses for the seam collar are *provable* by kernel `decide` (no axioms): $D = 16$, rank $= c = 8$, the range $8 \leq 8 \leq 16$, $c_- = 8 \neq 0$, and $\det K = 1$ vs $SO(16)$'s 4. The MMST scaling-limit theorem (CMP 2022) and the Adamo–Moriwaki–Tanimoto OS reconstruction (2024) are *named cited axioms*; the conclusion $(E_8)_1$ follows by a well-typed composition whose residual (`#print axioms`) is exactly those two published theorems plus the carrier gap — no hidden assumption, no `sorry`. Mirrors the S3 closure stack `v376–v379`; so `SEAM.EQUIV.01` is now "closed modulo a cited theorem", the honest formalisation ceiling. Cited in `tfpt_research_contracts`; `lean_manifest` refreshed.

2026-06-23 (S3 closure stack — the target of the seam scaling limit pinned at EVERY computable level: central charge $c=8$ from the lattice (v376), the $(E_8)_1$ character (v377), the genus-1 torus count = 1 (v378), and reflection positivity (v379); only the abstract continuum existence (cited MMST) stays external)

- **v376 (SEAM.S3.CENTRALCHARGE.01).** The Calabrese–Cardy finite-size entanglement-entropy scaling of the critical free-fermion content gives $c = 1.0000$ per complex mode (Peschel correlation-matrix EE; the $L=4m+2$ sizes avoid the half-filling zero-mode), so the 16-Majorana (= 8 complex) collar has $c = 8 = g_{car}+N_{fam}$, chiral ($c_-=8$, v367/v368). Numerical $c=8$ from the lattice. Python (numpy).

- **`v377 (SEAM.S3.E8CHARACTER.01)`.** The exact affine character $\chi = E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ gives $c = 8$ and 248 level-1 currents (one primary); the μ_4/GSO promotion $248 = 240 + 8 = 120 + 128$ ($240 = 16{\cdot}5{\cdot}3$) pins $(E_8)_1$ over the same-c rival $SO(16)_1$ (120 currents, 4 primaries). Python (sympy q-series).
- **`v378 (SEAM.S3.MODULAR.01)`.** The genus-1 discriminator: KLM condensation μ(carrier)=16 $\to$ $\mu(E_8)$=16/4^2=1, $|\det \mathrm{Cartan}(E_8)|$=1 vs D_8=4, and the single character's modular T-eigenvalue $e^{-2\pi i/3}$ give torus ground-state degeneracy 1 (holomorphic = $(E_8)_1$) — the "hard part" v344 isolated. Python (sympy).
- **`v379 (SEAM.S3.RP.01)`.** Reflection positivity (the OS axiom) of the explicit gapped collar: the positive spectral measure makes the Hankel/reflection Gram matrix PSD (a ghost mode breaks it). With c=8 + $(E_8)_1$ in hand, the OS inputs are complete and C7/`QG.AMB.01` is then discharged by the redundancy theorem v369. Python (numpy).
- **Net:** S3 is the master key — on the explicit lattice model the target net $(E_8)_1$ is pinned exactly (central charge, character, genus-1 count, RP); the only remaining residual is the abstract *continuum existence* of the scaling limit (the cited MMST theorem v336), which lives outside the computational suite. Cited in `tfpt_research_contracts`.

2026-06-23 (Track 4 — the prediction OBSERVATORY: a status-typed CI over the frozen prediction registry that makes the falsifiability surface machine-checkable, with a live data scorecard (v375))

- **`v375 (OBSERVATORY.REGISTRY.01)`.** A CI over `freeze_file.csv`: [E] every prediction carries a kill criterion (falsifiability complete, no orphan claims); [E] the headline values re-derive from $\{\varphi_0^{\rm ret}\}$ ($\sin^2\theta_{12}$=1/3$-\varphi_0^{\rm ret}$/2, $\sin^2\theta_{13}$=$\varphi_0^{\rm ret}e^{-5/6}$, $\beta_{\rm rad}$=$\varphi_0^{\rm ret}/(4\pi)$); [C] a live scorecard records θ_{12} vs the JUNO first run (0.28σ, compatible), θ_{13} vs NuFIT 6.0 NO (2.0σ, the flagged most-tensioned core prediction), $\beta_{\rm rad}$ vs ACT DR6 (0.37σ, a hint), $r < 0.036$ (compatible). Tooling that makes the closed pieces usable and the open ones honestly fenced; no new physics. Cited in `tfpt_5_redteam`; Python (stdlib).

2026-06-23 (Track 3 — the $F_{\rm transfer}$ functor promoted from contracts to TYPED runnable solvers with kill tests: Koide source→pole (v371), η_B Boltzmann (v372), axion finite-T relic (v373), m_p/m_e uncertainty budget (v374))

- **`v371 (FR.POLE.SOLVE.01)`.** A clean *negative*: standard QED/EW running cannot be the Koide source→pole transfer (it pushes Q above 2/3; the required ϵ is negative while QED's is positive), so the transfer is the non-QED 53/54 operator / $(2/3)^6$ Moebius generator (v183/v82). [C], plain running excluded as the kill. Python (numpy+scipy).
- **`v372 (FR.BOLTZMANN.SOLVE.01)`.** The integrated Buchmueller–Di Bari–Pluemacher Boltzmann ODE at the TFPT-frozen M_1=$M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda$ (v212) gives η_B within a factor $\sim$1.1 of $6.1{\times}10^{-10}$ with no M_R dial; replaces the v326 toy washout. [C]/[X].
- **`v373 (FR.RELIC.SOLVE.01)`.** The finite-T axion misalignment ODE decides the branch: the frozen spine angle $3\pi/5$ gives $\Omega_a h^2{\sim}0.12$ (in $[0.08, 0.16]$, lands on $\Omega_{\rm DM}$ untuned) while the hilltop 170° over-produces ($\sim$5.4×); replaces the v326 toy. [C]/[X](lattice $\chi(T)$ input).
- **`v374 (FR.QCD.BUDGET.01)`.** The m_p/m_e uncertainty budget: with $b_3 = -7$ [E] and the two declared externals ($\alpha_s(M_Z)$, C_p, $\sim$5.3% each), the central $\sim$1824 (v262) carries a propagated $\pm$7.5% band $[\sim 1690, \sim 1960]$ CONTAINING 1836.15; lattice tightening is the kill. The falsifiability companion to v262; [C].

2026-06-23 (Track 1 — the S3 input made CONCRETE: an explicit gapped chiral-Majorana ($p{+}ip$) lattice model with numerical Chern $|C|{=}1$ realises the $c_{-}{=}8$ edge (v367), and a strip shows the chiral edge exists by anomaly inflow (v368))

- `v367` (`SEAM.S3.LATTICE.01`) — **Track 1.** The abstract S3 residual of v356/v366 ("the collar realised as a genuine lattice chiral free-fermion invertible phase") is turned into a RUNNABLE model. [E] the $p{+}ip$ Bloch model is gapped at $M{=}1$ with a numerically computed Fukui–Hatsugai–Suzuki Chern number $|C|{=}1$ ($C{=}0$ trivially at $M{=}3$); [E] $N_{\mathrm{Maj}}{=}\dim S^{+}{=}16$ copies give $c_{-}{=}16/2{=}8{=}g_{\mathrm{car}}{+}N_{\mathrm{fam}}$, with the edge K-matrix $SO(16)_1$ ($\det 4$) $\to$ $(E_8)_1$ ($\det 1$) by the order-4 μ_4 clock (v351/v369). [C] the μ_4 condensation (v154/v351); [O] the continuum scaling-limit existence (MMST, v336) stays open. Cited in `tfpt_research_contracts`; Python (numpy + sympy).
- `v368` (`SEAM.S3.INFLOW.01`) — **Track 1.** The bulk-edge / anomaly-inflow leg (S4) made concrete on the v367 model: [E] a finite strip has an edge-localised in-gap chiral branch ($\min_{k_x}|E|\sim 0$) in the topological phase and a clean gap in the trivial phase (neg control); [E] bulk-edge correspondence gives one chiral Majorana per edge ($c_{-}{=}8$). [C] so by anomaly inflow the chiral edge EXISTS as a consequence of the invertible bulk, discharging the v336/v351 "does the edge exist?" residual on the lattice. [O] the continuum scaling limit stays open. Cited in `tfpt_research_contracts`; Python (numpy).

2026-06-23 (Track 2 — perturbative graviton unitarity sector-by-sector: the Barnes–Rivers spin decomposition localises the Stelle ghost in the spin-2 sector, cured by the entire KMS form factor; the spin-0 mode by the GHP contour (v370))

- `v370` (`GRAV.SPIN2.UNITARITY.01`) — **Track 2.** Extends v304 (the scalar $1/(p^2(p^2{+}M^2))$ pole algebra) to the actual tensor spin structure. [E] the Barnes–Rivers projectors are complete ($\mathrm{tr}\,P_2{=}(d{-}2)(d{+}1)/2$, $\mathrm{tr}\,P_1{=}d{-}1$, $\mathrm{tr}\,P_{0s}{=}\mathrm{tr}\,P_{0w}{=}1$ sum to $d(d{+}1)/2$; $d{=}4$ split $5{+}3{+}1{+}1{=}10$), the EH propagator is $P_2/p^2 - \frac{1}{d-2}P_{0s}/p^2$, and the Stelle ghost is a *spin-2* state. [E] the entire KMS form factor e^{p^2/M^2} (v304/v259) makes the spin-2 sector ghost-free; [C] the wrong-sign spin-0 conformal mode (v332) is cured separately by the GHP contour (v334). [C] so the full *perturbative* graviton is ghost-free sector-by-sector (conditional on v304's entire-analyticity assumption). [O] residual: perturbative-only + the open analyticity assumption; not the non-perturbative measure. [O] DECLINE: $\mathrm{tr}\,P_2{=}5$ is dimension counting, not g_{car} (v354/v355). Cited in `tfpt_5_redteam` (Target F); Python-only.

2026-06-23 (Directions 3 & 7 — the two hard functional-analysis residuals ADVANCED: the QG measure as one-loop fluctuations around the parameter-free saddle (v365), and the seam collar verified IN MMST's free-fermion class so the scaling limit is $(E_8)_1$ (v366))

- `v365` (`QGAMB.SADDLE.01`) — **Direction 3.** With the parameter-free saddle (v358/v359) the QG *measure* (gate C7/`QG.AMB.01`) organises as a one-loop Gaussian fluctuation determinant. [E] the fluctuation stiffness is $1/c_3 = 8\pi$ (no free dial, so the one-loop problem is now well-posed); [E] the fluctuation operator is gapped $M \geq \Delta_{\mathrm{eff}} = 6\log\frac{3}{2} - \frac{31}{4\pi^2} \approx 1.648 > 0$ (v76/v337) $\Rightarrow$ mode-by-mode convergence, finite model log-det $\mathrm{Tr}'\log M = 6\log\frac{9}{2}$; [C] the conformal mode converges on the GHP contour (v334); [E] the admissible projective limit exists ($\chi = 729/665$, v330) with 0 new dials (v364). [O] residual: the full non-perturbative projective limit on the ambient sector. Sharpens C7 from "diffuse full QG" to a one-loop Gaussian problem around the parameter-free saddle; does not close it. Cited in `tfpt_research_contracts`; Python-only.

- **`v366` (`SEAM.MMST.INCLASS.01`) — Direction 7.** The seam collar is verified IN MMST's free-lattice-fermion scaling-limit class hypothesis-by-hypothesis: **[E]** $D = \dim S^{+} = 16$ Majoranas, $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8$, $\operatorname{rank} E_8 = 8$ so rank $\leq c \leq D$ is $8 \leq 8 \leq 16$ (in range); **[E]** gapped ($\Delta = 6 \log \frac{3}{2} > 0$, derived v302); **[C]** quasi-free CAR class (v155/v160); **[E]** chiral ($c_{-} = D/2 = 8 \neq 0$). **[C]** the MMST scaling-limit theorem then applies, and **[E]** the order-4 μ_4 clock pins the target to $(E_8)_1$ ($\det K = 1$ vs $SO(16)$ $\det K = 4$, v351). **[O]** residual: the single S3 input (the collar as a genuine lattice chiral invertible phase, the seam one-sidedness, v297/v356). Reduces the continuum side of `SEAM.EQUIV.01` to one named input; ADVANCES, does not close. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Consistency pass: the parameter-free *full covariant* Einstein equation (`v358`/`v359`) propagated across ALL stale paper and website surfaces; new `v359` keybox; homepage gravity animation)

- **Deep-sync of the parameter-free gravity status.** A cross-surface audit found several documents and website components still framing gravity as "exact only in the $R + R^2$ regime" or the metric-sector field equation as "open". These were reconciled with the canonical post-v358/v359 status: the *local* covariant field equation $G_{ab} + \Lambda g_{ab} = c_3^{-1} T_{ab}$ is PARAMETER-FREE (both coefficients fixed); what remains open is the Jacobson equation-of-state status, the global ambient measure (`QG.AMB.01`/C7), and the single unit v_{geo}. Touched: `introduction`, `tfpt_2_standard_model`, `tfpt_3_e8_audit_bootstrap`, `tfpt_4_frontier` (summary table), `tfpt_research_contracts`, `origin_theory`.
- **New `v359` keybox** added to `tfpt_research_contracts` (full covariant equation: fixed-volume ⇒ Einstein tensor, Lovelock ⇒ conservation an output); the v358 residual corrected from "linear→nonlinear" (stale — v359 closed it) to the equation-of-state + C7 + v_{geo} residual, including the generated script index (`script_registry.csv`).
- **Website.** The `GravityEmergence` animation (three origins of c_3 converging to $1/(8\pi)$; the parameter-free Einstein equation) was updated to the full covariant result and surfaced as a dedicated homepage section (`/#gravity`); gravity-status wording aligned across `ReconstructionChain`, `VerificationDag`, `StatusMatrix`, `WhatTfptClaims`, `ThreeDecoderMap`, `StatusPyramid`, `papers.ts`, `glossary.ts` and the review page.

2026-06-23 (Direction 6, v_{geo} sharpened: after the parameter-free gravity, the *gravity* sector adds NO new dimensionful input either, so v_{geo} is the SINGLE dimensionful input of the complete theory — final tally $\{v_{\mathrm{geo}}, \pi\}$, ZERO dimensionless dials: `v364`)

- **`v364` (`VGEO.SHARPEN.01`) — Direction 6.** The No-Unit Theorem (v153) already made v_{geo} a forced metrology primitive. **[E]** The SHARPENING after the parameter-free gravity (v358/v359/v361): the Einstein coefficient $1/c_3 = 8\pi$ and the Λ prefactor $(8\pi)^2 \cdot 48 c_3^4 = 3/(4\pi^2)$ are dimensionless, so $1/G \sim v_{\mathrm{geo}}^2$ and $\Lambda \sim$ (dimensionless) v_{geo}^4 reduce to v_{geo} times atoms — the gravity sector adds no new scale. **[E]** FINAL TALLY: every dimensionful quantity = (a dimensionless ratio)×(a power of v_{geo}); TFPT's free content is $\{v_{\mathrm{geo}}, \pi\}$ with ZERO dimensionless dials, a $\sim 26 \to 1$ reduction vs the SM. **[O]** v_{geo} stays irreducibly primitive (No-Unit; $1/G$ UV-sensitive, Sakharov v68). Direction 6 is clarity — the last dimensionful anchor named and shown to be the only one. Cited in `tfpt_research_contracts`; Python-only.

2026-06-23 (Direction 5, the matter–gravity backreaction: the carrier's gravitational anomaly is forced c_{-}=8, and the matter vacuum-energy backreaction is finite ($\to$ the forced Λ from α, not $M_{\rm Pl}^4$) — the loop closes; no new SM-quantum-number read-out (declined): `v361`)

- **`v361` (`GRAV.BACKREACT.01`) — Direction 5.** With the parameter-free equation (v359) and the explicit J3 flux (v358), the backreaction of TFPT's carrier matter is computed. [E] the carrier's gravitational anomaly is forced: $c_{-} = 16/2 = 8 = g_{\rm car}+N_{\rm fam}$ (the chiral central charge of the 16-Majorana content). [E] the backreaction is FINITE $\to \Lambda$ from α: the matter vacuum energy through $G_{ab} = c_3^{-1}T_{ab}$ is gap-suppressed (Decoupling Thm v337) to $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$ (v60), the 123 orders $= 119.03 + 3.92$ — *not* the $M_{\rm Pl}^4$ catastrophe; the loop (v359+v60 + the carrier vacuum) closes. [E] J3 explicit: $\delta\langle K_B\rangle = (8\pi^2 R^4/15)\delta\langle T_{00}\rangle$, carrier T_{ab} = the SM's (v159). [C] the SM trace anomaly is reproduced (content = SM). [O] DECLINE: no new atom-forced read-out of individual SM quantum numbers beyond c_{-}=8 and Λ — per v354/v355 declined. The backreaction is finite and consistent, adding no new dial. Cited in `tfpt_4`; Python-only.

2026-06-23 (Direction 2, the disciplined result: the gap-induced GR correction is the known R^2 scalaron — already falsifiable via n_s/r; a *new* forced near-term deviation does NOT survive the discriminator and is declined: `v360`)

- **`v360` (`GRAV.GAPCORR.01`) — Direction 2, run honestly.** Does the seam transfer gap $\{1,(2/3)^6,(1/3)^6\}$ yield a NEW, calculable, FORCED deviation from GR (a near-term kill test) beyond the known R^2 scalaron? With the v354/v355 discriminator the answer is *no*, and this is recorded rather than manufacturing a coefficient. [E] the leading higher-curvature term IS the R^2 scalaron (spectral-action a_4=$R^2/72$, v36; $M_{\rm scal}^2/\bar{M}_{\rm Pl}^2 = c_3^7$, exponent 7=$g_{\rm car}$+$N_{\rm fam}$−1) — the calculable beyond-Einstein correction, already in TFPT. [C] it is the entanglement-equilibrium NEXT order (the Wald first law $\to f(R)$=R+R^2, v28/v359). [E] it is ALREADY falsifiable via n_s=1−2/$N_\star$, r=12/$N_\star^2$ (live CMB-S4 kill test). [E] the gap Δ=$6\ln\frac{3}{2}$ is a dimensionless rate, so a distinct gap-induced curvature^2 term sits at $\xi\, v_{\rm geo} \sim$ the seam/Planck scale (not near-term observable). [O] DECLINE: a new dimensionless coefficient is not atom-forced (the scalaron exponent 7=g+N−1 and the gap exponent 6=2$N_{\rm fam}$ are different; c_3^7 and $(2/3)^6$ unrelated) — per v354/v355 declined. Direction 2 yields no new near-term forced prediction; the answer is the existing scalaron. Cited in `tfpt_4`; Python-only.

2026-06-23 (the FULL covariant Einstein equation, parameter-free: the fixed-volume entanglement equilibrium gives $G_{ab}+\Lambda g_{ab} = c_3^{-1}T_{ab}$ with BOTH coefficients TFPT-fixed ($8\pi = 1/c_3$ from v358, Λ from α via v60); B6 closed at the local level: `v359`)

- **`v359` (`GRAV.NONLINEAR.01`) — Direction 1, the full covariant equation.** Extends v358 from the linearised (fixed-radius, Ricci-level) result to the FULL covariant equation. [C][E] Jacobson 2015: demanding stationarity of the small-ball entanglement entropy at fixed *volume* brings in the EINSTEIN TENSOR G_{ab} (not just R_{ab}) — trace $g^{ab}G_{ab} = (1-d/2)R = -R$ in d=4, the structure that carries Λ. [E] Lovelock: G_{ab} is the unique symmetric divergence-free second-order metric tensor, so $\nabla^a(G_{ab}+\Lambda g_{ab}) = 0$ forces $\nabla^a T_{ab} = 0$ — conservation is an output. [E] coefficient 1: $8\pi G \to 8\pi = 1/c_3$ fixed (v358). [E] coefficient 2: Jacobson leaves Λ free, TFPT fixes it — $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/(4\pi^2))e^{-2\alpha^{-1}}$, prefactor $(8\pi)^2{\cdot}48c_3^4 = 3/(4\pi^2)$ (v60); so BOTH coefficients are TFPT-fixed, not free integration constants. [C] all timelike directions $\to$ the covariant tensor equation; the matter side is the assembled CHM ball modular flux (v358/v323). NET: the original B6 "full covariant Einstein-side field equation" is now PARAMETER-FREE at the local level. [O] residual: the Jacobson equation-of-state status (not a from-action quantisation) + the global ambient measure (C7/`QG.AMB.01`, separate, gap-decoupled); $v_{\rm geo}$ the one unit. Cited in `tfpt_research_contracts`; mirrored in Wolfram (303/303). (`next-plan.md` Direction 4 folded in: the Λ here matches the α-readout.)

2026-06-23 (the classical gravity field equation, parameter-free: the entanglement first law $\delta S = \delta\langle K\rangle$ with TFPT's atoms gives the LINEARISED Einstein equation with G from c_3; the thermodynamic and geometric origins of c_3 coincide via $|\mu_4| = |\mathbb{Z}_2|\chi = 4$; the two formerly-open pieces (matter flux J3, entropy-density coefficient) closed/atom-fixed: v358)

- **v358 (GRAV.ENTROPY.EQUILIBRIUM.01) — the bridge to the dynamic world, instantiated on the gravity side.** Carrying out the Jacobson-2015 / Faulkner-et-al. "first law of entanglement ⇒ Einstein equation" with TFPT's atoms. [E] PARAMETER-FREE COEFFICIENT: $\delta S = \delta\langle K\rangle$ gives $G_{ab} = c_3^{-1}T_{ab}$ with $c_3^{-1} = 8\pi$ fixed; all four c_3-roles consistent (Unruh $1/(2\pi)$=$4c_3$, EH $1/(16\pi)$=$c_3/2$, Einstein 8π=$1/c_3$, Bekenstein $1/4$=$1/|\mu_4|$) — no free dimensionless Newton dial. [E] THERMO = GEO COINCIDENCE (new): the first law's coefficient $2\pi/\eta$ equals the geometric $|\mathbb{Z}_2|2\pi\chi$ iff $|\mu_4| = |\mathbb{Z}_2|\chi = 4$ (v216) — the thermodynamic and geometric origins of c_3 are ONE. [C] J3 SOLVED: the CHM ball modular Hamiltonian (boost via BW, v323) gives the matter flux $\delta\langle K_B\rangle = \frac{8\pi^2 R^4}{15}\delta\langle T_{00}\rangle$. [E] ENTROPY DENSITY ATOM-FIXED: $1/4$=$1/|\mu_4|$ (v57) + central charge $c = g_{\rm car}$+$N_{\rm fam} = 8$ (form v59). [O] residual: linear→full non-linear (the original B6) + the $v_{\rm geo}$ unit. Closes the LINEARISED Einstein equation parameter-free; cited in `tfpt_research_contracts`; mirrored in Wolfram (302/302).

2026-06-23 (the two frontier walls attacked directly: Direction A assembles the continuum chain of `SEAM.EQUIV.01` theorem-by-theorem and ties its residual to the seam one-sidedness; Direction B was found ALREADY closed by v40 (harmonic metric = finite linear algebra, not a PDE): v356)

- **v356 (SEAM.EQUIV.CONTINUUM.03) — Direction A, the continuum chain assembled.** After the reverse-audit/bandwidth work showed no hidden *discrete* lever remains, the only fundamental progress is to assemble the *continuum* chain. Link-by-link: **S1** the gapped (Δ=$6\ln(3/2)$>0, v302) quasi-free 16-Majorana collar (v155/v160) → **S2** invertible (gapped free fermions have no intrinsic topological order, Kitaev, v301) → **S3** the *nontrivial* invertible phase (c_-=8≠0) → **S4** a *non-gappable* anomalous chiral edge (anomaly inflow / bulk–edge), so edge *existence* is a CONSEQUENCE of S2+S3, not a separate assumption — this directly reduces the v336/v351 "does the collar have a chiral edge?" residual → **S5** its chiral-CFT scaling limit (MMST, v336; range rank $\le c \le D$, $8 \le 8 \le 16$) → **S6** the order-4 μ_4 clock + the AQFT stack pin $(E_8)_1$ (v351/v297). [E] the only non-theorem link is S3, and [C] S3 is the seam being *one-sided* (chiral, c_-≠0) — the *same* one-sidedness that makes c_3=$1/(8\pi)$ a one-sided Gauss–Bonnet ($|\mathbb{Z}_2|\oint K$, v58). [O] the one remaining input is "the abstract collar is realized as a genuine lattice chiral free-fermion invertible phase" (the v297 Flat-Away). NET: `SEAM.EQUIV.01` reduced to ONE realization input tied to the constant that *defines* the theory; it ADVANCES, it does not close.
- **Direction B — found ALREADY closed by v40.** The U-wall "transcendental Hitchin solve" (v31) is the worst case for a *generic* Higgs bundle; but at the *polystable* point TFPT selects, Mehta–Seshadri makes the bundle *unitary*, so the harmonic metric is the constant invariant Hermitian form (*finite linear algebra*, Higgs Φ=0) and det R=8 is read off — not a PDE. Only the absolute amplitude $U_{\rm point}$ (an anchor) remains. A redundant new B module was built, found to duplicate/contradict v40, and *deleted* (no padding). Honest outcome: B needs no new work.

2026-06-22 (bandwidth search of the unmapped E_8 region, with a strict forced-identity discriminator: the genuine overlooked structure is COLLECTIVE ($\sum \deg = 128 = \dim S^+$, $\prod \deg = |W(E_8)|$, exponents = totatives of 30); the per-degree atom matches are unforced and DECLINED; no new physical hit – reconciling `v66` and `v354`: `v355`)

- **`v355` (`E8.UNMAPPED.BANDWIDTH.01`) — the next step after v354, run honestly.** The obvious follow-up ("search those five degrees for hits we overlooked") is itself the numerology temptation v354 warns about: a rich object like E_8 returns a numerical match for almost anything. So the scan uses a STRICT discriminator – a candidate counts only if it is a FORCED identity (a theorem about E_8), never a coincidence. [E] FORCED, SET-LEVEL (the genuinely overlooked structure, collective not per-number): $\sum \deg = 128 = \dim S^+$, the spinor half of $248 = 120 + 128$ (theorem: $\sum \deg = \#\text{pos roots}+\text{rank} = 120 + 8$) – the invariant budget of E_8 equals the fermion content the carrier reads; $\sum \exp = 120 = \#\text{pos}$ roots; $\prod \deg = |W(E_8)| = 696{,}729{,}600$; exponents = the $\varphi(30){=}8$ totatives of 30 (v223), so the unmapped degrees' exponents are part of the forced set; $\max \deg = 30 = h(E_8)$. [C] SUGGESTIVE, not a theorem: $12{=}h(E_6)$, $18{=}h(E_7)$, $30{=}h(E_8)$ are all degrees of E_8 (the $2T/2O/2I$ McKay ladder, v219), but "deg(E_8) contains its subalgebras' Coxeter numbers" is not a general theorem – flagged, not claimed. [O] DECLINED (the numerology temptation): the per-degree atom matches $14{=}\dim G_2$, $20{=}\det L$, $24{=}|W(A_3)|$, $12{=}|R(A_3)|$, $18{=}p_4(a)$ (v66 already flagged $12, 18$ as fittable) each admit ≥ 2 readings, and the lone extra prime in $|W(E_8)| = 2^{14}3^5 5^2\, 7$ "=" the scalaron exponent is one of many readings of 7 – all declined. [E] RECONCILES v66 (the 8 degrees coincide with the load-bearing integers) and v354 (only $2, 8, 30$ are primary readouts): the search finds NO new forced physical hit, and the disciplined decline IS the anti-numerology result. (This also sharpens v354: the five are "no PRIMARY readout / hull overhead", not literally "unmapped" – they coincide with structural atoms, but unforced.) Cited in `tfpt_5_redteam` (with a full audited degree table); mirrored in Wolfram (301/301).

2026-06-22 (the REVERSE numerology audit, and what the golden ratio means: of E_8's 8 Casimir invariants only 3 feed a primary readout and 5 are hull overhead; the golden ratio is UNMAPPED structure, so it cannot be numerology: `v354`)

- **`v354` (`E8.REVERSE.AUDIT.01`) — the reverse complement of v305.** TFPT's forward discipline (v305) is one-directional: every physical READOUT must map onto an E_8 structure (anti-fitting). The sharp adversarial question runs the OTHER way — how much E_8 structure maps onto NOTHING, and is the recurring golden ratio numerology? [E] THE REVERSE AUDIT: of E_8's eight Casimir invariant degrees $\{2, 8, 12, 14, 18, 20, 24, 30\}$ (exponents +1), exactly THREE feed a readout — 2 (metric), 8 (rank $\to c_3{=}1/(8\pi)$), 30 (Coxeter $\to g_{\text{car}}{=}5$ and the order-30 cycle) — and FIVE carry no primary readout $\{12, 14, 18, 20, 24\}$ (they coincide with structural atoms per v66, but unforced – reconciled in v355); so 3/8 are primary readouts, 5/8 hull overhead. [E] ECONOMY, NOT PROMISCUITY: a small fixed set (rank 8, $h{=}30$, $\det K{=}1$, $D_5{\oplus}A_3$, the 16 half-spinor) generates the ENTIRE readout set, and the higher Casimirs are never reached into to fit — a numerological use would be PROMISCUOUS (mine the rich structure), TFPT is ECONOMICAL (few invariants, many readouts) and leaves most of E_8 untouched, the anti-numerology signature being precisely the large unused remainder. [E] THE GOLDEN RATIO IS UNMAPPED ⇒ NOT NUMEROLOGY: $\varphi = 2\cos(\pi/5)$ appears only in internal structure (the affine-E_8 attractor eigenvalue v312, the icosian lattice v348) and in NO physical readout; numerology means a number FITTED to an observation, φ is fitted to nothing — it is the algebraic signature of the 5-fold ($h{=}30{=}2\cdot3\cdot5$), the carrier announcing it is icosahedral. [O] the 5/8 unmapped is the HULL OVERHEAD (E_8 is the minimal even-unimodular hull, not the world group); where a future higher-invariant readout could live, or genuine over-capacity — stated plainly, not hidden. Cited in `tfpt_5_redteam`;

mirrored in Wolfram (`300/300`).

2026-06-22 (the bird's-eye rethink: TFPT is a unique self-consistent LOOP with ZERO free dials, not a linear axiom system; π is a fixed geometric constant, not a dial; the one residual is the continuum closure: `v353`)

- `v353` (`TFPT.SELFLOOP.01`) — **refining v352.** Stepping all the way back: TFPT is not a LINEAR theory (axioms → theorems) but a CLOSED SELF-CONSISTENT LOOP whose outputs fix its own inputs. [E] it is a loop: g_{car}=5= max prime of the output $h(E_8)$=30, and the 8 in c_3= rank E_8 – the outputs fix the inputs (v6), with a unique fixed point (v56); "which is the axiom?" is the wrong question for a loop. [E] ZERO free adjustable dimensionless dials: every load-bearing number is forced/over-determined. [E] π is a FIXED geometric constant (the boundary 2-sphere, $\oint K$=4π), not a dial. [C] this refines v352's "framework + π": π is not a free input and the framework is a SELECTION PRINCIPLE (the unique self-consistent loop), not a free meta-axiom – the honest characterisation is "a unique self-consistent loop with zero free dials". [O] the one residual is the CONTINUUM closure of the loop (the chiral-edge existence, v336/v351); structural-realism (loop complete) vs constructivism (continuum open) is a philosophical, not numerical, fork – no free numbers either way. A capstone synthesis.

2026-06-22 (two consequences: the c=8 which-net ambiguity resolved by the order-4 clock, and BOTH axioms shown bootstrap-forced so the only irreducibles are the framework + π: `v351`, `v352`)

- `v351` (`SEAM.EQUIV.CONTINUUM.02`) — **the which-net ambiguity falls.** The long-standing "c=8 ambiguity" (E_8 det 1 vs $SO(16)$=D_8 det 4, both c=8, flagged open in v277/v344) is RESOLVED, non-circularly, by the seam's ORDER-4 μ_4 clock: the index-4 simple-current extension of $D_5 \oplus A_3$ is E_8 (v154), the index-2 would be $SO(16)$, so the order-4 clock (the four Gauss–Bonnet marks, v216) selects E_8 (full μ_4 condensation 16→4→1) not $SO(16)$ (partial $\mathbb{Z}_2$ 16→4). The bootstrap agrees ($h(E_8)$=30 max prime 5=g_{car} vs $h(D_8)$=14 max prime 7). [E] so det K=1 is a consequence, not an open input. [O] the residual of `SEAM.EQUIV.01` is then PURELY the chiral-edge / scaling-limit existence (v336), not which-net.
- `v352` (`TFPT.IRREDUCIBLE.01`) — **the inputs are not axioms.** The deepest answer to "the inputs are back-determined": BOTH P1 and P2 are bootstrap-forced fixed points. [E] g_{car}=5 is forced three ways (v6); the 8 in c_3 four ways (rank E_8 = $h(D_5)$ = $\phi(30)$ = det R, v54); the E_8 hull is forced (unique even unimodular rank-8, v83/v154). [E] so the ONLY irreducibles are the FRAMEWORK (a discrete reflection-positive boundary with a finite carrier) and the transcendental π. [O] the framework is the META-AXIOM — not math-reducible, the foundational stance (like "spacetime is a manifold"), minimal, tested by its consequences not proved. TFPT's true input is ONE framework + π. Wolfram mirror +2 (297 → 299).

2026-06-22 (correction: the inputs are bootstrap-forced fixed points, not free axioms – g_{car}=5 IS the icosahedral 5 of $h(E_8)$=30, and the golden ratio is emergent from the μ_4-glue, not external: `v350`)

- `v350` (`SEAM.EQUIV.BOOTSTRAP.01`) — **correcting v349.** An honest correction prompted by the right objection that the inputs are not axioms but are back-determined by the theory. [E] the inputs ARE bootstrap-forced (v6): g_{car}=5 is over-determined three ways – rank-fill (g_{car}+N_{fam}=8), Coxeter-match (g_{car}= max prime of $h(E_8)$=30), Pascal (2^4=$C(5,0)$+$C(5,1)$+$C(5,2)$). [E] so g_{car}=5 IS the icosahedral 5-fold (the 5 in $h(E_8)$=30=2·3·5), not the D_5 rank; the atoms (2, 3, 5) ARE the primes of the icosahedral Coxeter number 30. [E] the golden ratio is EMERGENT from the glue: D_5 (h=8)/A_3 (h=4) are non-golden alone, but $D_5 \oplus A_3 \to E_8$ (μ_4 index-4, v154) gives $h(E_8)$=30 (golden) – the seam's own μ_4 produces the 5-fold, so v349 checked the un-glued pieces, the wrong object, and its pentagon-vs-count dichotomy was false. [O] the residual relocates: the golden is bootstrap-forced, the only open thing is the physical continuum realisation (v336), NOT the golden ratio. Honest caveat: self-consistency within the framework, not from-nothing. Wolfram mirror +1 (296 → 297).

2026-06-22 (the honest test – does the raw seam carry φ? – answer NO: the bare carrier gives $\sqrt{2}$, golden is the icosahedral input; the keystone reduces to "is $g_{\rm car}$=5 a pentagon or a count?": v349)

- **v349 (SEAM.EQUIV.GOLDEN.01).** The honest, computed test of the v348 question, and a clarifying NEGATIVE: the raw seam does *not* carry the golden ratio. [E] the golden ratio $2\cos(\pi/5)$ is the signature of a 5-fold rotation ($5 \mid h$); the carrier D_5 has h=8 (eigenvalue $\sqrt{2}$) and A_3 has h=4 – no 5-fold; the dynamical data (cusp $\{0, 1/3, 2/3\}$, recovery $(2/3)^6$, seed φ_0) are rational/transcendental. [E] $5 \mid h$ holds only for A_4=$SU(5)$ (h=5), H_3 (h=10), E_8 (h=30) – the output/icosahedral side, so φ is the icosahedral INPUT. [E] φ alone would not suffice ($2I \neq C_5 \times \mu_4$, $20 \neq 120$). [O] the resolution: the keystone reduces to one question – is $g_{\rm car}$=5 a PENTAGON (a 5-fold rotation ⇒ golden ⇒ icosahedral $2I = E_8$) or just a COUNT (the rank of $D_5 \Rightarrow \sqrt{2}$)? The "absurdly simple solution" is real but is a *reading* of axiom P2 (the golden ratio is the pentagon content of $g_{\rm car}$=5, not derivable from $g_{\rm car}$=5-as-a-count); there is no hidden φ. L2 is foundational: P2-as-pentagon closes it (mode C), else a rigidity theorem must produce the 5-fold (mode B); not closed. Wolfram mirror +1 (295 → 296). A NEGATIVE/clarification; prevents a false closure.

2026-06-22 (Route B carried out: rigidity + the icosian-ring identity reduce the whole keystone to ONE question – does the raw seam carry the golden ratio? – the simplest form of L2: v348)

- **v348 (SEAM.EQUIV.RIGID.01).** Executes the rigidity route (mode B of v347) and lands on a strikingly simple statement. [E] rigidity closes the crystallographic part: the order-4 clock has the unique normal form $z \mapsto iz$ (Nielsen/Kerckhoff, v180), Troyanov gives the unique flat pillowcase metric (v284). [E] the downstream is a LATTICE identity, not just a graph: by Conway–Sloane the ring of icosians (120 unit icosians = $2I$, golden-ratio quaternion norm) IS the rank-8 E_8 lattice. [E] the single bridge is the golden ratio: the crystallographic restriction allows only orders $\{1, 2, 3, 4, 6\}$ (the 5-fold absent), so $\varphi = 2\cos(\pi/5)$ is exactly what extends the crystallographic μ_4 to the non-crystallographic icosahedral $2I$ (the quasicrystal). [O] so the entire keystone reduces to ONE question: *does the raw seam carry the golden ratio?* – if yes, μ_4+$\varphi \Rightarrow 2I \Rightarrow E_8$ and rigidity fixes the rest. [O] honest subtlety: TFPT's φ is currently E_8-side (v312/v313); the raw-seam data so far are not manifestly golden, so showing the RAW seam carries φ non-circularly is the residual (= L2, sharpest form). Wolfram mirror +1 (294 → 295). An analysis/reduction; closes nothing.

2026-06-22 (can the one open arrow be *fully* solved? the closure-mode classification: the flat→spherical base locus, four modes, no free theorem – only rigidity or axiom P_3: v347)

- **v347 (SEAM.EQUIV.CLOSURE.01).** A direct, honest answer to "is there any other way to fully solve the one open arrow L2?". [E] the precise locus: the seam gives the *flat* pillowcase base $S^2(2, 2, 2, 2)$ ($\chi_{\rm orb} = 2 - 4(1 - \frac{1}{2}) = 0$, v216) while the $2I/E_8$ structure is the Seifert base over the *spherical* $S^2(2, 3, 5)$ ($\chi_{\rm orb} = 2 - (\frac{1}{2} + \frac{2}{3} + \frac{4}{5}) = \frac{1}{30}$), so L2 bridges two geometric types – why it needs the carrier (2, 3, 5) input. [E] as a "physics-realises-geometry" statement L2 has exactly four closure modes: A construct (the analytic wall), B rigidity (a unique-realisation theorem – the only theorem-route w/o new input, pursued v284/v300/v331), C axiom P_3 (then TFPT is complete relative to 3 axioms), D empirical. [E] the "no new state" discipline (v246) blocks deriving L2 from extra physics, so no route makes it a free theorem. [E] verdict: full closure is either a rigidity theorem (B) or axiom status (C), validated either way by (D); only B closes it as a theorem. [O] NOT closed – classifies the modes, locates the difficulty. Wolfram mirror +1 (293 → 294). An analysis/roadmap; closes nothing.

2026-06-22 (the geometric bridge made explicit: the keystone is ONE arrow – raw-seam ⇒ $\det K=1$ is a six-link chain, five theorems + one open orbifold realisation, the group forced by the $(2,3,5)$ atoms: v346)

- **v346 (SEAM.EQUIV.GEOM.01).** The honest capstone of the `SEAM.EQUIV.01` investigation: it lays out the full path raw-seam ⇒ $\det K=1$ as six links and shows exactly one is open. [E]/[C] L1 raw seam → μ_4 + four marks (Gauss–Bonnet, v216); [O] L2 μ_4 + the $(2,3,5)$ atoms → the $2I$ orbifold $\mathbb{C}^2/\Gamma$ (THE ONE OPEN ARROW = the geometric content of `SEAM.EQUIV.01`); [E] L3 $2I$ → affine E_8 McKay (v219); [E] L4 $2I$ → E_8 du Val (v232); [E] L5 $\mathbb{C}^2/2I$ → Poincaré link H_1=0 (v345); [E] L6 H_1=0 → $\det K$=1. [E] five of six links are theorems; [E] the group is FORCED by the seam's own atoms (the exceptional $SU(2)$ axis signatures $(2,3,3)/(2,3,4)/(2,3,5) \to 2T/2O/2I$; $(2,3,5)$ unique to $2I$, and $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}})=(2,3,5)$ ARE those axes); [E] everything downstream of L2 is then forced. [O] the irreducible open content is L2 alone ("the raw seam normal bundle is an ADE orbifold point $\mathbb{C}^2/\Gamma$") = `SEAM.EQUIV.01`; not closed, but the bridge is now demonstrably ONE arrow with the group pinned and all downstream proven.

2026-06-22 (executing route R3: the $(2,3,5)$-rewrite attractor link computed to be the Poincaré homology sphere ($\det K=1$), the combinatorial core complete, only the geometric bridge open: v345)

- **v345 (SEAM.DETK.02).** Carries out the combinatorial core of the hypergraph-homotopy route (R3, the most promising fresh angle from v344) as far as it computably goes. By plumbing topology (Brieskorn/Hirzebruch) the link of a tree plumbing with the ADE Cartan intersection form has H_1 = coker(Cartan). [E] the Smith normal form of the E_8 Cartan matrix is the identity, so coker(E_8) = 0 – the E_8-graph plumbing boundary is an INTEGRAL HOMOLOGY SPHERE (the genus-1 meaning of $\det E_8=1$); among ADE only E_8 does this ($A_n \to \mathbb{Z}/(n+1)$, $D_n \to \mathbb{Z}/4$, $E_6 \to \mathbb{Z}/3$, $E_7 \to \mathbb{Z}/2$). [E] $\pi_1 = 2I = SL(2,5)$ is PERFECT (commutator subgroup = the whole order-120 group, computed directly), so $H_1 = 0$ and the link is THE Poincaré sphere, $|2I| = 120 = |\mu_4|\, h(E_8)$. So R3's combinatorial half is [E]-complete: the E_8 attractor link is forced to be the Poincaré sphere ($\det K$=1), uniquely among ADE. [O] what stays open is only the geometric realisation bridge (the raw rewrite seam attractor *is* this E_8 du Val plumbing) = the `SEAM.EQUIV.01` content; not closed, but the keystone is now reduced to that single geometric identification. Wolfram mirror +1 (292 → 293).

2026-06-22 (the one keystone bit $\det K=1$ mapped: six equivalent faces, the genus-1 obstruction, three access routes, the hypergraph-homotopy route the most promising fresh angle: v344)

- **v344 (SEAM.DETK.01).** A bird's-eye, full-spectrum map of `SEAM.EQUIV.01`'s only open residual, the holomorphy bit $\det K$=1. [E] it is ONE statement with six equivalent faces (holomorphy / homology-sphere link H_1=0 / one 1-dim irrep / det Cartan = 1 / full μ_4 condensation / τ=i CM), unified by $|\det \text{Cartan}| = |H_1(\text{link})| = \#$(1-dim irreps) across ADE ($A_n \to n+1$, $D_n \to 4$, $E_6 \to 3$, $E_7 \to 2$, $E_8 \to 1$) – only E_8 gives 1. [E] the GENUS-1 OBSTRUCTION explains why it is hard: every easy argument (plane vacuum, gap, free-fermion) is genus-0, necessary but not sufficient (v87); $\det K$ is the torus degeneracy. [E] THREE genus-1 access routes: R1 continuum-modular (v336), R2 anyon-condensation (v281), R3 hypergraph-homotopy. [C] the HYPERGRAPH route is the most promising fresh angle: the $(2, 3, 5)$ rewrite attractor is the affine-E_8 graph (v312) and the E_8 du Val link is the Poincaré sphere (v232), so det K=1 becomes a finite combinatorial statement (no continuum limit). [O] none closes it; the gap is the combinatorial-to-geometric bridge = the `SEAM.EQUIV.01` content. A synthesis/roadmap; closes nothing.

2026-06-22 (the four black-hole-birth completion routes investigated: none closes all or is 100% provable alone; all four converge on `SEAM.EQUIV.01` + the decoupled QG contour: `v343`)

- **`v343` (`FOUR.ROUTES.01`).** A direct, honest answer to "which black-hole-birth route closes ALL problems and is 100% provable": *none individually* — but the four converge on the same two residuals as v335. [E] **(A)** the finite causal diamond reframes `QG.AMB` to a finite thermal trace (de Sitter $S_{\mathrm{dS}} = 32\pi^4 e^{2\alpha^{-1}}$ finite, $\chi = 729/665$) but inherits the GHP contour (v334, [O]); does not touch the seam. [C] **(B)** Carlip–Cardy gives a second route to existence+c (c=8 consistent, but the Carlip ambiguity and no holomorphy det K=1) — inherits `SEAM.EQUIV.01`. [E] **(C)** the modular/thermal flow (v239) is well-defined but its geometric content is `QGEO.SYM.01`, a corollary of `SEAM.EQUIV.01` (v335) — a reduction. [E]/[O] **(D)** the unique attractor (v56) gives parameter-freeness but the cyclic dynamics is [C] interpretation. [E] **Verdict:** independent closures = 0; irreducible residuals after all four = 2 (unchanged from v335); the horizon premise makes both more natural but eliminates neither — focus = `SEAM.EQUIV.01`. An analysis module; closes nothing, fabricates nothing.

2026-06-22 (the heat-kernel origin of the α terms: the Calderón quadratic is the Gilkey gauge-curvature coefficient; the cubic Maxwell moment stays the `EM.WARD.01` residual: `v342`)

- **`v342` (`EM.WARD.02`) — the heat-kernel sharpening of `EM.WARD.01`.** Derives the ORIGIN and STRUCTURE of the $F_{U(1)}$ determinant-line terms from textbook Seeley–DeWitt / Gilkey coefficients, advancing the EM-Ward origin from [C] (conjectured) to [E] for the term structure; no new gate. [E] $c_3 = 1/(8\pi)$ *is* the boundary Gauss–Bonnet coefficient ($\oint_{S^2} K = 2\pi\chi = 4\pi$, χ=2, v216). [E] the α^2 (Calderón) term *is* the Gilkey gauge-curvature coefficient ($30\,\Omega^2/360 = \frac{1}{12}\Omega^2$, Ω=$i\alpha F \Rightarrow -(\alpha^2/12)F^2$), so it is forced, not an ansatz. [E] the c_3-orders $\{0, 3, 6\}$ are an arithmetic ladder and $2c_3^3 = 1/(256\pi^3)$ carries π^3 (three boundary insertions). [C] the exact coefficient needs the seam F-normalisation. [O] the cubic α^3 Maxwell moment (a topological/Chern level) + the full "this is the seam $U(1)$ ζ-determinant" identification stay the `EM.WARD.01` residual, tied to `SEAM.EQUIV.01`. α^{-1} stays [E] (v3). Wolfram mirror +1 (290 → 291).

2026-06-22 (α as the stationarity of a $U(1)$ determinant line: every ingredient an index / heat-kernel coefficient / discriminant form, the open EM-Ward obligation named: `v341`)

- **`v341` (`ALPHA.QUILLEN.01`).** Answers the external review's sharpest valid point — "derive the *form* of α, not more decimals". The EM closure $F_{U(1)} = \alpha^3 - 2c_3^3\alpha^2 - \frac{4}{5}c_3^6 M \log(1/\varphi_{\mathrm{seam}}) = 0$ is rewritten as a *stationarity* of the boundary $U(1)$ determinant line: $[\alpha^3 - 2c_3^3\alpha^2]_{\delta \log \det \Delta_{U(1)}} = [8b_1 c_3^6]_{\text{counterterm}} \cdot [\log(1/\varphi_{\mathrm{seam}})]_{\text{seam response}}$. [E] every ingredient is a NAMED object, not a knob: M=41 is a $U(1)$ index ($\frac{4}{5}M = 8b_1$, b_1=41/10 the SM $U(1)_Y$ beta, EM Ward v48); c_3=$1/(8\pi)$ the Gauss–Bonnet boundary coefficient (8=rank E_8); $-5/4 = -q(D_5)$ a discriminant form and φ_{base}=$1/(6\pi)$=$\frac{4}{3}c_3$ carries the other, $c_3/\varphi_{\mathrm{base}}$=3/4=$q(A_3)$; δ_{top}=$\Omega_{\mathrm{adm}} c_3^4$. The Quillen split is verified at the α^{-1}=137.0359992 root. **No new gate:** α^{-1} stays closed [E]; v341 adds no open item — it *sharpens* the pre-existing [O]/physical-origin obligation `EM.WARD.01` ("why *this* function", already conjectural). [O] the one still-open part *is* that residual: the from-first-principles AQFT proof that $F_{U(1)}$ is the EXACT Quillen functional (the variational principle derived, not assembled), unchanged in scope. Wolfram mirror +1 ($289 \to 290$).

2026-06-22 (A_s reconciled with the archived derivation: predicted & dimensionless (so it does not define a mass), the tension is the reheating channel: `v340`)

- **`v340 (AS.RECONCILE.01)`.** Answers "why don't we have A_s / the old versions derived it" and "can a ratio define mass itself". **[E]** A_s *is* predicted and *dimensionless*: $A_s = N_\star^2(M_{\rm scal}/\bar{M}_{\rm Pl})^2/(24\pi^2) = N_\star^2 c_3^7/(24\pi^2)$ with $M_{\rm scal}/\bar{M}_{\rm Pl} = c_3^{7/2} = 1.26\times10^{-5}$ a pure ratio, so A_s pins the e-fold count $N_\star$, NOT $\bar{M}_{\rm Pl}$ — it cannot define an absolute mass (No-Unit, v153, stands). [C] the -11.4σ at the slow Higgs channel ($N_\star$=51.4) is the slow-vs-fast reheating dichotomy; the A_s-preferred $N_\star$=56.2 $\sim$ the instantaneous bound. [C] the archived TFPT v4.5/v4.6 reported a clean $A_s = 2.124\times10^{-9}$ using the algebraic $N_\star = 3/\varphi_0^{\rm ret} - 1 = 55.42$; at that $N_\star$ the current normalisation gives 2.047×10^{-9} (-1.9σ) — the old clean A_s is near-instantaneous reheating, not the slow channel. **[E]** nothing lost: the current theory replaced the posited algebraic $N_\star$ with one derived from reheating physics and exposed honestly that A_s requires fast (pre)heating. Record band $N_\star \in [50, 60]$ (v84) untouched.

2026-06-22 (the first-principles boundary, mapped: can $M_{\rm Planck}$ be computed? can $F_{\rm transfer}$ be first-principles? both NO, with the exact reason: `v339`)

- **`v339 (FIRSTPRINCIPLES.BOUNDARY.01)`.** An honest boundary audit answering two head-on questions and fabricating nothing. **[E]** *The Planck mass cannot be computed* — $v_{\rm geo}/M_{\rm Planck}$ is forbidden by theorem (No-Unit, v153): a mass is mass-dimension 1 while every TFPT datum is dimension 0, so no dimensionless map outputs an absolute scale. It is not a gap but over-determined (gravity = dark energy to 0.11%, v274); every dimensionless *ratio* is derived, only the one unit is not. **[E]** $F_{\rm transfer}$ *cannot be made first-principles* either: for m_p/m_e the structure is in place (v262, carrier b_3=−7 + closed electron) but two inputs stay external. **[X]** $\alpha_s(M_Z)$ is external *because of a tension* — the SM (= TFPT content, no new states) does not unify (1-loop spread $\sim$8.8 at 2×10^{16} GeV, v246). The lattice $C_p\sim2.83$ is parameter-free but lattice-only (v35); η_B/axion ride on cosmological initial conditions; Koide Q=2/3 is structural **[E]**. **[E]** The residuals classify into four honest kinds: forbidden-by-theorem, external-and-tensioned, external-but-parameter-free, cosmological. A classification, not a solution.

2026-06-22 (next-steps round: the one open lemma literature-anchored, the Decoupling Theorem, the holomorphy discriminator hardened in Lean, and the θ_{13} budget: `v336`, `v337`, `v338`, `FORM.CARTAN.DET.01`)

- **`v336 (SEAM.EQUIV.CONTINUUM.01)` — the one open lemma, literature-anchored.** SEAM.EQUIV.01's only open input (the continuum OS reconstruction of the gapped quasi-free collar) is split into a CITABLE part and a narrow open part. [C] continuum leg: Morinelli–Morsella–Stottmeister–Tanimoto (*CFT from Lattice Fermions*, doi:10.1007/s00220-022-04521-8; CMP 387 (2021) 299; PRL 127 (2021) 230601) give, by operator-algebraic (wavelet) renormalization, the chiral-CFT scaling limit of free lattice fermions (Koo–Saleur → Virasoro, correct c); WZW range rank $\le c \le D$, and $(E_8)_1$ (c=8, rank 8, D=16 Majoranas, $SO(16)_1 \to (E_8)_1$) is inside it. [C] OS leg: Adamo–Moriwaki–Tanimoto (arXiv:2407.18222, 2024) prove the conformal OS axioms for unitary lattice VOAs. **[E]** target pinned by c=$g_{\rm car}$+$N_{\rm fam}$=8, $\det \mathrm{Cartan}(E_8)$=1. **[O]** narrowed residual = "the raw collar's massless scaling limit is the holomorphic $(E_8)_1$ net" ($\det K$=1), not a from-scratch construction.
- **`v337 (DECOUPLING.THEOREM.01)` — the Decoupling Theorem.** **[E]** every dimensionless readout factors through the gapped admissible transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$; finite

susceptibility $\chi = 1/(1-(2/3)^6) = 729/665$ gap-suppresses the ambient contribution (margin $6\ln\frac{3}{2} - 31/(4\pi^2) \approx 1.648 > 0$, with $2\dim(E_8)c_3^2 = 31/(4\pi^2)$ exactly), so no readout depends on the ambient measure ⇒ `QG.AMB.01` decoupled; negative control (gap closed ⇒ χ diverges) makes it non-vacuous. This is the theorem that makes the `QG.AMB.01` reclassification (v335) honest; consolidates v76/v311/v330/v332.

- **v338 (THETA13.BUDGET.01) — the θ_{13} budget, fenced.** The $+2\sigma$ pressure point (v328) quantified against *both* NuFIT 6.0 NO variants: $+2.0\sigma$ vs the no-SK central (0.02195) but $+1.7\sigma$ vs the with-SK best fit (0.02215 ± 0.00057), a $\sim 0.3\sigma$ systematic spread; **[E]** 0.023108 lies inside the with-SK 3σ band (upper 0.02388), an upper-edge pressure point not a falsification; **[X]** JUNO's sub-1% precision turns it into a $\sim 6\sigma$ kill-or-confirm.
- **Lean (FORM.CARTAN.DET.01).** `CartanDeterminants.lean` extended with the explicit $D_8 = SO(16)$ Cartan matrix and a fully-proven (kernel-only) holomorphy discriminator: D_8 is even and symmetric just like E_8 (same c=8) but $4 = \det \mathrm{Cartan}(D_8)$ is genuinely not a unit while $\det \mathrm{Cartan}(E_8)$ is — so unimodularity (one anyon) is the load-bearing selector. `lake build` green.

2026-06-22 (keystone unification + status reframe: ONE open theorem, ONE decoupled foreign problem: v335, FORM.QGEO.BW.01 qgeoSymIsCorollary)

- **v335 (SEAM.EQUIV.UNIFY.01) — the keystone unification.** A bird's-eye consolidation of the v308/v323/v325/v329/v333 arc that collapses the "two open structural items" narrative. **[E] QGEO.SYM.01 is a COROLLARY of SEAM.EQUIV.01** (no extra premise): a chiral conformal net has, by axiom, a Möbius-covariant vacuum (the unique invariant positive-energy vector); rotations $U(1)$ are the compact subgroup of the Möbius group, so the vacuum is rotation-invariant, and the μ_4 clock = rotation by $\pi/2$ ($(e^{i\pi/2})^4$=1) fixes it ⇒ $\omega \circ \rho = \omega$. So `SEAM.EQUIV.01` ⇒ `QGEO.SYM.01` with NO extra premise (the v323 "rotation-invariant vacuum" is a net axiom) — the two bedrock items collapse to ONE. **[E] the one keystone theorem**: `SEAM.EQUIV.01` = a construction theorem (c=g_{car}+N_{fam}=8, $\det \mathrm{Cartan}(E_8)$=1, gap $6\ln\frac{3}{2} > 0$ derived) with EXACTLY one open continuum lemma (the rigorous continuum OS reconstruction + invertibility of the raw collar). **[E] QG.AMB.01 decoupled, not a gate**: margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76/v311/v330/v332) ⇒ no TFPT prediction depends on the ambient measure; it is the *general* Euclidean-QG conformal-factor problem (Gibbons–Hawking–Perry 1978), inherited not TFPT-specific — reclassified from "TFPT structural frontier" to "decoupled general QG problem". **[E] collapsed status**: open structural items $2 \to 1$; live residual = $v_{\mathrm{geo}} + G_{\mathrm{net}}$(=`SEAM.EQUIV.01`) + F_{transfer}. **[O]** the one residual = `SEAM.EQUIV.01`'s continuum OS reconstruction.
- **Lean (FORM.QGEO.BW.01, qgeoSymIsCorollary).** `BWKeystone.lean` extended with the cited conformal-net vacuum axiom `conformal_net_vacuum_rotation_invariant`; with it, `StateInvariance` (=`QGEO.SYM.01`) follows from `SeamIsE8` alone, so `QGEO.SYM.01` is a COROLLARY of `SEAM.EQUIV.01` with no independent open premise (`#print axioms` confirms only the chain + cited BW/net axioms). `lake build` green (3362 jobs).
- **Status deep-sync.** The ledger (`QG.AMB.01` reclassified to "decoupled general QG; non-TFPT"; `QGEO.SYM.01` marked a corollary; new `SEAM.EQUIV.UNIFY.01` row), the canonical status card (`tfpt_status.tex`), `introduction` (status core + gate table), `origin_theory`, `tfpt_4_frontier`, `tfpt_5_redteam`, `tfpt_research_contracts`, `README`, `next.txt` and the website (`OpenGates`, FAQ, `papers.ts`) all moved to "one solvable theorem (`SEAM.EQUIV.01`) + one decoupled foreign problem (`QG.AMB.01`)". No falsifiable prediction changed; this is a status-framing sharpening backed by `v335` + the Lean corollary.

2026-06-22 (closing round: the CM-selection mechanism (the last keystone residual), the conformal-mode resolution, and the Conway–Sloane even-unimodular side in Lean: v333, v334, FORM.CARTAN.DET.01)

- **`v333_cm_selection.py` (`QGEO.CMSELECT.01`, [E]/[O]).** The CM-selection mechanism for the last keystone residual: $\tau=i$ (the square) is the *unique* order-4 modulus — $S=\left(\begin{smallmatrix}0 & -1\\ 1 & 0\end{smallmatrix}\right)$ has $S^4=I$ and fixes i, and an order-4 deck selects $j=1728$ (vs the order-3 hexagon $j=0$) — AND the square is the Fekete attractor of symmetric mark-equilibration (four points minimizing the log-energy converge to equal $\pi/2$ spacing). So given the μ_4 deck, $\tau=i$ is forced; the residual sharpens from "why the square" to "the deck acts geometrically" (= `QGEO.SYM.01`). NOT a closure.

- **`v334_conformal_resolution.py` (`QGAMB.CONFORMAL.01`, [E]/[C]/[O]).** The conformal-mode resolution for the QG metric sector: the Gibbons–Hawking–Perry contour rotation $\phi \to i\phi$ flips the wrong-sign kinetic term, making the divergent conformal Gaussian $\int e^{+|c|\phi^2}$ into the convergent $\int e^{-|c|\phi^2}=\sqrt{\pi/|c|}$; the entire IDG form factor e^{p^2/M^2} dresses the propagator pole-free (the polynomial $R+R^2$ ghost as control). Together they give a *conditional* mode-by-mode construction of the metric-sector measure; the residual is only the contour's nonperturbative validity (gap-decoupled, blocks no test). NOT a closure.

- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The Conway–Sloane lattice side: the E_8 Cartan matrix is proved to be an EVEN (diagonal = 2), symmetric, unimodular ($|\det|=1$) integer Gram matrix — the defining data of an even unimodular rank-8 lattice — all by kernel `decide` / the integer-inverse argument. The remaining content (Witt's uniqueness: E_8 is the only even unimodular lattice in dim < 16) is the cited classification residual.

2026-06-22 (endgame round: the necessity of H reduced to the order-4 CM selection, the QG metric sector characterized as the conformal-mode problem, $|\det E_8|=1$ and the μ_4 commutation grounded in Lean: v331, v332, FORM.MU4COMM.01)

- **`v331_necessity_of_H.py` (`QGEO.NECESS.01`, [E]/[O]).** The necessity companion to `v329`: a concrete Steklov (DtN) computation shows the seam $\Lambda=|k|+M_f$ commutes with the μ_4 clock *iff* the four marks sit at the square ($\tau=i$) — moving a mark off the square breaks $\mathbb{Z}_4$ and $[\rho,\Lambda]\neq 0$. The square has cross-ratio 2 ($j=1728$), the unique 4-config with an order-4 automorphism (`v214`/`v267`), so $H \Leftrightarrow$ square $\Leftrightarrow \mu_4$ and "necessity of H" sharpens to: the raw dynamics places the four Gauss–Bonnet marks (`v216`) at the order-4 CM point. Residual = that dynamical selection.

- **`v332_qgamb_metric_sector.py` (`QGAMB.METRIC.01`, [E]/[C]/[O]).** The open metric sector of `QG.AMB.01` is characterized precisely as the Gibbons–Hawking–Perry conformal-factor problem: the conformal mode has a negative kinetic coefficient $-(D-1)(D-2)/4$ ($=-\frac{3}{2}$ at $D=4$), so the bare Euclidean measure diverges over it. The obstruction is in the conformal/gauge direction, gap-insulated from the admissible sector (`v330`/`v76`); the only route is the conditional IDG taming (`v304`), with the polynomial $R+R^2$ ghost as the control. Open, but pinned to one named obstruction.

- **`CartanDeterminants.lean` extended (`FORM.CARTAN.DET.01`, [E]).** The rank-8 holomorphy discriminator $|\det \mathrm{Cartan}(E_8)|=1$ (E_8 has one anyon, vs D_8's four) is now proved principally and kernel-only: `cartanE8 * cartanE8inv = 1` by `decide` on the 8×8 PRODUCT gives `IsUnit (det E8)` via $\det E_8 \cdot \det E_8^{-1}=\det 1=1$ — no 8! determinant, no `native_decide` (Mathlib itself leaves `E₈_det = 1` a `proof_wanted`).

- **`Mu4Commutation.lean` (`FORM.MU4COMM.01`, Lean 4, [E]).** The v201/v323 mark-locality criterion as exact linear algebra over the Gaussian integers $\mathbb{Z}[i]$: $i^4=1$, $i^2=-1$, and (on concrete 6×6 matrices) an offset-4 coupling commutes with the μ_4 clock while an offset-2 coupling does not — all by kernel `decide`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (frontier round: the OS step discharged at the many-body level, the ambient measure built on the gap-decoupled sector, and the carrier Cartan determinants grounded in Lean: `v329`, `v330`, `FORM.CARTAN.DET.01`)

- **`v329_os_gap_reduction.py` (`QGEO.OSGAP.01`, [E]/[O]).** The one analytic input the v308 keystone chain still cited — "the OS-reconstructed bulk gap = the transfer gap" — is discharged at the many-body level: the OS/Euclidean Hamiltonian is the second quantization $d\Gamma(-\log T)$, whose spectrum is the set of sums of single-mode energies, so the bulk gap = the smallest positive single-mode energy $= 6\ln\frac{3}{2}$, *independent of system size.* Together with $H \Rightarrow \omega \circ \rho{=}\omega$ and $H \Rightarrow$ SRE $\Rightarrow (E_8)_1$, this exhibits the rotation-invariant flat-pillowcase premise H as the *single sufficient premise* of both bedrock IDs; residual = the necessity of H on the raw collar.

- **`v330_qgamb_admissible_measure.py` (`QGAMB.MEASURE.01`, [E]/[O]).** The constructive step `v275`'s roadmap calls for: on the gap-decoupled *admissible* sector the ambient measure is built as a bona fide OS object — reflection-positive (symmetric PSD transfer), exponentially clustering ($C(n)/C(n{-}1) \to (2/3)^6$), with finite susceptibility $\chi = 729/665$ so $\mathrm{Var}(S_\Lambda)/\Lambda \to \chi$ converges $\Rightarrow$ uniform tightness (Prokhorov) $\Rightarrow$ the projective limit exists; OS reconstruction gives $H_{\mathrm{OS}} \geq 0$ with gap $6\ln\frac{3}{2}$. The full $\mathrm{QG}_{\mathrm{amb}}$ (the metric sector) stays open, gap-decoupled — a genuine *partial* construction, not a closure.

- **`CartanDeterminants.lean` (`FORM.CARTAN.DET.01`, Lean 4, [E]).** The carrier piece of the holomorphy/anyon discriminator is grounded in real `Matrix.det` computations (kernel `decide`, no extra axiom): $|\det \mathrm{Cartan}(A_3)|{=}4{=}|\mu_4|$ ($3{\times}3$, `det_fin_three`) and $|\det \mathrm{Cartan}(D_5)|{=}4$ ($5{\times}5$, `decide`), giving the carrier $|\det \mathrm{Gram}|{=}4 \cdot 4{=}16$ (the `v281` anyon count). The rank-8 E_8/D_8 case stays the Nat discriminator — Mathlib's own `Data.Matrix.Cartan` leaves `E₈_det = 1` as a `proof_wanted`. Built `lake build` on Lean v4.29.1 + Mathlib.

2026-06-22 (next-steps round: the keystone collapsed to one shared premise (Lean + one citable theorem), the F_{transfer} solver suite, the cusp weight derived from a minimal rewrite, and the θ_{13} pressure point: `FORM.QGEO.BW.01`, `v325`, `v326`, `v327`, `v328`)

- **`BWKeystone.lean` (`FORM.QGEO.BW.01`, Lean 4, [E]/[O]).** The `v323` Bisognano–Wichmann linkage formalised: the nodes RotationInvariantVacuum $\to$ GeometricModularFlow $\to$ Mu4ModularSymmetry $\to$ StateInvariance (=`QGEO.SYM.01`) composed over the `v308` chain. The theorem `bedrockReducesToRotationInvariance` shows the WHOLE bedrock `QGEO.SYM.01` reduces to the SINGLE shared open premise `RotationInvariantVacuum` — the two open bedrock items collapse to one — with `#print axioms` confirming the residual = the cited chain steps + recovery gap + the named BW steps + that one premise. Plus a `decide`-proven discriminator $(1 \mid 4) \wedge \neg(4 \mid 1)$ (BW geometric flow strictly stronger than `v309` clock invariance). Built `lake build` on Lean v4.29.1 + Mathlib.

- **`v325_pillowcase_keystone.py` (`QGEO.KEYSTONE.01`, [E]/[O]).** The keystone as ONE citable theorem: IF the raw seam state is the flat $\tau{=}i$ pillowcase (rotation-invariant) state THEN the whole bedrock closes (4 square marks, μ_4 deck, $\omega \circ \rho{=}\omega$, holomorphic $c{=}8 \Rightarrow (E_8)_1$, with the recovery gap as input); all pieces machine-checked (incl. the pillowcase $\chi_{\mathrm{orb}}{=}0$), the single open lemma shared by both bedrock IDs and Lean-pinned. An assembly certificate, not a closure.

- **`v326_ftransfer_suite.py` (`FR.SUITE.01`, [E]/[C]).** The four F_{transfer} readouts productised into one runnable harness, each a gapped relaxation to a unique attractor: F_{pole} (Koide, seam rate $(2/3)^6$), $F_{\text{Boltzmann}}$ (η_B, H-theorem), F_{relic} (axion adiabatic invariant), F_{QCD} (m_p/m_e, RG to the UV fixed point), plus the F_{RareKaon} bridge. Only F_{pole} carries the seam rate; the four external rates are honestly fenced (`v187`/`v303`).

- **`v327_hypergraph_rewrite.py` (`HYP.REWRITE.01`, [E]/[O]).** The cusp weight 2/3 is DERIVED from a minimal rewrite: a 3-channel/1-absorbing family rule has spectrum $\{1, 2/3, 0\}$, so $2/3{=}(N_{\text{fam}}{-}1)/N_{\text{fam}}{=}|\mathbb{Z}_2|/N_{\text{fam}}$ emerges from the rule arity and $(2/3)^6{=}64/729$ is the recovery gap; with a PROOF that 2/3 is not a root of the affine-E_8 charpoly ($p(2/3) = -3520/19683 \neq 0$, sharpening `v312`). `v324`'s injected datum is reduced to the anchor arity $\{2, 3\}$.
- **`v328_theta13_pressure.py` (`FLAV.TH13.PRESSURE.01`, [C]/[X]) + `v307` hardening.** The honest weak spot named and quantified: $\sin^2\theta_{13}{=}\varphi_0^{\text{ret}} e^{-5/6}{=}0.023108$ is $+2.0\sigma$ above NuFIT 6.0 NO, a SEPARATE carrier-trace channel (`v268`), not relieved by any freeze-respecting correction (RG $< 1\%$, exponent 5/6 exact), pre-registered as the kill. The `v307` watchdog gains a data provenance/date check and the rare-kaon F_{transfer} bridge row.
- **Wolfram mirror 285 → 288.** `v325` ($\chi_{\text{orb}}{=}0$) and `v327` (rewrite spectrum $\{0, 2/3, 1\}$, $(2/3)^6{=}64/729$, and the non-graph-spectral proof) added to `tfpt_readouts_extension.wl`; `GATE.WOLFRAM.02` and the Wolfram README updated to 288/288.

2026-06-22 (consistency + de-accretion editorial pass: current-state alignment across papers, website and ledger; no new modules)

- **Keystone naming reconciled to `SEAM.EQUIV.01`.** Following `v323`/`QGEO.BW.01` (the deck premise is downstream of the seam being a chiral net), the single named open keystone is now `SEAM.EQUIV.01` *everywhere*, with `QGEO.SYM.01` stated as its *downstream* conformal-deck face. The ledger row `QGEO.SYM.01` was updated (it no longer calls itself "the single structural residual of the whole theory"; it now depends on `SEAM.EQUIV.01`/`QGEO.BW.01`), and the superseded "`GATE.QGEO` open gate" / "two research gates (U_{wall}) and G_{metric}" framing was replaced by the current Rest $= v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ with the two structural items `SEAM.EQUIV.01` + `QG.AMB.01` (intro status card, `tfpt_4_frontier`, `origin_theory`, `tfpt_5_redteam`, `tfpt_research_contracts`, and the website).
- **De-accretion: the sprint-by-sprint reduction history was moved to this changelog, where it belongs.** Several reader-facing sections had grown by accretion (each round appended a clause). They were rewritten to state the *current* result once, with the round-by-round narrative left here: the introduction closure ledger ("honest bottom line") and hard-reduction subsection; the `origin_theory` "residual inventory" (the five "Update/…/Terminal" passes collapsed to one current inventory); the `tfpt_5_redteam` Target-A Outcome; and on the website the FAQ "What is still open?", the `OpenGates` closing-condition block, the `papers.ts` research-contracts/red-team entries, the verification-DAG `qft` node, the glossary `G_net` entry, the Rosetta lexicon, and the verification/review pages. No scientific content or `\veri` citation was dropped; only the embedded version-stamp logs were removed.
- **Stale facts corrected.** Version label `5.0`→`v5.2` (intro, single-sourced via `\TFPTversion`); Lean build `1886`→`3357` jobs (`FORM.SEAMEQUIV.01`); README Python suite `v213`/ 212→`v324`/ 322 modules; Wolfram extension `269/269` / `249/249`→`285/285` (README, website replication page, `zenodo_description.html`); residual "Paper-3" old-paper attributions in `tfpt_2` neutralised.

(Original 2026-06-22 entry follows.)

2026-06-22 (external-review pass: a registry bug fix, the NA62 2026 rare-kaon update, CP-phase prose hygiene, and three firewall/visualisation enhancements: v202, v203, v305, v307, v321)

- **Bug fix — $\delta_{\rm CKM}$ display value.** The predictions table (`tex-artefacts/predictions.tex`) showed $\delta = \frac{\pi}{3} + 3\lambda^2 = 66.96°$, but that formula evaluates to 68.65° (the canonical frozen `DELTA_CKM_RAD`= 1.198232 rad, used by `v88/v202/FLAV.CP.01` everywhere else). Corrected to 68.65°. An external review correctly flagged the inconsistency; it was an isolated display typo (the website was already clean).
- **`v202_rare_kaon.py` — NA62 2016–2024 update ([E]/[C]).** The charged-mode comparison is moved from the historical NA62 2016–2022 observation $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ ($\sim 1.2\sigma$ above) to the full 2016–2024 combination $\mathrm{BR}(K^+) = (9.6^{+1.9}_{-1.8}) \times 10^{-11}$ (La Thuile 2026), on which the prediction 9.45×10^{-11} lands at $+0.08\sigma$. Framed honestly: a strong *consistency* hit for the closed CKM point, but an $F_{\rm transfer}$ bridge — *not* a unique TFPT-vs-SM discriminator (the SM $(8.6 \pm 0.4) \times 10^{-11}$ is also inside the NA62 error). Propagated to the ledger (`FR.RAREKAON.01`), `freeze_file` (new `rare_kaon` kill row), the watchdog (`v307`, a new $F_{\rm transfer}$ bridge row), the forward board (`v321`, rank 9), and `tfpt_2_standard_model` + the website.
- **CP-phase prose hygiene.** The Galois CP lock is tightened from "a relation to the *measured* quark phase" to "a relation to the *leading* ($\pi/3$) component of the quark phase": the lock is to the structural $\pi/3$, not the full measured $\gamma = \delta^{\rm lead}_{\rm CKM} + 3\lambda_C^2 \approx 68.7°$, so $\delta_{\rm PMNS} = 240°$ (not 248.7°). Fixed in `tfpt_2_standard_model`, `origin_theory`, `papers.ts` and `predictions.ts`.
- **`v305_witness_independence.py` — the No-Golden-Dynamics rule ([E]).** The number-field split is made an explicit anti-numerology firewall rule: $\phi = 2\cos(\pi/5) \in \mathbb{Q}(\sqrt{5})$ is the *static* carrier ($\mathbb{Z}/5$) field, every *dynamic* rate is rational ($\mathbb{Q}$, the $\mathbb{Z}/6$ hand), so a dynamic rate explained by ϕ is a cross-field false friend (`v314/v315`).
- **`v203_eht_achromatic.py` — the μ_4 Fourier fourth null ([X]).** The EHT residual pre-registration gains a fourth null: the azimuthal residual must carry power *only* at $m \equiv 0 \bmod 4$ (the μ_4 fingerprint of the four marks, `v201`); a μ_4-orbit profile has ~ 0 off-mode power while a $\mathbb{Z}_2$ control leaks at $m = 2$.
- **Seed-hyperplane figure (Fig. ??).** A new figure visualises `v306`: every scorecard observable inverts *independently* to the one latent seed φ_0, the error-weighted mean matching the axiom to 0.04%, with $\sin^2\theta_{13}$ the $\sim 2\sigma$ pressure point. Added to `make_figures.py`, `tfpt_5_redteam` and the manifest.

2026-06-22 (Wolfram mirror of the arithmetic arc + three sharpening steps: the CP-lock made quantitative, the Bisognano–Wichmann keystone, the minimal hypergraph fibre: v322, v323, v324)

- **Wolfram mirror of the cyclotomic/Galois arithmetic arc (the independent second path).** The exact algebraic results of the arc are now mirrored in `verification/wolfram/tfpt_readouts_extension.wl` (ten new `checkExact` blocks, 275 → 285, *all passing*): `v313` (the affine-E_8 characteristic polynomial with the golden factor $x^2 - x - 1$; the $(2, 3, 5)$ atoms as $2\cos(\pi/k)$; the 31/30 selection), `v314` (the rational kernel vs $\mathbb{Q}(\sqrt{5})$ dynamic-rate split), `v315` (30=5·6, $\varphi(30)$=8, the Galois group $\mu_4 \times \mathbb{Z}_2$, the $\sqrt{5}$ Gauss sum), `v316` ($\rho = \zeta_6 = \zeta_{30}^5$, $\rho^4 = -\rho$, Gal=$\mathbb{Z}_2$= CP conjugation), `v317` ($N_{\rm fam}$=3 as the μ_3 orbit, the 1+2 Galois split), `v318` ($\varphi_0^{\rm ret} = (4/3)c_3 + 48c_3^4$ a pure function of π) and `v320` (the CP lock $\delta_{\rm PMNS}$=240°). Ledger `GATE.WOLFRAM.02` and the Wolfram README updated to 285/285.
- **`v322_cp_lock_sharpened.py` (`GALOIS.CP.SHARPEN.01`, [E]/[C]/[X]).** The `v320` CP-lock sharpened into a quantitative, pre-registered kill test. [E] the leading phase sits on

the six hexagonal nodes $\arg \rho^k = k \cdot 60^\circ$, and *which* one is selected is the deck order: $\delta^{\rm lead}_{\rm PMNS} = |\mu_4|\, \delta^{\rm lead}_{\rm CKM} = 4(\pi/3)$ (node 4); [C] the sub-leading correction is bounded by the quark analogue $3\lambda^2 \approx 8.7^\circ$, so the prediction is the band $240^\circ \pm \sim 9^\circ$; [C] against NuFIT 6.0 (NO, $\delta_{CP} = 212^{+26}_{-41}{}^\circ$) the leading value is $+1.08\sigma$, the band edge $+0.74\sigma$ – compatible today; [X] the nearest wrong node is 60° away, far beyond the budget, so DUNE/Hyper-K (~ 5–15°) discriminate cleanly. Cited in `origin_theory`; ledger `GALOIS.CP.SHARPEN.01`.

- **`v323_bw_geometric_modular.py` (`QGEO.BW.01`, [E]/[C]/[O]).** The keystone pushed one honest step via Bisognano–Wichmann. [E] the μ_4 clock is a geometric rotation $\rho = \mathrm{diag}(i^n) = \exp(i\frac{\pi}{2}L)$, $\mu_4 \subset U(1)_{\rm rot}$; [E] for a rotation-covariant covariance $C = f(L)$ the modular Hamiltonian $K = \log((1-C)/C) = g(L)$, so $[K, L] = 0$ (the modular flow is geometric) and the clock is a modular symmetry *for free* (the discriminator: a period-4 curvature preserves the clock but its flow is not geometric, $[K, L] \neq 0$); [C] given the seam is the $(E_8)_1$ chiral net (`v308`), BW/Hislop–Longo make the vacuum modular flow geometric, so `QGEO.SYM.01` ($\omega \circ \rho = \omega$) is *downstream* of `SEAM.EQUIV.01` + a rotation-invariant vacuum, not an independent axiom; [O] residual: the raw collar vacuum is rotation-invariant (cleaner than `v309`, shared with `v308`). Cited in `origin_theory`; ledger `QGEO.BW.01`.
- **`v324_hypergraph_fiber.py` (`HYP.FIBER.01`, [E]/[C]).** The constructive `v312` follow-up: the *minimal* substrate that carries both the E_8 skeleton and the recovery gap is a *product*. [E] the carrier network $T_{\rm net} = (A + 2I)/4$ (attractor = Kac marks = skeleton) fibred by the 3-node family cusp $T_{\rm cusp} = \mathrm{diag}((1-w)^{2N_{\rm fam}})$, $w \in \{0, \frac{1}{3}, \frac{2}{3}\}$ (spectrum $\{1, (2/3)^6, (1/3)^6\}$); the fibred $T_{\rm net} \otimes T_{\rm cusp}$ has top eigenvalue 1 (eigenvector marks $\otimes$ $(w=0)$) and $(2/3)^6$ as a genuine eigenvalue (marks $\otimes$ $(w=\frac{1}{3})$) – the $N_{\rm fam}=3$ cusp is the minimal fibre and the substrate = carrier $\times$ family = the $30 = g_{\rm car}(2N_{\rm fam}) = 5\times 6$ of `v315`; [C] honestly (`v312`) the cusp weight is still *injected*, not derived from the graph – a consistent realisation, not a derivation of 2/3 from a pure rewrite. Cited in `origin_theory`; ledger `HYP.FIBER.01`.

2026-06-22 (the forward kill-test board: the decisive upcoming measurements ranked, each locked to a freeze row — incl. the new Galois-CP relation: v321)

- **`v321_killtest_board.py` (`KILL.BOARD.01`, [E]).** The forward companion to `v307` (the current-data board): the decisive *upcoming* measurements, ranked and each locked to a `freeze_file.csv` kill row – #1 JUNO $\sin^2\theta_{12} = 0.306747$ (fastest), #2 CMB-S4/LiteBIRD r (kill $r > 0.01$), #3 DESI/Euclid w (kill $w \neq -1$), #4 DUNE/Hyper-K $\delta_{\rm PMNS} = 240^\circ$ (the new `v320` Galois-CP relation), #5 DUNE/LEGEND/nEXO ordering+$m_{\beta\beta}$, #6 PSI n2EDM, #7 JUNO $\sin^2\theta_{13}$ ($\sim 2\sigma$ now), #8 LiteBIRD β. A forward decision ledger (publicly-stated experimental reach), complementing `v307`; no new physics. Cited in `origin_theory`; ledger row `KILL.BOARD.01`.

2026-06-22 (Lean: the SEAM.EQUIV.01 composition chain formalised — the keystone residual machine-pinned to the named cited steps: FORM.SEAMEQUIV.01)

- **`SeamEquivChain.lean` (`FORM.SEAMEQUIV.01`, Lean 4, [E]/[O]).** A Lean 4 mirror of the `v308` closing chain: the four nodes GapPositive → InvertibleSRE → HolomorphicC8 → SeamIsE8 assembled as a well-typed composition theorem `seamEquivChain`, with the three cited literature steps (Kitaev free-fermion invertibility; Müger / Kawahigashi–Longo–Müger holomorphy; Conway–Sloane E_8 uniqueness) as named `axiom`s and the carrier recovery gap (the OS step) as the single open input. [E] `#print axioms seamEquivChain` confirms the residual is exactly those named steps + `recovery_gap` (no hidden assumption); [E] a `decide`-proven discriminator primariesE8 $= 1 \neq$ primariesD8 $= 4$ (the $|\det \mathrm{Cartan}|$ that separates the holomorphic E_8 from the same-c rival $D_8 = SO(16)$). Built end-to-end (`lake build`, 3357 jobs) on Lean v4.29.1 + Mathlib. [O] The value is the machine-checked composition + axiom audit (pinning the keystone residual), *not* a from-scratch proof — it does NOT close `SEAM.EQUIV.01`. Ledger row `FORM.SEAMEQUIV.01`; mirrors `v308`.

2026-06-22 (a NEW falsifiable prediction from the Galois structure: the two CP phases are locked, $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM}}^{\mathrm{lead}} + \pi$: `v320`)

- **`v320_galois_cp_relation.py` (`GALOIS.CP.PREDICT.01`, [E]/[C]/[X]).** The `v316` follow-up turned into a testable cross-prediction. [E] both leading CP phases are powers of the *one* hexagonal unit $\rho = \zeta_6$ of the family Galois factor: $\delta_{\mathrm{CKM}}^{\mathrm{lead}} = \arg\rho = \pi/3$ $(60°)$, $\delta_{\mathrm{PMNS}} = \arg\rho^4 = 4\pi/3$ $(240°)$, and $\rho^4 = -\rho$ locks them as $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM}}^{\mathrm{lead}} + \pi$ (the $\mathbb{Z}_2$ sheet flip); [C] this upgrades the previously assigned $\delta_{\mathrm{PMNS}} = 240°$ to a Galois-forced relation to the measured quark phase (the quark and lepton leading CP phases are not independent); [X] kill test: a δ_{PMNS} robustly incompatible with $240°$ ($> 3\sigma$ at DUNE/Hyper-K/JUNO) falsifies the Galois-CP organisation; $240°$ is currently within the broad NuFIT 6.0 range. A genuine new falsifiable consequence of the `v313`–`v319` arithmetic arc (also a `freeze_file.csv` `cp_phase_relation` kill row). Cited in `origin_theory`; ledger row `GALOIS.CP.PREDICT.01`.

2026-06-22 (the translation clock: the discrete↔dynamic bridge *is* the order-30 Coxeter element, a clock with two coprime hands 5×6, read law-inclusive $(0..5)$ or live-only $(1..5)$: `v319`)

- **`v319_translation_clock.py` (`TRANSLATE.CLOCK.01`, [E]/[C]).** A synthesis of the `v313`–`v318` arc with `v124`/`v55`, answering the question "is the static↔dynamic translation a clock running $0, 1, 2, 3, 4, 5$ or $1, 2, 3, 4, 5$?". [E] the only object holding both the static carrier $(\mathbb{Q}(\sqrt{5}))$ and the dynamic family (rational) data is the order-30 Coxeter element (`v314`); $30 = h(E_8) = g_{\mathrm{car}}(2N_{\mathrm{fam}}) = 5 \cdot 6$ and $\gcd(5, 6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (two coprime hands, `v315`). [E] the *dynamic* hand is $\mathbb{Z}/6 = 2N_{\mathrm{fam}}$ (ticks $0, \ldots, 5$): the recovery exponent is exactly $6 = 2N_{\mathrm{fam}}$, rate $(2/3)^6 = (|\mathbb{Z}_2|/N_{\mathrm{fam}})^{2N_{\mathrm{fam}}}$, order-6 generator c^5 ($\zeta_6 = \zeta_{30}^5$, `v316`). [E] position 0 is the *conserved law* — the resummed clock $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$ (`v124`) has $\mathrm{rate}(0) = 0$ (the attractor, eigenvalue 1, `v56`), the live decaying modes from n=1, and the E_8 live phases (totatives of 30) start at 1 ($\{1, 7, \ldots, 29\}$), the 0 phase not a totative — so "$0, 1, 2, 3, 4, 5$" is law-inclusive, "$1, 2, 3, 4, 5$" live-only. [E] the *static* hand is $\mathbb{Z}/5 = g_{\mathrm{car}}$ (ticks $1, \ldots, 5$): the golden field $\mathbb{Q}(\sqrt{5})$ ($\varphi = 2\cos(\pi/5)$), field-obstructed (no rational dynamic image, `v314`) — static-only; $\zeta_5 = \zeta_{30}^6$ carries no phase. [E] one full turn $= \mathrm{lcm}(5, 6) = 30 = h(E_8)$, rank $E_8 = 8 = \varphi(30)$ live phases pairing into $|\mu_4| = 4$ invariant planes (`v55`). [C] verdict: the discrete→dynamic translation *is* the order-30 clock = (static 5-ring) $\times$ (dynamic 6-ring), the dynamic hand carrying the rate (exponent $6 = 2N_{\mathrm{fam}}$), the static hand the golden carrier (no rate), and "0 vs 1" = law-inclusive vs live-only; the arithmetic is [E], the "it is one clock" reading [C]. Python-only (`sympy`), like the rest of the arc. New figure `figures/translation_clock.pdf` (concentric 5- and 6-hands). Cited in `origin_theory`; ledger row `TRANSLATE.CLOCK.01`.

2026-06-21 (capstone: the SM structural sector is $\mathbb{Q}(\zeta_{30})$+ Galois $\mu_4 \times \mathbb{Z}_2$, and the complete input is $\{a, \pi, v_{\mathrm{geo}}\}$ — zero dimensionless free parameters: `v318`)

- **`v318_arithmetic_capstone.py` (`ARITH.CAPSTONE.01`, [E]/[O]).** The synthesis of the `v313`–`v317` arc. [E] the SM *structural* sector (the 3 generations, the 2 CP phases, their orbit/hierarchy) is the cyclotomic field $\mathbb{Q}(\zeta_{30})$ with Galois group $\mu_4 \times \mathbb{Z}_2$ (degree $\varphi(30) = 8 = \mathrm{rank}\, E_8$, $30 = |\mathbb{Z}_2| N_{\mathrm{fam}} g_{\mathrm{car}} = h(E_8)$), forced by the anchor atoms $\{2, 3, 5\} = e(a)$; [E] the magnitude seed $\varphi_0 = (|\mu_4|/N_{\mathrm{fam}})c_3 + \Omega_{\mathrm{adm}} c_3^4 = \frac{4}{3}c_3 + 48c_3^4$ ($c_3 = 1/(8\pi)$) is a pure function of π (no free parameters, anchor-derived coefficients), so the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$) are $\{a, \pi\}$-determined, not free; [E] free inventory: *zero* dimensionless free parameters, π the unique transcendental primitive (`ARCH.CORE.01`), v_{geo} the unique dimensionful input (No-Unit

Theorem) — complete input $\{a, \pi, v_{\rm geo}\}$. [O] honest residual: this rigidity holds *given* the axioms and does not realise the structure on the raw seam (`QGEO.SYM.01`/`SEAM.EQUIV.01` stay [O]). Cited in `origin_theory`; ledger row `ARITH.CAPSTONE.01`.

2026-06-21 (are the 3 generations a μ_3/Galois orbit? Yes (μ_3), Galois-refined 1+2 with the fixed generation = the recovery attractor: v317)

- **`v317_galois_family.py` (`GALOIS.FAMILY.01`, [E]/[C]).** The `v316` follow-up to the generation structure. [E] the three cusp weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (with $\mathrm{Spec}(Q_+) = \{1,2,3\}$, the A_3 exponents) are the cube roots of unity $\{1, \zeta_3, \zeta_3^2\}$ under $w \mapsto e^{2\pi i w}$, and the cyclic family symmetry $w \mapsto w+\frac{1}{3}$ is a *transitive* μ_3 orbit (the family IS the cube-root orbit); [E] the transfer eigenvalues $(1-w)^6 = \{1, (2/3)^6, (1/3)^6\}$ attach to the weights (attractor = eigenvalue 1 at w=0); [E] the family Galois $\mathbb{Z}_2$ ($\zeta_3 \mapsto \zeta_3^2 = \bar{\zeta}_3$, = the `v316` CP conjugation) refines the orbit to 1+2 — fixing w=0 (rational) and swapping w=$\frac{1}{3} \leftrightarrow \frac{2}{3}$; [E] so the Galois-FIXED generation is the recovery ATTRACTOR (the law) and the Galois-conjugate pair the decaying hierarchy. [C] verdict: the family STRUCTURE is the same arithmetic as the CP phase (a μ_3 orbit, Galois-refined 1+2), the mass MAGNITUDES remaining the analytic seed. Cited in `origin_theory`; ledger row `GALOIS.FAMILY.01`.

2026-06-21 (does the Galois $\mu_4 \times \mathbb{Z}_2$ organize the readouts? Yes — the CP phases are the family-factor cyclotomic data, Galois $\mathbb{Z}_2$ = CP conjugation: v316)

- **`v316_galois_readout.py` (`GALOIS.READOUT.01`, [E]/[C]).** The `v315` follow-up to phenomenology: does the Galois coupling $\mu_4 \times \mathbb{Z}_2$ reach the physics? [E] it does, through the *family* factor — the hexagonal CP unit $\rho = \zeta_6 = \zeta_{30}^5$ is exactly the order-6 = $2N_{\rm fam}$ factor c^5 of `v315` (the carrier being $\zeta_5 = \zeta_{30}^6$); [C] the leading CP phases are its cyclotomic arguments $\delta_{\rm CKM}^{\rm lead} = \pi/3 = \arg\zeta_6$ and $\delta_{\rm PMNS} = 4\pi/3 = \arg\zeta_6^4$ (power $4 = |\mu_4|$), with $\zeta_6^4 = -\zeta_6$ the $\mathbb{Z}_2$ sheet flip and $\mu_6 = \mu_3 \times \mu_2$ (inherits `v231`/`v233`); [E] the family Galois $\mathrm{Gal}(\mathbb{Q}(\zeta_6)/\mathbb{Q}) = \mathbb{Z}_2$ ($\zeta_6 \mapsto \zeta_6^5 = \bar{\zeta}_6$) *is* CP conjugation $\delta \mapsto -\delta$; [E] the carrier factor ζ_5 (golden $\sqrt{5}$) carries no phase (μ_4 static), and the magnitudes ($\sin^2\theta_{12} = \frac{1}{3} - \varphi_0/2$, transcendental φ_0) are not cyclotomic. [C] verdict: the Galois bridge reaches phenomenology in the phase sector predicted by the `v314` field split. Cited in `origin_theory`; ledger row `GALOIS.READOUT.01`.

2026-06-21 (couple or factorize? the order-30 Coxeter element couples the carrier and family sectors as a cyclotomic compositum, Galois = $\mu_4 \times \mathbb{Z}_2$: v315)

- **`v315_coxeter_coupling.py` (`COX.COUPLE.01`, [E]/[C]).** Answers the `v314` question — does $30 = g_{\rm car}(2N_{\rm fam}) = 5{\cdot}6$ couple the carrier (5-fold, golden, $\mathbb{Q}(\sqrt{5})$) and family ($6 = 2{\cdot}3$, rational) sectors, or just factorize? [E] the cyclic group *factorizes*: $\gcd(5,6) = 1 \Rightarrow \mathbb{Z}/30 = \mathbb{Z}/5 \times \mathbb{Z}/6$ (direct product, group-decoupled, in line with the `v314` field obstruction); [E] the E_8 exponents = totatives(30) split as totatives(5)$\times$ totatives(6), $\varphi(30) = 4 \cdot 2 = 8 = \mathrm{rank}\ E_8$; [E] but the *field* couples: $\mathbb{Q}(\zeta_{30})$ is the compositum of the carrier $\mathbb{Q}(\zeta_5) \ni \sqrt{5}$ and the family $\mathbb{Q}(\zeta_3)$, degree 8 = rank E_8; [E] the Galois group is exactly $(\mathbb{Z}/5)^\times \times (\mathbb{Z}/3)^\times = \mathbb{Z}/4 \times \mathbb{Z}/2 = \mu_4 \times \mathbb{Z}_2$ (order 8, max element order 4), so the μ_4 seam clock *is* the Galois group of the carrier's golden field and $\mathbb{Z}_2$ that of the family field (refining `v223`); [E] $\sqrt{5} = \phi + 1/\phi$ is the quadratic Gauss sum in $\mathbb{Q}(\zeta_5)$. [C] verdict: $30 = 5 \cdot 6$ does not dynamically entangle the sectors — it couples them as the cyclotomic compositum, the static↔dynamic bridge being *Galois*, not a dynamical map. Cited in `origin_theory`; ledger row `COX.COUPLE.01`.

2026-06-21 (do the discrete-kernel and dynamic rates translate? An exact number-field split $\mathbb{Q}$ vs $\mathbb{Q}(\sqrt{5})$: v314)

- **`v314_rate_translation.py` (`RATE.TRANSLATE.01`, [E]/[C]).** Answers, exactly, whether the discrete kernel (the affine-E_8 network, carrying the golden ratio) and the dynamic part (the seam transfer, carrying the recovery rate $(2/3)^6$) translate. [E] FIELD SPLIT: the adjacency spectrum is the rational part $\{0,\pm1,\pm2\}\subset\mathbb{Q}$ plus the golden part $\{\pm\phi,\pm1/\phi\}\subset\mathbb{Q}(\sqrt{5})$, while the transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ is *entirely* rational; [E] the family/sheet rates DO translate (the rational angles $2\cos(\pi/2){=}0 = |\mathbb{Z}_2|$ and $2\cos(\pi/3){=}1 = N_{\rm fam}$ build $(2/3)^6 = (|\mathbb{Z}_2|/N_{\rm fam})^{2N_{\rm fam}}\in\mathbb{Q}$, same field, same atoms), but [E] the carrier 5-fold (golden, $\mathbb{Q}(\sqrt{5})$) has NO rational dynamic image – a field obstruction (2/3 is neither an adjacency nor a lazy-update eigenvalue; heat-kernel $t=\Delta/(2-\phi)\approx 6.37$ not clean), so the carrier is *static-only*; [E] the only object holding both is the order-30 Coxeter element $30 = g_{\rm car}(2N_{\rm fam}) = 5\cdot 6$, and the ϕ-quasicrystal $E_8 = H_4{+}\phi H_4$ $(240 = 2\cdot 120)$ carries the carrier but not the (rational, 3-fold) cusp-weight family dynamics. Cited in `origin_theory`; ledger row `RATE.TRANSLATE.01`.

2026-06-21 (the golden ratio made precise: the $(2,3,5)$ atoms are the affine-E_8 network spectral angles, $\phi = 2\cos(\pi/5)$ the $g_{\rm car}{=}5$ signature: v313)

- **`v313_golden_atoms_spectral.py` (`GOLD.ATOMS.01`, [E]/[C]).** A follow-up to `v312`'s observation, made exact. [E] the affine-E_8 adjacency characteristic polynomial factors as $x(x^2{-}4)(x^2{-}1)(x^2{-}x{-}1)(x^2{+}x{-}1)$, so the golden minimal polynomial $x^2{-}x{-}1$ divides it and $\phi = 2\cos(\pi/5) = (1{+}\sqrt{5})/2$ is an *exact* eigenvalue; [E] the non-trivial eigenvalues are $2\cos(\pi/k)$ for exactly $k\in\{2,3,5\} = \{|\mathbb{Z}_2|, N_{\rm fam}, g_{\rm car}\}$ $(2\cos(\pi/2){=}0,\ 2\cos(\pi/3){=}1,\ 2\cos(\pi/5){=}\phi,\ 2\cos(2\pi/5){=}1/\phi)$, so the golden ratio is *specifically* the $g_{\rm car}{=}5$ (5-fold) signature and $g_{\rm car}{=}5$ gains a new spectral witness distinct from Pascal/Riemann–Roch (raising the `v305` multiplicity); [E] the $(2,3,5)$ spherical excess $1/2{+}1/3{+}1/5 = 31/30 > 1$ (the condition selecting the finite $2I \Rightarrow E_8$) ties the `v63`/`v76` gap-margin numerator $31 = 2^{g_{\rm car}}{-}1 = 248/8 = 1{+}h(E_8)$ to the Coxeter $30 = h(E_8)$ — a unification of already-anchored quantities, NOT new evidence (honest anti-numerology, `v305`); [E] honest null: ϕ is a structural lattice/network quantity, not a phenomenological readout. [C] direction: ϕ is the E_8-quasicrystal (Elser–Sloane cut-and-project) ratio, a candidate aperiodic substrate for the `v312` question. Cited in `origin_theory`; ledger row `GOLD.ATOMS.01`.

2026-06-21 (TOE-roadmap solver round: the `SEAM.EQUIV.01` keystone certificate, the modular bedrock, the carrier→SM closure, the gap-decoupled measure, and the sharpened hypergraph substrate: `v308`–`v312`)

- **`v308_seam_equiv_chain.py` (`SEAM.EQUIV.CHAIN.01`, [E]/[O]).** TOE attack point 1: the keystone "the raw seam-Calderón net is the holomorphic $c{=}8$ $(E_8)_1$ net" assembled into ONE well-typed logical chain gap>0 $\to$ invertible/SRE (Kitaev) $\to$ holomorphic $c{=}8$ (Müger/KLM) $\to (E_8)_1$ (Conway–Sloane), with the exact discriminators $|\det\mathrm{Cartan}(E_8)|{=}1$ vs $|\det\mathrm{Cartan}(D_8{=}SO(16))|{=}4$, the carrier tower $4{\cdot}4{=}16\to4\to1$ and $c{=}g_{\rm car}{+}N_{\rm fam}{=}8$ all machine-checked; the input gap is the derived recovery gap $6\ln(3/2){>}0$. The only undischarged premise is the cited OS step — an assembly/reduction certificate (a Lean target), not an end-to-end closure. Cited in `tfpt_research_contracts`; ledger row `SEAM.EQUIV.CHAIN.01`.
- **`v309_modular_bedrock.py` (`QGEO.MODHAM.01`, [E]/[O]).** TOE attack point 2: the bedrock $\omega{\circ}\rho{=}\omega$ via an explicit modular Hamiltonian, non-circular — the four marks come from Gauss–Bonnet (`v216`) and the order-4 clock from cross-ratio 2 (`v214`), so μ_4 is derived; the quasi-free modular Hamiltonian $K = \log((1{-}C)/C)$ round-trips to the seam operator and $[\rho,K]{=}0 \Leftrightarrow \omega{\circ}\rho{=}\omega$ (a $\mathbb{Z}_2$ profile breaks it). Reduces the bedrock to the Bisognano–Wichmann intrinsicality of the raw collar; not a closure of `QGEO.SYM.01`. Cited in `tfpt_research_contracts`; ledger row `QGEO.MODHAM.01`.
- **`v310_carrier_sm_anomaly.py` (`CAR.SM.ANOM.01`, [E]).** TOE attack point 3: the carrier half-spinor $16 = \Lambda^{\rm even}(\mathbb{C}^5) = 1{+}10{+}5$ is exactly one anomaly-free Standard-Model generation

($SU(5) \subset SO(10)$: $10+\bar{5}+1$), with $\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$ (exact) and the one-loop $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$; a fourth family shifts b_1, so $N_{\rm fam}$=3 pins the RG. Closes the spectrum/anomaly/β sub-question; the interacting Epstein–Glaser S-matrix and the Yukawa sector stay [C]/[O]. Cited in `tfpt_research_contracts`; ledger row `CAR.SM.ANOM.01`.

- **`v311_gap_decoupled_measure.py` (`QGAMB.CLUSTER.01`, [E]/[C]/[O]).** TOE attack point 4: the physical QG measure as a gap-decoupled, exponentially-clustering object — the transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ gives geometric clustering at rate $(2/3)^6$ (correlation length $1/(6\ln(3/2))$, finite susceptibility 729/665), and the `v76` margin $6\ln(3/2) - 31/(4\pi^2) \approx 1.648 > 0$ decouples the physical sector from the un-built ambient. `QG.AMB.01` stays [O] (gap-decoupled, inert). Cited in `tfpt_research_contracts`; ledger row `QGAMB.CLUSTER.01`.
- **`v312_hypergraph_substrate.py` (`HYP.INJECT.01`, [E]/[O]).** TOE attack point 5: the hypergraph substrate sharpened — the affine-E_8 Kac marks are the Perron eigenvector ($A\,m$=$2m$) and the subleading eigenvalue is exactly $2\cos(\pi/5)$ = the golden ratio, but the recovery rate $(2/3)^6$ is *not* any eigenvalue and φ_0 is not a rational edge fraction; so a full-structure rewrite must inject exactly two non-graph-spectral data — the cusp weight $2/3$=$|\mathbb{Z}_2|/N_{\rm fam}$ and the analytic seed φ_0. The open question is precisely delimited, not closed. Cited in `introduction`; ledger row `HYP.INJECT.01`.

2026-06-21 (adversarial-review follow-ups: a structural anti-numerology firewall, an out-of-sample seed cross-validation, an operational data watchdog, and the Target-A holomorphy kill test: `v305`–`v307`)

- **`v305_witness_independence.py` (`WITNESS.ECON.01`, [E]).** The *structural* anti-numerology firewall complementing the `v100` phenomenological null test, against the sharpest self-deception worry ("the same integer counted many times as if independent"). Every headline integer ($|\mu_4|$=4, $g_{\rm car}$=5, $|\mathbb{Z}_2|$=2, $N_{\rm fam}$=3, $\dim S^+$=16, $\operatorname{rank} E_8$=8, 240, 248, $|2I|$=120, $h(E_8)$=30, $|W(D_5)|$=1920, $\Omega_{\rm adm}$=48, the gap exponent 6) is a documented image of the single anchor a=$(1,1,2)$ with π the only transcendental primitive (13 atoms from 2 free primitives, 6.5× compression made explicit), so the recurring $2/3$=$|\mathbb{Z}_2|/N_{\rm fam}$ and $(2/3)^6$ are *one* ratio; the E_8 spine is overdetermined (≥ 3 structurally independent witnesses) while the pure-seed readouts are single-witness (scored jointly by `v100`, never multiplied); the negative control shows α^{-1}'s "137" is foreign to the anchor family (an independent witness via the $F_{U(1)}$ cubic, `v3`). Cited in the `tfpt_5_redteam` body; ledger row `WITNESS.ECON.01`. Python-only.
- **`v306_seed_crossval.py` (`SEED.CROSSVAL.01`, [E]).** A leave-one-out cross-validation turning the "many observables, one seed φ_0" worry into a falsifiable out-of-sample test: back-solving φ_0 from each of the five seed-grammar observables ($\sin^2\theta_{12}$, $\sin^2\theta_{13}$, β, Ω_b, λ_C; NuFIT 6.0 / ACT DR6 / Planck / PDG) and error-weighted-averaging reproduces the axiom seed $\varphi_0 = 1/(6\pi) + 48c_3^4 = 0.0531720$ to 0.04%; leave-one-out predicts every held-out observable within 3σ out of sample (most-tensioned = $\sin^2\theta_{13}$, $\sim 2\sigma$); a data↔formula shuffle explodes the χ^2 by $> 100\times$. Cited in `tfpt_5_redteam`; ledger row `SEED.CROSSVAL.01`. Python-only.
- **`v307_data_watchdog.py` (`DATA.WATCHDOG.01`, [E]).** The operational decision pipeline over the frozen registry (`v84`): it classifies all twenty registry entries PASS/WATCH/TENSION/KILL by live n_σ against the repo-documented current bests (CODATA 2022, NuFIT 6.0, ACT DR6, Planck 2018, PDG 2024, BK18). No decidable prediction is in KILL; the watch items are α^{-1} (1.9σ), s_{23} (1.8σ) and the most-tensioned core prediction $\sin^2\theta_{13}$ (2.0σ, the honest current pressure point); $\sin^2\theta_{12} = 0.306747$ is compatible (0.02σ, the JUNO falsifier), the $r \in [0.0033, 0.0048]$ band is below the BK18 limit and the kill, and w=-1 is flagged as the DESI dark-energy watchdog. Cited in `origin_theory`; ledger row `DATA.WATCHDOG.01`. Python-only.

- **Target-A holomorphy kill test (`freeze_file.csv`).** A new `seam_holomorphy` freeze row records the sharpened *physical* kill criterion for Target A: a raw quasi-free seam bulk showing topological ground-state degeneracy on the torus ($|\det K| > 1$, the same-c rival $SO(16)_1$) would break holomorphy and kill "seam net $= (E_8)_1$" — since holomorphic $c{=}8 \Leftrightarrow E_8$ is machine-proven (`v83`/`v143`) and $\det K{=}1 \Leftrightarrow$ invertible/SRE (`v237`/`v301`/`v302`).

2026-06-19 (an infinite-derivative narrowing of the Stelle-ghost attack, the double-pushout figure, and the "constancy of c" framing of `SEAM.EQUIV.01`: `v304`)

- **`v304_idg_nonlocal_ghost.py` (`QGAMB.IDG.01`, [C]/[O]).** A conditional, parameter-free narrowing of the red-team `rt_F`/v278 Stelle-ghost attack on the perturbative gravity sector. [E] the local $R{+}R^2/\mathrm{Weyl}^2$ four-derivative propagator $1/(p^2(p^2{+}M^2))$ carries a negative-residue massive spin-2 mode (the Stelle ghost) only as a *truncation artefact*; [E] for an *entire* graviton form factor $a(p^2) = e^{p^2/M^2}$ (nowhere zero) the dressed propagator $1/(p^2 a)$ has no extra pole (Tomboulis 1997; Biswas–Mazumdar–Siegel 2006), whereas any *polynomial* truncation has a real zero = the ghost; [C] the spectral-action cutoff is the seam KMS weight $f(u) = e^{-u}$ (v259), so the un-truncated trace points to a nonlocal form factor with *no new parameter*. [O] It does *not* close `QG.AMB.01`: the trace regulator is not yet shown to resum to an entire $a(\Box)$, and a ghost-free *perturbative* graviton is not the nonperturbative ambient measure. Cited in the red-team Target F body; ledger row `QGAMB.IDG.01`.
- **Double-pushout figure (`tfpt_1_architecture_e8`).** A new shared TikZ macro (`TFPTdoublepushout` in `tex-artefacts/tfpt_figures.tex`) draws the commuting square showing that the order-$|\mu_4|{=}4$ glue is *one* operation in two categories: the lattice pushout $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ (v1) and the conformal-net / anyon-condensation extension $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1$ (v92/v281/v234/v277/v260). Expository only; no new claim.
- **Framing (`introduction`, `tfpt_research_contracts`).** `SEAM.EQUIV.01` is now stated explicitly as the *one* irreducible structural postulate of the theory — the role the constancy of c plays in relativity (cf. v267). Prose sharpening; no status change.

2026-06-19 (the discrete→dynamic lens on `F_transfer`: one shape, one seam-rate, `v303`)

- **`v303_ftransfer_dynamics.py`** asks whether the main-branch mechanism — a gapped, positivity-preserving relaxation to a *unique* attractor (the local-averaging update → the E_8 marks, gap $6\ln(3/2)$) — is also the shape of the four `F_transfer` instances, or an unrelated bolt-on. It is the *same* shape: [E] F_pole (Koide) *is* the seam dynamics (Möbius multiplier $F'{=}(2/3)^6{=}\lambda_2$, the seam-clock eigenvalue ⇒ a unique attractor at the seam rate, v82/v54); [C] F_Boltzmann (η_B) is a gapped positive contraction (washout $\kappa_f \in (0,1)$, H-theorem); [C] F_relic (axion) freezes at an adiabatic fixed point; [E] F_QCD (m_p/m_e) is the 1-loop RG flow with the carrier $b_3{=}-7$ (asymptotic freedom, Gaussian UV attractor). [O] Verdict: one shape underlies all four; F_pole has the seam rate $(2/3)^6$ *exactly*, the other three share only the shape with external rates (thermal/cosmological/RG). So `F_transfer` is the readout end of the one discrete→dynamic principle, *not* a bolt-on; the theory's simplicity is preserved because the external rates are honestly fenced (v187), not reduced to the seam. [E] suite `v303` 5/5, `AUDIT OK`.

2026-06-19 (the last lever: the seam mass gap *is* the derived Recovery gap $6\ln(3/2)$, `v302`)

- **`v302_seam_gap.py`** closes the chain on the single statement v301 had left: "the quasi-free seam bulk is gapped". That gap is *not* a free input — it is the established *Recovery gap* of the seam transfer operator, derived from the carrier. [E] the frozen transfer spectrum is spec(T) $= \{1, (2/3)^6, (1/3)^6\}$ (v160/v162), a simple Perron eigenvalue 1 separated from the rest; **[E]** the gap is $\Delta = -\ln((2/3)^6) = 6\ln(3/2) = 2N_{\mathrm{fam}}\ln(3/2) \approx 2.4328 > 0$ (multiplicative gap $1-(2/3)^6 = 665/729$; $(2/3)^6$ the μ_4-deck transfer eigenvalue forced by the carrier); **[E]** with the decoupling margin $\Delta - 31/(4\pi^2) \approx 1.648 > 0$ (v76) it is robust, and Perron–Frobenius primitivity makes the leading mode simple; **[C]** by Osterwalder–Schrader / quasi-free clustering a transfer-operator gap $\Delta > 0$ *is* a mass gap of the reconstructed Hamiltonian. **[O]** **Bottom line:** with v300/v301, `SEAM.EQUIV.01`'s entire residual is now a composition of *standard cited theorems* (OS/quasi-free clustering + Kitaev free-fermion invertibility + the v297 AQFT stack) over *established* TFPT facts (16 quasi-free Majoranas; transfer gap $6\ln(3/2)$) — *no undischarged TFPT-internal assumption remains.* It stays **[O]** (not machine-proved end-to-end), but the open front is reduced to citable machinery over a spectral gap that is provably positive. **[E]** suite `v302` 5/5, `AUDIT OK`.

2026-06-19 (Route A's last hypothesis discharged: the gapped quasi-free bulk is invertible by the free-fermion classification, `v301`)

- **`v301_route_a_invertible.py`** attacks the single remaining *unconditional* gap of `SEAM.EQUIV.01`. Since v300 collapsed Route B into Route A's rationality, that gap is Route A's open hypothesis (v297 LIT-A): "the raw quasi-free seam bulk is invertible (SRE)". A *gapped* free-fermion (Gaussian) 2+1D bulk cannot carry intrinsic topological order — free fermions give the integer (IQHE/Chern) phases, all *invertible*; intrinsic topological order needs interactions (Kitaev's periodic table, class D). **[E]** the carrier is 16 Majoranas, $c = g_{\mathrm{car}} + N_{\mathrm{fam}} = 8 = 16/2$ (v148); **[E]** $SO(16)_1 \supset (E_8)_1$ ($\dim \mathfrak{so}(16) = 120$ currents $+\,128$ spinor $= 248$, v154); **[C]** so a gapped quasi-free bulk is invertible automatically; **[E]** the invariant $|\det K| = \#$anyons is 1 for E_8 (invertible) vs 4 for the gauged $D_8{=}SO(16)$ (anyons), which free fermions cannot produce. **[O]** LIT-A's 'invertible bulk' now *follows* from 'gapped 16-Majorana quasi-free bulk' (v148/v160+the mass gap), so the residual changes from a topological-order statement to a spectral one. **[O]** **Bottom line:** with v300, the whole open content of `SEAM.EQUIV.01` reduces to the *single spectral statement* "the quasi-free seam bulk is gapped" — the topological-order obstruction (hidden anyons) is removed; the gap is not manufactured. **[E]** suite `v301` 6/6, `AUDIT OK`.

2026-06-19 (the Flat-Away attack: a hard discrete obstruction + the pin derived from $(E_8)_1$, `v300`)

- **`v300_flataway_rigid.py`** attacks the single external fact the three Flat-Away routes (v291/v293/v295) left open, on two fronts. *Harden:* the flat seam was only a *soft* minimum ("the spectral mismatch grows with ε"); here the obstruction becomes a *discrete* on/off invariant. **[E]** the flat fingerprint — spec($|D_\theta|$) is the integer ladder with every level $n \geq 1$ of multiplicity exactly 2; **[E]** the resonant mode $4k$ splits level $2k$ by gap $= g_k$ (sympy 2×2 $[[2k, g/2], [g/2, 2k]] \Rightarrow 2k \pm g/2$, + an $O(g^2)$ common shift), validated against the full Toeplitz operator; **[E]** any $g_k \neq 0$ lifts the resonant degeneracy ($2 \to 1{+}1$) and changes the spectral *multiset*, so preserving the flat ladder *with* its mult-2 degeneracies forces every $g_k = 0 \Rightarrow f = 0$ — strictly stronger than the "deviation grows" scan; **[E]** with the v295 convexity, flat is isolated analytically *and* discretely. *Derive the pin:* **[C]** the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + \dots)$ has integer q-coefficients (248 at level 1; E_8 even unimodular $\Rightarrow$ integer weights), so by 2d Steklov conformal rigidity the integer ladder $\Leftrightarrow$ flat seam, i.e. the external pin *is* the $(E_8)_1$ rationality (Route A). **[O]** verdict: Flat-Away carries *zero* new open content beyond Route A — the two `SEAM.EQUIV.01` routes converge on one rationality fact. **[E]** suite `v300` 6/6, `AUDIT OK`.

2026-06-19 (the hypergraph/Wolfram bridge hardened: the $(2, 3, r)$ seed reaches E_8 as the last finite point, v299)

- **`v299_hypergraph_0d_autonomous.py`** hardens the Wolfram/hypergraph reading into a precisely typed bridge (builds on v219/v298): **[E]** the icosahedron is a 3-uniform hypergraph ($|\mathrm{Aut(graph)}| = 120 = |I_h| = |A_5 \times \mathbb{Z}_2|$; the McKay group is the $SU(2)$ binary lift $2I$, a *different* order-120 group whose McKay graph is affine E_8; 30 edges $= h(E_8)$); **[E]** the $1 \to 4$ subdivision rewrite is equivariant (all 120 symmetries at every scale); **[E]** the $2I$ fusion tensor is a weighted 3-uniform hypergraph whose 2-dim slice is the affine E_8 McKay graph; **[E]** Smith ($\rho \le 2 \Leftrightarrow$ affine ADE) + Collatz–Wielandt ($\rho \le 2 \Leftrightarrow$ a local witness $Am \le 2m$); **[E]*** the autonomous rule (diffusion-generated witness + *local* gate $Am \le 2m$, Perron only as audit) runs $E_6 \to E_7 \to E_8 \to \hat{E}_8$; **[E]** the negative controls $\rho(A_n) = 2\cos\frac{\pi}{n+1}$ and $\rho(D_n) = 2\cos\frac{\pi}{2n-2}$ are < 2 for *all* n (closed form), so neither family ever reaches E_8. Honest type: **[O]** the $(2, 3, \cdot)$ seed is the single irreducible input = P2 — *the rule reaches E_8 as the last finite point on this seed; it is not a derivation of P2.* **[E]** build green, `AUDIT OK`.

2026-06-18 (E8 as a network attractor — the Wolfram-program reading, v298)

- **`v298_e8_network_attractor.py`** is an honest audit of the "TFPT as a hypergraph/rewrite system" idea, on the v219 McKay bedrock. **[E]** the affine-E_8 Kac marks $(1, 2, 3, 4, 5, 6, 4, 2, 3)$ are the *unique dynamical attractor* of a minimal local-averaging update $m \leftarrow (A+2I)/4\, m$ on the $(2, 3, 5)$ McKay graph (converges from any positive start; $A\, m = 2m$, top eigenvalue 2); the spherical tower terminates at E_8 and the cascade is nested-subgraph coarse-graining. **[C]** the Wolfram reading "E_8 is the universal attractor of a minimal $(2, 3, 5)$ network" holds at the Coxeter-skeleton level. **[O]** a minimal hypergraph *rewrite* reproducing the *full* structure stays open; honest negatives: φ_0^{ret} is the analytic seed $\frac{4}{3}c_3 = 1/(6\pi)$, not an edge fraction, and recovery is a 1-D Möbius fixed point, not a rewrite. **[E]** build green, `AUDIT OK`.

2026-06-18 (a_2 closed form + Route A literature stack, v296–v297)

- **`v296_flataway_a2_closed.py`** (**`FLATAWAY.A2.CLOSED.01`**) evaluates the exact a_2 coefficient in *closed form*: the operator tails telescope geometrically ($C_k + D_k e^{-4kt} = 4kt(1 - e^{-4kt})$) to $W_k^{\mathrm{tail}} = t(1 - e^{-4kt})/(8k(1 - e^{-t}))$, leaving a finite $(4k-1)$-term middle; e.g. $W_1(t) = t(2te^t + e^{3t} + 3e^{2t} + e^t - 1)e^{-3t}/8$. Manifestly positive, small-t law $W_k = t/2 + O(t^3)$ (the t^2 term cancels). Reproduces v295 to machine precision.

- **`v297_route_a_literature_stack.py`** (**`SEAM.EQUIV.A.LIT.01`**) writes Route A's one import as a citable stack: LIT-A (Kitaev; Freed–Hopkins) → LIT-B (Müger; Kawahigashi–Longo–Müger) → LIT-C (Conway–Sloane; Dong–Xu) compose into "invertible bulk $\Rightarrow (E_8)_1$", all legs established; the single open hypothesis (seam-bulk invertibility) coincides with Route B's Flat-Away, and the stack is non-circular with Route B (the v286 firewall holds). **[E]** build green, `AUDIT OK`.

2026-06-18 (Flat-Away heat route made exact + Lean-formalised, v295 / `FORM.FLATAWAY.01`)

- **`v295_flataway_a2_exact.py`** (**`FLATAWAY.A2.01`**) upgrades the v292 positive-definiteness from a numerical scan to a *proof*: the first variation of $\mathrm{Tr}\, e^{-t\Lambda}$ vanishes (zero-mean off-mark curvature, Gauss-Bonnet), and since $\Lambda \mapsto \mathrm{Tr}\, e^{-t\Lambda}$ is convex (Klein/Peierls), ε=0 is the *global* minimum — so $\Delta\mathrm{Tr} \ge 0$ for all deformations, $= 0$ iff $f = 0$. The exact second-order coefficient is the divided-difference kernel of e^{-tx} (Duhamel), validated to rel. err $< 10^{-6}$.

- **Lean `FORM.FLATAWAY.01`** (`SeamDeckClosure.lean` Part 5): the formal Δ=0 $\Leftrightarrow$ flat step — a positive-weighted sum of squares is positive-definite (`heatDeviation_eq_zero_iff`); built and axiom-checked (only `propext`, `Classical.choice`, `Quot.sound`). So the heat route's positive-definiteness is now both analytically proved and machine-formalised; only the single external fact (the seam fixes a_2) remains. [E] build green, `AUDIT OK`.

2026-06-18 (Flat-Away attacked on all three routes, v292–v294)

- **`v292_flataway_heat_reduction.py` (`FLATAWAY.HEAT.01`)**: the heat route, made precise. The heat-trace deviation $\Delta\mathrm{Tr}(e^{-t\Lambda}) = c(t)\varepsilon^2 + O(\varepsilon^4)$ is a *positive-definite* quadratic form in the off-mark curvature ($c_k(t) > 0$ for every $\mathbb{Z}_4$ mode, positive on all combinations), leading invariant $\|f\|_{L^2}$ — so Flat-Away reduces to the single fact "the seam fixes the a_2 heat coefficient".

- **`v293_flataway_spectral_hessian.py` (`FLATAWAY.SPEC.01`)**: the spectral route, upgraded from one direction to the *full* smooth-$\mathbb{Z}_4$ space. Mode $4k$ splits the degenerate Steklov level $|2k|$ at first order; the Hessian of the spectral mismatch over modes $\{4, 8, 12\}$ is positive-definite (eigenvalues $\{0.497, 0.500, 0.503\}$), so the flat seam is a strict isolated minimum — the "only one direction" objection is answered.

- **`v294_flataway_rp_energy.py` (`FLATAWAY.ENERGY.01`)**: the variational route. With the rigid mark divisor (4 cone angles π, Gauss-Bonnet 4π), Troyanov uniqueness gives a unique flat cone metric, and the curvature energy $\int f^2$ is strictly convex (Hessian $2\pi I$) with a unique minimiser — Flat-Away reduces to "RP+gap realises the energy-minimiser". The three routes converge on one selection principle for the flat metric; closing any one closes Route B. [E] build green, `AUDIT OK`.

2026-06-18 (Route-B red-team: the smooth $\mathbb{Z}_4$ adversary + Flat-Away as a mini-theorem, v290–v291)

- **`v290_z4_smooth_curvature_adversary.py` (`SEAM.ADVERSARY.01`)** is the honest red-team that $\mathbb{Z}_4$ block-diagonality is *not* mark-locality: a smooth $\mathbb{Z}_4$ off-mark curvature $f = \varepsilon\cos(4\theta)$ [E] passes the v288 commutator ($\|[\rho, \Lambda]\| = 0$, same as mark-local) yet [E] shifts the DtN spectrum ($\max|\Delta\lambda| = 0.155$) and the heat trace ($\Delta = 0.019$). So the commutator is necessary, not sufficient; the modulus is excluded by the spectral data, not $[\rho, \Lambda]{=}0$ — which is exactly why Flat-Away cannot be read off the commutator.

- **`v291_flataway_contract.py` (`FLATAWAY.RP.01`)** states Flat-Away as its own named mini-theorem and types three independent proof routes: [E] spectral-rigidity and [E] heat-kernel (the flat seam is the *unique* spectral/heat match of the smooth $\mathbb{Z}_4$ family — deviation zero only at $\varepsilon{=}0$) and [C] RP-energy/Troyanov (constant-curvature minimiser). Closing any one closes Route B. [E] build green, `AUDIT OK`.

2026-06-18 (Route B sharpened: raw mark-locality decomposed + reviewer firewalls, v289 / v287–v288)

- **`v289_marklocal_raw.py` (`MARKLOCAL.RAW.01`)** decomposes the one remaining Route-B residual ('why is the raw seam sub-principal term mark-local?') into a 5-lemma chain and isolates the *single* open analytic lemma: [C] L1 conic parametrix (b-calculus/Melrose, imported); [O] L2 **Flat-Away** — RP+gap+4 marks ⇒ curvature vanishes away from the marks; [E] L3 orbit-source (v195/v216), L4 Fourier-support on $4\mathbb{Z}$ (v264/v288), L5 full-operator $[\rho, \Lambda]{=}0$ (v288). 4/5 discharged — Route B reduces to exactly L2.

- **v288 reviewer firewall**: a SCOPE FIREWALL check makes the exact-label split explicit (Fourier lemma Formal/Exact; numerical test Numerical Exact; transfer to the raw conic DtN Conditional, v264; mark-locality from the RP seam Open Selection, v289) — so it can never be read as 'full-L^2 QGEO proved'.

- **v287 literature matrix**: L3 ('invertible bulk ⇒ single-sector') is typed as a composition of citable results (LIT-A Kitaev/Freed–Hopkins, LIT-B Müger/Kawahigashi–Longo–Müger, LIT-C Kawahigashi–Longo) conditional on one open hypothesis — seam-bulk invertibility, which coincides with Route B. [E] build green, `AUDIT OK`.

2026-06-18 (closing sprint: the Seam Equivalence Theorem named + both routes attacked, v286–v288)

- **`v286_seam_equivalence_contract.py` (`SEAM.EQUIV.01`)** names the one open theorem — *the raw RP seam state is the holomorphic* $(E_8)_1$ *net at* $\tau{=}i$ — and concentrates the whole open structure on it: four objects, six arrows (2 closed, 1 conditional v282, 1 open Gral), an **import firewall** proving the two routes (v276/v280/v284 vs v277/v281/v285) never import each other (no 'E_8 into the geometry and back' circularity), and an exact-label taxonomy. Writes `seam_equivalence.json`.
- **`v287_free_rp_bulk_to_holomorphic_boundary.py` (`SEAM.EQUIV.A01`)**: Route A (AQFT) dependency checker — the 5-lemma chain RP+gap+chirality+Gaussian+invertible ⇒ $\mu{=}1$ ⇒ holomorphic ⇒ $(E_8)_1$ is logically complete *modulo one* standard theorem: *invertible (SRE) Gaussian bulk ⇒ single-sector boundary* (the AQFT form of $\det K{=}1$). 4/5 discharged.
- **`v288_full_l2_subprincipal_z4.py` (`SEAM.EQUIV.B01`)**: Route B (DtN) — *proves* the full-L^2 lift of the sub-principal $\mathbb{Z}_4$ block-diagonality (a mark-sourced curvature has Fourier support only on $m \equiv 0 \bmod 4$, so $\Lambda = |n| + M_f$ commutes with $\rho = \mathrm{diag}(i^n)$ on the full operator, $\|[\rho, \Lambda]\| \sim 2{\times}10^{-15}$; generic curvature → 4.44), lifting v201/v284 to L^2. The residual shrinks to one sharper question: why is the raw seam sub-principal term mark-local? [E] build green, `AUDIT OK`.

2026-06-18 (the two proof routes decomposed: Route (i) and Route (ii) into lemma chains, v284/v285)

- **`v284_route_i_rp_uniqueness.py` (`QGEO.ROUTEI.01`)** turns Route (i) (RP-state uniqueness) into a 6-lemma chain: [E] 5/6 lemmas are already discharged — L1 RP→DtN (v194), L2 marks→pillowcase (v195/v216/v214), L3 flat Steklov = $\sqrt{-\Delta_{\text{flat}}}$ (verified here), L4 covariance from DtN (v258), L5 μ_4-invariance (v276/Lean/v280); [C] Troyanov ($\chi_{\text{orb}}{=}0$ ⇒ unique flat $\tau{=}i$) reduces the residual to one lemma. [O] the one open lemma: *the raw seam is the constant-curvature geometric state.*
- **`v285_route_ii_seam_condensation.py` (`QGAMB.ROUTEII.01`)** turns Route (ii) (seam-net condensation) into a 4-lemma chain: [E] 3/4 discharged (no abelian sector ⇔ $\det K{=}1$ ⇔ holomorphic ⇔ E_8; the tower 16→4→1), and **its open lemma coincides with Route (i)'s** — both are "the raw seam is the $(E_8)_1$-at-$\tau{=}i$ object" (v282), so proving *either* route closes *both*. [O] the unified open lemma: a canonical equivalence between the raw seam KMS/DtN state and the holomorphic $(E_8)_1$ net at $\tau{=}i$. Build green, `AUDIT OK`.

2026-06-18 (self-investigation round: pillowcase DtN, modular data, the E8-at-$\tau{=}i$ unification, live scorecard, v280–v283)

- **`v280_pillowcase_steklov.py` (`QGEO.STEKLOV.01`)** runs the QGEO chain on the actual flat $\tau{=}i$ pillowcase orbifold: the order-4 deck permutes the 4 cone points (2 fixed + 1 swap) and the geometric chain $[\rho, \Delta]{=}0 \Rightarrow [\rho, H]{=}0 \Rightarrow [\rho, C]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ is numerically exact — route-(i) evidence for `QGEO.OBLIG.01`; the premise stays [O].
- **`v281_holomorphy_modular_data.py` (`QGAMB.MODULAR.01`)**: #anyons = $|\det \mathrm{Gram}|$ (E8→1, SO(16)→4), the anyon-condensation tower $16 \to 4 \to 1$, Gauss–Milgram → $c{=}8$; holomorphy ($\det K{=}1$) is the single Tier-B discriminator.

- **`v282_e8_tau_i_unification.py` (`QGAMB.UNIFY.01`)**: the candidate *simplification* — $\chi_{E_8}{=}j^{1/3}$ gives $\chi_{E_8}(i){=}1728^{1/3}{=}12$, so $\tau{=}i$ is both the order-4 CM point (QGEO) and the $(E_8)_1$ modulus (QG.AMB): the two obligations are two faces of one object (the seam is $(E_8)_1$ at $\tau{=}i$), **obligation count** $2 \to 1$.
- **`v283_live_killtest_scorecard.py` (`PRED.LIVE.01`)**: the recent round's computable predictions vs current data — all consistent $< 2\sigma$, with two sharp near-term kill tests (Σm_ν floor 0.0586 eV; proton decay $1.55{\times}10^{35}$ yr).
- Wolfram 275/275 (v281/v282 exact-mirrored); build green, `AUDIT OK`.

2026-06-18 (the Gräl: QGEO.SYM.01 as one precise lemma + the all-orders step Lean-formalised, v279/`FORM.QGEO.03`)

- **`v279_qgeo_obligation_lemma.py` (`QGEO.OBLIG.01`) writes the bedrock as ONE precise constructive-QFT lemma.** Given the premise *the raw seam carries the flat $\tau{=}i$ pillowcase metric* (Troyanov-unique, $\Lambda_\Sigma{=}\sqrt{-\Delta_{\mathrm{flat}}}$), then $\omega_\Sigma \circ \rho = \omega_\Sigma \to$ mark-locality $\to E_8$. [E] the reduction DAG from 'free RP seam' has exactly *one* open leaf — 'the raw seam state is the flat state'; [E] the geometric side is fixed (flat DtN diagonal, ρ commutes to all orders), so the obligation is a state-identification, with [C] two proof routes (RP-state uniqueness, or Troyanov + OS).
- **`FORM.QGEO.03`: the all-orders step is now machine-proved in Lean.** `SeamDeckClosure.lean` gains `diagonal_commutesClock` and `flat_all_orders_clock`: any spectral function $g(\Delta_{\mathrm{flat}})$ of the Fourier-diagonal flat Laplacian commutes with the carrier clock $\rho{=}\mathrm{diag}(i^n)$ — so the v276 all-orders closure is a Lean theorem (only `propext`, `Classical.choice`, `Quot.sound`; no `sorry`; full `lake build` green, 3356 jobs). The premise (the seam *is* flat) stays the [O] `QGEO.SYM.01` input. Build green, `AUDIT OK`.

2026-06-18 (4D QFT: the $S_{\mathrm{pert}} \to S_{\mathrm{phys}}$ LSZ bridge + one-loop unitarity, v278)

- **`v278_lsz_bridge_unitarity.py` (`QFT4D.SPERT.04`) connects the perturbative S-matrix to the physical asymptotic one and verifies one-loop unitarity.** [E] LSZ bridge: the EG T-products of S_{pert} (v269) *are* the OS Wick-rotated correlators (v240), so amputate+on-shell-residue gives $S_{\mathrm{pert}} \xrightarrow{\mathrm{LSZ}} S_{\mathrm{phys}}$; [C] the gap $\Delta{=}6\log(3/2){>}0$ gives Haag–Ruelle asymptotic states. [E] *one-loop optical theorem*: the bubble discontinuity $\mathrm{Im}\,B(s)/\pi = x_+{-}x_- = \sqrt{1-4m^2/s}$ *exactly* = the two-body phase space, so $2\,\mathrm{Im}\,M = \sum\int d\Pi|M|^2$ (cutting rule) holds — the matter+gauge sector is unitary, with the cut turning on at the physical two-particle threshold. [O] the gravity sector's Stelle ghost (v269/`rt_F`) is non-unitary $\to$ `QG.AMB.01`. Exact core mirrored in Wolfram (273/273). Build green, `AUDIT OK`.

2026-06-18 (QG.AMB.01 Tier-B made concrete: the seam-Calderón $\to (E_8)_1$ match, v277)

- **`v277_seam_calderon_e8_match.py` (`QGAMB.TIERB.01`) reduces Tier-B to ONE bit.** [E] the full $(E_8)_1$ matching target is computed and matches: $c{=}8$, $\det\mathrm{Cartan}(E_8){=}1$, the holomorphic character $E_4/\eta^8{=}j^{1/3}{=}q^{-1/3}(1{+}248q{+}\dots)$ with a single primary and exactly $248{=}\dim E_8$ weight-1 currents (240 roots). [E] the discriminator is holomorphy — the $c{=}8$ counterexample $(D_8){=}SO(16)_1$ has $\det\mathrm{Cartan}{=}4$ (non-holomorphic), so the one condition is $|\det K|{=}1$. [C] the seam side matches via v276 (flat $\tau{=}i$) + v260 (Kummer/K3). [O] residual = the single holomorphy bit; reduced, not closed. Exact core mirrored in Wolfram (272/272). Build green, `AUDIT OK`.

2026-06-18 (QGEO.SYM.01 narrowed: the flat pillowcase closes the commutator to all orders, v276)

- **`v276_qgeo_flat_closes_commutator.py` (`QGEO.SYM.03`) collapses the bedrock to ONE geometric postulate.** The state-level residual the whole bedrock rests on is the operator equation $[\rho, H]$=0. [E] if $H = f(\Delta_{\text{flat}})$ and the order-4 deck ρ ($z\mapsto iz$) is a flat-τ=i isometry, then $[\rho, \Delta]$=0 ⇒ $[\rho, H]$=0 *exactly*, to all orders (verified on the discretised square torus: ρ^4=I, $[\rho, \Delta]$=0, $[\rho, \sqrt{-\Delta}]$=0; generic non-isometry fails) — upgrading the leading-order v198/v201 to the full H. [C] τ=i is forced by order-4 (j=1728; hexagonal j=0 is order-6, excluded). [O] `QGEO.SYM.01` therefore collapses to the single statement "the raw seam IS the flat τ=i pillowcase quasi-free state" — one human constructive-QFT proof, or an honest axiom. Narrows, does not close. Build green, `AUDIT OK`.

2026-06-18 (Red Team Target F — adversarial audit of the v269–v275 round, rt_F)

- **`redteam/rt_F_qft4d.py` (`REDTEAM.F.01`) stress-tests the just-published round** (the perturbative S_{pert} layer, the PMNS Jarlskog, the ν-scale, the over-determination, the `QG.AMB.01` roadmap). Survives (narrowed): two attacks land and are folded in. **(1)** the gravity R^2/Weyl^2 sub-sector is power-counting renormalizable but *non-unitary* (the Stelle ghost: the four-derivative propagator $1/(p^2(p^2 + M^2))$ has a negative-residue massive spin-2 mode), so S_{pert} is unitary for the matter+gauge sector only — which is exactly why the nonperturbative `QG.AMB.01` is the real frontier (caveat added to v269). **(2)** the 0.11% anchor over-determination is conditional on the Λ-branch readout (`LAMBDA.BRANCH.01`, [C]), the gravity route being unconditional (caveat added to v274). Added as Target F in `tfpt_5_redteam` + the red-team table; `run_redteam.py` green.

2026-06-18 (Zenodo release 5.2 (rev 191) — the perturbative 4D-QFT layer + scale closure, v269–v275)

- **Published to Zenodo.** Version `5.2 (rev 191)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20743347`, DOI `10.5281/zenodo.20743347`. This release carries the perturbative 4D-QFT layer (the Epstein–Glaser S_{pert} skeleton v269, the concrete one-loop quartic v271 and gauge β-coefficients v273), the complete complex PMNS matrix and leptonic CP strength v270, the honest absolute ν-mass-scale typing v272, the over-determination of the single mass anchor v274, and the `QG.AMB.01` obligation roadmap v275. Suite `273` modules (highest ID `v275`; v186/v226 skipped), Wolfram `271/271`, `run_all.py` green, `AUDIT OK`, `npm` build clean. The Zenodo description (`zenodo_description.html`) and the public website were synced in the same release.

2026-06-18 (scale + frontier round: ν-mass scale, EG gauge running, over-determination, QG roadmap, v272–v275)

- **`v272_nu_mass_scale.py` (`FLAV.NUSCALE.01`) types the absolute neutrino-mass scale honestly** (no fabricated Σm_ν — would contradict the v184 firewall). [E] the absolute scale reduces to *one* seesaw ratio $m_3 = (y_\nu v_{\text{EW}})^2/M_R$ (one UV input, like v_{geo}, v153); [C] the observed m_3=0.050 eV with the TFPT PS→GUT window $M_R \in [4.2{\times}10^{13}, 2.4{\times}10^{15}]$ GeV (v249) needs $y_\nu \in [0.26, 2.0]$ — $O(1)$, no tuning; [O] M_R is not a clean $\bar{M}_{\text{Pl}}$ power, so the absolute value stays open; [X] the NO floor Σm_ν=0.0586 eV is the cosmological kill test. Closes the PMNS bookkeeping (the last [O] is one seesaw ratio).
- **`v273_eg_gauge_running.py` (`QFT4D.SPERT.03`) derives the SM one-loop b_i via the EG gauge self-energy.** [E] the gauge 2-point is the same scaling-degree-4 extension as the quartic (v271), giving $(b_1, b_2, b_3) = (41/10, -19/6, -7)$ from the carrier/SM content (the same

41 as the carrier algebra, v159); [O] two-loop + all-order stay open. Exact b_i mirrored in Wolfram.

- **`v274_scale_overdetermination.py` (`SCALE.OVERDET.01`) shows the single mass anchor is over-determined.** [E]/[C] two independent routes to $\bar{M}_{\rm Pl}$ agree — gravity $(8\pi G)^{-1/2} = 2.4353\times10^{18}$ GeV and cosmology (the compiler prediction $\rho_\Lambda/\bar{M}_{\rm Pl}^4 = (3/4\pi^2)e^{-2\alpha^{-1}}$ with $\rho_\Lambda^{1/4} = 2.24$ meV) $= 2.438\times10^{18}$ GeV, to 0.11%; the seam being a conformal c=8 CFT makes "no derivable absolute scale" a theorem, not a gap. Sharpens v153/v78.
- **`v275_qgamb_roadmap.py` (`QGAMB.ROADMAP.01`) gives the open gate `QG.AMB.01` a module.** [E] Tier A gap decoupling ($\Delta_{\rm eff} = 1.648 > 0$, v36/v76); [E]/[C] Tier B reduction to the holomorphic $(E_8)_1$ boundary net (v77); [C] the perturbative layer in hand (v269/v271/v273); [O] the nonperturbative ambient measure stays the one structural frontier.
- Suite `271/271` Wolfram, `run_all` green, `AUDIT OK`, build green.

2026-06-18 (4D QFT: a concrete Epstein–Glaser one-loop quartic renormalization, v271)

- **`v271_eg_oneloop_quartic.py` (`QFT4D.SPERT.02`) takes the v269 $S_{\rm pert}$ skeleton from "exists" to an actual EG number** (no path integral). [E] the first renormalization is the order-2 product T_2; the massless propagator has scaling degree sd=2 in d=4, the one-loop bubble has sd=4=d, so ω=sd$-d$=0 $\Rightarrow$ exactly one logarithmic local counterterm ($C(\omega{+}d, d)$=1, finite, no infinite tower); the loop factor $\Omega_3/(2(2\pi)^4) = 1/(16\pi^2)$ exactly — the *same* factor that normalises the spectral-action a_4 geometric quartic. [C] the ϕ^4 one-loop $\beta_\lambda = 3\lambda^2/(16\pi^2)$ (symmetry $3 = s, t, u$ channels), EG initial condition $\lambda_\Phi(\mathrm{UV}) = 1/(16\pi^2)$. [O] order-2/one-loop only; the all-order EG construction and the nonperturbative measure `QG.AMB.01` stay open. Exact core mirrored in Wolfram (270/270). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the complete complex matrix + the leptonic Jarlskog CP strength, v270)

- **`v270_pmns_jarlskog_assembly.py` (`FLAV.PMNS.03`) assembles the four TFPT channels** ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\sin^2\theta_{23}{=}\frac{1}{2}$, $\sin^2\theta_{13}{=}\varphi_0^{\rm ret}e^{-5/6}$, $\delta{=}4\pi/3$) **into one exactly-unitary complex PMNS matrix and derives the leptonic CP strength.** [E] $U = R_{23}U_{13}(\delta)R_{12}$ is unitary; [C] the Jarlskog $J_{\rm PMNS} = \Im(U_{e1}U_{\mu2}U_{e2}^*U_{\mu1}^*) = -0.02965$ ($J_{\max} = 0.03424$) is a *derived* number, the leptonic analogue of the CKM J (v88); [C] data: $J_{\max}$ vs NuFIT 5.2 0.0332 ($\sim 3\%$), J vs best fit -0.026, the negative CP-violating sign reproduced. [O] δ is the μ_6/triality phase (assembled, not from M_ν/D_F) and the absolute ν-mass scale stays open — closes the CP-*strength* readout, not the phase origin nor the scale. [E] build green, `AUDIT OK`.

2026-06-18 (4D QFT: the perturbative S-matrix as an Epstein–Glaser / pAQFT skeleton, v269)

- **`v269_spert_paqft_skeleton.py` (`QFT4D.SPERT.01`) builds the honest middle layer of the three-layer S-matrix** (the first concrete 4D-QFT step beyond the spectral-action contract), *not* a nonperturbative closure. [E] three distinct layers: $S_{\rm top}$ (2d DHR braiding $|M|$=1, statistics, v243), $S_{\rm pert}$ (4d Epstein–Glaser/BV S-matrix of the spectral action), $S_{\rm phys}$ (LSZ on the OS Wightman functions, v240). [E] the spectral-action a_4 interaction is power-counting renormalizable (all operators dim ≤ 4; 7 marginal + 3 relevant, a finite counterterm basis), so [C] by the Epstein–Glaser theorem the perturbative S-matrix exists order by order (causal factorisation + finite local extension; no path integral, no UV divergences). [C] the gap $\Delta = 6\log(3/2)$ gives an IR-safe adiabatic limit ($S_{\rm pert} \to S_{\rm phys}$ via LSZ). [O] perturbative only — the ambient measure `QG.AMB.01` stays open, the canonical 4d reading remains boundary-only (v265). [E] build green, `AUDIT OK`.

2026-06-18 (PMNS: the reactor-angle exponent closed as the carrier hypercharge trace, v268)

- **`v268_theta13_carrier_trace.py` (`FLAV.TH13.01`) closes the θ_{13} exponent origin.** The value $\sin^2\theta_{13} = \varphi_0^{\rm ret} e^{-5/6} = 0.0231$ was already checked (v16); now the exponent is exact: [E] $\gamma = 5/6 = \mathrm{tr}_E Y^2$, the carrier hypercharge-squared trace ($Y = \mathrm{diag}(-\frac{1}{3}, -\frac{1}{3}, -\frac{1}{3}, \frac{1}{2}, \frac{1}{2})$, roots of $6Y^2 - Y - 1 = 0$; complement $\frac{1}{6}$ the neutrino ratio). [E] own channel: the $\mu\tau$-breaking $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret}$ induces only $\sim 1.6 \times 10^{-3} \ll 0.023$, so θ_{13} is a separate carrier-trace channel — correcting the tempting (and wrong) "θ_{13} from M_ν" route flagged by the new script atlas. [O] the CP magnitude and the absolute ν-mass scale remain the open PMNS items. Exact core mirrored in Wolfram. [E] build green, `AUDIT OK`.

2026-06-18 (Infrastructure: a generated causal "script atlas" so the suite is navigable at a glance)

- **`verification/make_script_atlas.py` generates `verification/script_atlas.md`** — a single grouped, dependency-aware view fusing the existing single sources (`status_ledger.csv` for claim→status→dependencies→supersedes, `script_registry.csv`/`script_clusters.csv` for the theme grouping, `docs_map.csv` for where each script is cited). It lists every registered script by cluster with its claim(s), four-class marker, one-liner, resolved dependencies and citing papers, plus a *supersede map* (which old claim each new one replaces) and a dependency/most-depended-on overview — the antidote to duplicating work or forgetting superseded results. No new data, a re-projection only; wired into `bash build.sh gen`. [E] build green, `AUDIT OK`.

2026-06-18 (The bedrock in its minimal-axiom form: a rigidity theorem for `QGEO.SYM.01`, v267)

- **`v267_qgeo_rigidity_minimal_axiom.py` (`QGEO.SYM.02`) sharpens the one bedrock postulate to its irreducible kernel** (it does *not* close it). [E] the *single* bare statement "the seam admits the carrier order-4 clock as a conformal symmetry permuting its four marks" ($= \omega \circ \rho = \omega$) forces *everything*: an order-4 Möbius orbit $\{a, ia, -a, -ia\}$ has cross-ratio 2 independent of a; cross-ratio 2 $\Leftrightarrow$ j=1728 (only $\lambda \in \{-1, \frac{1}{2}, 2\}$) $\Leftrightarrow$ the order-4 CM automorphism on y^2=x^3−x; and the unique flat pillowcase metric, mark-locality, $[\rho, \Lambda]$=0 and $\omega \circ \rho$=ω all follow (v214/v201/v264/v198). [E] neg controls: generic config $j \neq 1728$ ($\mathbb{Z}_2$ only), hexagonal j=0 ($\mathbb{Z}_6$, order 6) — only the square (order-4) realises the μ_4 clock. [C] second, independent reason: $\{1, i, -1, -i\}$ over $\mathbb{Q}$ with j=1728 (CM by $\mathbb{Z}[i]$), Belyi/dessins Galois-rigidity. [O] the bare order-4 symmetry stays the one foundational postulate (like c=const) — the *reduction* is closed (axiom ⇒ everything), the axiom is not. [E] build green, `AUDIT OK`.

2026-06-18 (Branch B decided: the explicit PS threshold model + proton-decay window — minimal content killed, $+(15, 1, 1)$ survives, Hyper-K decisive, v266)

- **`v266_ps_threshold_proton.py` (`PS.PROTON.02`/`PS.THRESHOLD.02`) decides branch B of the v265 fork.** The explicit two-step Pati–Salam thresholds give d=6 $\tau_p(p \to e^+\pi^0)$ (writes `ps_threshold_proton.json`): [X] the minimal 16-Higgs content (M_{GUT}∼2.4×10^{15}) gives τ_p∼4.4×10^{33} yr < Super-K and is *killed* even at the high hadronic band; [C] the E_8-allowed $(15, 1, 1)$ [the single 45] raises M_{GUT}∼5.9×10^{15} ⇒ τ_p∼1.6×10^{35} yr, proton-safe *now*; [X] but that sits at the Hyper-K reach (∼1.4×10^{35} yr) — a *dated* kill test decisive within the decade. So branch B is promoted from "well-mannered candidate" to *content-selected (must be* $+(15, 1, 1)$*), Hyper-K-decisive UV branch*; [O] structurally marginal (only one 45 in the hull, v247) with an $O(3)$ τ_p band. Branch A (boundary-only) is unchanged and canonical. [E] build green, `AUDIT OK`.

2026-06-18 (The 4D-QFT fork frozen as a branch policy: boundary-only canonical, Pati–Salam a falsifiable UV branch, SM-only GUT killed, v265)

- **`v265_qft4d_fork_freeze.py` (`QFT4D.FORK.01`) turns the 4D-QFT fork from an open ambiguity into a frozen decision tree** (`qft4d_fork_freeze.json`), no new physics. **A (canonical):** boundary-only is the default 4D reading — a 4D-GUT is *not claimed by default* (E_8 is the audit hull). [X] SM-only 4D-GUT unification is killed (1-loop: at the $\alpha_1{=}\alpha_2$ scale α_3^{-1} off by $\sim$5.6, spread $\sim$8.8 at $2{\times}10^{16}$ GeV, re-deriving v246) — the canonical expectation, not a failure. **B (conditional):** carrier-native Pati–Salam is a falsifiable UV branch only with the E_8-allowed content $\{1, 10, 16, 45\}$ (no **126**, v247), $\sin^2\theta_W{=}3/8$ (v248), the $O(1)$ $\kappa{=}M_{PS}/M_s$ band (v249/v253), a threshold model and a proton-decay kill test ([X], no fake τ_p window; carrier gauging stays the contract). [E] a prose guard (`QFT4D.NO_OVERCLAIM.01`) forbids bare 4D over-claims and requires the scoped policy wording. New ledger rows `QFT4D.BOUNDARY.DEFAULT.01`, `QFT4D.SMONLYGUT.01`, `PS.UVBRANCH.01`, `PS.THRESHOLD.01`, `PS.PROTON.KILL.01`. A fork-policy box added to `tfpt_research_contracts`; [E] build green, `AUDIT OK`.

2026-06-18 (Three open-residual advances: the F_{QCD} solver, M_ν from D_F, and the raw-seam⇒mark-locality reduction, v262–v264)

- **`v262_fqcd_mp_me.py` (`FR.MPME.02`) implements the one contract-only F_{transfer} pipeline.** A real two-loop α_s solver runs $\alpha_s(M_Z){=}0.1179$ with n_f thresholds (carrier $b_3{=}-7$, [E]) to $\alpha_s(2\,\mathrm{GeV}){\sim}0.30$ and $\Lambda^{(3)}_{\overline{\mathrm{MS}}}{\sim}0.33\,\mathrm{GeV}$; with the lattice bridge C_p (external $O(1)$) and the closed m_e it reproduces $m_p/m_e{\sim}1824$ vs the measured 1836.15 within the $\sim 7\%$ external budget. A [C] transfer reproduction (firewall: no compiler integer for 1836), advancing `FR.MPME.01` ([O]) to `FR.MPME.02` ([C]).
- **`v263_mnu_from_df.py` (`FLAV.PMNS.02`) makes M_ν a seesaw operator of D_F.** [E] $M_\nu = -m_D M_R^{-1} m_D^\top$ is symmetric, seesaw-suppressed and (for $\mu\tau$-symmetric inputs) $\mu\tau$-symmetric; [C] it reproduces $\theta_{23}{=}45°$ and $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\mathrm{ret}}}{2}$ (v9) under the seam misalignment; [O] $\theta_{13}{=}0$ in the leading texture, the CP *magnitude* (240°) is the μ_6/triality phase (v231/v233), and M_R is the free seesaw scale. An operator advance, not a closure of full PMNS dynamics.
- **`v264_raw_seam_marklocal.py` (`QGEO.ENERGY.03`) sharpens the bedrock.** Composing v216+v201+v198 on the canonical pillowcase: the flat-pillowcase curvature lives only at the 4 marks, so the DtN sub-principal symbol is $\mathbb{Z}_4$-invariant (Fourier on $4\mathbb{Z}$, FFT-verified), block-diagonal, and $[\rho, \Lambda]{=}0 \Rightarrow \omega \circ \rho{=}\omega$ on this metric. So the whole bedrock collapses to the single metric statement "the raw RP seam carries the uniformising flat pillowcase metric" (=`QGEO.SYM.01`). A reduction/sharpening, [O] — it does *not* prove the seam must carry that metric (negative control: curvature off the μ_4 orbit breaks the chain). Paper citations added in `tfpt_4_frontier`, `tfpt_2_standard_model` and `tfpt_research_contracts`. [E] `AUDIT OK`, build green.

2026-06-18 (Paper consistency pass: the closure ledger and the status/QFT sections updated for the Modular Spectral Closure, with an explicit "what is *not* closed" line)

- **Five paper sections that predated the `v238`–`v261` round are brought up to date** (additive, no status marker moved). The `introduction` closure-ledger table and the "five questions" item now carry a *boundary QFT (Modular Spectral Closure)* row — one relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ closed modulo `QGEO.SYM.01` (v258–v261) — plus an explicit list of what the closure does *not* reach: the full measured pole masses, the absolute QCD scales (m_p/m_e), the complete PMNS dynamics, and the real 4D interacting QFT / ambient measure. The `tfpt_2` "ambient closure" section now states that the cutoff of its $S_{\rm rel}$ *is* the seam KMS weight (f_2/f_0=1, v259) and $D_{\rm rel}$ the covariance induction (v258); the `tfpt_2` "(4) Constructive QFT closure" section adds the relative-object pointer; `tfpt_4_frontier`'s full-quantum-gravity section records that the cutoff moment f_0 is pinned by the seam state (v255/v259); and `origin_theory`'s honest-status keybox notes that the whole boundary QFT collapses onto the same `QGEO.SYM.01` postulate (no sixth structural class). An editorial consistency pass so every status surface agrees with the ledger. [E] build green (`10 ok, 0 failed`), `AUDIT OK`.

2026-06-18 (Consistency pass: the Modular Spectral Closure surfaced structurally across the website and the intro status card)

- **The `v258`–`v261` closure is now reflected on every "what is open" surface, not just the research-contracts paper.** The homepage open-gates section and compiler-pipeline diagram, the hostile-referee FAQ, the glossary $G_{\rm net}$ entry, the reviewer page and the verification residual-chain caption now state that the *entire* emergent-QFT layer (Dirac, cutoff, gauging, glue, orientability) collapses onto the *same* premise `QGEO.SYM.01` via the Modular Spectral Closure — the boundary QFT is one relative object that adds *no new open item*, ambient QG separate. The introduction status card ("the structural residual is one condition") carries the same sentence. No status marker moved; this is an editorial consistency pass so the public surfaces agree with the ledger and the paper. [E] build green, `AUDIT OK`.

2026-06-17 (The Modular Spectral Closure: Dirac as covariance induction, the modular cutoff, the Kummer/K3 unification, and the assembly certificate, `v258`–`v261`)

- **`v258_dirac_covariance_induction.py` (`PS.DIRAC.03`) closes the "posited Yukawa vs derived operator" gap of v250/v252.** The finite Dirac operator D_F is the modular/covariance *induction* of the boundary seam quasi-free (KMS, β=1) state: a Gaussian state is fixed by its covariance C (a positive contraction) with one-particle modular Hamiltonian $H = \log((1-C)C^{-1})$ (Araki; Casini–Huerta), inverse to the KMS occupation $C = (1+e^H)^{-1}$. [E] the inversion identity holds exactly; [E] C_F is a positive contraction; [E] the induced operator is chirality-odd; [E] it is unitarily equivalent to the v252 D_F (same mass spectrum); [E] the Majorana/seesaw m_D^2/M_R is an off-diagonal covariance entry. [C] $C_F = P_F C_\Sigma P_F$, so D_F is the Kasparov shadow $[D_\Sigma]\widehat{\otimes}_{(E_8)_1}[\mathcal{K}_{\rm car}]$ of the boundary geometry and the v18 Yukawas are the readout of C_Σ.
- **`v259_modular_cutoff_kappa.py` (`PS.SPECACT.02`) fixes the last open spectral-action input.** The cutoff f is the seam's own modular/KMS weight: by Tomita–Takesaki/Bisognano–Wichmann the seam is KMS at β=1 (v239) and 2π=$1/(4c_3)$ (v58) fixes β, so the heat-kernel cutoff $f(u) = e^{-u}$ is *derived*. [E] its moments give $f_2/f_0 = f_4/f_2 = 1$ exactly; [E] a generic Gaussian cutoff gives $\sqrt{\pi}/2 \neq 1$ (the choice is non-vacuous); [E] hence $\kappa = \sqrt{(f_2/f_0)\, c_{PS}/c_{\rm grav}} = \sqrt{c_{PS}/c_{\rm grav}}$ loses its scheme factor. [C] the open [O] cutoff freedom of v253/v255 becomes a finite trace ratio over the 96-dim H_F (the RG κ≈1.3 pins $c_{PS}/c_{\rm grav}$≈1.7).
- **`v260_k3_kummer_unification.py` (`ARCH.K3.01`) unifies seam, carrier and lattice on one surface.** The pillowcase is the elliptic-involution quotient $T^2_{\tau=i}/(z\mapsto -z)$ (v214); squaring it gives the Kummer/K3 surface, whose three facets are three TFPT objects. [E] the E_8 Cartan is even/det 1/positive-definite; [E] $H^2(\mathrm{K3},\mathbb{Z}) = U^3 \oplus E_8(-1)^2$ (rank 22, det -1, even, signature $(3,19)$); [E] the seam is $T^2/(z\mapsto -z)$ with 4 marks (orbifold Euler char 0); [E] the carrier-16 are the $|A[2]|$=2^4=16 Kummer nodes = $\dim S^+$ (v197), 16=4×4; [E] honest note — the rank-16 Kummer lattice is *not* the $E_8(-1)^2$ summand. [C] so the v1 "glue $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$" is the lattice shadow of a single object; E_8 appears both locally (du Val $\mathbb{C}^2/2I$, v232/v236) and globally (H^2 summand). A unification, not a closure.

- **v261_modular_spectral_closure.py (QFT.MSC.01) is the assembly capstone.** The whole QFT round (v238–v260) is certified as *one* relative object $\mathsf{TFPT}_{\mathrm{QFT}} = (\mathcal{A}_\Sigma, \omega_\Sigma, \Delta_\Sigma, \rho, A_F, H_F, D_F, J, \gamma, S_{\mathrm{rel}})$ reduced to *one* open premise. [E] the cross-consistency invariants agree: $4 = [B{:}A] = |\mu_4| = 2\chi = |(\mathbb{Z}/2)^2|$ (index = marks = fixed points = glue order), one carrier-16 (spinor = lattice = triple), one gap $6\log(3/2)$ (OS = recovery = Lindblad); [E] the three closures re-derived; [E] the ten-component manifest present. [O] the QFT layer reduces to QGEO.SYM.01 (seam realisation), the ambient QG measure QG.AMB.01 kept separate. [C] the *Modular Spectral Universality* master statement: the unique RP holomorphic index-4 KMS seam state induces the 96-dim KO-6 triple whose inner fluctuations give the gauged Pati–Salam spectral action — the honest sense in which the boundary QFT is complete.

2026-06-17 (Infrastructure: the changelog is now a public website page, generated from changelog.tex and audit-gated)

- **The canonical changelog now also drives a public /changelog website page.** A new generator `verification/make_changelog_web.py` parses this file (dated `\subsection*` entries, `itemize` items, and the inline markup `\textbf`/`\emph`/`\texttt`/`\vref`, the status markers [E]/[C]/[O]/[X], inline math and accents) into the typed data file `website/lib/changelog.ts`, rendered server-side (math via KaTeX) by `website/app/changelog/page.tsx` + `website/components/Changelog.tsx`. The generator is wired into `bash build.sh gen`; its freshness is enforced by `audit_sync.py` (section A.generated), so the page can never drift from this file – editing the changelog and re-running `gen` is the only supported path (the `.ts` mirror is never hand-edited). `/changelog` is linked from the navbar, footer and sitemap. The `tfpt-workflow`, `website-sync` and `sync-maps` cursor rules were updated to record the new single-source generation step. [E] build (`npm run build` green, `/changelog` prerendered static) and `bash build.sh audit` (`AUDIT OK`).

2026-06-17 (Orientability + Poincaré duality: the honest, literature-confirmed status — strict fail, weakened hold, v257)

- **v257_quasiorient_modpd.py (PS.NCG.ORIENT.02) corrects the slightly too-generous v256 typing.** The published result (Cacic–Stephan arXiv:0902.2068; CCM07; SM-vacuum hep-th/0601192) is that the Chamseddine–Connes–Marcolli Standard-Model finite triple (KO-dimension 6, with three right-handed neutrinos — exactly the TFPT seesaw case) *genuinely fails* strict orientability and strict Poincaré duality, and instead satisfies the physically correct weakened versions. Verified on the explicit v252 triple: [E] strict orientability fails ($\gamma \notin \mathrm{span}\{\lambda(a)\rho(b)\}$); [E] *quasi-orientability* holds (the left/right A-irrep boxes are disjoint by chirality: $L = H$ weak-doublet, $R = \mathbb{C}$ singlet, so $L^{\mathrm{LR}}(\mathcal{H}^{\mathrm{even}}, \mathcal{H}^{\mathrm{odd}}) = \{0\}$); [E] strict Poincaré duality fails (KO-6 skew intersection form, corank 1, the equal-L/R-neutrino case); [C] modified Poincaré duality holds (CCM), Stephan's alternative triple (St06) restoring strict PD with identical physics. A known property of the NCG-SM model class, not a TFPT defect — no faked closure.

2026-06-17 (Remaining NCG obligations discharged or honestly typed: inner fluctuations, spectral action, orientability/Poincaré, v254–v256)

- **v254_inner_fluctuations.py (PS.NCG.FLUCT.01) closes v250 #5 + the no-junk obligation.** The inner fluctuations $\Omega^1_D(A_F)$ of the finite Dirac operator are computed explicitly: [E] they are purely chirality-odd (scalars), and for the SM algebra sit *exactly* in the one Higgs doublet $(1,2)_{1/2}$ — no junk, no triplet, no $(10,1,3)/\mathbf{126}$. [E] the σ (ν_R Majorana) direction is absent for the SM algebra but present for Pati–Salam (the CCvS beyond-first-order scalar), and [E] σ is an SM singlet $(1,1,0)$ with $B{-}L = 2$. [C] σ = scalaron.

- **`v255_spectral_action_expansion.py` (`PS.SPECACT.01`) closes the spectral-action expansion (v252 #11).** The Seeley–DeWitt expansion $S = 2f_4\Lambda^4 a_0 + 2f_2\Lambda^2 a_2 + f_0 a_4$: [E] structure ($a_0$ cosmological, a_2 Einstein–Hilbert+Higgs-mass, a_4 Yang–Mills+Weyl^2 + R^2+Higgs-quartic, $N = 96$); [E] $b/a^2 = 1/3$ (top-dominated); [E] $\sin^2\theta_W = 3/8$; [E] the R^2 spin-0 mode = the Starobinsky scalaron; [C] Higgs quartic → ∼125 GeV with σ; [C] $\kappa = \sqrt{(f_2/f_0)c_{PS}/c_{\rm grav}}$ explicit; [O] exact decimal needs the cutoff moments.
- **`v256_orientability_poincare.py` (`PS.NCG.ORIENT.01`) — honest status, no faked proof.** KO-dimension 6 forces the intersection form $\mathrm{Tr}(\gamma\,\pi(p)J\pi(q)J^{-1})$ to be *antisymmetric*: [E] skew form (verified), [E] rank 2, corank 1, null $\sim (1,1,-1)$ on the SM K-theory $\mathbb{Z}^3$; [E] the orientability local condition $[\gamma_F, \pi(a)] = 0$. [C] the canonical hypercharge-faithful SM/PS triple satisfies orientability and Poincaré duality (van Suijlekom; CCM); [O] a self-contained machine proof needs the complete canonical bimodule.

2026-06-17 (Full 96-dim finite spectral triple built + NCG axioms verified, and κ pinned to $O(1)$, `v252`–`v253`)

- **`v252_full_finite_triple.py` (`PS.DIRAC.02`) advances the Dirac contract v250 from a 7-obligation contract to a concrete object discharging 5 to [E].** The full 96-dim finite spectral triple $(A_F, H_F, D_F, J, \gamma)$ — H_F the particle⊕antiparticle doubling of three $SO(10)$ **16**-plets — is built explicitly and the NCG axioms are checked by direct computation: [E] reality ($J^2{=}{+}1$, $JD{=}DJ$), grading ($\gamma^2{=}1$, $[\gamma,a]{=}0$, $J\gamma{=}{-}\gamma J$), KO-dimension 6, order-zero, the first-order condition (HOLDS for the SM algebra including the Majorana; VIOLATED for the Pati–Salam algebra exactly by the Majorana, localized to $\{\nu_R, e_R\}$ — the CCvS beyond-first-order mechanism on the full triple), and the spectrum of D_F reproducing the fermion masses and the type-I seesaw $m_{\rm light}{=}m_D^2/M_R$. [C] Poincaré duality; [O] orientability, no-junk, spectral-action expansion.
- **`v253_kappa_scalaron_ps.py` (`PS.KAPPA.01`) discharges residual 6(b) of v249.** κ in $M_{PS} = \kappa\, M_s$ ($M_s = c_3^{7/2}\bar{M} \approx 3.06{\times}10^{13}$ GeV, exact) is pinned by an RG scan (PDG errors × the two E_8-allowed contents) to a tight $O(1)$ interval (≈ 1.3). [C] heat-kernel origin: $\kappa = \sqrt{(f_2/f_0)\,c_{PS}/c_{\rm grav}}$ is a scale-free ratio of heat-kernel traces over the 96-dim H_F (Λ cancels). [X] the **126**-content drives κ out of the $O(1)$ window (content-selective). [O] the exact decimal needs f_2/f_0 fixed.

2026-06-17 (Spectral triple, first step: the first-order condition is violated exactly by the Majorana/σ — the CCvS Pati–Salam mechanism, `v251`)

- **`v251_first_order_sigma.py` (`PS.DIRAC.01`) advances the Dirac contract v250 from a contract to an exhibited mechanism.** On a faithful lepton-block model ($H = H_F \oplus H_F^c$, $H_F = \mathbb{C}^3$), the real even structure is checked ($J^2{=}{+}1$, $\gamma^2{=}1$, $D\gamma{=}{-}\gamma D$, KO-dim-6 signs), order-zero holds, the first-order condition HOLDS for the Yukawa block, and is VIOLATED *exactly* by the Majorana ($\nu_R\nu_R^c = \sigma$) block — the Chamseddine–Connes–van Suijlekom "beyond first order" mechanism by which the inner fluctuations promote the SM to Pati–Salam and generate σ. Honest correction to the review: σ does *not* rescue the first-order condition; its controlled violation *is* the mechanism. [C] σ = scalaron (v249); [O] the full 96-dim triple + spectral action.

2026-06-17 (Carrier-native Pati–Salam program: E_8 forbids the 126, PS algebra, unification kill-test, Dirac contract, `v247`–`v250`)

- **Four scripts turning the gauge-unification finding into typed claims with negative controls.** `v247_e8_branching_no126.py` (`PS.E8BRANCH.01`): the E_8 hull branches under $SO(10){\times}SU(4)$ as $(45,1){+}(1,15){+}(10,6){+}(16,4){+}(\overline{16},\bar{4})$, so the $SO(10)$ content is exactly

$\{1, 10, 16, 45\}$ and the **126** is *absent* — the renormalisable 126_H is algebraically forbidden and 10+16+45 is E_8-supplied (only one 45 $\Rightarrow$ proton-marginal). `v248_ps_algebra.py` (`PS.ALGEBRA.01`): $A_F = \mathbb{H}_L \oplus \mathbb{H}_R \oplus M_4(\mathbb{C}) \to SU(2)_L \times SU(2)_R \times SU(4)_c$, exact hypercharges $Y = T_{3R} + (B-L)/2$, matching $\alpha_1^{-1} = \frac{3}{5}\alpha_{2R}^{-1} + \frac{2}{5}\alpha_4^{-1}$, $\sin^2\theta_W = 3/8$ (colour $SU(4)_c \subset D_5$, distinct from the family A_3). `v249_ps_unification.py` (`PS.RGTEST.01`): the two-step PS$\to$ $SO(10)$ unification kill-test with negative controls (SM-only fails; 126 breaks the scalaron match; proton decay selects content), $M_{PS}/M_s = 1.0$–1.7. `v250_dirac_triple_contract.py` (`CONTRACT.QFT4D.DIRAC.01`): the spectral-triple contract ($H_F = 3\times16$, the seven NCG axioms, σ =scalaron) that would turn "carrier is gauged" from a fork into a theorem. The root-level `qft.txt` carries the full analysis.

2026-06-17 (Honest data cross-check + emergent-QFT figures: the unification tension, v246)

- **`v246_unification_data_crosscheck.py` (`QFT4D.RGTEST.01`) — the first hard data confrontation of the spectral-action prediction.** TFPT's gauge-charged content *is* the Standard Model (v159) and the theory adds no new states; the E_8 cascade is an arithmetic spine (v5), not a threshold tower. So the run *is* the SM run: at one loop the couplings spread $\sim 10^{13}$–10^{17} GeV (residual $\Delta\alpha^{-1} \approx 9$ at 2×10^{16} GeV), and at two loops (RK4) the minimal spread is $\Delta\alpha^{-1} \approx 3.2$ at 1.7×10^{14} GeV — the three never meet. Typed [X]/[O]: with "no new state" there is *no* admissible threshold source, so the spectral-action 4d-GUT route is in genuine tension (same status as NCG-SM), likely falsified unless extended — never a confirmation, and *not* rescued by a cascade. Two new figures (`figures/qft_skeleton.pdf`, `figures/qft_unification.pdf`) added via `make_figures.py` (PDF + website PNG) and embedded in `tfpt_research_contracts.tex`; a root-level `qft.txt` summarises the emergent-QFT skeleton, what each script proves, the data status and the open questions.

2026-06-17 (Spectral-action matter content: the $SO(10)$ 16 = one anomaly-free SM generation, v245)

- **`v245_spectral_matter_content.py` (`QFT4D.MATTER.01`) discharges the computable core of the 4d contract v244.** The carrier half-spinor (the $SO(10)$ **16**) is verified, in exact rational arithmetic, to be one Standard-Model generation: $\mathbf{16} = SU(5)(\mathbf{10}+\bar{\mathbf{5}}+\mathbf{1})$, the SM multiplet dimensions sum to $16 = \dim S^+$ (15 Weyl fermions $+\,\nu_R$), the electric charges $Q = T_3 + Y$ are exactly the SM values, all four gauge anomalies cancel ($\sum Y = \sum Y^3 = [SU(2)]^2U(1) = [SU(3)]^2U(1) = 0$), the spectral-action normalisation gives $\sin^2\theta_W = 3/8$ ($g_3 = g_2 = \sqrt{5/3}\,g_1$), and the Higgs is the $(1,2)_{1/2}$ inner fluctuation. So the matter-content + tree-relation half of `CONTRACT.QFT4D.01` is now [E]; only the seam$\to$Dirac realisation and the run-down couplings stay [O]. Wired into `run_all.py`, the ledger, the registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The QFT skeleton on the seam: thermal time, GNS/OS reconstruction, DHR sectors, the braiding S-matrix, and the 4d spectral-action contract, v239–v244)

- **Six emergent-QFT skeleton scripts (v238).** `v239_kms_thermal_time.py` (`DYN.KMS.01`): the modular flow is KMS at $\beta = 1$, and $2\pi = 1/(4c_3)$ fixes the modular temperature as $T_H = c_3/M$ — thermal time = horizon time. `v240_gns_os_reconstruction.py` (`DYN.GNS.01`): GNS reconstructs the Hilbert space from $(\mathcal{M}, \omega)$ and the OS Hamiltonian $H_{\rm OS} = -\log T = -L$ is positive with gap Δ (the v64 mass gap). `v241_dhr_sectors.py` (`DHR.SECTORS.01`): particles = DHR sectors = the discriminant anyons of the Lean glue form, with the condensation tower $16 \to 4 \to 1$ (v235/v237). `v242_spin_statistics_modular.py`

(DHR.MODULAR.01): spin-statistics, the modular (S, T) data, Verlinde fusion, and Gauss–Milgram $c = 8$. `v243_haag_ruelle_braiding.py` (DHR.SMATRIX.01): the 2-particle S-matrix = the braiding monodromy (factorised, crossing), with the v238 mass gap supplying Haag–Ruelle asymptotic states. `v244_spectral_action_contract.py` (CONTRACT.QFT4D.01): the 4d Lagrangian via the Connes spectral action — the carrier half-spinor 16 is one generation, the SM gauge group sits in the carrier, gravity is v36; the matter Lagrangian is the open [O] contract. Net: the residual to a full 4d QFT is *one premise* (QGEO.SYM.01) plus *one contract* (CONTRACT.QFT4D.01). All wired into `run_all.py`, the ledger and registry, cited in `tfpt_research_contracts.tex`, mirrored to the website.

2026-06-17 (The static→dynamic flip: modular flow + a recovery Lindblad generator, v238)

- **New `v238_modular_lindblad_dynamics.py` (DYN.SEMIGROUP.01, cluster registry).** A companion to the modular round (v196/v198) and the recovery code (v221/v64) that reads the *same* seam objects *dynamically* rather than statically. [E] the modular flow $\sigma_t = e^{it\Lambda_\Sigma}$ is a unitary one-parameter group (Tomita–Takesaki) whose conservation of the μ_4 clock, $\rho\sigma_t\rho^{-1} = \sigma_t \Leftrightarrow [\rho, \Lambda_\Sigma] = 0$, IS the v198 state-invariance — so the static commutator is the conservation law of modular time. [E] the recovery channel's generator $L = \log T$ is a doubly-stochastic/CPTP-classical Markov generator ($e^L = T$, $e^{L\tau} \to$ the democratic projector = the arrow of time). [E] keystone: the dissipative gap $= -\log((2/3)^6) = 6\log(3/2) =$ the OS mass gap Δ (v64) — the static recovery bound (v221) and the dynamical mass gap (v64) are one number, one exponentiated. [E] the GKSL split $G = i\Lambda_\Sigma + L$ (imaginary + real≤ 0 spectrum). [O] the residual is the *geometric* modular flow (Connes–Rovelli "thermal time = physical time") on the full seam algebra = QGEO.SYM.01 (v198) — a target, not a closure. Wired into `run_all.py`, `status_ledger.csv`, `script_registry.csv` and cited in `tfpt_research_contracts.tex`; mirrored to `website/public/verification`.

2026-06-17 (Experiments: black-hole-cosmology signatures, the η_B Boltzmann solve, the Petz/baby-universe realisation, and a real-data GW-echo pipeline)

- **Three new standalone black-hole-cosmology experiments (`experiments/`, all `data_limited`, not load-bearing).** From the `problem_b` black-hole-cosmology survey: (i) `ccbh-dark-energy` — the de Sitter seam interior (w_{in}=−1) forces the Croker–Weiner cosmological-coupling index $k = -3w_{\text{in}} = 3$, hence a population $w = -1$; vs. Farrah+2023 $k = 3.11 \pm 0.79$ (-0.14σ), a *contested* downstream bridge and an *alternative reading* of the same $w = -1$ the DESI watchdog tests (`alternative_group w_de_eos`, never double-counted). (ii) `gravastar-compactness` — the Nariai Q_{geom}=3/8 equals the Jampolski–Rezzolla (2026) maximum horizonless compactness $\mathcal{C}$=3/8 (shared de Sitter endpoint 1/2); since 1/3<3/8<4/9<1/2 this is a light-trapping ECO, yielding an echo template (round-trip delay ∼0.7 ms at 62 $M_\odot$, amplitude $\leq (2/3)^6$) — a [C] structural echo, no $\mathcal{C} \leftrightarrow Q_{\text{geom}}$ map proven. (iii) `cosmic-handedness` — a parity watchdog (Shamir JADES ∼3.3σ spin monopole, most likely a Milky-Way-aberration dipole), explicitly *not* promoted (frontier).

- **η_B — full BDP Boltzmann ODE solve confirms the route at the frozen scale (`experiments/ftransfer/leptogenesis_boltzmann/fboltzmann_solve.py`; `tfpt_4_frontier`, [C]).** Integrating the Buchmüller–Di Bari–Plümacher Boltzmann network (integrated efficiency $\kappa_f = 0.092$, validating the analytic fit) at the *frozen* heavy scale $M_1 = M_{\text{scal}}(\varphi_0^{\text{ret}})^2/A_\Lambda \approx 8.65 \times 10^9$ GeV with $\delta_{\text{CP}} = 4\pi/3$ gives $\eta_B = 6.5 \times 10^{-10}$ — a factor 1.07 from the observed 6.1×10^{-10}, with *no* free M_R dial. This moves the leptogenesis route from "solve pending" to a consistent [C] readout (the full *flavored* density-matrix solve is the next refinement); `tfpt_4_frontier` Route C synced.

- **The Petz recovery map and the rank-one baby universe, realised numerically (`experiments/recovery-channel`; `tfpt_horizon_readouts`, [C]; companion to v221).** The gapped transport contracts to a *rank-one* projector at exactly $\|T^n - P_\infty\| = (2/3)^{6n}$ — the one-dimensional baby-universe Hilbert space (Engelhardt 2025) — and the explicit Petz map is CPTP, recovers the reference, and equals the identity only on the protected λ=1 (Knill–Laflamme) mode; free-ratio and degenerate-spectrum negative controls hold. So the Petz identification v221 deferred is now verified at the channel level (`internal_consistency`, no new datum); the full holographic reading stays gated on the Seam–Horizon theorem.
- **GW ringdown echo — Stage-1 matched filter validated, then run on real GWOSC strain (`experiments/gw-ringdown-echo`; `tfpt_horizon_readouts search-targets`).** The Stage-1 machinery (Kerr-ringdown subtraction → matched filter on the residual, ratio $(2/3)^6$ frozen, lag free, + free-ratio control) classifies 3/3 synthetic injections; run on *real* GWOSC strain (GW150914, GW190521; PSD-whitening + off-source background): **no kernel-ratio echo coincident in ≥ 2 detectors**, consistent with the $(2/3)^6$ *upper* bound (a single loud H1 excess has $\hat{q} \approx 2$, far from $(2/3)^6$, so the free-ratio control flags it as residual ringdown, not an echo). First pass: dominant-QNM subtraction, incoherent combination — full multi-mode + coherent stacking is the next step; no detection claim.
- **Scorecard + website synced.** `experiments/evidence_scorecard.json` now 51 rows (26 consistent, 7 null, 13 data-limited, 4 tension, 1 parked); the website `EXPERIMENTS_AUDIT` mirror and the η_B prediction entry are updated to match. These are standalone `experiments/` surfaces (not in the verification suite or ledger); the `tfpt_4_frontier/tfpt_horizon_readouts` and `papers.ts/predictions.ts` edits are the in-same-change sync.

2026-06-17 (Red-team layer re-synced to the paper; shadow export mirrors `figures/`)

- **Red Team Target A re-synced to `tfpt_5_redteam` (`rt_A_e8net.py`, `redteam_table.txt`, `redteam/README.md`, infrastructure).** The Python red-team surface still emitted the historical *three-residual* "A reduced, not closed" form, lagging the reader-facing note. It now encodes the sharper narrowing already in the paper: the residual collapses to the *single* statement "the seam–Calderón boundary net is holomorphic with $c = 8$" ($\Leftrightarrow$ the index-4 simple-current extension), with the index-4 $(D5)_1 \times (A3)_1 \to (E8)_1$ glue realised at Lie level (v143) and its odd sectors twisted/Ramond ($E_8 = 120_{\mathrm{NS}} + 128_{\mathrm{R}}$, v148), while the $q(A_3)$ normalisation collapses into the one declared anchor (v152). Two exact adversarial checks added (120+128 = 248; glue index = $|\mu_4| = N_{\mathrm{fam}}+1 = 4$); status stays [C]/[O] (reduced, not closed).
- **Shadow export now mirrors `figures/` but not `website/` (`scripts/export-shadow.sh`, `verification/make_script_index.py`, infrastructure).** `export-shadow.sh` ships `figures/*.pdf` (so `make_manifest.py --check` passes literally on the exported tree) while still excluding `website/` and the compiled paper PDFs; the `SHADOW_MIRROR.md` table is corrected accordingly. `make_script_index.py` now detects a missing `website/` and skips its `ScriptIndex.tsx` mirror, so `build.sh notes` runs on the subset without crashing.

2026-06-17 (CP phases as μ_6 powers — a reduction of red-team Target D)

- **Both CP phases are μ_6 powers of one hexagonal CM unit, split by the sheet (`v231_cp_mu6_phases`, `tfpt_5_redteam`, [E]/[C]).** Sharper than v220/v225 (which only *located* the CP residual). With $\rho = e^{i\pi/3}$ the primitive 6th root (the j=0 Eisenstein CM unit): δ_{CKM} leading $= \arg(\rho^1) = \pi/3$ (quark) and $\delta_{\mathrm{PMNS}} = \arg(\rho^4) = 4\pi/3$ (lepton, the phase lattice), with $\rho^4 = -\rho$, so $\delta_{\mathrm{PMNS}} = \delta_{\mathrm{CKM,lead}} + \pi =$ (CM unit) $\times$ ($\mathbb{Z}_2$ sheet, ρ^3=−1). The two CP phases are *one* hexagonal unit on the two sheets, not two independent inputs; the dual-frame Jarlskog orientations are sheet-flipped ($\pm 21 \sin(\pi/3)$). A structural reduction of

Target D (removes one of its two phase inputs); the quark seam correction $3\lambda_C^2$ stays the v88 misalignment, the deck stays $\mathbb{Z}/4$, and the physical CP derivation stays [C] (`CP.MU6.01`).

- **The seam as the E_8 Kleinian singularity (`v232_e8_kleinian_seam`, `origin_theory`, [E]/[C]).** The du Val side of the McKay correspondence (v219) gives the open seam-realisation premise `QGEO.REALIZE.01` a canonical algebraic-geometric model: dropping the trivial-rep node of the affine-E_8 McKay graph of $2I$ leaves the finite E_8 Dynkin = the dual intersection graph of the eight exceptional $\mathbb{P}^1$'s of the minimal resolution of $\mathbb{C}^2/2I$; the eight curves = rank $E_8 = g_{\text{car}}+N_{\text{fam}}$ (a fourth reading of the 8), their negated intersection form is the E_8 Cartan (det 1, even, (-2)-curves), and the link is the Poincaré homology 3-sphere $S^3/2I$ ($2I$ perfect $\Rightarrow H_1{=}0$, $\pi_1{=}2I$ order 120). The raw seam ($\mathbb{P}^1$ with marks) is canonically a du Val exceptional curve — a model for the bedrock, not a P2 proof (it presupposes E_8; `TOPO.E8.01`, bedrock stays [O]).
- **CP is the universal family/triality phase, only sheet-split (`v233_cp_triality_phase`, `tfpt_5_redteam`, [E]/[C]).** One step past v231 in the $\mu_6 = \mu_3$(family/triality)$\times\mu_2$(sheet) factorisation: $\rho = \omega^2(-1)$ with $\omega = e^{2\pi i/3}$ the $\mathbb{Z}_3$ triality centre (the 2/3 cusp weight). Since $\rho^4 = \rho^1\rho^3$ with $\rho^3 \in \mu_2$, the quark (ρ^1) and lepton (ρ^4) CP phases share the *same* family/triality class and differ only by the sheet — so CP *is* the universal triality phase, sheet-split, not a free power choice (`CP.TRIALITY.01`).
- **The Seam-Holomorphy selection certificate: the structural residual is one gate (`v234_seam_holomorphy_selection`, `tfpt_research_contracts`, [E]/[O]).** The whole structural residual is *one* condition — "the seam carries no nontrivial abelian sector" — with three provably-equivalent faces, all forcing E_8: (i) holomorphy (μ-index 1; Target A), (ii) the seam link is a homology 3-sphere (Γ perfect $\Leftrightarrow 2I$; v232), (iii) exactly one 1-dim irrep (v219). All equal #(mark-1) $= |\Gamma^{\text{ab}}| = |H_1|$, which is 1 only for E_8; and holomorphic $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ gives the unique even unimodular rank-8 lattice E_8. So P2, G_{net}, Target A and the Kleinian seam are one gate. The stated closing theorem ([O]): RP+gap+chirality $\Rightarrow$ quasi-free bulk (v160) $\Rightarrow$ holomorphic boundary $\Rightarrow E_8$; the single residual analytic step is "free bulk $\Rightarrow$ holomorphic boundary" (`GATE.HOLO.01`).
- **The closing step in Chern–Simons language: holomorphic $\Leftrightarrow$ $\det K{=}1$ (`v235_seam_chern_simons`, `tfpt_research_contracts`, [E]/[O]).** A genuine reduction (not a closure) of `GATE.HOLO.01` into standard anyon-condensation physics: a free gapped bosonic 2+1d bulk is an even integer K-matrix with #anyons $= |\det K|$, edge $c = \text{signature}(K)$, and the edge net holomorphic $\Leftrightarrow |\det K|{=}1$. The TFPT extension tower (v92) *is* the condensation tower read by determinant: carrier $D_5{\oplus}A_3$ (det 16, 16 anyons) $\to SO(16)_1{=}D_8$ (det 4) $\to (E_8)_1$ (det 1, holomorphic) — all rank 8, $c{=}8{=}g_{\text{car}}{+}N_{\text{fam}}$ — so holomorphic $= E_8 =$ the Kitaev E_8 quantum-Hall state. Holomorphy $\Leftrightarrow$ condensing the order-$|\mu_4|$ Lagrangian glue (det $16{\to}1$, vs. a partial order-2 isotropic $\to 4{=}D_8$). The residual narrows to one integer condition "the free RP seam condenses the Lagrangian glue (det$\,{=}1$)" = the sheet/QGEO selection, now with holomorphy$\Leftrightarrow$ det 1 a theorem (`GATE.HOLO.02`).
- **The $(2,3,5)$ Brieskorn singularity is the one generator of the skeleton (`v236_brieskorn_capstone`, `origin_theory`, [E]).** Capstone of the icosahedral round: the singularity $x^2{+}y^3{+}z^5$, whose exponents are the atoms $(|\mathbb{Z}_2|, N_{\text{fam}}, g_{\text{car}}){=}(2,3,5)$, has Milnor number $\mu{=}(2{-}1)(3{-}1)(5{-}1){=}8{=}\operatorname{rank}E_8$ (a *fifth* origin of the "8"), Milnor monodromy = the order-30 E_8 Coxeter element (eigenvalues the primitive 30th roots = the E_8 exponents, charpoly Φ_{30}), Milnor lattice E_8, link the Poincaré sphere; the two clocks are facets of this one monodromy — the μ_3 triality (CP, v233) is h^{10} in $\langle h\rangle{=}\mathbb{Z}/30$, the μ_4 clock (v223) is the order-4 Galois automorphism of $\mathbb{Q}(\zeta_{30})$ (`TOPO.BRIESKORN.01`).
- **The closing step as a physical condition: no topological degeneracy $\Leftrightarrow$ $\det K{=}1$ $\Leftrightarrow$ the seam is SRE (`v237_seam_sre_closure`, `tfpt_research_contracts`, [E]/[O]).** Sharpens `GATE.HOLO.02`'s residual from definitional to falsifiable: the K-matrix ground-state

degeneracy on a genus-g surface is $|\det K|^g$ (torus: carrier 16, D_8 4, E_8 1), so "no topological degeneracy on any closed surface" $\Leftrightarrow$ $\det K{=}1$; among rank-8 even positive-definite K ($c{=}8{=}g_{\rm car}{+}N_{\rm fam}$) only E_8 is short-range-entangled (the Kitaev E_8 phase). A unique vacuum on the plane (RP+gap+cluster) is necessary but not sufficient (the plane cannot see $\det K$); so the one open step is the physical statement "the free RP seam is SRE (no topological order)" (`GATE.HOLO.03`).

2026-06-16 (the icosahedral bedrock, CM-norm duality, and a structural-finds round)

- **McKay bedrock: why the atoms are $\{2,3,5\}$ (`v219_icosahedral_mckay`, `origin_theory`, [E]).** E_8 is the exceptional *top* of the McKay tower of finite $SU(2)$ subgroups ($2T{\to}\hat{E}_6$, $2O{\to}\hat{E}_7$, $2I{\to}\hat{E}_8$). Built *from the group*: the 120 unit-quaternion icosians close to the binary icosahedral group $2I$ (element orders $\{1,2,3,4,5,6,10\}$ — it contains the $2,3,5$ rotation-axis orders), 9 conjugacy classes; the 9 irreducible character degrees (Dixon, from the class algebra) are $\{1,2,2,3,3,4,4,5,6\}$, sum $30{=}h(E_8)$, sum of squares $120{=}|R^+(E_8)|$; the McKay adjacency from the natural 2-dim rep *is* the affine E_8 graph with Kac marks = the degrees. A backward certificate of the closed E_8, not a P2 proof (`MCKAY.E8.01`).
- **CM-norm duality: 41 (square) and 7 (hex) (`v222_cm_norm_duality`, `origin_theory`, [E]).** The square seam (Gaussian $\mathbb{Z}[i]$, $j{=}1728$) gives $N(g_{\rm car}{+}i|\mu_4|){=}5^2{+}4^2{=}41{=}10b_1$ (the EM index *is* the Gaussian norm of the carrier-glue vector) and $N(3{+}2i){=}13{=}\Delta_Q$; the hexagonal partner (Eisenstein $\mathbb{Z}[\omega]$, $j{=}0$) gives $N(3{+}2\omega){=}7{=}$ scalaron. The $(3,2)$ split generates $(5,6,7,13)$ by sum/product/Eisenstein/Gauss norm; rings rigid (`CMNORM.DUAL.01`).
- **The μ_4 clock is the order-4 character of the E_8 Coxeter cycle (`v223_coxeter_totative_clock`, `origin_theory`, [E]).** $(\mathbb{Z}/30)^\times$ are the E_8 exponents; the order-4 generator 7 ($\langle 7\rangle{=}\{1,7,13,19\}$) is a literal μ_4 clock permuting the four conjugate planes; only E_8 has $\varphi(h){=}$rank and $h{=}2{\cdot}3{\cdot}5$ (`COX.CLOCK.01`).
- **$248{=}120{+}128$ as a magnitude/phase channel typing (`v227_degree_exponent_channel_split`, `tfpt_5_redteam`, [E]).** Exponents sum $120{=}|R^+(E_8)|$ (magnitude channel), degrees sum $128{=}\text{rank}\cdot\dim S^+$ (phase/glue channel); the red-team Target D magnitude bijection lives in 120, the un-covered CP phases in 128; the dark-sector reading of 128 stays Frontier (replaces the over-strong S^- reading; `E8.CHAN.01`).
- **P2 as the mode space of a degree-4 seam divisor (`v228_rr_index_gate`, `origin_theory`, [E]/[C]/[O]).** A Riemann–Roch index gate: $\deg D{=}4{=}|\mu_4|$ on $\mathbb{P}^1$ forces $h^0{=}5{=}g_{\rm car}$, rank $H_1{=}3{=}N_{\rm fam}$, $h^0{+}H_1{=}8{=}\text{rank}\,E_8$, $\Lambda^{\rm even}(\mathbb{C}^5){=}16{=}\dim S^+$ — bedrock language for P2, conditional on the seam producing the degree-4 divisor (`QGEO.RR.01`).
- **The seam as a finite recoverability code (`v221_seam_qecc`, `origin_theory`, [E]/[C]).** Code dimension $\dim S^+{=}16$; the gapped transport is a CPTP doubly-stochastic contraction with spectrum $\{1,(2/3)^6,(1/3)^6\}$, recovery bound $\|T^n\delta\|{\le}(2/3)^{6n}\|\delta\|$ (Knill–Laflamme type); free ratio breaks it. Holographic/Petz reading [C], gated on the Seam–Horizon theorem (`QEC.SEAM.01`).
- **CP lives in the hexagonal phase fiber (`v220_cp_hexagonal_modulus`, `tfpt_5_redteam`, [E]/[C]).** The seam deck is the square modulus $j{=}1728$ ($\mathbb{Z}/4$); its partner is the hexagonal $j{=}0$ ($\mathbb{Z}/6$, $\arg\rho{=}\pi/3$). The frozen $\delta{=}\pi/3{+}3\lambda_C^2{=}68.654^\circ$ has leading term $\pi/3{=}\arg\rho$; the deck stays $\mathbb{Z}/4$, CP is the $\mathbb{Z}/6$ fiber (`CP.MOD.01`).
- **The dual normal frame (d,n) with CP as orientation (`v225_dual_normal_frame`, `tfpt_5_redteam`, [E]/[C]).** $d{=}a^\top R^{-1}{=}(-\tfrac12,-\tfrac12,1){=}{-}\tfrac12$ Nariai; $n{=}(5,-9,6)$ reads det on the first-generation column; $\det(\mathbf{1},d,n){=}21{=}3{\cdot}7$; $\operatorname{Im}\det(\mathbf{1},d,\rho n){=}21\sin(\pi/3)$ — CP as orientation (`DUAL.FRAME.01`).

- F_{transfer} **as one path on the Sheet Diamond (`v224_diamond_ftransfer_path`, `tfpt_2_standard_model`, [E]/[C]).** $M(s,t)=R+Q\,\mathrm{diag}(s,t,t)$: K, C, F on the sheet axis (curved, 2nd diff 8), the winding axis flat (slope 6), Plücker steps $(1,8,10),(1,8,16)$; the Koide source→pole is the 1-D projection (`FTR.PATH.01`).

- **The charged leptons as an étale Frobenius algebra (`v229_lepton_frobenius_algebra`, `tfpt_2_standard_model`, [E]/[C]).** $(c_e, c_\mu, c_\tau)=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$, ring closure $c_e c_\tau=\frac{8}{3}=|\mathbb{Z}_2|c_\mu$, product $\frac{32}{9}$; $A=\mathbb{Q}[t]/(m)$ is étale (trace-form Gram nondegenerate), a commutative Frobenius algebra over the μ_6 resolvent (`LEP.FROB.01`).

- **The center budget $(7, 11, 13)$ as three local norms (`v230_center_budget_norms`, `tfpt_2_standard_model`, [E]).** $C=R+Q\,\mathrm{diag}(1,0,0)$ row sums $(7,11,13)$: $7=N_{\mathbb{Z}[\omega]}(3+2\omega)$ (hex), $13=N_{\mathbb{Z}[i]}(3+2i)$ (square), $11=\binom{4}{0}+\binom{4}{1}+\binom{4}{2}$ (QBL Fock count); the two CM rings and the boundary count meet in one operator center (`CENTER.NORM.01`).

2026-06-15 (the diamond axis geometry — two axes + the transfer corner)

- **The Sheet Diamond is a discrete geometry with two axes (`v218_diamond_axis_geometry`, `tfpt_2_standard_model`, [E]).** Sharpens `v94`/`v95` with three exact [E] lemmas (no new numbers). **(1) Axis curvature:** along the centered cross the determinant is linear on the winding axis $\det(C+xU) = 14+6x$ (slope $6=|R^+(A_3)|$) and quadratic on the sheet axis $\det(C+yV) = 14+14y+4y^2$ (2nd difference $8=\mathrm{rank}\,E_8$); the anchor block has $\det B(C+xU) = 2\det(C+xU)$ and sheet curvature $6=|R^+(A_3)|$ — two curvatures $(8,6) = (\mathrm{rank}\,E_8, |R^+(A_3)|)$. **(2) Plücker transfer ladder:** $\mathrm{Pl}(K) \to \mathrm{Pl}(C) \to \mathrm{Pl}(F)$ lifts in steps $(1,8,10) = (N_\Phi, \mathrm{rank}\,E_8, A_\Lambda)$ then $(1,8,16) = (N_\Phi, \mathrm{rank}\,E_8, \dim S^+)$ — the decuple then the full generation (identity [E], source→pole reading [C]). **(3) Spectral ramification:** for Q, K, C, F the cubic discriminant factors as $q(r)^2\,\mathrm{Disc}(q)$ with squares $(1,3,4,6)$ and kernels $(13,48,65,105)$, F carrying $105 = 3{\cdot}5{\cdot}7$. The anchor-defect spectrum and pair-sum/staircase blocks are honest *audit* fingerprints ($28{=}2\dim G_2$, $52{=}\dim F_4$ stay audit-only, not spine, as in `v94`/`v95`); F as the transfer-completion corner is a *heuristic*, not a hard filter. Ledger `DIAMOND.AXIS.01`, `DIAMOND.PLUCKER.01`, `DIAMOND.SPECTRAL.01`; Wolfram-mirrored (249/249).

2026-06-15 (QGEO: the four marks emerge from Gauss–Bonnet — K1 executed)

- **The four marks emerge from Gauss–Bonnet — count derived, not assumed (`v216_marks_gauss_bonnet`, `tfpt_research_contracts`, [E]; ledger `QGEO.MARKS.03`).** The K1 lever of the `v215` kill-test ("exactly four marks") is *executed*: if the marks are $\mathbb{Z}_2$ branch points (deficit π) on the flat seam sphere ($\chi = 2$, the same χ as the 8 in c_3), Gauss–Bonnet forces $n\pi = 2\pi\chi = 4\pi$, so $n = 2\chi = 4 = |\mu_4| = N_{\text{fam}} + 1$. The closed Euclidean sphere 2-orbifolds are exactly $(2,3,6)$, $(2,4,4)$, $(3,3,3)$, $(2,2,2,2)$, and three independent criteria — all cone orders $= 2$ (the $|\mathbb{Z}_2|$ sheet branch), $\mathrm{rank}\,H^1 = \#\text{marks} - 1 = 3 = N_{\text{fam}}$ (the three generations pick the 4-mark square over the 3-mark hexagonal), and a free order-4 deck — converge on the pillowcase $(2,2,2,2)$. The numerical sibling `v217_marks_emergence_scan` ([C], ledger `QGEO.EMERGE.01`) confirms it on the actual DtN/state (number n and positions free, the 4-square config is the unique joint minimiser of clock-invariance + 3-family cohomology + gap $(2/3)^6$). The mark *count* and *equality* move from input to derived; only the square *modulus* (cross-ratio 2, the order-4 clock) stays the one carrier input, and the from-zero raw-seam emergence `QGEO.REALIZE.01` stays [O]. `v216` exact (Wolfram extension → 248/248); `v217` numerical (Python-only). Suite → 217 modules.

2026-06-15 (QGEO: the bedrock recast as a falsifiable kill-test)

- **`QGEO.SYM.01` as a kill-test, not a proof-promise (`v215_seam_deck_killtest`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.KILL.01`).** A foundational symmetry cannot be derived from nothing (Bisognano–Wichmann would presuppose the covariance it should produce), so the bedrock $\omega \circ \rho = \omega$ is recast as four independently-falsifiable predictions about the raw seam DtN. The non-circular reduction (`v198/v199/v201`) gives $\omega \circ \rho = \omega \Leftrightarrow [\rho, C] = 0$ with $|k|$ commuting exactly, so the residual is the bounded sub-principal M_f (equivalently the raw Gaussian bulk form A is μ_4-deck-invariant). Levers: **K1** exactly four marks ($b_1 = \#\text{marks} - 1 = 3 = N_{\text{fam}} \Rightarrow \#\text{marks} = 4 = |\mu_4|$); **K2** the square config (cross-ratio $2 \Rightarrow j = 1728$, Aut $= \mathbb{Z}/4$; order 4 selects $\mathbb{Z}/4$ not the $j = 0$ $\mathbb{Z}/6$; generic $\Rightarrow \mathbb{Z}/2$); **K3** mod-4 sub-principal support ($[\rho, M_f] = 0 \Leftrightarrow f$ $\mathbb{Z}_4$-invariant); **K4** transfer gap $(2/3)^6 = 64/729$. Freeze-bound via `freeze_file.csv` (`seam_deck_symmetry`), carrier-side [E] (regression guards), with the *decisive* kill deferred to `QGEO.REALIZE.01` [O] (the raw collar DtN must *produce* four square marks without inputting them — an emergence test). A kill-test, not a closure. Python-only (falsification layer; the exact sub-parts $j = 1728$ and $(2/3)^6$ are Wolfram-mirrored via `v214/v54/v56`). Suite → 215 modules.

2026-06-15 (QGEO: the pillowcase reduction — two residuals become one canonical metric)

- **Pillowcase reduction of `QGEO.SYM.01` (`v214_seam_pillowcase`, `tfpt_research_contracts`, [E]/[O]; ledger `QGEO.PILLOW.01`).** The two separately-tracked open QGEO residuals — `QGEO.ISO.01` (`v180`: the carrier clock is an order-4 *isometry* of the seam metric) and `QGEO.SUBPRIN.01` (`v201`: the DtN sub-principal symbol is *mark-local*, the seam flat away from the μ_4 marks) — are shown to be *one* statement: the seam carries the uniformising flat orbifold (*pillowcase*) metric. **(1)** The four marks make the Euclidean orbifold $S^2(2,2,2,2)$ ($\chi_{\text{orb}} = 2 - 4(1 - \frac{1}{2}) = 0$), so by Troyanov uniformisation the metric is flat (unique up to scale) with curvature only at the cone points (Gauss–Bonnet $4\pi = 2\pi\chi$) — this *is* the `v201` "flat away from the marks". **(2)** New link: cross-ratio(μ_4) $= 2$ (`v168`) $\Rightarrow j = 1728$ ($j(\lambda) = 256(\lambda^2 - \lambda + 1)^3/(\lambda^2(\lambda - 1)^2)$; all six harmonic cross-ratios give 1728) $\Rightarrow$ the *square* modulus $\tau = i$, whose CM by $\mathbb{Z}[i]$ *derives* the order-4 clock $z \mapsto iz$ (explicit $(x, y) \mapsto (-x, iy)$ on $y^2 = x^3 - x$); generic cross-ratio gives $j \notin \{0, 1728\}$ and only $\mathbb{Z}/2$. **(3)** Unification: "seam = uniformising flat pillowcase metric" implies *both* `v201` and `v180`, so two residuals collapse to one canonical-metric premise. [O] It does *not* close `QGEO.SYM.01`: the choice "RP seam metric = the constant-curvature representative" stays open, milder and more canonical than the bare isometry premise. Wolfram-mirrored (the exact orbifold/j-invariant/CM arithmetic); Klein-J values mpmath-numerical. Suite → 214 modules.

2026-06-15 (the F_transfer functor contract CONTRACT.F.01)

- **F_{transfer} formalised as a typed functor with four axioms (`v213_ftransfer_functor`, `tfpt_research_contracts`, [C]).** The four frontier transfers are consolidated into ONE functor $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$, each instance a discrete compiler kernel [E] composed with an external continuous solver [C]: **(1)** μ_4-deck equivariance (the Koide multiplier λ_2=$(2/3)^6$=$64/729$ is the deck transfer eigenvalue); **(2)** Plücker preservation (53=$a^T(R$+$Q)\mathbf{1}$, $\text{Pl}(F)$=$(1, 22, 30)$, $\|\text{Pl}(K)\|_1$=11); **(3)** positivity/stochasticity (spec T positive, $\kappa_f \in (0, 1)$); **(4)** external modules explicit (b_3=-7 a carrier output). The four instances are F_{pole} (`v183`), $F_{\text{Boltzmann}}$ (`v212`), F_{relic} (`v211`), F_{QCD} (`v164`). `CONTRACT.F.01` is the third research contract alongside (U_{wall}) and (G_{metric}), the structural complement of the `v187` typing guard — an architectural consolidation, *not* a closure of any frontier number. Ledger `CONTRACT.F.01`.

2026-06-15 (frontier: two sharper [C] scenario branches — axion angle + leptogenesis)

- **Axion DM spine-angle branch $\theta_i = \pi N_{\rm fam}/g_{\rm car} = 3\pi/5$ (`v211_axion_spine_angle`, `tfpt_4_frontier`, [C]).** A more *robust* misalignment angle than the determinant-line hilltop $\theta_i \approx 170°$ (`v185`, which over-produces $\Omega_a h^2 \approx 0.66$): the central spine quotient gives $\theta_i = 3\pi/5 = 108°$ exactly (no fit), the finite-T solver reaches $\Omega_{\rm DM}$ at $\approx$106°, and 108° sits in the *mild*-anharmonic regime (62° below the hill-top) — not exponentially sensitive. An alternative ansatz to $\theta_i = \pi(1-\varphi_\Sigma)$ (mutually exclusive, the full solver decides, `DM.AXION.SPINE.01`); [X] $\Omega_a h^2$ outside $\sim$[0.08, 0.16] demotes it. A sharper scenario, not a derivation.
- **Leptogenesis scalaron-decuple branch (`v212_leptogenesis_decuple`, `tfpt_4_frontier`, [C]).** Both Boltzmann inputs share the decuple $A_\Lambda = 10 = |E(K_5)|$: $M_1 = M_{\rm scal}(\varphi_0^{\rm ret})^2/A_\Lambda \approx 8.65\times10^9$ GeV and $\tilde m_1 = m_3/A_\Lambda \approx 5$ meV. M_1 uses *only* $\{M_{\rm scal}, \varphi_0^{\rm ret}, A_\Lambda\}$ — no hidden seesaw scale, cleaner than `v184`'s $M_R\varphi_0^4$ — and lands in the thermal window where $\eta_B \approx 1.2\times10^{-9}$ brackets the observed 6.1×10^{-10}. The coupling mechanism is posited not derived, so η_B stays [C]; [X] a precise Boltzmann/RGE solve outside the band falls the route, not the theory (`FR.ETAB.04`; Route A independent).

2026-06-15 (Lean: QGEO.COHOM.01 character grading + MODULE parity machine-proved)

- **The cohomology character grading is now Lean-formalised** (`CohomologyGrading.lean`, ledger `FORM.QGEO.03`, [O]). A new module machine-proves the Cohom node of the QGEO proof tree (`v177`; audit: pass, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the three $H^1(\mathbb{P}\setminus\mu_4)$ eigenforms $\omega_k = z^{k-1}/(z^4-1)$ (k=1, 2, 3) have pullback character $\rho^*\omega_k = i^k\omega_k$ under the clock $\rho : z \mapsto iz$ (each an exact rational-function identity, the denominator clock-invariant since i^4=1), so the grading is $(i, -1, -i)$ with weights $(1, 2, 3)$ = the A_3 exponents = Spec(Q_+), rank 3 = $N_{\rm fam}$. The Module *parity* is also proved: the reflection $\sigma : z \mapsto 1/z$ satisfies $\sigma^*\omega_1 = \omega_3$, $\sigma^*\omega_2 = \omega_2$, $\sigma^*\omega_3 = \omega_1$ (swap $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed). Signature-locked in `AuditContract.lean`. The geometric Cohom/Module-parity nodes are now [E]; the Marks/Kernel obligations and the Module *uniqueness* stay [O].

2026-06-15 (Lean: QGEO.UNIFORM.01 geometric normal form machine-proved)

- **The Möbius uniformisation normal form is now Lean-formalised** (`MobiusUniformisation.lean`, ledger `FORM.QGEO.02`, [O]). A new module machine-proves the constructive Uniform node of the QGEO proof tree (`v177`; audit: pass, no `sorry/admit`, axioms `propext, Classical.choice, Quot.sound` only): the clock $\rho : z \mapsto iz$ has ρ^4=id and $\rho^2\neq$id (order exactly 4); the reflection $\sigma : z \mapsto 1/z$ is an involution with $\sigma\rho\sigma = \rho^{-1}$ (so $\langle\rho,\sigma\rangle$ is the dihedral D_4 of order 8); a non-fixed orbit $\{a, ia, -a, -ia\}$ scales to $\mu_4 = \{1, i, -1, -i\}$; σ permutes μ_4 (fixes $1, -1$, swaps $i, -i$); and the multiplier classification "$z \mapsto \zeta z$ has order exactly 4 $\Leftrightarrow \zeta = \pm i$". Signature-locked in `AuditContract.lean`. This formalises the geometric half of the seam realisation *given* the four marks and the clock; the raw-seam marking obligation `QGEO.MARKS.01` stays [O], *not* closed.

2026-06-15 (Lean: the QGEO.SYM.01 conditional theorem machine-proved)

- **The mark-local $\Rightarrow \omega\circ\rho = \omega$ implication is now Lean-formalised** (`SeamDeckClosure.lean`, ledger `FORM.QGEO.01`, [O]). A new module in the carrier-rigidity Lean project machine-proves the algebraic core of `v201/v210` (audit: pass, no `sorry/admit`, no domain axioms — `#print axioms = propext, Classical.choice, Quot.sound` only): (i) the 4th-root character orthogonality $\sum_{j<4}\zeta^j$ = (if ζ=1 then 4 else 0) for ζ^4=1 (so a μ_4-mark sum is supported only on modes $\equiv 0$ mod 4); (ii) the clock generator $-i$ has order 4; (iii) `markLocal_blockDiagonal`:

a mark-local Toeplitz symbol connects only equal clock-characters, $[\rho, M_f] = 0$; (iv) the premise is a typed `structure SeamDeckPremise` (the physical seam DtN *is* mark-local) — consumed, *not* proved, exactly as `CalderonProjector` encodes the Paper-1 analytic input; (v) `SeamDeckPremise.clock_invariant`: *given* the premise, the clock commutes with the DtN, whence $\omega \circ \rho = \omega$. So the *implication* is now [E] (formally proved); the *premise* (the physical seam being mark-local) stays **[O]** — the one fundamental seam-identification postulate, *not* closed.

2026-06-15 (QGEO sharpening: mark-local DtN certified on realistic profiles)

- **Mark-local DtN $\Rightarrow \omega \circ \rho = \omega$, certified on realistic Steklov profiles and at the state level (`v210_mark_local_dtn`, `tfpt_research_contracts`, [O]).** A numerical sharpening of `QGEO.SUBPRIN.01/v201`: a real von-Mises curvature bump summed over the four μ_4 marks, $f(\theta) = \sum_{j=0}^{3} g(\theta - j\pi/2)$, builds the Steklov DtN $\Lambda = |D_\theta| + M_f$ whose off-(mod 4) Fourier support vanishes to $< 10^{-16}$, $\|[\rho, \Lambda]\| < 10^{-15}$, and — the genuinely new step — the quasi-free covariance $C = \frac{1}{2}(1 + \operatorname{sgn} H_1)$ is clock-invariant ($\rho C \rho^\dagger = C$ to $< 10^{-13}$), i.e. the actual $\omega \circ \rho = \omega$. Negative controls (single off-centre bump, $\mathbb{Z}_3$ source, four generic marks) break both by $O(1)$; convergence stable for $N \in \{16, 32, 64\}$. This certifies the *implication*; the premise that the physical seam *is* mark-local stays **[O]** (it does *not* close `QGEO.SYM.01`; net existence and full-cone RP remain the [E] of `v175`). Ledger `QGEO.MARKLOCAL.01`. Python-only (the exact mark-sum identity is Wolfram-mirrored via `v201`).

2026-06-15 (archive integration #5: six dormant results revived from the old papers)

- **Muon $g-2$ as a seam vertex (`v204_muon_seam_g2`, `tfpt_4_frontier`, [C]).** The carrier's second-order topological defect $\delta_2 = \frac{5}{4}\delta_{\text{top}}^2$ projected through the seam-loop phase gives $a_\mu^{\text{seam}} = \delta_2/(2\pi) = 45/(524288\,\pi^9) \approx 2.879 \times 10^{-9}$ (0.81σ vs the dispersive Δa_μ; $\sim 1.5\sigma$ vs lattice/CMD-3). The value is exact [E]; the vertex identification is the [C] bridge. Ledger `FR.MUONG2.01`; Wolfram-mirrored.

- **An independent gravitational $3/4$ (`v205_xi_threequarter`, `tfpt_4_frontier`, [E]/[C]).** $\xi = c_3/\varphi_{\text{tree}} = 3/4$ exactly (Einstein limit of $\kappa^2 = \xi\varphi_0^{\text{ret}}/c_3^2$) — the same $3/4 = q(A_3) = \ln(m/\mu)$ as the seam-determinant replica (`v152`), an over-determination. Ledger `GRAV.XI.01`; Wolfram-mirrored.

- **Quantitative H_0 from the Λ branch (`v206_h0_lambda_branch`, `tfpt_4_frontier`, [C]).** $\delta_\Sigma = 48\,e^{-2\pi\alpha^{-1}/3} \approx 1.085 \times 10^{-123}$, the $(8\pi)^2$ Planck-convention split, and $H_0 = 66.5$–67.1 km/s/Mpc (below Planck, far below SH0ES — the Hubble tension is *not* relieved). Ledger `GRAV.H0.01`.

- **Asymptotic Safety as a second QG route (`v207_asymptotic_safety`, `tfpt_4_frontier`, [O]).** A complementary continuum-FRG argument (a non-Gaussian UV fixed point from the same axioms; $y^2 = 16c_3^2 = 1/(4\pi^2)$) — a plausibility cross-check, *not* load-bearing, does *not* close G_{metric}. Ledger `GRAV.ASYMP.01`.

- **Black-hole thermodynamics from the seam** (`v208_bh_thermodynamics`, `tfpt_horizon_readouts`, [E]/[C]). Modular $T_H = \kappa/(2\pi)$ (the $2\pi = 1/(4c_3)$ seam unit, ties to `v198`/`v199`) and the scalaron-corrected Wald entropy $S_W = (A/4G)(1 + R_h/3M_s^2)$ (exact from `v28`). Ledger `HOR.BHTHERMO.01`; Wolfram-mirrored.

- **The black hole as a seam fixed point** (`v209_bh_regular_core`, `tfpt_horizon_readouts`, [C]). No singularity ($\varphi \to \varphi_\star$ at the gapped attractor $(2/3)^6$); the maximal BH = the anchor (Nariai roots $(1, 1, -2)$); information via Page recovery; a Planck-scale holographic floor. The superseded RN/torsion metric is *not* resurrected. Ledger `HOR.BHCORE.01`.

2026-06-15 (archive integration #4: Möbius $\cong$ Lorentz foundation citation)

- **The Möbius seam is the boundary Lorentz structure (`tfpt_3`, [O]).** Added a paragraph to the "Möbius origin of π" section: $\mathrm{Möb}(\hat{\mathbb{C}}) \cong \mathrm{PSL}(2,\mathbb{C}) \cong \mathrm{SO}^+(3,1)$ (Penrose & Rindler, *Spinors and Space-time*, Vol. 1, §1.2–1.4), the celestial sphere of null directions via stereographic projection. This gives the self-consistency loop a spacetime meaning — the carrier clock $z \mapsto iz$ is an elliptic $\mathrm{PSL}(2,\mathbb{C})$ rotation (v180) and the RP→OS causal cone is the same Lorentz structure — so "Möbius origin of π" and "one Lorentzian cone" are two faces of one boundary $\mathrm{PSL}(2,\mathbb{C})$. A foundation citation, not a new claim (π stays primitive).

2026-06-15 (archive integration #3: axion coupling $g_{a\gamma\gamma} = -4c_3$ + retired small-angle reading)

- **Haloscope coupling tied to c_3 (`tfpt_4_frontier`, [C]).** Made explicit: in the determinant-line normalization the axion–photon anomaly coefficient is $g_{a\gamma\gamma} = -4c_3 = -1/(2\pi)$ ($y^2 = 16c_3^2 = 1/(4\pi^2) \approx 0.0253$), so the haloscope $F\tilde{F}$ coupling is set by the *same* c_3 that fixes α and $\beta_{\mathrm{rad}} = \varphi_0/4\pi$ — a [C] structural relation (the coefficient, not a parameter-free physical $g_{a\gamma\gamma}$). The earlier small-angle $\theta_i = \varphi_0$ reading (old determinant-line note, a much higher axion mass) is *superseded* by the closed near-hilltop $\theta_i = \pi(1 - \varphi_{\mathrm{seam}}) = 170.4°$, recorded as a retired reading and guarded by v188 (a new sentinel forbids that stale axion mass from re-entering the active docs).

2026-06-15 (archive integration #2: the EHT achromatic polarization intercept — v203)

- **`HOR.EHT.01` (v203, [E]/[C]/[X]).** The surviving core of the old UFE/black-hole notes (*not* the RN-type metric — superseded by the Nariai/seam=horizon readouts v101–v104/v190; only the polarization signature survives): a local achromatic polarization-rotation intercept around a horizon-scale black hole, $\beta_{\mathrm{BH}}(r) = 16c_3^4\, Q_e^{\mathrm{eff}} Q_m^{\mathrm{eff}}/r^2$. The coupling $16c_3^4 = 1/(256\pi^4) = \delta_{\mathrm{top}}/3$ is *exact* ([E], Wolfram-mirrored) — the *same* top-form coefficient $\delta_{\mathrm{top}} = 48c_3^4$ that fixes the α-kernel precision-zone correction, one row from the cosmic-birefringence seed $\beta_{\mathrm{rad}} = \varphi_0/4\pi$. TFPT fixes the $1/r^2$ shape, achromaticity and sign-flip ([C]); the amplitude $Q_e Q_m$ is an MHD/GR weight, never a pixel prediction. **Dated kill test** [X]: the residual intercept must pass three nulls (frequency, spatial $1/r^2$, sign-flip) after honest GRMHD subtraction. Integrated into `tfpt_horizon_readouts` (new EHT section), the ledger, registry, Wolfram (243/243), and the website (`predictions.ts`). Suite 202 modules.

2026-06-15 (archive integration #1: the rare kaon sector $K \to \pi\nu\bar{\nu}$ — v202)

- **`FR.RAREKAON.01` (v202, [C]).** The closed TFPT CKM point ($s_{12}{=}\lambda_C$, $s_{23}{=}\varphi_0/(1{+}\lambda_C){=}|V_{cb}|$, $s_{13}{=}\lambda_C^3/3$, $\delta{=}\pi/3{+}3\lambda_C^2{=}1.19823$ rad; v18/v88, no flavor fit) feeds the cleanest FCNC probe. With standard external short-distance input (Brod–Gorbahn–Stamou $X_t, P_c, \kappa_\pm$) it gives $\mathrm{BR}(K^+ \to \pi^+\nu\bar{\nu}) = 9.45 \times 10^{-11}$ and $\mathrm{BR}(K_L \to \pi^0\nu\bar{\nu}) = 3.33 \times 10^{-11}$ — a [C] downstream flavor readout, *not* a compiler power (the SD functions are external). Recomputed from the *current* CKM point (an earlier archive scaffold quoted $\mathrm{BR}(K_L) = 3.47 \times 10^{-11}$ from an $\Im(\lambda_t)/\lambda^5$ slip, corrected here). Grossman–Nir ratio $0.352 \ll 4.3$; NA62 $(13.0^{+3.3}_{-3.0}) \times 10^{-11}$ is $\sim 1.2\sigma$ above. **Dated kill test** [X]: a stable NA62 $\mathrm{BR}(K^+)$ outside $[7, 12] \times 10^{-11}$, or a KOTO-II $\mathrm{BR}(K_L)$ off the GN point, breaks the flavor bridge (core untouched). Integrated into `tfpt_2_standard_model` (§rare kaon), the ledger, v187 guard (new `FR.RAREKAON.` family), registry, and the website (`predictions.ts`). Python-only. Suite 201 modules.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (v201, [E]/[O]).** The v199 bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([E]). So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* v185's optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\mathrm{seam}})$ only.

- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5× above $\Omega_{\mathrm{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements" — a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.

- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\mathrm{fail}}(\theta)$ has its unique minimum at

$\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i$ = the $(1,2,3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.

- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID `v200`; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \operatorname{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \operatorname{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4-invariant (v177). So the entire residual collapses to the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **`v195_marks_lefschetz_character.py` [E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\operatorname{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and $\operatorname{rank} H^1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So Marks reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **`v196_seam_energy_functional.py` [E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **`v197_rr_carrier_clifford_d5.py` [E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\text{car}} \to D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.
- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **`v194_raw_seam_dtn_rp.py` [O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \backslash \mu_4$. (3) [O] the residual relocates: the full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **`v193_qgeo_energy_commutator.py` [O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)=S_4$, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1,2,3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **`v189_riemann_roch_carrier.py` [E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\text{car}}, N_{\text{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \backslash \mu_4) = \deg -1 = 3 = N_{\text{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5 = g_{\text{car}}$, matched by $\operatorname{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\text{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **`v190_nariai_entropy_bound.py` [E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\text{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v191_universal_branch_line.py` [C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\text{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\text{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly typed [C]* — an exact *relabeling*, not a theorem (a decoy pair $\{3, 4\}$ admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.

- **`v192_qgeo_energy_clock.py` [O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).
- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **`v184_etaB_anchored_boltzmann.py` [C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R\varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure*. The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (c_3-/φ_0-powers off ∼75%/∼40%), so $M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **`v185_axion_relic_solver.py` [C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces (∼21% at $\theta \sim O(1)$), so the $\theta_i \approx 170°$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **`v187_ftransfer_laws.py` [E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **`v188_frontier_wording_guard.py` [E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording → v184 [C]; the old "open relic scale" dark-matter line → v185; the old "near, not exact" Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest

verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).

- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **`v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\rm pole}/u_{\rm source} = a^T(R{+}Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78{+}8{+}2{\cdot}27{\cdot}3$) and $a^T(R{+}Q)\mathbf{1} = 53 = 52{+}1$, where $F = R{+}Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from "near miss + conjecture" to "operator-sourced source→pole transfer": the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an $F_{\rm transfer}$ special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, exponent ratio $|\mathbb{Z}_2|$, "one clock, two geometries") is exactly `HOR.TRISECT.01`/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **`v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{$`QGEO.SYM.01`$, F_{\rm transfer}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B (≡ `QGEO.SYM.01`), D (Möbius forced by $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$ on the deck; the missing generator = $F_{\rm transfer}$). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a "Rosetta" translation lexicon** (website `RosettaLexicon`): TFPT's compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, $F_{\rm transfer}$, $v_{\rm geo}$, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \le 10^{-30.7}$) are now a prominent strip directly under the hero.
- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what

is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- G_{net} **typed in three explicit levels** (website "What is open" card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1,1)\rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- **`QGEO.SYM.01` now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\mathrm{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".
- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ∼181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.

- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\text{car}} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01`/v23, `ARCH.CORE.01`/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01`/`AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\text{car}} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE`(v175)$\to \cdots \to$`SYM.01`(v181) ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, G_{net}), the `glossary` G_{net} entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **`v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry$\Rightarrow$ conformal$\Rightarrow$Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **`v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\, 2\pi\, \chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). KERNEL then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\mathrm{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **`v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\mathrm{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the v176 statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET and closes *four* nodes *constructively* (all validated symbolically before adoption): UNIFORM [E] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening v168's cross-ratio to the full uniformisation); COHOM [E] — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; MODULE [E] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); NET [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c=8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4-equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1} v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54}u$ source→pole transfer is already typed [C] (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **`v175_net_existence_full_cone.py` [E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of spec(t), so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every m* (not only $m \leq 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$

character $E_4/\eta^8 = 1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120+128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **`v174_seam_fock_readings.py` [E] (claim `QFT.FOCK.01`).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL ($\deg \le 2$) keeps $\sum_{k\le 2}\binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5+3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **`v170_diamond_slice_compression.py` [E] (claim `E8.SLICE.01`).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3,4,6,10)$ ($p_n = 2+2^n$) and Sheet-Diamond determinants $(3,4,8,14,20,32)$ summing to $81 = N^4_{\text{fam}}$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7,11,13)+s(3,3,3)+t(1,2,3)$ follow; $78 = \dim E_6 = p_2(p_0+p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).
- **`v171_os_moment_cluster.py` [E]/[C] (claim `QFT.OSMOMENT.01`).** Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster 729/665, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1+h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6\log\frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).
- **`v172_trace_anomaly_seed.py` [E] (claim `SEED.ANOMALY.01`).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 =$ scalaron exponent $48-41$ (confinement ↔ inflation), and the Volkov–Wipf $-98/45 = -2\cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).
- **`v173_pfaffian_cp_car.py` [E] (claim `FLAV.STRONGCP.02`).** Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D,\Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).
- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical

reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **`v168_qgeo_rigidity.py` [E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\mathrm{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$ $(k = 1, 2, 3)$ are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = \mathrm{A}_3$ exponents $= \mathrm{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \backslash \mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, [E]). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).

- **`v169_etaB_boltzmann_interface.py` [C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2}\varepsilon_1\kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **`v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\mathrm{seam}}) = 0$ and $\beta_\lambda(M_{\mathrm{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\mathrm{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\mathrm{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\mathrm{Pl}})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim 129$–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.

- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **`v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2}Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\text{scal}}/M_Z) \sim 1.8 \times 10^{-5}$ and $|\Delta\sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **`v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "[O] — not forced" frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}{+}1 = 41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_transfer + irreducible core, v167)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P}\setminus\mu_4$, cohomology $b_1{=}N_{\text{fam}}{=}3$); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer}, one functor with four typed conditional interfaces (Koide, η_B, axion,

m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).

- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)$=16/16=1, 248=120+128, character E_4/η^8=$j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (v156/v157). R_1 **(seam clock) reduces to `GATE.QGEO`** [C]: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification "seam boundary = $\mathbb{P}\setminus\mu_4$" = `GATE.QGEO` (cohomology established, v137, b_1=3=N_{fam}). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\mathrm{geo}}\}$; premise (A) = ($A_2$ assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\mathrm{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module `v162_seam_transport_identification.py` [E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content.* The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4|\,\mathrm{diag}\,X$ for U=$\mathrm{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star)$=$\{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6\log\frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of ω_k=$|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, v161)

- **New module `v161_one_particle_reduction.py` [E]/[O].** *Discharges* the scope premise of v160 ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is T_Σ=$\Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\mathrm{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all m=0..6); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation ⇔ one vacuum); (4) quasi-freeness is *forced* — the 120= $\dim \mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160's three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 — verified by the *signed* set-partition/Pfaffian moment↔cumulant inversion on a concrete pure Majorana covariance (the v113 "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray ⇒ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56's gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection` × `{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral c=8 fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (v157, not (1,1)-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).

- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1, 1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}). With $(E_8)_1$ unique (v83), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).

- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN = $|k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).

- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **G_{net}'s last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net}, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **$v_{\rm geo}$ re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. $v_{\rm geo}$ is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\rm point} \sim v_{\rm geo}$, $1/G \sim v_{\rm geo}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay {one unit, π} (ledger `ANCHOR.VGEO.02`).
- **$G_{\rm net}$ stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads $v_{\rm geo}$ as a metrology *primitive*, $G_{\rm net}$ as the simple-current extension $\cong (E_8)_1$ ([E]/[C]), and $F_{\rm transfer}$ as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\rm geo}\}$; one seam-net theorem; frontier = $F_{\rm transfer}$) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts ↔ papers ↔ ledger ↔ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in d=2 that ratio *is* the induced Newton constant $1/G$ (v68). So $v_{\rm geo}$ (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay {one scale, π}, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).

- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory`/`next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo`/`zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI `10.5281/zenodo.20579275`), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts ("Why the interpretation is internally consistent", "Dependency chain of the birefringence test", "The precise reformulation", "The three theses honestly graded", "The whole story", "One-line summary") become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a handwritten, *stale* copy of the document-set card ("five short documents", missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box "clock" wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon "Scope/typing" marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.

- **Build hardening.** `build.sh` now removes a document's `.aux`/`.toc`/`.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the "Calderón version" demanded by v150 needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory`/`next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4′: both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6,3,5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2,8,121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on `SEAM.THEOREM.01`.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log\det' = 2C(\gamma)\ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi)\int\sqrt{g}R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4′: the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4|e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N^2_{\text{fam}}$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\text{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z\mapsto iz,\ z\mapsto 1/z\rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity $+1$ on the self-conjugate line ($= T_A$ in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (v147).** Variance ratio $(1-\alpha)$ (forced by norm = area) gives the Born-squared ratio $(1-\alpha)^{p_2}$, the ln series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).

- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\mathrm{NS}}+128_{\mathrm{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum q_D+q_A (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.

- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\mathrm{Pl}}$ vs M_{Pl} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.

- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (v86); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes **[E]**/[**C**]/[**O**] instead of the legacy [*A*]/[*I*]/[*L*]/[*B*]/[*P*]/[*R*] badges.

- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + c=8 ⇔ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.

- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never "Paper 5–7"); the marker system on every website surface migrated to [*E*]/[*C*]/[*O*]/[*X*] (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).

- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, "one seam clock" framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (v97/v98) pairs the Q_+=1 cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \text{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. `GATE.QGEO` keeps only its realisation premise — no discrete freedom left (ledger `FLAV.QGEO.06`).
- **R4′ narrowed: three pairings → two (v142, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x-3y+z \equiv 0$ mod 11; the index *is* $\|\text{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \text{rank}\, E_8)$ the σ-pairing is forced $\equiv 0$ mod 11, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1's det-ratio step derived in-family (v144).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays [C] with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (`SEAM.THEOREM.01`); the gate keeps its [O] typing untouched.
- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the `vN` scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri`/`\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the `vN` scripts,

and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.

- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves. . . ", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the v86 scalaron-reheating point ($N_\star$=51.4 ⇒ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly,

matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` → [E], `[N]/[P]` → [E]/[C]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with v86/`COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [E] → [C](conditional on M_{Star}=M_{scal} + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and `v5.4 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion

documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.

- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to `tex-artefacts/`.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh notes` deliverable. The `make_manifest.py TEX` list and the workflow rule were updated to the new paths.
- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long `vN`→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven `keybox`es are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible*/*follows next* `winbox`es become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winbox`es, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction's document map gained the missing `tfpt_5_redteam` row; "seven/eight companion documents" and the ledger "six documents" wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types [E] (exact algebra), [E] (exact number), [C] (readout / standard physics in TFPT units) to the existing [C] (bridge) and [X] (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain [E] where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are [E] closed on the derived stratum, only the absolute scale stays [O], with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ (historical $U_{\mathrm{wall}}/G_{\mathrm{metric}}/F_{\mathrm{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\mathrm{geo}} \to G_{\mathrm{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4′ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights (spec $A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9}$ = the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).

- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law → zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\text{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1,1,-2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8, 14, 20)$, $\det F = 32$, row-budget axes with $\operatorname{rowsum}(V) = \operatorname{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6, 8, 7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2, -9, 5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1, 2, 3) = \operatorname{Spec}(Q_+)$ = the A_3 exponents and cusp weights = $\operatorname{spec} A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2, 8, 121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5, -9, 6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1, 1, -2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\text{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).

- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank = c) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- U_{wall} **made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad ($\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$); the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.